THIRD EDITION

CHEMICAL ANALYSIS

MODERN INSTRUMENTATION METHODS AND TECHNIQUES

FRANCIS ROUESSAC
ANNICK ROUESSAC

TRANSLATOR:
JOHN TOWEY

with website

WILEY

Chemical Analysis

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Modern Instrumentation Methods and Techniques

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Foreword

This book is intended to provide basic knowledge regarding the most common methods encountered in qualitative, quantitative, and structural chemical analysis. These methods are used in fields as varied as the chemical, pharmaceutical, and food industries, as well as environmental issues and various other regulations.

The methods reviewed in this book are classified as *separation methods*, *spectral methods* or *other methods*. Each of these is examined, firstly with a focus on basic concepts and then with a look at the main corresponding instrumental techniques. The book is illustrated with explanatory diagrams, drawings, and photographs, many of which are inspired from real instruments and documents obtained from manufacturers. To keep this book to a reasonable size, methods that are rarely used or have fallen into disuse are not discussed.

Clearly written, this text is addressed to a wide range of students in technical colleges (chemistry, physical measurements, applied biology, etc.) or advanced technician courses or even to students in Bachelor's and Master's programmes who wish to complement or review basic knowledge initially learned in a more fragmented manner. This book should also be useful for continuing education students and for technicians working in industry who are faced with problems of chemical analysis or who want to prepare for competitive examinations. The need for chemical analysis in many industries that had not previously used such methods, combined with the growing array of available techniques and instruments, is a further reason for many people to learn or refresh their knowledge about these techniques.

The knowledge required to approach this book corresponds to that of first-year students at undergraduate level. Hence, the authors have limited themselves to reviewing fundamental principles and taken into consideration the varying levels of students' knowledge about physical phenomena and mathematics. The text includes some theoretical reviews about the phenomena in question, in order not to lose some of the intended readers. Any readers who want more in-depth treatment of a subject may then read more specialized works, after having acquired a solid overview of the current methods and their practical aspects from this book.

It may seem like a challenge to present over 20 methods, with exercises (and their answers), in about 500 pages. That is why the authors have chosen to limit themselves to the presentation of the tools, rather than describing everything those tools can do. Only the methods have a universal character. Applications have thus been selected for purposes of illustration only. The final chapter of this book deals with basic statistical considerations.

This book originates from the lectures and laboratory work offered to IUT (technical college) students in Le Mans. This new edition has been updated and enhanced with respect to the previous ones. The third and fifth editions of this publication were translated into English under the title *Chemical Analysis, Modern Instrumentation Methods and Techniques*, John Wiley & Sons Ltd. (Chichester, 2002 and 2007). The fifth edition was translated into Spanish. Its title is *Anàlisis Quimico. Methodos y Técnicas Instrumentales Modernas*, McGraw-Hill Interamericana de Espana, S.A.U (2003). The sixth edition was translated into Korean (Dunod, 2009).

We would like to thank Daniel Cruché, Professor Emeritus, who collaborated in the proofreading of certain chapters and helped in writing some sections in this new edition. We would also like to thank all the French and foreign companies we contacted, who always kindly answered our requests to provide practical information. Their help was precious, since the field of analytical instrumentation is constantly integrating technological advances from a great variety of areas.

Lastly, we also thank our editorial team at Dunod Editions, with whom it was a pleasure to work, and more specifically, Laetitia Hérin, who launched this new edition, and Johan Dillar for his work on the production of this book.

The authors also express their gratitude to the late Professor Guy Ourisson, former President of the French Academy of Sciences, who followed the evolution of this publication throughout its consecutive editions beginning in 1992. He also gave us the great honour of writing the preface to our earliest editions. Finally, we would like to express our deepest gratitude to the late Professor Férey, member of the French Academy of Sciences and CNRS gold medal winner, for his kind preface to the seventh edition in which he announced the forthcoming arrival of the eighth edition!

Le Mans, February 2019 F. Rouessac & A. Rouessac

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About the companion website

This book is accompanied by a companion website.

www.wiley.com/go/Rouessac/Analysis3e

This website includes:

- Figures of the book

Introduction

Analytical chemistry is a close cousin to physical chemistry. It relates to the study of the chemical and physical behaviour of pure compounds or compounds in solution subject to various conditions.

It is often viewed solely in its applied aspect, with the purpose of identifying, characterizing, and quantifying chemical substances as well as developing the methods necessary for this analysis. This reductive aspect of analytical chemistry is none other than *chemical analysis*, which is the subject of this book.

Any study of chemical analysis involves addressing varied fields of knowledge. To reach the desired goal, it calls on many concepts, including some that range very far away from chemistry, in the usual sense of the word. It is a multidisciplinary science whose repercussions are felt in all experimental sciences.

In chemical analysis, it is common to distinguish between two categories of methods. Firstly, there are chemical methods, based on a specific reaction of the *analyte* (the compound being assayed) with reagents, and then there are physical techniques, which make use of the physico-chemical properties of analytes.

This second category of methods, now in the forefront, has replaced the traditional 'wet' methods, which are very limited in their performance but which were at the origin of the field of *analytical chemistry*. Most of this book is therefore dedicated to modern techniques, which have been greatly developed over these past few decades with the rise of miniaturized sensors, digitization, and computing power. These improvements have thus enabled the use of very efficient mathematical tools and less bulky equipment. As an example, we can compare the imposing Raman spectrophotometers from the 1960s with the portable Raman spectrometers that are now used for quality control in numerous fields.

And so was born instrumental analysis with its incredible arsenal of processes and devices which we now often find installed outside of traditional analytical laboratories.

The evolution of technologies has led to the creation of very efficient instruments that open up new possibilities. This is especially true with the introduction of coupled methods, nondestructive testing, and the concept of *speciation* (distinction between the various structures in which an element may be present in the sample). These methods use small samples that require little or no preparation prior to measurement. Current analytical trends also include an increase in miniaturized, portable instruments, as well as specific or automatic analysers.

In general, the basic principle consists in measuring a physical value, followed by establishing a simple relationship between this measurement – the device's

response – and the composition of the analyte in the test sample. The search for this relationship often consists in a mathematical linearization between the device's response and the analyte concentration. This relationship may be simple and direct (the response is proportional to concentration), or it may be more complex.

Whatever the tool used to establish the relationship between the signal and the concentration, we also have to determine the limits of quantification. Below that limit, any attempt at quantification is bound to fail, while above it the relationship no longer applies.

To conduct these studies, analysts must not only be trained in the various techniques, but they must also know the basic concepts of chemistry, particularly since a compound can often be assayed by various methods. Choosing a good method, and if possible the best method, requires knowing many parameters. Therefore, when a new analytical objective has been defined, the problem must be addressed methodically. We must first *choose the right method*: spectroscopic, electrochemical, separation, etc. Then the *choice of technique* is made, followed by the *choice of process* in terms of sampling and prior treatment of the sample. Finally, a *protocol*, the recipe so to speak, is established. In general, this is a process governed by established national or international standards, which leads to the chosen operating procedure. This approach focuses on standardizing all the steps: from sample preparation all the way to conducting measurements. Finally, results must be established according to existing standards. Additionally, the raw analytical data conserved as computer files cannot be changed. This entire approach is the subject of official texts known as Good Laboratory Practice (GLP).

To recommend the best method in order to resolve an analytical problem, there is a science called *chemometrics*. Its purpose is to help the analyst, as a function of specific requirements and aims: minimum sampling plan, appropriate methodology, data processing and interpretation of results. Thanks to the use of IT tools, it seeks to provide a correct answer by exploiting the results with statistical methods in order to reduce the number of tests for long or costly analyses.

In this way, users may acquire devices that meet the standard of precision and quality necessary to acquire certification or to have a laboratory officially recognized for the quality of its results. These accreditation procedures are now imposed by many control organizations around the world.

Analytical chemistry is therefore essential in many fields other than the traditional ones of chemistry or chemical engineering. It can be found in fields as varied as medicine, food, biochemistry, the environment (pollution), safety (explosives, drugs and chemicals), and art. Now more widely used in human activities than ever, analytical chemistry can be of benefit to us all.

Chapter 1

General Aspects of Chromatography

INTRODUCTION

Chromatography, the process by which the components of a mixture can be separated, has become one of the primary analytical methods for the identification and quantification of compounds. The basic principle is founded on the concentration equilibrium of the components of interest between two immiscible phases. One is called the stationary phase, because it is immobilized within a column or fixed upon a support, while the second, called the mobile phase, is forced through the first. The phases are chosen such that the components of the sample have differing solubilities in each phase. The differential migration of compounds leads to their separation. This hydrodynamic process, which has been constantly evolving since its discovery, is an analytical method that no laboratory involved in molecular analysis can ignore, as its applications are so numerous.

Objectives

Review the principle of chromatography
Distinguish between separation and analysis by chromatography
Explain the protocol of a chromatographic analysis
Use a chromatogram and model its signals
Describe the various retention parameters
Discuss the hydrodynamic aspects of a separation
Define parameters influencing the efficiency of a separation
Classify the various chromatography techniques
Describe the steps of an assay using chromatography

1.1 GENERAL CONCEPTS OF ANALYTICAL CHROMATOGRAPHY

Chromatography is a physico-chemical method of separation of components within mixtures, liquid or gaseous, in the same vein as distillation, crystallization or fractionated extraction. The applications of this process are therefore potentially numerous, since many heterogeneous mixtures, or those in solid form, can be dissolved by a suitable solvent (which becomes, of course, a supplementary component of the mixture).

A basic chromatographic process may be described as (Figure 1.1):

1. A vertical, hollow glass tube (the *column*) is filled with a suitable finely powdered solid, the *stationary phase*.
2. At the top of this column is placed a small volume of the *sample* to be separated into individual components.
3. The sample is then forced through the column from inlet to outlet by continuous addition of the *mobile phase*, carrying the various constituents of the sample along with it. If the components migrate at different velocities, they will become separated from each other and can be recovered, each in solution with the mobile phase.

While this use of chromatography has continued since its origins, this process became a method of analysis with the idea of measuring the retention time of compounds through the column in order to identify them. To do so, it became essential to control certain parameters (flow rate, temperature, etc.) and a detector had to be placed at the column's outlet to identify compositional changes in the mobile phase. This form of chromatography, whose goal is not simply to recover the components but to measure their retention time, has developed slowly.

The identification of a compound by chromatography is achieved by comparison. To identify a compound, which may be either A or B, using chromatography, we compare

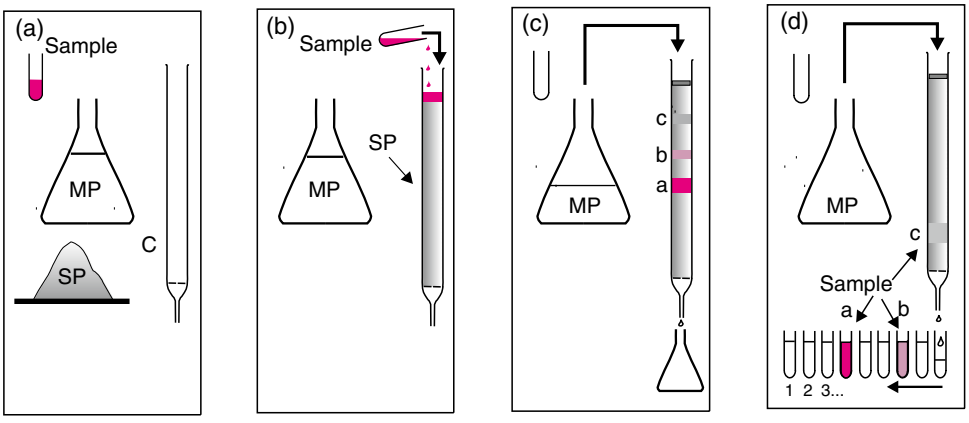


Figure 1.1 A basic experiment in chromatography. (a) The necessary ingredients (C, column; SP, stationary phase; MP, mobile phase and sample); (b) introduction of the sample; (c) start of elution; (d) recovery of the products following separation.

its *retention time* with those for the two reference compounds A and B previously recorded using the same apparatus and the same experimental conditions.

In this experiment, there was no true separation (A and B were pure products), only a comparison of the products' retention times. However, this method does have three weaknesses: the procedure is fairly slow; absolute identification is unattainable; and the physical contact between the sample and the stationary phase could modify the sample's properties, in particular the retention times.

This specific method of separation, in its modern form, was first undertaken at the beginning of the twentieth century by the botanist Mikhail Tswett (or Tsvet), who is credited with inventing the terms *chromatography* and *chromatogram*.

The technique has improved considerably since its beginnings. Nowadays, chromatographs are piloted by software programs that run highly efficient miniature columns able to separate nano-quantities of sample. These instruments comprise a complete range of accessories designed to ensure repeatability of successive experiments by the perfect control of the different parameters of separation. Thus it is possible to obtain, during successive analyses of the same sample conducted several hours apart, recordings that are reproducible to within a second (Figure 1.2).

The specific recording that is obtained for each separation is called a *chromatogram*. It corresponds to a two-dimensional diagram that reveals the variations of composition of the eluting mobile phase as it exits the column. To obtain this read-out, a sensor, or *detector*, of which there exists a great variety, needs to be placed at the outlet of the column.

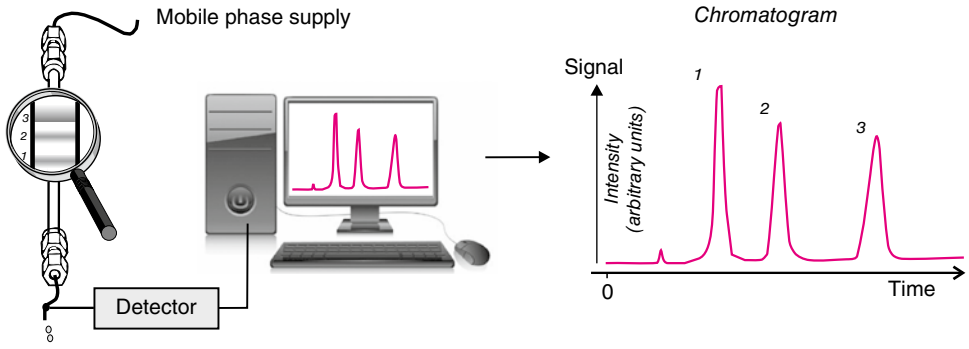


Figure 1.2 The principle of analysis by chromatography. The chromatogram, the essential graph of every chromatographic analysis, is obtained from variations, as a function of time, of an electrical signal emitted by the detector. It is either produced in real time or reconstructed at a later time from values that have been digitized and stored. The chromatography software recalculates these values and puts them in the desired format. This chromatogram illustrates the separation of a mixture of three principal components. Note that the order of appearance of the compounds corresponds to the relative position of each constituent on the column.

The identification of a molecular compound from the chromatogram can sometimes be risky. A better method consists in associating two different complementary methods, for example, a chromatograph coupled with a second instrument, such as a mass spectrometer or an infrared spectrometer. These *coupled* (or *two-dimensional)* techniques provide two independent types of information (retention times and the spectrum). Therefore, it is possible to determine without ambiguity the composition of complex mixtures or the concentration of certain compounds on the nanogram level (confirmation analyses).

1.2 THE CHROMATOGRAM

The *chromatogram* is a curve representing the variation over time of a parameter related to the concentration or quantity of the solute at the column outlet (Figure 1.3). Time (or very rarely the *elution volume*) is found on the horizontal axis, where the time origin coincides with the introduction of the sample in the injection system. The detector response is found on the vertical axis. The *baseline* corresponds to the detector response in the absence of any solute. The separation is complete between two compounds when the chromatogram shows two *chromatographic peaks* that start from and return to the baseline.

A component is characterized by its *retention time* t_R, which represents the time elapsed between sample introduction and the detection of its peak maximum on the

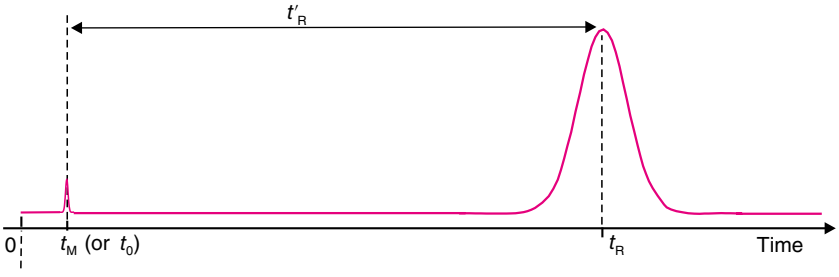


Figure 1.3 Chromatographic elution curve. Example of a graph of Eq. (1.1).

chromatogram. In an ideal case, t_R is independent of the quantity injected. The longer the retention time, the wider the peak is.

A component that is not retained will elute out of the column at time t_M, called the *hold-up time* or *dead time*[1] (also designated t_0). The difference between the retention time and the hold-up is referred to as the adjusted retention time of the compound t'_R.

In quantitative analysis, we often simply separate the mixture from the compound(s) to be assayed. If the signal sent by the sensor varies linearly with the concentration of a compound, then the same variation will occur for the area under the corresponding peak on the chromatogram.

1.3 GAUSSIAN PEAKS AND REAL PEAKS

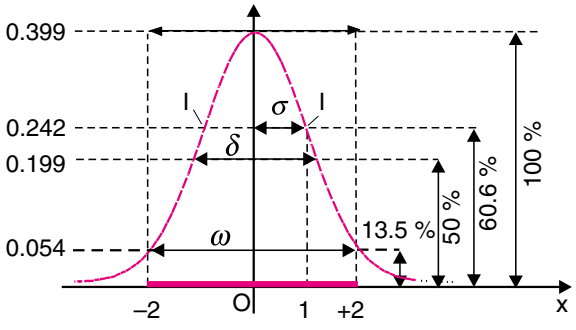
On a chromatogram, the ideal elution peak would have the same form as the graphical representation of the normal distribution of random errors (Gaussian curve). In keeping with the classic notation, μ corresponds to the retention time of the eluting peak and σ to the standard deviation of the peak (σ^2 represents the *variance*). y represents the signal as a function of time x from the detector located at the outlet of the column (Figure 1.4).

This is why ideal elution peak signals of a compound are usually described by the probability density function (Eq. (1.2)).

Equation (1.1) is a mathematical relationship describing a Gaussian function, whatever the x variable. In this expression, σ represents the width unit to describe the peak and μ corresponds to the horizontal axis of the Gaussian curve (in this case, retention time t_R). If we make the peak symmetry axis correspond with the new time origin (μ or $t_R = 0$), we obtain Eq. (1.2)).

$$y = \frac{1}{\sigma\sqrt{2\pi}} \cdot \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \qquad (1.1)$$

[1] The symbols used follow IUPAC recommendations – ***Pure and Applied Chemistry***, 65(4), 819 (1993).



$\delta = 2.35\ \sigma$
$\omega = 4\ \sigma$
$\omega = 1.7\ \delta$

The area between –2 and +2 accounts for 95.4% of the total area under the curve and over the x-axis

Figure 1.4 Characteristics of an ideal chromatographic peak. Meaning of the three classic parameters and summary of characteristics of a Gaussian curve.

$$y = \frac{1}{\sqrt{2\pi}} \cdot \exp\left[-\frac{x^2}{2}\right] \tag{1.2}$$

This function is characterized by a symmetrical curve (maximum at $x = 0$, $y = 0.399$) possessing two inflection points at $x = \pm 1$ (Figure 1.4), whose y-value is 0.242 (i.e. 60.6% of the maximum value). The width of the curve at the inflection points is equal to 2σ ($\sigma = 1$).

In chromatography, δ represents the full width at half-maximum (FWHM, $\delta = 2.35\sigma$) and σ^2 the *variance* of the peak. The width of the peak 'at the base' is labelled ω and corresponds to the base of the triangle formed from the tangents to the inflection point *I* of the Gaussian curve. It is measured at 13.5% of the peak height. At this position, for a Gaussian curve, $\omega = 4\sigma$ by definition.

Real chromatographic peaks often deviate significantly from the ideal Gaussian form. There are several reasons for this. In particular, the peak's half-width at the inflection point is not only due to elution in the column but also to injection and detection, which we summarize with the following expression:

$$\sigma^2_{tot} = \sigma^2_{inj} + \sigma^2_{col} + \sigma^2_{\text{det}} \tag{1.3}$$

where σ^2_{tot}, σ^2_{inj}, σ^2_{col}, σ^2_{det} are, respectively, the total variance (as observed experimentally), the variance due to *injection* (injection time, time for the sample to penetrate into the column), the variance due to the *column* (elution), and the variance due to *detection* (dead volume between column outlet and detector, detector response time, etc.).

1.4 PLATE THEORY

For more than half a century, different theories have been and continue to be proposed to model chromatography. The best known are those employing a statistical approach (stochastic theory), the theoretical plate model or a molecular dynamics approach.

To explain the mechanism of migration and separation of compounds on the column, the oldest model, known as Craig's *theoretical plate model*, is a static approach now judged to be obsolete, but which once offered a simple description of the separation of constituents.

Although chromatography is a continuous phenomenon, Craig's model considered that each solute moves progressively along a sequence of distinct static steps. This elementary process is represented by a cycle of adsorption/desorption. The continuity of these steps reproduces the migration of the compounds on the column, in a similar fashion to that achieved by an animated film, giving the illusion of movement through a sequence of fixed images. Each step corresponds to a new state of equilibrium for the *entire* column.

These successive equilibria provide the basis of *plate theory* according to which a column of length L is sliced horizontally into N fictitious, small plate-like discs of the same height H and numbered from 1 to n. For each of them, the concentration of the solute in the mobile phase is in equilibrium with the concentration of this solute in the stationary phase. At each new equilibrium, the solute has progressed through the column by a distance of one disc (or plate), hence the name *theoretical plate theory*.

The *height equivalent to a theoretical plate* (HETP or H) will be given by:

$$H = \frac{L}{N} \tag{1.4}$$

This approach employs the rules of polynomial expansion to calculate, for a given plate, the masses distributed between the two phases present. This theory has a major fault in that it does not take into account the dispersion in the column due to the diffusion of the compounds.

The term *theoretical plate* comes from an early approach to describe chromatography by analogy with the distillation model proposed by Martin and Synge (Nobel laureates in Chemistry, 1952). This term, though widely used in chromatography for historical reasons, has no physical significance, in contrast to its homonym which serves to measure the performance of a distillation column. It may have been better to name it after Tswett, for example!

The total retention time t_R of the solute on the column can be subdivided into two terms: t_M (hold-up time), which includes the times during which the solute is dissolved in the mobile phase and travels at the same speed as this phase, and t_S, the time during which it is attached to the stationary phase. Between two successive transfers from one phase to the other, it is accepted that the concentrations have the time to re-equilibrate.

Chromatography involves at least three sets of equilibria: solute/mobile phase, solute/stationary phase, and mobile phase/stationary phase. In a more recent theory of chromatography, it is no longer claimed that the molecules are immobilized by the stationary phase, but rather that they are simply slowed down when passing in close proximity to it.

Various physical phenomena may reveal an asymmetry in the chromatographic peak. More specifically, there is a concentration irregularity in the substance deposit zone at the column head. Moreover, the speed of the mobile phase is null on the wall and at a maximum in the centre of the column. The observed asymmetry of a peak is represented by two parameters: one named the *peak asymmetry factor* (F_a) (Eq. (1.5)), measured at 10% of the peak height, and the other named *tailing factor* (F_t) (Eq. (1.6)) measured at 5% of peak height (for the meaning of a and b in these equations, see Figure 1.5).

$$F_a = \frac{b}{a} \tag{1.5}$$

$$F_t = \frac{a+b}{2a} \tag{1.6}$$

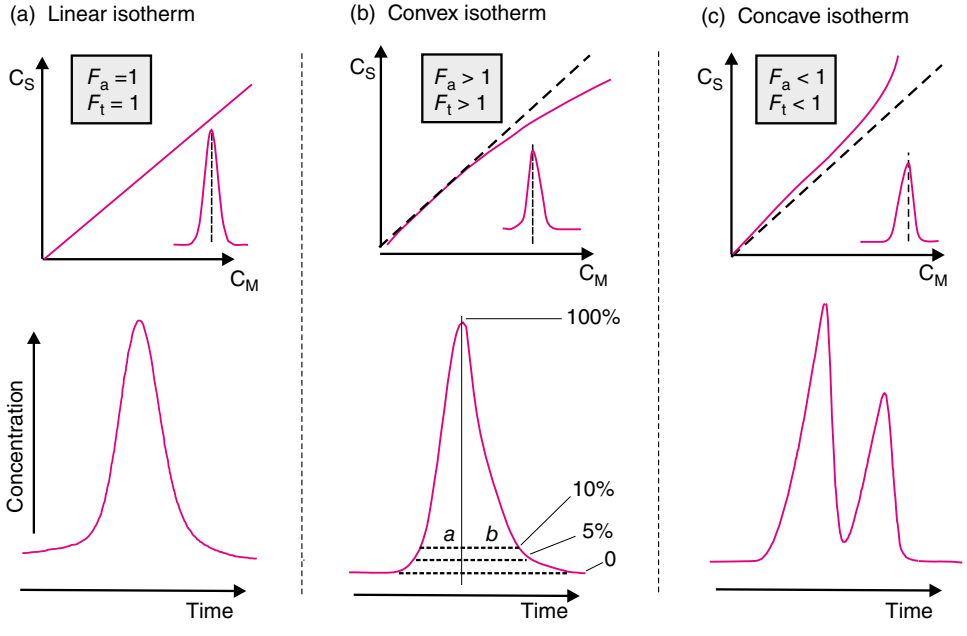


Figure 1.5 Distribution isotherms. (a) The ideal situation corresponding to the invariance of the concentration isotherm. (b) Situation in which the stationary phase is saturated – as a result, the ascent of the peak is faster than the descent. (c) The opposite situation: the component is retained too long by the stationary phase, the retention time is therefore extended, and the ascent of the peak is slower than the descent. For each type of column, the manufacturers indicate its capacity limit, expressed in ng/compound, prior to a potential deformation of the corresponding peak. The three situations are illustrated by authentic chromatograms.

1.5 NERNST PARTITION COEFFICIENT (*K*)

The fundamental physico-chemical parameter of chromatography is the equilibrium constant *K*, termed the *partition coefficient*, quantifying the ratio of the concentrations of each compound in the two phases.

$$K = \frac{C_S}{C_M} \quad \frac{\text{Molar concentration of the solute in the stationary phase}}{\text{Molar concentration of the solute in the mobile phase}} \tag{1.7}$$

Values of *K* are quite variable. The higher they are, the more solute is retained. At least for liquid chromatography, they depend on the intensity of three types of interactions: stationary phase/solute, mobile phase/solute, and mobile phase/stationary phase.

Like any equilibrium constant, it is a function of temperature. It is related to variations in free enthalpy, standard enthalpy, and entropy of the exchange reaction by means of the following relationships:

$$\Delta G^0_T = -RT \ln K_T = \Delta H^0_T - T\,\Delta S^0_T \tag{1.8}$$

The experimental determination of K_T at two different temperatures enables us to calculate these variations, if we assume that the standard enthalpy and entropy variations remain virtually the same between these two temperatures. We can write:

$$lnK_{T_1} = -\frac{\Delta H^0_T}{RT_1} + \frac{\Delta S^0_T}{R} \approx -\frac{a}{T_1} + b \tag{1.9}$$

If we know K_T at two temperatures, Eq. (1.9) helps us calculate the terms *a* and *b*.

$$a = \frac{\Delta H^0_T}{R} \tag{1.10}$$

and

$$b = \frac{\Delta S^0_T}{R} \tag{1.11}$$

In general, the standard enthalpy variation is negative. The same goes for the standard entropy variation, which corresponds to an increase in order when the solute is fixed on the stationary phase.

1.6 COLUMN EFFICIENCY

1.6.1 Theoretical Efficiency (Number of Theoretical Plates)

As the analyte migrates through the column, it occupies a continually expanding zone (Figure 1.6). This linear dispersion σ_l, measured by the variance σ_l^2, increases with the distance of migration. When this distance reaches L, the total column length, the variance will be:

$$\sigma_L^2 = H \cdot L \tag{1.12}$$

In line with the plate theory model of distillation, this approach also leads to the value of the *height equivalent to one theoretical plate H* and to the number *N* of theoretical plates ($N = L/H$).

Therefore, for any chromatogram that shows an elution peak of a compound with the temporal variance σ^2 ($\sigma = \sigma_L / \upsilon$ where υ is the mean elution speed of a solute), we can determine the *theoretical efficiency N* for this compound (Eq. (1.13)) as a function of retention time t_R and, by deduction, we can also get the value of H, knowing that $H = L/N$.

$$N = \frac{L^2}{\sigma_L^2} = \frac{L^2}{\sigma^2 \upsilon^2} = \frac{t_R^2}{\sigma^2} \tag{1.13}$$

These two parameters are indirectly accessible from the elution peak of the compound. We measure t_R and σ, whose ratio is identical to that of L over σ_L (Eq. (1.13)).

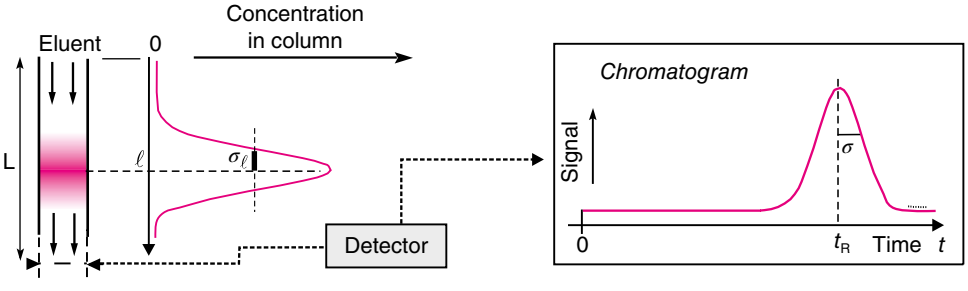


Figure 1.6 Dispersion of a solute in a column. Left, a graph corresponding to the isochronic image of the concentration of an eluted compound at a particular instant. Right, a chromatogram revealing the variation of the concentration at the column outlet, as a function of time. t_R and σ have the same ratio as L and σ_L. The efficiency N can therefore be calculated from the chromatogram by measuring σ directly. On the graph, we find about 100 theoretical plates.

On the chromatogram, σ represents the half-width of the peak at 60.6% of its height and t_R the retention time of the compound. t_R and σ should be measured in the same units (time, distance or eluted volume if the flow rate is constant). If σ is expressed in units of volume (using flow rate), then 4σ corresponds to the volume of the peak, which corresponds to the eluent volume containing 95% of the initially injected compound. By consequence of the properties of the Gaussian curve ($\omega = 4\sigma$), Eq. (1.14) results. However, because of the distortion of most peaks at their base, that equation is rarely used and Eq. (1.15) is preferred.

N is a relative parameter, since it depends upon both the solute chosen and the operating conditions adopted. Generally, a compound that appears towards the end of the chromatogram is selected in order to get a reference value, when it is unknown whether the column will successfully achieve a given separation.

$$N = 16\frac{t_R^2}{\omega^2} \tag{1.14}$$

$$N = 5.54\frac{t_R^2}{\delta^2} \tag{1.15}$$

For asymmetric peaks (which is often the case), we sometimes encounter the empirical Eq. (1.16) for calculating the column's efficiency:

$$N = 41.7\frac{\left(\frac{t_R}{\omega_{0.1}}\right)^2}{\left(\frac{a}{b}+1.25\right)} \tag{1.16}$$

where $\omega_{0.1}$ designates the width of the peak measured at 10% of its height (Figure 1.5).

1.6.2 Number of Effective Plates (Real Efficiency)

To compare the performance of columns of different design for a given compound – or to compare, in gas chromatography, the performances between a capillary column and a packed column – more realistic values are obtained by replacing the *total retention time* t_R, which appears in Eqs. (1.13)–(1.15), with the *adjusted retention time* t'_R (Figure 1.7), which does not take into account the hold-up time t_M spent by the compound in the mobile phase. The three equations then become:

$$N_{eff} = \frac{t'^2_R}{\sigma^2} \tag{1.17}$$

$$N_{eff} = 16\frac{t'^2_R}{\omega^2} \tag{1.18}$$

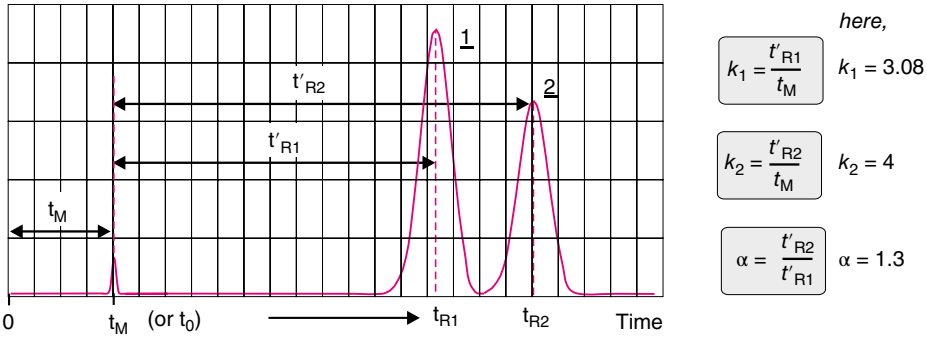


Figure 1.7 Retention factors and separation factor (or selectivity factor) between two adjacent compounds. Each compound has its own retention factor. On this figure, the separation factor is around 1.3. α alone is not enough to determine whether separation is really possible.

$$N_{eff} = 5.54 \frac{t'^{2}_{R}}{\delta^{2}} \qquad (1.19)$$

Currently, it is considered that these last three equations are not very useful.

1.6.3 Plate Height

The *height equivalent to a theoretical plate H*, as already defined (Eq. (1.4)), is calculated for reference compounds to permit a comparison of columns of different lengths. However, *H* does not behave as a constant; its value depends upon the compound chosen and upon the experimental conditions.

For a long time in gas chromatography, an adjustment value called the *effective plate height* H_{eff} was calculated using the true efficiency, instead of the theoretical efficiency.

The calculation of H_{eff} from the real efficiency uses Eq. (1.20):

$$H_{eff} = \frac{L}{N_{eff}} \qquad (1.20)$$

Reduced plate height

In liquid chromatography, when the column is filled with spherical particles, a parameter known as the reduced plate height *h* is often encountered. This parameter takes into account the mean diameter d_m of the particles. This eliminates the effect of the particle size for better comparison of columns with different mean diameter values. Columns with the same reduced plate height *h* will yield similar performance.

$$h = \frac{H}{d_m} \quad \frac{L}{Nd_m} \qquad (1.21)$$

1.7 RETENTION PARAMETERS

1.7.1 Retention Times

The definition of retention times has been given previously (Section 1.2).

1.7.2 Retention Volume (or Elution Volume) V_R

The *retention volume* V_R of an analyte represents the volume of mobile phase necessary to enable its migration from one end of the column to the other. To estimate this volume, different methods (direct or indirect) that depend of the physical state of the mobile phase may be used. On the chromatogram, it corresponds to the volume of mobile phase that flows through between the time of injection and the time when the peak reaches its maximum point. If the flow rate F is constant, then:

$$V_R = t_R \cdot F \tag{1.22}$$

The **volume of a peak** V_{peak} corresponds to that volume of the mobile phase in which 95% of the solute is diluted when leaving the column. It is defined by:

$$V_{peak} = \omega \cdot F \tag{1.23}$$

1.7.3 Hold-up Volume (or Dead Volume) V_M

The volume of the mobile phase in the column (known as the dead volume), V_M, corresponds to the accessible interstitial volume. It can be calculated from a chromatogram, provided a solute not retained by the stationary phase is present. The dead volume is deduced from t_M and the flow rate F:

$$V_M = t_M \cdot F \tag{1.24}$$

1.7.4 Stationary Phase Volume

This volume designated by V_s is not directly accessible from the chromatogram. In simple cases, we calculate it by subtracting the volume of the mobile phase from the total internal volume of the empty column. A column, whatever its design, may always be characterized by its *phase ratio* β defined as:

$$\beta = \frac{V_M}{V_s} \tag{1.25}$$

1.7.5 Retention (or Capacity) Factor k

When a compound of total mass m_T is introduced onto the column, it separates into two quantities: m_M, the mass in the mobile phase, and m_S, the mass in the stationary phase. During the solute's migration down the column, these two quantities remain constant. Their ratio, called the *retention factor k*, is constant and independent of m_T:

$$k = \frac{m_S}{m_M} = \frac{C_S}{C_M} \cdot \frac{V_S}{V_M} = K \frac{V_S}{V_M} = K/\beta \tag{1.26}$$

The retention factor *k*, also known as the *capacity factor*, is a very important parameter in chromatography for defining column performance. Though it does not vary with the flow rate or the column length, *k* is not a constant, as it depends upon the experimental conditions. *k* is dependent via K (Nernst distribution law) on the intensity of solute–stationary phase interactions, and via β on the column's design. For this reason, it is sometimes designated by k' rather than by *k* alone.

This parameter takes into account the ability, great or small, of the column to retain each compound (*capacity*). When separations are being developed, *k* should not exceed 10. Ideally, *k* should be around five, otherwise the time of analysis is unduly long.

> The expression **capacity factor** presents a possible confusion with the **capacity** of a column, which is the maximum solute mass the column may retain without being saturated. This factor is given by manufacturers when describing a column.

An experimental approach to the retention factor *k*

On the basis of Craig's model, each molecule is considered as passing alternatively from the mobile phase (in which it progresses down the column) to the stationary phase (in which it is immobilized). The average speed of the progression down the column is slowed if the time periods spent in the stationary phase are long. Extrapolate now to a case that supposes *n* molecules of this same compound (a sample of mass m_T). If we accept that, at each instant, the ratio of the n_S molecules fixed upon the stationary phase (mass m_S) and of the n_M molecules present in the mobile phase (mass m_M) is the same as that of the times (t_S and t_M) spent in each phase for a single molecule, the three ratios will therefore have the same value:

$$\frac{n_S}{n_M} = \frac{m_S}{m_M} \quad \frac{t_S}{t_M} = k$$

Take the case of a molecule that spends 75% of its time in the stationary phase. Its average speed will be four times slower than if it stayed permanently in the mobile phase. As a consequence, if 4 μg of such a compound has been introduced onto the column, there will be an average of 1 μg at all times in the mobile phase and 3 μg in the stationary phase.

Since the retention time of a compound t_R is such that $t_R = t_M + t_S$, the value of k is therefore accessible from the chromatogram $\left(t_S = t'_R\right)$ (Figure 1.7):

$$k = \frac{t'_R}{t_M} = \frac{t_R - t_M}{t_M} \tag{1.27}$$

This important relation can also be written:

$$t_R = t_M\left(1 + k\right) \tag{1.28}$$

In light of Eqs. (1.16) and (1.18), the retention volume V_R of a solute can be written:

$$V_R = V_M\left(1 + k\right) \tag{1.29}$$

or

$$V_R = V_M + KV_S \tag{1.30}$$

This last expression linking the experimental parameters to the thermodynamic coefficient of distribution K is valid for ideal chromatography.

1.8 SEPARATION (OR SELECTIVITY) FACTOR

The separation factor α (Eq. (1.31)) enables the comparison of two adjacent peaks 1 and 2 present in the same chromatogram (Figure 1.7). It is defined by Eqs. (1.31) and (1.32).

By definition α is greater than unity:

$$\alpha = \frac{t'_{R(2)}}{t'_{R(1)}} \tag{1.31}$$

or

$$\alpha = \frac{k_2}{k_1} = \frac{K_2}{K_1} \tag{1.32}$$

The expression, connecting α to the Nernst distribution coefficients of the two solutes, shows that selectivity is dependent only on the value of these constants (intensity of interactions, temperatures) and does not depend on the column's geometry (length and diameter) or its packing (diameter of particles and quantity of stationary phase). For nonadjacent peaks, the *relative retention factor r* is calculated in a similar manner to α, and cannot be less than 1.

1.9 RESOLUTION FACTOR

To quantify the separation between two compounds, the *resolution factor R* is calculated from the chromatogram (Figure 1.8):

$$R = 2\frac{t_{R(2)} - t_{R(1)}}{\omega_1 + \omega_2} \tag{1.33}$$

For two adjacent peaks, this relationship involves two parameters: firstly, the difference between their retention times, $t_{R(2)} - t_{R(1)}$, which corresponds to the distance between the two peaks, and, secondly, their half-width at the base, ½ ($\omega_2 + \omega_1$) if we assume that each peak corresponds to an isosceles triangle (Figure 1.8).

Other expressions derived from the preceding ones and established with a view to replacing one parameter by another or to accommodating simplifications may also be employed to express the resolution. Therefore, Eqs. (1.34)–(1.36) are used quite often.

Equation (1.35) shows how resolution is affected by the efficiency, capacity factor, and selectivity factor. The chromatograms in Figure 1.9 present an experimental verification of this.

$$R = 1.177\frac{t_{R(2)} - t_{R(1)}}{\delta_1 + \delta_2} \tag{1.34}$$

$$R = \frac{1}{4}\sqrt{N_2}\cdot\frac{\alpha - 1}{\alpha}\cdot\frac{k_2}{1 + k_2} \tag{1.35}$$

$$R = \frac{\sqrt{N}}{2}\cdot\frac{k_2 - k_1}{k_1 + k_2 + 2} \tag{1.36}$$

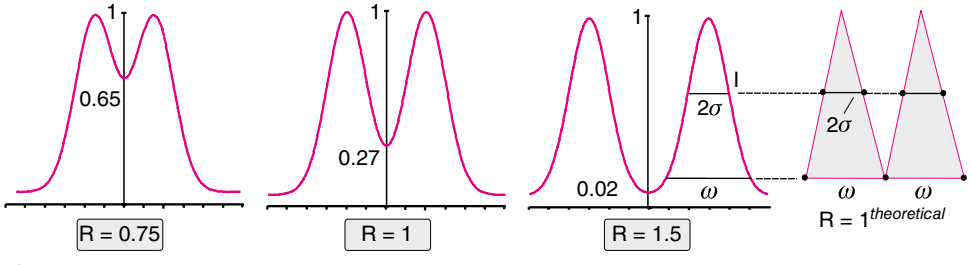


Figure 1.8 Resolution factor. A simulation of chromatographic peaks by juxtaposition of two identical Gaussian curves to a greater or lesser extent. The visual aspects corresponding to the values of R are indicated on the diagrams. From a value of R = 1.5 the peaks can be considered to be baseline resolved, the valley between them being around 2%.

1.10 INFLUENCE OF SPEED OF THE MOBILE PHASE

In all of the previous discussion and particularly in the various equations that characterize separations, the velocity (a function of flow rate) of the mobile phase in the column is not taken into account. However, if it becomes too high, the speed, which has an influence upon the progression of the analytes down the column, disturbs the equilibrium kinetics $(Solute)_{MP}$ / $(Solute)_{SP}$, and hence it acts on their dispersion, in other words on the quality of the analysis undertaken (compare Figure 1.9).

The influence of the speed of the mobile phase was demonstrated in the case of packed columns in gas chromatography and then modelled by Van Deemter, who proposed the first kinetic equation.

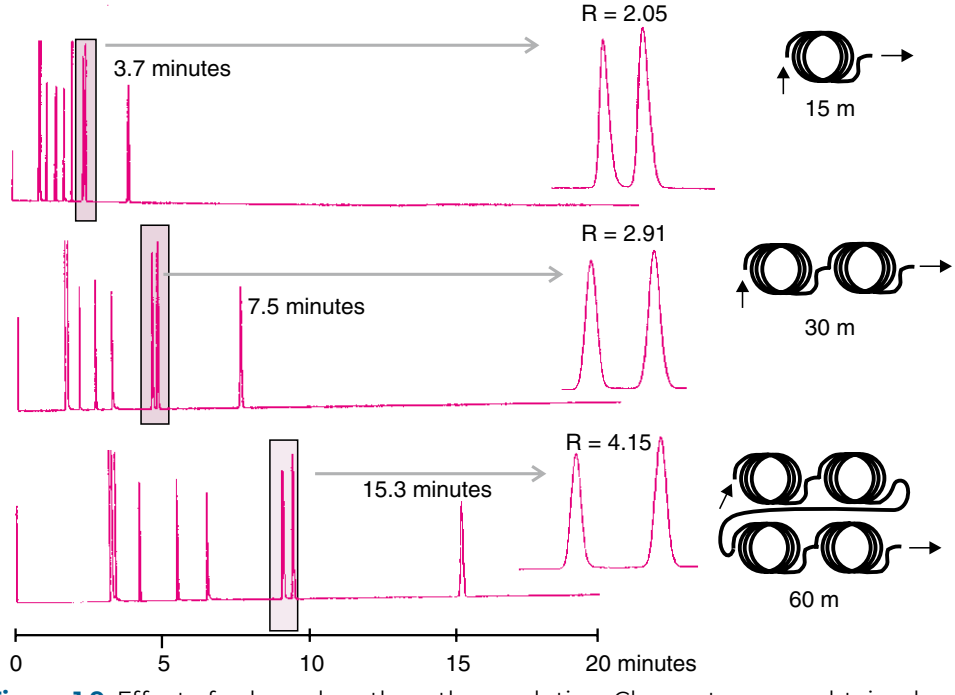


Figure 1.9 Effect of column length on the resolution. Chromatograms obtained with a GC instrument illustrating that by doubling the length of the capillary column, the resolution is multiplied by a factor of 1.41. (Source: Adapted from a document provided by the Waters company.)

1.10.1 Van Deemter Equation

The simplified equation proposed by Van Deemter in 1956, is well known for packed GC columns (Eq. (1.37)). The expression links the plate height H (HETP) to the average linear velocity of the mobile phase $\bar{u}$ in the column (Figure 1.10):

$$H = A + \frac{B}{\bar{u}} + C\bar{u} \tag{1.37}$$

This equation reveals that there exists an *optimal flow rate* for each column, corresponding to the minimum value of H, as shown by the curve of this equation. The loss in efficiency as the flow rate increases is obvious and represents what occurs when an attempt is made to rush the chromatographic separation by increasing the mobile phase flow rate. However, the loss in efficiency that occurs when the flow rate is too slow is less intuitive. To explain this phenomenon, the origins of the terms A, B, and C must be reviewed. Each of these parameters represents a domain of influence that can be perceived on the graph (Figure 1.10).

The three basic experimental coefficients A, B, and C are related to diverse physico-chemical parameters of the column and to the experimental conditions. If H is expressed in cm, A will also be in cm, B in cm²/s and C in s (where velocity is measured in cm/s). The curve of the Van Deemter equation is a hyperbola that goes through a minimum (H_{min}) when:

$$\bar{u}_{opt} = \sqrt{\frac{B}{C}} \tag{1.38}$$

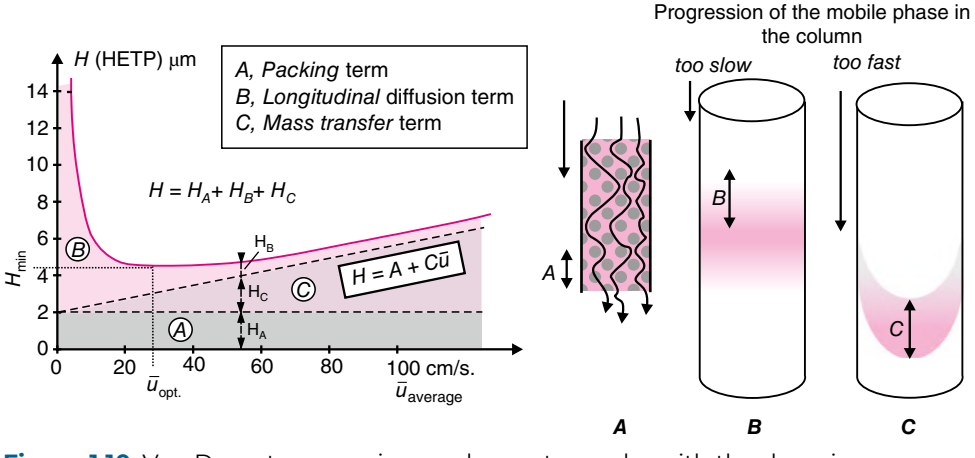


Figure 1.10 Van Deemter curve in gas chromatography with the domains of parameters A, B, and C indicated. There exists an equation similar to Van Deemter's that considers temperature: $H = A + B/T + CT$.

Packing-related term $A = 2\lambda.d_p$

Term *A* is related to the flow profile of the mobile phase passing through the stationary phase. The size of the particles (diameter d_p), their size distribution, and the uniformity of the packing (packing factor λ) can all create preferential paths, potentially causing imperfect exchanges between the two phases. This is known as the turbulent or Eddy diffusion factor, which is considered to have little importance in liquid chromatography and to be absent by nature for wall-coated open tubular (WCOT) capillary columns in GC (Golay equation without term *A*, see Section 1.10.2).

Gas (mobile phase) diffusion term $B = 2\gamma D_G$

Term *B*, which can be expressed from D_G, the diffusion coefficient of the analyte in the mobile phase, and λ, the above packing factor, is taken into consideration above all when the mobile phase is a gas. The longitudinal diffusion in the column is in effect quite fast.

This term is a consequence of entropy, which states that a system will tend spontaneously towards the greatest degree of disorder, just as a drop of ink diffuses into a glass of water into which it has fallen. Consequently, if the flow rate is too slow, the compounds undergoing separation will mix faster than they will migrate. This is why one must never interrupt a chromatographic analysis, even temporarily, once it is underway, as this could negatively impact efficiency.

Liquid (stationary phase) term $C = C_G + C_L$

Term *C*, which is related to the resistance to mass transfer of the solute between the two phases, becomes dominant when the flow rate is too high for an equilibrium to be attained. Local turbulence within the mobile phase and concentration gradients slow the equilibrium process ($C_S \Leftrightarrow C_M$). The diffusion of solute between the two phases is not instantaneous, and thus the solute will be carried along out of equilibrium. No simple formula exists which takes into account the different factors involved in term *C*. The parameter C_G is dependent upon the diffusion coefficient of the solute in the gaseous mobile phase, while the term C_L depends upon the diffusion coefficient in the liquid stationary phase.

In practice, the values for the coefficients *A*, *B*, and *C* can be obtained by making several measurements of efficiency for the same compound undergoing chromatography at different flow rates, since flow rate and average linear speed are related. Next, the hyperbolic function that best satisfies the experimental values can be calculated, preferably by using the multiple linear regression method.

1.10.2 Golay Equation

In 1958, Golay proposed a modified relationship reserved for capillary columns used in gas chromatography:

$$H = \frac{B}{\bar{u}} + C_L\bar{u} + C_G\bar{u} \tag{1.39}$$

Equation (1.39) leads to the minimum value of HETP for a column of radius *r*, if the retention factor of the test compound is known. The *coating efficiency* can then be calculated as the ratio between the value found and that deduced from the efficiency ($H = L/N$) obtained from the chromatogram, multiplied by 100.

$$H_{\text{theo.min}} = r\sqrt{\frac{1+6k+11k^2}{3(1+k)^2}} \tag{1.40}$$

1.10.3 Knox Equation

Another, more recent equation (1977), the Knox equation, is applicable to various types of liquid chromatography and includes the adjusted height *h* (see Eq. (1.21)):

$$h = A\bar{u}^{1/3} + \frac{B}{\bar{u}} + C\bar{u} \tag{1.41}$$

1.11 OPTIMIZATION OF A CHROMATOGRAPHIC ANALYSIS

The utility of analytical chromatography depends on its ability to separate analytes, whether to identify or quantify them. Particularly in the latter case, peak areas must be defined with precision. Since many parameters affect the quality of results, there are a certain number of requirements regarding the *resolution* between solute peaks, the *speed* of analysis (elution time, then the time needed to return to the initial conditions of analysis if different from the final conditions), and even the column's *capacity* (to avoid saturation). To do so, we often use software to simulate the separation by adjusting various physico-chemical factors such as temperature, flow rate, and composition of the mobile phase.

Due to the various parameters interacting together, the chromatographer must always find a compromise between *resolution*, *speed*, and *capacity*, which we often represent with the image of a triangle whose corners correspond to these three opposing parameters (Figure 1.11). In any optimization, the goal is to ensure sufficient separation of the compound(s) of interest in a minimum amount of time. The *resolution* and the *elution time* are thus the two most important dependent variables to consider. However, chromatography is a slow type of analysis, since we have to wait for the elution of the most retained compounds and then for the column to return to the initial conditions before we can start the next analysis. If the resolution is very good, then optimization consists in saving time, for example by choosing a shorter column.

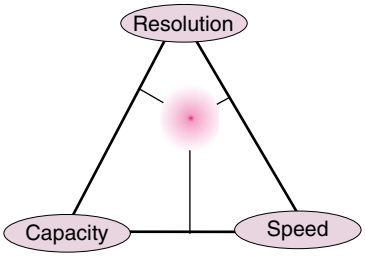


Figure 1.11 The chromatographer's compromise triangle between resolution, speed, and capacity. Any chromatographic analysis is subject to three contradictory criteria. If we favour one of these three criteria, the two others are necessarily disadvantaged. When we look for good selectivity, we must fall close to the resolution corner of this triangle. The shaded area indicates the domain corresponding to analytical chromatography based principally upon the five parameters K, N, k, α, and R.

Remember that resolution varies according to the square root of column length (see parameter N in Equation (1.35) and Figure 1.9).

Often, in a chromatographic assay, we are only interested in one or two compounds present in the sample. If we have a selective detector adapted to the analytes in question, we can facilitate the optimization stage, since the resulting chromatogram will be greatly simplified. In other cases, however, we try to separate the greatest possible number of compounds from the initial mixture.

1.12 CLASSIFICATION OF CHROMATOGRAPHIC TECHNIQUES

Chromatographic techniques can be classified according to various criteria: the *physical nature* of the phases, the *process* used, or *physico-chemical phenomena* occurring during separation, etc. The following classification has been established by consideration of the physical nature of the two phases involved.

1.12.1 Liquid Chromatography (LC)

This type, in which the mobile phase is a liquid, is the oldest known form of chromatography, with the earliest applications involving preparative methods of separation (for which it is still used today). Depending on the nature of the stationary phase (nature of interactions determining the separation of solutes), we can distinguish between *liquid/solid chromatography* (LSC), *liquid/liquid chromatography* (LLC), *ion chromatography* (IC), and *size exclusion chromatography* (SEC) (also known as gel *permeation* or gel *filtration* chromatography depending on the nature of the stationary phase).

Liquid/solid chromatography (LSC)

The stationary phase is a permeable solid medium to which the solute molecules adhere through the dual effect of physisorption and chemisorption. The physico-chemical parameter involved here is the *adsorption coefficient*. It can be considered as the ancestor of LSC. We can include thin-layer chromatography, which differs from LSC in the distribution of the stationary phase but which proceeds from the same separation principle.

Liquid/liquid chromatography (LLC)

The stationary phase and the mobile phase are both liquids. The stationary phase is most often a polymer fixed on an inert support, generally by grafting active species with covalent bonds, ensuring a long lifespan for these columns. The separation relies upon the *partition coefficient* of the solutes between the two phases (a phenomenon comparable to a liquid–liquid extraction in a separatory funnel, a well-known process in laboratories). This category also includes centrifugal partition chromatography, which is used in preparative chromatography.

Ion chromatography (IC)

Even though the stationary phase is considered to be a solid, the separation principle is different from LSC. Reserved for ionic species, this technique is not based on the adsorption phenomenon, as in LSC, but rather on exchanges between the ions to be separated and those in the stationary phase (anion or cation exchange resin). This type of separation relies on ion exchange equilibria characterized by *ionic distribution coefficients*.

Size exclusion chromatography (SEC)

The stationary phase here is a material containing pores whose dimensions are selected as a function of the size of the species to be separated. This technique is used to separate synthetic or natural macromolecules. It is known as *gel filtration* (e.g. separation of natural macromolecules such as proteins) when the mobile phase is aqueous, or *gel permeation* (e.g. separation of synthetic macromolecules) when the mobile phase is an organic solvent. Thus, it involves a sort of selective *permeation* at the molecular scale. For this technique, the distribution coefficient is called the *diffusion coefficient*.

Affinity chromatography

Halfway between chromatography and extraction, this technique, though not often employed for analytical purposes, is used as a purification method in biochemistry. It consists of preparing a stationary phase (carrier resin) on which a specific ligand for

the molecule we wish to separate is grafted to a maximum number of sites. As it has affinity for the ligand, the molecule is retained on the column. After elution of the other compounds present with the binding liquid phase, the latter is transformed into an elution buffer to retrieve the molecule of interest. The yield of this operation is related to the number of sites available on the resin, and the selectivity of the separation depends on the choice of ligand, which must be pure.

1.12.2 Gas Chromatography (GC)

The mobile phase is an inert gas with regards to the stationary phase and to the solutes to be separated. Three gases are used in most applications: nitrogen (not very efficient, see the Van Deemter curve), helium (more expensive and more efficient), and hydrogen (more efficient but also more dangerous). Depending on the physical state of the stationary phase, we shall distinguish between *gas/solid chromatography* (GSC) and *gas/liquid chromatography (*GLC).

Gas/solid chromatography (GSC)

This is the oldest application of GC. As with LSC, the stationary phase is a solid (such as graphite, silica gel or alumina) and the separation occurs via selective adsorption on the stationary phase. This type of gas chromatography is very effective for the separation of volatile species (gas or liquids that have a low boiling point).

Gas/liquid chromatography (GLC)

It was Martin and Synge who suggested the replacement of the liquid mobile phase in LLC with a gas. As indicated above, the stationary phase is a liquid immobilized on a support (inert silica particles for packed columns, inner wall of the column for capillary molecules) generally by grafting. Most often used for molecules in solution, the sample must be brought to its vapour state beforehand or at the time of its introduction. Therefore, this technique can be used for compounds whose boiling points are not very high, as well as for compounds which have been subject to a prior pyrolysis stage.

1.12.3 Supercritical Fluid Chromatography (SFC)

Here the mobile phase is a fluid in its supercritical state, such as carbon dioxide at about 50°C and more than 150 bar (15 MPa). The stationary phase can be a liquid or a solid. This technique combines the advantages of gas chromatography and liquid chromatography. It has many applications, even though it requires more complex equipment.

Quantitative Analysis by Chromatography

The significant development of chromatography in quantitative analysis is essentially due to its reliability and its precision. The relationship between the **injected mass** of the analyte on the column and the **area of the corresponding peak** on the resultant chromatogram leads to a comparative method used in many standardized assay protocols. The reproducibility of separations, allied with data processing software, makes it possible to automate all the calculations associated with these analyses. Trace and ultratrace analyses by chromatography are used, particularly in EPA methods for environmental analysis, although their costs are rather high. The three most widely used methods are described here, along with their simplest configuration.

1.13 PRINCIPLE AND BASIC RELATIONSHIP

To calculate the composition of a sample or to assay an analyte responsible for a peak on a chromatogram, two basic conditions must be met. Firstly, an *authentic sample of the analyte* to be measured must be available, as a reference, to determine the detector's sensitivity to this compound. Secondly, software calculating the *areas* (or failing that, the *heights*) of the different eluting peaks of interest is also required. All of the quantitative methods in chromatography are comparative methods (which is very often the case in quantitative analysis).

For a given tuning of the instrument, it is assumed that a linear relationship (linearization) exists for each peak of the chromatogram, over the entire concentration range, between its area (or failing that, its height) and the quantity of the analyte responsible for this peak in the injected sample. This relationship can be used for concentration ranges bounded by the *detectable quantity* limit for lower concentrations and the *linearity* limit for higher concentrations. This hypothesis is translated into the following equation:

$$m_i = K_i \cdot A_i \tag{1.42}$$

where m_i is the mass of compound i injected on the column, K_i is the *absolute response factor* for compound i, and A_i is the area of the eluting peak for compound i.

The absolute response factor K_i (not to be confused with the partition coefficient) is not an intrinsic parameter of the compound, as it depends upon the tuning of the chromatograph. To calculate the response factor K_i of an analyte, according to the above expression, it is essential that both the area A_i and the mass m_i, of compound i injected on the column, are known. However, this injected mass is difficult to

determine with precision. This is why most chromatographic methods used for quantitative analysis, preprogrammed into the software, do not make use of the absolute response factor, K_i.

1.14 CHROMATOGRAPHY SOFTWARE

Equation (1.42) highlights the need to determine peak areas. To do so, chromatography software has been developed to ensure control of the chromatogram, the acquisition of chromatograms and the determination of certain information related to the observed peaks, such as the FWHM δ, the width at the base ω, and the peak height or its area. These software programs help correct the baseline, select the start and end, if necessary, of each peak for integration, and, most importantly for quantitative analysis, select the quantification method. Lastly, by taking into account compositions of reference solutions (calibration solutions), they calculate sample compositions.

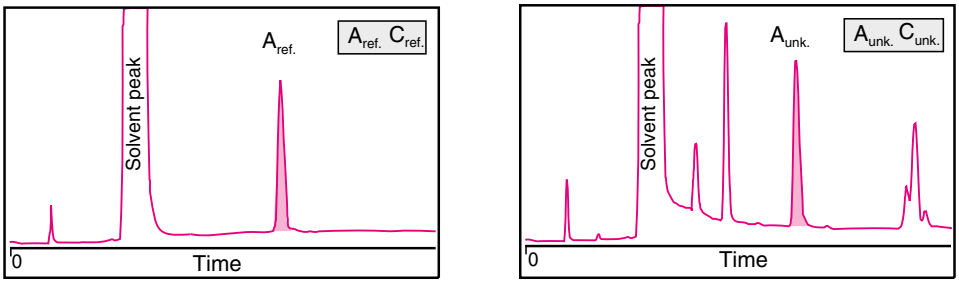
The three quantification methods used in chromatography (and sometimes in other techniques using proportionality between composition and response of a measuring instrument) are described here.

1.15 EXTERNAL STANDARD METHOD

This method allows the measurement of the concentration (or percentage by mass) of one or more components that appear as resolved peaks on the chromatogram, even in the presence of other compounds yielding unresolved peaks whose concentrations are not of interest to the experimenter.

The process is based upon the comparison of two chromatograms obtained successively *without changing the control settings* of the chromatograph (Figure 1.12). The first chromatogram is acquired from a standard solution (reference solution) with a known concentration C_{ref} of the analyte in a solvent. A volume V of this solution is injected. On the resulting chromatogram, the area A_{ref} of the corresponding peak is measured. The second chromatogram results from the injection of the *same volume V* of the sample in solution, containing an unknown concentration of the compound to be measured (conc. C_{unk}). The area of the corresponding peak is A_{unk}. Since an identical volume of both samples has been injected, the *ratio of the areas* is proportional to the *ratio of concentrations*, which depend upon the masses injected ($m_i = C_i \cdot V$). Applied to the two chromatograms, Eq. (1.42) leads to Eq. (1.43), which characterizes this method:

$$m_{ref} = C_{ref} \cdot V = K \cdot A_{ref} \quad \text{and} \quad m_{unk} = C_{unk} \cdot V = k \cdot A_{unk} C_{unk} = C_{ref} \frac{A_{unk}}{A_{ref}} \tag{1.43}$$



Chromatogram of the standard ($C_{ref.}$)

Chromatogram of the solution ($C_{unk.}$) *to be measured*

Figure 1.12 Analysis by the external standard method. The precision of this basic method is improved when several solutions of varying concentrations are used in order to create a calibration curve. For trace analyses by liquid chromatography, it is sometimes advisable to replace the areas of the peaks by their heights, as the latter are less sensitive to variations in the mobile phase flow rate.

The single-point calibration method, as depicted in Figure 1.12, assumes that the calibration line goes through the origin. Precision will be improved if the concentrations of the reference solution and of the sample solution are similar. That means the device settings do not need to be changed between injections.

This technique, employing the *absolute response factors*, yields very reliable results with current high-performance chromatographs equipped with an autosampler: a combination of a carousel sample holder and an automatic injector. This gives numerous measurements without human intervention. A single reference solution compensates for a potential drift of the instrument with scheduled control re-injections.

Precision of the assay can obviously be improved if several injections of the sample and the reference solutions are made, always using equal volumes. The average peak area is then calculated; however, unless several measurements are made, it is preferable to conduct a multilevel calibration, in which case equal volumes of a series of standard solutions are injected. The analytical results are obtained directly from the calibration curve, $A = f(C)$.

This method, the only one adapted to gas samples but also applicable in LC, has the added advantage of its ease of execution and its rapidity. However, it requires perfect reproducibility of injected volumes, which automatic injectors can do nowadays.

1.16 INTERNAL STANDARD METHOD

This second method relies on the *relative response factor* of each compound to be measured against a marker added as a reference. This compensates for any imprecision in the injected volumes, which is the main drawback of the previous method.

As above, this more reliable method requires two chromatograms, one to calculate the relative response factors of the compounds of interest, and the other to analyse the sample.

The areas of the peaks to be quantified are compared with that of an *internal standard* (designated by IS), added at a known concentration to the sample solution.

1.16.1 Calculation of the Relative Response Factors

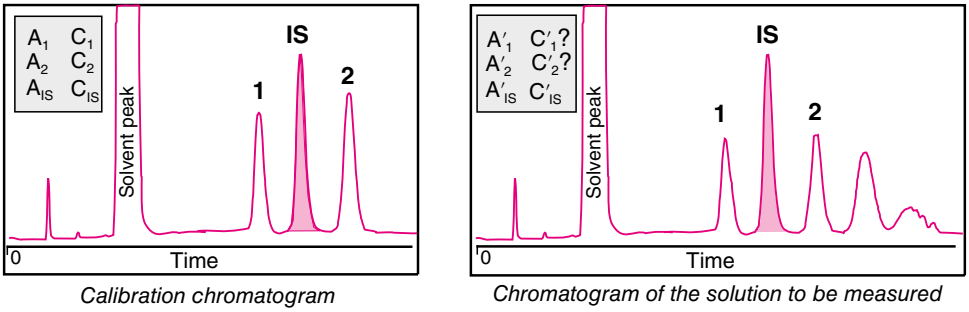
Let us suppose that a sample contains two compounds **1** and **2** to be measured and that compound ***IS*** represents an additional compound for use as an internal standard (Figure 1.13).

In the first stage, a solution containing compound **1** at known concentration C_1, compound **2** at known concentration C_2, and the internal standard **IS** at known concentration C_{IS} is prepared and then injected onto the chromatograph. A_1, A_2, and A_{IS} are the areas of the elution peaks in the chromatogram due to the three compounds. If m_1, m_2, and m_{IS} represent the real quantities of these three substances introduced onto the column, then three equations of the type of Eq. (1.42) can be derived:

$$m_1 = K_1 \cdot A_1$$
$$m_2 = K_2 \cdot A_2$$
$$m_{IS} = K_{IS} \cdot A_{IS}$$
$$\frac{m_1}{m_{IS}} = \frac{K_1 \cdot A_1}{K_{IS} \cdot A_{IS}} \text{ and } \frac{m_2}{m_{IS}} = \frac{K_2 \cdot A_2}{K_{IS} \cdot A_{IS}}$$

These ratios enable the calculation of the *relative response factors* of **1** and **2**, with respect to ***IS***. They are designated by $K_{1/IS}$ and $K_{2/IS}$:

$$K_{1/IS} = \frac{K_1}{K_{IS}} = \frac{m_1 \cdot A_{IS}}{m_{IS} \cdot A_1} \quad \text{and} \quad K_{2/IS} = \frac{K_2}{K_{IS}} = \frac{m_2 \cdot A_{IS}}{m_{IS} \cdot A_2}$$



Calibration chromatogram *Chromatogram of the solution to be measured*

Figure 1.13 Method of analysis by internal standard.

Since the injected masses m_i are proportional to the corresponding mass concentrations C_i ($m_i = C_i V$), the above equations can be rewritten as:

$$K_{1/IS} = \frac{C_1 \cdot A_{IS}}{C_{IS} \cdot A_1} \quad \text{and} \quad K_{2/IS} = \frac{C_2 \cdot A_{IS}}{C_{IS} \cdot A_2}$$

1.16.2 Chromatogram of the Sample – Calculation of the Concentrations

The second step of the analysis is to obtain a chromatogram for a given volume of a solution containing the sample to quantify and to which has been added a known quantity of internal standard ***IS***. This will yield A'_1, A'_2, and A'_{IS}, the areas of this new chromatogram obtained under the same operating conditions as previously. If m'_1, m'_2, and m'_{IS} represent the quantities of **1**, **2**, and ***IS*** introduced into the column, then:

$$\frac{m'_1}{m_{IS}} = K_{1/IS} \cdot \frac{A'_1}{A'_{IS}} \quad \text{and} \quad \frac{m'_2}{m'_{IS}} = K_{2/IS} \cdot \frac{A'_2}{A'_{IS}}$$

From the relative response factors calculated in the first experiment as well as from the known concentration of the internal standard in the sample, C'_{IS}, this leads to:

$$C'_1 = C'_{IS} K_{1/IS} \cdot \frac{A'_1}{A'_{IS}} \quad \text{and} \quad C_2 = C'_{IS} K_{2/IS} \cdot \frac{A'_2}{A'_{IS}}$$

If we expand to n components, it is easy to calculate the mass concentration of the solute i using Eq. (1.44):

$$C'_i = C'_{IS} K_{i/IS} \cdot \frac{A'_1}{A'_{IS}} \tag{1.44}$$

In addition, the percentage concentration of i can be expressed using Eq. (1.45):

$$X_i\% = \left(C'_i / \text{Mass of sample taken}\right) \times 100 \tag{1.45}$$

This method becomes even more precise if several injections of the solution and of the sample are carried out. In conclusion, this general and reproducible method nevertheless demands a good choice of internal standard, which should have the following characteristics:

- it must be stable, pure, and not exist in the initial sample;
- it must be measurable, giving a well-resolved elution peak on the chromatogram;
- its retention time must be close to that (or those) of the solute(s) to be quantified;

- its concentration must be close to or above that of the analytes in order to be in the detector's linear response range;
- it must be inert with respect to the compounds in the sample.

The advantage of this method is not needing perfect reproducibility of injections, which makes manual injections possible if the device is not equipped with an automatic injector. The disadvantage, however, is the choice of internal standard, which extends the development time of the analysis. However, with mass detectors that do not require perfect resolution, coelution of the internal standard and of the solute to quantify is possible, if we can recognize peaks caused by each of the two compounds on the mass spectrum of the mixture.

1.17 INTERNAL NORMALIZATION METHOD

This method, also called percentage normalization, is used for mixtures in which all components have been identified and have been assigned a well-resolved peak on the chromatogram. This method uses relative response factors, as in the internal standard method. The biggest difference here is that the solute used as a reference to calculate the relative response factor is part of the mixture to quantify.

Let us suppose that we are trying to find the mass concentrations of three compounds **1**, **2**, **3** in a mixture (Figure 1.14). The analysis is again carried out in two steps.

1.17.1 Calculation of the Relative Response Factors

A standard solution containing the three compounds **1**, **2**, and **3** at known concentrations C_1, C_2, and C_3 is prepared. The chromatogram corresponding to the injection of a volume V of this standard solution shows three peaks of area A_1, A_2, and A_3. These areas

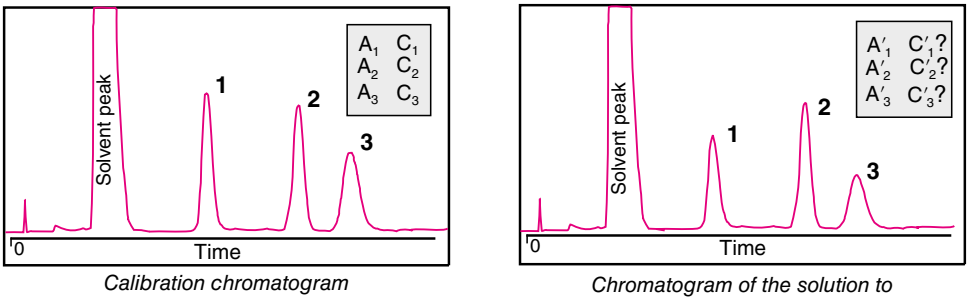


Figure 1.14 Analysis by internal normalization method.

will be related to the masses m_1, m_2, and m_3 of the compounds in volume V, by three expressions of the Eq. (1.42) type.

One of the compounds, **3** for example, is chosen as the substance for internal normalization. This compound **3** will serve to calculate the relative response factors $K_{1/3}$ and $K_{2/3}$ for compounds **1** and **2** with respect to **3**. As previously deduced:

$$K_{1/3} = \frac{K_1}{K_3} = \frac{m_1 \cdot A_3}{m_3 \cdot A_1} \quad \text{and} \quad K_{2/3} = \frac{K_2}{K_3} = \frac{m_2 \cdot A_3}{m_3 \cdot A_2}$$

Given that $m_i = C_i.V$, then the following equations for $K_{1/3}$ and $K_{2/3}$ are obtained:

$$K_{1/3} = \frac{C_1 \cdot A_3}{C_3 \cdot A_1} \quad \text{and} \quad K_{2/3} = \frac{C_2 \cdot A_3}{C_3 \cdot A_2}$$

1.17.2 Chromatogram of the Sample – Calculation of the Concentrations

The next step consists in injecting a specimen of the mixture to be measured containing compounds **1**, **2**, and **3**. Labelling the elution peaks as A'_1, A'_2, and A'_3 will give direct access to the percentage mass composition of the mixture represented by x_1, x_2, and x_3 via three equations of the following form:

$$x_i\% = \frac{K_{i/3} \cdot A'_i}{K_{1/3} \cdot A'_1 + K_{2/3} \cdot A'_2 + A'_3} \times 100 \text{ with } i = 1 \text{ or } 2 \text{ or } 3.$$

The condition of normalization is that $x_1 + x_2 + x_3 = 100$

If the procedure is extrapolated to n components normalized to compound j (internal reference), a general expression for the response factor of a given compound i can be obtained:

$$K_{i/j} = \frac{C_i \cdot A_j}{C_j \cdot A_i} \tag{1.46}$$

It is also possible to determine $K_{i/j}$ by plotting a concentration–response curve for each of the solutes.

In a mixture containing n solutes, if A'_i designates the area of the elution peak of compound i, and if the internal reference is j, then the content of compound i will obey the following equation:

$$x_i\% = \frac{K_{i/j} \cdot A'_i}{\sum_{i=1}^{n} K_{i/j} \cdot A'_1} \times 100 \tag{1.47}$$

This method is ideal to monitor rates of advancement of a chemical reaction and, at the same time, to follow the conversion kinetics.

KEY POINTS OF THE CHAPTER

1. Chromatography is a separation method and also an analytical method. It is a very reliable method, with continued innovation in order to separate, purify, identify, and quantify chemical species.
2. It is based on the adsorption or partition, depending on various parameters and interactions, of chemical species between two immiscible phases, one fixed or *stationary* (SP) and the other *mobile* (MP).
3. There are many chromatographic techniques that differ in the process used, the nature of phases, physico-chemical equilibria, the physical state, etc., and just as many classifications are possible depending on the criterion retained!
4. To each compound, we assign a distribution factor $K = C_s/C_M$ as a function of its concentration in the two phases. For each type of chromatography, thermodynamic relationships apply. From K (which is not a constant), we can calculate ΔH, ΔS, and ΔG for the separation in question.
5. A document inseparable from any chromatographic analysis, the *chromatogram*, or elution curve, represents the graphic recording of concentration over time. It provides information on the analysed sample and on the quality of the separation in progress.
6. One peak = one compound is the ideal situation. At column outlet, we treat the ideal peak as a Gaussian peak (hence the parameters σ, δ, ω), but the reality may be different. An ideal chromatogram has peaks that do not overlap and which are at least somewhat spaced out. A peak alone does not enable the absolute identification of a compound.
7. A number of parameters can be used to better characterize each separation: *efficiency, retention, separation*, and *resolution*. The flow rate of the mobile phase has an influence on efficiency. Many empirical equations, not reported in this chapter, are offered to mathematically define the behaviour of analytes during elutions with a gradient.
8. The speed of progression of the mobile phase with respect to the stationary phase has an essential influence on separation quality. A hyperbolic function, formed from the addition of three independent terms, known as the *Van Deemter equation*, demonstrates the existence of an optimal linear speed, i.e. leading to a minimum value of HETP.
9. First established for GC, and later adapted to HPLC, the three terms of the previous equation refer to preferential paths (turbulent diffusion), to longitudinal diffusion, and to resistance to mass transfer.
10. The quantification of analytes using chromatography is founded on the relationship between the mass and the area of the corresponding peak on the chromatogram. There are three methods, called the external standard method, internal standard method, and internal normalization method. The first is by far the most used because it is the quickest when it comes to processing a large number of analyses. Chromatograms equipped with a sample holder carousel and an automatic injection device make this method very reliable.

PROBLEMS

1. For a given solute, show that the analysis time (which we can assume is equal to the retention time of the most retained compound) depends, amongst other things, on the column length L, on the average linear speed $\bar{u}$ of the mobile phase, and on volumes V_S and V_M, which respectively designate the volume of the stationary phase and the volume of the mobile phase.

2. Calculate the separation factor (or selectivity factor) between two compounds, 1 and 2, whose retention volumes are 6 ml and 7 ml, respectively. The dead volume of the column is 1 ml. Show that this factor is equal to the ratio of the distribution coefficients K_2/K_1 of these compounds ($t_{R(1)} < t_{R(2)}$).

3. Equation (P1.2) is sometimes employed to calculate N_{eff}. Show that this relation is equivalent to the more classical Eq. (P1.1):

$$N_{eff} = 5.54 \frac{\left(t_R - t_M\right)^2}{\delta^2} \qquad \text{(P1.1)}$$

$$N_{eff} = N \frac{k^2}{\left(1+k^2\right)} \qquad \text{(P1.2)}$$

4. The resolution factor R for two solutes 1 and 2, whose elution peaks are adjacent, is sometimes expressed by Eq. (P1.3):

$$R = 2 \frac{t_{R(2)} - t_{R(1)}}{\omega_1 + \omega_2} \qquad \text{(P1.3)}$$

$$R = \frac{1}{4} \sqrt{N_2} \frac{\alpha - 1}{\alpha} \frac{k_2}{1+k_2} \qquad \text{(P1.4)}$$

a. If it is assumed that the two adjacent peaks have the same width at the baseline ($\omega_1 = \omega_2$), then show that Eq. (P1.4) is equivalent to Eq. (P1.3) for resolution.

b. The effective plate number N_{eff} may be calculated as a function of the separation factor α for a given value of the resolution, R. Derive this relationship.

$$N_{eff} = 16R^2 \frac{\alpha^2}{\left(\alpha - 1\right)^2}.$$

5. Show that if the number of theoretical plates N is the same for two neighbouring compounds 1 and 2, then the classical expression is thus:

$$R = \frac{\sqrt{N}}{2} \frac{k_2 - k_1}{k_1 + k_2 + 2} \qquad \text{(P1.5)}$$

Then show that if $\bar{k} = (k_1 + k_2)/2$, Eqs. (P1.5) and (P1.6) are equivalent:

$$R = \frac{1}{2}\sqrt{N}\,\frac{\alpha - 1}{\alpha + 1}\,\frac{1}{1}\,\frac{\bar{k}}{1+\bar{k}} \qquad \text{(P1.6)}$$

6. The characteristics of a C-18 column, offered by a manufacturer for ultra-rapid HPLC, are:
Length: 50 mm, diameter: 2 mm, particle diameter: 2 μm. The test chromatogram accompanying the column shows six well-spaced peaks in less than 2 min.

Solute No.	**0**	**1**	**2**	**3**	**4**	**5**
t_R (min)	0.21	0.31	0.42	0.76	1.14	1.58

On the chromatography report, the following information can also be found:
For solute No. 5: N = 106,000 plates per metre; asymmetry at 10% of height A = 1.07. Remember that for an asymmetrical peak, the number of theoretical plates *N* is calculated from the following equation:

$$N = 41.7\left(\frac{t_R}{\omega_{0.1}}\right)^2 / \left(\frac{a}{b} + 1.25\right) \quad \text{and that} \quad A = \frac{a+b}{2a}$$

a. For peak No. 5, calculate width $\omega_{0.1}$ measured at 10% of its height (from the baseline) and values *a* and *b*.

b. Knowing that peak No. 0 corresponds to a nonretained solute, calculate for solutes 4 and 5, the retention factors k_4 *and* k_5, the separation factor α, and the resolution *R* between these two peaks. Assume that the efficiency N is the same for these two solutes, i.e. the value given by the manufacturer.

c. Knowing that the phase ratio for such a column may be expressed as:

$$\beta = 0.6\,\frac{SA}{SV + 0.45}$$

where *SA* is the specific area of the material contained in the column and *SV* is the specific volume of the pores, calculate the phase ratio in the case where *SA* = 330 $m^2\ g^{-1}$ and *SV* = 1 ml g^{-1} (units adapted to the direct application of the relationship giving β).

Comment. The factor 0.6 corresponds to the column's packing rate with silica and 0.45 corresponds to the silica volume in ml for 1 gram.

d. Knowing the phase ratio, calculate, for solutes 4 and 5, the Nernst distribution factor. Do these two values correspond to the elution order obtained?

7. Using the internal normalization method, we want to determine the mass composition of a sample made of four butyric esters. A reference solution of these four esters (with known mass concentrations) leads to the following relative response factors for methyl ester (ME), ethyl ester (EE), and propyl ester (PE) as compared to butyl ester (BE):

$$k_{ME/BE} = 0.919; \quad k_{EE/BE} = 0.913; \quad k_{PE/BE} = 1.06.$$

From the information provided by the chromatogram of the sample being assayed, find the mass composition of this mixture.

Peak *No.*	t_R	Compound	Area (arbitrary units)
1	2.54	Methyl ester (ME)	2,340.1
2	3.47	Ethyl ester (EE)	2,359.0
3	5.57	Propyl ester (PE)	4,077.3
4	7.34	Butyl ester (BE)	4,320.7

8. Assay of serotonin *S* (5-hydroxytryptamine) using the internal standard method. We sample 1 ml of the solution to assay, to which we add 1 ml of a solution containing 3 ng of *N*-methylserotonin (*NMS*). We treat this mixture to remove all other intrusive compounds from it. We perform solid-phase extraction to isolate serotonin and its methyl derivative, diluted in a suitable phase.

a. Why do we add the internal standard compound before the extraction stage?

b. Calculate the response factor of serotonin with respect to *N*-methylserotonin, knowing that the calibration chromatogram leads to the following results:

- serotonin area	30,885,982 μV s	quantity injected: 5 ng
- *N*-methylserotonin area	30,956,727 μV s	quantity injected: 5 ng

Based on the chromatogram of the sample solution, find what the serotonin concentration is in the starting sample, knowing that the serotonin peak area is equal to 2,573,832 μV s and the *N*-methylserotonin peak area is 1,719,818 μV s.

SOLUTIONS

1. We have $t_R = t_M(1+k)$. And $t_M = L/\bar{u}$ and $k = KV_S/V_M$, hence $t_R = (L/\bar{u})(1+KV_S/V_M)$.

2. Knowing that $V_R = t_R \cdot D$, we get: $\alpha = (V_{R(2)} - V_M)/(V_{R(1)} - V_M)$, i.e. $\alpha = 1.2$.

3. To transform Eq. (P1.1) into Eq. (P1.2), we add k, from $t_M = t_R/(1+k)$.

4. **a.** When peaks are neighbours, base widths are generally comparable. By posing $\omega_1 = \omega_2$ and by expressing ω_2 with its value as a function of *N2*, we obtain $R = ¼ \sqrt{N_2}\,(t_{R(2)} - t_{R(1)})/t_{R(2)}$. Then, we add k, by using $t_R = t_M(1 + k)$; finally, we introduce $\alpha = k_2/k_{1;}$

b. The relationship given in the problem is obtained by rearranging the relationship for R (as a function of k and α) and substituting the relationship for N_{eff} from Eq. (P1.2) of problem 1.3.

5. **1.** From the base relationship $R = 2(t_{R(2)} - t_{R(1)})/(\omega_1 + \omega_2)$, we can replace ω with its value depending on N, $\omega = 4t_R/\sqrt{N}$ and since $t_R = t_M(1 + k)$, we obtain Eq. (P1.5);

2. We multiply the second member of (1), numerator and denominator by $(k_1 + k_2)$ and we obtain α ($\alpha = k_2/k_1$).

6. **a.** $\omega_{0.1} = a + b = 2A \bullet a$; $A = (a + b)/2a \Rightarrow b = a(2A - 1) \Rightarrow a/b = 1/(2A - 1) \Rightarrow$

$$N = 41.7\left(\frac{t_R}{2A \cdot a}\right)^2 \Big/ \left(\frac{1}{2A-1} + 1.25\right)$$

$$a = \sqrt{41.7\left(\frac{t_R^2}{4A^2 N}\right)} \Big/ \left(\frac{1}{2A-1} + 1.25\right)^{1/2}$$

where N = 53,000 plates, t_R = 1.58 min and A = 1.07, and thus we get:

a = 0.0142 min.

b = 0.0162 min.

$\omega_{0.1}$ = 0.0304 min.

b. $k = (t_R - t_M)/t_M$

therefore, k_4 = 4.43 and k_5 = 6.62; $\alpha_{4\text{-}5} = k_5/k_4$ = 1.47;

$$R = 2\frac{t_{R5} - t_{R4}}{\omega_5 + \omega_4} = 2\frac{t_{R5} - t_{R4}}{\frac{4}{\sqrt{N}}(t_{R5} + t_{R4})} = 18.6$$

(this assumes that the peak is Gaussian)

c. β = 136.6;

d. $K = k\beta$; K_4 = 605 and K_5 = 884; $K_5 > K_4$ means that solute 5 has more affinity for the stationary phase than solute 4; therefore, it is more retained and exits later.

7. %(ME) = 16.6; %(EE) = 16.6; %(PE) = 33.4; %(BE) = 33.4.

8. **a.** By adding *N*-methylserotonin before extraction, we do not have to take into consideration any potential loss of product due to the various manipulations. We suppose that the extraction yield is the same for these two compounds, which are very similar;

b. $k_{S/NMS}$ = 1.002;

c. serotonin mass: 45 ng/ml.

Chapter 2

Gas Chromatography

Gas chromatography (GC) separates compounds that may be vaporized without decomposition when heated. To do so, analytes, in contact with the gaseous mobile phase, are brought to high temperatures. The same goes for the stationary phase caught in the column. GC can be paired with many types of detection, especially mass spectrometry, which often helps in positive identification of analytes. This versatile and very sensitive technique is known for its quick optimization of

analytical conditions, and thanks to current advances, such as high-speed or multidimensional gas chromatography, it is a very attractive and essential resource when studying volatile compounds.

Objectives

Represent a GC device
Choose the carrier gas
Choose the column
Compare the stationary phases
Know the injection methods
List the main detectors
Optimize a separation
Address new orientations of micro GC and fast GC
Explain retention indexes and constants of stationary phases

2.1 COMPONENTS OF A GC INSTALLATION

A gas chromatograph is composed of three components within a single surround. These components include the injector, the column, and the detector associated with a temperature-controlled oven that enables the column to attain high temperatures (Figure 2.1). The mobile phase that transports the analytes through the column is a gas referred to as the *carrier gas*. The carrier gas flow, which is precisely controlled, enables great reproducibility of the retention times.

Analysis starts when a very small quantity of sample is introduced in either liquid or gas form into the injector, which has the dual function of vaporizing the sample and mixing it with the gas flow at the head of the column. The column is usually a narrow-bore tube that coils around itself with a length that can vary from 1 m to over 100 m, depending upon the type and the contents of the stationary phase. The column, which can serve for thousands of successive injections, is housed in a temperature-controlled oven. At the end of the column, the mobile phase (carrier gas) passes through a detector before it exits to the atmosphere. Some gas chromatograph models of reduced size have their own electrical supply, enabling them to operate in the field (Figure 2.1).

In GC, there are four operational parameters for a given stationary phase: L, length of the column; u, velocity of the mobile phase (which affects the theoretical efficiency N); T, temperature of the column; and β, phase ratio, which affects the retention factor k. The settings of the chromatograph enable modifications in terms of T and u, and therefore both the efficiency of the column and the retention factors can be adjusted as well.

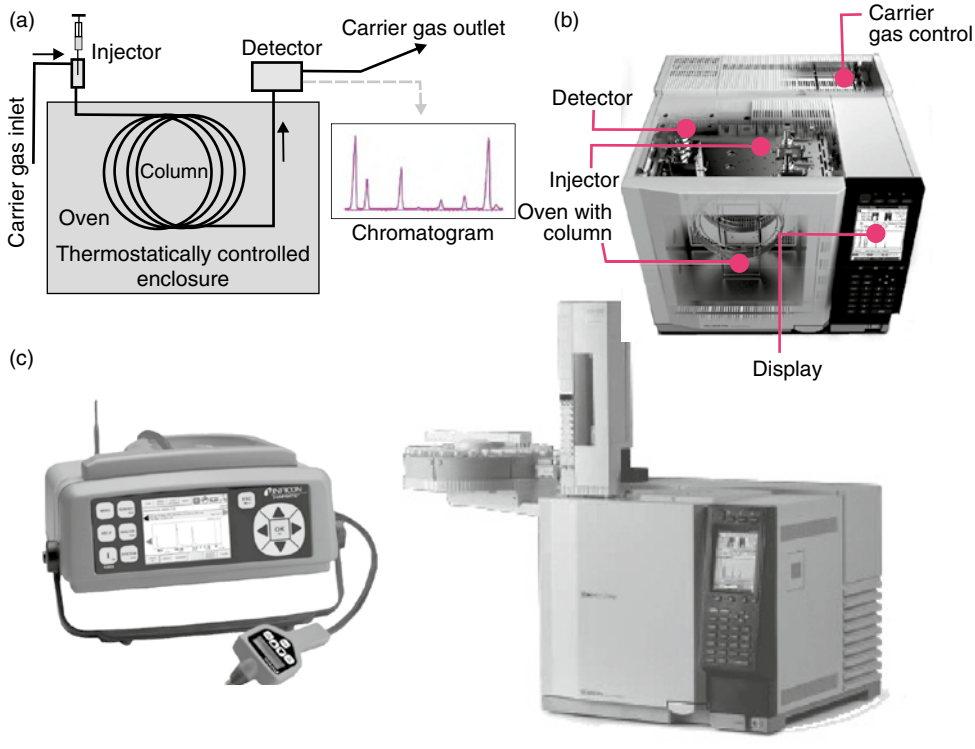


Figure 2.1 Operational diagram of a gas chromatograph and practical uses. a, b) Versatile analytical chromatogram (Source: GC-2010 model from Shimadzu). The instrument represented is equipped with a sample holder (carousel), an injector, an automatic sampler, and a mass spectrometry detection system (GC-MS). c) Portable model for analyses made in the field (volatile organic compounds, toxic industrial materials) (Source: Hapsite ER, information courtesy of Inficon).

2.2 CARRIER GAS AND FLOW REGULATION

The mobile phase is a gas (helium, hydrogen or nitrogen), either drawn from a commercially available gas cylinder or obtained, in the case of hydrogen or nitrogen, from an on-site generator (water electrolysis for H_2 and air separation for N_2), dedicated to the installation. To eliminate all harmful traces of water vapour and oxygen from polar stationary phases and detectors, a double filter, for drying and reducing, is placed before the injector.

The nature of the carrier gas has no significant influence upon the values of the partition coefficients *K* of the solutes between the stationary and mobile phases,

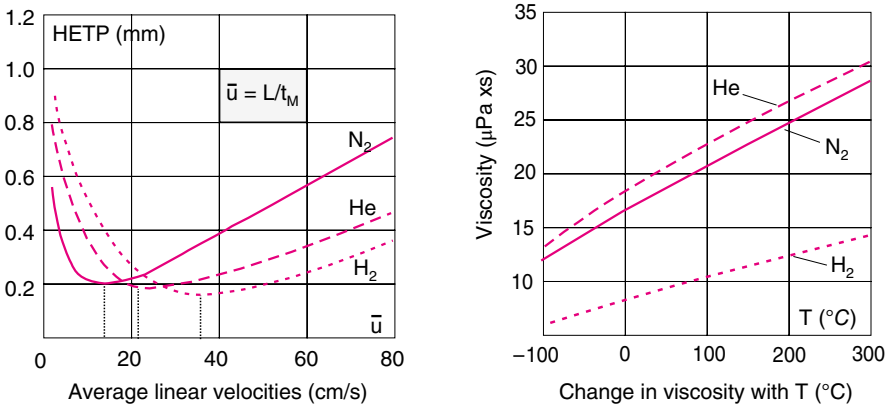


Figure 2.2 Efficiency as a function of the nature and linear velocity of the carrier gas. The Van Deemter curves show the relationship between HETP and linear velocity of the carrier gas for a given compound. Comparison of the viscosities of these three gases. Note the increase in the viscosity of these gases with temperature.

owing to an almost total absence of interaction between the gas and the solutes. Temperature is the only significant modification factor. By contrast, the viscosity and speed of the carrier gas, related to its flow rate or pressure at the head of the column, have an effect on the analytes' dispersion in the stationary phase and on their diffusion in the mobile phase (see the Van Deemter equation, Chapter 1, Section 1.10). These two factors have a direct impact on column efficiency *N* (Figure 2.2).

The pressure at the head of the column (several tens to hundreds of kPa – a few tenths to a few bar) is stabilized by *electronic pressure control* (*EPC*), so that the flow rate remains constant at its optimal value. This device is valuable because if the analysis is performed with ascending temperature programming (temperature gradient), the viscosity of the stationary phase and, by consequence, the pressure drop in the column, increase with time. By controlling the pressure, we conserve a constant and optimal speed of the carrier gas. The result is a faster analysis with the same efficiency.

The comparison of the three Van Deemter curves shows that the minimum of each is obtained for various linear velocities of the carrier gas: low for nitrogen, higher for helium and even higher for hydrogen. This means that the latter gas reduces the analysis time. For hydrogen also, the growth of the curve after the peak is less quick than for the other two gases, which gives more latitude in choosing the carrier gas speed without impacting the column's efficiency (Figure 2.2). The three curves plotting the viscosity of these gases versus temperature T again show that hydrogen has a lower viscosity than the other two carrier gases. Lower viscosity means a lower pressure drop and an increased column life.

2.3 INJECTION CHAMBER

2.3.1 Sample Introduction

The sample to be analysed is never introduced into the chromatogram as is, whether it is a liquid or a solid, but rather in a highly diluted solution. We use either a microsyringe (or loop injector) or a device such as a *headspace* sampler for volatile compounds, which both concentrates the sample and introduces it into the chromatograph.

Microsyringe and septum

The most common injection method is where a microsyringe is used (Figure 2.3) to inject a very small quantity of sample in solution (1 µl or less) through a rubber *septum* sealing the injection chamber. Some very elaborate septa have been developed (Figure 2.4) for microsyringes, which are an integral part of the automatic injectors found in most current instruments (Figure 2.1).

Automatic injectors with a sample holder carousel are called *autosamplers*. With some of them, we can select the injection method (liquid, headspace, SPME (see Chapter 21)) and, if necessary, conduct a short preparation of each sample beforehand

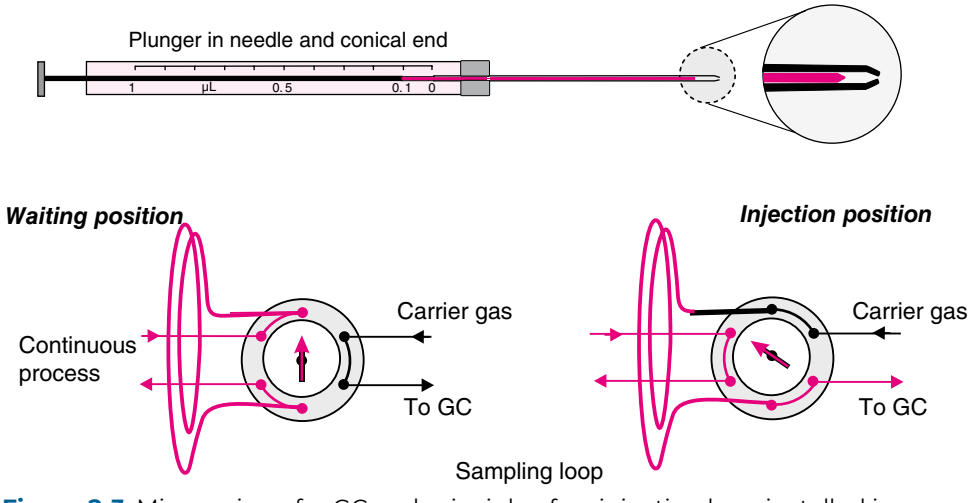


Figure 2.3 Microsyringe for GC and principle of an injection loop installed in a continuous process. The model chosen for this illustration has a cone shape, adapted to several septa or automatic injectors. In this model, the piston enters the needle to deliver the entire sample and prevent any dead volume. At the bottom, there is an example of an injection loop for gas or liquids (also see Figure 3.5).

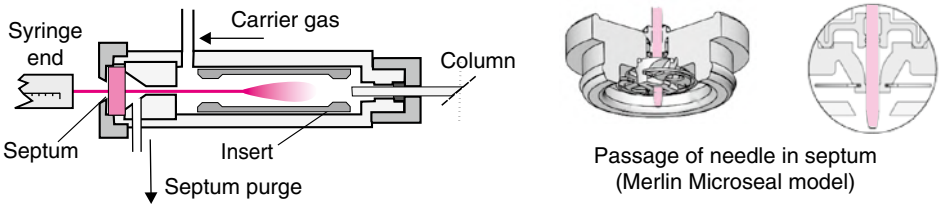


Figure 2.4 Direct vaporization injector used for packed columns. The typical septum is an elastomer disc but there are also more sophisticated versions, including the Merlin Microseal, which can be used thousands of times. (Source: Courtesy of Agilent Technologies.)

(dilution, addition of an internal standard, derivatization). These devices complete the chromatogram, improve results, and shorten analysis times. They have become essential for multi-analyte methods in environmental analysis (search for PAHs, mycotoxins, pesticides).

When there is too little analyte to be detected (detectability limit) or to be quantified (quantification limit), we must make a preconcentration. To do so, we have such techniques as SPME (solid phase micro-extraction) or SBSE (stir bar sorptive extraction). These techniques use the possibility of adsorbing an analyte on a solid phase, and inversely to desorb in the injector when subjected to heat (see Chapter 21).

Injection loops

For some applications, such as process control, there are injectors for gases or for liquids (Figure 2.3), which include a loop, i.e. a small volume waiting and ready to be injected by rotation of a valve. We find this same principle in liquid chromatography (see Chapter 3, Section 3.3).

2.3.2 Injectors

The injector, which is the sample's entrance to the chromatograph, has different functions: to *vaporize* and *entrain* the sample mixed with the carrier gas at the head of the column. Depending on the way injection is conducted and on its rapidity, it can impact the quality of analysis.

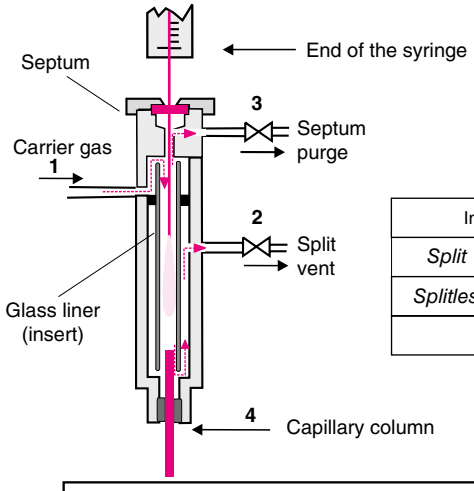
Injectors differ depending on the nature of samples and types of analyses. Historically, the first injectors, called *direct vaporization* injectors, were intended for packed columns or macrobores, in which the carrier gas flow rate must be at least 6–7 ml/min. Any model of this type comprises a metal tube with a glass sleeve (called the *insert*). It is swept by the carrier gas and heated to a sufficient temperature so that the less volatile solutes of the sample vaporize. The sample is introduced with a microsyringe, piercing the *septum* placed at the entrance of the injector (Figure 2.4).

The other end is connected directly to the column. Once all of the sample has been introduced with a syringe, it is immediately volatilized and enters the column, swept along by the carrier gas.

Split/splitless injector

For *capillary columns* able to handle only a small capacity of sample, even the smallest volume that it is possible to inject with a microsyringe (0.1 µl) can saturate the column. Special injectors are used that can operate in two modes, *with* or *without* flow splitting (also called *split* or *splitless*). This helps to adjust the sample fraction sent into the column as compared to the total quantity introduced in the chromatogram.

In *split mode,* a flow of carrier gas (e.g. 100 ml/min) arrives in the vaporization chamber where it mixes with the vapours of the injected sample (Figure 2.5). A vent valve separates this flow into two fractions, of which the largest portion is vented from the injector, taking with it the majority of the sample introduced. The *split ratio*



examples of flow rates in Split or Splitless mode

Inlets/outlets		1	2	3	4
Split		104	100	3	1
Splitless	PURGE OFF	54	0	53	1
	PURGE ON	54	50	3	1

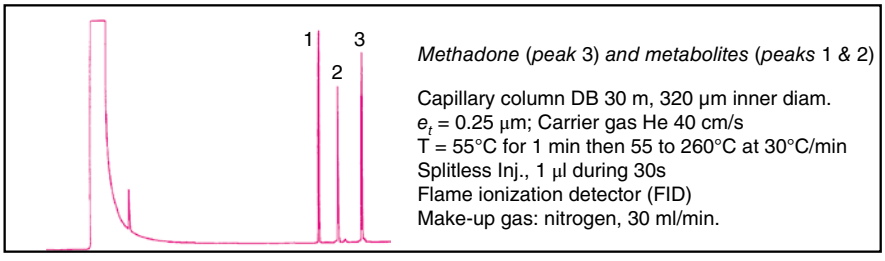


Figure 2.5 Injectors. Above left, injection chamber with splitter (vent 2 regulates the *split*). Below, a typical chromatogram obtained in *splitless* mode. For solvent peaks that are superimposed upon those of the compounds, a selective detector which does not "see" the solvent is recommended.

typically varies between 20:1 and 500:1. Only the smallest fraction, containing an amount of sample equal to the split ratio, will penetrate into the column.

$$Split\ ratio = (split \text{ vent flow rate}) / (\text{column flow rate})$$

Splitless mode is used only for very dilute samples. The contents of the microsyringe are injected very slowly from the microsyringe, during which bleeding valve 2 (Figure 2.5) is maintained in a closed position for 0.5–1 minute in order for the vaporized mixture of compounds and the carrier solvent to be concentrated in the first few decimetres of the column, where the temperature is lower than the boiling point of the solvent. The opening of valve 2 provides an outlet for an excess of sample. The solvent precedes the compounds in the column and on the chromatogram. For either one of these injection techniques, chromatographs with software-controlled parameters (flow rates and pressures) have a *gas saver,* which reduces the *split* ratio after the injection phase.

The split injector may lead to a poor evaluation of mixture composition, following a discrimination between compounds with varying volatilities: the composition of the fraction entering the column is not the same as the composition of the eliminated fraction. Therefore, it is a problem in quantitative analysis, where external standards are not advised.

Cold on-column injection (COC)

The sample is injected directly into the capillary column (*cold on-column* or *COC*). It is vaporized just after introduction. A special microsyringe, whose needle (steel or silica) is just 0.15 mm in diameter, is necessary for penetrating the column, which is cooled to 40°C before being allowed to return to its normal operating temperature. This process, useful for fragile compounds, is difficult to master in order to correctly eliminate the solvent(s). It is known not to discriminate between compounds of different volatilities (Figure 2.6). The injector has a special *duckbill* septum. While it is useful for thermally labile compounds, there is a risk of column pollution if nonvolatile compounds are present.

Programmed temperature vaporization (PTV) injector

This PTV injector is considered to be the most universal. It can be used successively as a cold on-column injector and then as a split/splitless injector. The temperature of the injection chamber can be programmed to create a gradient, e.g. from 20 up to 300°C, in a few tens of seconds (Figure 2.6). So, the advantages of split/splitless injection are combined with those of cold injection onto the column. The advantages are:

- Absence of discrimination due to the needle.
- Use of classic syringes.

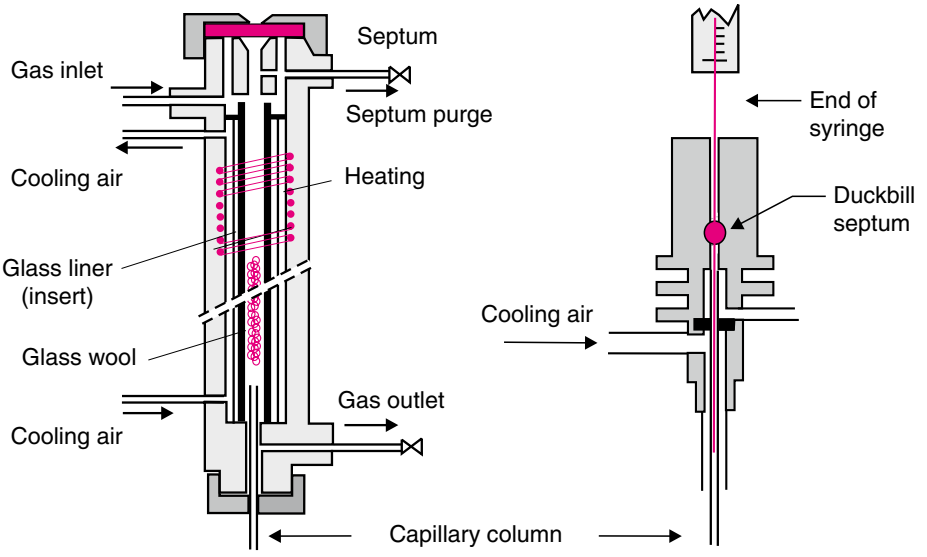


Figure 2.6 PTV injector, with programmable temperature and cold on-column injector. To make fast temperature gradients, the injection chamber is surrounded by a heating element or cooled by the circulation of a cold gas.

- Elimination of the solvent or low-boiling point compounds.
- Larger injection volume.

The three principal modes of operation are named split cold injection, splitless cold injection and injection with elimination of solvent.

Split cold injection: the sample is introduced into the cold vaporization chamber. The vent valve is then immediately opened and the injector is heated. As the sample is not instantaneously vaporized, the solvent and the different compounds penetrate the column in the order of their boiling points. In this way, the column is never overloaded.

Splitless cold injection: this mode is employed for trace analysis. The vent valve is closed during injection. The injection chamber is then heated in order to transfer the sample into the column, which is maintained cold.

Injection with elimination of solvent: the sample is introduced into the cold injector, after which the vent valve is opened. The vent flow rate is very high and can reach 1,000 ml/min in order to eliminate all of the solvent. The injector is then heated to permit transfer of the less volatile compounds onto the column, with the vent valve now being closed (*splitless* mode). In this way, it is possible to inject up to 50 µl in a single injection or up to 500 µl of sample solution in a volatile solvent, over several injections. This method eliminates the preliminary concentration step of the sample prior to injection.

Injection after pyrolysis. Solid samples that are not soluble in solvents may nevertheless be the subject of a GC analysis by placing a device upstream from the injector. This device brings the sample to 600°C to decompose it into smaller molecules. The composition of these molecules will be used to identify the starting sample. The related injector is generally of the PTV type. This method is used for polymers, paints, rubber, additives or textiles.

2.4 TEMPERATURE-CONTROLLED OVEN

The gas chromatograph comprises a temperature-controlled enclosure, or ventilated oven, in which the column is placed. This column can be heated uniformly to more than 400°C, if necessary. The enclosure must have low thermal inertia, enabling a rapid but controlled temperature rise when a temperature gradient is included in the analysis (gradient can reach 130–140°C/min. with excellent stabilization at 0.05°C). By installation of a cryogenic valve fed with nitrogen or carbon dioxide in the liquid state, the oven can be very quickly regulated at low temperature. This valve is essential when we work with an autosampler and a temperature gradient, as temperature cycles must succeed each other at a few minutes' interval between two injections.

2.5 COLUMNS

There are two column types, which differ in their performance: *packed* columns, which are being progressively abandoned, and *capillary*, or *open tube*, columns, which have a very small diameter (Figure 2.7). Their performance levels are different. For packed columns, the stationary phase is deposited by impregnation or by covalent bonds between the stationary phase and the porous support. For capillary columns, a thin layer of stationary phase is deposited onto, or grafted to, the inner surface of the column.

2.5.1 Packed Columns

These columns, less commonly used today, have diameters of 1/8 or 1/4 inch (3.18 or 6.35 mm) and a length of 1–3 m. Manufactured from steel, the internal wall of the tube is treated to avoid catalytic effects with the sample. The carrier gas flow rate ranges from 10 to 40 ml/min.

- In gas–liquid chromatography, the column contains an inert porous support onto which the stationary phase can be impregnated or grafted (between 3 and 25%), depending on the expected application.
- In gas–solid chromatography, the stationary phase is composed of an adsorbent solid (molecular sieve, glassy carbon or porous polymer).

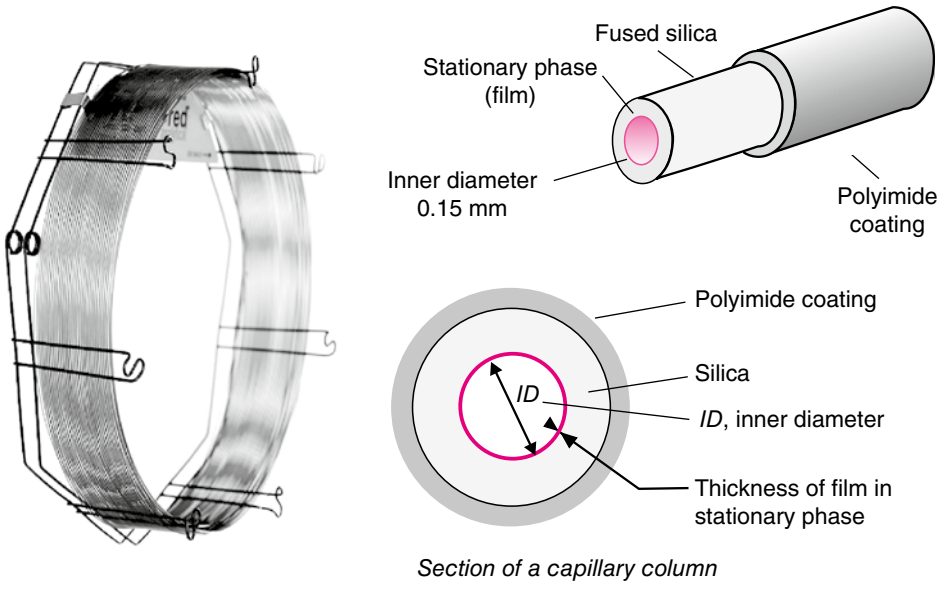


Figure 2.7 Capillary columns. Representation of a 50 m long commercial capillary column coiled around its metal support (Document from the Alltech Associates Inc.) and close-up of a column. At this scale, the thickness of the stationary phase will scarcely be visible.

While less efficient than capillary columns, owing to preferential paths (Eddy factor of the Van Deemter curve), they are still used for some standardized routine analyses. Nevertheless, these are not adapted to trace analyses, and the progressive generalization of GC-MS coupling has greatly reduced their use.

Porous supports are made of spherical particles of around 0.2 mm in diameter. These are obtained from fossilized silicates (kieselguhr, tripoli) whose backbone is chemically comparable to that of amorphous silica. One of the most common goes by the name of Chromosorb®. Their specific surface area is variable (between 2 and 8 m^2/g).

2.5.2 Capillary Columns (Open Tubular)

They are usually made of the highest purity *fused silica* obtained by the combustion of tetrachlorosilane ($SiCl_4$) in an oxygen-rich atmosphere. The internal diameter of the tube used for these columns varies from 100 to 530 µm. The technology is particularly delicate in order to obtain perfectly cylindrical columns with a thickness of 50 µm and lengths going up to 100 m (Figure 2.7). These columns have a polyimide outer coating, which is a thermally stable polymer (T_{max} = 370°C), to make them less fragile, both chemically and mechanically. They can be wound into coils around a lightweight

metal circular support. Some manufacturers offer columns made from a metal capillary (aluminium, nickel or steel) which tolerates high operating temperatures on the order of 450°C, providing that the stationary phase is stable enough. The internal surface of the column is usually treated to promote good bonding for the stationary phase. This could be a chemical treatment or the deposit of a thin layer of alumina or silica gel.

The regular thickness of the stationary phase can vary between 0.05 and 5 μm. It is either simply *deposited* or better yet *grafted* with covalent bonds, possibly followed by a polymerization with cross-linking on the wall. This deposit is obtained by evaporating a solution or by polymerization *in situ* in contact with the wall. These are WCOT (*wall-coated open tubular*) or PLOT (*porous layer open tubular*) columns, depending upon the nature of the stationary phase employed. Columns are particularly stable and can be rinsed periodically with solvents, which enable them to recover their initial performance levels.

To compare or anticipate the behaviour of capillary columns, it is useful to know the *phase ratio* $\beta = V_M / V_S$. By designating *ID* as the internal diameter of the column and d_f as the thickness of the deposited film, an approximate calculation leads to:

$$\beta = \frac{ID}{4d_f} \tag{2.1}$$

The column's capacity is related to the phase ratio but also, for each solute, to its retention time, since *k* is inversely proportional to it. The phase ratio β, accessible from the physical characteristics of the column, and *k (retention factor)* from the chromatogram, help us calculate the partition coefficient *K* of a solute, whose value is generally quite high (1,000 for example) owing to the nature of the mobile phase (gas).

WCOT columns are well suited to mass spectrometry detection and are thus used often. To deposit a film of known thickness, a method consists in filling the column with a solution of stationary phase of known concentration (e.g. 0.2% in ether), so that the desired thickness is obtained after solvent evaporation. This layer can then be cross-linked by a peroxide or by γ irradiation. The process is similar to the application of paint on a surface that has been pretreated to obtain good adherence.

2.6 STATIONARY PHASES

For packed columns, impregnation or deposit techniques can lead to many different stationary phases from a large selection of low-volatility organic compounds. On the other hand, for capillary columns, manufacturing constraints require a much more limited selection of compounds. The current phases correspond in principle to three families: *polysiloxanes, polyethylene glycols*, and *ionic liquids*. Each category can have many structural variants. For the study of optically active compounds, specific phases are used.

Each of these phases can be used between a minimum temperature, below which concentration equilibria occur too slowly, and a maximum temperature, above which degradation of the polymer occurs. The high limit depends on the film thickness and the nature of the polymer.

Squalane is used as a reference phase, since it is the only one that is perfectly defined. On the McReynolds scale, squalane has a polarity of zero (Section 2.10.3). This saturated hydrocarbon ($C_{30}H_{62}$) is derived from squalene, a natural terpene extracted from shark's liver. On this stationary phase, which can be used between 20 and 120°C (using either deposition or impregnation), the compounds are eluted in increasing order of their boiling points (retention time being inversely proportional to vapour pressure). Various grafted phases based upon polyalkylsiloxanes are also almost apolar.

2.6.1 Polysiloxanes

For capillary columns, polysiloxanes (also known as silicone oils and gums) are the mostly commonly used stationary phases. They are based upon a repetitive backbone that consists of two hydrocarbon chains per silicon atom (Figure 2.8). Dozens of different compositions of alkyl or aryl chains (methyl or phenyl), onto which can be incorporated further functional groups (e.g. cyanopropyl, trifluropropyl), are available on the market. Monomers combined in various proportions also convey changes in the properties of stationary phases (polarity, stability range). As such, dimethylpolysiloxanes have a very large temperature range, from –50 to 300/325°C.

2.6.2 Polyethylene Glycols (PEG)

The best known representative of this family is Carbowax®. These polar polymers (M_r = 1,500–20,000 – for Carbowax 20M) can be used for deposition, impregnation or as grafted phases, for temperatures not exceeding 260°C, depending on the type (Figure 2.8).

Column catalogues present optimized phases for special applications: separation of sulfur products, chlorinated pesticides, permanent gases, triglycerides, PAHs, petroleum products or fatty acid esters. Some columns accept high temperatures up to 450°C (e.g. DEXSIL 400 or PETROCOL). The original applications include simulated distillation in the oil industry, replacing conventional distillation, which can take up to 100 hours per analysis.

2.6.3 Ionic Liquids (IL)

These stationary phases, developed more recently than the previous ones, are made of di- or tri-cationic salts (ammonium or phosphonium) associated with sulfonate- or imide-type organic anions (Figure 2.8). For GC, we use viscous ionic

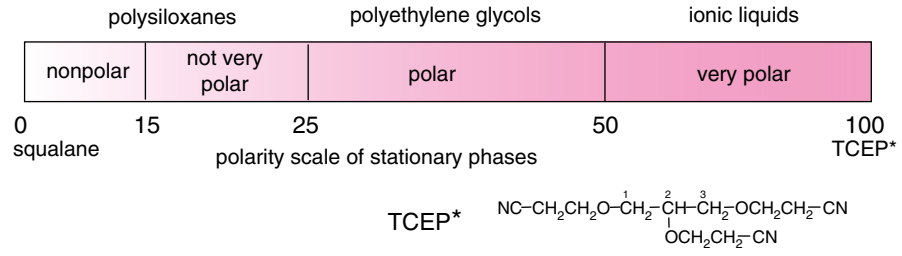


Figure 2.8 Polarity scale of stationary phases in GC and summary of the three types of stationary phase. The polarity scale goes from squalane (polarity 0 by definition) to polarity 100 for TCEP (tricyanoethoxypropane). (a) *Polysiloxane structure (silicones),* (b) *polyethylene glycols*. Many phase compositions of this type, used in impregnation or grafting. (c) e.g. a dicationic tetraalkylphosphonium type ionic liquid associated with an imide (here, IL60); max. temperature 300°C.

liquids with a melting point lower than ambient temperature, due to the fact that these are stable from a heating standpoint and not very sensitive to oxygen. These polar phases, deposited and not grafted on the column surface, are suitable for the analysis of neutral, basic or diverse compounds (fatty acid esters). Elution diagrams are different from those obtained with Carbowax, which makes them useful in two-dimensional GC. Their near-zero vapour pressure makes them useful for GC-Mass Spectrometry.

2.6.4 Chiral Stationary Phases

An organic molecular compound, whose formula demonstrates a centre of asymmetry, leads at our scale, i.e. macroscopic, to a mixture of the two possible enantiomers *R* and *S*, in equal quantities or not. Their physical properties are identical, but their behaviour regarding polarized light is what distinguishes them, hence the name of optical activity. In space, these two forms cannot be superimposed. One is the mirror image of the other.

The separation of optical isomers is an important application of chromatography. GC optimizes separations more quickly than other chromatographic techniques (see Chapters 4, 5, and 6). Therefore, it holds a privileged spot in this field, even though the selectivity factor for enantiomer pairs remains close to 1.

Two separations methods exist: *direct process* and *indirect process*.

Direct process

The analyte, supposedly composed of two enantiomers (*R* and *S*), is chromatographed on a chiral stationary phase composed of a single enantiomer (e.g. *R*-type). Between the analyte enantiomers and the stationary phase, reversible interactions of the *R* (analyte)/*R* (stationary phase) and *S* (analyte)/*R* (stationary phase) are created, with slightly different stabilities. This leads to distinct migration speeds for the two enantiomers. Finally, this leads to two peaks on the chromatogram for this single analyte (Figure 2.9), whose areas reflect the abundances of the *R* and *S* forms, from which the analyte's optical purity is calculated.

The *optical purity* of the analyte refers to its enantiomeric excess (*e.e.*), calculated from the following relationship, where S_R and S_S refer to the areas of the two enantiomer peaks:

$$Optical\ Purity\left(e.e.\%\right) = 100\frac{\left|S_R - S_S\right|}{S_R + S_S} \tag{2.2}$$

According to this formula, a pure chemical compound present as a racemic mixture will yield two peaks equal in size, each corresponding to an enantiomer. Its optical purity will thus be equal to 0.

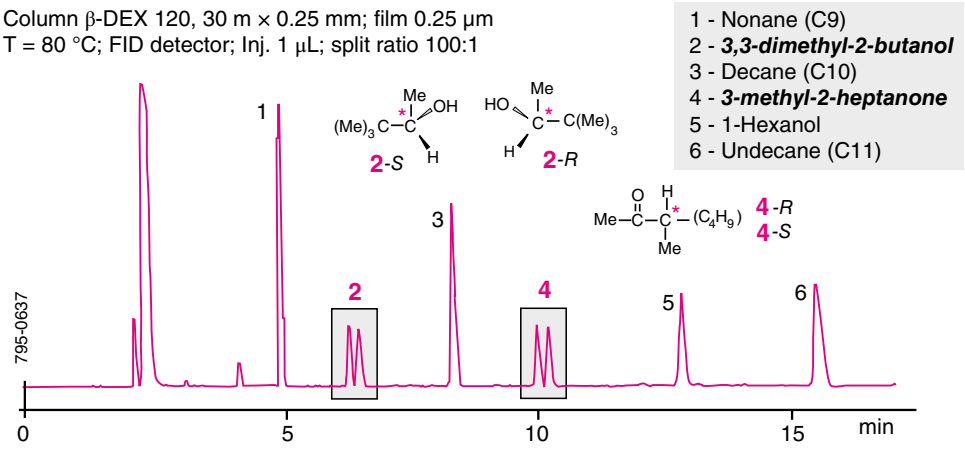


Figure 2.9 Example of a separation with a chiral phase which contains grafted cyclodextrins. The use of a chiral column to separate a racemic mixture of compounds (alcohols 2 and 4). This chromatogram in isothermal mode enables the calculation of retention indexes for the separated compounds.

Indirect process

Unlike the direct process, this process consists in creating a chemical reaction between the analyte and a chiral reagent, prior to the injection, to form diastereomers. As these have different physical properties, they should be separable with conventional phases. This longer process is not used as often as the direct one. It may cause a partial racemization of the analyte.

Chiral stationary phases for GC are mainly obtained from cyclodextrins (Figure 2.10), ring-shaped macromolecules composed of six, seven or eight units of *D*(+)-glucose. They look like cones, with a hydrophobic central cavity, which enables the selective and reversible inclusion of a large variety of compounds, such as diastereomeric complexes.

After possible chemical transformation, cyclodextrins are either deposited on the internal surface of the capillary column or incorporated into a poly(dimethylsiloxane)-type polymer or grafted onto the silanol functions of silica via a short carbonated chain. These can be used up to 200°C. Above this temperature, analytes can racemize.

As a reminder, the other chiral vectors for GC, much less in use, are crown ethers and some asymmetrical diamides, which are progressively being abandoned (Figure 2.10).

2.6.5 Solid Stationary Phases

These phases are made from a variety of adsorbent materials: silica or alumina deactivated by mineral salts, molecular sieves 5 Å (0.5 nm), porous glass or polymers, or graphite (e.g. Chromosorb® 100, Porapak®). Capillary columns made by

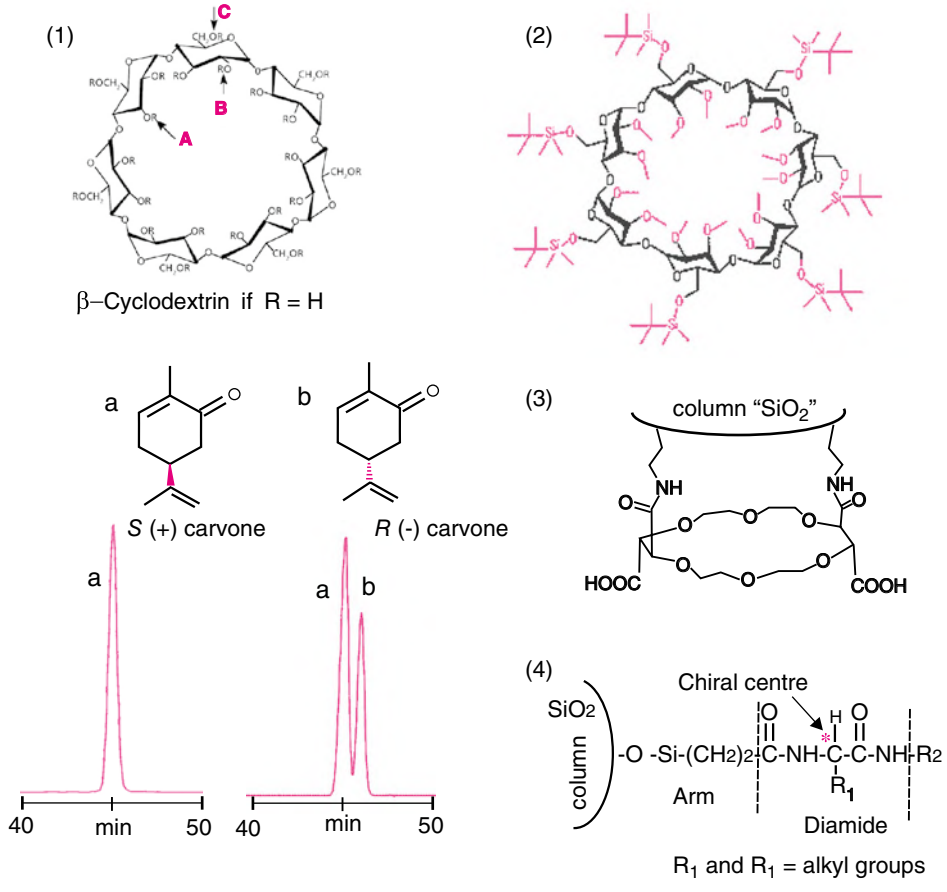


Figure 2.10 Chiral stationary phase in GC. Among the three chiral vectors encountered in GC (**1 and 2**, β-cyclodextrins, **3**, crown ethers, **4**, diamides), cyclodextrins are by far the most commonly used. They include three types of sites: **A**, axial hydroxyl, **B**, equatorial hydroxyl and **C,** hydroxymethyl. Their reactivities are sufficiently different to enable selective reactions and thus obtain some 50 phases (e.g. **2**, ungrafted cycloSil-B phase from Chromoptic). Partial chromatograms of natural extracts, demonstrating the separation of optical isomers of carvone.

deposition of these materials in the form of a fine porous uniform layer are called PLOT. They are employed to separate gaseous or highly volatile samples. Columns containing graphitized carbon black have been developed for the separation of N_2, CO, CO_2, and very light hydrocarbons. The efficiency of these columns is very high (Figure 2.11).

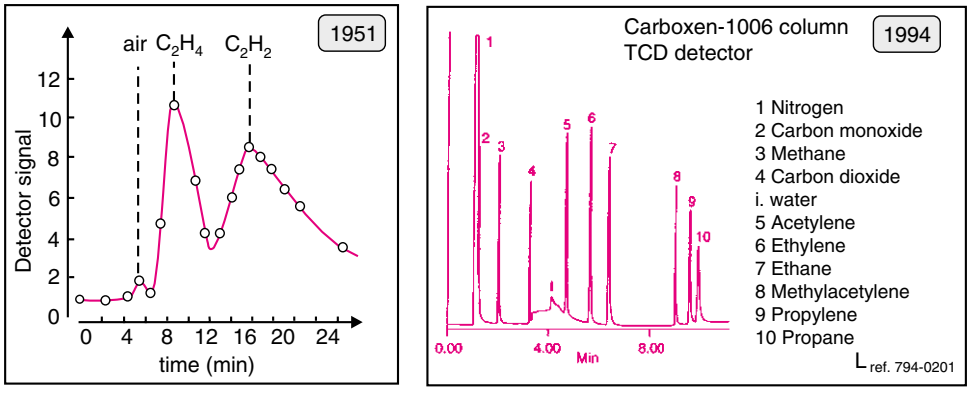


Figure 2.11 Gas analyses. Left, one of the earliest ever chromatograms, obtained point by point and representing a mixture of air, ethylene and acetylene separated on silica gel (E. Cremer and F. Prior, *Z. Elektrochem.* 1951, 55, 66). Right, an analysis of gas on a modern PLOT column (reproduced courtesy of Supelco).

Historically, silica gel, a thermostable material that is insensitive to oxygen, was one of the first compounds to serve as a stationary phase for GC columns (Figure 2.11). Today, solid phases have become much more elaborate.

2.7 MAIN DETECTORS

Downstream from the column, the chromatograph has one last device: the detector, which provides a signal with the passage of each analyte. All of the signals will form the chromatogram. This detector uses variations in physical amplitude related to the presence of a solute, in addition to that of the carrier gas. Some detectors are *universal*; that is they are sensitive to practically every solute, while others are *selective* detectors that are sensitive only to specific compounds (Figure 2.14). The most effective rely on the adaptation of spectral methods, or on mass spectrometry, which not only helps generate the conventional chromatogram but also helps identify each solute (see the Chapter 16 on Mass Spectrometry). Nevertheless, this is expensive, and we always find a number of conventional detectors in laboratories.

2.7.1 Universal or Near-Universal Detectors

Flame ionization detector (FID)

Considered as almost universal for organic compounds, this is the detector *par excellence* for GC. The gas flow issuing from the column passes through the flame of a small burner fed by a mixture of hydrogen and air. When a compound is eluted, its combustion results

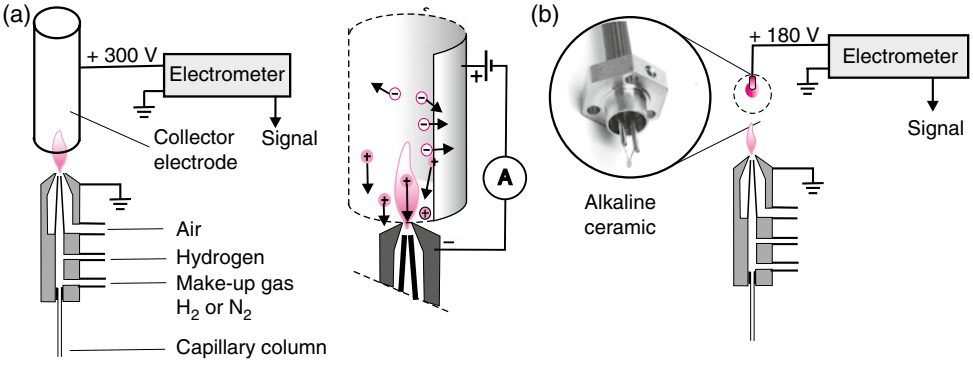


Figure 2.12 FID detector (a) and NPD detector (b). The make-up gas (or filling gas) is useful when the flow rate of the capillary column is too small. The electrometers used with detectors enable the measurement of intensities that would too small for a galvanometer. Many variants exist depending on the manufacturer. (Source (a): Modified from Cremer, E. and Prior, F. (1951), Anwendung der chromatographischen Methode zur Trennung von Gasen und zur Bestimmung von Adsorptionsenergien. *Zeitschrift für Elektrochemie und angewandte physikalische Chemie*, **55**, 66–70. https://doi.org/10.1002/bbpc.19510550115. (b): Courtesy of Supelco.)

in the release of ions and charged particles responsible for the passage of a very weak current (10^{-12} μA) between two electrodes (p.d. of 100–300 V). The signal is transformed by an electrometer into a measurable voltage (Figure 2.12). One end of the burner, held at ground potential, acts as a polarization (ground) electrode, while the ring-shaped second electrode surrounds the flame.

For organic compounds, the intensity of the signal is sensitive to the *mass flow* of the sample, except in the presence of heteroelements, such as halogens. The latter may change the response and several simple compounds, such as water, carbon dioxide or ammonia, do not give any response. Thus, the area under the peak reflects the mass *m* of the compound eluted (dm/dt integrated between the beginning and end of the peak). An FID detector is not affected by variations in flow rate, which can lead to errors with some types of detectors. The sensitivity of this detector is expressed in Coulombs/g of carbon, and the dead volume of the detector is null. The detection limit is in the order of 2 or 3 pg/s, and the linear dynamic range reaches 10^8; however, concentrated solutions do not lead to the best resolution.

To evaluate the overall quantity of volatile organic compounds (VOCs) in polluted air, there exist small portable instruments housing a flame ionization detector that allows the measurement of the *carbon factor* of the atmosphere examined, without prior chromatographic separation.

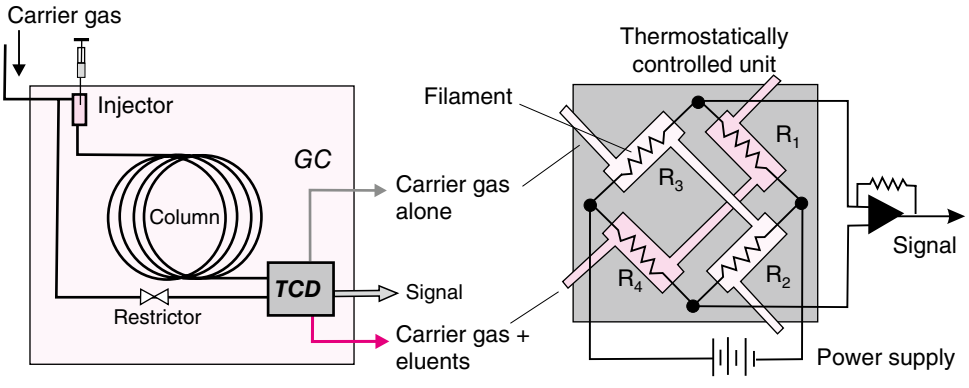


Figure 2.13 Thermal conductivity detector. Left, layout demonstrating the dual circulation of the carrier gas. Right, a katharometer unit with the principle of its electrical connections in a Wheatstone bridge type assembly.

Thermal conductivity detector (TCD)

This universal detector, developed in the early days of GC for packed columns, is still in use now. Easy to build, it exists in a number of variants (Figure 2.13), including miniaturized forms (μ-TCD) for capillary columns.

Its operating principle is based on the thermal conductivity of gas mixtures as a function of their composition. The main part of this detector is the *katharometer*, a thermostatted metal unit that is brought to a temperature slightly higher than that of the column and which includes thermistors located in tiny cavities. In the given example, the katharometer includes four thermistors, placed two-by-two and fed as indicated either with carrier gas sourced upstream from the injector or with the mobile phase downstream from the column. When a solute elutes, the conductivity of the mixture (carrier gas + compound) decreases with respect to that of the carrier gas alone. The thermal equilibrium is disrupted and this results in a variation in the resistance of one of the filaments, which is proportional to the *concentration* of the compound in the carrier gas. The dynamic range of this detector extends over some six orders of magnitude, and while its sensitivity is quite average (from ng to mg), it is being used more frequently thanks to the rise of micro-GC (see Section 2.9.2).

Mass spectrometry detector (MSD)

For several years now, the rise in mass spectrometry has favoured the use of the coupled GC/MS technique, which consists in connecting the chromatograph to a *mass detector* (a low-resolution mass spectrometer, see Chapter 16 for more details). Its use as a universal detector for capillary GC has the advantage of leading to a fragmentation spectrum of each eluted compound and, therefore, to their identification when using the spectral libraries. There are several types

of interfaces between the chromatograph and the mass spectrometer. The most developed ionization mode is electron impact (EI). From the *total ion current* (TIC), a chromatogram can be plotted which represents all of the compounds eluted. By choosing a particular ion (*selective ion monitoring*, or SIM), a selective chromatogram can be produced. This method has become essential, notably in the context of environmental studies. Nevertheless, it requires the use of high-performance columns (*ID* = 0.1–0.2 mm) with no bleeding.

2.7.2 Selective Detectors

Nitrogen phosphorus detector (NPD)

This specific detector is sensitive to nitrogen (N) or phosphorus (P) compounds. It comprises a small ceramic cylinder doped with an alkaline salt (e.g. rubidium sulfate). A voltage is applied to maintain a small plasma (800°C) through the combustion of an air/hydrogen mixture (Figure 2.12). Compared with FID, the flame is much smaller. Compounds containing nitrogen or phosphorus give, fairly specifically, decomposition fragments transformed into negative ions. These ions are then received by a collector electrode. The nitrogen present in air is inactive. Detector sensitivity is typically 0.1 pg/s for nitrogen- or phosphorus-containing analytes, with a linear range of five orders of magnitude. However, it varies a lot with settings.

Electron capture detector (ECD)

This detector is considered to be selective because it is much more sensitive to halocarbon groups. A flow of nitrogen gas that has been ionized by electrons generated from a low-energy β^- radioactive source (a few mCi of ^{63}Ni) passes between two electrodes maintained at a potential differential of around 100 V (Figure 2.14). At rest, a base current I_0 is generated, mainly due to free and very mobile electrons. If molecules (*M*) containing a halogen (F, Cl, Br) cross the zone between the two electrodes, they capture thermally excited electrons to form heavy negative ions, which by consequence are much less mobile.

$$N_2 \xrightarrow{\beta^-} N_2^+ + e^-$$
$$M + e^- \rightarrow M^-$$
$$M^- + N_2^+ \rightarrow M + N_2$$

The measured intensity decreases exponentially by following a law of the type $I = I_0 \exp[-kc]$. The linear range is about four orders of magnitude with nitrogen as the make-up gas. The presence of a radioactive source in this detector means that it is subject to special regulations (inspection, location and maintenance visits). This detector is often used for analyses of chlorinated pesticides and polychlorinated biphenyls.

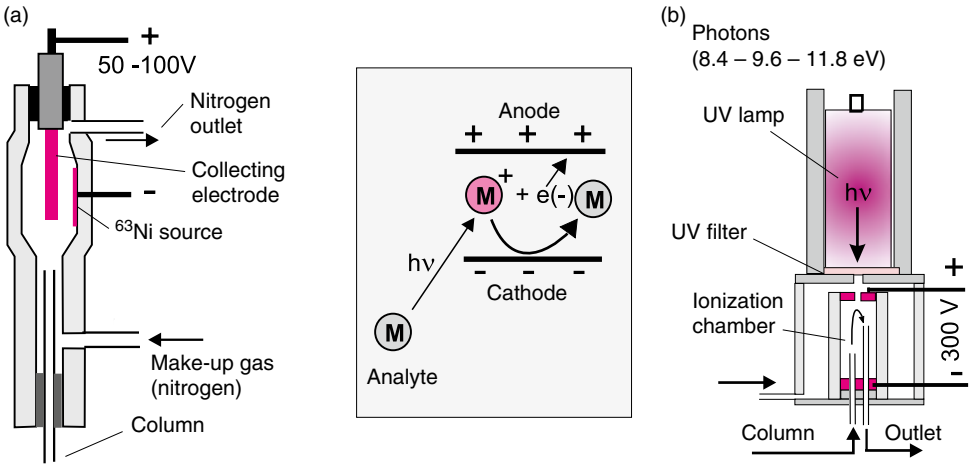


Figure 2.14 (a) Electron capture detector (ECD) and (b) photo-ionization detector (PID). The PID contains a filter, used to select the photon energy in order for analytes to be the only ionized species apart from carrier gas molecules (LiF at 11.8 eV, or sapphire at 8.4 eV). The ionization mechanism is reversible.

Make-up gas. To provide satisfactory response and increase their sensitivity, the above two detectors described above should be supplied with a gas flow of at least 20 ml/min, which is far greater than that in capillary columns. This flow rate is attained by mixing a make-up gas with the gas coming out of the column outlet. This make-up gas is either identical to the carrier gas or different from it (inexpensive nitrogen gas).

Photo-ionization detector (PID)

This detector is fairly selective but it has only a narrow range of applications. It is suitable for hydrocarbons as well as for sulfur or phosphorus derivatives. The operating principle consists in provoking ionization of the analytes by irradiation with a UV lamp emitting high-energy photons (8.4–11.8 eV). Photo-ionization occurs when the energy of the photon is greater than the first ionization energy of the compound (Figure 2.14). A photon of 9.6 eV can, for example, ionize benzene (PI_1 = 9.2 eV) but not isopropanol (PI_1 = 10.2 eV), which will therefore not be very visible on the chromatogram. Electrons released by an electrode connected to the terminal of an electrometer are collected for concentration measurements.

This detector can function at more than 400°C and is not destructive, as the ionization is reversible and affects only a small fraction of the molecules of each compound.

2.7.3 Detectors Providing Structural Data

None of the detectors previously described yield any information as to the nature of the eluted analytes. At most, they are selective for a certain category of compounds. Identification involves the use of an internal calibration based on retention times or requires the knowledge of retention indexes (see Section 2.10). When the chromatogram has peaks that are close together, a confusion of identity could occur.

To counteract this, several complementary detectors could be combined with each other (Figure 2.15), or we could select a detector able to provide structural information based on spectroscopic data or about the elemental composition of the analytes. The

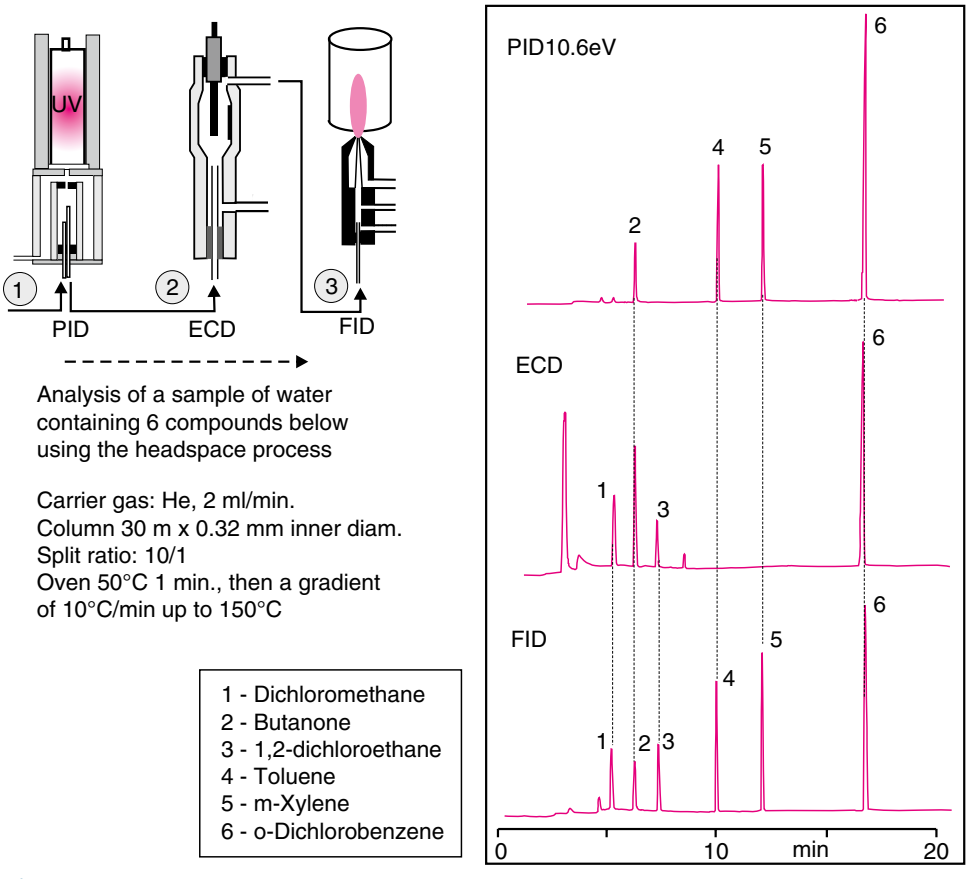


Figure 2.15 Three detectors connected in series. At the outlet of a capillary column, either in series or in parallel and depending upon whether any of them destroy the sample, several detectors can be installed. Here, three chromatograms of an injected mixture as obtained from each detector. Note that the sensitivity varies significantly from one detector to another.

retention time and specific characteristics for each compound can then be determined. The mass detector is the best example, yet other combinations of techniques have been tested. Such is the case for atomic emission detection with a plasma torch: compounds at the column outlet flow into a plasma (see Chapter 14) where the temperature is sufficient to increase atomic or ion populations in the excited state. When returning from the excited state, these atoms or ions emit specific radiations, which help to identify them. Tests have been conducted to couple GC with an infrared spectrometer (GC-IR). This pairing leads in principle to mid-infrared spectra of the eluted analytes. An entire set of technological adaptations have made the existence of coupled methods possible. Therefore, potential applications are more numerous than for each of these when taken alone.

2.8 OPTIMIZATION OF A SEPARATION

If we assume that the optimal stationary phase has been chosen, the length and internal diameter *ID* of a capillary column as well as its stationary phase film thickness (d_f) must also be taken into account. The conditions for a good separation of the analytes need to be found without increasing the analysis time. From a practical standpoint, in GC, we can only change the temperature and the carrier gas flow rate. In both cases, retention factors k and selectivities α are not much changed. For the carrier gas, we choose a flow rate such that its speed $\bar{u}$ is close to the optimal value of the Van Deemter curve. For volatile compounds, we choose a column with a weak phase ratio (β < 100), hence with a thick film. Inversely, we choose a thin-film column for less volatile compounds.

However, we must not forget that coupled GC-MS does not necessarily require greatly optimized separation of chromatographic peaks (sufficient resolution), which is a significant time gain for the chromatographer. Moreover, in a series of analyses, if we can avoid the use of a temperature gradient, we can thus eliminate the time necessary to return to initial conditions in order to conduct the following analysis.

2D GC. Optimization sometimes involves dual chromatography on two different stationary phases (for example, polar and nonpolar) with a single set-up including a carrier gas inlet shared by the two columns. The first chromatograph is used to isolate a peak, possibly corresponding to an unresolved mixture, which then goes through a second column for another separation.

2.9 FAST CHROMATOGRAPHY AND MICROCHROMATOGRAPHY

2.9.1 'Fast' and 'Ultra-Fast' Chromatography

In general, conventional chromatography is a slow method of analysis. For application reasons (such as control analysis, reaction monitoring, etc.), it may be beneficial to shorten analysis times. To reduce these times, use a shorter column, as retention times

are proportional to column length (Table 2.1). Due to decreased efficiency, the phase ratio β must be increased by choosing a thinner stationary phase film (*ID* = 100 µm, film = 0.1 µm, β = 250). Hydrogen must be used as the carrier gas (see Van Deemter curve). The last factor that we can easily regulate is the oven temperature (the higher it is, the shorter the analysis). Be aware, however, of the first peaks appearing, since good separation requires a lower temperature. Therefore, the solution is a temperature gradient. Devices allow for ramps that can go up to 100°C/min. For volatile compounds and with various column designs, covered with a heat-resistant outer sheath, the temperature can be increased more sharply (200°C/20 s). As a result, retention times are reduced significantly (Figure 2.16). This type of *fast chromatography*, sometimes called *high-speed GC,* finds its main use in control analyses.

Table 2.1 Comparison of conventional, fast and ultra-fast GC set-ups.

GC type	Ramp (°C/min)	Analysis time (min)	Peak width (s)	Column length (m)	Internal diameter (µm)
Conventional	Conventional oven (1–20)	~30	5–10	15–100	250–320
Fast chromatography	Conventional oven (20–100)	5–10	0.5–5	5–15	100–250
Ultra-fast chromatography	Resistive heating (60–1,200)	~1	0.05–0.2	2–5	50–100

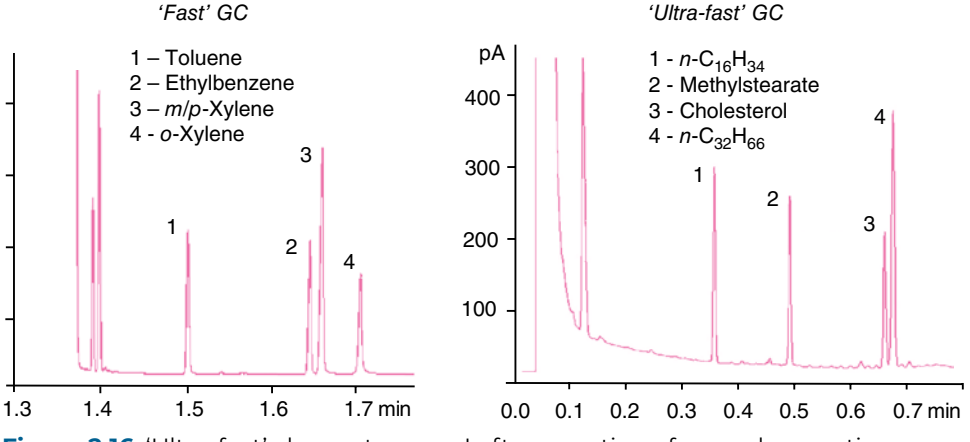


Figure 2.16 'Ultra-fast' chromatogram. Left, separation of several aromatic compounds ('fast' chromatography according to a document from Thermo Electron Corp.); right, example of a chromatogram obtained under 'ultra-fast' chromatography conditions according to a document from Aviv Analytical.

The detector must be able to follow the rapid variations in concentration almost immediately, i.e. at the moment of each analyte's elution. For detection by mass spectrometry, there is good reason to be attentive to the sweep speed of the *m/z* ratio; a slow sequential sweep may lead to a situation in which the concentration in the ionization chamber is not the same from one end of the recording to the other. TOF-MS (Time-of-Flight Mass Spectrometry) does not suffer from this drawback.

2.9.2 Micro Gas Chromatography

In parallel with laboratory equipment, portable devices (μ-GC) have been developed for fast analyses in the field (Figure 2.1). These devices, sometimes referred to as 'noses,' must be light and small, despite a carrier gas reserve enabling their stand-alone use. The detector used most often is the katharometer with its detection and quantification limits, or more specific variants (μ-katharometer). The flame ionization detector, requiring an additional gas source, or the mass detector would make heavier and more complex devices. These portable devices may include several analytical modules, operating in parallel for multiple, simultaneous analyses. For each module, simply choose a column with a different polarity and different temperature conditions. In fact, each column is covered with a metal envelope supplied with an electrical current, which enables its temperature programming.

2.10 RETENTION INDEXES AND STATIONARY PHASE CONSTANTS

These parameters have been developed to pursue at least three objectives:

- To identify a compound by a more general characteristic than its retention time under predefined conditions. As a result, a *system of retention indexes* has been developed; it is an efficient and cheap means by which to avoid certain identification errors.
- To follow the evolution of a column's performance over time.
- To classify all stationary phases in order to simplify the choice of the column best adapted to a particular kind of separation problem. The polarity or chemical nature of a stationary phase does not allow prediction of which column will be optimal for a given separation. For this, the behaviour of stationary phases with respect to several reference compounds should be examined, in order to determine *stationary phase constants*.

2.10.1 Kovats Straight-Line Relationship

Retention indexes are determined using the following general principle. When a mixture containing compounds belonging to a homologous series of *n*-alkanes is injected onto a column maintained in isothermal mode, the resulting chromatogram is

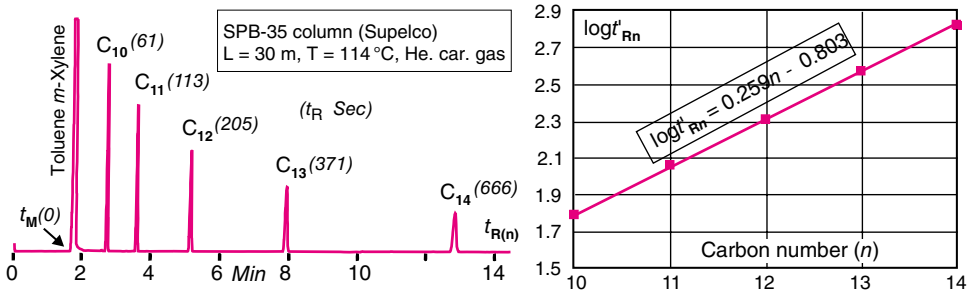


Figure 2.17 Kovats straight line graph. Left, isothermal chromatogram for a series of five *n*-alkanes (C_{10}–C_{14}). Right, Kovats' relationship for the chosen stationary phase and for the predetermined analytical conditions.

such that the logarithm of the adjusted retention times $t'_{R(n)}$ increases linearly with the number *n* of carbon atoms present in the corresponding *n*-alkane (Figure 2.17).

On a graphical representation, the carbon number *n* versus log $t'_{R(n)}$ usually yields a series of well-aligned points:

$$\log\left(t_{R(n)} - t_M\right) = \log t'_{R(n)} = a \cdot n + b \qquad (2.3)$$

The adjusted retention time $t'_{R(n)}$ corresponds to the retention time t_R of an alkane having *n* atoms of carbon, minus the dead time t_M; *a* and *b* are numerical coefficients. The slope of the graph obtained depends on the overall performance of the column and the operating conditions of the chromatograph.

This expression follows on from another linear relation seen in thermodynamics linking the variation in free energy and the equilibrium constant *K* ($\Delta G = -RT \ln K$), for a homologous family of compounds in which each term differs from the preceding one by an additional CH_2 unit. Since $K = k\beta$, $t'_R = Kt_M/\beta$; thus log t'_R will increase with ln *K* for the homologous family: ln $t'_R = \ln K + \ln\left(t_M/\beta\right)$.

2.10.2 Kovats Retention Index

A compound (X) is now injected onto the column *without changing the settings of the instrument*. The resulting chromatogram will enable I_x, the *Kovats retention index*, to be calculated for X on the specific column employed: this is equal to 100 times the equivalent number of carbon atoms n_x of the 'theoretical alkane' having the same adjusted retention time as X. Two methods can be used to find the number n_x of equivalent carbons of X.

The first is based on the Kovats relationship obtained above (Figure 2.17). This leads to a calculation of n_x (therefore I_x), using a spreadsheet, for example.

The second can be used to calculate I_x directly from the adjusted retention times of the two *n*-alkanes (*n* and *n* + 1 C) that bracket compound X on the chromatogram:

$$I_x = 100n + 100\frac{\log t'_{R(X)} - \log t'_{R(n)}}{\log t'_{R(n+1)} - \log t'_{R(n)}} \qquad (2.4)$$

In contrast to the Kovats regression line, the retention indexes depend only on the stationary phase and not on the settings of the chromatogram. In particular, they do not depend on retention times.

In practice, to ensure that the experimental conditions for the two injections are uniform, compound X and the *n*–alkanes mixture are co-injected (Figures 2.9 and 2.18).

The chromatogram that gives the Kovats relationship for a given stationary phase can also serve to evaluate expected column performance. For this, the *separation number*, also known as the *trennzahl number* (TZ), is calculated from Eqs. (2.5) or (2.6). The two retention times occurring in these relationships relate to two successive alkanes differing by one carbon number (*n* and *n* + 1 atoms) or to two compounds of the same type. The separation number indicates how many compounds could be separated reasonably well by the column in the arbitrary elution interval of these two compounds. The alkanes whose elution times are on either side of that of the analyte are chosen. For the chromatogram in Figure 2.18, *TZ* is around 30.

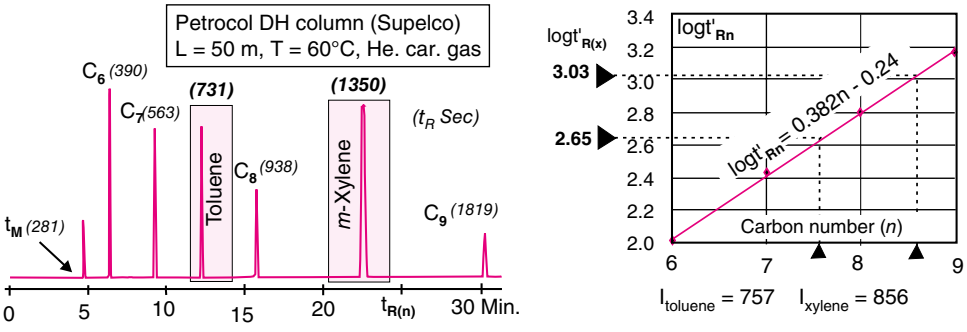


Figure 2.18 Kovats retention index (I = 100n_x) on a column in isothermal mode. The equivalent number of carbons n_x is found from the logarithm of the adjusted retention time $t'_{R(X)}$ of *X*. The chromatogram corresponds to the injection of a mixture of four *n*-alkanes and two aromatic hydrocarbons. The values in italics indicate the retention times given in seconds. By injecting this mixture periodically, the modifications to the Kovats indexes of these hydrocarbons enable us to track the column's performance. With temperature programming, we can still plot this relationship using an adjusted formula, though this entails a reduction in precision.

$$TZ = \frac{t_{R2} - t_{R1}}{\delta_1 + \delta_2} - 1 \tag{2.5}$$

or

$$TZ = \frac{R}{1.18} - 1 \tag{2.6}$$

There are tables of retention indexes of compounds currently in general use on the most common stationary phases. If several retention indexes of the same compound obtained on different stationary phases are available, then this unique collection of values could then characterize the compound with greater certainty. However, identification by retention index is not as reliable as using coupled techniques such as GC/MS (see Section 2.7.1), which require more expensive equipment.

It should be remembered that the above calculations for retention indexes imply that the measurements were made under isothermal conditions. With temperature programming, they still give good results when substituting retention times for the corresponding logarithms in Eq. (2.4).

Retention time locking. It is obviously difficult to identify compounds whose retention times are very close and whose mass spectra are almost identical (certain forms of isomers). A current method consists in selecting an internal standard or a compound known to be present in all of the samples to be analysed. Through the use of computer software, the value of its retention time is locked for the different analyses, even if these are undertaken on different apparatuses. The effect of this is to conserve the retention times of the other compounds of the mixture, facilitating their identification. This approach, which avoids use of retention indexes, is possible with modern GC instruments and is known as *Retention Time Locking* (RTL).

2.10.3 McReynolds Constants for Stationary Phases

To evaluate the behaviour of a stationary phase, a comparison of the Kovats indexes for five reference compounds belonging to different structural classes is made on the studied phase as well as on squalane, chosen as the reference phase for this calculation. The five indexes on a column using squalane, the only reproducible nonpolar phase since it is formed from a pure material, have been established once and for all (Table 2.2).

The five McReynolds constants for a given stationary phase are obtained (Eq. (2.7)) by calculating the differences observed for each of the substances tested between their Kovats indexes on squalane (I_{Squalane}) and on the stationary phase being studied (I_{Phase}):

$$\text{MCReynolds constant} = \Delta I = \left(I_{\text{Phase}} - I_{\text{Squalane}} \right) \tag{2.7}$$

Table 2.2 McReynolds constants (ΔI) for several stationary phases normalized to squalane.

Stationary phase	Benzene X′	1-butanol Y′	2-pentanone Z′	Nitropropane U′	Pyridine S′
Squalane	0	0	0	0	0
SPB-Octyl	3	14	11	12	11
SE-30 (OV-1)	16	55	44	65	42
Carbowax 20M	322	536	368	572	510
OV-210	146	238	358	468	310
Kovats index for the five reference compounds above (X′, Y′, Z′, U′, S′) on squalane					
$I_{squalane}$	653	590	627	652	699

The sum of the five calculated values, using Eq. (2.7), has been used to define the overall polarity of the phase under study.

Each of the test compounds yields specific information regarding the stationary phase: benzene for the inductive effect, pyridine for H^+ proton accepting, butanol for hydrogen bonding, nitropropane for dipolar interactions, etc.

These constants, which are related to molecular structures, allow an appreciation of the interactive forces between stationary phase and solute as a function of compound class. A high index suggests that the stationary phase strongly retains the compounds that contain the corresponding organic functions. This generally leads to improved selectivity for this type of compound. Thus, to separate an aromatic hydrocarbon from a mixture of ketones, we would select a stationary phase whose McReynolds constant for benzene is sufficiently different from that for butanone. These differences in retention indexes are provided by manufacturers for use by chromatographers (Table 2.2). McReynolds constants have replaced Rohrschneider constants, which were based upon the same principle but used certain different reference compounds.

KEY POINTS OF THE CHAPTER

1. As long as settings remain unchanged, GC chromatographs are able to reproduce the retention time of a compound to the nearest second, in the case of several successive injections. This can only be obtained by perfect control of all parameters: temperatures, flow rates, pressures, and carrier gas purity.
2. The range of use of GC depends on the volatility of compounds. The upper limit of this range is reached if molecular weight exceeds the 500 Da boundary, or if hydrogen bonds or dipole–dipole interactions are created between compounds.

3. At equal volatility, compounds elute from the column by following the order of their distribution coefficients in the stationary phase. The carrier gas does not participate in concentration equilibria. The two main factors determining the behaviour of an analyte are its volatility and analyte-stationary phase interactions.
4. The carrier gas must be free from oxygen in order not to alter the analytes, which are weakened when they are brought to a high temperature in the instrument's oven. In general, hydrogen is chosen as the carrier gas. It enables faster analyses than nitrogen or helium, without altering the *efficiency* (*N*) of the separation.
5. An abundance of capillary columns for GC is on the market, either for general use or for specific separations. They are classified according to their retention index and their polarity, as defined by their McReynolds constants. For the GC-MS technique, we choose grafted, cross-linked, and low-bleed columns, which are able to interact with the molecules that require separation.
6. A new column is accompanied by a document specifying its *efficiency* (*N*), with the conditions of acquisition of that value, as well as its retention indexes for five test compounds, with different, universally used chemical properties.
7. Detectors are either general, such as FID, which is by far the most common, owing to its sensitivity and its linearity, or they are adapted to a category of compounds or are even specific to a single compound, thus simplifying the chromatograms when the matrix is complex and enabling a better quantification of the analytes in question.
8. The Kovats index of a compound is calculated from the retention times of adjacent *n*-alkanes. It is of interest because it depends only on the stationary phase and not on other characteristics of the column or apparatus. Index tables help to identify compounds by comparison of their retention indexes, without worrying about retention times, which are variable.

PROBLEMS

1. Show that for a capillary column, the average flow can be calculated from the following formula:

$$D_{\text{mL/min}} = \bar{u}_{\text{cm/s}} \times 0.47\ \text{ID}^2_{\text{(mm)}}$$

(where $\bar{u}$ represents the average linear velocity in a column of internal diameter *ID*).

2. A comparative study of the evolution of the retention factors of *n*-undecane and *n*-tridecane, as a function of the temperature of the GC column, at a constant carrier gas flow rate of 3 ml/min, gave the following results:
n-decane: $\log k_1 = -6.58 + 2{,}450 \cdot T^{-1}$
n-tridecane: $\log k_2 = -7.91 + 3{,}010 \cdot T^{-1}$

a. Justify the general form: $\log k = -A + B/T$

b. At what temperature T_1 would these two solutes coelute? Which of the two would elute first if we work at a temperature T below T_1? Same question if T is over T_1?

c. At what temperature T_2 will the separation factor be equal to 2?

d. Knowing that the column phase ratio is equal to 250, calculate the Nernst distribution factors K_1 and K_2 of the two solutes when we work at 150°C.

3. We propose determining the maximum efficiency of a capillary column, with the following characteristics:

L = 12 m, ID = 200 µm, stationary phase: methyl-phenyl polysiloxane, d_f = 0.33 µm
Operating conditions: carrier gas: H_2; injector temperature: 250°C; oven temp.: 100°C; FID temp.: 250°C; *split* flow rate: 30 ml/min.

We conduct several injections of an *n*-undecane solution in pentane (injected volume: 0.5 µl) while changing the carrier gas flow rate. Calculation software gives the Golay curve equation:

$H = 5.44/u + 0.004u$ where H is in mm and u in mm/s.

a. For what value of u does the height equivalent of a theoretical plate go through a minimum? What is the value of H_{min}?

b. Calculate the maximum efficiency value N_{max}.

c. Between what values of u can we work, if we tolerate efficiency being greater than 0.95 N_{max}?

4. The table below contains values of the retention factor k for four refinery gases, studied at three different temperatures on the same capillary column (length L = 30 cm, internal diameter = 250 µm), whose stationary phase is of the SE-30 type. The chromatograph is equipped with a cryogenic accessory.

		***k* values**		
Compound	**B.P. (°C)**	−35°C	25°C	40°C
ethene	−104	*0.249*	*0.102*	*0.0833*
ethane	−89	*0.408*	*0.148*	*0.117*
propene	−47	*1.899*	*0.432*	*0.324*
propane	−42	*2.123*	*0.481*	*0.352*

a. Can the polarity or nonpolarity of the SE-30 phase be deduced from the elution order of the compounds?

b. Calculate the selectivity factor α for the propene–propane pair at the three temperatures indicated.

c. For a given compound, why does k decrease in response to an increase in temperature?

d. What is the number of theoretical plates of the column for propane at 40°C, if it is known that at this temperature the resolution factor for the propene–propane pair is 2? Calculate the corresponding HETP.

e. What would be the minimum theoretical value of HETP for propane at 40°C?

5. In a GC analysis series, we seek to determine the influence of the column's length on several parameters of the chromatogram. All experiments are conducted under the same temperature and carrier gas flow rate conditions.

L (m)	a = √L	t_R (min)	t_R/L	R	R/a
15		3.7		2.05	
30		7.5		2.91	
60		15.3		4.15	

a. Complete the table.

b. To the nearest measurement uncertainties, what simple relationship can we consider between retention times and column length?

c. To the nearest measurement uncertainties, what simple relationship can we consider between the resolution and the square root of the column length?

d. If we assume that the two peaks used to determine the resolution have practically the same full width at half-maximum (FWHM), deduce from the previous questions a relationship between the FWHM and the column length as well as a relationship between the theoretical efficiency of the column and its length.

e. The two peaks used to calculate R have retention times respectively of 8.3 min and 9.7 min with the column length equal to 60 m. Calculate the FWHM value of these peaks (same approximation as in question d), as well as the theoretical efficiency of the column for the solute whose retention time is 8.3 min.

f. Calculate the theoretical efficiency of 15 and 30 m columns for the same solute.

6. The chromatographic analysis of an unleaded gas is conducted under the following conditions:

Column: L = 150 m, ID = 0.25 mm, d_f = 0.25 μm, nonpolar stationary phase 'PDH 150'. Injector, 100:1 *split* ratio, T = 200°C; FID detector, T = 200°C; oven T = 35°C; carrier gas, He; u = 20 cm/s; column head pressure, 2.5 bar; injected volume = 1 μL.

In the following table, the variations in the free enthalpy of dissolution are shown, at the experimental temperature, for the solutes observed on the chromatogram.

Solute	2,3-dimethyl pentane	2,4-dimethyl pentane	2-methyl hexane	3-methyl hexane	benzene
ΔG^0_{308} (kJ/mol)	–17.79	–16.80	–17.73	–18.05	v17.27

The retention times of the various solutes are given according to peak No.

Peak No.	10	11	12	13	14
t_R (min)	48.00	55.17	63.90	64.59	68.28

a. Calculate the hold-up time t_M associated with this analysis. Since methane is not retained by the stationary phase, determine its free enthalpy variation at the temperature of the experiment.
b. Assign each peak to its corresponding solute and justify your answer.
c. From the 2,3-dimethyl pentane peak, whose FWHM is 25 s, determine the theoretical plate number of the column.
d. Calculate the phase ratio of this column.
e. Calculate the distribution coefficient *K* of benzene in two different ways:
i. By using its variation of free enthalpy of dissolution.
ii. From its retention factor.

7. The best-known method for estimating the hold-up time, t_M, consists of measuring the retention time of a compound not retained on the column. Described here is another method for calculating the hold-up time, which involves the relationship used in establishing retention factors. Knowing that for a homologous series of organic compounds if the column temperature is constant, we can write:

$$\log\left(t_R - t_M\right) = a \cdot n + b$$

(where t_R represents the total retention time of a compound having *n* atoms of carbon, while *a* and *b* are constants that depend upon the type of solute and the stationary phase chosen).

a. Recall the chromatographic parameters for which it is essential to know the hold-up time t_M. Which compound is used in general to determine t_M?
b. Calculate t_M from the following experiment, employing the method above: a mixture of linear alkanes, possessing six, seven and eight atoms of carbon, is injected into the chromatograph. The retention times for these compounds are respectively, 271 s, 311 s, and 399 s, at a constant temperature of 80°C (column length 25 m, *ID* = 0.2 mm, d_f = 0.2 µm and a polysiloxane-based stationary phase).
c. If the Kovats index for pyridine on squalane is 695, what is the McReynolds constant of this compound on the column in question, if it is known that under the conditions of the experiment, its retention time is 346 s?

8. In a GC experiment, a mixture of *n*-alkanes (up to *n* carbon atoms, where *n* represents a variable number) and 1-butanol ($CH_3CH_2CH_2CH_2OH$) is injected onto a column maintained at a constant temperature and whose stationary phase is of the

dimethylpolysiloxane type. The equation of the Kovats straight line derived from the chromatogram is:

$$\log t'_R = 0.39n - 0.29 \quad \text{(where } t'_R \text{ is expressed in seconds).}$$

The adjusted retention time of butanol is 168 s. If it is known that the retention index for butanol on a squalane column is 590 s, then deduce its corresponding McReynolds constant on this column.

9. We study a chromatogram obtained under the following conditions: Column: DB-Wax, L= 30 m; *ID* = 0.321 mm; d_f = 0.25 µm; oven *T* = 210°C; mobile phase: H_2, *u* = 50 cm/s; volume injected 0.2 µl; detector: FID, *T* = 220°C.
The chromatogram has three peaks with the following characteristics:
peak 1: t_{R1} = 1.01 min.; δ = 0.151 min.
peak 2: t_{R2} = 1.95 min.; δ = 0.0277 min.
peak 3: t_{R3} = 2.01 min.; δ = 0.0284 min.

- **a.** Calculate the dead time of the column. What can we conclude from this?
- **b.** From this, deduce the mobile phase flow rate in ml/min.
- **c.** Calculate the column phase ratio.
- **d.** Calculate the HETP (*H*) for peak No.2.
- **e.** Calculate retention factors *k* for peaks 2 and 3.
- **f.** From this, deduce the distribution coefficients K of the two corresponding compounds.
- **g.** Calculate the selectivity factor α between compounds 2 and 3.
- **h.** Calculate the resolution between these two peaks. Is it enough? What could be a solution to improve the resolution?
- **i.** Kovats retention indexes for octane, toluene, and methyl pentanoate on this same column are respectively 800, 850, and 985. Classify these three solutes by increasing retention times on this stationary phase.
- **j.** What is the polarity of this stationary phase? Justify your answer.

SOLUTIONS

1. We consider that volume *V*, leaving the column every second, is equal to the internal volume of the column over length $\bar{u}$. The section of the column is $\pi ID^2/4$ and thus V = $\bar{u}\pi ID^2/4$. Flow rate $\mathfrak{D}$ expressed in ml/min (if $\bar{u}$ and *ID* are in cm) is then $\mathfrak{D}$ = 60V. If we choose to express *ID* in mm, we write $\mathfrak{D} = 60\bar{u}\pi(0.1)^2 ID^2/4$, or $\mathfrak{D} = 0.47\bar{u}ID^2$.

2. **a.** The thermodynamic study of a chemical equilibrium leads to the formula:

$$\Delta G^0_T = -RT \ln K_T = -RT \ln k \cdot \beta = \Delta H^0_T - T\Delta S^0_T$$

$$\text{or} \quad \ln k \cdot \beta = -\frac{\Delta H^0_T}{RT} + \frac{\Delta S^0_T}{R} \quad \text{or} \quad \ln k = -\frac{\Delta H^0_T}{RT} + \frac{\Delta S^0_T}{R} - \ln \beta$$

If we assume that enthalpy and entropy variations are constant in the temperature range in question and if we switch to log, we can write $\log k = -A + B/T$ if we assume the entropy variation < 0 (loss of disorder when the solute is fixed on the stationary phase) and the enthalpy variation < 0 (exothermic reaction).

b. Coelution => $k_1 = k_2$, hence $\log k_1 = \log k_2$ and thus $T_1 = 421$ K or $t = 148$°C
For $T < T_1$, $k_2 > k_1$, therefore **2** elutes after **1**.

c. If $\alpha = 2$, this means that $k_2 / k_1 = 2$, or $\log k_2 - \log k_1 = \log 2$, and thus $T = 343$ K.

d. $T = 423$ K thus $k_1 = 0.163$ and $k_2 = 0.161$; $K_1 = 40.73$ and $K_2 = 40.16$.

3. a. The cancelation of the first derivative of $H = f(u)$ enables us to find u corresponding to H_{min}, thus $u = 36.9$ mm/s; $H_{min} = 0.295$ mm.

b. $N_{max} = L/H_{min} = 40{,}674$ plates.

c. We are looking for the value of u that gives $N = 0.95N_{max}$ or $N = 38{,}641$ theoretical plates, thus $H = 0.310$ mm; $u = 50.2$ mm/s or $u = 27.3$ mm/s.

4. a. The order of elution follows that of the increasing boiling points. The alkenes, which are more polar than the corresponding alkanes, are less retained, which is expected from a nonpolar column.

b. $\alpha = k_2/k$: $\alpha = 1.12$ (at –35°C), $\alpha = 1.11$ (at 25°C) and $\alpha = 1.09$ (at 40°C).

c. Since the column is the same, the retention time decreases following the reduction in the partition factor $K = C_S/C_M$ with temperature.

d. N = 138,493 theoretical plates.

e. $H_{min\ theo} = 0.113$ mm.

5. a.

L (m)	a = √L	t_R (min)	t_R/L	R	R/a
15	3.87	3.7	0.25	2.05	0.53
30	5.48	7.5	0.25	2.91	0.53
60	7.75	15.3	0.25	4.15	0.53

b. t_R is proportional to length.

c. R is proportional to $\sqrt{L}$.

d. δ proportional to $\sqrt{L}$ and N is proportional to L.

e. $\delta = 0.0568$ min, $N_{60} = 148{,}166$ plates.

f. $N_{30} = 74{,}083$ plates and $N_{15} = 37{,}041$ plates.

6. a. $t_M = L/u = 750$ s $= 12.5$ min. For methane, $k = 0$, therefore $K = 0$ and $\Delta G_T^0 = -RT \ln K$ cannot be calculated.

b. The higher K is, the more solute is retained and the larger the absolute value of ΔG_T^0 is. Therefore, peak 10 → 2,4 dimethyl pentane, 11 → benzene, 12 → 2-methylhexane, 13 → 2,3 dimethyl pentane and 14 → 3-methyl hexane.

c. $N = 5.54\,(t_R/\delta)^2 = 133{,}126$ plates.

d. $\beta = ID/4d_f = 250$.

e. $\Delta G_T^0 = -RT \ln K \Rightarrow K = \exp\left(-\Delta G_T^0 / RT\right) = 849$, $K = k \cdot \beta = 853$. These are very close values.

7. a. N_{eff}, k, α; GC: methane.

b. The experiment described helps us write three equations for three unknowns, t_M, a, and b: (1) $\log(271 - t_M) = 6a + b$; (2) $\log(311 - t_M) = 7a + b$; (3) $\log(399 - t_M) = 8a + b$, thus $t_M = 237.7$ s.

c. $Ik = 749$. The McReynolds constant for pyridine on this stationary phase is 54.

8. The Kovats index of butanol is thus 645. Its McReynolds constant is: 645 – 590 = 55.

9. a. $t_M = L/u = 3{,}000/50 = 60$ s, i.e. 1 minute. Peak No.1 probably corresponds to the solvent peak.

b. **D** $= V_M/t_M = \pi \cdot L \cdot (ID)^2/4$ $t_M = 40.46$ mm$^3 \cdot$s^{-1}, i.e. 2.4 ml/min.

c. $\beta = (ID)/4d_f = 321/(4 \times 0.25) = 321$.

d. HETP for peak No.2: $H = L/N$ with $N = 5.54 t_R{}^2/\delta^2 = 27{,}555$ plates => $H = 1.09$ mm.

e. $k = (t_R - t_M)/t_M k_2 = 0.93$ and $k_3 = 0.99$.

f. $K = k \cdot \beta$, thus $K_2 = 0.93 \times 321 = 298.5$; $K_3 = 0.99 \times 321 = 317.8$.

g. $\alpha_{2\text{-}3} = k_3/k_2 = 1.06$.

h. $R_{2\text{-}3} = 1.18(t_{R3} - t_{R2})/(\delta_2 + \delta_3) = 1.2$. The resolution between two peaks for a good separation must be at least 1.5. The two peaks are therefore not well separated. To improve this separation, retention times must be increased by reducing either the carrier gas flow rate or the oven temperature.

i. The higher the Kovats index is, the longer the retention time. Therefore, t_R values are in the following order: octane < toluene < methyl pentanoate.

ii. These retention times are in the order of growing polarity of the analytes; therefore, this is a polar column.

Chapter 3

High-Performance Liquid Chromatography

Of all the chromatographic separation techniques whose mobile phase is a liquid, high-performance liquid chromatography (HPLC) is perhaps the best known. Its field of applications overlaps a large section of gas chromatography, to which can be added the analyses of many thermolabile compounds or high-molecular-weight biomolecules. Stationary phases composed of very small particles or porous monoliths necessitate the use of high-pressure pumps to make the mobile

phase migrate. *Efficiency* is much lower than in GC but the possible interactions between stationary phase, mobile phase, and solutes improve *selectivity*. Progress regarding columns, which have become narrower and now even include capillaries, has led to ultra-HPLC (or *UPLC*) and to nano-HPLC for greater rapidity and savings on the mobile phase. Finally, HPLC coupled with mass spectrometry (HPLC-MS/MS) has become one of the most essential tools for a number of analytical, research or biochemical laboratories.

Objectives

Describe the composition of an HPLC apparatus
Understand the role of the mobile phase
Compare stationary phases
Choose the stationary phase adapted to the separation
Demonstrate the principle of an injection loop
Describe the main detectors
Optimize a separation
Present the current orientations of HPLC

3.1 DESIGN OF AN HPLC APPARATUS

HPLC is derived from the oldest form of preparative liquid chromatography on columns. Performance has been progressively enhanced through miniaturization, computer control of the entire process, and, above all, the progress made in producing stationary phases. These phases are most often composed of spherical microparticles with diameters of 2–10 μm, or porous monolithic structures that lead to very compact columns requiring the use of high-pressure pumps to obtain a sufficient flow rate of the mobile phase, despite counterpressure of several hundred bar at the column head.

Due to these specificities, a liquid chromatography system includes several specialized modules, which may be present either in stand-alone form or integrated in a single surround in order to reduce the path length and thus the mobile phase hold-up volumes, or for a smaller footprint, or even for presentation purposes (Figure 3.1).

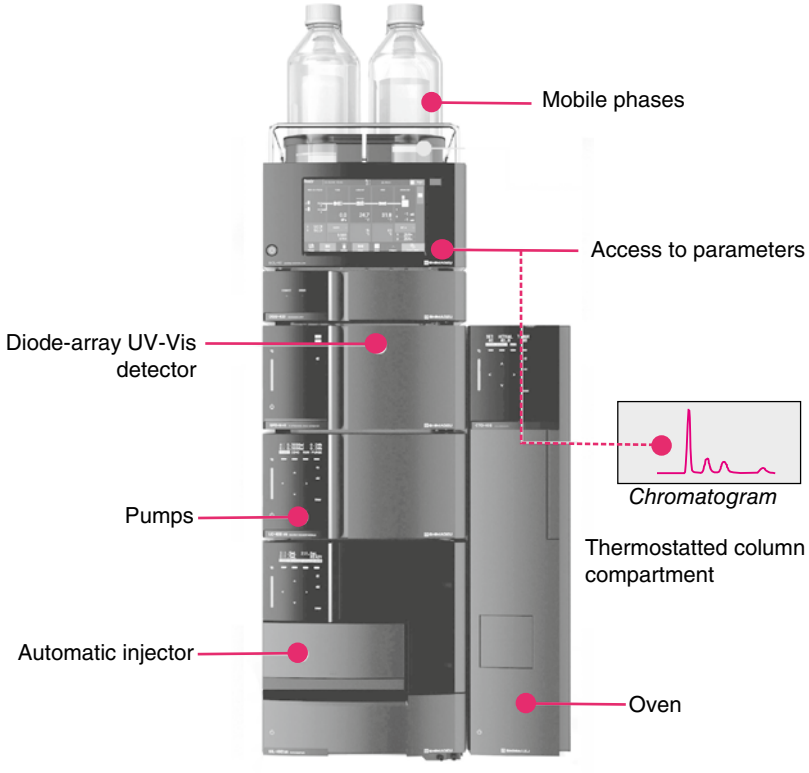


Figure 3.1 Diagram of a modular HPLC instrument. The vertical assembly of the different modules affords an economy of space. Here the chromatograph, model Nexera, includes an autosampler for continuous operation and a thermostatted column compartment to improve the reproducibility of the separations. (Source: Reproduced courtesy of Shimadzu.)

A tubing system of very small internal diameter (0.1 mm) ensures the circulation of the mobile phase between the modules. These transfer tubes are made of stainless steel or PEEK® (polyether ether ketone), a flexible polymer capable of resisting the solvents commonly used in HPLC, even under high pressure (up to 350 bar).

The low flow rates obey Poiseuille's law. The maximum velocity of the mobile phase is found at the centre of the tubing, while in contact with the walls it is zero. This produces a dispersion of the compounds and thus an inevitable widening of the peaks. To improve the separations, the volume of the mobile phase outside the column is maintained as low as possible (10% of the column's hold-up volume).

3.2 PUMPS AND ELUTION GRADIENTS

3.2.1 Eluent Pumps

All HPLC systems include at least one pump to force the mobile phase through the tightly packed column. The result of this is a pressure increase at the injector that can reach 40 MPa (400 bar) depending upon the flow rate imposed upon the mobile phase, its viscosity, and the nature of the stationary phase.

These various factors are found in Eq. (3.1), which calculates with good approximation the pressure drop (ΔP) between the HPLC system inlet and outlet (the latter is most often at atmospheric pressure):

$$\Delta P = \Phi \frac{\eta \cdot L \cdot \bar{u}}{d_p^2} \tag{3.1}$$

where Φ is the flow resistance factor depending on the column packing, η is the viscosity of the mobile phase, L is the length of the column, $\bar{u}$ is the average velocity of the mobile phase (directly related to its flow rate), and d_p is the diameter of the particles packing the column.

Pumps are designed in order to maintain a stable flow rate, avoiding pulsations even when the composition of the mobile phase varies. These reciprocating pumps contain, in general, two pistons in series, working in opposition, to avoid interruptions to the flow rate (Figure 3.2). The movement of pistons is controlled by a stepper motor associated with a specific-shaped cam.

The presence of a non-negligible quantity of ambient gases (N_2, O_2, CO_2, etc.), dissolved in the solvents, can perturb the separations by modification of the compressibility of the mobile phases, which leads to the possible formation of bubbles. In particular, oxygen can shorten the lifetime of the column while hindering the working of both electrochemical and photometric UV detectors. The solvents are therefore degassed either by ultrasound, by rapid helium bubbling, or by diffusion, in which they pass through a small-diameter polymer tube that is permeable to gases (Figure 3.2).

Perfecting flow rate regulation is achieved by the insertion of a pressure (or pulse) damper between the pump(s) and the injector. It operates on the mechanical principle of ballasting. The simplest set-up consists of inserting a large coil of narrow-bore tubing, several metres long, in the liquid phase circuit. Under the influence of a pressure wave of solvent, the tubing unwinds slightly, thus increasing its internal volume and counteracting the variation in pressure.

To obtain microflow rates (for example, 1 µl/min) necessary for a packed capillary column, the same pumps are used, though with the addition of a bypass at the outlet. This bypass divides the flow into two fractions, of which only the smaller one is directed towards the column. To resist the low pHs of many elution mixtures, which can be more

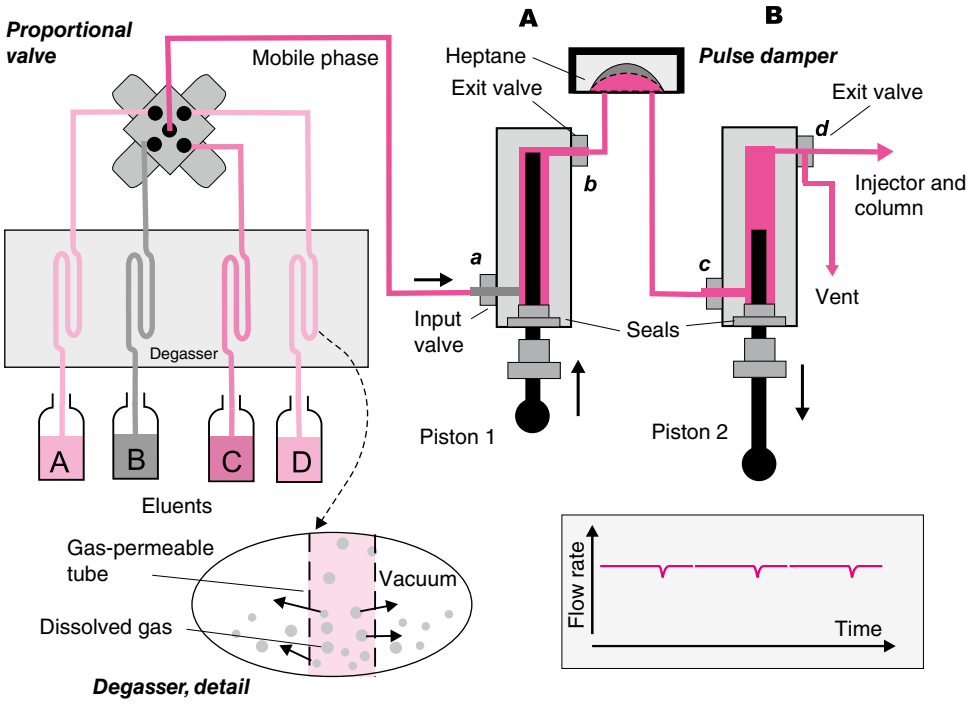


Figure 3.2 Simplified diagram of a dual-headed reciprocating pump. From the moment when the exit valve *b* of cylinder A closes and its input valve *a* opens, piston A moves backwards and the chamber fills. Meanwhile, *c* is closed, and *d* is open: piston B moves forward to force the mobile phase towards the column. The volume displaced by piston B is half of that aspirated by piston A. At the end of its stroke, *a* closes, and *b* and *c* open. Piston A now advances and pushes out the contents of cylinder A (as shown in the figure). Half of this volume is expelled directly towards the column, while the other half serves to fill cylinder B as piston B retracts. A pulse damper is located between the two cylinders. Below, graph showing variations in the flow rate as a function of cycles and the principle of a degasser.

corrosive when pressure is high, the parts and surfaces that come into contact with the mobile phase need to be inert. The pistons and valves of the pumps are thus made of Teflon or special alloys or often even precious gems, sapphire or agate.

Depending upon the design, an HPLC instrument contains one or more pumps. These pumps are associated with a mixing chamber, either upstream or downstream. They are capable of delivering an eluent of fixed (*isocratic mode*) or variable composition during analysis (to create an *elution gradient*). For the latter case, the system must compensate for differences in solvent compressibility in order for the composition to be respected at the pressure of use.

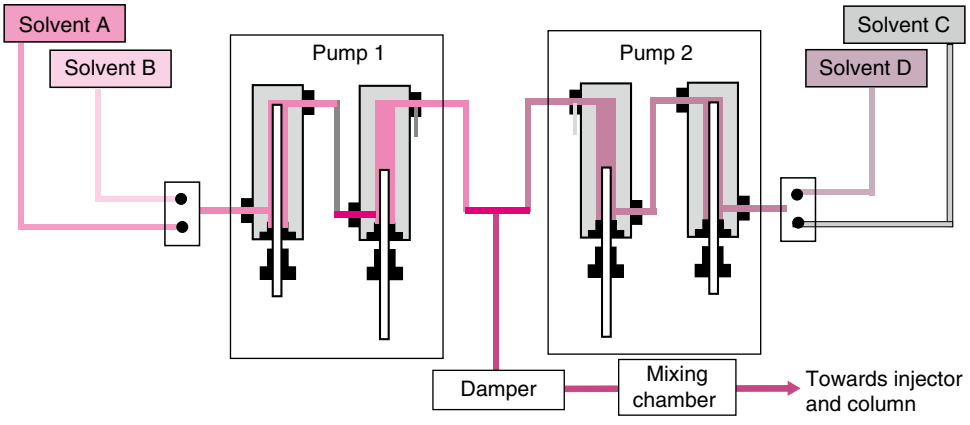


Figure 3.3 Example of a high-pressure gradient system. The pumps are called binary, ternary or quaternary depending upon the number of solvents that can be mixed together (quaternary in this example).

3.2.2 Low-Pressure and High-Pressure Gradients

To conduct mobile phase gradients, several solvents of various polarities are required. When the system contains a single pump, it should be preceded by a *low-pressure* mixing chamber into which solenoid valves make it possible to feed the solvents at a programmed composition. Alternately, other systems contain more than one pump, depending on the number of solvents (e.g. binary pumps). The final composition is obtained under *high pressure* with a tee located after the pump but before the column (Figure 3.3).

> When several analyses are to be done successively, gradients should be avoided, if possible, by seeking a practical separation compromise with a single eluent of fixed composition. This reduces postanalysis time between successive analyses. With a gradient, the reequilibration of the two phases to their initial composition requires the passage of a volume corresponding to at least 10 times the hold-up volume of the system.

3.3 INJECTORS

In HPLC, the injection of a precise volume of sample at the head of the column must be done as fast as possible in order to cause the minimum disturbance to the dynamic regime of the mobile phase, whose flow must be stable from column to detector. This is done by a special high-pressure valve, either manual or motorized, possessing several flow paths, and which is situated just prior to the column (Figure 3.4). It is a precision component that can resist pressures greater than 30,000 kPa. The valve functions in two positions:

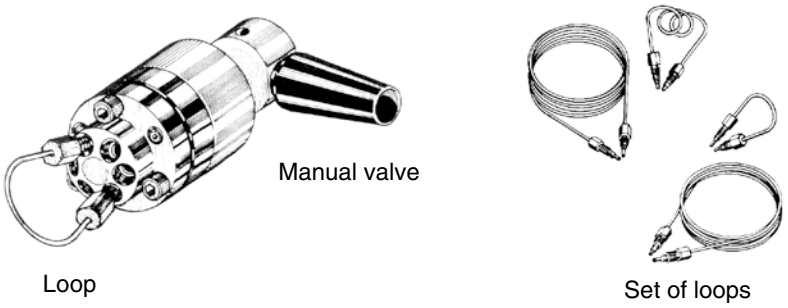


Figure 3.4 Injection valve for HPLC and an assortment of loops. Left, rear view of an injection valve (valve with six inlets/outlets and a loop attached); Right, an assortment of loops of different volumes. (Source: Courtesy of Rheodyne Inc.)

- In the *load* position, only communication between the pump and the column is assured (Figure 3.5). The sample, contained in a solution, is introduced at atmospheric pressure with the aid of a syringe into a small tubular curved section called a *loop*. Loops have a range of volumes and are either external or are integrated into the valve body.
- In the *inject* position, the sample is inserted into the flow of the mobile phase by the 60° rotation of a lever, which reverses the flow direction in the loop. Highly reproducible injections are attained only if the loop has been completely filled with the sample. The volume sampled with the syringe is thus always largely superior to that of the loop.

3.4 COLUMNS

While HPLC is constantly evolving, the stationary phase is always held in the steel column between two sintered discs, or for nano-HPLC in a silica column that has a much shorter length than in GC (Figure 3.6).

These columns are designated by their effective volume, i.e. the volume reserved for the stationary phase they can contain. Thus, since the origins of HPLC, the standard column has an internal diameter of 4.6 mm and a length of 10–25 cm. Progressively, the diameter of these columns has become smaller. As a result, we have a large selection of dimensions going all the way to a format that is 1.5 mm in diameter and 5 cm long, not to mention the capillary columns reserved for nano-HPLC. This evolution, made possible by progress in the stationary phase field, presents several advantages:

- *Faster separation*. With a shorter column, we reduce the analysis time, but to conserve the same efficiency N, the stationary phase particles must have a smaller diameter d_p in order for the L/d_p ratio to remain the same. Currently, the

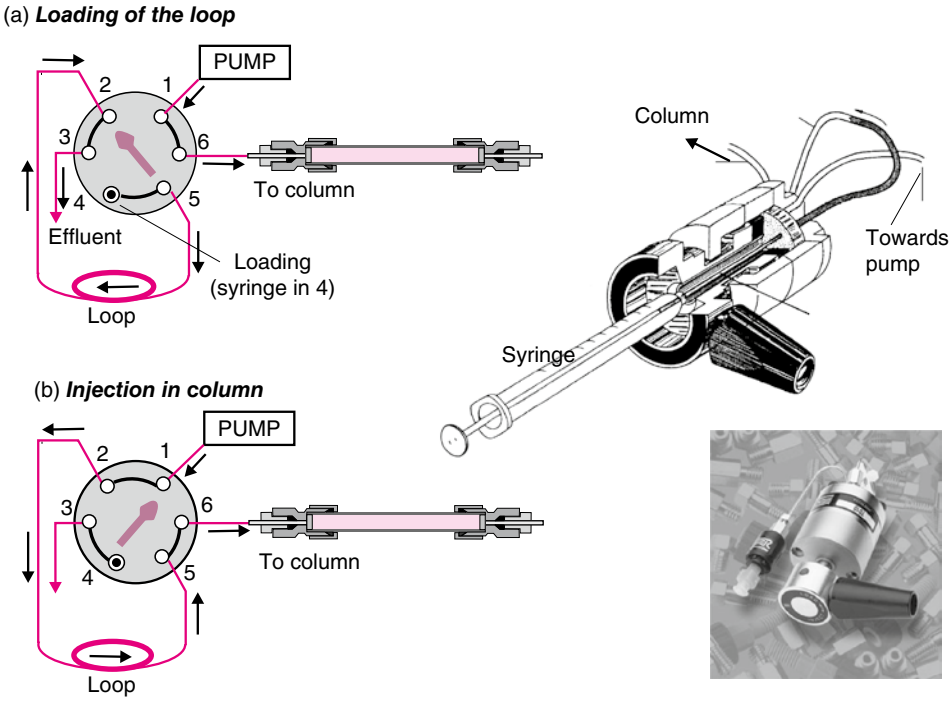
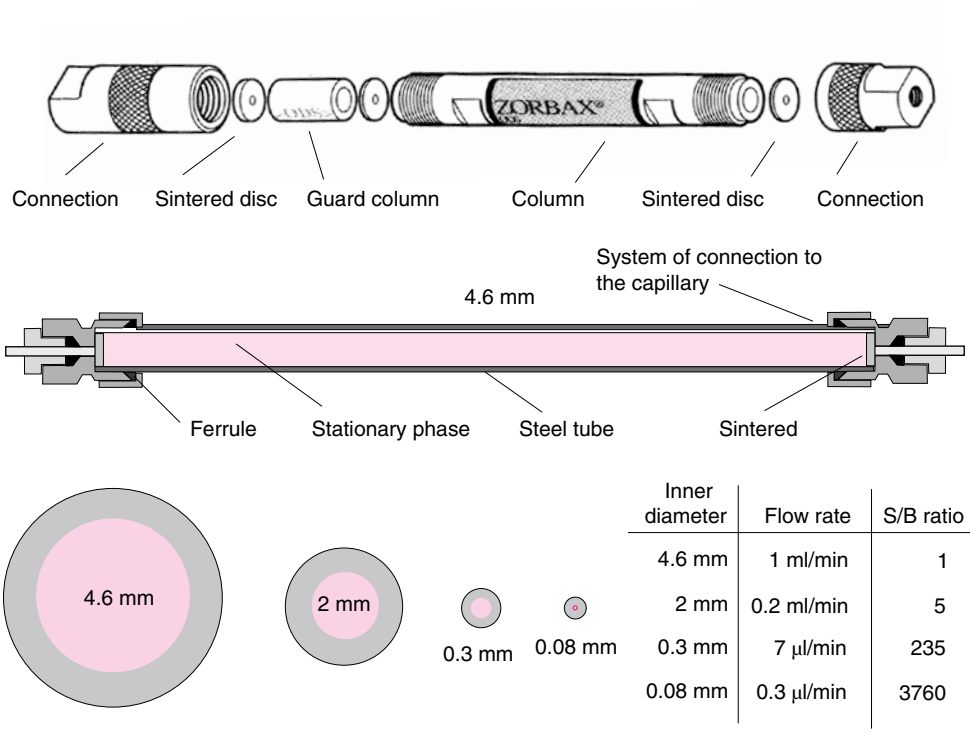


Figure 3.5 Injection with a loop. (a) Loading of the loop. In this step, the syringe is introduced in position number 4; (b) Injection onto the column (note the new position of the handle). Model 7125 valve. Valves can be motorized. (Source: Reproduced courtesy of Rheodyne Inc.)

smallest diameters d_p are 1.5 µm. In return, the pressure drop ΔP, which is inversely proportional to d_p^2 (Eq. (3.1)), is greatly increased. Thus, current pumps are able to reach pressures of 800 bar. We can also increase speed by heating the column, which decreases the viscosity of the mobile phase and makes it possible to reach concentration equilibria much faster.

- *Better efficiency N*. Narrower columns improve sensitivity. They limit the required mobile phase volume, which increases the analyte concentration in the column and thus the detector's response. Sensitivity is therefore increased; chromatographic peaks obtained in U-HPLC (Ultra-High-Performance Liquid Chromatography) are narrower than in conventional HPLC, and the signal/background noise ratio is improved (Figure 3.6).
- *Less consumption of mobile phases*. The use of a smaller quantity of eluent is both a source of savings and an objective: to help HPLC enter into the 'green chemistry' field. In return, for very narrow columns in capillary chromatog-



Inner diameter	Flow rate	S/B ratio
4.6 mm	1 ml/min	1
2 mm	0.2 ml/min	5
0.3 mm	7 µl/min	235
0.08 mm	0.3 µl/min	3760

Figure 3.6 Standard column and guard column for HPLC. View of the ZORBAX® column assembly and its exploded view. The stationary phase is maintained between two porous discs. The internal surface of the tube is rendered inert by a passivation treatment or by an appropriate lining. The guard column is changed periodically to prevent clogging of the column. In this diagram, the four types of columns with different diameters have been outlined. If we suppose that the packing of these four columns is of the same nature and of the same porosity, we can see the advantage of switching from conventional HPLC (on the left) to capillary HPLC or nano-HPLC (on the right). The narrower the column diameter is, the lower the consumption of mobile phase will be, and the more the sensitivity (in this case, theoretical) increases.

raphy or nanochromatography, the equipment must be adapted in order to manage microflow rates of the mobile phase and to minimize the hold-up volumes on the entire path, from the gradient mixing chamber to the detection cell volume.

- *Better detection by mass spectrometry.* A lower flow rate of the mobile phase is better adapted to coupling with mass spectrometry. Moreover, the use of compatible mobile phases has greatly helped the coupled LC/MS technique, which has entered new fields, such as proteomic applications.

The column is often preceded by a precolumn, or *guard column*, which is short (0.4–1 cm) and packed with the same stationary phase (Figure 3.6). It retains certain impurities, which otherwise would clog the main column, and therefore it extends column lifetime and preserves column performance.

3.5 STATIONARY PHASES

The stationary phase, contained in the column, is composed of spherical microparticles or a porous solid, whose surface area in contact with the mobile phase reaches several hundred m^2/gram. Therefore, it favours partition mechanisms with the various solutes present.

3.5.1 Silica Gel, the Basic Material for Stationary Phases

Among all materials that have been or are currently used to design stationary phases, such as alumina, zirconium oxide or organic polymers, *silica gel* plays a dominant role. Its intrinsic qualities are the reason for its significant commercial development.

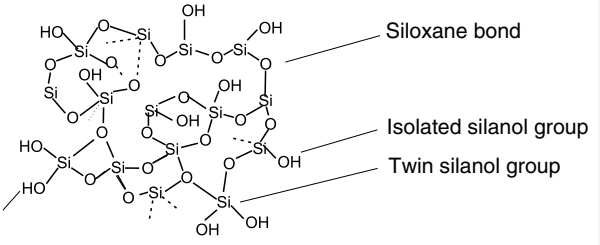
It is a rigid, amorphous solid with a formula of $SiO_2(H_2O)_n$ (where *n* is very close to 0). It is quite different from natural crystalline silica (SiO_2), which is only a distant precursor of it. The latter involves *sol–gel* polymerization processes of a tetraethoxysilane.

Though it does not possess the well-ordered structure of crystalline silica, silica gel does maintain the tetrahedral arrangement of four bonds around the silicon atom (Figure 3.7). This is a cross-linked inorganic polymer containing a variable number of *silanol (Si-OH) groups*, which have resisted the final phase of thermal dehydration. These silanol groups are responsible for the acidic catalytic properties of this very polar material, because Si–OH has a pK_a of 10, comparable to that of phenol. To measure the concentration of silanol groups, ^{29}Si nuclear magnetic resonance (NMR) can be used, or alternatively the percentage carbon content can be determined for grafted phases (see below).

Silica gel has various sized pores. To homogeneously pack the column, it is prepared either as spherical microparticles or a porous monolith (Figure 3.7). It is in fact necessary to avoid the formation of preferential pathways for the mobile phase and later on for the compounds dissolved in it.

- Microspheres have a generally constant diameter in a given column, but there are several types ranging from 1 to 12 μm.
- Monoliths, which have appeared more recently, are thus named because they involve a silica gel formed of a single piece in the same column (Figure 3.7). They offer an interesting alternative: since they are permeable, the pressure drop due to the column is less significant (Figure 3.8).

(1) Chemical structure of silica gel



(1) Silica gel

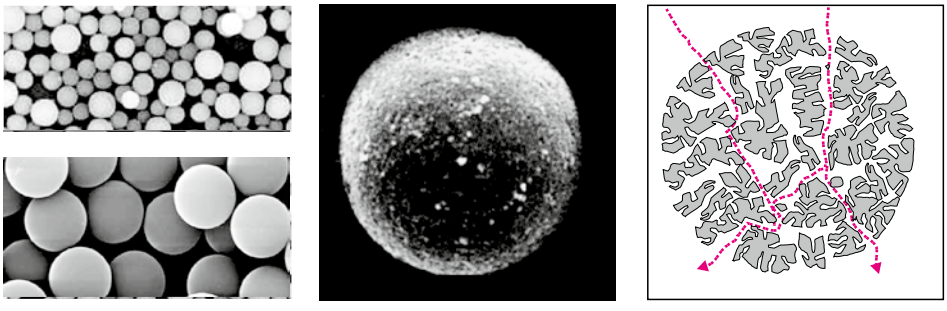
(2) Fully porous particle (70% porosity)

(3) Particle porous on the surface (hard core – 55% porosity)

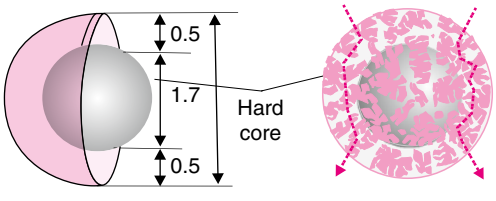
(4) Monolithic phase (80% porosity, very low pressure drop)

(5) Van Deemter curve to compare phases 2 and 3

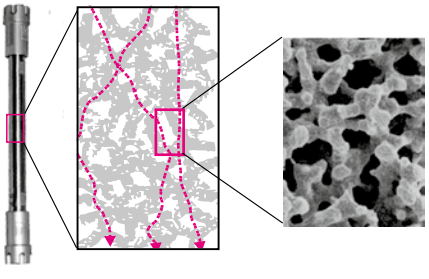
(2) Microspheres 5 µm



(3) Coreshell
Ultracore
Solid-core
Fused-core 1.5 to 2.7 µm



(4) Silica monolith



Mesopores diam. 12 nm and macropores diam 3 µm

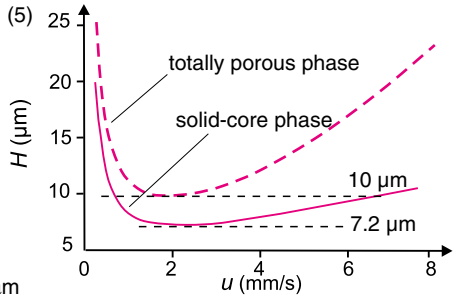


Figure 3.7 Silica gels for HPLC. Solid-core phases lead to notable improvements, as the two Van Deemter curves demonstrate, comparing phases 2 and 3 (Source: Adapted from Perform Rugged, Fast LC with Confidence: Agilent InfinityLab Poroshell 120 columns for HPLC and UHPLC). Macropores and mesopores of monolithic columns ensure flow of the mobile phase, and micropores ensure the concentration equilibria.

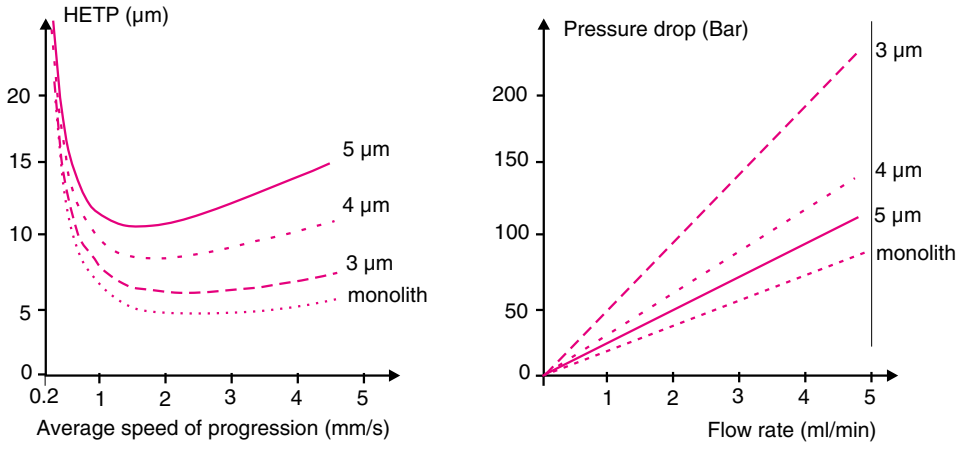


Figure 3.8 Nature of the stationary phase and efficiency. HETP decreases with particle size, but at an identical flow rate, the pressure drop increases. Monolithic columns improve mass transfers between phases and thus appear superior. Tests conducted with 5-cm columns packed with RP-18 stationary phase from the same manufacturer, using naphthalene as the test compound.

The characteristics of a silica gel depend upon several parameters: *internal structure*, *open porosity* (dimension and distribution of the pores), *specific surface area*, *resistance to crushing*, and *polarity*. Common silica gels used in HPLC contain around five silanol groups per μm^2. The specific surface area is on the order of 350 m^2/g, with a pore size of around 10 nm. For spherical particles, the measured porosity per volume of mobile phase in the column varies from 30 to 70% of the column's total volume.

Monolithic stationary phases enable higher elution flow rates than with traditional columns. Macropores, with a pore diameter of around 3 µm, ensure column permeability (total porosity over 80%), and mesopores, with an average diameter of 12 nm, form a fine porous structure, used to obtain a significant specific surface area on which molecules are adsorbed (Figure 3.7).

Things have come a long way since the phases used at the beginning of the last century by Tswett, which were made of chalk dust or powdered sugar. The treatment to which the silica is subjected makes it something rather like magic sand!

The preparation of silica gel in the form of irregular grains used in preparative chromatography, involves a hydrothermal process. Firstly, orthosilicic acid 1, an unstable compound [$Si(OH)_4$], is formed by acidification of sodium silicate [Na_2SiO_3] (formerly known as *water glass*) obtained from highly pure sands by alkaline fusion. 1 first gives a dimer 2, which then

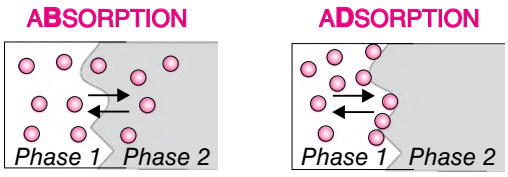


Figure 3.9 The phenomena of adsorption and partition. Adsorption is an interface phenomenon, in contrast with absorption.

polycondenses progressively to colloidal particles having a hydroxylated surface. Through aggregation, a *hydrogel* of gelatinous silica is obtained and its calcination yields dense grains of silica gel (*xerogel*):

$$SiO_2 + NaOH \rightarrow Na_2SiO_3 + H_2O$$

$$Na_2SiO_3 \xrightarrow{H_3O^+} \underset{\text{orthosilicic acid 1}}{\left[Si_i(OH)_4\right]} \xleftarrow{-H_2O} \underset{\text{orthosilicic acid 2}}{\left[(HO)_3SiOSi(OH)_2\right]} \xrightarrow{-nH_2O} \underset{\text{'silica gel'}}{(SiO_2)_n \cdot (H_2O)_x}$$

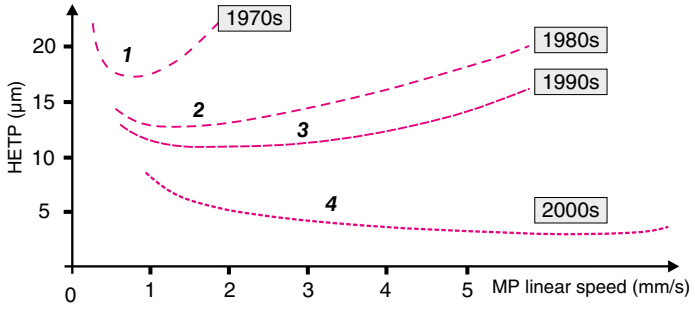
The mode of action of silica gel is founded upon adsorption (Figure 3.9), a phenomenon that leads to the accumulation of a compound at the interface between the stationary and mobile phases. In the simplest cases, a monolayer is formed (known as a Langmuir isotherm), though often some attraction occurs between the already adsorbed molecules and those still in solution. This contributes to the asymmetry of the elution peaks. Other theories explain the separation phenomenon of components by a simple slowing down (without immobilization), at the MP/SP interface, which is different for each analyte.

3.5.2 Chemical Changes of Silica Gels

The various silica gels previously described are not used as such in analytical chromatography, except for hydrophilic interaction chromatography (see Section 3.7.3). Their characteristics are not very stable and evolve over time, resulting in a lack of reproducibility of separations.

To reduce the excessive adsorption capacity and polarity of these gels, their surface is chemically changed by essentially making it hydrophobic. Over the past decades, a technology and specific chemistry have been developed, aiming to improve separation performance (Figure 3.10).

Chemical changes use the reactivity of silanol functions present on the surface (about 5 silanols per nm^2) to graft organic molecules with covalent bonds. Only about half of the sites are accessible. It is common to calculate the percentage of fixed carbon for each transformed phase. This percentage is between 10 and 25%. Modified in this way, silica gel behaves as an immobilized liquid, in that the separation mechanism now depends on *partition coefficients* instead of *adsorption coefficients*. These grafted



	Particle diameter	Column diameter
1	10 µm	4.6 mm
2	5 µm	4.6 mm
3	3.5 µm	2.5 mm
4	1.7 µm	1.5 mm

Figure 3.10 Evolution of liquid chromatography performance. The four Van Deemter curves reproduced here illustrate the technological progress of stationary phases over the last few decades. Columns have gained in efficiency, while enabling quicker analyses with less consumption of MP. (Source: Courtesy of the Waters Corporation regarding Acquity columns).

phases, whose polarity can be adjusted with precision, constitute the basis of *reversed-phase partition chromatography* (or RP-HPLC), which is used in 90% of separations. These changes of the gel surface lead to monomeric or polymeric grafted phases.

- **Monomeric grafted phases (10–15 µm thickness**). These are obtained by reaction of a monochlorosilane in the presence of an alkaline agent with surface silanol groups (Figure 3.11). The conventional phases RP-8 (dimethyloctylsilane) and RP-18 (dimethyloctadecylsilane groups, or ODS) are prepared in this way. However, some of the Si–OH groups remain intact and can be the cause of interfering polar interactions. A more complete reaction is obtained with other silyl reagents such as chlorotrimethylsilane ($ClSiMe_3$) or hexamethyldisilazane ($Me_3SiNHSiMe_3$). The few sites that remain non-transformed, because they are inaccessible to the reagent, are also inaccessible to the analytes.
- **Polymeric grafted phases (25 µm or greater thickness).** Here, a di- or trichlorosilane is used in the presence of water vapour, which provokes a polymerization of the reagent in solution prior to deposit and grafting on the silica. A cross-linked polymer layer is obtained. At the molecular scale, the final architecture of the coating is difficult to represent: we can only speculate whether it is mono- or multilayer.
- **Other grafted phases.** The silica gels discussed previously contain grafted alkyl chains with 8–18 carbon atoms. They are versatile and therefore very useful (80% of separations are conducted with C18 grafts). Yet, for improving the separation of certain classes of compound, the tendency is now to use one of the growing number of specific stationary phases. To the silica are grafted linear chains bearing aminopropyl, cyanopropyl, benzyl groups or zwitterions, which

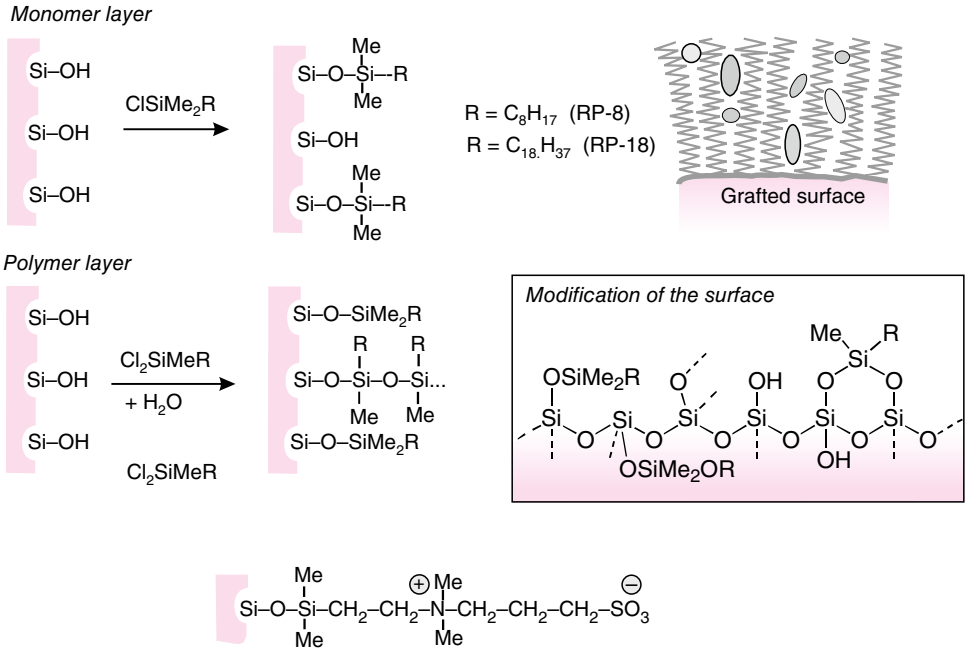


Figure 3.11 Formation of organosilanes grafted at the interface of silica gel. Diagram of organic monomers and polymers at the surface of silica gel. Close-up of a 'long-haired' particle. The Si–O–Si–C arrangement is more stable than Si–O–C. For polymer coatings, a carbon content exceeding 15% can be attained after grafting on accessible silanol groups. Bottom, an example of a phase containing a zwitterion (HILIC phase). Other reactions are also used (hydrosilylation in particular).

confer an intermediate polarity on the stationary phase. An improvement is noted in the separation of mixtures containing both polar and nonpolar molecules, which require mobile phases rich in water. Sugars, peptides, and other hydrophilic compounds become separable under these conditions.

3.5.3 Stationary Phases Without Silica

Aluminium oxide (Al_2O_3) or zirconium oxide (ZrO_2) are also used as supports for cross-linked deposits based on various polymers: polybutadiene, styrene/divinylbenzene or hydroxymethylstyrene.

These stationary phases remain stable in both acidic and basic media, under conditions that Si–O–C bonds would not normally resist. Porous graphite, in the form of spheres whose surface is completely hydrophobic, has been used in applications for compounds with high retention factors.

3.5.4 Mixed Phases for Hydrophilic Interaction Chromatography

In this case, spherical silica gels (diameter between 2 and 5 μm) grafted by various ligands are used to reach a controlled polarity (Figure 3.11). This can involve an amine, amide, diol or more polar functions, such as a quaternary ammonium, phosphate or sulfate (see Section 3.5.7). These phases combine two (or more) retention mechanisms in a single hydrophobic, polar graft.

3.5.5 Stationary Phases for Chiral HPLC

Like in GC, HPLC directly separates enantiomers on columns containing a chiral selector (see Chapter 2, Section 2.6.4). In contrast with GC, these columns do not have to resist high temperatures.

Considering the choice of chiral selectors available, it is difficult to anticipate which one will be most effective for the analysis in question. The nature of interactions between each of the enantiomers and the selector present in the column varies as a function of column choice. They combine steric, chemical, and electron aspects of the donor/acceptor-type and have weak interactions. In general, diastereoisomers, resulting from combinations, form unstable, reversible complexes. These complexes have different solubilities, which finally lead to a separation over time, with the less stable diastereoisomer being eluted first.

The chiral columns available are divided into several categories depending on the selector used:

- Proteins (albumin serum) and glycopeptides, immobilized on silica gel, which requires sophisticated chemistry (choice of 'spacers' and anchor points).
- Polysaccharides, derived from cellulose or amylose.
- Small molecules, such as camphor, ethyl lactate, amino acid derivatives (dinitrobenzoyl-phenyl) and carbamates (this is the oldest category – Pirkle-type columns, Figure 3.12).
- Cyclodextrins (with six, seven or eight glucopyranose units) grafted on a silica gel using an 'arm' with several carbon atoms (Figure 3.12).

To separate enantiomers using HPLC, there is another, less commonly used method, which consists of introducing an appropriate chiral additive into the mobile phase, in order to form labile complexes within the column. These species, with various physical properties, yield two distinct peaks in principle, with areas in the same ratio as the starting enantiomers. Valid for ionic enantiomers, the chiral vector is in very large excess compared to the analyte. These reagents include (S) (+)-ethyl lactate and (+)-10-camphorsulfonic acid.

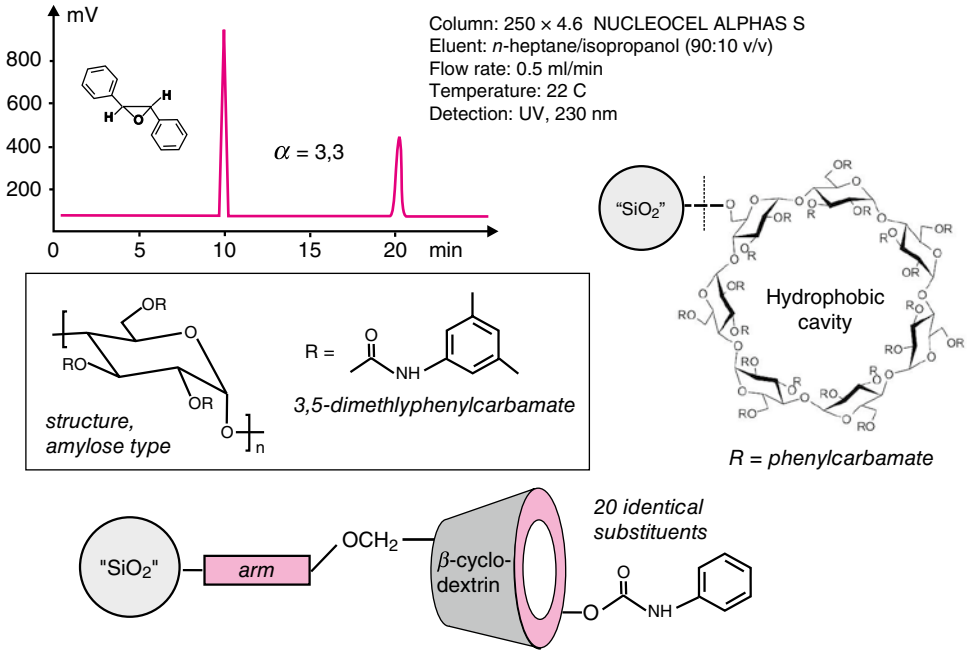


Figure 3.12 Examples of chiral stationary phases used in HPLC. Separation of trans-stilbene oxide enantiomers (Source: Courtesy Macherey-Nagel). Amylose- or β-cyclodextrin-based chiral selector, substituted in the form of carbamates.

These specific analyses have become routine for authenticity checks, in toxicology, in the flavouring industry, and especially in the pharmaceutical industry, where chiral chromatography is also used at the preparation scale, as a separation method that has been described as 'elegant' for its simplicity.

3.6 MOBILE PHASES

In principle, a polar stationary phase is coupled with a not very polar mobile phase and vice versa. Chromatography is said to be *normal-phase chromatography* in the first case and *reversed-phase chromatography (RP-HPLC)* in the second.

Since most current applications use transformed, not very polar, rather hydrophobic silica gels, a polar mobile phase is selected (most commonly water with a modifying solvent such as methanol or acetonitrile). By changing the composition of the mobile phase and therefore its polarity, this affects the distribution coefficients K (C_S/C_M) and therefore the retention factors k of the analytes (Figure 3.13). The main difficulty for

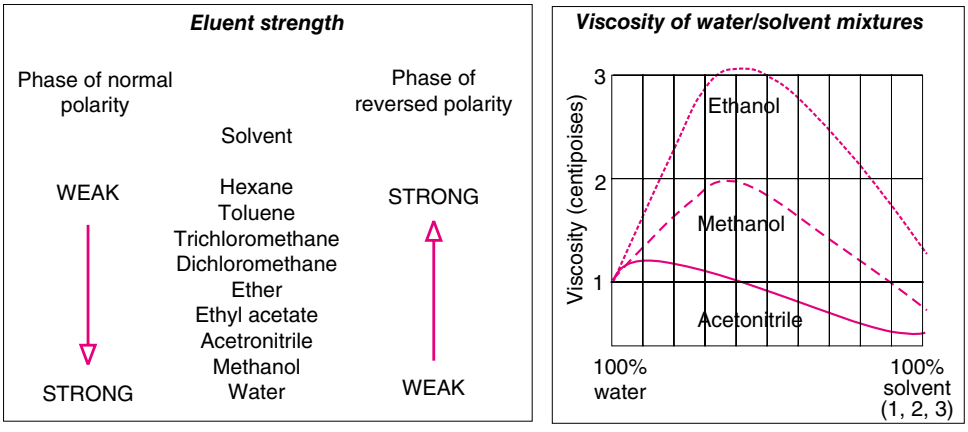


Figure 3.13 Strength of solvents used as mobile phases. By mixing several solvents, the eluent strength of the mobile phase can be adjusted. Note that the viscosity and, by consequence, the pressure at the head of the column varies according to the composition of the mobile phase.

the chromatographer is, as a function of the compounds to be separated, to make the best choice of mobile phase (Figure 3.14).

With a phase of the RP type, the order of elution will be the opposite of that encountered with normal phases. With a polar eluent, a polar compound migrates faster than a nonpolar compound. Under these conditions, hydrocarbons are strongly retained. However, polar compounds are fairly difficult to separate. To solve this, an elution gradient must be used by progressively reducing the water concentration (polar) and thus increasing that of the modifying solvent (less polar). For example, the elution begins with an 80/20 mixture of water/acetonitrile and ends at 40/60. This is the domain of hydrophilic interaction chromatography (HILIC).

Four different types of interaction are possible between solvent molecules and the analyte:

- *Ion* interactions when analyte and solvent both possess dipole moments.
- *Dispersion* due to the attraction between neighbouring molecules.
- *Dielectric*, favouring the solubility of ionic solutes in polar solvents.
- *Hydrogen bonding* between solute and solvent, where one is a proton donor and the other a proton acceptor.

The separation of very polar compounds on stationary phases of the RP-18 type requires the use of a mobile phase rich in water. Under these conditions, the hydrophobic stationary phase becomes suddenly nonwettable and the separation becomes poor. For this reason, phases with residual polarity are preferred in order to maintain the interaction between the analytes and the water in the mobile phase (Figure 3.15).

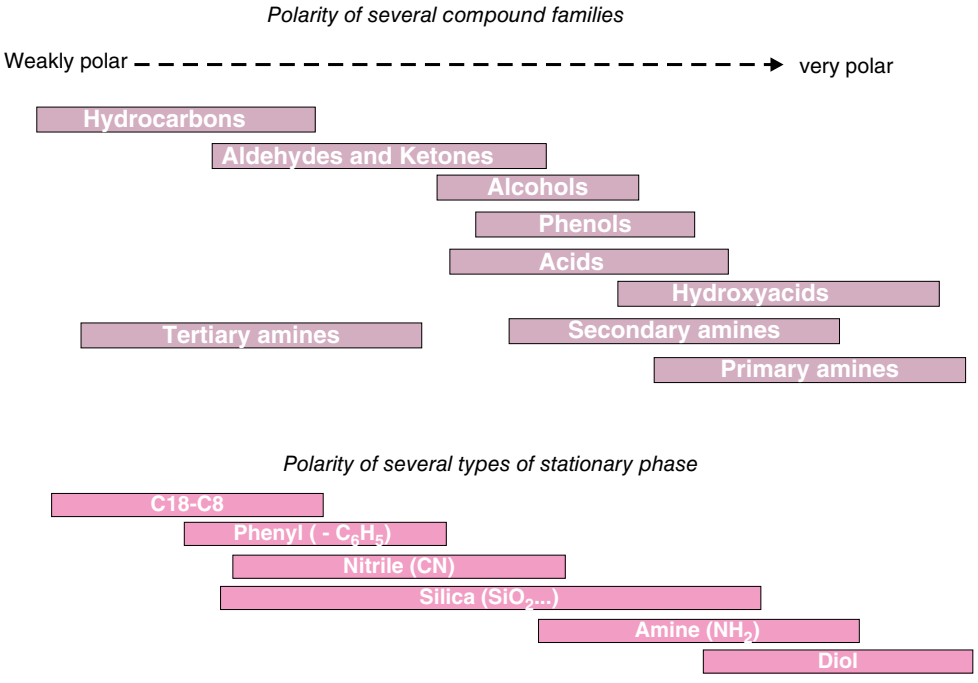


Figure 3.14 Relative polarities of several organic compound families along with some principal types of stationary phases with ligands in current use.

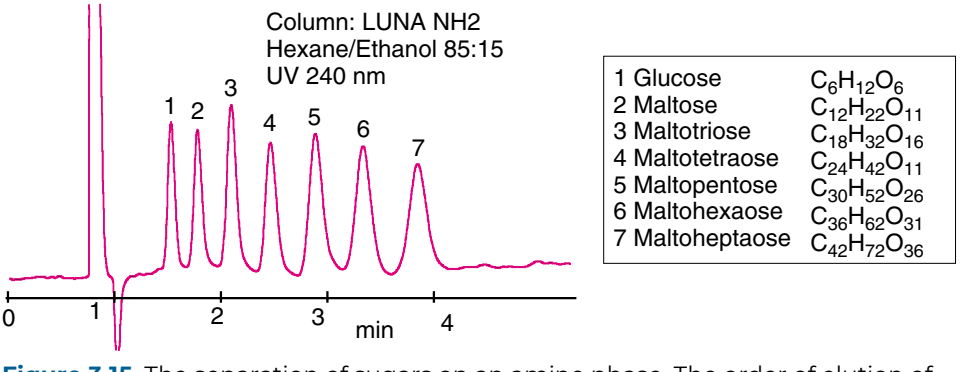


Figure 3.15 The separation of sugars on an amine phase. The order of elution of seven homologous sugars indicates that the stationary phase is polar and that the eluent strength of the mobile phase diminishes as the molecular weight increases.

3.7 SPECIFIC COLUMNS

To improve the separation of specific categories of compounds (acid, proteins, etc.) or to determine the optical purity of analytes, we modify the liquid phase with additives, or we choose specific columns. The following three techniques are examples.

3.7.1 Paired-Ion Chromatography

Paired-ion chromatography (PIC) affords an improvement in the separation of some ionic or highly polar compounds such as amino acids and organic acids that have little affinity for a nonpolar column (Figure 3.16). The ionic or polar character must be reduced to increase retention and thus improve separation. This is usually done with a compound containing a carbon chain (weakly polar) and a functional group whose charge is opposite to that of the analyte being separated (alkylammonium or alkylsulfonate). This forms a neutralized *ion pair* that is fairly stable. This principle enables the separation of inorganic cations or anions on an RP-type phase.

3.7.2 Hydrophobic Interaction Chromatography *(HIC)*

Hydrophobic interaction chromatography (HIC) is an adaptation of HPLC principally employed for the separation, without denaturation, of hydrophobic amino acids, proteins, and peptides. It is not really used for analysis. It is based on the reversible interaction, in the aqueous phase, between hydrophobic sites located on the biomolecule surface and a nonpolar stationary phase, such as an RP-8 phase (Figure 3.17). Fixation is done in a highly saline environment, and progressive elution is obtained with an aqueous mobile phase whose ionic strength is progressively reduced (reverse salt gradient).

In a highly saline environment, hydrophobic zones on the protein surface favour their retention by the nonpolar stationary phase. The elution reduces the saline

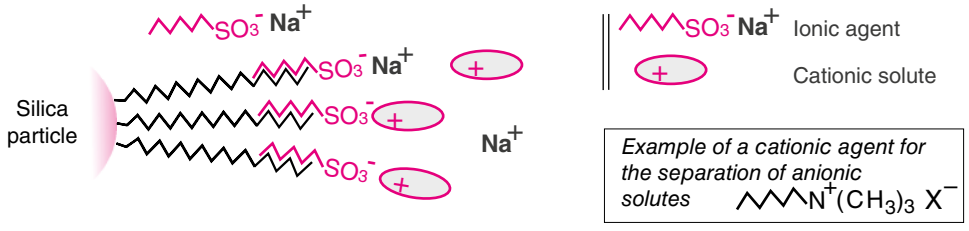


Figure 3.16 Ion pairing effects on a separation with an RP-18 type column. Acting like a soap, the hydrophobic part of hexane sulfonate pairs either with the C18 chains of the stationary phase, whose surface thus becomes an anionic charge carrier, or with the solute. The separation of charged species is thus improved.

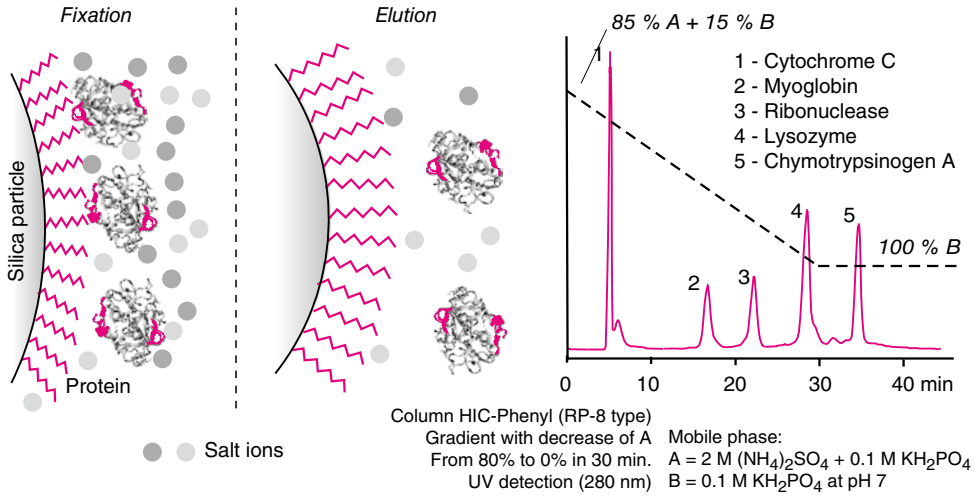


Figure 3.17 Separation on the principle of hydrophobic interaction. The chromatogram of a mixture of proteins obtained by gradually reducing the saline concentration. The most retained analytes are the most hydrophilic proteins. (Source: Courtesy of Supelco.)

concentration of the MP, which produces conformational changes and thus reveals hydrophilic domains. The analytes return to the MP and are eluted in increasing order of their hydrophobic character.

3.7.3 Hydrophilic Interaction Chromatography *(HILIC)*

The previous chromatographic technique should not be confused with *hydrophilic interaction chromatography* (HILIC). HILIC is used to separate very polar molecules. To do so, it uses a silica gel-based SP that is modified by grafting on a hydrophilic ligand. The MP is an aqueous buffer mixture with an organic modifier (e.g. acetonitrile). Polar sites of the SP fix water in a near-immobile layer, enabling several types of interactions (ionic, polar, partition) with analytes (Figure 3.18). Their separation is obtained by changing these equilibria via an increase in the water concentration of the MP (increasing polarity gradient). The main applications concern the separation of organic acids, nucleosides, sugars, peptides, and fat-soluble vitamins.

An HILIC-type phase is suitable for polar analytes. It poorly separates hydrophobic compounds. We can look at the partition factor K_{ow} of the analyte in question between octanol and water ($K_{octanol/water}$). An HILIC phase should be favoured over a reverse polarity phase if log K_{ow} is < -1. The drawback of this method is that it separates neighbouring structural compounds rather poorly, since the hydrophilic interactions are not very different in this case.

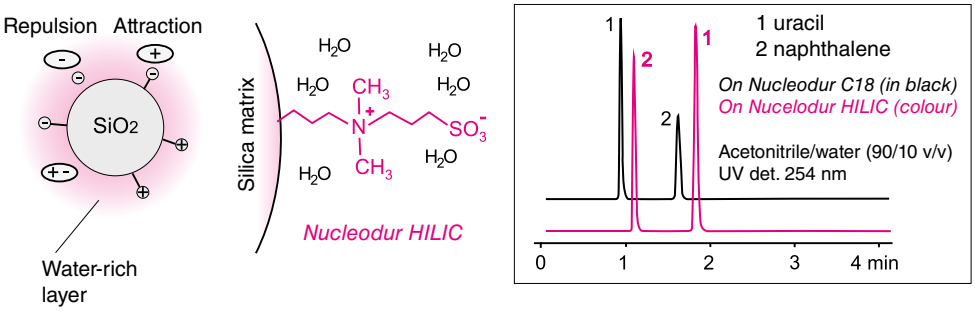


Figure 3.18 Separation using the hydrophilic interaction principle (HILIC). Chromatogram of a mixture of two compounds used as hold-up time markers on a C18 phase (uracil, very polar) and on an HILIC phase (naphthalene, not very polar) (Source: Courtesy Macherey-Nagel). Note the reversed elution order of the two compounds. H_2O indicates the existence of the aqueous layer on the stationary phase surface. In HILIC-type chromatography, toluene is also used as a hold-up time marker.

3.8 PRINCIPAL DETECTORS

For routine analyses, the most common detection modes rely on the optical properties of analytes: light absorption (UV-Visible), luminescence (fluorimetry or phosphorimetry), refraction index (refractometry), and light diffusion. In addition to these general detectors, other specific detectors based on polarimetry (measurement of optical rotation) or on radioactivity (detection of α or β emissions) can be found. Please note that none of these techniques can be used in GC, because they are not sensitive enough to track analytes present at the trace level.

For complex mixtures, coupling a mass spectrometer to the chromatograph has become rather common. Thanks to its extreme sensitivity and its ability to positively and immediately identify analytes, this *coupled technique* has become essential.

3.8.1 Spectrophotometric Detectors

Detection is based upon the Lambert–Beer Law ($A_\lambda = \varepsilon_\lambda lC$): the absorbance A of the mobile phase is measured at the column outlet, at one or more wavelengths λ in the UV or visible spectrum (see Chapter 9). It is essential for the mobile phase to have little or no absorption (Figure 3.19). The intensity of the absorption depends upon the molar absorption coefficient ε_λ of the species detected. Thus, by the simple observation of a chromatogram, it is impossible to get an idea of the concentration of the species found, even very approximately.

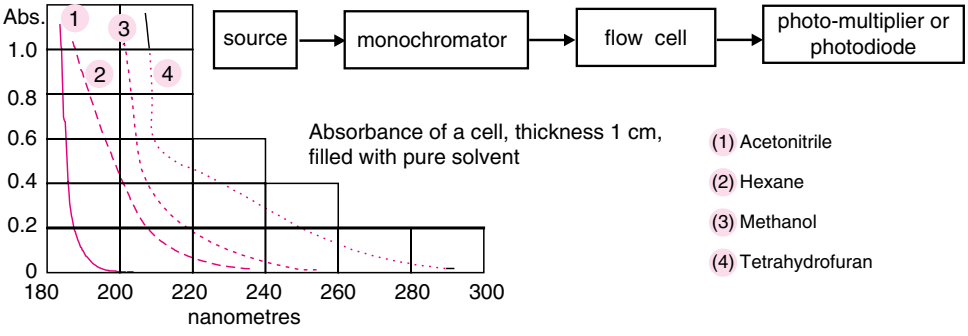


Figure 3.19 Photometric detection. Principle of a photometric detector along with the absorption spectra of several solvents used in liquid chromatography. Here the transparency limit of a solvent corresponds to an absorbance of 0.2 for optical path length of 1 cm in the cell.

For a certain wavelength, the solute may also absorb less than the mobile phase in which it is diluted. In that case, the absorbance decreases as the compound passes through, thus leading to a reverse peak (negative detection).

UV detection thus corresponds to a selective detection. For compounds without a useable absorption spectrum, we use post-column derivitization of the analytes.

The corresponding detectors may be used in elution gradient mode.

Monochromatic detection

The basic model comprises a deuterium or mercury vapour source, a monochromator for isolating a narrow bandwidth (10 nm) or a characteristic spectral line (e.g. 254 nm if the source is a mercury lamp), a flow cell with a volume of a few µl (optical path of 0.1–1 cm) and a means of optical detection.

> The current rise in biotechnologies, such as biochemistry, has led to increased demand for analysis of amino acids (protein hydrolysates). We can use photometric detectors as long as a postcolumn reaction (with ninhydrin, for example) is operated before passage into the measurement cell (see Chapter 8).

Polychromatic detection

More advanced detectors are able either to record the absorbance at several wavelengths quasi-simultaneously, or to capture in a fraction of a second a whole range of wavelengths without interrupting the circulation in the column (Figures 3.20–3.22). The diode array detector (DAD) leads not only to a chromatogram but also provides spectral information which can be used to identify the separated compounds (Figure 3.21). This is called *certainty analysis* (see Chapter 9).

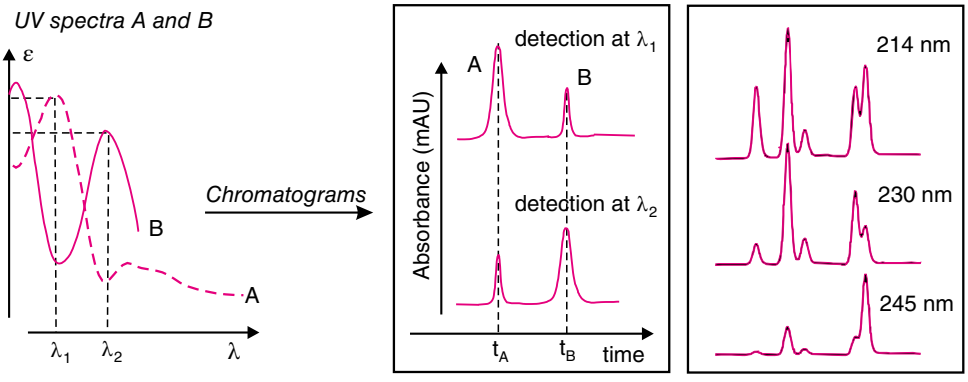


Figure 3.20 Chromatograms of a sample containing two compounds A and B whose UV spectra are different. As a function of the choice of detection wavelength, the chromatogram will not have the same appearance. On the right, the chromatograms represent a mixture of several pesticides recorded at three different wavelengths, thus illustrating this phenomenon. In quantitative analysis, the response factors of each of the compounds must therefore be determined prior to the analysis (see Quantitative Analysis, Chapter 1).

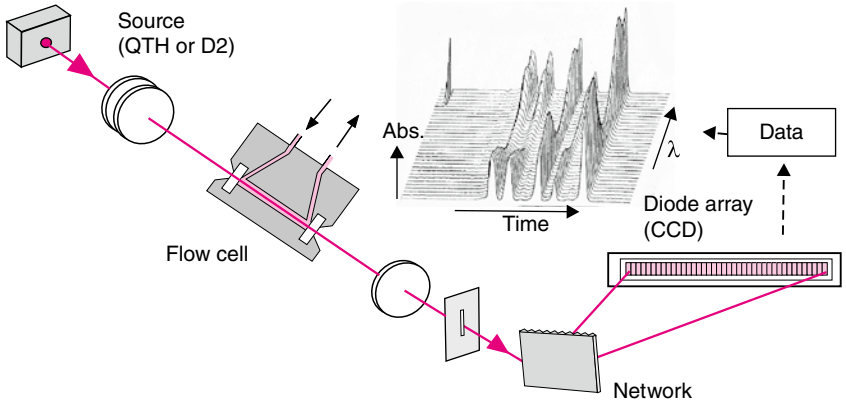


Figure 3.21 The diode array detector. Unlike the monochromatic detector, the above is a reverse optical assembly, the cell receives at each instant all of the light resulting from the source. Throughout the elution, the *y*-axis of the chromatogram corresponds to the sum of intensities sent by each diode (e.g. 1,024 diodes). Data recorded are then used to calculate the UV spectrum corresponding to each separated analyte.

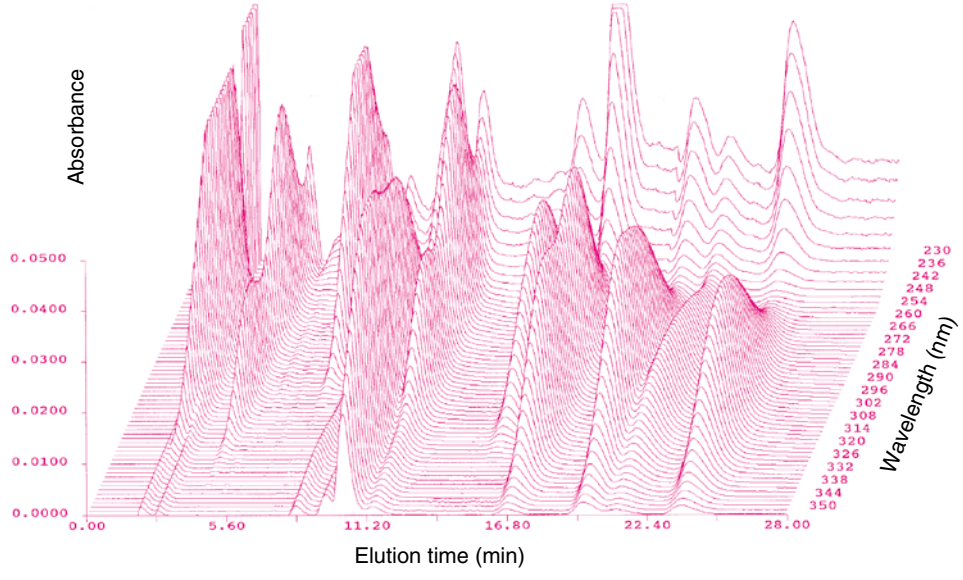


Figure 3.22 Three-dimensional representation, i = f(t, λ), of a chromatographic separation obtained by a rapid recording method. (Source: Courtesy of TSP Instruments.)

All recorded data can be processed subsequently, for example to create spectrochromatograms (Figure 3.21). The ability to record thousands of spectra during a single analysis increases the potential of these detector systems, which thus have multiple usages, such as a topographic representation of the separation, $A = f(\lambda, t)$ (iso-absorption diagrams).

3.8.2 Spectrofluorometric Detector

Some compounds re-emit a more or else large fraction of radiation from the source to which they are subjected as fluorescence radiation of a greater wavelength (see Chapter 11). The intensity of this fluorescence is proportional to the concentration of the analyte, as long as this concentration is kept low. This detection mode, both very sensitive and selective, is used for naturally fluorescent compounds and for all compounds that can become naturally fluorescent with a treatment prior to detection (Figure 3.23). In the latter case, pre- or postcolumn derivatization processes are used: an automatic reagent dispenser is placed either prior to the column or between the column and the detector to effect one or more reactions on the sample analytes in order to make them fluorescent.

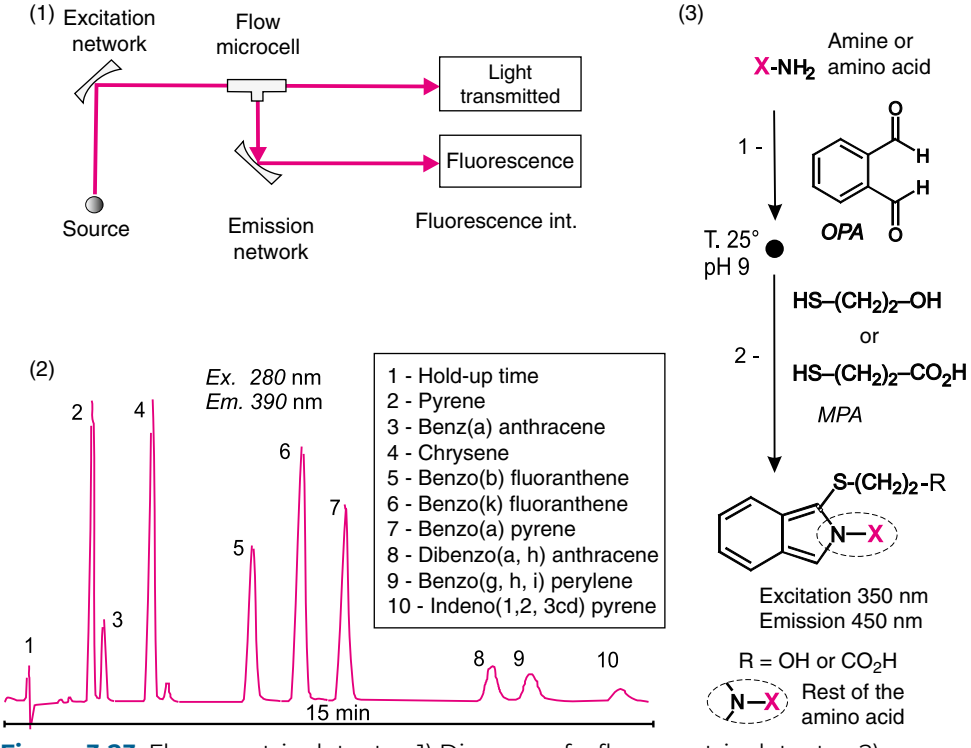


Figure 3.23 Fluorometric detector. 1) Diagram of a fluorometric detector. 2) Chromatogram of a mixture of several polycyclic aromatic hydrocarbons (PAH). The intensity of fluorescence (390 nm) varies from one compound to another, because it depends on the structure and on the chosen excitation wavelength which may not be optimal for all compounds. 3) Primary amines or amino acids can be chemically transformed into fluorescent compounds (see Chapter 11, Section 11.6). Through the reaction of *o*-phthalaldehyde (OPA) and then of monothioglycol or 3-mercaptopropionic acid (MPA), the heterocycle obtained is fluorescent. *MPA* leads to a more polar compound (R = CO_2H) migrating more rapidly on an RP-type column. This derivatization may be automated with autosamplers.

3.8.3 Refractive Index Detector (RI Detector)

This type of detector has a differential refractometer, its purpose being to continuously measure the difference in refractive index *n* between the mobile phase and the column effluent. Schematically, a beam of light travels through a cell that has two compartments: one is filled with the pure mobile phase and the other with the mobile phase eluting from the column (Figure 3.24). The difference in refractive index between the two liquids, which appears when a compound is mixed with the eluent,

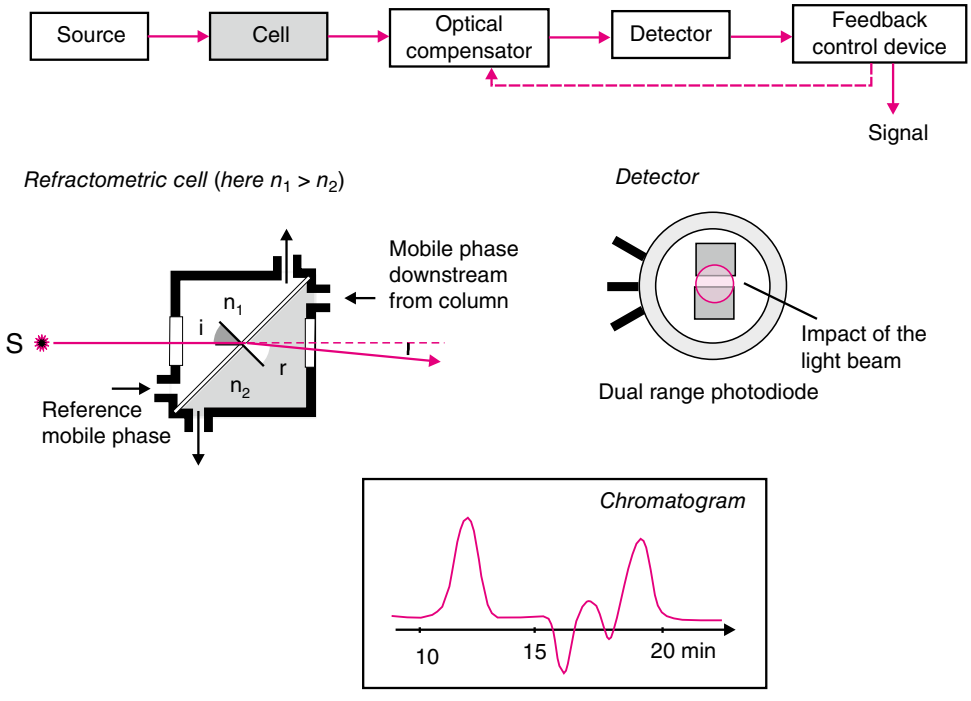


Figure 3.24 Diagram of a differential refractive index detector. Optical path at the cell level. The principle relies on the Fresnel law of light transmission through a transparent medium with refractive index *n*. The position of the refracted beam is determined with a dual-range photodiode. The responses of the two areas are maintained equal by the optical assembly (not displayed). A chromatogram of a mixture of sugars obtained with this type of detector.

is visualized as an angular displacement of the refracted beam. In practice, the signal corresponds to a continuous measurement of the feedback that must be provided to the optical element in order to compensate for the deviation of the refracted beam.

While almost universal, the refractive index detector is much less sensitive than the UV detector and is very sensitive to temperature variations. The temperature of the detector must be regulated with precision (to 0.001°C) and the column must be thermostatted. This detector leads to both positive and negative peaks (Figure 3.24). Also, this detector can only be used in isocratic mode because in gradient elution the composition of the mobile phase evolves with time, and thus so does its refractive index *n*, which leads to baseline drift. The compensation, easily obtained in the case of a mobile phase of constant composition, is no longer attainable when the composition of the eluent at the column outlet differs from that at the inlet. Consequently, it is reserved for compounds not absorbing in the UV-Visible range or when arranged in series with other detectors.

3.8.4 Evaporative Light Scattering Detector (ELSD)

This detector relies on the scattering of light by solid or liquid particles (Mie scattering). For that purpose, the mobile phase at column outlet is first nebulized into small droplets of homogenous size, with an inert gas, generally nitrogen (Figure 3.25). After this first stage, the spray thus formed crosses a tubular space heated to an adjustable temperature. In the absence of analytes, mobile phase droplets (volatile solvent mixture) entirely pass into the vapour phase. However, when there is an analyte that is difficult or impossible to evaporate, these droplets remain in a more concentrated form by creating a mixed solid/liquid aerosol made of submicron particles. Following this second stage, this aerosol penetrates into a cell, where it is subjected to the light flow of a laser or LED source. A photomultiplier, located on an optical axis angularly offset from the incident beam (often 45°), detects the light scattered by the particles.

This nearly universal detection model is nevertheless limited to analytes that are less volatile than the MP solvents. It has many advantages: no solvent peak, great sensitivity (a few nanograms of analyte) that is independent of spectral properties,

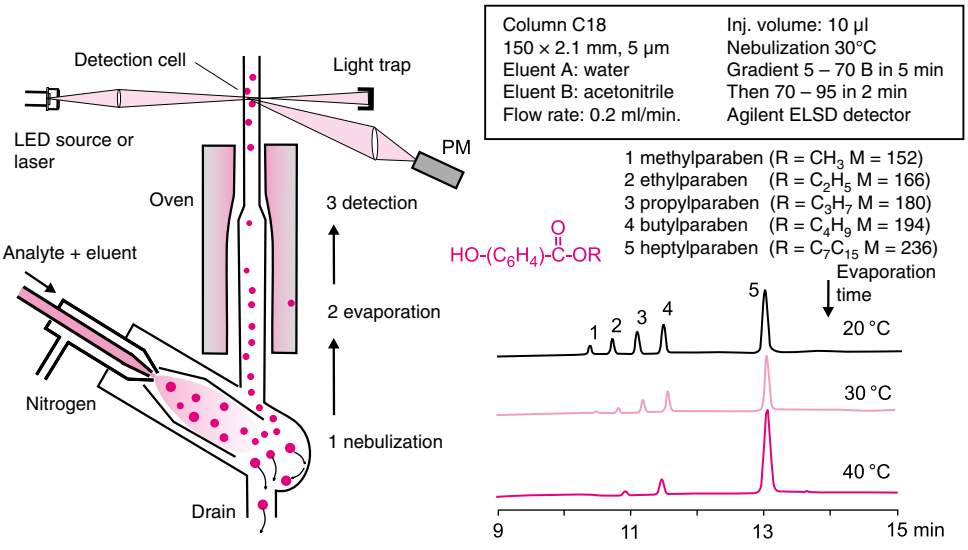


Figure 3.25 Evaporative light scattering detector. In three successive stages, *nebulization*, *evaporation*, and light *detection*, an analyte present in the mobile phase is transformed into an aerosol able to scatter the light coming from a light source. The chromatograms show that the sensitivity greatly depends on the temperature of the hot area. The comparison between peak molecular mass and intensity is very clear. (Source: Courtesy of Agilent.)

linear logarithmic response in log [c], and use in isocratic mode and with an elution gradient. However, since the response is not directly proportional to the solute concentration, use in quantitative analysis requires very precise calibration.

3.8.5 Polarimetric Detector

This selective detector presents the advantage of being specific to chiral species, for which it can determine the activity or optical purity without requiring a chiral column (see Chapter 2, Section 2.6.4). Consequently, it can be used to identify enantiomers or to remove the interference of compounds with a plane of symmetry or containing racemic mixtures.

This detector includes a polarimeter, whose principle is based on the specific rotation of the polarized light that any chiral compound produces in enriched enantiomer form (Figure 3.26).

> Polarimeters are widely used for the analysis of chiral compounds and the determination of their specific optical rotation. For a long time, the measurement of optical rotation alone was the method of choice used to determine the optical purity of a compound. However, this technique has now mostly been replaced by chiral chromatography (see Section 3.5.4) for the determination of the proportions of each enantiomer of an enantiomerically enriched compound.

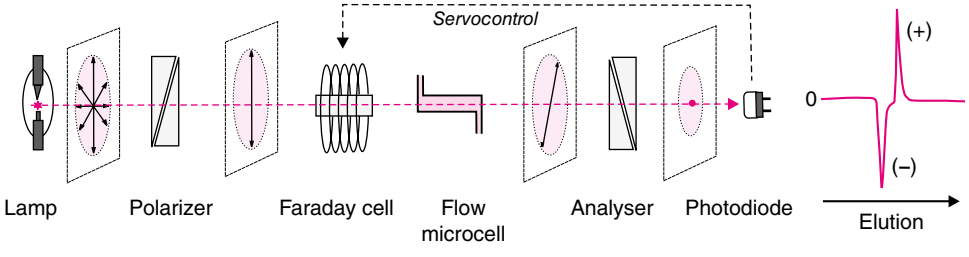


Figure 3.26 Polarimeter detection. Diagram representing the main components of a polarimetry detector and the effect on light between the source and the photodiode detector. The light emanating from the source is first polarized and then crosses through the measurement microcell placed upstream from the column. Then, a second polarizer, or analyser, has a polarization axis perpendicular to the first, thus ensuring the extinction of a light signal controlled by the photodiode. During passage through the measurement cell, the angle of rotation of the light will change by a value α. The presence of a Faraday rotator in the set-up will enable us, by the induction of a magnetic field on the optical material located in the coil axis, to correct the α angle in order to return to the extinction of the light signal. The intensity of this current constitutes the detector signal (model OR-2090). (Source: Courtesy of Jasco.)

The main limitation of this detector is its low sensitivity, even though technological progress has made it possible to evaluate rotations of the plane of polarized light at the microdegree level. This may lower the detection threshold down to the microgram level for some compounds.

Circular dichroism detector. The polarized light following an axis is in fact the result of two components of left and right circular polarized light. However, a chiral compound has a molar extinction coefficient of the left component ε_g that is different from that of the right component ε_d. The detector based on this property records this difference as a function of wavelength, thus leading to symmetrical spectra with respect to the *x*-axis for each enantiomer. This rarely used detector thus provides additional information to identify an enantiomer.

3.8.6 Other Detectors

The identification of a compound from its retention time alone can be uncertain, unless these are repetitive controls, for example in process analysis. In general, it requires an authentic sample in order to make a comparison. However, as has already been explained for GC, another approach is possible if there is a detector giving complementary information about the compounds being eluted.

When the detector itself provides a second method of analysis, qualifying it as two-dimensional, identification becomes more certain. These can be a spectrophotometer used simultaneously as a classic detector (obtaining the chromatogram) and as a more effective tool (such as a mass spectrometer) for identification of the separated species (see Chapter 16, Section 16.5). Coupling an HPLC instrument with a mass spectrometer (HPLC-MS), using the implementation of new interfaces and the development of ionization techniques in MS, has become essential in many domains (environment, drug screening, pharmaceutical research, etc.). This is currently the best technique to identify compounds, at least when we have an appropriate spectral library in which they can be found. Coupling HPLC with a proton-NMR spectrometer (HPLC-^{1}H NMR) greatly facilitates the study of mixtures containing unknown compounds (see Chapters 15 and 16).

3.9 HPLC OPTIMIZATION

As in GC, the optimization of a separation consists of finding a compromise between a good separation and a relatively short analysis time.

The parameters related to separation are the resolution and the selectivity factor. Optimization depends on:

1. **The column and its characteristics**
   Firstly, this involves the selection of the stationary phase, depending on the solutes that need separating. However, as most current separations are

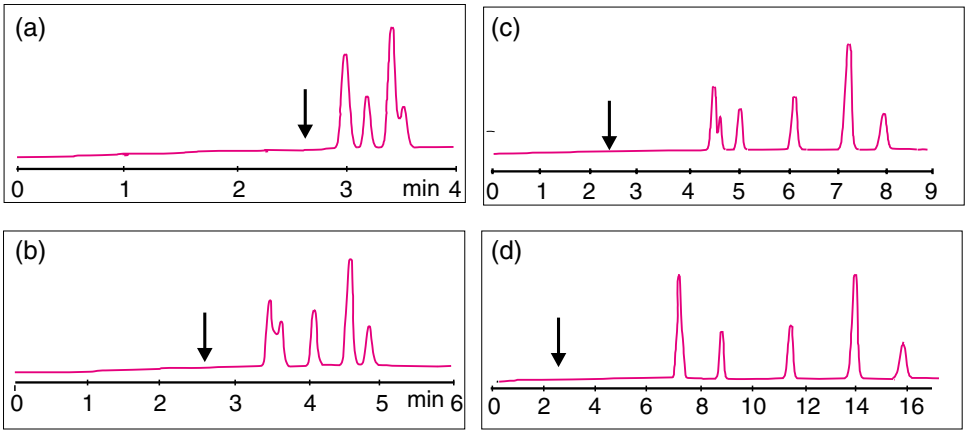


Figure 3.27 Chromatograms of a separation. The mobile phase is a binary water/acetonitrile mixture: (a) 50/50; (b) 55/45; (c) 60/40; (d) 65/35. The arrow indicates the hold-up time t_M (min). (Source: J.W. Dolan, *LC-GC Int.*, 1994, **7**(6), p. 333.)

conducted on RP-18 type columns, length *L*, diameter *ID*, and particle diameter d_p remain the criteria of choice.

2. **The mobile phase and parameters related to chromatograph settings**
   Unlike in GC, the mobile phase plays a significant role. Its composition (therefore, its polarity and its pH) may be constant (*isocratic mode*) or vary during an analysis (*elution gradient*). The column's efficiency also depends on the choice of proper flow rate.

   Figure 3.27 illustrates the optimization of a separation in liquid chromatography, with an aromatic hydrocarbon mixture by changing the composition of the mobile phase. We notice that this optimization is accompanied by a significant increase in the analysis time.

3. **Temperature**

   While it is well known in GC that temperature is a significant factor in improving a separation without changing the column, the effect of this factor remained unknown for a long time in HPLC, probably because grafted silica-based stationary phases degraded above 60°C. At most, if we brought the column to 35°C, this was to be free from temperature variations in the laboratory and therefore improve the separation reproducibility. Yet, for any solute, the retention factor *k* varies with temperature, which is logical, knowing that the equilibrium constant, or partition factor *K*, varies with temperature (Eq. (1.8)) and that *k* is related to *K* by the phase ratio β of the column in question (Eq. (1.26)). By comparing chromatograms of a given solute at two different temperatures, the corresponding Van Deemter curves highlight that term

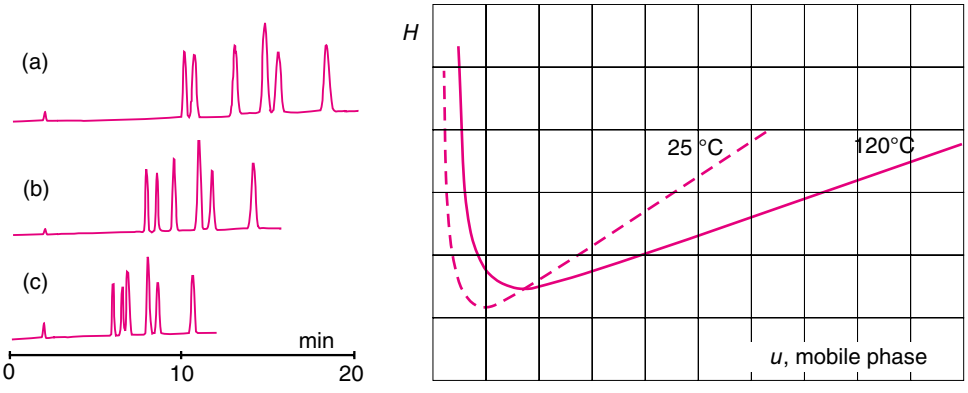


Figure 3.28 Effect of column temperature on a separation. Example showing three tests conducted on the same mixture and with the same mobile phase flow rate at different temperatures (a) 25°C, (b) 35°C, and (c) 45°C. Hold-up time is independent of temperature.

B (which reflects the diffusion phenomenon in the column) increases with temperature, while term *C* (related to the mass transfer) decreases (Figure 3.28). Working at high temperature therefore improves separation, because the diffusion of solute between the two phases is favoured.

Many liquid chromatographs can conduct separations at 120°C or above if the stationary phase as well as solutes are resistant to these temperatures and do not chemically interact with each other. The benefits include time savings and more symmetrical peaks. It also becomes possible to replace a solvent gradient with temperature programming.

3.10 NEW DEVELOPMENTS IN HPLC

Reducing analysis times and the mobile phase consumption, working on very small samples, improving separations and identification of eluted compounds are all drivers helping to move conventional HPLC forward and increase opportunities. Thanks to manufacturers' efforts in designing these instruments, a succession of improvements has given rise to more efficient devices. While this is not a new form of chromatography, new acronyms, originally commercial, have entered the chromatographer's jargon to designate these advances in instrumentation. Such is the case for U-HPLC (*Ultra-High-Performance Liquid Chromatography*) and UFLC (*Ultra-Fast Liquid Chromatography*).

An inexpensive improvement of HPLC performance levels first consists of determining whether there is a dedicated column and an operating procedure defined for the analysis in question. This is the case for several current assays, such as sugars, PAH, amines or nucleotides.

3.10.1 Reduction of Analysis Times

To reduce the analysis time, time needs to be saved on the separation stage, without losing the desired efficiency and selectivity. We do this by using shorter columns (2–5 cm) as long as the support phases selected are either monolithic or made up of small diameter, nonporous particles (2 μm or less). However, the latter cause a greater pressure drop. Moreover, pumps must be able to reach pressures exceeding 80 MPa. We can also increase speed of analysis if we conduct the separation by heating the column, which helps to more quickly reach concentration equilibria by reducing mobile phase viscosity. This progress may lead to retention factors that are 10 times shorter. That is the domain of UFLC (Figure 3.29).

3.10.2 Reduction of Mobile Phase Volumes

To reduce mobile phase consumption, and hence the cost of separations, and increase sensitivity, making it possible to work on very small samples, we use columns with diameters as low as 75 μm packed with traditional support phases or monolithic phases (Figure 3.30).

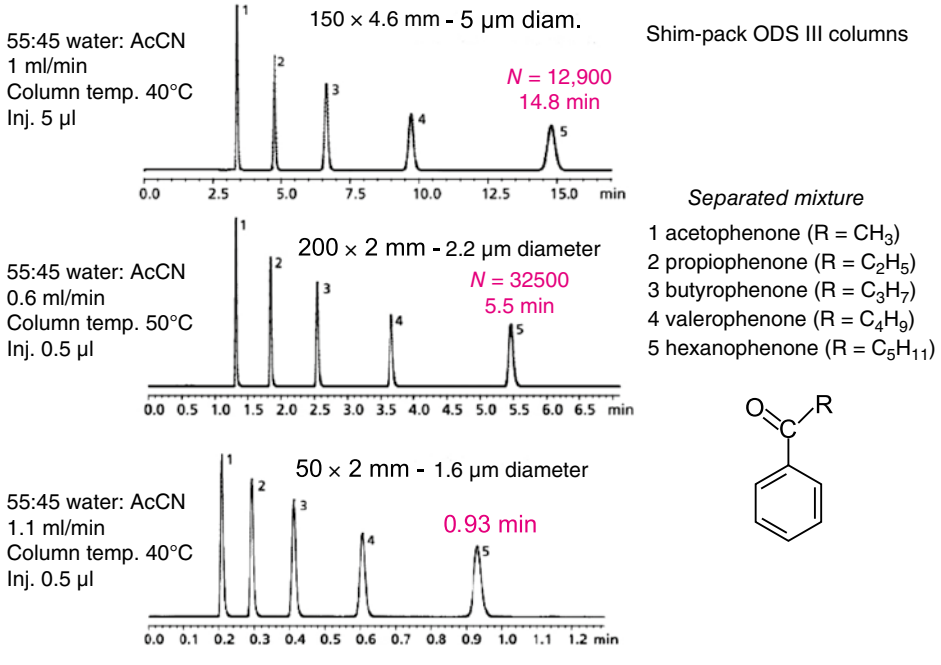


Figure 3.29 Fast chromatography. This example shows that the choice of column and operating conditions enables significant time savings without altering resolution and efficiency (courtesy of Shimadzu).

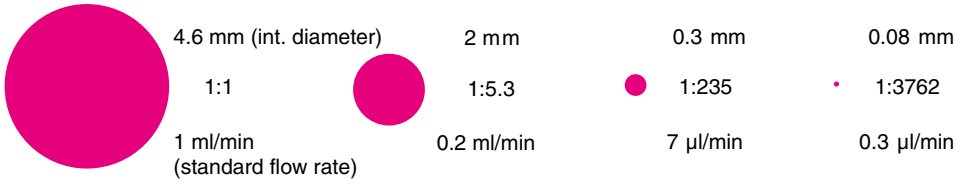


Figure 3.30 Comparison of flow rates in columns with different internal diameters. While the packing in these four columns is of a similar nature, there are advantages in switching from classical HPLC (left) to capillary HPLC or to nano-HCLP (right). The narrower the column, the greater is the decrease in mobile phase consumption. However, gradient reproducibility is difficult at this level.

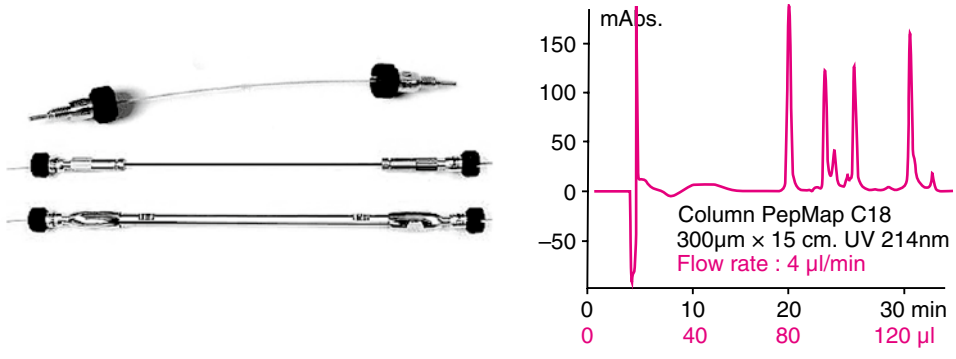


Figure 3.31 Columns for U-HPLC. Nanochromatography. Capillary columns, 75 µm, 300 µm, and 2 mm in diameter (Waters, Acclaim PepMap columns). Example of separation of 40 ng of a protein mixture: ribonuclease, insulin, cytochrome C and myoglobin with only 0.12 ml of mobile phase (document courtesy of Waters).

However, moving capillary liquid chromatography or nanochromatography into the kingdom of Lilliput is rather demanding to conduct. Mobile phase microflow rates and the smallest possible hold-up volumes must be managed throughout the process from the mixing chamber for gradients to the volume of the detection cell. The equipment must again be adapted. This is the U-HPLC domain.

3.10.3 Technology Related to HPLC Miniaturization

Nano-HPLC is used in fields where available samples are very small, such as in biochemistry for the identification of peptides and proteins. The compounds, which are less diluted by the mobile phase, lead to more intense signals, which facilitate their detection. Narrow (<2 mm), short columns, packed with particles with a diameter under 2 µm or monolithic type, are used to reach resolutions of the same order as with standard columns of the same length, packed with 5 µm particles, but in 10 times less time (Figure 3.31).

Although mobile phase flow rates are very low, the coupling of nano-HPLC with mass spectrometry cannot be done directly. Once again, the set-up must involve an additional miniaturized atmospheric pressure ionization with mobile phase elimination, inserted between the two devices.

KEY POINTS OF THE CHAPTER

1. An HPLC installation includes five components: a pump system, an injection device, a temperature-controlled column, a detector, and a data acquisition system.
2. HPLC separates all types of molecules as long as they are soluble in the mobile phase. The possibility of operating at ambient temperature helps avoid the breakdown of thermolabile molecules (which is especially necessary in biology: proteome, biomolecules).
3. The column is the heart of any HPLC installation. The chemical nature of the analytes guides the choice of column (length, diameter, etc.) and especially the stationary phase.
4. *Efficiency* of columns increases when the particle diameter decreases, as shown in the graph of the Van Deemter equation. Efficiency is lower than in GC, which we observe by the presence of wider peaks on chromatograms.
5. The interest in monolithic columns is due to the fact that the macroporous channels in them produce a lower pressure drop, which then enables a higher flow rate without altering performance and thus reduces analysis time.
6. Some GC detectors are nearly universal (for example, FID with a signal which depends on the mass flow rate), while others lead to a unique response, which depends on the physico-chemical characteristics of the compound.
7. HPLC-MS coupling requires the elimination of the mobile phase between the two instruments, which can be done using atmospheric pressure ionization (API) techniques. These gentle methods produce few fragments and therefore not much structural information about the analytes.
8. The success of HPLC is due to the optimization of analytical conditions using the many interactions between the *eluent*, *stationary phase* and *analytes*.
9. Two major drawbacks of HPLC are the duration of an analysis (considering re-equilibration time of the phases between two analyses) and the consumption of mobile phase, which is expensive.
10. A good choice of column alone can improve analyses. The three most interactive parameters are: column length, internal diameter, and the size and nature of the solid-core spherical particles constituting the stationary phase.

PROBLEMS

1. Current HPLC devices can use columns with an internal diameter (written as ID in the text) of 300 μm, and with a recommended optimum flow rate of 4 μl/min.

a. Show by simple calculation that this flow rate will lead to practically the same linear speed of the mobile phase as in a column of the same type but with a standard diameter of 4.6 mm and whose recommended flow rate is 1 ml/min.

b. The separation of a sample containing 16 PAHs (polyaromatic hydrocarbons) was conducted on an RP-18 type column (L_C = 25 cm, ID = 300 μm). The mobile phase flow rate (acetonitrile/water) is 4 μl/min. One of these PAHs has a 48-minute retention time. Calculate the retention volume of this compound. What can we conclude from that?

c. Two columns selected for an experiment are packed with the same stationary phase. One column has an internal diameter of 4.6 mm, the other 300 μm, while both have the same length and the same packing rate (V_S/V_M). It is decided to use them in succession with the same chromatograph and under identical conditions with the flow rates for each as advised above. An equal amount of the same compound is injected into each of them in turn.

i. When passing from one column to the other, would a difference in the retention volume (or elution volume) of the compound be expected?

ii. If the sensitivity of the detector has not been adjusted between the two experiments, will the intensity of the corresponding elution peaks be different?

2. What is the order of elution of the following acids from an HPLC column whose stationary phase is of the C18 type, while the mobile phase is a formate buffer, C = 200 mM, pH 9.

Mixture:

a. Linoleic acid	$CH_3(CH_2)_4CH = CHCH_2CH = CH(CH_2)_7CO_2H$
b. Arachidic acid	$CH_3(CH_2)_{18}CO_2H$
c. Oleic acid	$CH_3(CH_2)_7CH = CH(CH_2)_7CO_2H$

3. Indicate for each of the chromatographic techniques below the term which best expresses the essential analyte interaction with the stationary phase

1. Reverse phase	a. Molecular mass
2. Gel permeation	b. Hydrophilicity
3. Ion chromatography	c. Hydrophobicity
4. Normal phase	d. Protonation/ionization

4. The separation of two compounds A and B is studied by HPLC on a column of RP-18 type. The mobile phase is a binary mixture of water and acetonitrile. A linear relationship exists between the logarithm of the retention factor and the % of acetonitrile in the binary mixture (H_2O/CH_3CN).

Two chromatograms were obtained, one in which the mixture of water and acetonitrile was 70/30 v/v, respectively, and the other where the same two components were mixed 30/70 v/v. The equations of the straight lines are, for compound A:

$$\log k_A = -6.075 \times 10^{-3} \left(\%CH_3CN\right) + 1.3283$$

and for compound B:

$$\log k_B = -0.0107 \left(\%CH_3CN\right) + 1.5235$$

a. Find the composition of the binary phase which leads to a selectivity factor of 1.

b. Suppose that for each compound the corresponding full width at half-maximum (FWHM) of the peak on the chromatogram is the same and the column efficiency does not change with the composition of the mobile phase. Is the resolution between the two peaks better for the mobile phase containing 70% water or 30% water? Discuss the practicalities of your choice.

5. Our goal is to determine the simazine and atrazine composition of surface water sampled from a river. We sample one litre of this water which we treat as follows:
Extraction of simazine and atrazine with dichloromethane,
Evaporation of dichloromethane to dryness in the rotating evaporator,
Dissolution of the dry residue with 10 ml of methanol. We then obtain the unknown solution S_{unk}.
We analyse this HPLC solution on an RP-type column by successively conducting three injections, under identical conditions. Results are recorded in the following table:

Solute	Peak area	Solute	Peak area
Atrazine	3,021	simazine	2,505
	2,989		2,432
	2,897		2,317

We prepare three standard solutions respectively containing 10 mg (solution S1), 15 mg (solution S2), and 20 mg (solution S3) of atrazine and simazine in 100 ml of methanol. Each standard solution is injected twice. The peak areas of atrazine and simazine are:

Solution	Injection No.1	Injection No.2
S1	875 (Atrazine)	880 (Atrazine)
	912 (Simazine)	924 (Simazine)
S2	1,308 (Atrazine)	1,319 (Atrazine)
	1,362 (Simazine)	1,371 (Simazine)
S3	1,698 (Atrazine)	1,730 (Atrazine)
	1,792 (Simazine)	1,852 (Simazine)

a. For each standard injection, calculate the response factors of the detector to atrazine and simazine in mg/area unit.

b. For each solute, calculate the average response factor of the detector to atrazine and simazine in mg/area unit.

c. For each injection of the unknown solution $S_{unk.}$, deduce the atrazine and simazine mass concentration of the alcohol solution. Give the average mass concentration of atrazine and simazine in the sampled river water.

6. According to the elution curve theory, the concentration of a solute at the maximum point of the chromatogram peak is given by the expression:

$$C_{max} = \frac{C_0 V_{inj}}{V_R} \sqrt{\frac{N}{2\pi}}$$

Where C_0 is the solute concentration in the injected solution, V_{inj} is the injected solution volume and N is the column efficiency.
Characteristics of the column: L = 20 cm, N = 3,000 plates under the conditions of use.
Conditions of use: isocratic regime, flow rate D = 0.5 ml/min, hold-up time t_M = 0.8 min.
Injected solution: C_0 = 0.5 g/l, V_{inj} = 20 µl

a. Calculate the volume and the retention time of the solute in question if we want the concentration C_{max} at the peak summit to be equal to 0.75 C_0.

b. Under these conditions, deduce the value of the solute's retention factor.

c. What happens to the retention time if the length of the column is multiplied by two? What happens to C_{max}? The HETP of the column remains constant.

d. If we assume that the detectability limit of the detector for the solute in question is equal to 0.5 mg/l, what will be the maximum length of the column (we maintain the same injection volume)?

7. A silica gel for chromatography has a specific surface area of 370 m^2/g. To assess the average number of silanol groups accessible per nm^2, we sample 0.2 g of silica gel, to which we add an excess of methyl lithium (CH_3Li), in an appropriate set-up. We collect 12.5 ml of methane (normal T and P conditions).
Calculate the average concentration of the residual silanol groups per nm^2.

8. Trans-anethole (TA) (trans-1-methoxy-4-(1E)-1-propenyl-benzene) is used as an additive in anise-flavoured spirits, such as pastis. Toxic at high doses, its content is regularly checked, usually by GC. However, some of these analyses may be conducted by HPLC. The following analysis was conducted by HPLC with a Lichosorb CN column (205 × 7 mm, 5 µm), with n-hexane as the mobile phase, with the temperature being maintained at 20°C and UV detection at 254 nm.

a. What are the polarities of the stationary phase and of the mobile phase? What is the working mode used?

b. To quantify trans-anethole (TA) present in an anise-flavoured spirit, we proceed as follows: we start by preparing a range of standards from a mother solution obtained by dissolving 2.5 g of TA in 250 ml of *n*-hexane.

i. What is the concentration of the mother solution?
Standard solutions of 0.1, 0.2, 0.3, 0.4, and 0.5 g/l are prepared from the mother solution by using 100 ml graduated flasks.

ii. What volumes must be sampled to prepare this range of standards?
We inject 20 µl of each solution of the range of standards. Peak areas of TA, as determined from the various chromatograms obtained, are gathered in the table below:

Standard solutions (g/l)	Peak area
0.1	6,110
0.2	12,257
0.3	18,404
0.4	24,551
0.5	30,698

iii. Determine the equation of the calibration curve A = f(C). Give the correlation factor.

c. To assay TA from the anise-flavoured beverage marketed in a 0.7-l bottle, we sample 10 ml and put it into a 100 ml flask. We fill to the graduated line with n-hexane. We obtain solution S1. We inject 20 µl of this solution. The TA peak area obtained is equal to 15,662.

Determine the mass concentration of TA in S1. Deduce the TA content of the anise-flavoured beverage.

SOLUTIONS

1. a. For identical packing (same porosity ε_c), we get $\bar{u}_1/\bar{u}_2 = 1.06$.

b. $V_R = 48 \times 4 = 192$ µl or 0.192 ml (*solvent savings*).

c. i. Hold-up volumes are in the same ratio as the cross-sections of the two columns:

$$V_{M(0.3)} / V_{M(4.6)} = 0.0043.$$

The retention volume on the column (0.3 mm) is therefore equal to the retention volume of the column (4.6 mm) multiplied by 0.0043 (*i.e. 235 times less*).

ii. As the quantity of product injected is the same, the product will be 235 times less diluted in the mobile phase, and the signal will thus be theoretically 235 times more intense (*increased sensitivity for detection*).

2. Compounds will be eluted in the order 1, 3 and 2.
3. (1,c) reversed phase; (2,a) gel permeation; (3,d) ion chromatography; (4,b) normal phase.
4. **a.** If α is equal to 1, the peaks overlap, and retention times are identical, which means that k, therefore $\log k_A = \log k_B$. The MeCN percentage must verify both expressions:

$$-0.0107\left[\%\text{MeCN}\right]+1.5235=-6.075\times10^{-3}\left[\%\text{MeCN}\right]+1.3283.\ \text{We find MeCN}=42.2\%.$$

   b. Compound A: $k_{30} = 14$; $k_{70} = 8$; compound B: $k_{30} = 16$; $k_{70} = 6$. For 30% MeCN: $\alpha = 1.143$ and for 70% MeCN: $\alpha = 1.333$. By applying the formula giving the resolution according to k and α, we find that the term in α and k is 0.222 for 70% and 0.118 for 30%. Therefore, it is preferable to use the mobile phase at 70% of MeCN (much quicker separation).
5. **a.** For S1: $k_{at(1)} = 2.28 \times 10^{-6}$; $k_{at(2)} = 2.27 \times 10^{-6}$; $k_{sim(1)} = 2.19 \times 10^{-6}$; $k_{sim(2)} = 2.16 \times 10^{-6}$; for S2, $k_{at(1)} = 2.29 \times 10^{-6}$; $k_{at(2)} = 2.27 \times 10^{-6}$; $k_{sim(1)} = 2.20 \times 10^{-6}$; $k_{sim(2)} = 2.18 \times 10^{-6}$; for S3, $k_{at(1)} = 2.35 \times 10^{-6}$; $k_{at(2)} = 2.31 \times 10^{-6}$; $k_{sim(1)} = 2.23 \times 10^{-6}$; $k_{sim(2)} = 2.16 \times 10^{-6}$.
   b. $k_{at} = 2.29 \times 10^{-6}$ and $k_{sim} = 2.19 \times 10^{-6}$
   c. For the injected solution, $C_{at} = 340$ mg/l, $C_{sim} = 265$ mg/l; for river water, $C_{at} = 3.4$ mg/l, $C_{sim} = 2.65$ mg/l.
6. **a.** $V_R = 0.582$ mL, $t_R = 1.164$ min;
   b. $k = 0.455$;
   c. The retention time is multiplied by two, C_{max} is divided by $\sqrt{2}$;
   d. Maximum column length $L_{max} = 1.125 \times 10^7$ cm because the concentration at the peak summit is inversely proportional to the square root of the length: solution not possible, therefore we will not be under the detection limit.
7. During this treatment, silanol functions at the surface of the silica gel generate methane according to the reaction Gel-SiOH + $CH_3Li \rightarrow$ Gel-SiOLi + CH_4
   12.5 mL of methane (resulting from 0.2 g of silica gel, therefore 370/5 = 74 m^2) corresponds for 1 m^2 to 12.5/(22,400 × 74) = 7.5×10^{-6} moles, and for 1 nm^2, i.e. a surface area that is 10^{18} times lower, to 7.5×10^{-24} mole. Considering Avogadro's number, the average silanol number is thus $7.5 \times 10^{-24} \times 6.02 \times 10^{23} = 4.5$ per nm^2.

   Note. During silica gel grafting, one or two nontransformed silanol groups subsist. They are eliminated in an additional step called '*end capping*' using specific reagents.
8. **a.** HPLC in normal phase: nonpolar solvent and polar Lichrosorb CN stationary phase. Isocratic mode.

b. Concentration of the mother solution: C = m/V => C = 10 g/l; like for any other dilution, we use the expression $C_m V_m = C_d V_d$ (index m for mother solution, index d for the daughter solution).

Solution (g/l)	Vm in ml
0.1	1
0.2	2
0.3	3
0.4	4
0.5	5

Equation of the calibration curve: A = 61,220 C.

c. C_{S1} = 0.256 g/l. Anise-flavoured beverage C = 2.56 g/l.

Chapter 4

Ion Chromatography

Ion-exchange chromatography (IEC) is the main technique in ion chromatography (IC), which also includes ion-pair chromatography (IPC) and ion-exclusion chromatography. IEC enables the study of organic compounds as long as they are polar. It shares numerous common features with HPLC, yet can be differentiated by the principle of separation. The mobile phase is composed of an aqueous ionic medium and the stationary phase is a solid that behaves as an ion exchanger. Among the many detection methods available, those based on

the conductivity of solutions are preferred. However, the current applications of IC (environmental and biochemistry analyses) are far broader than the analysis of simple cations or anions by which the technique first gained renown.

Objectives

Understand the role of stationary phases (cationic and anionic)
Understand the role of the mobile phase and its additives
Explain the interest in detection by conductometer
Find the origin of the water peak and the system peak
Describe the role of the eluent regenerator
Compare chemical and electrolytic ion suppressors
Differentiate IEC from ion-exclusion chromatography

4.1 BASICS OF ION CHROMATOGRAPHY

Ion-exchange chromatography is mostly used for the separation of polar compounds, whether organic or mineral, depending on their charge. Stationary phases are made up of polymers or organic resins or silica gel, on the surface of which ionized functional groups are covalently bound, capable of reversibly exchanging charged species present in the analytes. If a compound has a high charge density, it will be retained a longer time by the stationary phase. This exchange process is much slower when compared with those found in other types of chromatography.

IEC instruments have the same modules as those found in HPLC (Figure 4.1), with suitable characteristics to manage the aggressiveness of solutions and acidic or basic mobile phases.

The progression and separation of compounds within the sample is founded upon ion-exchange phenomena, for which two classic situations are:

- If cationic species (M^+ type) are to be separated, a *cationic* column (to separate cations) with a stationary phase capable of exchanging cations will be employed. Such a phase is constituted, for example, of a polymer containing grafted sulfonate ($-SO_3H$) groups. Consequently, the stationary phase is the equivalent of a polyanion.
- Alternately, if anionic species (A^- type) are to be separated, an *anionic* column is selected. This is achieved, for example, by employing a polymer containing ammonium groups.

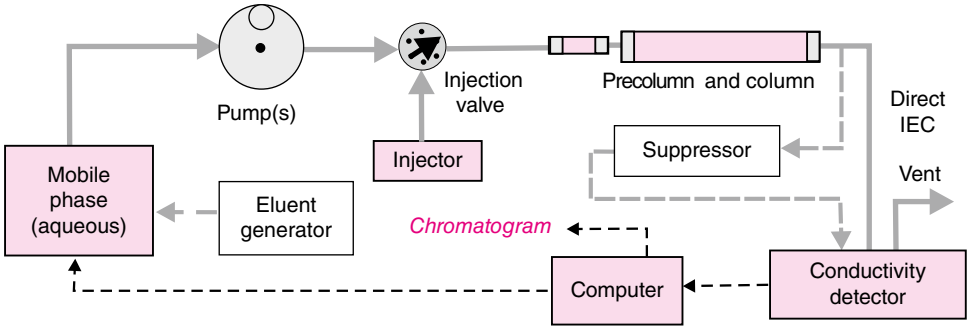


Figure 4.1 Diagram of an ion-exchange chromatograph. The classic modular architecture, whether integrated or not, with here, two optional modules (eluent generator and suppressor) and a detector based on the conductivity of the solutions. The configuration includes a 'suppressor' device installed between the column and the detector. The suppressor serves to eliminate the ions arising from the eluent by acid–base reaction. It is recommended for mobile phases that are highly concentrated in ions.

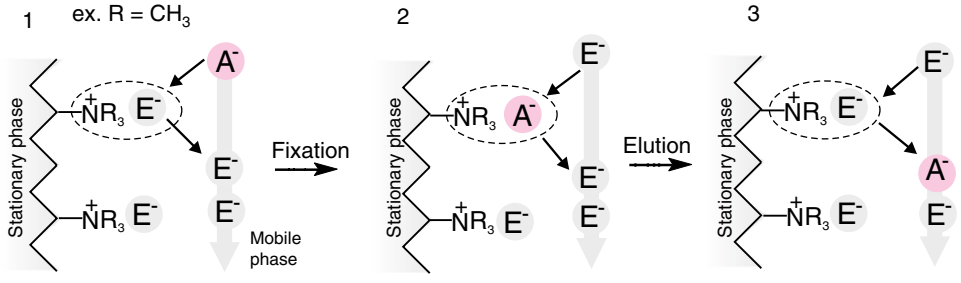


Figure 4.2 Diagram showing the progression of an anion A^- in contact with an ammonium stationary phase. Ion E^- (generally a bicarbonate ion) is the counteranion of the mobile phase. It is bound to the stationary phase. The anionic species A^- (chloride ion, for example), which is the analyte present in the mobile phase, takes the place of the ion E^-. Elution reverses the phenomenon by regenerating, on the site in question, the initial state (binding of an ion E^- or another ion of the same type). The progression of A^- through the column will depend on its affinity with ionic sites of the column.

To understand the mechanism of separation, take, for example, an anionic column containing quaternary ammonium groups in equilibrium with a mobile phase composed of a solution of bicarbonate anions (e.g. counteranions). All of the cationic sites of the stationary phase find themselves paired with anions from the mobile phase (Figure 4.2).

When an anion A^- within the sample is entrained by the eluent, a series of reversible equilibria are produced, governed by an exchange equation giving the ion's distribution between the mobile phase (MP) and the stationary phase (SP).

Arrow 1 in Eq. (4.1) corresponds to the attachment of the anion A^- to the SP and arrow 2 to its return to the mobile phase and therefore to its progression down the column.

$$A^-_{MP} + \left[HCO_3^-\right]_{SP} \underset{2}{\overset{1}{\rightleftarrows}} \left[HCO_3^-\right]_{MP} + A^-_{SP}$$

$$\frac{\left[A^-_{SP}\right] \cdot \left[\left[HCO_3^-\right]_{MP}\right]}{\left[A^-_{MP}\right] \cdot \left[\left[HCO_3^-\right]_{SP}\right]} = Cste \qquad (4.1)$$

This equilibrium represents the selectivity α between two anions regarding the stationary phase cation.

Using a cationic column, a similar situation can be described (the stationary phase, SP, corresponds, for example, to strongly acidic Polym-SO_3H,):

$$M^+_{MP} + H^+_{SP} \underset{2}{\overset{1}{\rightleftarrows}} M^+_{SP} + H^+_{MP}$$

This exchange phenomenon, which allows polar species to be retained on the resin, is known as *solid-phase extraction* (Figure 4.3). If the sample contains two ions X and Y and if $K_Y > K_X$, Y will be retained more than X on the column.

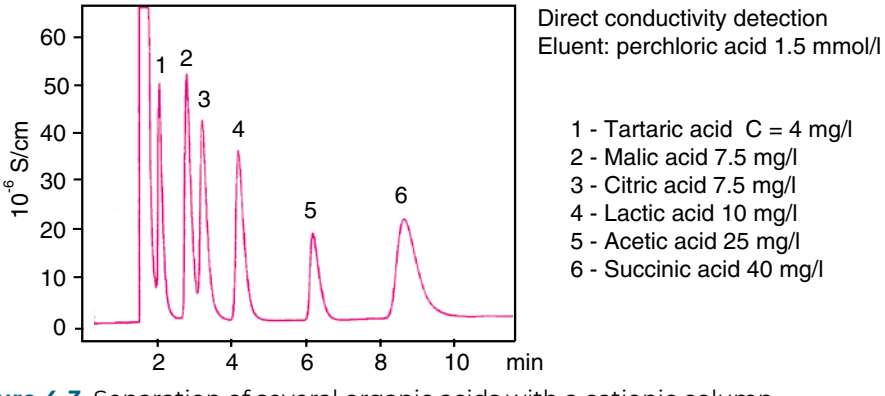


Figure 4.3 Separation of several organic acids with a cationic column.

4.2 STATIONARY PHASES FOR IEC

There is a large selection of stationary phases dedicated to analyses of anionic or cationic compounds and depending on whether the device used has a suppressor or not. These phases are either versatile or adapted to an analyte category: sugars, organic acids, proteins, etc. Stationary phases must satisfy a number of requirements, such as narrow particle-size distribution (monodisperse), large specific surface area, mechanical resistance, stability under acid and basic pHs, and rapid ion transfer.

4.2.1 Synthesis Copolymers

The most common stationary phases come from copolymerization of styrene and divinylbenzene. We can thus obtain crush-resistant, cross-linked phases. They are made of spherical particles with diameters of a few micrometres (Figure 4.4). These particles are then grafted to make polyanions or polycations.

Thus, concentrated sulfuric acid is reacted with the accessible aromatic rings of the copolymer surface to graft on SO_3H functional groups. A highly acidic (strongly cationic) phase is obtained, on which the anion is bound to the copolymer, while the cation can be reversibly exchanged with other cationic species present in the eluent. The exchange capacity is a few mmol/g and is independent of solution pH.

Starting from the same copolymer, an anion exchange resin can be synthesized, firstly by chloromethylation, which binds $-CH_2Cl$ (*Merrifield's resin*), followed by reaction with a secondary or tertiary amine depending on the desired basicity for the stationary phase.

> On contact with water, a mildly basic stationary phase such as Polym-NMe_2 yields a weakly ionized phase $(Polym\text{-}NMe_2H)^+OH^-$, especially when the medium is basic. Alternatively, in an acidic medium, it will appear as a strongly basic phase whose active surface will be strongly ionized: $(Polym\text{-}NMe_2H)^+Cl^-$. The exchange capacity of these resins varies with the pH.

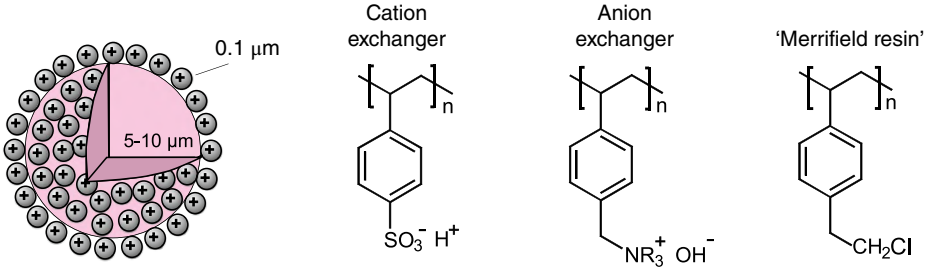


Figure 4.4 Stationary phases in IC. Cross-section of a spherical particle of polystyrene used as a cation exchanger. The polystyrene matrix is transformed either into a cation (e.g. DOWEX® 4) or an anion (e.g. DOWEX® MSA-1, or Permutite® if *R* = Me) exchange resin.

4.2.2 Grafted Silicas

Silica gel can also serve to support, through covalent bonding, alkyl chains carrying sulfonated groups or quaternary ammonium groups (Figure 4.5). This step is similar to that used to obtain grafted silica phases in HPLC. Some of these phases combine the properties of ion chromatography with those of HPLC. Separations depend simultaneously on both ionic coefficients and partition coefficients.

4.2.3 Polysaccharides

Two polysaccharides, cellulose and agarose (or sepharose), are used as matrices to graft on various acid or basic groups, in order to make them useable in IEC (Figure 4.6). Thus, these two types of linear polymer chains are chemically changed to be transformed

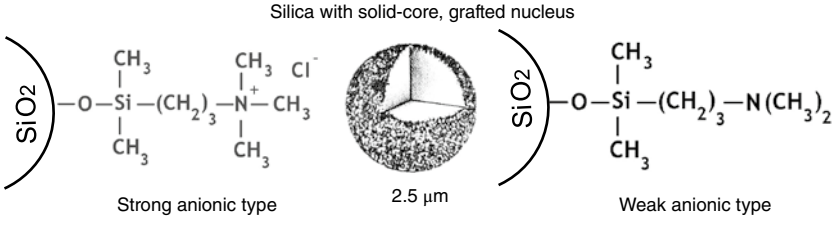


Figure 4.5 Anionic phases obtained by solid-core silica grafting. Nonporous monodisperse silica gel.

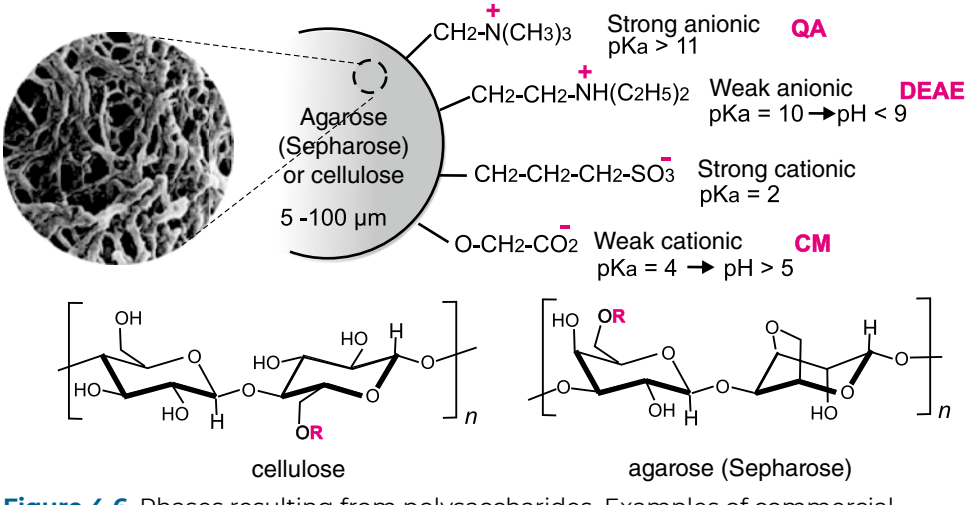


Figure 4.6 Phases resulting from polysaccharides. Examples of commercial resins made from cellulose or agarose. Micrography (50,000+). (Source: Reproduced courtesy of Interchim.)

into 'resins'. They come in the form of spheres with a diameter ranging from 5 to 100 µm. These phases, which are hydrophilic and not very resistant to crushing under pressure, are often reserved for separations and purifications of proteins containing both cationic and anionic sites and which are thus very sensitive to solution pH.

4.3 MOBILE PHASES

Eluents used as mobile phases are charged aqueous solutions containing saline or organic ions and, when necessary, a small amount of methanol or acetone used to dissolve certain samples. Depending on column type (cationic or anionic), ions in the eluent are provided either by mineral or organic acids (perchloric, benzoic, phthalic, methane sulfonic, etc.) or by bases (sodium or potassium hydroxide, carbonates/bicarbonates, etc.). In the latter case, elution solutions have the unfortunate tendency to be carbonated by absorption of carbon dioxide from the ambient air, which progressively disturbs retention times.

To avoid this contamination, a potassium hydroxide (KOH) generator can be used (either acidic or basic). This is inserted, as a supplementary module, between the pump and the injector of the ion chromatograph (Figure 4.7). If the flow rate of water and the electrolytic current intensity are known, then the KOH concentration of the eluent can be determined with precision and concentration gradients can be conducted.

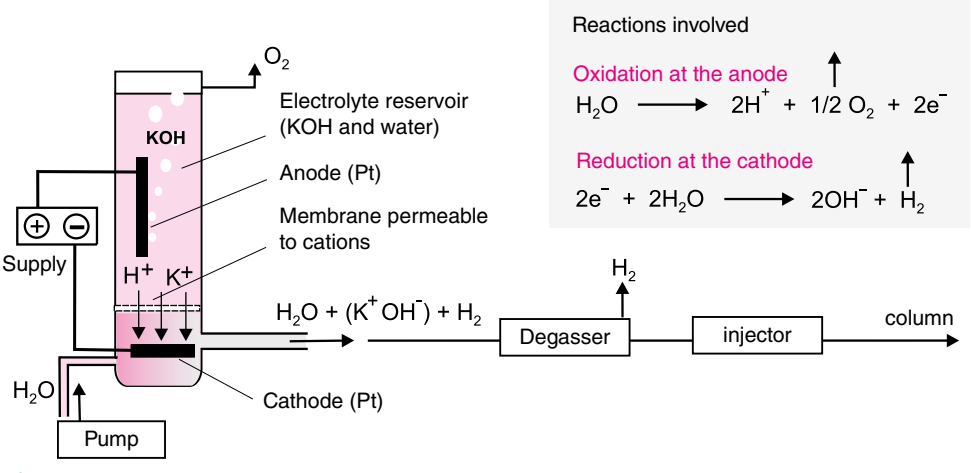


Figure 4.7 Ion chromatograph containing a hydroxide ion (OH^-) generator. Through electrolysis of water, we generate hydroxide ions in contact with cathodes, accompanied by hydrogen eliminated by a capillary degasser. KOH may be replaced by NaOH or LiOH. This generator is located upstream from the injection loop. (Source: Based on a document from Dionex.)

4.4 CONDUCTIVITY DETECTORS

Besides the spectrophotometric detectors based on absorbance or fluorescence of UV/visible radiation, used when the mobile phase does not absorb appreciably in the measurement range, another mode of detection exists based upon the property of an electrolyte to conduct an electrical current. Thus, at the outlet of the column, the conductance (the inverse of resistance) of the mobile phase is measured between two microelectrodes. To do direct measurements, the ionic charge of the eluent has to be as low as possible and the measuring cell requires strict temperature control to within 0.01°C because of the high dependence of conductance on temperature (~ 5%/°C). The measuring cell, downstream from the column, should have a very small volume (approx. 2 µl).

The sensitivity of the detector to an ion X (valence *z* and molar concentration *C*) can be predicted if its equivalent conductivity (Λ_X) and that of the eluent ion E (Λ_E) are known. The ΔK value can be calculated according to Eq. (4.2), knowing that, depending on the sign, the peak will be either positive or negative.

$$\Delta K = C \cdot z \cdot \left(\Lambda_X - \Lambda_E\right) \tag{4.2}$$

Conductance of solutions

Conductance G = 1/R corresponds to the reciprocal of the resistance *R*, measured between two electrodes that are plunged into the conducting solution and across which is maintained an electric potential difference. *G* is expressed in Siemens (S). For a given ion, the conductance of the solution varies with the concentration of the electrolyte. This relationship is linear for very dilute solutions. The *specific conductance* (in S/mol) or *conductivity k*, permits the measure to be independent from the detection cell settings:

$$k = GK_{cell} \tag{4.3}$$

$K_{cell} = d/A$ represents the *cell constant* (area *A* and spacing *d*). Its value cannot be obtained by direct measurement but is determined from a standard solution whose conductivity k is known.

Lastly, the *equivalent ionic conductance* ($S \cdot m^2/mol$) represents the conductivity of an ion with a valence z, in an aqueous solution at 25°*C*, when the molar concentration *C* (mol/l) tends towards zero in water (Table 4.1).

$$\Lambda_0 = 1000k/Cz \tag{4.4}$$

Table 4.1 Equivalent conductivities of ions at infinite dilution in water at 25°C.

Cations	Λ_0^+ (mS · m²/mol)	Anions	Λ_0^- (mS · m²/mol)
H^+	35	OH^-	19.8
Na^+	5.0	$½SO^{2-}_4$	8.0
K^+	7.3	Cl^-	7.6
NH^+_4	7.3	HCO^-_3	4.5
$½Ca^{2+}$	6.0	NO^-_3	7.1

4.5 WATER PEAK AND SYSTEM PEAK

Two specific peaks often appear on the chromatogram of the sample analysed, whatever the nature and in the absence of a suppressor.

4.5.1 Water Peak

The first peak appearing on a chromatogram originates from the species present in the sample injected and not retained on the column. In GC and HPLC alike, these may be traces of solvent or air. In IEC, since the column is supposed to be in a state of dynamic equilibrium, the injected sample produces a sudden introduction of ions, a part of which is exchanged by binding on the stationary phase. The other part remains in the mobile phase. For example, with a column to separate cations, only the latter bind to the stationary phase by displacing an equivalent number of counterions, which re-enter the solution and accompany anions in the sample. The whole migrates through the column at the velocity of the mobile phase and leads to the appearance of a signal on the chromatogram called an injection peak or *water peak*. With a conductivity meter, this peak will be positive or negative because it depends on the conductivity and the quantity of these ions. This peak marks the column's *hold-up time* (Figure 4.8).

4.5.2 System Peak

When we use a weak organic acid to elute anions, and if the chromatograph is not equipped with an ion suppressor, a peak appears, called the *system peak*, which corresponds to the elution of the nondissociated form of this acid (Figure 4.8). The more hydrophobic it is, the longer is its retention time. The mechanism of its migration, in competition with that of ions, is governed, as in HPLC, by its SP/MP partition

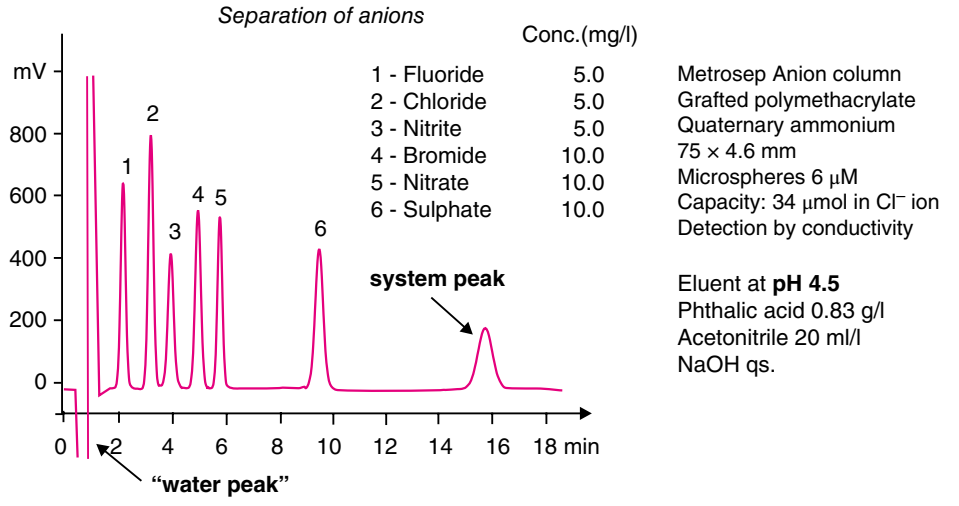


Figure 4.8 Chromatogram resulting in the water peak (1 min) and the system peak (16 min) and example of anion separation. Phthalic acid (the eluent) is an organic diacid (pKa_1 = 2.9 and pKa_2 = 5.5). Since the pH value selected is intermediate between those two values (pH < 4.5), the nondissociated form of this diacid subsists. (Source: Courtesy of Metrohm.)

coefficient. It is useful to locate it in order not to confuse it with an analyte, knowing that it does not always constitute the last peak. The existence of this peak also proves that the SP may be exploited in order to separate molecules which are polar but not ionized (see Section 4.7).

4.6 ION SUPPRESSORS OF THE ELECTROLYTE

In the highly ionic eluent, it may be difficult to track the signal made by low concentrations of analyte ions. To improve the sensitivity of this type of detection, we insert a device, called a suppressor, between the column and the detector to exchange by capturing the ions brought by the electrolyte. This suppressor plays a dual role. The first is to significantly reduce the background conductivity of the eluent, while decreasing the background noise by neutralizing the acid used in cationic analysis (methanesulfonic acid, CH_3SO_3H, or sulfuric acid, H_2SO_4) or the base used in anionic analysis (potassium hydroxide, KOH, or sodium bicarbonate, $NaHCO_3$). The second role is to replace cations with H^+ ions, whose ionic conductivity is much higher (Table 4.1), thus improving the signal-to-noise ratio and consequently the sensitivity.

Two suppression techniques have been developed: *chemical suppression* and *electrolytic suppression*.

4.6.1 Chemical Suppressor

The simplest model of a suppressor contains one or more anion exchange resins. This technique is currently used successfully to analyse anions and organic acids. The eluent used in this case for the analysis of anions is classically a diluted solution of sodium bicarbonate. The suppressor, constituted of a sulfonic acid resin, will first react with bicarbonate ions to form carbon dioxide, which is eliminated from the suppressor *through* a permeable membrane. This first reaction eliminates the background conductivity of the eluent:

$$Na^+HCO_3^- + RSO_3H \underset{\leftarrow}{\overset{\rightarrow}{}} RSO_3^-Na^+ + H_2CO_3 \underset{\leftarrow}{\overset{\rightarrow}{}} H_2O + CO_2g$$

The ion-exchange resin will then exchange cations with H^+ ions, which have a much higher conductivity, thus improving sensitivity. In the following example, Na^+ ions associated with the Cl^- ions being analysed are exchanged with H^+ ions:

$$Na^+Cl^- + RSO_3H \underset{\leftarrow}{\overset{\rightarrow}{}} RSO_3^-Na^+ + H^+Cl^-$$

Recent systems may combine several resin cartridges, which are regenerated one after the other by a dilute solution of sulfuric acid.

Inversely, at the end of a column to separate cations, we can add a suppressor containing an anion resin in order to neutralize cations carried by the eluent (Figure 4.9).

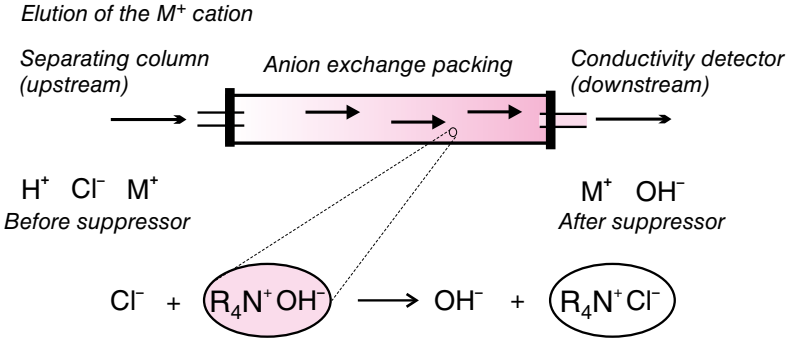


Figure 4.9 Chemical suppressor for a cation exchange column. In this example, the anion suppressor purges the mobile phase of H^+ ions and almost all of the Cl^- ions, facilitating the detection of the cation M^+. This suppressor contains an anion exchange resin (e.g. $ArCH2(NR)_3^+OH^-$), which will release hydroxyl ions and therefore neutralize a certain number of H^+ ions to form water molecules.

The drawback with this type of suppressor is its significant hold-up volume, which decreases separation efficiency via a remixing of ions before their detection.

Micromembrane suppressors have thus been the response. They have very small hold-up volumes of 50 µl and enable elution with a concentration gradient.

Therefore, Figure 4.10a corresponds to the passage of an anion A^- in solution in a typical electrolyte of an anionic column and then through a cationic membrane suppressor with a sodium bicarbonate-based eluent.

4.6.2 Electrolytic suppressors

These suppressors are self-regenerated by *in situ* electrolysis of water via permeable membranes that are either anionic or cationic.

Figure 4.10b illustrates the principle of action of such a suppressor installed downstream from a column to separate cations. Here, we see the passage of a cation eluted with dilute hydrochloric acid, using a membrane that is permeable to anions. Cl^- ions cross the membrane to the anode side. In the meanwhile, OH^- ions arrive from the cathode compartment to conserve the electroneutrality of the solution and react with H^+ ions to form water. The background conductivity of the eluent is significantly reduced, which improves the detection of analytes and/or cations.

Diagram ***a***, which illustrates the case of a membrane-based chemical suppressor designed to separate anions, can be transposed to the electrolytic method. The only difference is the formation of H^+ ions by electrolysis of water.

These reactions in the suppressor lower the conductance of the solution circulating in the conductivity meter to less than 1 µS, thus improving the detection of ions separated by the column and then analysed.

> For certain samples, we install an additional module on the circuit followed by the analytes in order to eliminate the carbonates that are often present. This module, named CRD (*Carbonate Removal Device*), is a self-regenerated membrane suppressor, which operates on the principle described in Figure 4.10 (by replacing ions and the membrane in diagram ***b*** with those from diagram ***a***).

4.7 ION-EXCLUSION CHROMATOGRAPHY

Ion chromatography separates or purifies compounds that have a polar nature (organic bases or acids, sugars, etc.) when they are in the presence of mineral ions or strong acids.

Therefore, to purify an organic acid, we choose a cation exchanger, such as a sulfonated resin and an aqueous acidic mobile phase, whose protons will play the role of counterions. Under these conditions, the anionic functional groups of the column catch oxonium ions (H_3O^+) and form a sort of film on the surface of the stationary phase, which can be distinguished from the rest of the solution and is called *Donnan membrane*

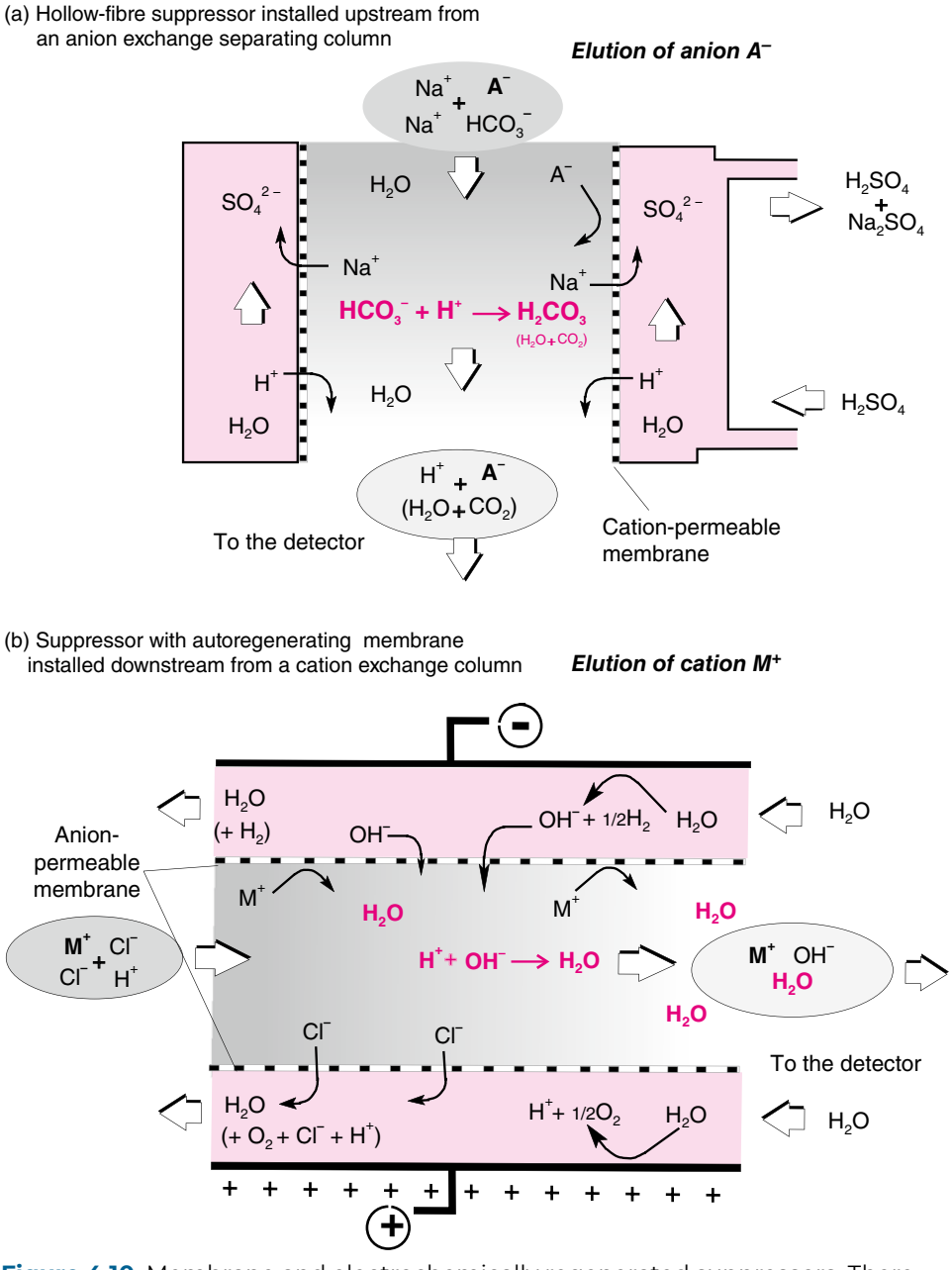


Figure 4.10 Membrane and electrochemically regenerated suppressors. There are two types of membranes, one is permeable to cations (H^+ and, in this example, Na^+), the other permeable to anions (OH^- and here Cl^-). (a) The *cationic membrane* model is adapted to the elution of anions. The membrane is a stationary polyanionic wall which keeps away the anions from the solution. (b) Electrolytic model (electrodeionization) with an *anionic membrane*, adapted for cation elution (in contrast with the previous example). Ions are regenerated by the electrolysis of water. Note in both cases the counterflow circulation between the eluted phase and the solution of the suppressor.

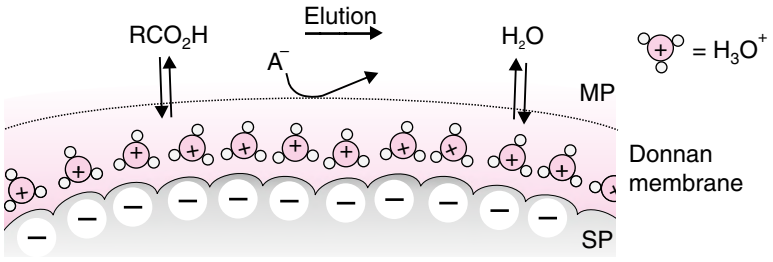


Figure 4.11 Ion-exclusion chromatography. Ions, subject to Donnan exclusion, remain in the hold-up volume of the column, while weak acids may attach to the resin and even enter into its cavities. This method for separating weak acids may be transposed to sugars or weak bases with a quaternary ammonium resin and a dilute sodium hydroxide or sodium carbonate mobile phase. The use of columns specially adapted to this type of separation is recommended.

(Figure 4.11). It pushes away anions present in the solution. However, water or weak acids that are not dissociated in this medium, with a pH that is much more acidic than their pKA, will be able to cross this space. If the sample contains several weak acids, they will separate in increasing order of their pKA. Therefore, methanoic acid (HCOOH, pKa = 3.75) will be eluted before ethanoic acid (CH_3COOH, pKa = 4.76).

4.8 AMINO ACID ANALYSERS

The separation and assay of amino acid mixtures is a classic application of ion chromatography coupled with a colorimetric method. As amino acids cannot be directly identified by UV absorption, they are made to react at the column outlet with ninhydrin to make the same coloured derivative appear, whatever the amino acid. This coloured derivative is easily detected by spectrophotometry.

To separate amino acids from each other, we use a specialized device, which includes a cation exchange column (sulfonated polystyrene) equilibrated with a solution of lithium hydroxide in order for sulfonic acid groups to be in the lithium salt form (Figure 4.12). Since the sample to be assayed has a pH of 2, the amino acids, in cationic form, displace Li^+ ions from the stationary phase and attach more or less solidly depending on their degree of ionization. Elution is done by progressively increasing the pH and saline concentrations of the mobile phase. At column outlet, the amino acids are mixed with a reagent containing ninhydrin and then heated. Except for secondary amino acids (R–NH–R' type), which give off a yellow colour (detected at 440 nm), all primary amino acids (R–NH_2 type) lead to the same purple colour (detected at 570 nm). The colour intensity is proportional to the quantity of amino acid present in the reaction medium. Under defined conditions, the optical absorption ratio at 570 nm/440 nm is characteristic for each amino acid. The detection limit is about 30 picomoles.

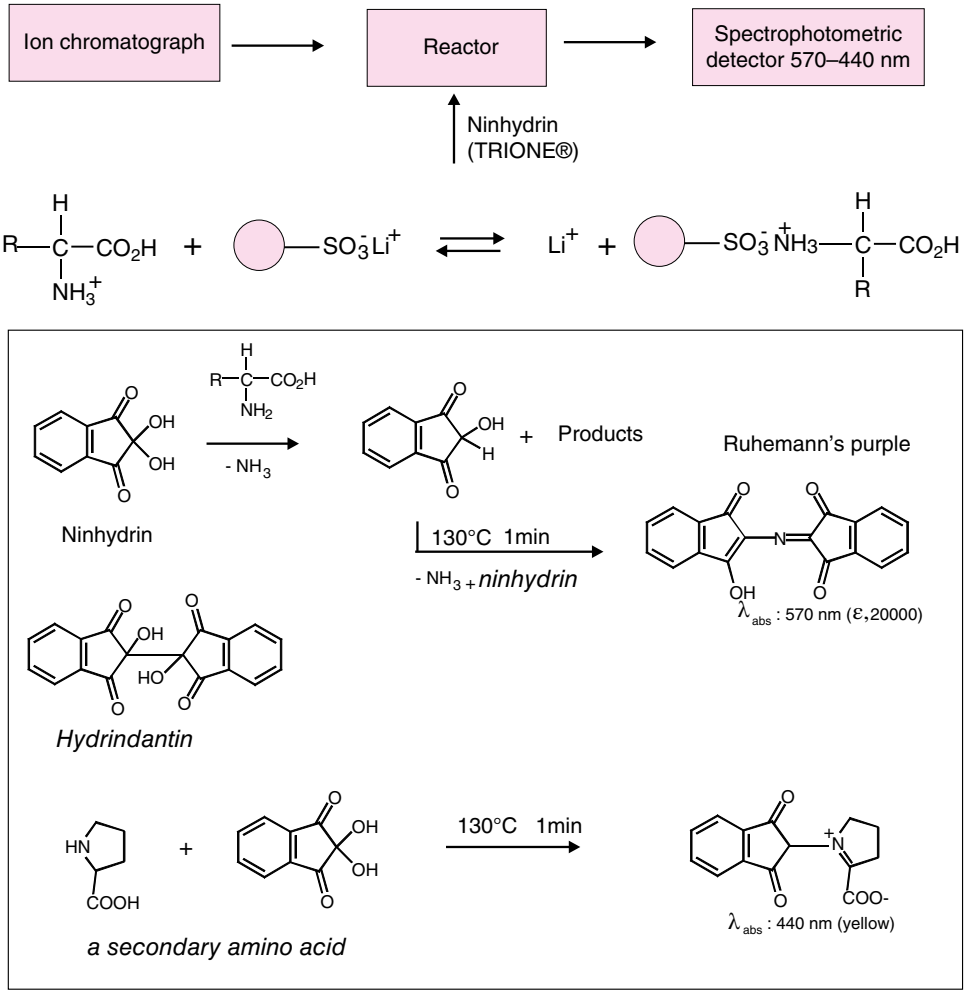


Figure 4.12 Analysis of amino acids. Exchange reaction on the column and derivative reactions with ninhydrin. Whatever the starting amino acid is, with R designating the rest of the molecule in a nonexplicit manner, Ruhemann's purple forms. Its absorption maximum is at 570 nm. Hydrindantin, which is formed by the combination of two ninhydrin molecules, is a catalyst of the reaction. An alternative consists in transforming each amino acid into a fluorescent derivative of indole in the presence of *o*-phthalic aldehyde and a thiol derivative.

An amino acid assay is generally conducted on protein and polypeptide samples taken from living species, whether animal or plant. Before the chromatographic analysis, they must be released by hydrolysis. The very stable peptide bond imposes rather drastic conditions. The effect of this process is to racemize the amino acids as well as destroy some, such as cysteine or tryptophan.

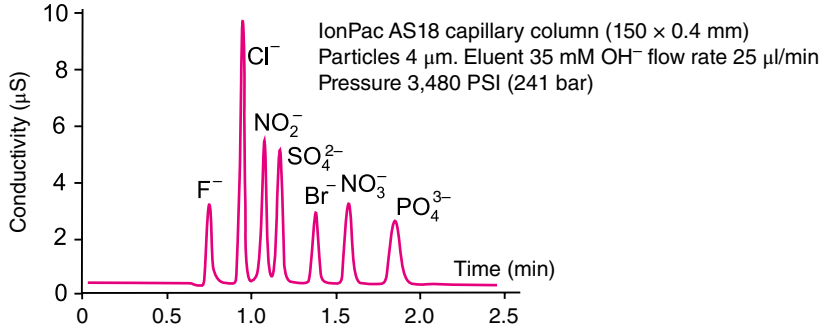


Figure 4.13 Separation of seven anions in less than two minutes.

4.9 PASSAGE FROM IEC TO ULTRA-IEC

The development of IC is essentially focused on two objectives: increasing speed and improving separation performance. From this basis, Ultra-IEC or HP-IEC was born.

Between two successive analyses, the stationary phase must return to its initial state of equilibrium. In IEC, equilibria are slower to develop than in HPLC, even more so if the analysis is done with an elution gradient. The solution resides in the development of shorter columns (e.g. 5 cm) packed with finer particles (4 µm) to exploit the entire potential of small particles. However, the pressure drop due to the resistance that the mobile phase encounters requires hardware changes, especially pumps and the development of crush-resistant particles. It is therefore important to know the maximum pressure the column can withstand (Figure 4.13).

KEY POINTS OF THE CHAPTER

1. In IEC, the stationary phase is composed of a polymer, either synthetic (polystyrene), natural (polysaccharide) or mineral (silica). This polymer is functionalized by means of chemically bonded polar groups. The mobile phase is an aqueous ion solution.
2. In the column in equilibrium, i.e. electrically neutral, all sites are associated with counterions provided by the mobile phase. The migration speed of an ion depends on its affinity, which itself depends on its own selectivity coefficient.
3. The greater the ion affinity is for the reactive sites of the column, the slower it migrates. As a result, a doubly charged ion will appear on the chromatogram after most ions carrying a single charge.
4. In anion-exchange chromatography, the eluent must contain an anion E^-. Conversely, for cation-exchange chromatography, the eluent contains a cation E^+. In both cases, the anion or cation must be compatible with the detection method used.

5. The role of the suppressor is to reduce the background conductivity of the eluent and to increase that of the analyte ions. Thus for analysis of anions, we use an eluent containing anions, and the suppressor will be an H^+ cation-exchange column, which will eliminate the conductivity due to anions.
6. The most common detector is based on the conductivity of the solutions. Spectrophotometric detectors (visible/UV, fluorescence) are the same as in HPLC.
7. In ion-exchange chromatography, the eluent pH determines the degree of dissociation of the analytes and thus their retention. The eluent used most often is dilute sulfuric acid (transparent in UV), which enables the two detection methods (conductometry and photometry).
8. In biochemistry, IEC is used to separate proteins. A protein must have a net charge of the same sign as the counterion it will displace on the resin. Its affinity with the resin will depend on the pH of the solution, as the net charge of proteins varies depending on pH.

PROBLEMS

1. A mixture of proteins is separated on a column with a stationary phase of carboxymethylated cellulose. The internal diameter *ID* of the column is 0.75 cm and its length is 20 cm. The pH of the mobile phase is adjusted to 4.8. The flow rate of the mobile phase is 1 ml/min. The hold-up volume is 3 ml. Three peaks appear on the chromatogram corresponding to the elution volumes V_1, V_2 and V_3 at 12 ml, 18 ml, and 34 ml, respectively.
   a. Is this an anionic or cationic phase? Give reasons for your answer.
   b. Why, when increasing the pH of the mobile phase, are the elution times of the three compounds subject to modification? Predict whether these times are increased or decreased.
2. Potassium and calcium ions present in sodium-free salt are quantified by ion chromatography.

   1.007 g of sodium-free salt is sampled and dissolved in ultra-pure water in a 200-ml graduated flask. Solution S_1 is obtained; it will be used to assay calcium ions. To assay potassium ions, solution S_2 is prepared by a 1:25 dilution of solution S_1.

   A series of standards containing potassium and calcium ions are prepared from a mother solution containing 1 g/l of each of these cations.

   a. Calculate the masses of potassium and calcium nitrate to be weighed in order to prepare one litre of this mother solution.
      The standard solutions as well as solutions S_1 and S_2 are chromatographed under identical conditions. On the various chromatograms obtained, peak areas corresponding to Ca^{2+} and K^+ are recorded.

All results are collected in the following table:

K^+			Ca^{2+}		
Solution	**Conc. mg/l**	**Area**	**Solution**	**Conc. mg/l**	**Area**
Standard 1	10	2.51	Standard 1	10	4.22
Standard 2	20	4.88	Standard 2	20	8.64
Standard 3	50	12.31	Standard 3	50	21.48
Standard 4	100	24.48	Standard 4	100	43.19
S_2		19.62	S_1		25.87

b. Use linear regression to establish the calibration equation for each ion.

c. Deduce the mass concentrations of solutions S_1 and S_2, and mass percentages of potassium and calcium in sodium-free salt.

3. Three amino acids (H_2N-CHR-COOH type) are separated by IEC with a polystyrene-based column containing HSO_3 groups. If we know the iso-electric pHs (pH_i) of these three compounds (see the table below), answer the following questions:

a. What do we call pH_i?

b. For pH < pH_i, what is the major form present?

c. For pH > pH_i, what is the major form present?

d. Fill out the table below.

e. Which pH (2, 7 or 11) must be selected to obtain a good separation of these compounds?

f. What is the predicted order of elution?

Amino acid	**pH_i**	**Charge at pH = 2**	**Charge at pH = 7**	**Charge at pH = 11**
Glutamic acid (Glu)	3.22			
Leucine (Leu)	5.98			
Lysine (Lys)	9.74			

SOLUTIONS

1. a. Such a phase corresponds to a polymer (250,000 < *M* < 900,000) that includes acid groups $-CH_2COOH$. It is weakly cationic. Since the separation pH is lower than classic pI values of proteins, these proteins are protonated on the basic functions ($-NH_3^+$);

b. If we increase the pH, we decrease the ionic nature, and the proteins will be eluted faster. Here $pI_1 < pI_2 < pI_3$.

2. a. Potassium nitrate: m = 2.5857 g; calcium nitrate: m = 4.0938 g.

b. Calibration line equation for K: $A = 0.2445C + 0.042$ (where C is in mg/l), Calibration line equation for Ca: A = 0.4325C – 0.079 (where C is in mg/l);

c. S_2: 80.07 mg/l for potassium; S_1: 59.99 mg/l for calcium; $\%_{m/m}$ for potassium: 40%; $\%_{m/m}$ for calcium: 1.2%.

3. a. The iso-electric point defined by pHi = (½(pK1 + pK2)) is the pH for which the acid is in zwitterion form, which corresponds to the species $^{+}H_3N\text{-}CHR\text{-}COO^{-}$.

b. For pH < pHi, the major form is $^{+}H_3N\text{-}CHR\text{-}COOH$.

c. For pH > pHi, the major form is $H_2N\text{-}CHR\text{-}COO^{-}$.

d. The table must be completed as follows:

Amino acid	pHi	Charge at pH = 2	Charge at pH = 7	Charge at pH = 11
Glutamic (Glu)	3.22	+	–	–
Leucine (Leu)	5.98	+	–	–
Lysine (Lys)	9.74	+	+	–

e. For a column intended to separate cations, the three amino acids must also be in cation form. It must be placed at pH 2.

f. The elution order will be the same as the pHi order, i.e. Glu, then Leu and then Lys.

Chapter 5

Thin-Layer Chromatography

Thin-layer chromatography (TLC), also known as planar chromatography, is a liquid chromatography technique, in which the stationary phase covers a flat support surface. The principle of separation and the nature of the phases are the same as in HPLC, though implementation is different. The mobile phase moves through the stationary phase under the effect of capillarity forces. Quick and easy to implement, it also enables several simultaneous analyses without prior treatment

and, therefore, the selection of those that must be the subject of a more precise quantification. Two-dimensional TLC, *nano*-TLC, and coupling with mass spectrometry or spectroscopic methods are just some examples of current advances in TLC.

Objectives

Explain the steps of TLC
Describe the stationary and mobile phases
Choose a detection mode
Define separation parameters of TLC
Compare these parameters to those of HPLC
Address quantitative TLC
Know two-dimensional TLC

5.1 PRINCIPLE OF THIN-LAYER CHROMATOGRAPHY (TLC)

The separation of the sample components between phases is conducted on a thin layer (100–200 μm) of stationary phase, which is usually based upon silica gel deposited on a rectangular plate made out of glass, plastic or aluminium with dimensions of a few centimetres per side. To maintain the stationary phase on the support and to ensure the cohesion of the particles, an organic binder is mixed into the stationary phase during the manufacture of the plate.

The principle of solute partition between the two phases (stationary and mobile) is similar to that of HPLC; however, a TLC analysis is conducted differently. There are three steps to a separation with this technique: sample application, development of the plate, and post-chromatographic revelation.

5.1.1 Sample Application

A small volume of sample (between a few nanolitres to a few microlitres) is deposited by capillary action or spraying, in a suitable solvent, near the lower edge of the plate. This *application by capillary action*, as a 1–3 mm diameter spot, is done either manually or automatically with a flat-ended capillary tube (Figure 5.1). This method is accompanied by diffusion of the sample around the spot, which prevents any complex separation.

Spray application, under nitrogen flow, helps to eliminate the dilution solvent from the sample and to deposit it as a small horizontal band of homogenous concentration along its entire length. This method of application is easily reproducible, which is indispensable in quantitative analysis.



Figure 5.1 Automatic sample applicator for TLC and a system to 'read' the plate. Biostep AS30 (Bionis) applicator and densitometer to measure radioactivity (thin-layer chromatography radioactivity detector). (Source: model Rita from Elysia-raytest SA Belgium.)

5.1.2 Developing the Plate

Once prepared, the plate is placed in a developing chamber (which comes in many models). Ordinarily, this chamber comes with a cover and contains a small amount of the mobile phase used as an eluent (Figure 5.2). The place where the sample was deposited must be located above the immersion level. The mobile phase climbs up the stationary phase vertically by capillary action, in equilibrium with the vapours of the eluent, entraining the components of the sample at various speeds because of their different degrees of interaction with the matrix and solubility in the solvent. Their separation may be complete in a few minutes and depends on various parameters. When the solvent front has travelled a sufficient distance (several centimetres), the plate is withdrawn from the chamber, the position attained by the mobile phase is immediately noted, and then it is evaporated.

When using a plate of reversed polarity (RP-18 type), the mobile phase will generally contain water. In this case, it can be useful to add a salt such as lithium chloride, which limits diffusion phenomena and thereby increases the resolution.

5.1.3 Post-Chromatographic Revelation

After migration, analytes are located on the plate, either visually or using a densitometer (Figure 5.1). For that purpose, the stationary phase generally contains zinc silicate (or another inorganic pigment) that emits a bright green or blue fluorescence when the plate is irradiated with a mercury vapour lamp equipped with a filter to isolate light

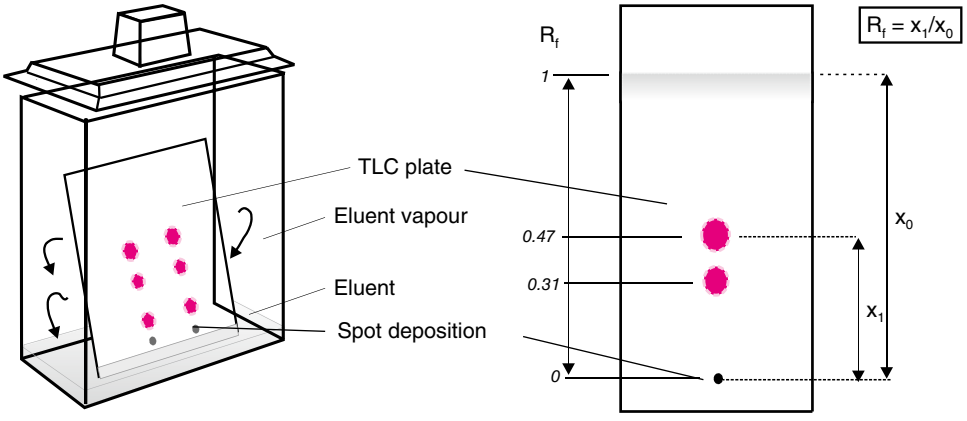


Figure 5.2 Vertical developing chamber and TLC plate. Left: available in a variety of dimensions depending on the size of the plates (from 5 × 5 to 20 × 20 cm), the chambers are made of glass and equipped with a tight fitting cover. Other tanks exist for horizontal migration of the mobile phase. Right: typical appearance of a plate after the revelation of migration spots. Calculation of R_f (see Chapter 4, Section 4.4). x_0 represents the migration distance of the mobile phase.

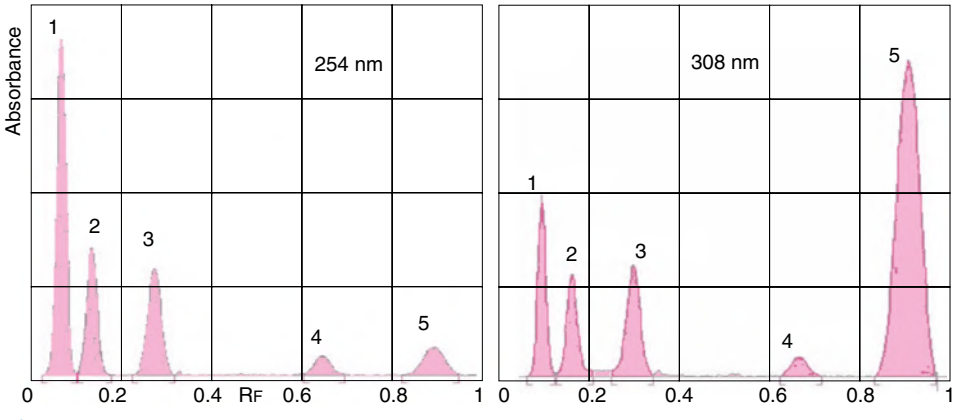


Figure 5.3 Scanning of a TLC plate. The absorbance of a given TLC plate, depending on whether it is recorded at 254 nm or 308 nm, demonstrates that the measurement sensitivity depends on the choice of wavelength (separation of five sulfonamides, document courtesy of Camag).

emitted at 254 nm or 366 nm. The fluorescence emitted by a virgin area presents the base signal, whose intensity decreases at the place where a compound appears in the form of a dark spot. The densitometer produces a densitogram, whose appearance is like that of a chromatogram (Figure 5.3). Fluorescent analytes may be revealed by shining light on the plate with a 366 nm wavelength.

Among the many other methods used to reveal compounds after migration, the plate is sprayed or immersed using reagents, either general (phosphomolybdic acid, vanillin) or specific (e.g. ninhydrin for amino acids). More than a hundred reagents have also been described that serve to introduce chromophore or fluorophore groups before revelation.

A specific technique for pollutant research in water consists of taking a TLC plate after migration and drying, and projecting on it a solution containing genetically modified yeast for example, selected to react with bio-active pollutants only. The line-up ends with a densitometer or fluorescence measurement. This technique, which includes both separation and an activity test, is known as TLC-EDA (*Thin-Layer Chromatography-Effect Directed Analysis*).

The use of square TLC plates enables two-dimensional chromatography to be carried out, using two successive elutions with two different mobile phases (Figure 5.4). A typical application of this method is the separation of amino acids.

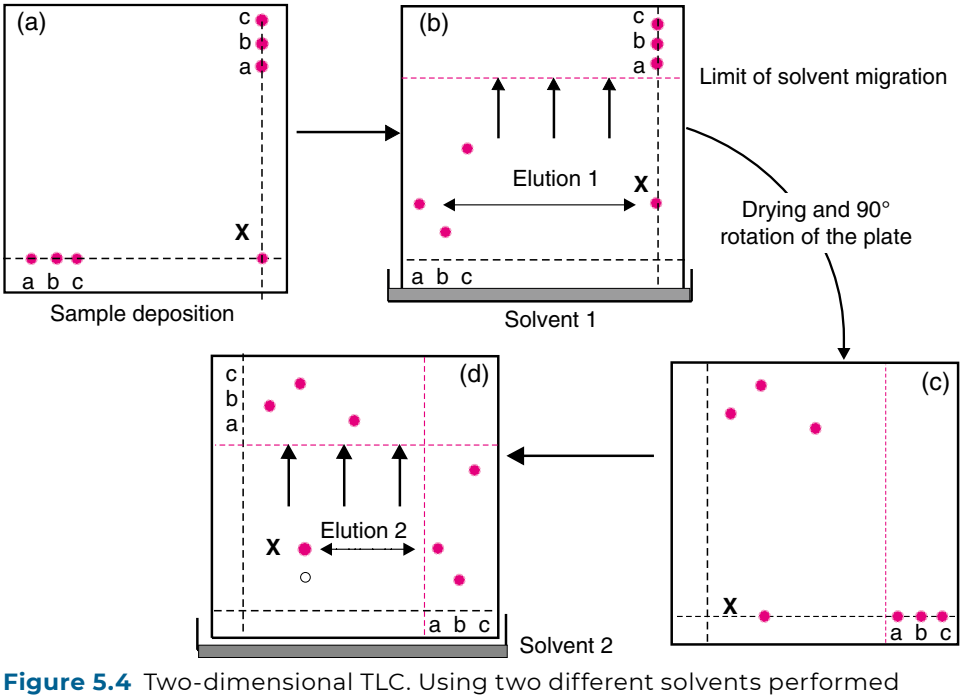


Figure 5.4 Two-dimensional TLC. Using two different solvents performed in two directions, we can conclude that compound X is a mixture of at least two compounds, including the reference compound a (the same R_f in both solvents), but the second compound is not b, even though it has the same R_f in elution 2.

5.1.4 Identification of Analytes through Mass Spectrometry

After migration, if the analytes are located in distinct areas of the plate, these areas will be the subject of an analysis via mass spectrometry, even though they contain only a minute quantity of compounds. Plates that are free of any impurities are marketed for this purpose. The retrieval of analytes may be done manually or using an automatic extraction system. The manual method consists in detaching the analyte area from the support in order to extract it with a solvent (Figure 5.5). This solution will be used as a sample for mass spectrometry. The automatic extraction system combines the extraction with introduction of the resulting sample into the mass spectrometer. The coupling of TLC and mass spectrometry enables the absolute identification of analytes.

> Nuclear Magnetic Resonance (NMR) (see Chapter 15) is not sensitive enough to be directly coupled with analytical TLC. However, by using preparative TLC plates, with a stationary phase thickness up to 2 mm, it is possible to deposit more significant quantities (e.g. 100 mg) of the sample. It is then possible to gather sufficient quantities of analytes after migration, thus making NMR possible.

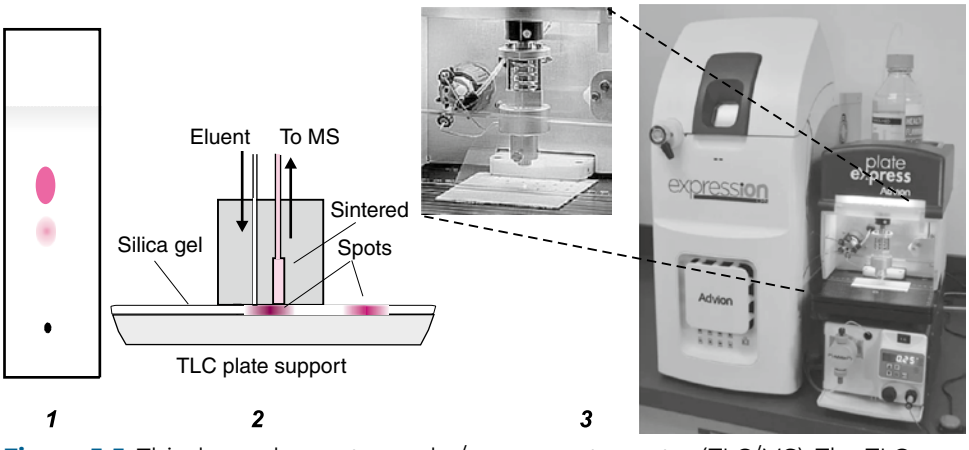


Figure 5.5 Thin-layer chromatography/mass spectrometry (TLC/MS). The TLC plate (**1**) after migration of the analyte is placed in an extractor (**2**), which helps retrieve it by elution with a solvent and to transfer it to MS for analysis (**3**). Several companies sell these devices (Source: S *Expression* model from Advion, without its operating panel, but with its *Plate Express* plate extractor). In the centre, diagram of an extractor.

5.2 CHARACTERISTICS OF TLC

TLC involves physico-chemical phenomena that are more complex than HPLC. In fact, three distinct phases are present between which equilibria are established: the stationary phase, the liquid mobile phase, and the vapour mobile phase.

- The stationary phase is only partially equilibrated with the liquid phase before the migration of the compounds. Depending upon the manner in which the separation is obtained, the mobile phase may or may not be in equilibrium with the vapour phase. Migration speeds depend on this.
- The surface adsorption capacity decreases substantially once a large part of the adsorption sites are occupied. This leads to elongation of the spots. As a result, the R_f (retardation factor) of a compound, whether in the pure state or present in a mixture, is thus slightly different. It will be reduced even more as saturation is increased during separation.
- The flow rate of the mobile phase cannot be modified in order to improve the efficiency of a separation. A remedy to this problem is the multidevelopment technique, by drying the plate before each new migration cycle. In fact, there are automatic devices that involve several successive migrations with eluents of various compositions and therefore reproduce the effect of an elution gradient in HPLC.
- The migration speed of the solvent front is not constant. It follows a complex function in which the size of the particles of the stationary phase plays a part. Migration speed can be described by a quadratic law: $x^2 = kt$, where x represents the height of the migration front, t is time, and k is a constant. As a result, the resolution between two spots depends greatly on the R_f values of the compounds. Resolution attains a maximum for an R_f value generally around 0.3.

To sum up, the efficiency N of a TLC plate is quite variable. The height equivalent to a theoretical plate has, as in HPLC, an optimal value.

5.3 STATIONARY PHASES

About 80% of stationary phases used are made up of nonmodified silica gel. The remaining 20% are shared between microcrystalline cellulose (modified or not) and treated alumina (Al_2O_3, which is less acidic than silica). The main shared characteristics, defining the properties of these materials, are particle size distribution (dimensions, irregular or spherical grains, monodisperse, if possible), specific surface area (500 m^2/g), pore size (6 nm), and pore volume (0.75 ml/g). They are fixed on the support with a binder (polymer) and generally contain a fluorescent indicator (Figure 5.6).

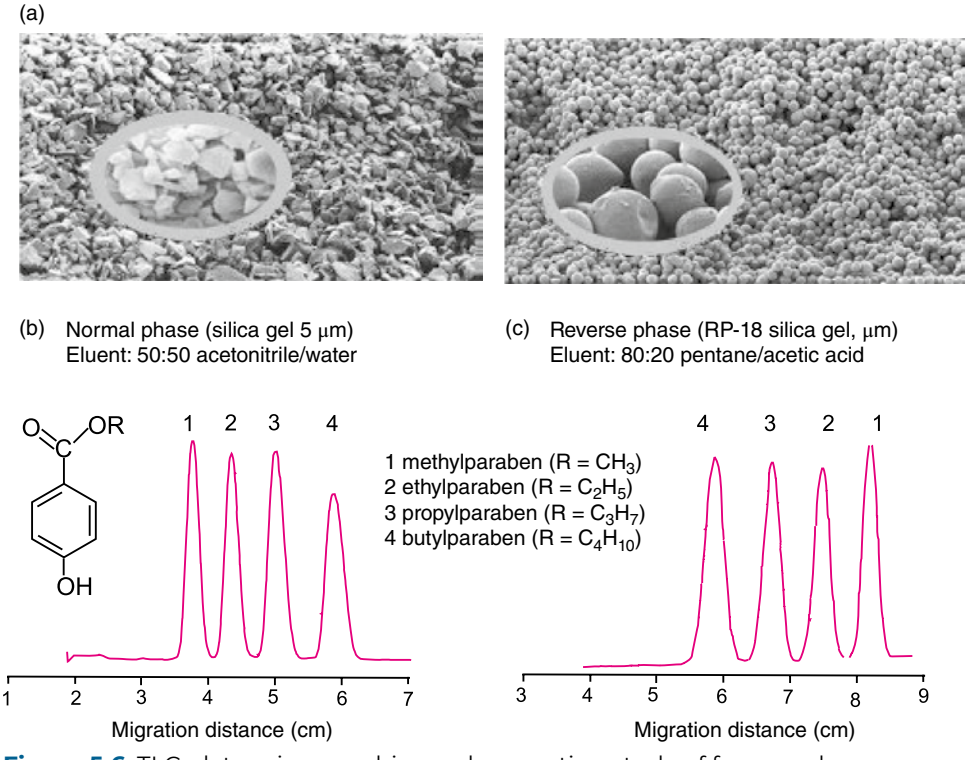


Figure 5.6 TLC plate micrographics and separation study of four parabens on various phases. (a) On the left, view of a standard TLC plate showing silica gel grains (10–15 µm) and on the right, *LiChrospher* plate from Merck (diameter 8–10 µm). (b) On a polar phase with a nonpolar eluent, the migration order (from the smallest displacement to the largest) corresponds to the decreasing order of polarity (the least polar compound, butylparaben, migrates faster). (c) However, on a nonpolar reversed phase, the migration order with a polar eluent corresponds to the order of increasing polarity.

The alumina used in chromatography comes from the calcination of aluminium hydroxide. As with silica gel, the alumina surface thus obtained is covered with hydroxyl (–OH) functions, which give it a hydrophilic character. Its rigid structure carries both *Lewis base*-type sites and *Lewis acid*-type sites. More basic than silica gel (the latter has a pKa of about 10), alumina retains acidic compounds and remains stable at high pHs. Its activity is reduced by neutralizing hydroxyl sites in the presence of water vapour.

For nonmodified silica gel, the ratio between the silanol and siloxane groups determines the more or less pronounced hydrophilic character of the phase.

As with HPLC, there are silicas to which chains of varied structures are grafted. Some phases incorporate simple alkyl chains (RP-2, RP-8, RP-18). Figure 5.6 helps to compare

results obtained with a given mixture studied on a nontransformed standard phase and on an R-18 phase. Many other grafts lead to more or less polar phases: nitrile (–CN), amine ($–NH_2$), or alcohol (–OH), allowing a large set of possibilities, without counting chiral selectors, for the separation of optical isomers.

Other phases with ion-exchange properties have a hydrophilic nature and can be employed for the separation of ampholytes. The most widely known is DEAE–cellulose, a fairly basic phase containing diethylaminoethyl groups.

HP-TLC, U-TLC. The improved performance of stationary phases intended for HPLC has also benefited TLC. Finer particle size distribution (average 6 μm) and thinner plates (150–200 μm) enable quicker analyses and improved resolution (reduced plate height) and sensitivity, while reducing diffusion. Marketing departments have created appropriate acronyms: U-TLC or HP-TLC (*Ultra-* or *High-Performance Thin-Layer Chromatography*) to refer to this domain of High-Performance TLC.

5.4 SEPARATION AND RETENTION PARAMETERS

The progression of the mobile phase within the stationary phase is governed by two contradictory constraints: capillary action and resistance to mass transfer. The distance travelled x_0 (Figure 5.2) follows a quadratic equation as a function of time t:

$$x_0 = \sqrt{kt} \tag{5.1}$$

where k characterizes the specific parameters of the plate. The solvent front therefore rises more slowly.

Each compound is defined by its *retardation factor* R_f (unitless), which corresponds to its relative migration compared to the solvent (Figure 5.2):

$$R_f = \frac{\text{Distance run by the solute}}{\text{Distance run by the solvent front}} = \frac{x}{x_0} \tag{5.2}$$

The efficiency N of the plate for a compound whose migration distance is x and whose spot diameter w is given by Eq. (5.3) and H (HETP) by Eq. (5.4):

$$N = 16\frac{x^2}{w^2} \tag{5.3}$$

$$H = \frac{x}{N} \tag{5.4}$$

To calculate the retention factor k of a compound or the selectivity coefficient between two compounds, the distances migrated on the plate are matched with the

migration times read on the chromatogram. Assuming that the *ratio* of the migration speeds $\bar{u}$ and $\bar{u}_0$ is the same on the plate as on the column (which is really only an approximation), then R_f and k can be related as follows:

$$R_f = \frac{x}{x_0} = \frac{\bar{u}}{\bar{u}_0} = \frac{t_0}{t} = \frac{1}{k+1}$$

such that

$$k = \frac{1}{R_f} - 1 \tag{5.5}$$

Finally, by comparison with Eq. (1.33), resolution can be given by the relation:

$$R = 2\frac{x_2 - x_1}{w_1 + w_2} \tag{5.6}$$

where x_1 and x_2 are the two migration distances. Resolution reaches a maximum when the R_f values are approximately 0.3.

5.5 QUANTITATIVE TLC

To validate an assay method by TLC, it is essential to quantify the spots (Figures 5.1 and 5.7) and also define all of the usual parameters (specificity, range of the domain of linearity, precision, etc.). This is done by placing the plate under the lens of a densitometer (or scanner) that can measure either absorption or fluorescence at one or more wavelengths. This instrument produces a pseudo-chromatogram that contains peaks whose areas can be measured (Figure 5.4). In fact, it is actually an isochronic image of the separation at the final instant. In TLC, a spot is usually detectable if it corresponds at least to a few ng of a compound that absorbs in the UV range.

To reveal compounds labelled by a β^- radioisotope, some densitometers are equipped with a video camera giving an image of the radioactive distribution on the plate. These densitometers, known as *Charpak machines*, which are also used in gel electrophoresis, have sufficient sensitivity for detecting activities on the order of a few Becquerels per mm^2.

For many applications, TLC yields results comparable in quality to those of HPLC. Although this technique currently requires more human intervention than the latter, the use of new tools perfected to deposit, develop using elution gradients, and read the plate give it the necessary reproducibility. Technological progress consists in making the eluent migrate at a constant velocity by means of an electrical field or by exerting excess gas pressure on a specially designed developing chamber.

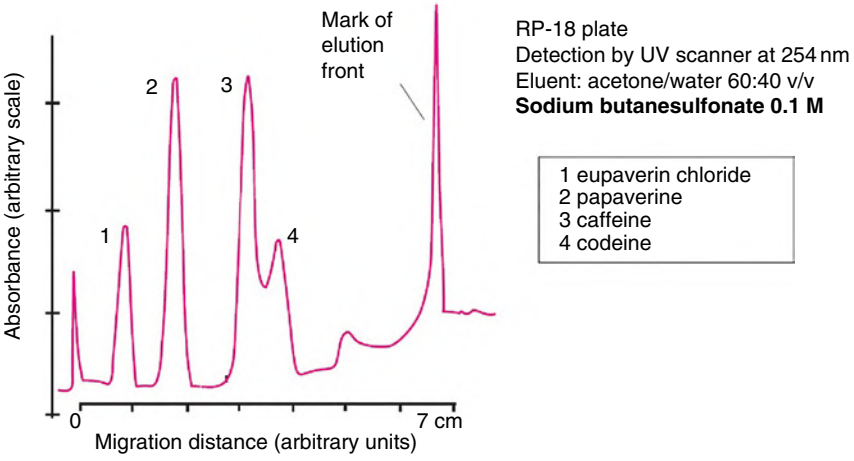


Figure 5.7 Separation of TLC catecholamines in TLC with the ion-pairing technique. Here, a reversed polarity phase is used. To improve the separation of these polar compounds, which may not be retained sufficiently, an alkyl sulfonate is added to the MP (see Chapter 3, Section 3.7.1). This pseudo-chromatogram was obtained by scanning the TLC plate (according to a Merck application note).

Multiple development. On an unmodified, polar silica plate, the test consists in making several successive migrations with more or less polar MPs, while making sure to dry the plate between each migration. If we assume that there are two compounds only slightly separated after the first development, the one with the smaller R_f will still migrate less in the second development, accentuating the difference between the two spots, which will thus be better resolved. This long and rather inefficient practice is equivalent to a gradient in HPLC.

When compared with HPLC, even in its latest developments (high pressure or high speed), TLC can treat more samples in the same time period by the set-up of analyses in parallel on the same plate. Sample preparation is less constraining than in HPLC, since the plate is disposable. The nature of the matrix is not very important, as it is adsorbed on the deposit, which is very useful for biological samples. The TLC plate on which the products have been separated is also a means, provisionally, to preserve very small samples after extraction, which can serve for other analyses (mass spectrometry, for example).

KEY POINTS OF THE CHAPTER

1. TLC is a very cost effective method, whose implementation involves four key steps: sample application, migration, revelation, and identification of the analytes on the plate. it does not require expensive equipment but is still facilitated with automatic applicators and densitometers.

2. The analyte identification step relies, in some well-targeted situations, on the comparison of migration distances with control compounds.
3. For a better identification, it is possible to retrieve analytes after migration by various processes in order to dispose of additional identification methods. For that purpose, there are plates with a thicker stationary phase layer, thus enabling them to accept larger samples.
4. With a sufficiently big plate, several analyses may be conducted simultaneously and therefore under the same conditions.
5. As in HPLC, the success of separations is ensured through the proper selection of the stationary phase and of the mobile phase used as an eluent.
6. The TLC plate is disposable, which enables the analysis of compounds dispersed in charged matrices (raw samples), and therefore avoids any prior treatment.
7. TLC may be used as a positive/negative test to quickly select samples with a specific analyte, which must be the subject of a deeper analysis. This translates into time and cost savings related to these analyses.
8. In organic synthesis, TLC is often used at regular intervals to monitor the evolution of a reaction: disappearance of starting products, appearance of new compounds, kinetics of the reaction, and related products formed.

PROBLEMS

1. A mixture of two compounds A and B migrates from the origin to leave two spots with the following characteristics (migration distance x and spot diameter w):

$$x_A = 27\,\text{mm} \quad w_A = 2.0\,\text{mm}$$
$$x_B = 33\,\text{mm} \quad w_B = 2.5\,\text{mm}$$

   In this experiment, the mobile phase front is 60 mm from the starting line.
   a. Calculate the retardation factor R_f, the efficiency N and the HETP H for each compound.
   b. Calculate the resolution factor between the two compounds A and B.
   c. Establish the relation between the selectivity factor and the R_f of the two compounds. Calculate its numerical value.
2. Figure P5.1 represents the results of scanning a TLC plate in normal phase (mobile phase: hexane/acetone 80/20). The three compounds have the structures A, B and C.
   a. Match compounds A, B, and C with the three peaks on the read-out.
   b. What would be the order of elution of these compounds if examined by an HPLC column containing the same type of stationary phase and the same mobile phase?

c. What would be the order of elution of these compounds on an HPLC column containing an RP-18-type phase with a binary mixture of acetonitrile/methanol (80/20) as eluent?

d. Calculate the R_f of the compound migrating the fastest on the plate.

e. From the read-out, calculate the efficiency of the TLC plate and the corresponding HETP (use the formula transposed from column chromatography, in particular that giving efficiency, according to the FWHM δ, with x as distance of migration).

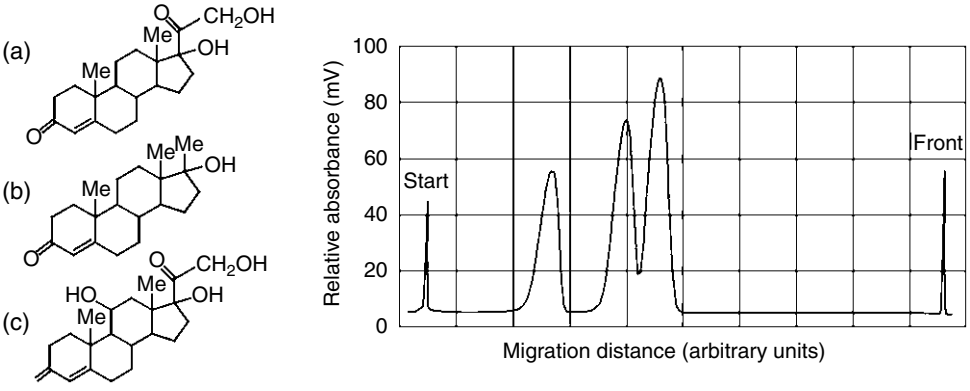


Figure P5.1

3. To anticipate the influence of a composition change of the mobile phase (MP) on retention factors k, Snyder associates each solvent for MP use with a parameter P, which is used in two equations, depending on whether it is a normal phase or a reverse polarity phase. From the data in the table and indicated relationships, for MP No. 2, calculate the new k values in the two types of liquid chromatography (normal phase and reverse phase).

In normal phase, $k_2/k_1 = 10^{(P1-P2)/2}$ and in reverse phase $k_2/k_1 = 10^{(P2-P1)/2}$

	Acetonitrile	Water	Methanol	k
P	5.8	10.2	5.1	
MP No. 1 (%)	40 (%)	50 (%)	10 (%)	6
MP No. 2 (%)	55 (%)	40 (%)	5 (%)	?

4. Demonstrate that, if TLC and HPLC use the same phases (mobile and stationary), the hold-up volume necessary to elute an analyte is equal to the inverse of its R_f.

5. Answer the following statements with YES or NO, and justify your answer.

a. The selectivity factor between two compounds may be below 1.

b. The retention factor of a compound cannot be less than 1.

c. The retention factor is different from the capacity factor.

d. The mobile phase flow rate has an effect on the efficiency of a separation.
e. In HPLC, on a nonpolar column, the retention time of an organic acid increases if the pH of the mobile phase is increased sufficiently.
f. In GC, temperature has no effect on retention factors.
g. When an elution gradient is conducted with an RP-18 column, the mobile phase polarity is generally reduced over time.
h. A selectivity factor close to 1 means poor resolution.
i. In GC, if we compare two columns, where only the phase ratio β is different, all things being equal otherwise, will the retention factor of a compound be different on the two columns?
j. In GC, we inject 1 µl of a 1% solution (vol/vol) of a compound **A**. The split ratio is set to 100:1. The solution density is 0.9 g/ml. Mass **A** migrating in the column is thus 90 µg.
k. In a separation, all compounds spend the same amount of time in the mobile phase.
l. Flow rate variations have an effect on peak areas but not on peak heights.
m. In normal-phase TLC, a nonpolar compound is entrained by the mobile phase.

SOLUTIONS

1. a. $R_{f(A)} = 0.45$; $R_{f(B)} = 0.55$; $N_A = 2{,}916$; $N_B = 2{,}788$; $H_A = 9.26 \times 10^{-4}$ cm; $H_B = 1.18 \times 10^{-3}$ cm.
b. R = 2.67.
c. $\alpha = 1.49$.

2. a. As this is a normal phase, the most polar compound will be the most retained. In order of increasing migration, first there will be C, then A and finally B.
b. The elution order would be A, B, and C.
c. The reverse order.
d. $R_{f(A)} = 0.45$; $N = 635$; HETP = 0.1 mm.

3. First, weighted values of P are calculated, for the two phase MP1 and MP2.
P_1: 5.8 × 0.4 + 10.2 × 0.5 + 5.1 × 0.1 = 7.9
P_2: 5.8 × 0.55 + 10.2 × 0.4 + 5.1 × 0.05 = 7.5
Then, we calculate new values of *k:*
In normal phase (polar SP), $k_2 = 6 \times 10^{(7.9-7.5)/2} = 9.5$ (*k* increases because the MP is less polar, and the compound is therefore more retained on the SP).
In reverse phase, $k_2 = 6 \times 10^{(7.5-7.9)/2} = 3.8$ (*k* decreases because the MP is more polar and the compound is thus more quickly entrained).

4. On the HPLC column, if we call its hold-up volume V_M and the retention volume of the analyte V_R, $V_R = V_M (1 + K)$. Knowing the relationship between *k* and R_f, i.e. $k = \frac{1}{R_f} - 1$, the result will be $V_R = V_M \left(1 + \frac{1}{R_f} - 1\right)$, which leads to $V_R/V_M = 1/R_f$.
Practice shows that this is a simple indication rather than a specific calculation.

5. **a.** NO, by definition, α is always over 1.

b. YES, if the retention time of the analyte is close to the hold-up time of the column.

c. NO, these are two equivalent names.

d. YES, see Van Deemter curves for example.

e. NO, it decreases because in neutral or basic media, the acid is ionized, and therefore it has less affinity for a nonpolar column.

f. NO, a temperature increase accelerates the migration of analytes and therefore reduces their retention time. For a given column (phase ratio β), K and k are related ($K = k \cdot \beta$).

g. YES, in general this is the case. By reducing the proportion of water in the MP, we make the MP less polar and thus less polar analytes are more easily separated.

h. NO, if the resolution is very high, such as in GC, the peaks are very thin and can be easily separated (R is over 1.5) even if the value of α is very close to 1.

i. YES, this is the same SP/MP pair; therefore, the value of K will be the same for the tested compound. Since $k = K \times \beta$ and thus K = $\frac{k}{\beta}$, then k increases if β decreases for example.

j. NO, 0.9 mg of solution is injected, i.e. a mass of **A** close to 9 µg. After the split, there will only remain 90 ng (and not 90 µg).

k. YES, at least how we conceive of it in the plate model, in which each compound is subject to a succession of MP/SP equilibria.

l. YES. For example, if the flow rate is decreased during the elution of a compound, the area of its elution peak will increase, while its height depends only on the instantaneous concentration and the sensitivity of the detector. That is why for some assays, it is preferable to base concentration calculations on peak heights.

m. YES, on a normal, i.e. polar, phase, a nonpolar compound will tend to stay in the mobile phase and therefore migrate faster than a polar one.

Chapter 6

Supercritical Fluid Chromatography

Supercritical fluid chromatography (SFC) employs, as its name suggests, a fluid in the supercritical or subcritical state as its mobile phase. In practice, carbon dioxide (CO_2) is used, because its low toxicity and its specific viscosity and diffusivity enable the rapid separation of compounds with similar structures or of optical isomers using suitable columns. Instrumentation, which is conceptually close to that of HPLC, both in terms of columns and detectors, is adapted to the nature of the mobile phase.

Regarded as a green technique, using very small amounts of organic cosolvents, its specific characteristics make its coupling possible with mass spectrometry. Considered to be a hybrid between GC and HPLC, its rise has been rather slow, yet its properties make it complementary for the study of hydrophobic compounds, lipids, polymers and thermolabile compounds.

Objectives

Review what is represented in the phase diagram of a gas
Give reasons for choosing CO_2
Describe the useful characteristics of CO_2 in SFC
Represent a specific SFC set-up
Compare SFC, GC, and HPLC
Apply SFC to the separation of enantiomers

6.1 SUPERCRITICAL FLUIDS: A REVIEW

The transformation of a pure compound from a liquid to a gaseous state and vice versa corresponds to a phase change that can be induced over a limited range by changing the pressure or temperature. For example, a pure substance in the gaseous state cannot be liquefied above a given temperature, called the critical temperature T_C, irrespective of the pressure applied to it. The minimum pressure required to liquefy a gas at its critical temperature is called the critical pressure P_C. These points are the defining boundaries on a phase diagram for a pure substance. The curve, which limits the gas and liquid domains, stops at the critical point *C* (Figure 6.1). Under these conditions, gas and liquid states have the same density. Above these temperatures and pressures, the compound becomes a supercritical or subcritical fluid.

The use of supercritical fluids as mobile phases in chromatography provides several advantages; these are related to their physical properties, which are intermediate between those of liquids and gases. More specifically, their viscosity is 10–20 times lower than that of usual liquid phases. Furthermore, their solvating properties (governed by the distribution coefficient *K*) are similar to those of a low-polarity organic solvent. In this way, carbon dioxide, at 300 bar and at 40°C, is comparable to benzene for chromatographic applications. Depending upon the choice of temperature

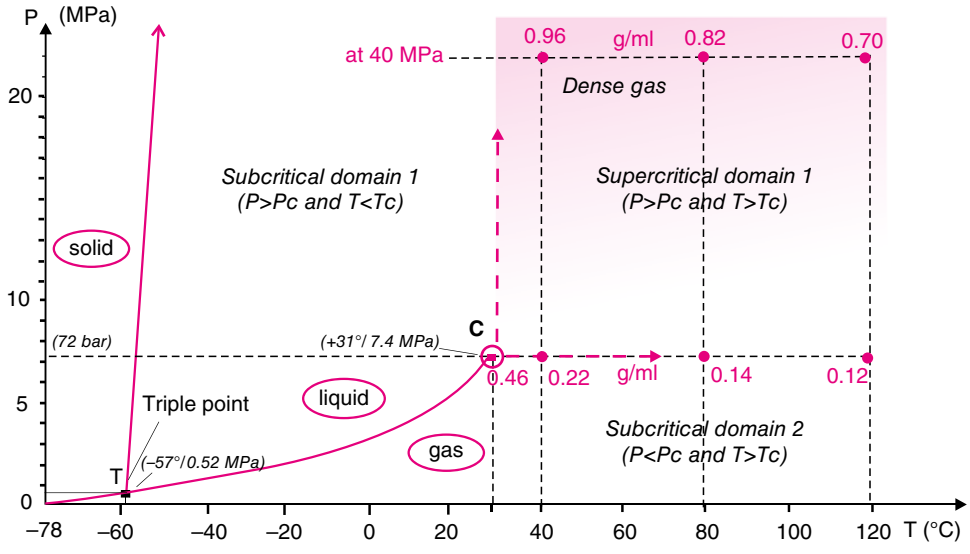


Figure 6.1 Phase diagram carbon dioxide. There exists for each pure substance a relation between three variables: temperature *T*, pressure *P*, and volume *V*, known as the equation of state. This illustration is the projection (P/T) for CO_2. The critical point is located at 31°C and 7.4 MPa (1 MPa = 10^6 Pa, or 10 bar). It is possible to go from the liquid state to the gaseous state without passing through the critical point and therefore without a discontinuity of phase.

and pressure, the behaviour of a supercritical fluid can sometimes look like that of a dense gas and sometimes like that of a liquid (Figure 6.1).

Carbon dioxide in its supercritical state can be used for the extraction, on a laboratory scale, of unstable compounds from complex matrices. Many industrial applications also use this methodology to extract food products (e.g. decaffeination, recovery of aromas and spices, elimination of fats) or as a dry cleaning agent for clothing.

6.2 CARBON DIOXIDE AS A MOBILE PHASE

Carbon dioxide (CO_2) is the only compound used in the supercritical state as a mobile phase. Considered to be part of 'green chemistry', this compound is low in toxicity, nonflammable and noncorrosive. The coordinates of its critical point are T_c = 31°C and P_c = 7.4 MPa (Figure 6.1). Beyond these values, the supercritical domain is attained, under conditions that are technically easy to reach. At 16 MPa and 60°C, carbon dioxide has a density of 0.7 g/ml. This is why the expression 'dense gas' is used. Some

separations of thermolabile compounds or enantiomers require temperatures under 31°C, at which CO_2 is in subcritical domain 1 (Figure 6.1).

Nitrous oxide (N_2O) and ammonia (NH_3), whose critical point coordinates are similar to those of carbon dioxide, were quickly abandoned.

Unlike gases, whose viscosity increases with temperature, supercritical fluids behave in the opposite manner. Therefore, for carbon dioxide, the diffusion factors (diffusivity) of analytes remain much higher, even at high pressures, than those observed for classic HPLC solvents, which leads to increased speed for separations. The density, and therefore the solvating power, of supercritical fluids varies according to the pressures to which they are subjected. Consequently, a pressure gradient in SFC is equivalent to an elution gradient in HPLC, or to a temperature gradient in GC.

As carbon dioxide does not have a dipole moment, its polarity is very weak in the supercritical state (at 60°C and 160 bar, it is close to that of hexane). As such, without a cosolvent, it cannot be used for samples containing polar analytes. This is why it is often combined with a *modifier*, up to a proportion of 30%, which makes analysis of these compounds possible. The most common modifiers are alcohols (methanol, ethanol or isopropanol). These are protic solvents, i.e. hydrogen donors, thus minimizing the interaction of polar solutes with polar groups in the SP. Acetonitrile, an aprotic solvent, is rarely used (Figure 6.2). When the mobile phase contains a modifier, viscosity and diffusivity are not affected very much but the critical point is moved, so much so that the separations are then conducted in the subcritical domain.

	Hexane	*Toluene*	*Isopropanol*	*Acetonitrile*	*Methanol*	*Water*
Dipole moment (D)	0	0.43	1.65	3.45	1.74	1.85
Snyder's polarity index	0.1	2.4	4.3	6.2	6.6	9

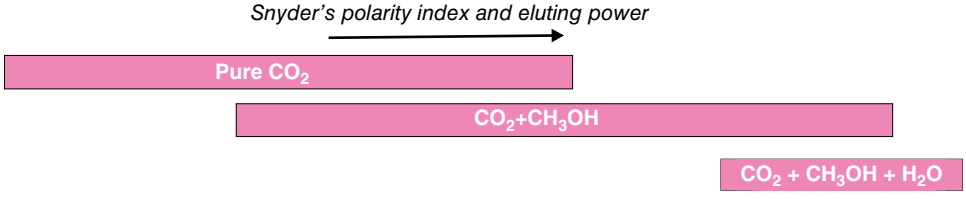


Figure 6.2 Dipole moments and Snyder's empirical scale for several modifiers. Modifiers act very precisely on the retention factors *k* of analytes, and therefore on their selectivities α. Depending on the quantity of modifiers (between 0 and 30%), the outlet pressure or the temperature, the eluting power of the MP will change. For that reason, for each modifier we define a polarity range. For very polar compounds, we add an additive to the modifier, such as ammonium formate or acetate.

6.3 INSTRUMENTATION IN SFC

The instrumentation for SFC is similar to that for HPLC (Figure 6.3). To ensure the flow rate of the supercritical fluid, either a syringe pump (if the composition of the mobile phase is fixed) or a reciprocal pump is used. The pump body is maintained below the critical temperature by means of a cryostat regulated at around 0°C. In instances where an organic modifier is added, we use either a second pump or a reciprocating pump with two chambers, one for the supercritical fluid and the other for the modifier. The liquid then passes through a heated coil maintained above the critical temperature in order to convert it to a supercritical or subcritical fluid.

The *back-pressure regulator,* often referred to as BPR, maintains the mobile phase in the supercritical state from the pump to the end of the column, or even downstream from the detector, depending on its type. This pressure restriction device must also be able to manage significant cooling as well as the gas volume formed when the MP expands when returning to atmospheric pressure.

Initially considered to be quite similar to GC, SFC used capillary columns as well as an FID for detection (see Chapter 2, Section 2.7.2). However, the latter cannot accommodate MPs containing modifiers whose abundance masks the analyte peaks.

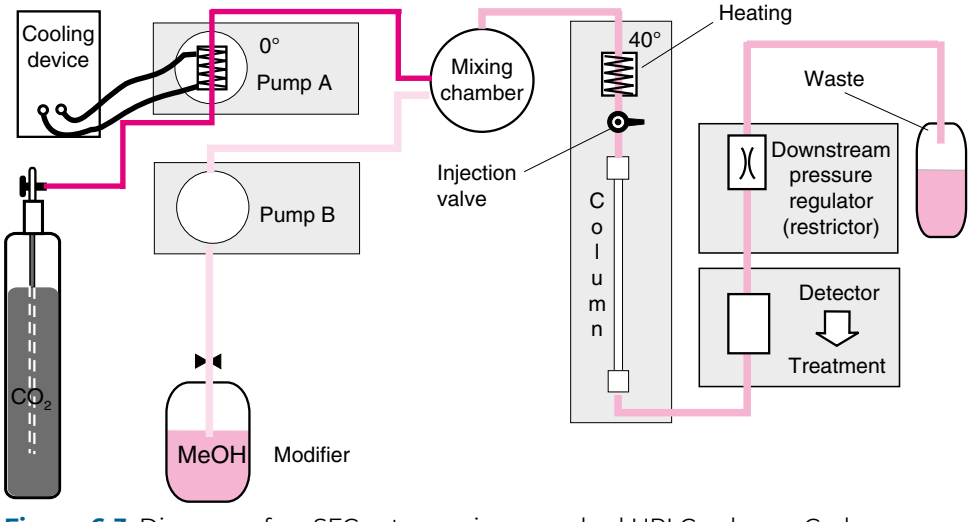


Figure 6.3 Diagram of an SFC set-up using a packed HPLC column. Carbon dioxide reaches a supercritical state between the pump and the injector. A pressure regulator (*restrictor*) is located after the column and either before or after the detector, depending upon its type. It maintains the mobile phase under supercritical conditions until the outlet of the column (diagram based upon a document from Vydac).

Furthermore, the now-common use of packed HPLC columns leads to MP flow rates that are too great for this type of detector.

With technical advances and the appearance of Ultra-HPLC, SFC adopted low-diameter, ultra-bore packed columns with detection based on UV absorption (see Chapter 3, Section 3.8.1) or fluorescence (see Chapter 3, Section 3.8.2) or even light diffusion (see Chapter 3, Section 3.8.4). The diffusion evaporation detector is more sensitive than in HPLC thanks to the high concentration of easy-to-evaporate carbon dioxide in the MP. However, just as for GC or HPLC, mass spectrometry, coupled downstream from the SFC, improves the specificity and the sensitivity of analyses (see Chapter 16, Section 16.1.1). This coupling is facilitated by the easy elimination of carbon dioxide and the low solvent amount. The various ionization sources lead to different ionization modes, making it possible to sweep a large range of analyte polarities.

6.4 COMPARISON OF SFC WITH HPLC AND GC

SFC is complementary to the other classic liquid or gas chromatography techniques. The migration of the solute results from a distribution mechanism between the nonpolar stationary phase and a slightly polar mobile phase. The retention phenomenon is therefore different from that of HPLC (Figure 6.4). The solvating power of the mobile phase is governed by both the temperature and pressure of the supercritical fluid. The resistance to mass transfer between the stationary and mobile phases is less than in HPLC because diffusion of solutes in the mobile phase is about 10 times faster.

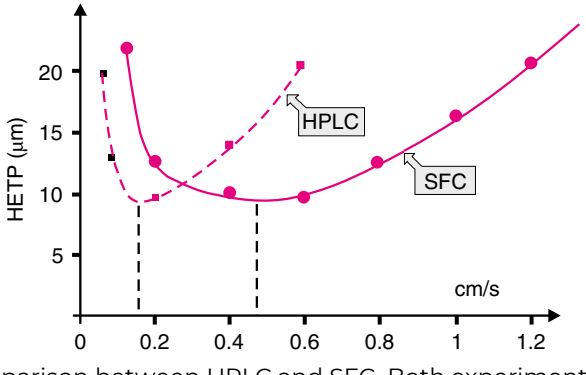


Figure 6.4 Comparison between HPLC and SFC. Both experimental curves have been obtained using the same compound on the same column, but with a classic liquid phase for HPLC and carbon dioxide in a supercritical state for SFC. The HETP values are comparable – but the separation can be up to three times faster with SFC, thus saving analysis time.

Since the *C* factor in the Van Deemter equation is smaller, the velocity of the mobile phase can therefore be increased without an appreciable loss of efficiency. Moreover, as the viscosity of the mobile phase is close to that of a gas, GC capillary columns can be used. However, the pressure drop across the column modifies the distribution coefficients of the compounds between the beginning and the end of their migration. This causes peak broadening. For this reason, the efficiencies obtained in capillary GC cannot be attained with SFC.

The restrictor maintains the selected pressure independently from the mobile phase (MP) flow rate. As the MP behaves like a compressible gas, the pressure influences its density and therefore its polarity. If the restrictor is controlled in order to conduct a pressure gradient, the pressure drop caused by the column has an impact on the MP speed. Therefore, if the pressure decreases, the density of the MP decreases, and the retention factors *k*, related to the retention times, increase, whatever the SP and solutes (Figure 6.5). The same is true if we increase the column temperature at constant pressure: reducing the density of the MP also increases the retention factors.

In HPLC, a flow rate is selected to generate a certain pressure at the column head. This pressure depends both on the MP and the column, which prevents its flow. As liquids are by nature not compressible, the density of the MP does not depend on the pressure and, consequently, its speed of advancement through the column is constant. In SFC, since the MP is compressible, this speed of advancement increases during elution.

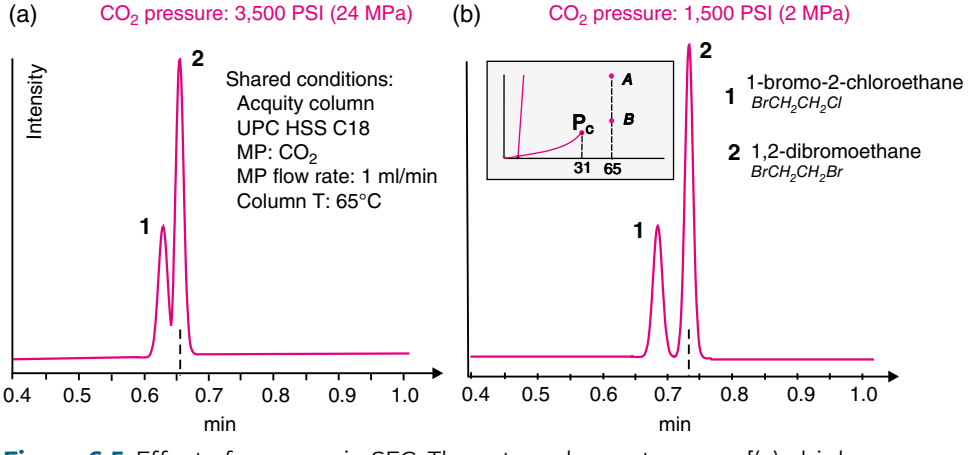


Figure 6.5 Effect of pressure in SFC. These two chromatograms [(a) - high pressure, (b) lower pressure], obtained with a nonpolar C18 column, demonstrate that by decreasing the outlet pressure using a restrictor, the polarity of the MP decreases too, which then increases the retention times and the separation factor (selectivity). Compound **2**, which is less polar than **1**, is a little more retained by the nonpolar SP. We can also note the speed of analysis, i.e. less than 50 seconds in total. (Source: Courtesy of A. Morand, Pharma-Physic.)

6.5 SEPARATION OF ENANTIOMERS BY SFC

The separation of enantiomers has always been hard to predict and difficult to implement. Like other chromatographic techniques, SFC can be used for this purpose (Figure 6.6).

Among the versatile chiral selectors, modified cellulose-based and amylose-based phases, fixed on a silica gel support, have various advantages:

- Quicker separations than in HPLC, thanks to shorter phase equilibration times.
- Possibility of conducting separations at lower temperatures than in GC, in order to avoid potential racemization of enantiomers during analysis.
- Lower emissions of polluting solvents (green chemistry).
- Easier coupling with mass spectrometry, thanks to the easy elimination of carbon dioxide.

A final advantage, considered as a priority in pharmacology, but which is outside the field of chemical analysis, is the possible extrapolation to the preparation scale on large columns. This enables the collection of small, optically pure quantities of compounds.

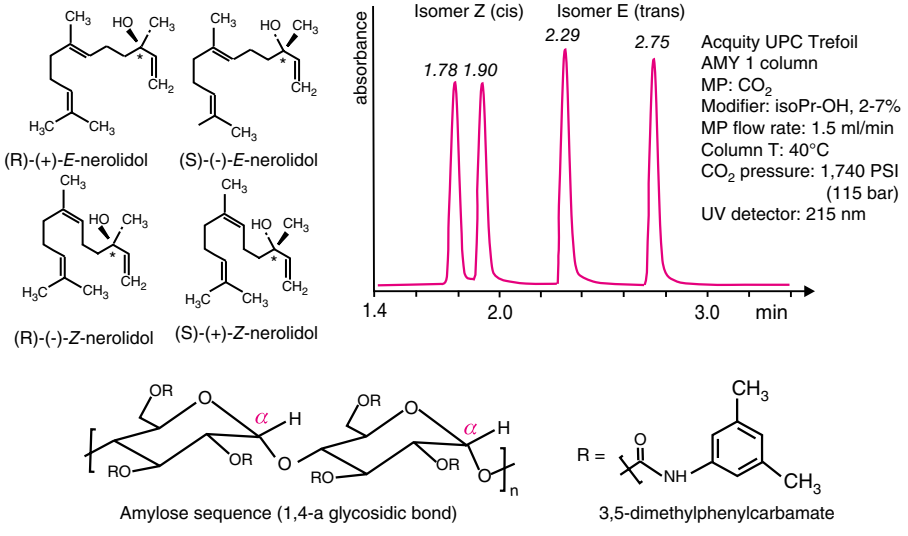


Figure 6.6 Example of separation on a chiral column. A blend of two *cis*/*trans* isomers of nerolidol leads to four peaks under the indicated conditions (figure composed according to a document from Waters). Again, it is demonstrated that the analysis lasts only three minutes, while in conventional GC on a 25-m cyclodextrin capillary column, about 30 minutes at 230°C would be needed. Below, molecular structure of an amylose-based phase derivatized in the form of carbamates, on a silica support.

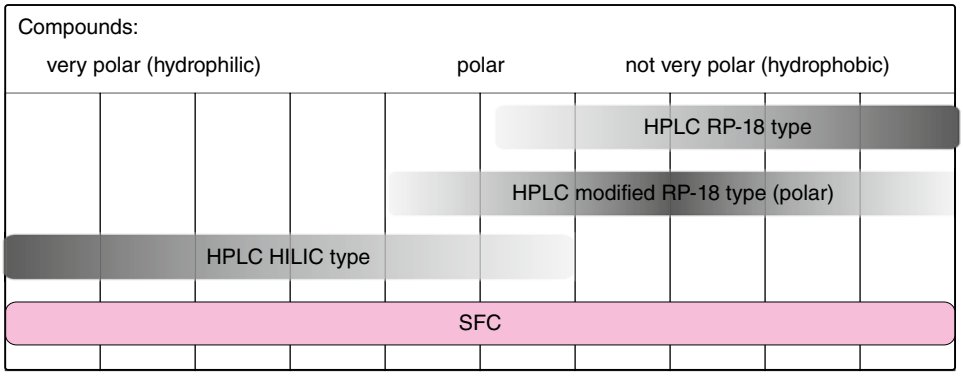


Figure 6.7 Comparison of SFC and HPLC variants. The SFC domain covers the full scope of HPLC and its variants, while enabling quicker separations and less consumption of mobile phases.

6.6 SFC IN CHROMATOGRAPHIC TECHNIQUES

SFC is potentially useful for a variety of applications. The major distinction of this technique lies in the ability to vary selectivity by programming the parameters *P* (pressure) and *T* (temperature) rather than by modifying the chemical composition of the eluent. The low viscosity of the mobile phase also permits an arrangement of several HPLC-type columns in series. It is possible to study compounds of higher molecular mass than in GC (>1,000), with a better resolution than in HPLC. The domain of SFC includes lipids and oils, emulsifiers, oligomers and polymers (Figure 6.7).

As separations are quicker, the technique can be used in the field. Finally, the use of carbon dioxide as a mobile phase facilitates detection modes involving an infrared spectrophotometer or even NMR (Chapter 15).

SFC becomes even more efficient with columns packed with particles under 3 µm in diameter, which requires higher pressures. *Ultra-High-Pressure Supercritical Fluid Chromatography* (UHPSFC) is characterized by quick separations of polar, labile or nonvolatile compounds.

KEY POINTS OF THE CHAPTER

1. SFC relies on the use of a fluid in the supercritical state or close to this state (subcritical) as a mobile phase. In practice, only carbon dioxide (a 'green' solvent that is low in toxicity, noncorrosive, T_c = 31°C) is used pure or blended with a polar cosolvent.

2. A supercritical fluid, used as a mobile phase, conducts separations rather similar to GC or HPLC with a number of advantages. Small pressure or temperature changes modify its density, which leads to significant disturbances in the *selectivity* of separations.
3. Choosing as a mobile phase a supercritical fluid whose density is similar to that of a liquid and whose viscosity is similar to that of a gas, reduces analysis times as compared to HPLC or leads to separations at lower temperatures than in GC.
4. The three interactions between the *solute*, *stationary phase*, and *mobile phase* are improved by adding a cosolvent to carbon dioxide, such as an alcohol providing sufficient polarity to the supercritical phase, which then increases *efficiency*.
5. With SFC, it is possible to change the eluting power of the mobile phase by using a pressure gradient in the column; this is equivalent to a concentration gradient in HPLC or a temperature gradient in GC.
6. In practice, longer columns than in GC are selected without increasing the pressure drop much. Moreover, carbon dioxide is compatible with GC (especially FID) and HPLC detectors. However, since it does not have a dipole moment, it does not dissolve polar molecules.
7. Unlike HPLC, SFC requires pressurization of the column. Depending on the choice of detector, the return to atmospheric pressure will be done before detection (if we use an FID) or after (if we choose a photometric detector).
8. In practice, the columns used in SFC are comparable to those used in HPLC, but separations are quicker because equilibria between phases are faster and the column re-equilibration time between two successive separations is shorter.
9. To separate enantiomers without fear of racemization, columns containing biopolymer-based chiral stationary phases are selected, and the separation is conducted at a moderate temperature.

PROBLEMS

1. A pure compound possessing one asymmetrical carbon is separated on a chiral column, leading to chromatogram **A** (Figure P6.1). A second compound (pure metolachlor) in the racemic state is separated on a chiral column, yielding chromatogram **B**.
   a. Calculate the optical purity or enantiomeric excess (e.e.) of compound **A**, by using Eq. (2.2) (Chapter 2) where S_R and S_S designate the peak areas of the two enantiomers. The calculation will be based on peak heights.
   b. Considering the structure of compound **B**, explain the presence of these four peaks.

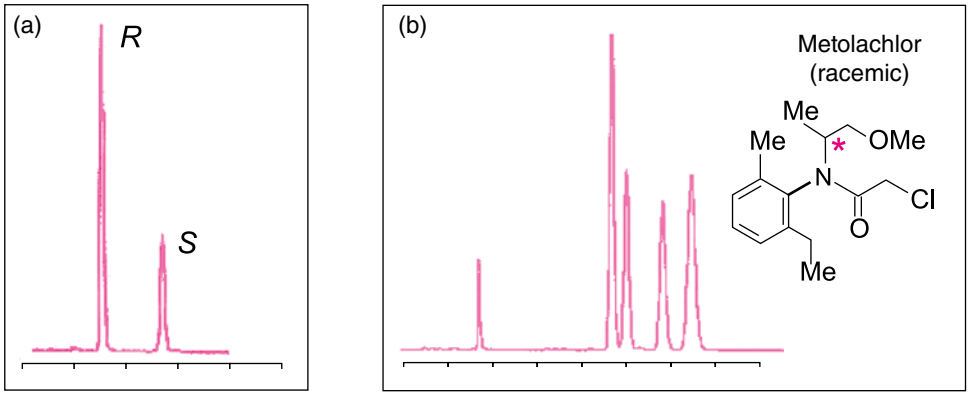


Figure P6.1

SOLUTIONS

1. a. By direct application of the equation for optical purity:

$$Optical\ Purity\left(e.e.\%\right)=100\frac{37-14}{37+14}=100\times\frac{23}{51}=45\%$$

Compound **A** has an optical purity of 45% (precision of a few %!) for the *R* enantiomer.

b. Metolachlor possesses one asymmetrical carbon (shown by (*) in Figure 6.8). If we examine a sample of racemic metolachlor with a chiral column, it should be possible to distinguish between the *R* and *S* isomers. However, for this molecule there is another type of isomer, called *atropisomers*, related to the existence of a steric barrier, which prevents rotation around the CN bond connecting the aromatic ring to the nitrogen atom. With a high-performance chiral column and under good analytical conditions, each enantiomer is doubled, which in the end gives four peaks on the chromatogram: Rα, Sα, Rβ and Sβ.

Chapter 7

Size-Exclusion Chromatography

Size-exclusion chromatography (SEC) is a separation method dedicated to macromolecules, both synthetic (gel permeation) and natural (gel filtration). This method is based on their size, without any solute/solvent/stationary phase interaction. To achieve this, stationary phases contain pores through which compounds are able to diffuse to a certain extent, depending on their volume in solution. Similar to HPLC, with which it shares basic equipment, SEC can, in a single test, separate mixtures

containing both small and large molecules. By combining several detection methods, SEC also provides information about the main physico-chemical characteristics of natural or synthetic macromolecules and polymers. This other aspect of SEC is discussed in the appendix of this chapter.

Objectives

Describe the separation principle used in SEC
List the parameters used
Distinguish between *gel permeation* and *gel filtration*
Classify the stationary phases used
Describe columns (variable or fixed porosities)
Compare total *exclusion* and *permeation*
Link detection methods with macromolecule characteristics

7.1 PRINCIPLE OF SEC

Size-exclusion chromatography (SEC) is based upon the ability of the sample molecules to penetrate into the stationary phase. The latter has pores with diameters comparable to those of the species present in solution in the mobile phase (Figure 7.1). This method is referred to as *gel filtration chromatography* (GFC) when the stationary phase is hydrophilic (the mobile phase being aqueous) and as *gel permeation chromatography* (GPC) when the stationary phase is hydrophobic (the mobile phase being an organic solvent).

The total volume V_M of the mobile phase in the column can be considered as the sum of two values $V_M = V_I + V_P$, where V_I is the interstitial volume (external to the pores) and V_P is the volume of the pores. V_I, also called the *void volume* in SEC, represents the volume of mobile phase necessary to transport a large molecule assumed to be excluded from the pores, and V_M is the volume accessible to a small molecule that can enter all the pores of the packing.

The elution volumes V_e are therefore between V_I and V_M:

$$V_e = V_I + KV_P \tag{7.1}$$

or

$$K = \frac{V_e - V_I}{V_M - V_I} \tag{7.2}$$

K, the diffusion coefficient, represents the degree of penetration of a species dissolved in the volume V_P ($0 < K < 1$). For the majority of modern packing materials, V_I and V_P are both in the order of 40% of the volume of the empty column.

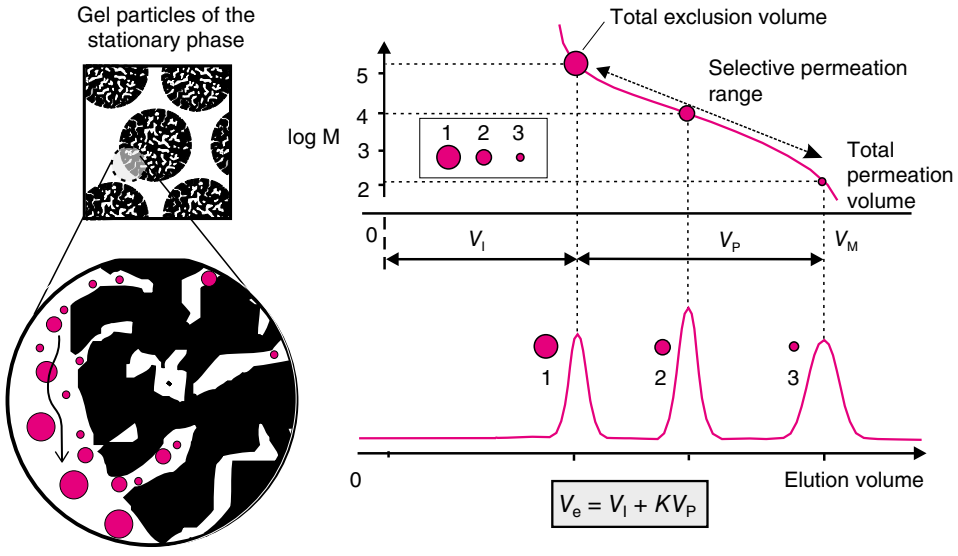


Figure 7.1 Migration across a stationary phase gel. A chromatogram displaying the separation of three species (**1, 2, 3**) of different sizes in solution. The largest molecules **1** are the first to elute, followed by the medium-sized molecules **2**, and finally by the smallest **3**. The elution volumes are located between V_I and $V_M = V_P + V_I$. Image of a porous gel to illustrate the notion of separation depending on the pore size.

When V_e / V_M is greater than one, the behaviour of the compound on the column no longer just follows the mechanism of size exclusion but, as in HPLC, it also undergoes physico-chemical interactions with the support. This situation must be avoided in SEC, as the latter must remain a purely entropic phenomenon.

In practice, molecules whose diameter is larger than those of the largest pores are excluded from the stationary phase ($K = 0$). This explains the expression *size exclusion* for this situation. They flow through the column without being retained. They appear as a single peak in the chromatogram at position V_I, which is known as the total exclusion volume (Figure 7.1). This is the origin of the name *size-exclusion chromatography*. In contrast, the elution volume of the very small molecules V_M is called the total permeation volume. Each stationary phase is thus adapted to a separation range defined by the upper and lower mass limits, above and below which there is no possible separation effect. To increase this range of separation, which is fixed by the difference between these two volumes, two or three columns with different porosities can be put in series in order to increase total column length.

Efficiency is optimal with low-viscosity solvents and when separations are done at high temperature. The diffusion coefficients K are independent of temperature.

7.2 STATIONARY AND MOBILE PHASES

SEC stationary phases are made of microbeads with diameters of 3–10 μm. Pore diameters are in the 4–200 nm range. These packings, usually called *gels*, have only moderate resistance to crushing, and thus the pressure at the head of the column is limited to about 250 bar. There are two types of such gels:

- *Macroporous polymers* for *gel permeation chromatography* (GPC). The packing material most often used is a *styrene-divinylbenzene* (PS-DVB) copolymer with an *organic mobile phase* (Figure 7.2) such as tetrahydrofuran, toluene or trichlorobenzene for low-solubility samples. Performances of each are provided for a given mobile phase as well as for the separation of a category of synthetic polymers: polystyrene, polyethylene oxide, starch, etc. (Figure 7.3 and Table 7.1).
- *Polyhydroxylated* or *polysaccharide matrices* for *gel filtration chromatography* (GFC), such as cross-linked dextran (Sephadex), agarose/polyacrylamide (Ultra-Gel) or *polyvinyl alcohols/polymethacrylates* or even porous silica particles. The surface of the latter is covered with hydroxylated groups. These materials are paired with *aqueous mobile phases*. The pores occupy almost all of the stationary phase volume. These phases are used to separate biopolymers (e.g. polysaccharides) or to analyse and study biological macromolecules (e.g. proteins, peptides or oligonucleotides) that require a buffered aqueous mobile phase.

Once the stationary phase has been selected, knowing that there is no chemical interaction between the mobile phase and the solutes, the improvement of separations depends on the temperature change in the column and on the available pore volume V_p. This is a reason for the large size of many commercial columns (typically 7.5 × 300 mm) and for the practice consisting of stacking two to four columns with different porosities. Efficiency N may thus reach more than 30,000 plates/m. The decrease in particle diameter, which leads to greater chromatographic efficiency, reduces interstitial passages and makes the migration of large excluded molecules more difficult.

7.3 INSTRUMENTATION

Instrumentation is similar to that employed in HPLC, except that the columns are bigger. To increase resolution, it is common to use two or three columns, with different porosities, in series.

Two detectors are used when the sole focus is classic chromatographic interpretation. They can be either a diode bar photometric detector (UV) or a refractive index detector (RI), identical to those used in HPLC (see Chapter 3, Sections 3.8.1 and 3.8.3). These potentiometric detectors lead to a recording

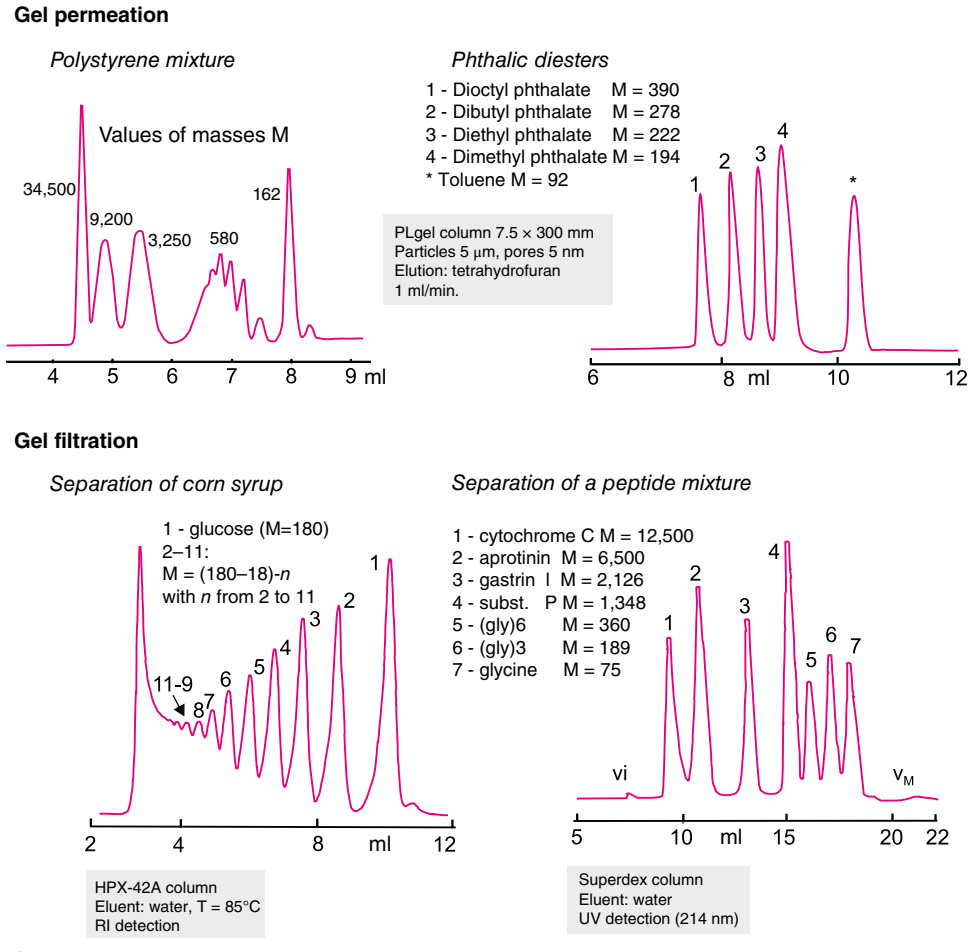


Figure 7.2 Comparison of gel permeation and gel filtration. By using a small-pore (5 nm) gel permeation SP, it is possible to separate low-molecular-weight molecules (the toluene peak makes it possible to calculate Vm). A gel filtration column with water as the solvent is adapted to the separation of sugars or dextrans, i.e. natural macromolecules. (Source: Courtesy of Varian Polymer Laboratories.)

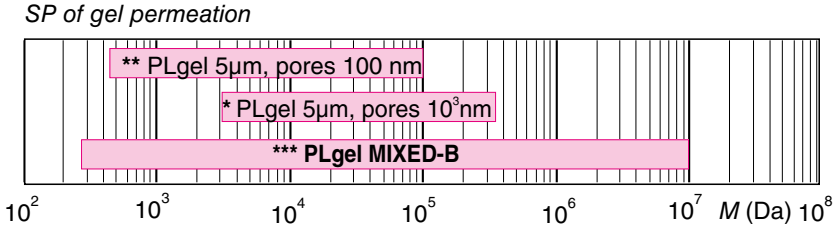


Figure 7.3 Comparison of three gel permeation phases.

Table 7.1 Permeation range (Da) of three gels for various standard compounds.

Standard	G2000: 12.5 nm pores	G3000: 25 nm pores	G4000: 45 nm pores
Globular protein	5,000–100,000	10,000–500,000	20,000–7,000,000
Dextran	1,000–30,000	2,000–70,000	4,000–500,000
Polyethylene glycol	500–15,000	1,000–35,000	2,000–250,000

representing hydrodynamic volumes of the compounds present. The refractive index detector, while not very sensitive, is often selected because the signal, generally independent from molecular weight, is proportional to the polymer concentration of the solution.

SEC can be used to determine various parameters of synthetic or natural polymers and other macromolecules, by coupling one of the above detectors with either a viscosity meter or a light scattering device (dynamic or static). These are continuously adapted to the measurements in order to be used as detectors. In fact, these are real coupled methods used with dedicated software to establish these physico-chemical parameters.

> Gel filtration (one of the main applications of which concerns the study of proteins) uses an aqueous mobile phase, in order not to denature the analytes. To neutralize the observed fixation of certain proteins on the column surface, arginine is added to the mobile phase.

7.4 APPLICATIONS OF SEC

Since SEC enables separations of molecular masses ranging from 200 to more than 10^7 Da, the main applications are found in the analysis of polymers (analysis or monitoring of polymerization) and other macromolecules, including biopolymers (simple proteins, conjugated proteins, or in the oligomer state, polysaccharides), and natural or synthetic nanoparticles. For organic compounds, which are usually analysed using HPLC or GC, there are not as many applications, because to separate low molecular masses with SEC the masses must differ by a factor of two.

The absence of any chemical interaction with the stationary phase, shorter retention times, and the possibility to recover all of the analytes convey many advantages. The development of analyses is quick, knowing that we do not use an elution gradient as there is no solute/mobile phase interaction.

SEC is useful in analysing samples of unknown composition, likely to contain both polymers and small molecules, which is the case of many commercial or industrial products, such as polysaccharides, starch or pulp, and pharmaceutical products. Finally, in addition to analytical applications, SEC is used for preparative separations with larger-diameter columns.

Characterization of Macromolecules

SEC makes it possible to determine several parameters pertaining to the physico-chemical properties of natural or synthetic macromolecules, including industrial polymers. Access to these sizes requires a combination of two, or even three, specific detectors, associated with dedicated software in order to determine the molecular distribution, the average masses (number average, weight average or absolute), as well as the radius of gyration and hydrodynamic radius of macromolecules.

Macromolecules, or very large molecules, whose molecular weights are very high, often result from the more or less coordinated repetition of one or more patterns. At the macroscopic level, macromolecular compounds, whether they are natural or artificial (synthetic polymers), are not composed of single structures (there are some exceptions), but rather of homogenous families of macromolecules whose molecular masses are distributed around an average value.

Consequently, unlike the previously studied chromatographic methods (GC, HPLC, ICE or SFC), a peak on an SEC chromatogram rarely corresponds to a single macromolecule of unique weight but rather to a set of macromolecules whose sizes vary between the start and the end of the peak. That is the case for synthetic polymers produced by the polymerization of a monomer and differing simply in their degree of polymerization or in the existence of branching.

Each retention volume or retention time corresponds to a molecular mass on the chromatogram. When we break down a peak into successive sections, each section corresponds to an average mass M_i; and on the surface of the section, there is a population of molecules N_i of mass M_i. Since these molecules are in the same solution, M_i can be replaced by C_i, or mass concentration. With software, the chromatographic peak can be associated with the *number-average molecular weight* $\bar{M}_n$ (total mass of macromolecules divided by their total number), with the *weight-average molecular weight* $\bar{M}_w$, and with the *polymolecularity index* $\bar{I}$, which demonstrates how the studied polymer is mass dispersed (for a perfect polymer where all molecules are identical, $\bar{I} = 1$). The relationships are expressed as follows:

$$\bar{M}_n = \frac{\sum C_i}{\sum C_i / M_i} = \frac{\sum N_i M_i}{\sum N_i} \tag{7.3}$$

$$\bar{M}_w = \frac{\sum M_i C_i}{\sum C_i} = \frac{\sum N_i M_i^2}{\sum N_i M_i} \tag{7.4}$$

$$\bar{I} = \frac{M_w}{\bar{M}_n} \tag{7.5}$$

7.5 POLYMER CHARACTERISTICS

7.5.1 Conventional Calibration

Easy to implement, it uses only *one detector* whose response is proportional to the concentration *C* (UV or more often RI).

We use a range of standards of monodisperse polymers of the same nature as the one present in the sample in order to find the molecular mass. This set of standards must be such that the retention volumes fall between the total exclusion and total permeation volumes. These standards are injected in order to find their retention volumes. By means of polynomial regression, an equation connecting log M_w to the retention volume is established (Figure 7.4). Then, a sample solution is injected. The elution volume at the peak maximum provides an evaluation of the average molecular mass in comparison with the calibration curve.

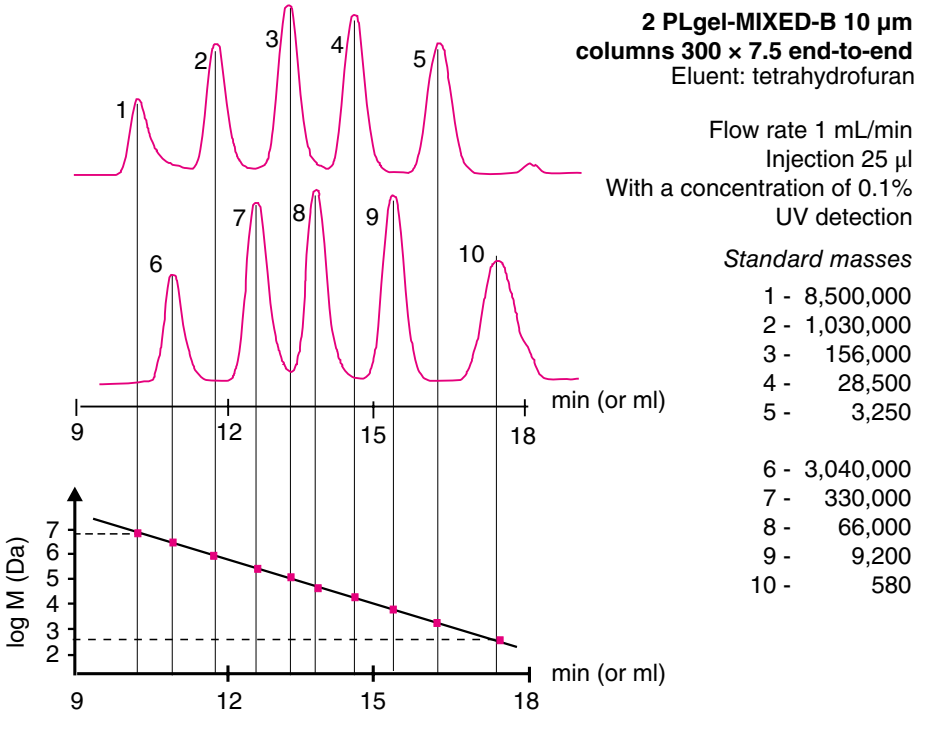


Figure 7.4 Linear calibration curve of a gel permeation column. The column can be calibrated by using two complementary mixtures of polystyrene standards with a sufficient number of masses. This curve is linear over a wide range of masses due to the use of a mixed stationary phase. This then helps determine the molecular mass of an unknown polystyrene.

The major drawback of conventional calibration is that it only provides a relative molecular mass. If, for example, the calibration is conducted with polystyrenes, it leads to the molecular mass of the unknown polymer in an equivalent mass of polystyrene. If the studied polymer changes structure, for example owing to the presence of branching, the given relative molecular mass is incorrect. Therefore, this calibration does not provide any information on the polymer structure.

If we use software and Eqs. (7.3) and (7.5), this calibration leads to the *number-average molecular weight* $\bar{M}_n$, the *weight-average molecular weight* $\bar{M}_w$, and the *polymolecularity index* $\bar{I}$.

7.5.2 Universal Calibration – Viscometric Detector

This calibration, which is independent from the chemical nature of the polymer, requires *two detectors*: a viscometric detector connected to a refractometric detector. The calibration phase follows the same principle as previously described. However, the graph of log ([η]M) is plotted as a function of the retention volume, knowing that the intrinsic viscosity (the viscosity without solvent) multiplied by the weight-average molecular weight is directly proportional to its volume in solution (called the *hydrodynamic volume*). Therefore, whatever the chemical nature and the structure of the polymers used as standards, all curves coincide for a given column (Figure 7.5).

This calibration method presents several advantages: for one, it leads to an *average molecular mass (viscometric mass)* and also to the intrinsic viscosity, making it possible to obtain the *a* and *K* coefficients of the empirical Mark–Houwink formula (Eq. (7.6)),

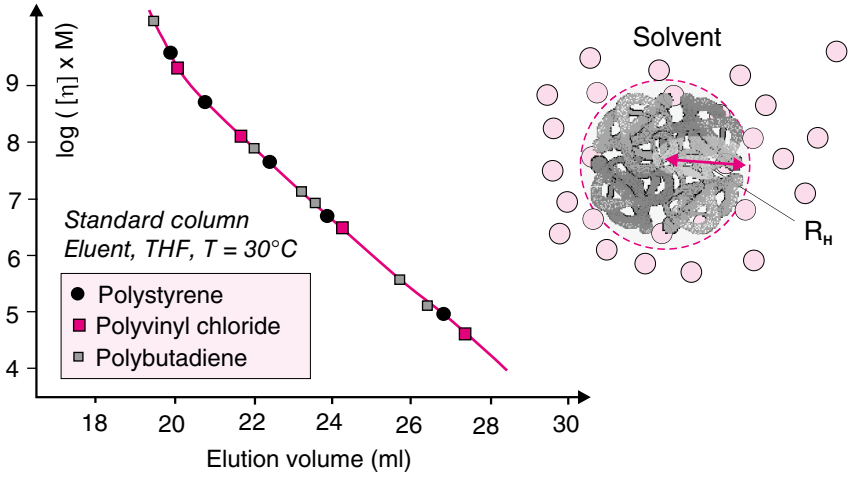


Figure 7.5 Universal calibration curve. Illustration of the hydrodynamic radius of a polymer in a nonmiscible solvent. In the case of a polymer ball, this radius corresponds to the radius of a sphere in which the solvent does not penetrate.

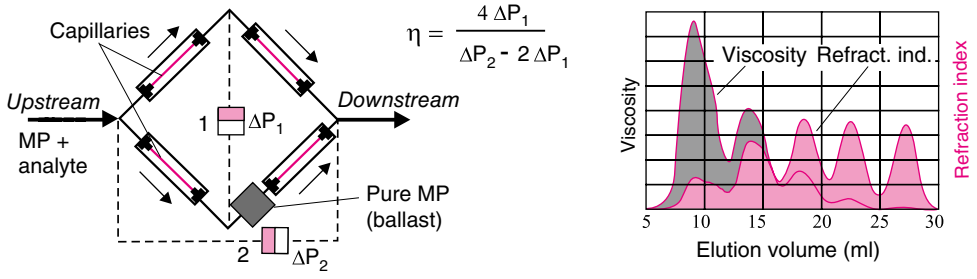


Figure 7.6 Viscometric detection. Diagram of a viscometric detector. Four capillaries are distributed into two paths with a pure mobile phase tank on the reference path. The instantaneous (and intrinsic) viscosity is deduced from pressure measurements by sensors 1 and 2. Comparison of two chromatograms (refractometric and viscometric detection) obtained from the same polymer mixture.

and therefore to provide information on the structure and branching of the polymer being studied.

$$[\eta] = K \cdot M_w^a \tag{7.6}$$

Viscometric detector

The viscosity, which depends on molecular weight, is measured by the viscometric detector (Figure 7.6), whose response may be written as:

$$Response = K \cdot [\eta] \cdot C \tag{7.7}$$

where K designates the detector constant and $[\eta]$ the intrinsic viscosity of the solute, inversely proportional to the density of particles in solution. Therefore, a ball-shaped macromolecule has a greater density than a macromolecule stretched into a rod. This detector coupled with a concentration detector leads to the weight-average molecular weight and to the hydrodynamic radius *R*h ('footprint' in the solvent). It is used for universal calibration.

7.5.3 Triple Detection – Light-Scattering Detector

By combining *three detectors*, i.e. following the refractometric (or UV) detector with a viscometric detector and a mono- or multi-angle light-scattering detector, we can obtain the hydrodynamic radius and the radius of gyration (*R*h and *R*g) as well as structural information (branching, conformation, ball or rod shape, and existence of aggregates). Triple detection, sometimes integrated into a single device, has become very useful in pharmaceutical research for better knowledge of biological macromolecules.

Light-scattering detector

The intensity of the light scattered by a particle depends on its size, its shape, its mass as well as the solvent and its refraction index. According to Rayleigh's theory, when subjected to a light ray, large molecules diffuse more light than smaller ones, and the *diffused intensity at various concentrations (elution curve) leads to their absolute molecular weights*.

There are two categories of light-scattering detector (which is different from the evaporation detector, see Chapter 3, Section 3.8.4): one is static (*Multi-Angle Light Scattering* or MALS) and the other is dynamic (*Dynamic Light Scattering*, DLS). The latter gives a proportional response to the solute concentration as well as its weight-average molecular weight M_w and to the d*n*/d*C* ratio using its refraction index.

For polymers whose hydrodynamic radius *R*h is under 15 nm, the scattering is practically uniform in all directions. A single measurement can thus be used, for example at a 90° angle (Figure 7.7).

For more voluminous polymers, the scattering is not uniform and, therefore, the intensity of the zero-angle scattering must be determined, which is experimentally impossible. The multi-angle detector avoids this difficulty. To do so, it measures the light scattered by the solution at several angles (between 10 and 20 in number) falling between about 10 to 150° (Figure 7.8). Appropriate software calculates a zero-angle

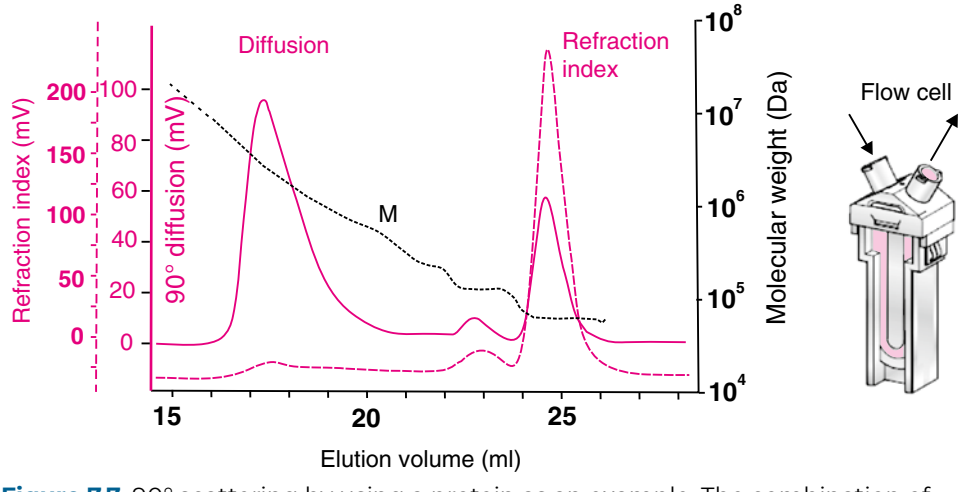


Figure 7.7 90° scattering by using a protein as an example. The combination of two detectors (refraction index and scattering at a right angle) helps to observe formation of aggregates and to calculate their absolute masses, without prior calibration (Source: Courtesy of Malvern). The peak located at 18 ml corresponds to a family of proteins with variable masses, while the peak at 25 ml is formed by a compound of a single mass.

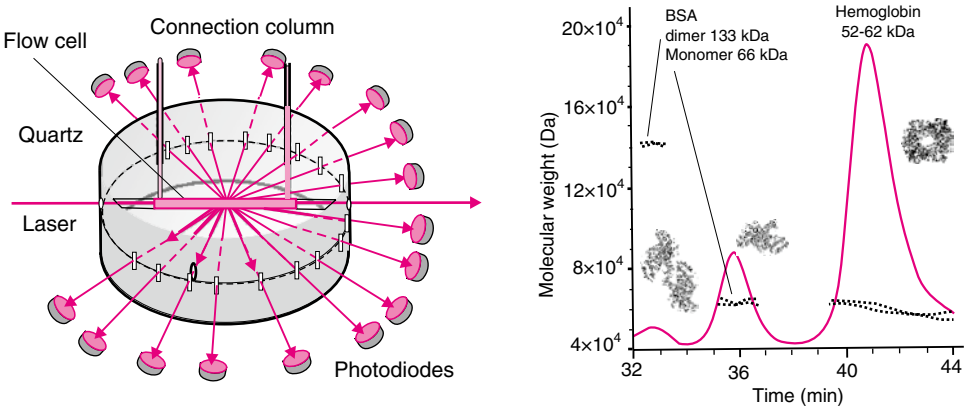


Figure 7.8 Measurement cell of a multi-angle light scattering (MALS) detector. The flow cell (40 µl) is made of a fine hollow channel in a quartz cylinder closed off by windows at its extremities. The laser beam crosses the cell following its axis. Photodiodes are placed all around, which retrieve the scattered light, each at a specific angle. Chromatogram of a mixture of two proteins, with superimposition of the plot of absolute masses calculated repetitively throughout the elution of each peak. (Source: Courtesy of Wyatt.)

scattering value by extrapolation. Therefore, the absolute molecular weight can be determined *without need for* a calibration method. Moreover, this detector also provides information about the 'footprint' of the macromolecule by calculating its radius of gyration *R*g (quadratic average of the distance separating the centre of the macromolecule mass from its extremities, without solvent).

The Dynamic Light Scattering (DLS) detector is less widely used. When a laser source lights up smaller particles than its wavelength, an elastic scattering in all directions occurs which the device measures at a given angle (e.g. 90°) over time. As particles move due to thermal agitation (Brownian movement), the distance separating them constantly varies and creates interferences from which information is obtained regarding their velocity and thus their size (*R*h calculation).

7.6 FIELD FLOW FRACTIONATION (FFF)

Field flow fractionation (FFF) refers to another method for separating macromolecules, which also makes it possible to obtain the various parameters described above. Since there is no stationary phase, it is often described as *single-phase chromatography*. There are several variants. For example, the classic column is replaced by the laminary space between two plates (Figure 7.9). It can be imagined as a hollow tape of several decimetres in length, 2–3 cm in width, and an internal thickness of 0.1–0.3 mm. In this space travelled

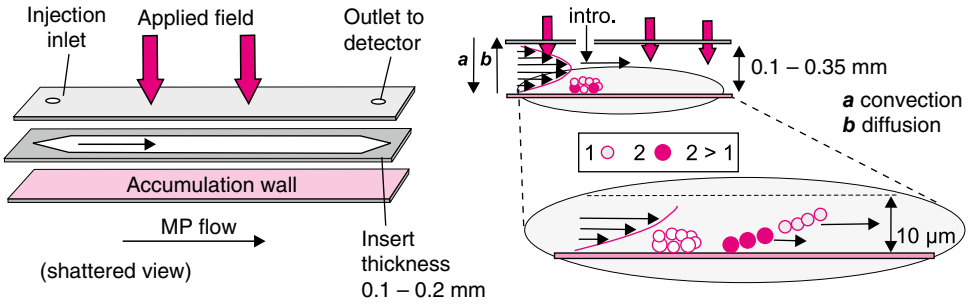


Figure 7.9 Migration of a mixture of two macromolecules based on the field flow fractionation (FFF) principle. The transverse field is produced by a thermal gradient obtained by bringing the upper wall to a higher temperature than the accumulation wall. FFF separation only acts on hydrodynamic properties and does not involve any interactions with a stationary phase.

by the longitudinal flow of the mobile phase, a transverse force is created by the production of a field (electrical, thermal, gravitational, etc.) to which the analytes must be sensitive. When the sample is injected and the mobile phase is stopped, the force created pushes the sample against the opposite wall, but under the effect of scattering (Brownian movement) the analyte molecules will leave this wall where they have accumulated and go back into the liquid flow over a thickness of ten micrometres. This will occur all the more easily when their weight is low. They will then be entrained by the mobile phase, whose flow has a parabolic profile. Molecules that are farthest from the wall will advance faster. The combination of these two transverse and longitudinal forces will continue throughout the laminar space, thus accentuating the separation throughout the path. In contrast with SEC, the lighter-weight species will reach the detector(s) first. Information collected about the analytes is the same as that previously described.

Several companies now market set-ups based on this principle, but resulting from various technological approaches.

KEY POINTS OF THE CHAPTER

1. Review of the basic concepts related to macromolecules: these are characterized by various parameters, including the average molar mass, or their absolute mass, their mass distribution and their shape (strand, ball, etc.). Their *hydrodynamic radius* and *radius of gyration* may change depending on the nature of the solvent used.
2. SEC is adapted to the separation of macromolecules (proteins, polysaccharides) or polymers, whether natural or synthetic. Stationary phases have pores that the analytes will be more or less able to penetrate. The higher their volume in solution (hydrodynamic volume), the faster they will be eluted.

3. In general, all molecules larger (in solution) than the pores of the stationary phase will be excluded and eluted without separation. They will arrive first at the detector (first peak on the chromatogram).
4. Inversely, below a certain size, the molecules may penetrate into the pores and remain longer on the column (last peak of the chromatogram). To improve separation, several columns of different porosities are placed in series, or columns with different porosity levels are used.
5. There are two variants of SEC: gel permeation (organic mobile phase) and gel filtration (aqueous mobile phase). They determine the relative molecular weight of analytes when we have standards of known mass and other structural information obtained by means of various detection methods.
6. There are many detection methods. The *photometric detector* (e.g. UV), whose response is proportional to concentration, is used for conventional calibration (relative mass). The *viscometer*, whose response is related to the viscometric index, is associated with universal calibration (it provides the average molecular mass). The *light-scattering* detector gives the absolute mass. By combining two or three detectors based on different principles, the knowledge of the studied polymers is improved.

PROBLEMS

1. A mixture of polystyrenes of known molecular masses in tetrahydrofuran (THF) is chromatographed on a column (ID = 7.5 mm, L = 300 mm) whose permeation range extends from 400 to 30,000 Da (Figure 7.2). The UV detection is conducted at 254 nm.

   Based on chromatogram *a* of Figure 7.2:

   a. Calculate the total exclusion volume (interstitial volume) and the volume of pores of the column used.
   b. Calculate the partition coefficient *K* for the compound weighing 3,250 Da.
   c. Sometimes, in SEC, *K* values over 1 are observed. Interpret this phenomenon.
2. Gel permeation chromatography is used to separate a mixture of four polystyrene standards whose molecular weights are: 9,200, 76,000, 1.1×10^6, and 3×10^6 Da. Three columns are available for this exercise.

   A: 70,000 to 4×10^5 Da
   B: 10^5 to 1.2×10^6 Da
   C: 10^6 to 4×10^6 Da

   How might these four polymers be separated in a single operation if it is permitted to use two of the above columns end to end?

3. In size-exclusion chromatography, after injection of polystyrene standards, a relationship between molecular weight and retention time is portrayed as follows:

$$\log M = 5.865 + 1.411 t_R - 0.333 t^2{}_R + 0.0163^2{}_R \qquad \text{(P7.1)}$$

where M is the molecular weight in Da and t_R the retention time in minutes.

a. From this equation, calculate the molar mass of a solute that would elute with a retention time equal to 7.48 minutes, if we assume that this polymer has a single mass.

b. In reality, single-mass polymers are almost nonexistent (except for standards). An average molecular mass M_N (Eq. (P7.2)) and a weighted-average mass M_W (Eq. (P7.3)) are defined:

$$M_N = \frac{\sum_i N_i M_i}{\sum_i N_i} \qquad \text{(P7.2)}$$

$$M_W = \frac{\sum_i N_i M_i^2}{\sum_i N_i M_i} \qquad \text{(P7.3)}$$

where M_i is the molar mass of a polymer eluting in the time interval corresponding to a chromatographic peak and N_i is the number of moles of this polymer. Calculation software cuts the chromatographic peak into sections and determines the related molar mass and area A_i for each section. Assuming that the response coefficient k of the detector remains constant during the chromatographic peak, and using the basic equation $m_i = N_i \times M_i = k \times A_i$, transform Eqs. (P7.2) and (P7.3) by only using A_i and, if needed, M_i.

c. A chromatographic report gave the following results:

t_R	6.95	7.05	7.15	7.25	7.35	7.45	7.55	7.65	7.75	7.85	7.95	8.05
A_i	0	0	1.77	8.53	17.74	40.36	19.44	9.32	2.03	0	0	0
M_i												

Complete this table by using Eq. (P7.1), and calculate molar masses M_N and M_W by using Eqs. (P7.2) and (P7.3).

4. Consider a mixture of two batches (1 and 2) of polyethylene glycol (polymer of ethylene oxide) and the corresponding tetramer (3), separated by gel filtration (Figure P7.1).

a. Explain why the last peak is narrower than the two others. Can the efficiency of the column be deduced under the conditions of this separation?

b. If we assume that V_M = 23 ml and V_I = 10 ml, calculate the volume of pores V_P as well as the partition coefficients of peaks 1 and 2 of the chromatogram.

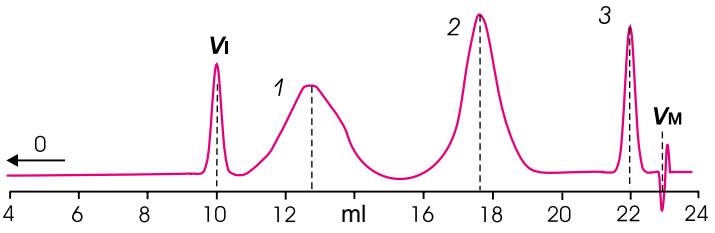


Figure P7.1

5. On a chromatogram (from a gel permeation column), the molecular masses and elution volumes of several macromolecules are noted: dextran 500–1,000 kDa, V(e) = 4.8–5 ml; fibrinogen 340 kDa, V(e) = 6 ml; catalase 60 kDa, V(e) = 9 ml; β-lactoglobulin 18 kDa, V(e) = 11 ml; dimer 36 kDa V(e) = 10.5 ml. Determine a value close to the mass (in Da) of the P-gp (glycoprotein P) for which V(e) = 7.2 ml.

SOLUTIONS

1. **a.** The exclusion volume is about 4.5 ml. The pore volume is about 3.75 ml.

b. For the mass of 3,250 Da, $K = 0.29$.

c. In size-exclusion chromatography $K \leq 1$, except if there are interactions between the solute and stationary phase, creating a classic partition phenomenon which is superimposed on the diffusion into the pores.

2. By putting columns C and A end-to-end, the four polystyrene standards will be separated: four distinct peaks will be obtained.

3. **a.** $M = 30{,}475$ Da;

b. (2) $M_N = \dfrac{\sum_i A_i}{\sum_i \dfrac{A_i}{M_i}}$ and (3) $M_w = \dfrac{\sum_i A_i M_i}{\sum_i A_i}$;

c. Table

M_i	90,772	73,800	60,016	48,830	39,756	32,398	26,431	21,593	17,668	14,482	11,895	9,792

$M_N = 31{,}210$ Da, $M_w = 33{,}134$ Da.

4. These industrial polymers result from a polyaddition mechanism of ethylene oxide. Their general formula is $H–[O–CH_2–CH_2]_n–OH$, where n varies from one batch to another. Peak V_I comes from all macromolecules for which $K = 0$. Peak **3** corresponds to a pure compound ($n = 4$, M = 194 Da), its full width at half-maximum δ thus provides us with the efficiency N of this separation, unlike the other peaks. By measuring δ and t_R (both in mm), $N = 5.54\dfrac{t^2R}{\delta^2} = 5.54 \times \dfrac{110^2}{2^2} = 16{,}760$ plates. Based on the previous formulas, $K_1 = \dfrac{12.9-10}{23-10} = 0.22$ and $K_2 = \dfrac{17.8-10}{23-10} = 0.60$. The pore volume is equal to $V_p = 23 - 10 = 13$ ml.

5. If we assume that in an intermediate range, between the extremes, the column used presents a normal straight-line plot where log(Mass) = f(V(e)), we can deduce that P-gp has a molecular mass of 170 kDa. Beyond a certain mass (very large or very small), molecules cannot be separated on this column. Either they cannot penetrate into the pores, or the small molecules may penetrate totally. To establish the linear part of the graph, extreme values must thus be rejected.

Chapter 8

High-Performance Capillary Electrophoresis

High-performance capillary electrophoresis (HPCE) is a very sensitive analytical separation method that has been developed largely through the experience acquired with HPLC. Developed at the request of biochemists in order to separate biomolecules, an increasing number of analytical protocols refer to it in pharmacopoeias. It has also found its place in chemical analysis to quantify low-molecular-weight compounds. By combining IEC and HPCE, a hybrid technique called *capillary*

electrochromatography (CEC) has been created with the benefits of each of these techniques. It can reach an *efficiency* potentially exceeding what we can encounter in HPLC.

Objectives

Review the principle of electrophoresis
Describe the migration of analytes in HPCE
Select the sample introduction method
Compare hydrostatic and electromigration injection
Explain the role of the capillary wall
Present the parameters used
Know the different HPCE techniques
Present the principle of electrochromatography

8.1 PRINCIPLE OF CAPILLARY ELECTROPHORESIS

Capillary electrophoresis is based upon the migration of the species present in the sample solution, whether or not they carry an overall electric charge, under the effect of an electric field and when supported by an appropriate medium. It corresponds to an adaptation of the more general methodology of electrophoresis.

8.1.1 Zone Electrophoresis

In the classic technique still widely used in bio-analysis, a small strip of polymer is used. It is covered by a porous substance impregnated with an electrolyte (in the beginning, paper strips were used). The two ends of the covered gel system are dipped into two independent tanks containing the same electrolyte. An electrode is immersed in each tank, one connected to the positive pole and the other to the negative pole of a direct-current generator (Figure 8.1).

The sample is deposited in the form of a transverse band on the strip, which can be cooled (to prevent heating by Joule effect) and then bedded between two isolating plates (for protection). Hydrated ionic species present in this deposit migrate at variable velocities, therefore, at variable times (from several seconds to several hours) towards either of the extremities of the strip (depending on their charge: sign, multiple or single charge, its volume, and therefore its mobility). The absence of a solvent front means that the distance of migration has to be estimated in relation to that of a substance used as an internal marker.

Following migration, the species are generally detected by transferring them to a membrane by contact. They are then revealed using silver salts with specific reagents such as *Coomassie Blue*®.

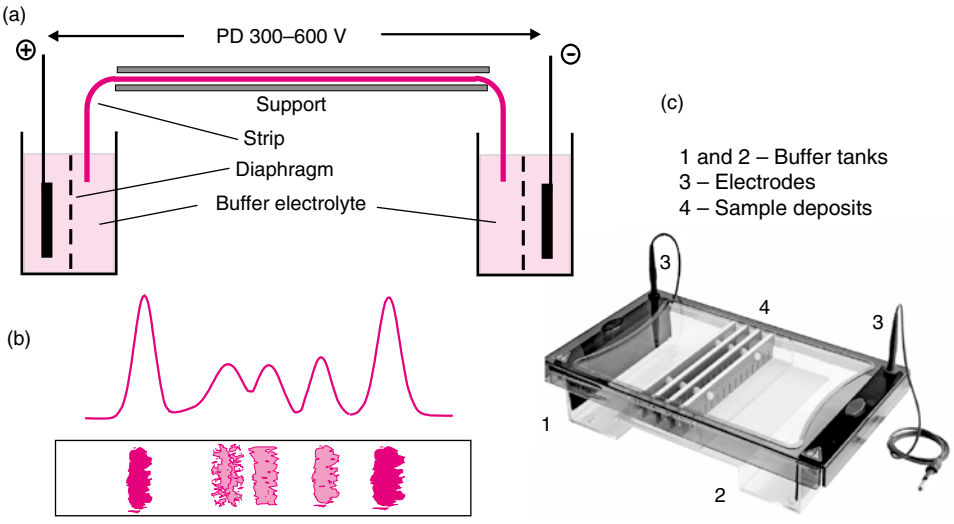


Figure 8.1 Zone electrophoresis: principle of a set-up. (a): each compartment is separated by a diaphragm in order to avoid contamination of the migration buffer by secondary products which are formed during contact with the electrodes. This technique is run either at constant current or at constant voltage or at constant power. (b): After migration, the strip can be examined by densitometry, in which the analytes are 'revealed' using an appropriate reagent. (c): Horizontal migration model.

8.1.2 Capillary Electrophoresis

In capillary electrophoresis, the flat slab of the classic technique (the strip) is replaced by a narrow-bore, open-ended, fused-silica capillary tube (15–150 μm in diameter). The capillary of length *L*, varying between 20 and 80 cm, is filled with an aqueous electrolyte buffer solution (Figure 8.2). The ends of this capillary are each submerged in a compartment, for example a flask, containing the electrolyte buffer: a source flask in which the 'inlet' of the capillary can be found and a destination flask in which the 'outlet' of the capillary is placed. An electrode connected to DC voltage supply terminals is plunged into each flask, creating an electrical field of up to 600 V/cm. However, the current should not exceed a few mA, in order to minimize the heating of the capillary by Joule effect. Since migration times depend on temperature, the capillary is installed in a temperature-controlled chamber.

A detector is placed near the cathode compartment, at a distance *l* from the head of the capillary. The signal obtained provides the baseline for the *electropherogram* (Figure 8.2), which yields information about the composition of the sample.

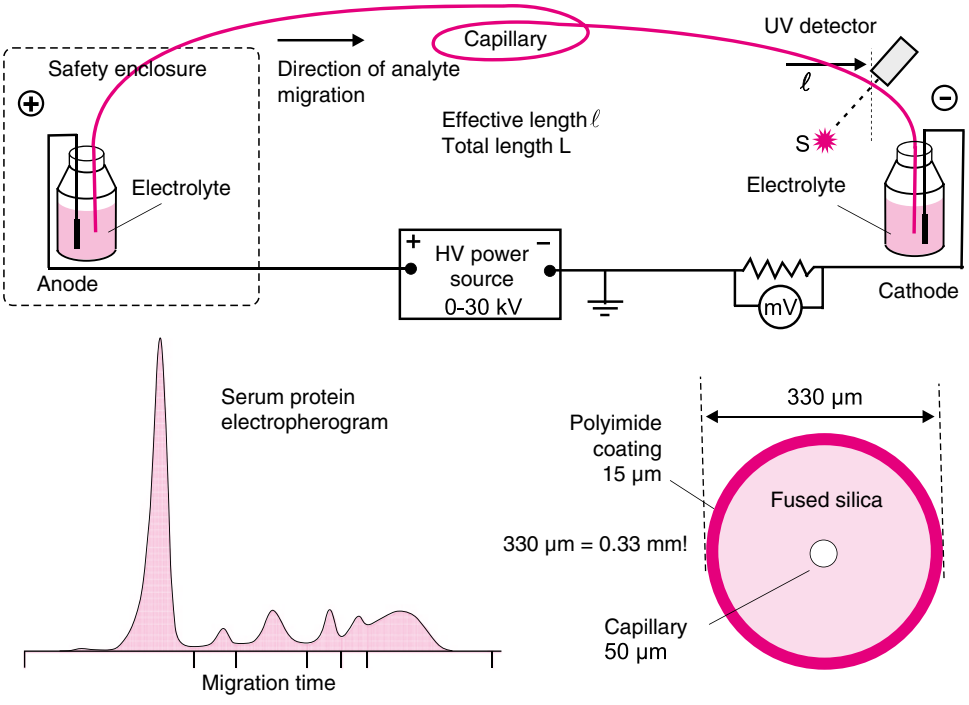


Figure 8.2 Capillary electrophoresis. The electrolyte is an aqueous ionic solution. A small volume of the sample is injected at the upstream part of the capillary (see Section 8.3). For a voltage greater than 500–600 V/cm (total potential difference = 30 kV if L = 50 cm), electrical insulation becomes necessary for safety reasons. Electropherogram routinely conducted in biology laboratories. Cross-section view of a common-size capillary. Note the narrowness of the channel through which the solutions circulate.

8.2 MIGRATION OF ANALYTES IN THE CAPILLARY

Chemical species present in the sample solution (anions, cations, neutral molecules) are subject to various interactions that will lead to their migration from the capillary inlet to the outlet. These interactions of various natures range from electrostatic forces to the movement of the entire electrolyte in the capillary and the friction due to its viscosity.

In general, the smaller and more charged the particle is, the faster the migration is. This phenomenon may be explained by two effects which combine to produce movement of the analyte: *electrophoretic mobility* and *electroosmotic flow*.

For each ion, the *limiting migration velocity* υ_{EP} results from the equilibrium between the electric force $\vec{F}$, which is exerted in the electric field $\vec{E}$ acting on the particle of charge q, and the forces resulting from the solution viscosity η.

8.2.1 Electrophoretic Mobility

Any charged particle placed in an electrical field $\vec{E}$ is subject to an electric force:

$$\vec{F} = q\vec{E} \tag{8.1}$$

For a cation, $\vec{F}$ and $\vec{E}$ have the same direction. For an anion, $\vec{F}$ and $\vec{E}$ have opposite directions. If only the electrical force $\vec{F}$ intervened, the movement velocity of the particle would continuously increase. However, owing to the friction forces due to the viscosity of the electrolyte, it will become uniform.

For a given electrolyte, the velocity reached, called the electrophoretic velocity ν_{EP}, depends on the size of the ion (ions are considered to be spheres) and on its charge. Therefore, large ions with a low charge will have a slower electrophoretic velocity than small, highly charged ions. In the case of an immobile electrolyte, Hückel proposed Eq. (8.2), taking into account the influence of these two factors:

$$\nu_{EP} = \frac{qE}{6\pi\eta r} \tag{8.2}$$

where η is the dynamic viscosity of the electrolyte and r is the radius of the ion considered to be a sphere.

Therefore, an ion will be characterized by its own electrophoretic mobility μ_{EP}. This parameter (see Section 8.2.3) is defined from the electrophoretic migration velocity of the compound and from the electrical field E:

$$\mu_{EP} = \frac{\nu_{EP}}{E} = \nu_{EP}\frac{L}{V} \tag{8.3}$$

where V is the difference in potential imposed between the two ends of the capillary of total length L, and E is the intensity of the electrical field resulting from the potential difference V.

This mobility has a + or – sign, depending on the cationic or anionic nature of the analyte. It is expressed in $cm^2/V{\cdot}s$. It is zero for an overall neutral species.

The migration of neutral or negatively charged species towards the cathode (Figure 8.3) results from the existence of another phenomenon: *electroosmotic flow*.

8.2.2 Electroosmotic Flow (EOF)

The second factor, the *electroosmotic flow,* produces general movement of the liquid vein in the capillary (particles, positive or negative ions, neutral species, and solvent). The direction of movement and its velocity vary according to the nature of the internal capillary wall.

In fact, a capillary made of untreated fused silica bears on its inner surface numerous silanol groups (Si–OH) which become ionized to silanoates ($Si–O^-$) when the pH of the electrolyte (or migration buffer) rises above three. Since they are an integral part of the wall, these fixed anionic sites attract cations present in the electrolyte and arrange them into a double layer: a fixed layer attached to the internal wall and

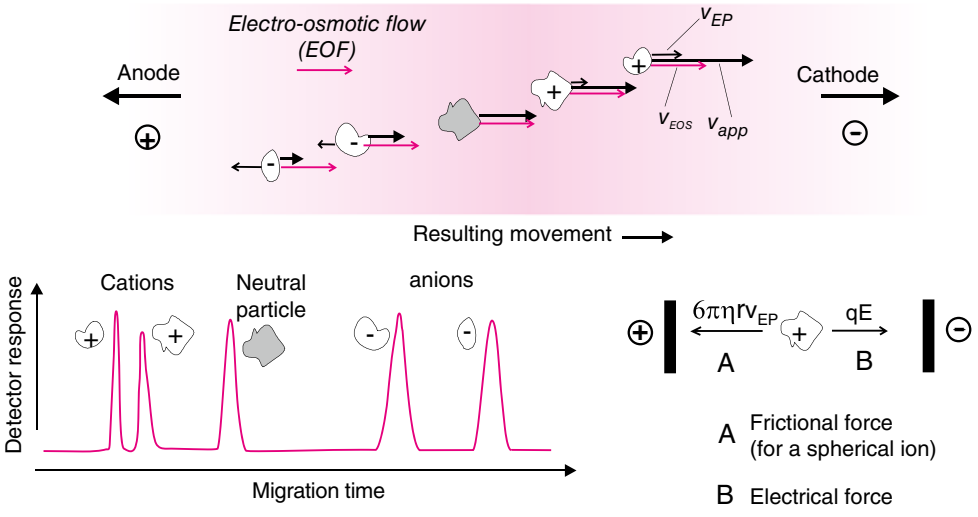


Figure 8.3 Movement of analytes in the capillary. Influence of the net charge, electric field, and hydrodynamic radius of the ion and the viscosity of the solution on the migration velocity in an electrolyte (see Section 8.2.2). The electropherogram shows that cations, neutral molecules, and anions migrate in the column in the direction of the electroosmotic flow (electrical field oriented from the inlet towards the outlet of the capillary). The separation depends approximately upon the mass/charge ratio of each species. Neutral species are not separated. The smaller anions arrive last since they would normally go towards the anode, without the electroosmotic flow.

a diffuse, mobile layer (Figure 8.4). Between these two layers a potential difference (Zeta potential) develops, the value of which depends upon the concentration and the pH of the electrolyte. When the electric field is applied, cations in the mobile phase are attracted towards the negative end of the capillary. Since these cations are solvated by water molecules, they entrain the electrolyte with them, which generates a general movement of the solution. This general movement towards the cathode is known as the electroosmotic flow. This flow is variable because it is proportional to the thickness of the double layer, which is a function of the electrolyte buffer concentration.

Usually, a capillary with a negative inner surface generates an electroosmotic flow directed towards the cathode. In contrast, if a surfactant is added, such as tetra-alkyl ammonium, capable of reversing the polarity of the inner wall, then the electroosmotic flow is directed towards the anode (Figure 8.4). For a given analyte, an *electroosmotic flow* μ_{EOS} is defined in the same way as electrophoretic mobility μ_{EP}:

$$\mu_{EOS} = \frac{v_{EOS}}{E} = v_{EOS}\frac{L}{V} \qquad (8.4)$$

In this equation, *E, L*, and *V* have the same meaning as in Eq. (8.3).

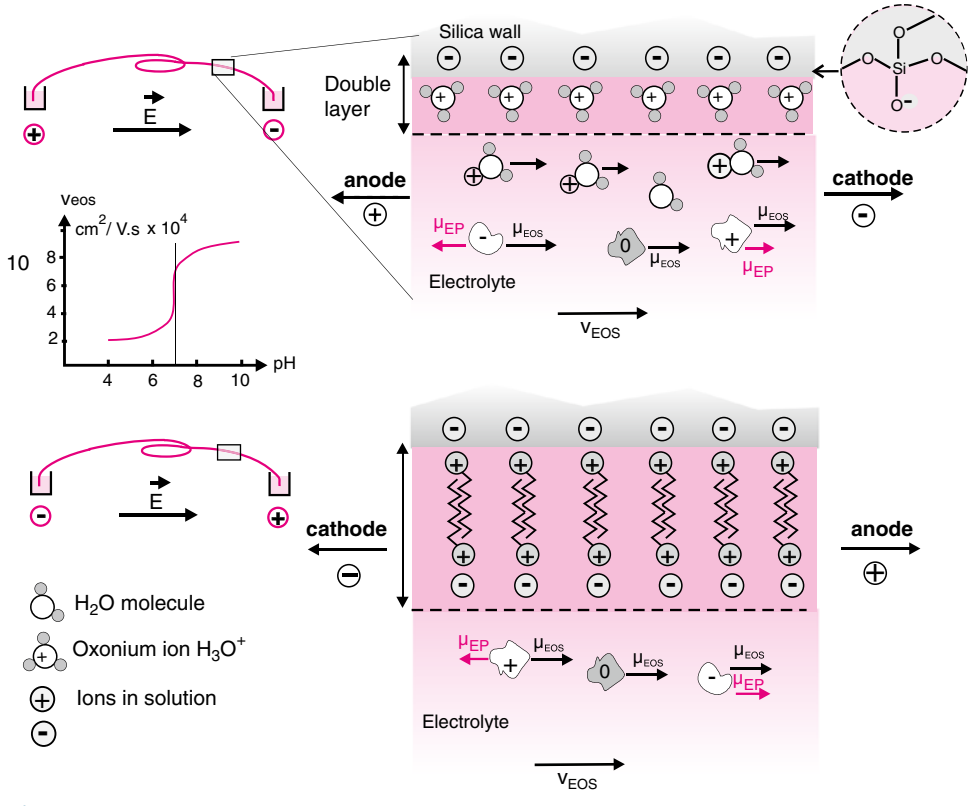


Figure 8.4 Electroosmotic flow in a capillary filled with an electrolyte. Top: If the silica wall has not been treated (polyanionic wall), then an electrolyte transfer arises from the anode to the cathode compartment: this is the electroosmotic flow. It grows with the applied potential and the pH (between pH 7 and 8, υ_{EOS} can increase by as much as 35%). It is due to the potential existing on the wall surface. If this potential is cancelled by covering the wall with a nonpolar film (e.g. hydrophilic polyhydroxyvinyl or octadecyl film), this flow disappears. Bottom: Dynamic modification of the silica wall by cationic surfactant effect. As migrations occur towards the end where the detector can be found, the poles of the apparatus must be reversed.

To calculate μ_{EOS}, the electroosmotic flow velocity v_{EOS} must first be determined. This corresponds to the velocity in the electrolyte of a species without any global charge. This is done by measuring the migration time t_{nm} for a *neutral marker* to migrate over the distance l from the capillary inlet to the detector:

$$v_{EOS} = l / t_{nm} \qquad (8.5)$$

As a neutral marker, an organic molecule that is not ionized at the pH of the electrolyte buffer used and easily detected by the detector is selected (e.g. acetone, mesityl oxide, phenol, etc.). Its detection may be negative.

By treating the wall with an alkylsilane to make it hydrophobic, proteins become separable, otherwise they tend to remain attached to the surface if it is untreated. At the very least, we can only recover one category of ions, depending on the direction of the applied field.

8.2.3 Apparent Mobility

According to the previous arguments, each ion has an apparent migration velocity ν_{app} that depends on both the electrophoretic velocity and the electroosmotic flow speed:

$$\nu_{app} = \nu_{EP} + \nu_{EOS} \tag{8.6}$$

The value of ν_{app} is easily calculated from the electropherogram. Using l, the effective length of the capillary between the injection and detection points, and t_m, the migration time, ν_{app} is given by the expression:

$$\nu_{app} = l / t_m$$

The apparent electrophoretic mobility μ_{app} is defined by an equation analogous to Eq. (8.3) or Eq. (8.4), such that:

$$\mu_{app} = \frac{\nu_{app}}{E} = \nu_{app} \frac{L}{V}$$

and, by consequence

$$\mu_{app} = \frac{l \cdot L}{t_m \cdot V} \tag{8.7}$$

By combining the electroosmotic flow of the electrolyte and the apparent mobility, it is possible to calculate the true electrophoretic mobility of charged species. From Eq. (8.6), it can be written as:

$$\mu_{EP} = \mu_{app} - \mu_{EOS} \tag{8.8}$$

or alternatively

$$\mu_{EP} = \frac{L \cdot l}{V} \left| \left(\frac{1}{t_m} - \frac{1}{t_{nm}} \right) \right| \tag{8.9}$$

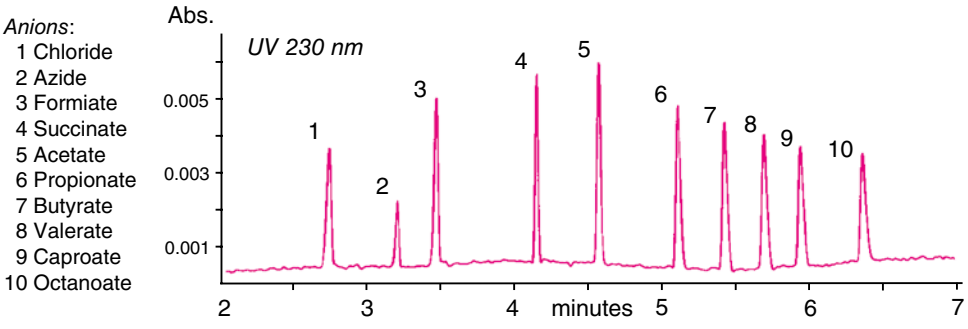


Figure 8.5 An electropherogram of an anion test mixture. Separation of a mixture of anions with the PAGE technique and detection by UV. (Source: Courtesy of Beckman-Coulter.)

HPCE is commonly used for the separation of anionic species. Generally, the polarity of the instrument is reversed in order to change the location of the detector from the cathode end to the anode end of the capillary. As a result, the electroosmotic flow direction is also reversed. Only anions whose μ_{EP} values are greater than those of μ_{EOS} will be detected (Figure 8.5).

8.3 INSTRUMENTATION

8.3.1 Different Modes of Injection

Two processes are used to introduce a microvolume of sample, which must not exceed 1% of the effective length of the capillary in order to maintain good resolution:

- **Hydrodynamic injection.** This action is achieved by dipping the end of the capillary into the solution (also an electrolyte) containing the sample, while creating a slight vacuum at the opposite end. The process can be improved by exerting a pressure of around 50 mbar upon the sample solution.
- **Electromigration injection.** This technique, used in gel electrophoresis, is achieved by putting the sample at a potential (50 mV/cm) of appropriate polarity compared to the other extremity and briefly dipping the capillary into it. In contrast to the previous procedure, this mode of injection induces a discriminatory action upon the compounds present, which leads to a nonrepresentative composition of the starting sample. This method is optimized for each mixture.

The hydrodynamic injection method is less precise than in HPLC, because injection loops do not exist for volumes of 5–50 nl. The quantity entering the capillary is dependent upon many of the parameters that appear in the well-known Poiseuille equation, which gives the flow rate in a tube for a liquid having a dynamic viscosity η. The application of this formula

results in an approximate value for what might be termed the 'input flow rate' F in the capillary.

$$F = \Delta P \frac{\pi r^4}{8\eta L}$$

This relationship reveals that if the radius r of a capillary is doubled, then the volume entering will be 16 times greater. This volume is also proportional to the difference in pressure ΔP. An increase in the length of the capillary L produces the reverse effect. An internal standard should be used to improve the quantitative performance of HPCE. Either a dilution solution containing an internal standard will be used to prepare all solutions, or a precise volume of a standard solution will be added to these same solutions.

8.3.2 Detection Methods

- **Direct UV/Vis detection** (see Chapter 9). The light intensity going through the capillary is measured, for which a small opaque coating area has been eliminated. The capillary cuts through the optical path going from the source to the photomultiplier (Figure 8.6). This avoids any dead volume. The very low optical path in the capillary makes this detection method not very sensitive if the solution does not absorb much, especially if the matrices are also absorbent.

  Reverse mode UV/Vis detection

  Inorganic ions, which absorb very little light, may nevertheless be detected by

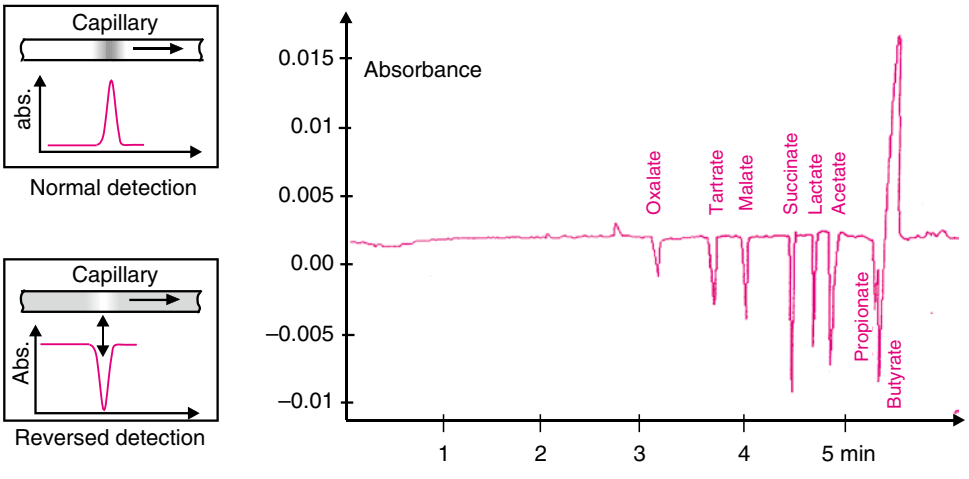


Figure 8.6 Separation of the principal organic acids in white wine by indirect UV detection. Malic acid and lactic acid are indicators for malolactic fermentation. (Source: Image courtesy of TSP.)

absorbent by adding a chromate or a phthalate. Under these conditions, when a sample ion arrives at the detector, the absorbance decreases, which leads to 'negative' peaks (Figure 8.6). This absorbance variation is linearly related to the concentration of the detected ion.

- **Fluorescence detection.** The sensitivity of this process is enhanced if a very intense laser source is employed. To extend the field of application of this form of detection, a prederivitization or postderivitization of analytes can be conducted to make them carry a fluorophore.
- **Electrochemical detection.** This detection can be conducted by inserting minuscule electrodes into the capillary.
- **Mass spectrometry detection.** A very low flow rate of the capillary suits the ionization methods at atmospheric pressure (see Chapter 16). Mostly used in pharmaceutical analysis, this coupling has gained in importance for the determination of protein and sugar structures. These biomolecules remain the major field of application of capillary electrophoresis.

  In-line coupling with a mass spectrometer imposes specific constraints: very low flow rate and presence of mineral salts in the migration buffer. Moreover, the end of the capillary must be deviated towards the detector, while ensuring the continuity of the electrical circuit (Figure 8.7). As a function of the ionization method (ESI, APCI, see Chapter 16, Section 16.4.2), an organic solvent is added to the circuit in order to increase and regularize the very low flow rate at the capillary outlet. Spray ionization is the most commonly used method.

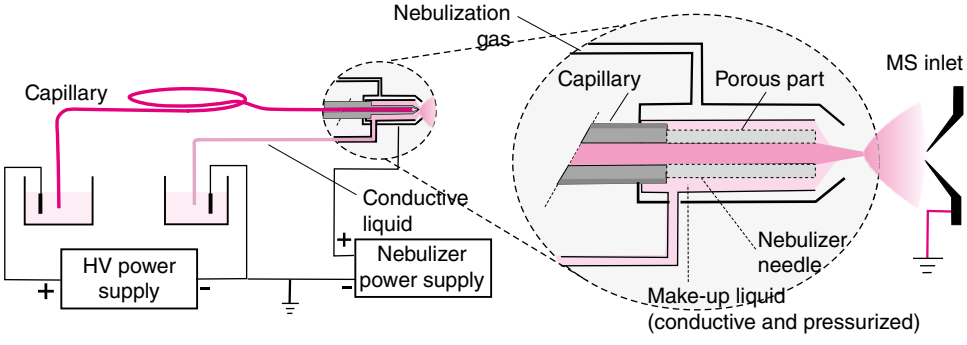


Figure 8.7 A model of the interface between HPCE and MS. The solutions differ depending on the ionization mode selected and on whether there is additional electrolyte or not. The capillary end is eroded to make it permeable to electrolyte ions in order to ensure the closing of the HPCE circuit. The metal point of the nebulizer must however be brought to a positive potential with regards to the inlet of the mass spectrometer.

8.4 ELECTROPHORETIC TECHNIQUES

8.4.1 Capillary Zone Electrophoresis (CZE)

This type of electrophoresis, in which the electrolyte migrates through the capillary, is the most widely used. Depending on the application, this electrolyte can be acidic (phosphate or citrate) or basic (borate) or an amphoteric substance (a molecule possessing both an acidic and a basic function). The electroosmotic flow increases with the pH of the liquid phase or can be rendered inexistent. This process is also called free-solution electrophoresis, in contrast to CGE (see Section 8.4.3).

8.4.2 Micellar Electrokinetic Capillary Chromatography (MEKC)

In this variant of CZE, a cationic or anionic surfactant, e.g. sodium dodecyl sulfate, is added in sufficient quantity to the migration buffer to form charged micelles in the aqueous phase. These charged microdroplets (Figure 8.8) form a pseudostationary phase that traps neutral compounds more or less efficiently. The separation of the analytes is done by means of a partition mechanism, just as in chromatography, via hydrophilic/hydrophobic affinity interactions. This type of electrophoresis is used for compounds that have a tendency to migrate without separating, such as hydrophobic molecules.

If the surfactant added to the aqueous phase is optically active, such as persulfated cyclodextrin, enantiomers will form complex inclusions of varying stability. This makes it possible to know their optical purity (see Chapter 2, Section 2.6.3 and Figure 8.10).

8.4.3 Capillary Gel Electrophoresis (CGE)

This technique represents the transposition of classic polyacrylamide gel electrophoresis (PAGE) or agarose gel electrophoresis into a capillary. The capillary is filled with an electrolyte containing a gel. This produces a filtration effect, which decreases the electroosmotic flow and minimizes convection and diffusion phenomena. Rather stable oligonucleotides can be separated in this way. The electroosmotic flow remains low.

> In plate electrophoresis, the support on which the migration takes place may contain a gel (starch or, better yet, polyacrylamide) impregnated with the electrolyte. If the latter contains sodium dodecyl sulfate (SDS), then the gel electrophoresis is named *SDS–PAGE*. Separations are improved by a filtering phenomenon that is added to the electric forces. These techniques are mainly used for the three large biomolecule categories: polypeptides (proteins), oligonucleotides (DNA or RNA fragments), and mono- and polysaccharides (carbohydrates).

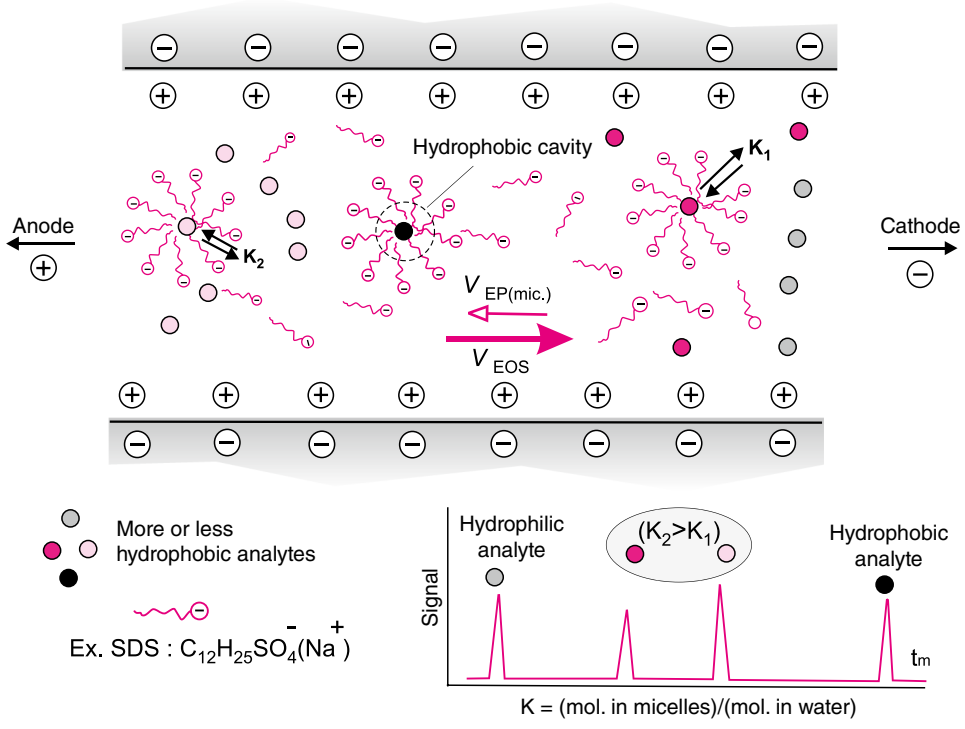


Figure 8.8 Separation of neutral analytes by using a surfactant (MEKC technique). In this illustration of an untreated capillary in which a polyanionic surfactant has been added to the migration buffer (e.g. sodium dodecyl sulfate), the various analytes are retained more or less efficiently by a succession of reversible equilibria (here $K_2 > K_1$). Polyanionic micelles, very slowly entrained towards the cathode, form a quasistationary phase. The inside of each micelle is hydrophobic. The theoretical electropherogram shows the effect of interactions between the analytes and the surfactant.

8.4.4 Capillary Isoelectric Focusing (CIEF)

This technique consists in creating a linear pH gradient in a surface-treated capillary containing an ampholyte. On the anode side, the capillary is immersed in a phosphoric acid (H_3PO_4) solution while on the cathode side, it is dipped into a sodium hydroxide (NaOH) solution. Each of the charged analytes migrates through the medium until it reaches the region where the pH has the same value as its isoelectric point (at its p*I*, the net charge is zero). Then, by maintaining the electric field and using a hydrostatic pressure, these separated species migrate towards the detector. The high resolutions obtained with this process enable the separation of peptides whose p*I*s may differ by only 0.02 pH units.

8.5 PERFORMANCE OF CE

The performance of CE for the separation of biomolecules, such as proteins, with varying sizes and low diffusion coefficients, is comparable to that of HPLC. The hydrodynamic flow profile is flat, almost perfect and not parabolic, as it is in HPLC (Figure 8.9). However, the heating of the capillary owing to the electrical field denatures the ideal profile. It is therefore important for the reproducibility of analyses to have a temperature-controlled system.

The sensitivity is very high: if we use laser-induced fluorescence (LIF) detection, a few thousand molecules can be observed after derivatization. The success of a separation depends on the choice of an appropriate buffer medium adapted to the analysis in question. Only a very small quantity of sample is required, and the consumption of reagents or solvents is negligible.

However, the reproducibility of the analyses is less certain than in HPLC because there are many subtle factors that can affect the repeatability of the injected volumes. With the hydrostatic introduction method, the relative standard deviation (CV) is on the order of 2%, if an internal standard is used.

8.5.1 Separation of Enantiomers

Two methods can be considered for the separation of enantiomers in HPCE: precolumn derivatization and the addition of a chiral selector to the migration buffer, such as a modified cyclodextrin. The latter is more commonly used (Figure 8.10). The necessary quantities of the selector are low, in light of the tiny flow rates in the capillary (a few μl only), when compared with HPLC, where consumption is a drawback.

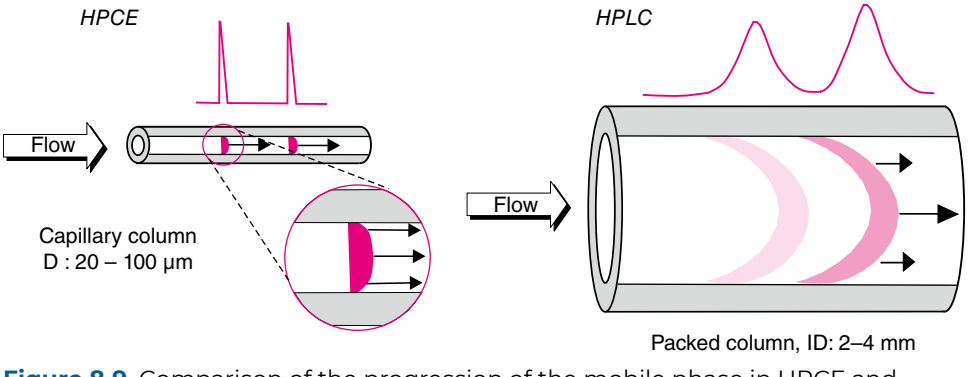
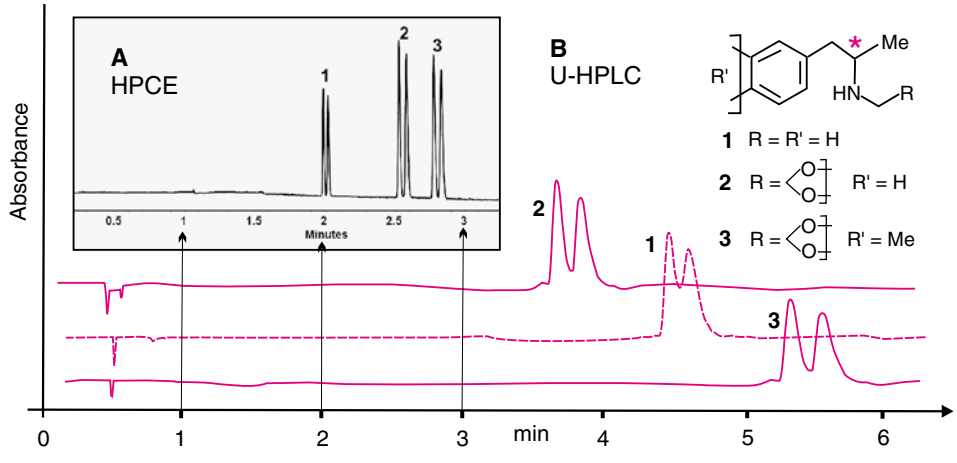


Figure 8.9 Comparison of the progression of the mobile phase in HPCE and in HPLC. In HPCE, the electrolyte and the analytes are pulled by the capillary wall, whereas they are pushed in HPLC. Diffusion is lower than in HPLC and thus *efficiency* (N) is greater. The diameter of the HPLC column must be at least 20 times greater than that of the capillary in electrophoresis.



A *HPCE*
Capillary 32 cm x 50 µm diam. 25 cm eff. length
Buffer electrolyte: Tris-phosphate 118 mM
pH 3.5 + 2-hydroxypropyl-β-cyclodextrin
17 mM. PD 25 kV; UV detection 195 nm.

B U-HPLC
ACQUITY Phenyl BEH column
100 x 2.1 mm diam. MP buffer sol. NH4 pH 10
+ MeCN (90:10 v/v) + 2-hydroxypropyl-β-cyclodextrin
20 mM. Inj. 2 µl of racemic amphetamines
Temp. 25°C. 500µl/min. UV det. 210 nm.

Figure 8.10 Separation of enantiomers. Comparative study between HPCE and U-HPLC in the separation of three racemic amphetamines (concentration: 25 µg/ml) with the same chiral selector added to the separation buffer for HPCE and to the mobile phase for U-HPLC. Electrophoresis enables a quick analysis, rivalling the most evolved forms of liquid chromatography. (Source; Based on a study by G. Bonvin, thesis at Geneva University, 2014.)

8.5.2 Separation Parameters

The separation parameters (efficiency, retention factor, selectivity, resolution) can be determined from electropherograms. The linear dispersion relationship and the concept of theoretical plates in chromatography (see Chapter 1, Section 1.6) are transposable via Einstein's law ($\sigma^2 = 2D \cdot t_m$). D is the diffusion coefficient expressed in m^2/s.

By transposing the equations used in chromatography to electrophoresis for the efficiency N and for dispersion, i.e. $N = L/H$ and $\sigma^2 = HL$ (see Chapter 1), we can use the effective length l to give the following:

$$N = \frac{l^2}{\sigma^2} = \frac{l^2}{2Dt_m} \tag{8.10}$$

By eliminating $t_m \left(v_{app} = \frac{l}{t_m} \right)$ and by using Eq. (8.7), finally we get:

$$N = \frac{\mu_{app}}{2D} \cdot V \cdot \frac{l}{L} \tag{8.11}$$

Or approximately:

$$N = \frac{\mu_{app}}{2D} \cdot V \tag{8.12}$$

This is a classic equation, based on the diffusion and giving efficiency *N*, which may reach 10^6 plates/m of capillary.

The efficiency is proportional to *V*, the applied potential difference. It is higher for macromolecules, because diffusion coefficients are smaller, than for small molecules (Figure 8.9).

The Van Deemter model, concerning the kinetic aspect of chromatographic separations (see Chapter 1, Section 1.10), is sometimes used in capillary electrophoresis. Among the three coefficients A, B, and C from the simplified form of Eq. (1.37), only term B (diffusion of solutes in the mobile phase) should be considered here.

Finally, in HPCE, as in chromatography, separations depend on several parameters, for which the selectivity (based on Eq. (8.9)) can be written as:

$$\alpha = \frac{\mu_{EP(2)}}{\mu_{EP(1)}} \tag{8.13}$$

The resolution R between two peaks may be calculated from the efficiency *N*, the difference in migration velocity between the two solutes and their average velocity (Eq. (8.14)) or their mobilities (Eq. (8.15)). Like *N*, resolution *R* depends on *V*.

$$R = \frac{\sqrt{N}}{4} \cdot \frac{\Delta v}{\bar{v}} \tag{8.14}$$

$$R = \frac{1}{4}\sqrt{N}\,\frac{\Delta\mu_{EP}}{\mu_{EPavg} + \mu_{EOS}} \tag{8.15}$$

8.6 CAPILLARY ELECTROCHROMATOGRAPHY

Capillary electrochromatography (CEC) is a miniaturized, hybrid separation method that is very efficient with biomolecules, by combining the electromigration of ions, typical of electrophoresis, with their partitioning between the phases, typical of chromatography.

The separation factor is very high, but a certain number of problems remain, including the effect of organic modifiers on the electroosmotic flow and the precise control of the sample volume introduced in the capillary.

The method requires a capillary filled with a stationary phase composed of particles that are very small in diameter (1–3 μm). These can be silica-based particles, polar polymers or various continuous porous structures (monolithic columns). The stationary phase plays two roles. Firstly, it acts as a material that more or less retains the compounds (e.g. RP-18) and, secondly, it helps the electrolyte migrate (Figure 8.11).

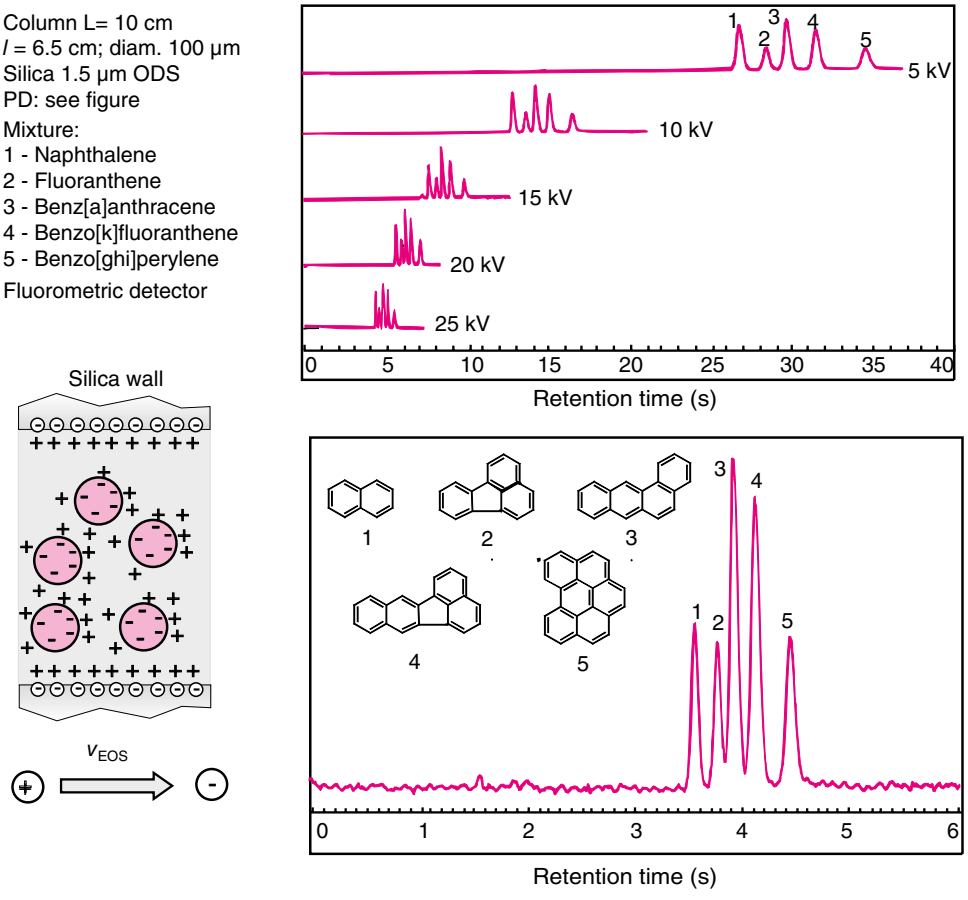


Figure 8.11 Separation of aromatic hydrocarbons by electrochromatography and representation of a packed capillary segment. The different attempts to separate a mixture of neutral compounds by varying the voltage enable us to verify, as a first approximation, the relationship between voltage and the velocity of electroosmotic flow. In this example, the use of a very short column and a high voltage enables separation of a mixture of nonionized compounds in a few seconds (for V = 28 kV), with good resolution. A high efficiency is attained with this procedure (Anex, D.S. *et al. Anal. Chem.* 1998, **70**, 4787–4792). The capillary packing materials include continuous polymer structures containing polymers of methacrylates or acrylamides.

Therefore, enough polar groups (e.g. silanols) must subsist on the external surface of the particles to generate an electroosmotic flow, in order for the packing to behave in a fashion analogous to an untreated capillary wall.

In electrochromatography, there is no pressure drop: each H_3O^+ ion migrates under the effect of an electric field, making a pump unnecessary. In liquid chromatography, a pressure close to 1,000 bar is needed to push the mobile phase through such a column.

A change in composition of the mobile phase will affect both the electroosmotic flow and the selectivity.

KEY POINTS OF THE CHAPTER

1. In capillary electrophoresis, the differential migration of analytes that occurs on a flat surface in conventional electrophoresis is reproduced inside a capillary, with the analytes being entrained in an electrolyte under the effect of an electrical field.
2. There are two specific processes for introducing the sample into the capillary: hydrostatic injection and electromigration. The latter may lead to a discrimination of the species introduced.
3. Unlike the parabolic progression of hydrodynamic flow in a tube (see HPCE), the traction force exerted by the capillary wall, which is the source of flow in HPCE, leads to a narrow flow profile, therefore with a lesser widening of the peak than in other chromatographic techniques. Consequently, there is a gain in *efficiency (N)*.
4. Most commercial devices include a detector based on UV absorption. The absorbance value is obtained 'in line' by a direct measurement through the capillary, with no measurement cell.
5. The electrophoretic mobility (μ_{ep}) and therefore the electrophoretic velocity (v_{ep}) depend on the pH and on the composition of the buffer (ionic strength, viscosity, etc.). For an ion considered to be spherical, Hückel suggested that the frictional force responds to a coefficient given by this equation: $6\pi\eta r$
6. Silanol functions covering the internal face of a nonmodified silica capillary are acidic (pKa = 4-5). Then, the electroosmotic flow (EOF) will be higher when the pH value is higher (basic pH).
7. The chemical modification of the internal surface of the capillary, for example, with a cationic surfactant, may cancel or reverse the EOF direction.
8. Several factors affect separations, including the *efficiency* (*N*) parameter, the main ones between temperature, the injection volume, and conductivity differences between the analytes and the support electrolyte.
9. The main separation parameters, as defined in chromatography, apply to electrophoresis.

PROBLEMS

1. A capillary zone electrophoresis (CZE) set-up involves the use of a capillary with total length L = 32 cm and effective length of separation l = 24.5 cm. The applied voltage is 30 kV. Under the conditions of the experiment, the migration time t_{nm} of a neutral marker appears on the electropherogram at three minutes.
   a. Calculate the electrophoretic mobility, μ_{EP}, of a compound whose migration time is 2.5 min. Specify the units for your response.
   b. Calculate the diffusion coefficient under these conditions for this compound, assuming that the calculated plate number is N = 80,000.
2. The set-up for an experiment uses an untreated, 'negative wall' fused silica capillary. The total capillary length L is 1 m, while the effective capillary length l is 90 cm (to the detector). The applied voltage is 20 kV. The detector is located at the cathode end of the capillary, and the electrolyte is a buffer of pH 5. A compound present in the sample has a migration time of t_m = 10 min.
   a. From the information given, can it be deduced whether the net charge carried by the compound is positive or negative?
   b. Calculate the apparent electrophoretic mobility μ_{app} of the compound.
   c. If a small uncharged molecule has a migration time t_{nm} = 5 min, deduce the value of the electroosmotic flow μ_{EOS}.
   d. Calculate the electrophoretic mobility μ_{EP} of the compound. What is the sign of the net charge of this compound?
   e. What would happen if the capillary wall was treated to make it neutral?
   f. If we suppose that the pI of the compound is 4, what would be the sign of its net charge if the pH of the electrolyte was lowered to 3?
   g. Calculate the number of theoretical plates N if the diffusion coefficient of the solute is $D = 2 \times 10^{-5}$ cm^2/s.
   h. From the relationship between the efficiency N and the diffusion D, explain why small molecules have poorer separations in capillary electrophoresis than larger molecules, and why separations are much better when the capillary is narrower.
3. During a capillary electrophoresis experiment, it is observed that if the capillary contains an acrylamide gel as well as an electrolyte, then the speed of migration is slowed by the mechanical effects of filtration through the gel. This phenomenon is particularly significant for larger molecules. The following relationship can be considered as proven:

$$\log M = \mathrm{a} \cdot v + \mathrm{b}$$

where a and b are both constants and M represents the molecular mass (Da) of a molecule migrating with a velocity v.

In an experiment designed to employ this relationship in order to calculate the molecular mass of an unknown protein, two known standards are used: ovalbumin

(M = 45,000 Da) and myoglobin (M = 17,200 Da), whose migration velocities are 1.5 cm/min and 5.5 cm/min respectively. In the same experiment, the unknown protein migrates at a velocity of 3.25 cm/min. Calculate its molecular mass in Daltons.

4. A capillary has a length L = 57 cm. The release of heat cannot exceed 0.04 W/cm, and we know that the intensity is limited to 100 μA.

a. Calculate the maximum admissible difference in potential.

To separate a mixture of three anions, Cl^-, SO_4^{2-} and NO_3^-, the electrolyte is a solution containing 5 mmol/l of $K_2Cr_2O_7^{2-}$ and 0.5 mmol/l of hexamethonium bromide. The pH is adjusted to 8 with a sodium hydroxide solution. The detection method employs UV absorbance at 254 nm.

b. Explain these choices.

5. The hydrodynamic injection method is used, obeying the following equation:

$$V = \frac{P \cdot d^4 \pi \cdot t}{128 \cdot \eta \cdot L}$$

V is the injected volume (m^3), P is the positive pressure (in Pa) applied outside the capillary, d is the internal diameter of the capillary (in m), t is the time during which the pressure is applied, η is the dynamic viscosity of the buffer (kg/m/s) and L is the total length (m) of the capillary.

a. Calculate the length l (in mm) of a capillary of a diameter d of 50 μm that can accept a volume V of 1 nl of a sample.

b. What positive pressure P (in Pa) must be exerted on a sample solution for t = 2 s to make a volume V of 1 nl enter into the capillary? Assume η = 0.001 kg · m^{-1} · s^{-1} and L = 0.75 m.

SOLUTIONS

1. **a.** $\mu_{EP} = 2.88 \times 10^{-5}$ $cm^2 \cdot s^{-1} \cdot V^{-1}$.

b. $D = l^2/(2N \cdot t_M) = 2.5 \times 10^{-5}$ cm^2/s.

2. **a.** No, because this is an untreated wall. An electroosmotic flow is therefore created. The compound migrates towards the cathode, even if it carries a negative charge.

b. $\mu_{app} = 7.5 \times 10^{-4}$ $cm^2 \cdot s^{-1} \cdot V^{-1}$.

c. $\mu_{EOS} = 1.5 \times 10^{-3}$ $cm^2 \cdot s^{-1} \cdot V^{-1}$.

d. $\mu_{EP} = -7.5 \times 10^{-4}$ $cm^2 \cdot s^{-1} \cdot V^{-1}$. The (negative) sign of μ_{EP} symbolizes there is a species carrying a negative net charge. The compound does not move as fast as a neutral marker.

e. If the capillary wall is made neutral, there will no longer be any electroosmotic flow and consequently, the compound will no longer migrate towards the cathode.

f. If the pI is 4, for all pH values under 4, the compound will be in cation form. In this case, the migration time will normally be shorter than for a neutral marker.

g. N = 337,500 plates.

h. A small molecule diffuses faster than a large molecule. Consequently, efficiency is better for heavier molecules.

3. M = 29,854 Da. The stationary phase does not behave quite like an SEC gel. It creates an obstacle to large molecules migrating slower than the smaller ones.

4. a. By applying the equation $P = E \times I$, the maximum admissible voltage will be: E = (57 x 0.04)/100 x 10^{-6} = 22,800 volts (i.e. 400 V/cm).

b. In the normal configuration, the cathode (–) is located on the detector side. While it is possible to make anions (–) migrate towards the cathode, by electroosmotic flow, the supply poles are reversed in order for the detector to be on the anode side (+) and, therefore, the migration direction of the electroosmotic flow must be conserved. Nevertheless, the polarity of the capillary surface must also be reversed, hence the addition of hexamethonium bromide, $[(CH_3)_3N^+(CH_2)_6N^+(CH_3)_3]$, $2Br^-$, to the electrolyte. One of the two ammonium groups in hexamethonium bromide is fixed on the silicate ions (pH 8), while the other remains free.

To track the passage of ions that are transparent in the UV domain with a UV detector, the inverse detection principle is used by adding potassium dichromate, which absorbs at λ_{max} = 254 nm, to the electrolyte. When a non-absorbent analyte reaches the detector, the dichromate concentration decreases, as does the absorbance of the solution, leading to a negative peak.

5. a. The inside of the capillary is a cylinder, $= \frac{V}{\pi \cdot r^2}$. Thus, $l = \frac{1 \times 10^{-12}}{314 \times 25^2 \times 10^{-12}} = 5 \times 10^{-4}$ m or 0.5 mm.

b. $P = \frac{V \cdot 128 \cdot \eta \cdot L}{d^4 \pi \cdot t}$ from the data: $P = \frac{1 \times 10^{-12} \times 128 \times 0.001 \times 0.75}{50^4 \times 10^{-24} \times 3.14 \times 2} = 2{,}450$ Pa

Chapter 9

Ultraviolet and Visible Absorption Spectroscopy

The interactions of electromagnetic radiation with matter in the range from the near ultraviolet to the very near infrared, between 180 and 1,100 nm, have been studied extensively via electron spectroscopy. This portion of the electromagnetic spectrum, which includes radiation perceptible to the human eye (400–700 nm), generally yields little structural information but is very useful for quantitative measurements. The concentration

of an analyte in solution can be determined by measuring the absorbance at some wavelength and applying the Beer–Lambert Law. The method is known as colorimetry, considered as the workhorse of many laboratories. Colorimetry applies not only to compounds possessing an absorption spectrum in the visible range but equally to all those compounds which, following modification by specific reagents, lead to derivatives that permit absorbance measurements. In this field, there is a wide selection of instruments, from the simplest colorimeters to high-resolution spectrophotometers.

Objectives

Define the spectral domain in question
Explain the origin of absorptions
Connect transitions and structures
Identify a molecule from its UV-Visible spectrum
Present the different device designs
Compare device performance levels
Review the Beer–Lambert Law in quantitative analysis
Demonstrate the interest of derivative spectroscopy
Provide an overview of colorimetric assays

9.1 SPECTRAL REGION FROM UV-VISIBLE TO VERY NEAR IR

This region of the spectrum is conventionally divided into three subdomains termed *near UV* (185–400 nm), *visible* (400–700 nm), and *very near infrared* (700–1,100 nm). This spectral domain has given rise to a great number of analytical measurements using spectrophotometers, which in most cases cover the entire range. Some also include the near infrared, which goes up to 2,500 nm. The lower limit of the instrument depends upon the nature of the optical components used (pushed down to 120 nm) and on the purging of ambient air along the optical pathway, since oxygen and water vapour absorb intensely below 190 nm. This is the domain known as *vacuum* or *far ultraviolet*, which has applications in fundamental research but not in routine analysis.

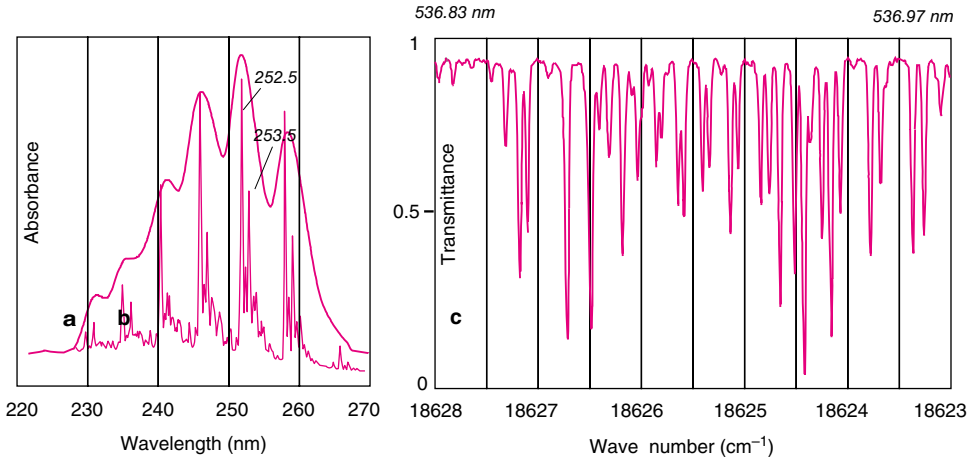


Figure 9.1 Three different types of UV/Vis spectra. Spectra of benzene (a) in solution (band spectrum); (b) in the vapour state (spectrum presenting a fine structure); (c) an expansion of a section from the high-resolution (0.14 nm total interval) spectrum of iodine vapour. This spectrum extends over 5 cm^{-1}, i.e. from 536.83 to 536.97 nm.

Spectrophotometers are designed to generate the *spectrum* (Figure 9.1) of the compound under analysis as a graph representing *transmittance* (or *absorbance*) as a function of wavelength along the *y*-axis, given in nanometres or sometimes as wave number in cm^{-1} (symbol $\bar{\nu}$ where $\bar{\nu} = 1/\lambda$, λ is expressed in cm).

In spectroscopy, the *transmittance* T is a measure of the attenuation of a beam of monochromatic light, based upon the comparison between the intensities of the transmitted light (I) and the incident light (I_0), depending on whether the sample is placed, or not, in the optical pathway between the source and the detector. T is expressed as a fraction or a percentage:

$$T = \frac{I}{I_0} \tag{9.1}$$

or

$$\%T = \frac{I}{I_0} \times 100$$

The *absorbance* is defined as:

$$A = -\log T \tag{9.2}$$

The recorded spectra of compounds in the condensed phase, whether pure or in solution, generally present few absorption bands, while those spectra obtained at low pressure from samples in the gaseous state and from compounds with very simple atomic composition reveal a 'fine structure' (Figure 9.1). If the spectrometer possesses a high enough resolution, the fundamental transitions (see Section 9.2) appear as if isolated. In extreme situations such as these, the positions of the absorptions are recorded in cm^{-1}, a more suitable unit than nm.

All compounds, whether in vapour form or in condensed liquid or solid phase, are made up of not one, but a considerable number of similar molecules. Following nearby interactions, these molecules are not all in the same energy state. Moreover, no matter how high its resolution, no spectrophotometer can generate a spectrum showing all the various individual transitions. The graph of the spectrum corresponds to an envelope of individual absorptions. The UV spectra are absorption-band spectra, with only a few, broad bands, except for simple gaseous molecules, as demonstrated by the example of iodine vapour.

9.2 THE ORIGIN OF ABSORPTIONS

Light absorption in the UV-Visible domain originates from the interaction between photons resulting from the source and electrons responsible for atomic bonds in the chemical species present in the sample, mostly made up of molecules.

Quantum mechanics teaches us that all molecules, taken individually, have a quantified internal energy, which corresponds to the sum of their electronic energy ($E_{elec.}$), their vibrational energy ($E_{vib.}$), and their rotational energy ($E_{rot.}$). The hierarchical arrangement is such that several vibrational levels *V* correspond to each electronic level *E*. These vibrational levels also have a very large number of rotational levels *J*.

In summary, there is a considerable, yet finite, number of possible energy levels subject to selection rules (Figure 9.2).

Therefore, when a photon, whose energy precisely corresponds to a permitted transition, is absorbed by a molecule, its internal energy increases.

$$\Delta E_{tot} = \Delta E_{rot} + \Delta E_{vib} + \Delta E_{elec} \tag{9.3}$$

(where $\Delta E_{elec} > \Delta E_{vib} > \Delta E_{rot}$.)

The quantum theory of molecular orbitals gives a more precise explanation of the absorption mechanism. In this model, *valence electrons* ensuring bonding at the periphery of the atoms are divided in pairs in bonding (called σ and π) and nonbonding (n) molecular orbitals (Figure 9.2). Each has its own energy level, which makes it possible to classify them.

According to this theory, bonding electrons, before absorption, are distributed in molecular orbitals classified by growing energy up to a frontier orbital called HOMO (*Highest Occupied Molecular Orbital*). Above that, there are other, higher-energy levels,

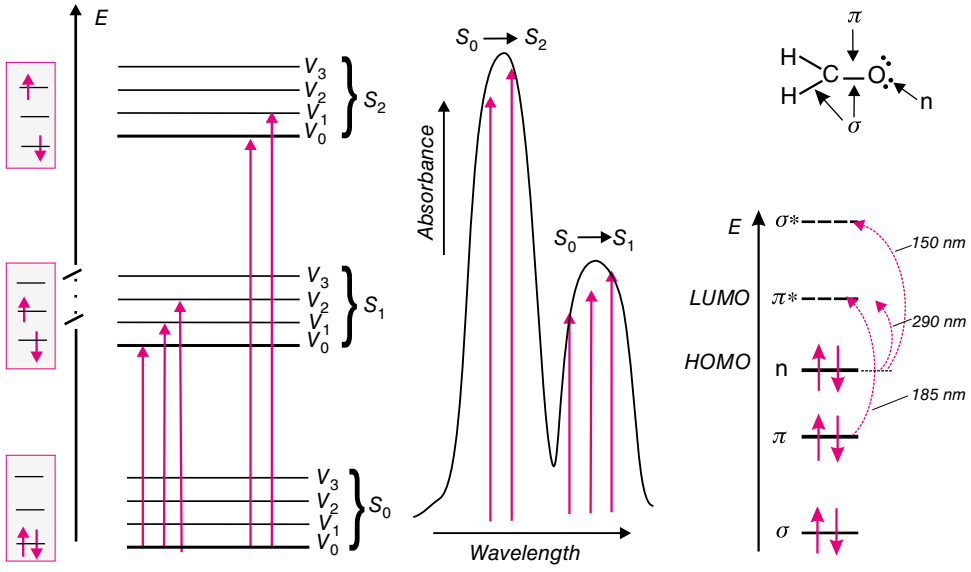


Figure 9.2 Energy diagram of a molecule and electronic transitions. Left: three electronic levels *S* with several related vibrational levels (the letter S is used to designate electronic states, rather than *E*, to remind us that these are singlet states). Each horizontal line corresponds to an energy level of the molecule. At rest, only the ground state S_0 is occupied, and the vibrational state V_0 is by far the most populated (in compliance with the Boltzmann distribution). Electronic transitions require about 20 times more energy than vibrational transitions. Passage from S_0 to S_2 corresponds to a harmonic of the transition from S_0 to S_1. Centre: an image of a UV spectrum, which demonstrates the presence of two types of transitions, to S_1 and to S_2. Right, a representation in the form of an energy diagram of the various molecular orbitals in a methanal (formaldehyde) molecule at rest. Each has two electrons with paired spins. Average values in nm of absorption bands of this molecule.

which are normally vacant, corresponding to excited states. The first one corresponds to the orbital called LUMO (*Lowest Unoccupied Molecular Orbital*). From there, we can see that electromagnetic radiation, in which the photons correspond to energy differences between these frontier orbitals, may be absorbed. In this case, the absorption corresponds to the jump of an electron from an occupied orbital to an initially vacant orbital (Figure 9.2).

Lightweight atoms, found in organic compounds, are the source of this type of transition, called *singlet–singlet*. The main ones are addressed below.

Electronic transitions are produced 1,000 times quicker than the vibrational movements of bonds, so much so that the vibrational energy remains the same after absorption (Franck–Condon principle). After the absorption of photons, a variety of things can happen to the

excited states. Various mechanisms coexist, which ensures the return to the fundamental state by liberating the energy received. In the absence of fluorescence or phosphorescence, which are other luminescence phenomena (see Chapter 11), all of the energy absorbed is returned to the system as heat via vibrational relaxation (10^{-12} s) or internal conversion relaxation (10^{-8} s). This very quick de-excitation (absorption and return to the original state in less than 10^{-6} s) explains why, at our scale, the absorption intensity seems stationary, without any saturation from the start.

9.3 ELECTRONIC TRANSITIONS OF ORGANIC COMPOUNDS

The atoms most frequently found in organic compounds, H, C, N, O, are the source of σ and π bonds, which with nonbonding doublets *n* lead to electronic transitions that are reflected in absorptions in the near UV range (Figures 9.2 and 9.3). Each is identified by its median position (in nm) and by its *molar absorption coefficient* ε ($L \cdot mol^{-1} \cdot cm^{-1}$) calculated for this position and with the solvent used (see Section 9.9).

- **The $\sigma \rightarrow \sigma^*$ transition** requires the most energy. This transition appears in the far UV, outside the traditional limits of devices (e.g. *hexane*: λ_{max} = 135 nm, ε = 10,000, in the vapour state). This is the reason why saturated hydrocarbons, such as cyclohexane and heptane, which contain only this type of bond, are used as solvents in the near UV. For heptane, at 200 nm, A = 1 when l = 1 cm). Unfortunately, the solvating power of these solvents is insufficient to dissolve many polar compounds.
- **The $\pi \rightarrow \pi^*$ transition**, characteristic of compounds that have a double bond, has a strong absorption band around 170 nm, in the case of an isolated double bond

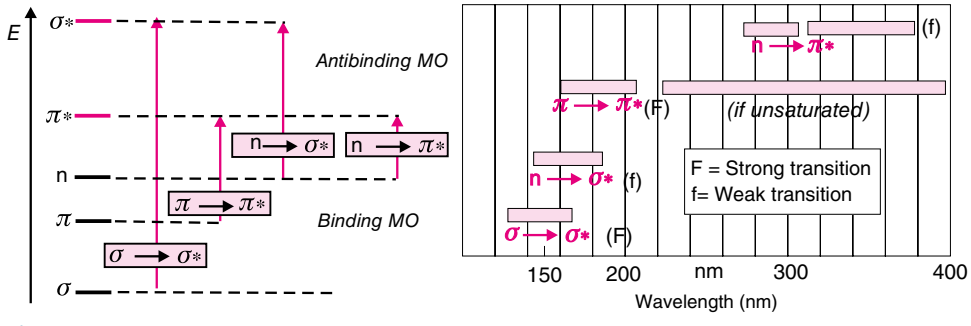


Figure 9.3 Transitions encountered most frequently in organic compounds. Among the four types of transitions shown on this energy diagram, the two most intense ones correspond to transitions permitted by the molecular orbital theory, while the weaker ones are said to be 'prohibited'. Average positions of corresponding absorption bands on a wavelength scale.

(e.g. *ethylene*: λ_{max} = 171 nm, ε = 15,000, gas; *ethanal*: λ_{max} = 180 nm, ε = 10,000, with hexane as solvent).

- **The $n \rightarrow \sigma^*$ transition** leads to an absorption of low intensity located around 180 nm for alcohols, near 190 nm for ethers or halogen derivatives and in the region of 220 nm for amines (Figure 9.4) (e.g. *methanol*: λ_{max} = 183 nm, ε = 50; *ether*: λ_{max} = 190 nm, ε = 2,000; *bromomethane:* λ_{max} = 205 nm, ε = 200; *ethylamine*: λ_{max} = 210 nm, ε = 800, with hexane as solvent for all of these compounds).

Water, whose only possible transitions are $\sigma \rightarrow \sigma^*$ and $n \rightarrow \sigma^*$, is transparent in the near UV range (A = 0.01 for l = 1 cm, at λ = 190 nm).

- **The n $\rightarrow \pi^*$ transition**, not very intense ('prohibited' by selection rules) is found from 270 to 295 nm, especially for C=O or N=O double bonds (e.g. *ethanal*: λ = 290 nm, ε = 15, in hexane; *nitromethane*, λ = 275 nm, ε = 17, with ethanol as solvent). For the C=O carbonyl group for example, a nonbonding electron *n* goes into a π^* orbital, half located on the carbon and the other half on the oxygen. As a result, there is an increase in its dipole moment (from 3 to 5 Debye). This phenomenon is the source of the *pressure transfer spectrum* equation provided for this type of electronic spectrum (Figure 9.5).

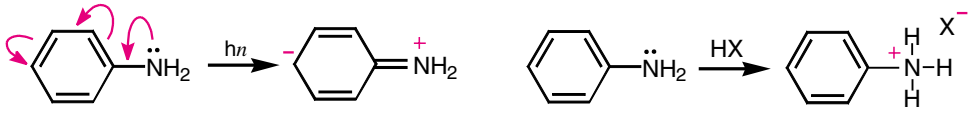


Figure 9.4 n → σ* transition of aniline (a primary amine). This transition corresponds to an increase in the 'weight' of the polar mesomeric form. The absorption peak of aniline representing this transition would disappear if an equivalent of one protonic acid of the HX type is added, since the formation of an ammonium salt would immobilize the free doublet of the nitrogen atom necessary for this transition.

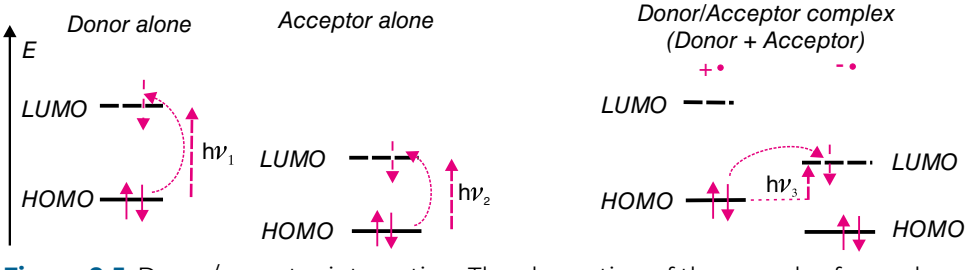


Figure 9.5 Donor/acceptor interaction. The absorption of the complex formed, corresponding to $h\nu_3$, is moved towards large wavelengths as compared to $h\nu_1$ and $h\nu_2$. Its position depends on the ionization potential of the donor and on the electronic affinity of the acceptor. This is an electronic transition between the MOs of different atoms.

When two compounds are mixed in a solution (one being a donor, the other an acceptor), an electron belonging to a bonding orbital of the donor passes into a vacant orbital of the acceptor of an attainable energy level (Figure 9.5). In this adduct formed between the donor (which becomes a radical cation) and the acceptor (which becomes a radical anion), the interval between frontier orbitals decreases with respect to products taken separately. The position of the absorption band of this *pressure transfer* complex is thus moved to greater wavelength and the coefficient of absorption ε is, in general, very high.

- **The *d* → *d* transition** is characteristic of many inorganic salts containing electrons engaged in *d* orbitals. This low-intensity transition leads to absorptions in the visible region. That is why aqueous solutions of metal salts of titanium $[Ti(H_2O)_6]^{3+}$ or of copper $[Cu(H_2O)_6]^{2+}$ are blue, while potassium permanganate $[MnO_4]$ yields violet solutions, chromate $[CrO_4]^{2-}$ yields yellow solutions and so on.

9.4 CHROMOPHORE GROUPS

The functional groups of organic compounds (ketones, amines, nitrogen derivatives, etc.), responsible for absorption in the UV/Vis domain are called *chromophores* (Table 9.1). A *chromogen* is a species formed from a carbon backbone that is transparent in the near UV and to which are attached one or several chromophores.

- *Isolated chromophores*: for a series of molecules having the same chromophore, the position and the intensity of the absorption bands remain mostly constant. When a molecule possesses several isolated chromophores, i.e. those which do not interact with each other because they are separated by at least two single bonds in the skeleton, then the overlapping of the effects of each individual chromophore is observed.
- *Conjugated chromophore systems*: when the chromophores interact with each other, the absorption spectrum is displaced towards longer wavelengths (*bathochromic effect*) with an increase in the absorption intensity (*hyperchromic effect*). A particular case is that of molecules with conjugated systems,

Table 9.1 Characteristic chromophores of some nitrogen-containing groups.

Name	Chromophore	λ_{max} (nm)	ε_{max} (L/mol/cm)
Amine	$-NH_2$	195	3,000
Oxime	= NOH	190	5,000
Nitro	$-NO_2$	210	3,000
Nitrite	−ONO	230	1,500
Nitrate	$-ONO_2$	270	12
Nitroso	−N = O	300	100

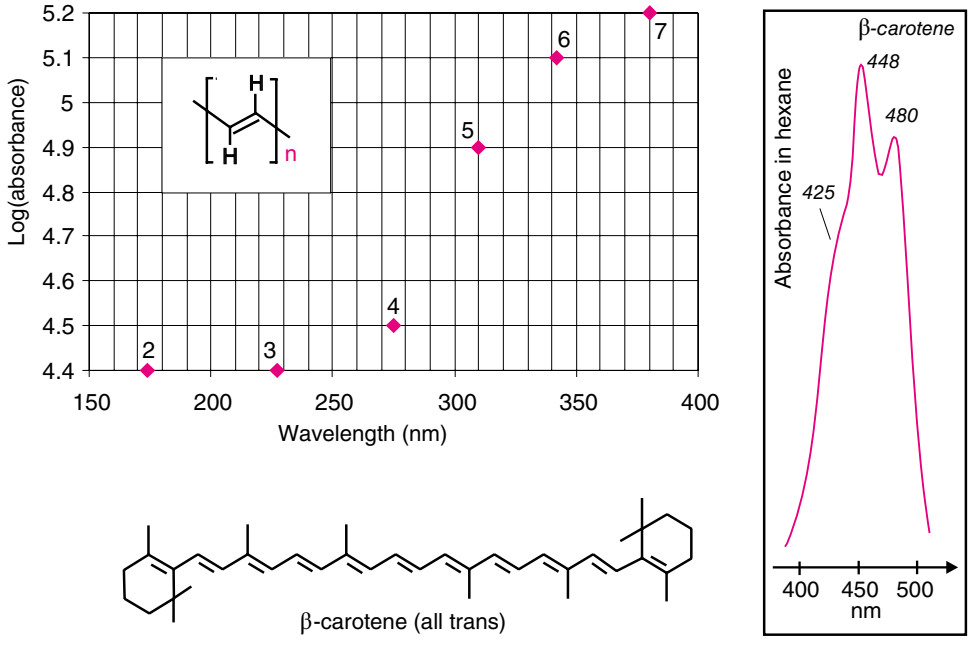
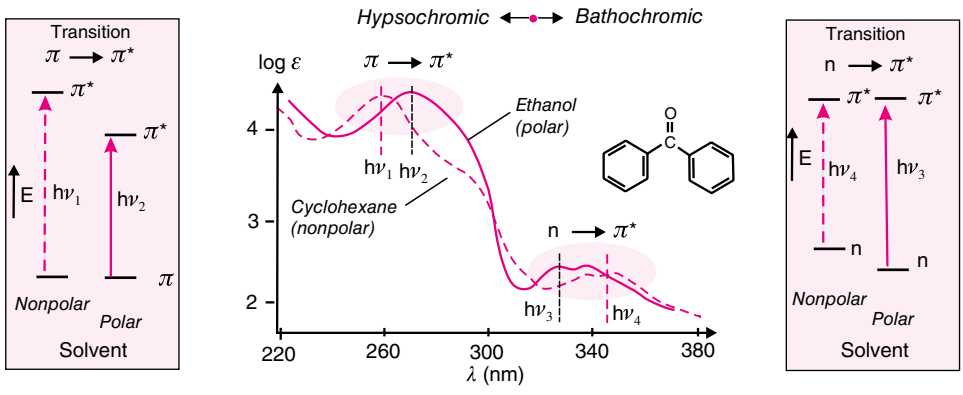


Figure 9.6 Values of λ_{max} for a family of *E*-disubstituted conjugated polyenes which differ by the number of conjugated double bonds. This effect is the source of the colour of many natural compounds, such as 'all-trans' β-carotene (11 conjugated double bonds).

i.e. organic structures containing several unsaturated chromophores separated by a single bond. The spectrum is then strongly disrupted with respect to the overlapping effects of isolated chromophores. The greater the number of carbon atoms upon which the conjugated system is extended, the smaller is the difference between frontier orbital levels. This results in a very large bathochromic effect (Figure 9.6).

9.5 SOLVENT EFFECTS

Each solvent has its own characteristic polarity. Since it is known that all electronic transitions modify the charge distribution of the compound in solution, it is obvious that the position and intensity of the absorption bands will vary a little with the nature of the solvent used. The solvent/solute interactions are sufficiently clear to be able to recognize the type of electronic transition involved. Two contrasting effects of solvatochromism can be distinguished (Figure 9.7).



Polar solvent
Bathochromic effect on transition $\pi \longrightarrow \pi^*$
Hypsochromic effect on transition n $\longrightarrow \pi^*$

Nonpolar solvent
Hypsochromic effect on transition $\pi \longrightarrow \pi^*$
Bathochromic effect on transition n $\longrightarrow \pi^*$

Figure 9.7 Spectra of benzophenone in cyclohexane (- - -) and in ethanol (–). The bathochromic and hypsochromic effects can be observed here with two different solvents and on the two types of transitions.

9.5.1 Hypsochromic Effect (Blue shift)

If the chromophore responsible for the observed transition is more polar in its ground state than when it is excited, then a polar solvent will stabilize, by solvation, the form prior to absorption of the photon. More energy would therefore be required to excite the electronic transition concerned, causing a displacement of the absorption maximum towards shorter wavelengths than would occur in a nonpolar solvent. This is the *hypsochromic effect*.

This is rather like the $n \rightarrow \pi^*$ transition of the ketone carbonyl in solution. The $C^+–O^-$ form (characterized by its dipole moment μ) will be more and more stabilized as the solvent becomes more polar. Since the excited state is rapidly attained, the solvent shell surrounding the carbonyl has no time to reorient itself and bring about stabilization following absorption of the photon. This same effect is observed for the $n \rightarrow \sigma^*$ transition and is accompanied by a variation of the coefficient ε.

Aromatic compounds lead to more complex spectra than compounds containing carbon-carbon double bonds. $\pi \rightarrow \pi^*$ transitions are the source of a 'fine structure'. Substitutions on the benzene ring will very significantly change the appearance of absorption bands.

9.5.2 Bathochromic Effect (Red Shift)

For less polar compounds, the solvent effect is weak. However, if the dipole moment of the chromophore increases during the transition, the final state will be more solvated. A polar solvent can thus stabilize the excited form, which favours the transition:

a displacement is observed towards longer wavelengths when compared with the spectrum obtained with a nonpolar solvent. This is the *bathochromic effect*. For example, this is the case for the $\pi \rightarrow \pi^*$ transition of alkene hydrocarbons, in which the double bond is only weakly polar.

9.5.3 Effect of pH

The pH of the solvent in which the solute is dissolved can have a significant effect on the spectrum. The compounds that display this effect in a spectacular fashion include chemical indicator strips, whose change in colour is used during acidimetric measurements (Figure 9.8). Equivalence points can be identified with this methodology.

Structural analysis from electron spectra can be rather problematic, in the sense that the spectra are relatively simple but also provide little information. However, before the

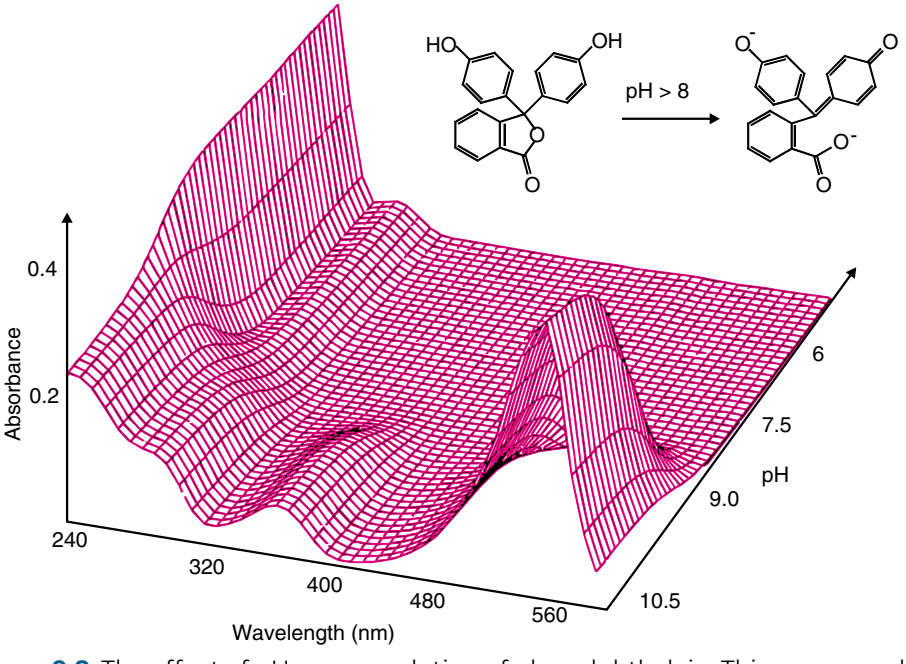


Figure 9.8 The effect of pH upon a solution of phenolphthalein. This compound is not coloured at pH values less than 8 though it is bright pink for those greater than 9.5. The graph presented here in 3D perspective reveals that for acidic pH values there is no absorption in the visible region of the spectrum. In contrast, absorptions towards 500 nm appear when the pH becomes alkaline and are responsible for the well-known colour of the compound. Notable in this example is the modification in the chemical bonds as a function of pH.

development of more powerful techniques of identification, now taken for granted, UV/Vis spectroscopy was employed for this purpose and led to the establishment of a correlation between structures and positions of the absorption maxima. The best known of these are the empirical rules developed by Fieser, Woodward, and Scott, which concern unsaturated carbonyl compounds, dienes and steroids.

9.6 INSTRUMENTATION IN THE UV-VISIBLE RANGE

A spectrophotometer contains three basic modules: the *source* and the *dispersive system* (often designed as a monochromator), which together constitute the optical section, and the *detection system* (Figure 9.9). These components are typically integrated in a single surround. A sample compartment is inserted into the optical path, either before or after the dispersive system, depending upon the design of the instrument.

Certain instruments are reserved for routine analyses when a high degree of resolution is not required. Many compounds in solution lead to spectra lacking in fine bands. However, it is essential that these instruments are able to give precise quantitative results to within several units of absorbance for an extended range of concentrations.

9.6.1 Light Sources

There are no continuous light sources intense enough to cover the entire region from the near UV to the very near infrared.

Many spectrometers therefore have two types of sources, one for near UV and the other for the part extending over the visible and even near infrared

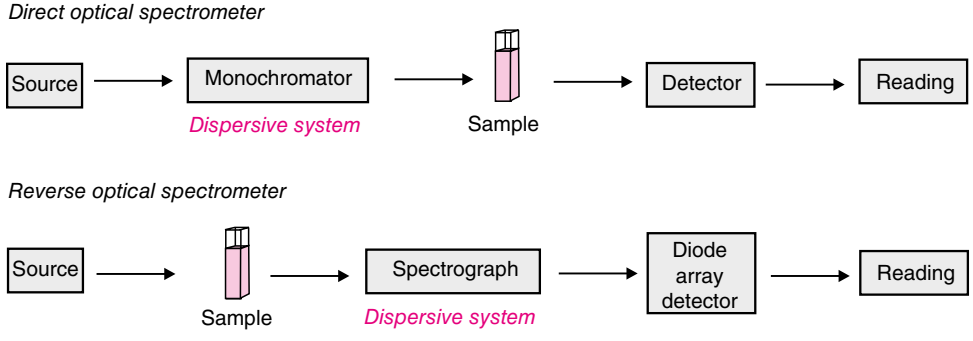


Figure 9.9 The two single-beam spectrometers configurations.

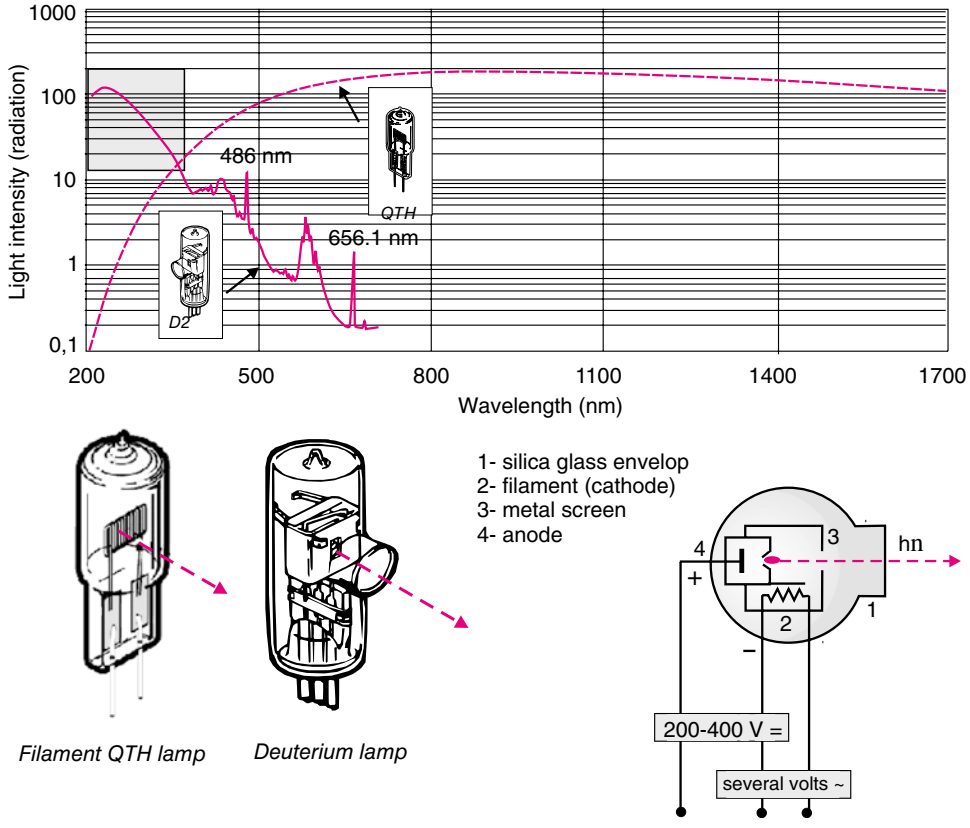


Figure 9.10 Emission curves of a Quartz Tungsten-Halogen (QTH) lamp and a deuterium source. The logarithmic scale takes into account the large irradiance differences as a function of wavelength. In grey, the effective domain for a deuterium source. General overview and top view of a deuterium lamp. The lamp has a firing voltage between 300 and 400 V. The cathode is a metal oxide filament emitting electrons, powered by a few volts of electricity. Emission peaks of deuterium at 486.0 and 656.1 nm are often used to adjust spectrometers.

regions (Figure 9.10). Therefore, the following are generally combined in one apparatus:

- For the UV region (<350 nm), a *deuterium arc lamp* working under medium pressure to maintain continuous radiation (but highly variable light intensity) with two characteristic narrow bands in the visible range.
- An *incandescent lamp* fitted with a tungsten filament and quartz envelope for the visible part of the spectrum, from 350 nm and above. It contains a small quantity of diiodine, to increase its lifespan (QTH lamp: *Quartz Tungsten*

Halogen). This source is sometimes replaced by a *xenon arc lamp*, more energetic and which then can be used a single source for routine apparatuses between 300 and 1,100 nm.

The deuterium lamp contains two electrodes bathed in a deuterium gas atmosphere (D_2 is preferred to hydrogen H_2, for technical reasons). Between the electrodes (near the anode), a metal screen is placed; it has a circular hole of approximately 1 mm in diameter (Figure 9.10). The migration of the electrons toward the anode creates a discharge current. Subjected to this electron bombardment, the molecules of deuterium D_2 dissociate, emitting a continuum of intense emission over the wavelength range between 160 and 700 nm. However, only the part below 350 nm, which is sufficiently intense, is used.

9.6.2 Dispersive Systems

We can distinguish between two categories:

- ***Direct assembly.*** Radiation emitted by the source is dispersed through either a flat or concave grating that forms part of a monochromator assembly. This device is intended to isolate a narrow band of the source's emission spectrum. This spectrum bandwidth depends on the *linear dispersion* of the grating and the *slit width* (Figure 9.11). To obtain a spectrum, a wavelength sweep is conducted by gradually pivoting the grating. Direct assembly devices are called *sequential instruments*. Monochromator assemblies with long focal lengths (0.2–0.5 m) and two or three gratings (each adapted to a specific wavelength range) yield the best resolution.
- ***Reversed assembly.*** In this category of instrument (known as *simultaneous*), the sample is directly placed after the source, and the monochromator is replaced by a device called a spectrograph. The latter has a grating but does not have an outlet slit, in order to let all of the radiation of the spectral domain from the source go through. It is connected to a photodiode array detector or a *charge coupled device (*CCD) detector (See Section 9.7.2 and Figure 9.14). The absence of moving parts ensures very good stability, and the signal/background noise ratio is improved by the accumulation of individual spectra.

9.6.3 Detectors

The detector converts the intensity of the light radiation reaching it into a proportional electrical signal (Figure 9.12). A simple scale factor then gives the photometric signal. Two types of detector are used for this spectral range, either a photomultiplier (PM) or photodiodes. The manufacturer makes its selection based on the device configuration (sequential or simultaneous, see Section 9.7). When a photodiode is used as a detector for a sequential spectrophotometer, it is used in photovoltaic mode (in the absence of polarization, it behaves like a battery).

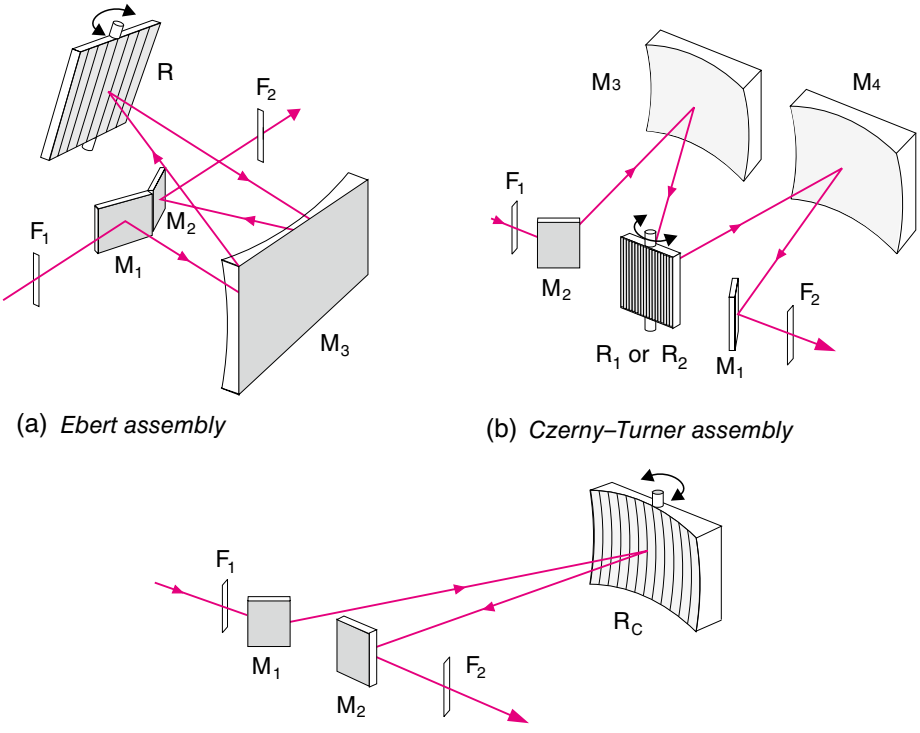


Figure 9.11 Monochromator gratings. (a) Ebert assembly incorporating a concave spherical mirror M_3 and a grating in offset configuration. This monochromator can compensate for aberrations and gives excellent image quality. (b) Of similar design to the Ebert, the Czerny–Turner assembly contains two spherical concave mirrors M_3 and M_4 and two gratings R_1 and R_2 for better spectral resolution; (c) The concave grating R_c in this design permits both dispersion and focusing of the radiation.

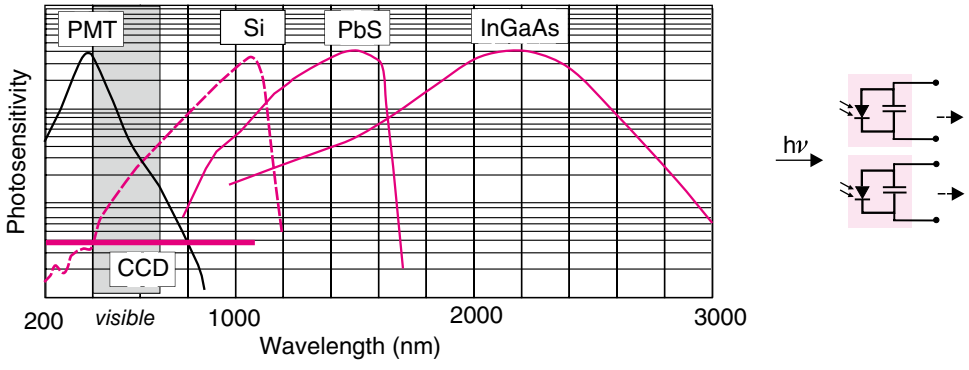


Figure 9.12 Response curves of several detectors used in the 200–3,000 nm range. Photomultiplier (PMT), silicon (Si), lead sulfur (PbS), gallium indium arsenide (for measurements in the near infrared), and silicon CCD. Right, photodiodes of a CCD bar used as pixels in photoamperic mode.

However, for simultaneous instruments which do not possess a monochromator but a dispersive system alone, like a spectrograph (see Section 9.7.2), a sensor is used with several hundred (or even up to 2,000–3,000) microscopic diodes, which, like pixels, are aligned in a *diode array*. Each diode (Si, PbS or InGaAs) is connected to a condenser to operate in photoamperic mode. The assembly forms a CCD sensor. For each diode, the measurement cycle (a few ms), managed by the electronic circuit, consists of charging the condenser then in exposing the diode to the radiation. Under the impact of photons, the diode, by becoming a conductor, progressively depletes the condenser's energy. From the quantity of current necessary to recharge it, the light energy (in J) accumulated throughout the cycle can be deduced and, therefore, the incident radiation intensity can be as well. Here, the diode operates as a light integrator. All wavelengths are practically observed at the same time, which considerably reduces the acquisition period of spectra.

The efficiency of a photomultiplier tube – whose sensitivity depends upon the yield of the photocathode – varies with the wavelength (e.g. 0.1 e^-/photon at 750 nm). The amperometric signal must be amplified (e.g. a gain of 6×10^5). The linearity of response can extend over seven decades. However, it is difficult for a photomultiplier, just as it would be for the eye, to compare two light intensities with precision, one arising from the reference beam and the other from the sample, when these are quite different. For an instrument with stray light of 0.01% (measured as a percentage of transmittance), an increase in solution concentration produces no significant variation in the signal beyond four units of absorbance. This is why it is desirable that the absorbance of the solutions does not exceed one (see Section 9.9).

9.7 CONFIGURATIONS OF UV/VIS SPECTROPHOTOMETERS

UV/Vis spectrometers are classified in two categories:

- single-beam spectrometers, subdivided into direct assembly and reversed assembly instruments;
- double-beam spectrometers.

9.7.1 Single-Beam, Single-Channel Optical Spectrometers

Many routine measurements are conducted at fixed wavelength by basic photometers fitted with interchangeable interference filters or simple monochromators. The measurement of the transmittance (or absorbance), at a given wavelength, requires the comparison of the source signal before and after it passes through the sample solution. A control corresponding to the solvent alone or a solution containing the reagents of the measurement (but without the compound to be measured, i.e. an *analytical*

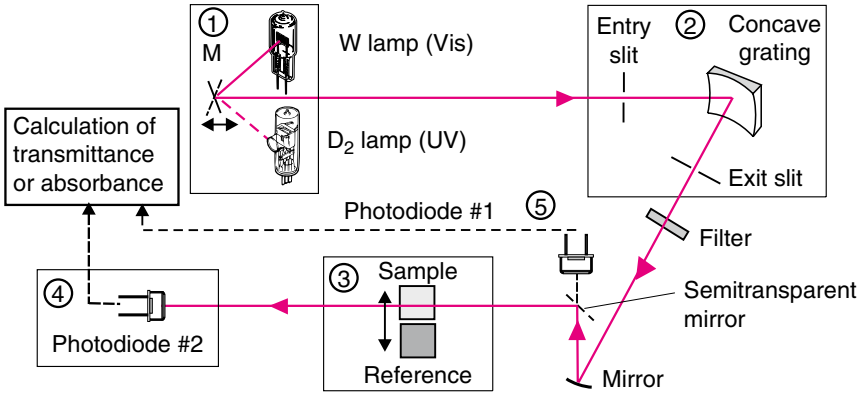


Figure 9.13 Simplified diagram of the optical path of a single-beam, sequential-mode spectrophotometer. 1. Two sources coexist, though only one is selected for the measurement. 2. The monochromator selects the measurement wavelength. 3. The measuring cell containing either the sample or a control blank is placed in the optical path. 4 and 5. Diode detector and control diode.

blank) is first placed in the optical path, then is replaced by the solution prepared from the sample of unknown concentration. These instruments sometimes have a built-in electronic compensation arrangement for variations in light source intensity (Figure 9.13) known as a *beam splitter*. A part of the light beam is diverted before it reaches the sample, permitting the stabilization of the source intensity (this is not a true reference beam). These instruments give the absorbance and usually calculate the analyte concentration.

9.7.2 Single-Beam, Multichannel Spectrophotometers

This type of instrument closely resembles a spectrograph, since it allows the simultaneous recording of all wavelengths of the spectrum by using an array (CCD sensor) that can contain 500–3,000 pixels (Figure 9.14). The spectral range is explored sequentially in a few milliseconds (see Section 9.6.3). The resolving power of these diode-array instruments, without a monochromator (thus more luminous), is limited by the size of the diodes.

9.7.3 Double-Beam Optical Spectrometers (Sequential Type)

The double-beam design, where one beam passes through the sample while the other passes through the reference solution, remains the best in this domain. Two rotating mirrors, synchronized with the alternating movement of the grating, enable

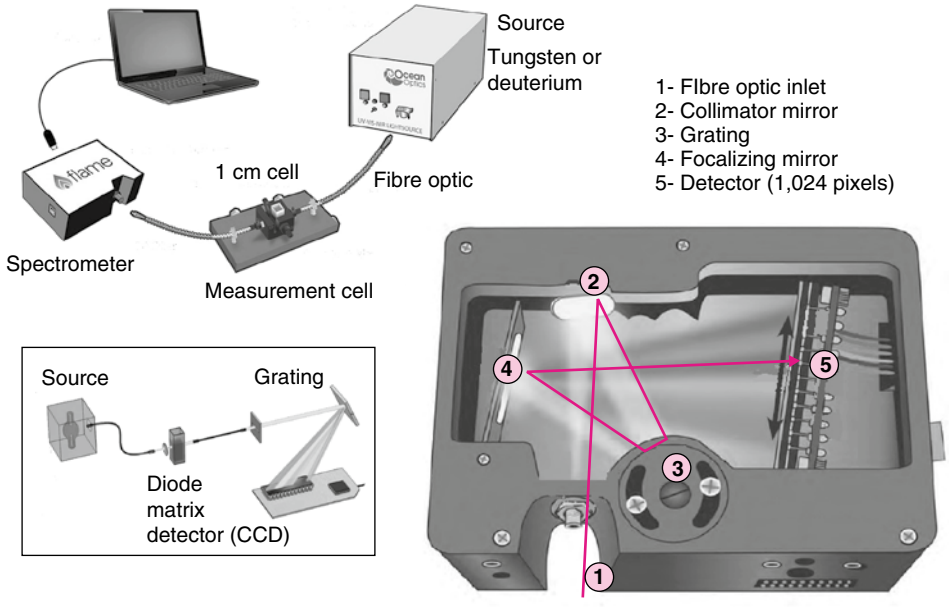


Figure 9.14 Single-beam spectrometer with detection by CCD array. Module layout of a portable instrument (Ocean Insight), which contains a source (tungsten or xenon), a diode array spectrograph, and a cell inserted on the fibre optic path to measure the absorbance remotely. Many instruments of this type are used for routine applications. (Source: Courtesy of Ocean Insight.)

the detector to compare the two intensities transmitted by the reference or sample solutions for the same wavelength (Figure 9.15). The amplification of the single modulated signal allows the elimination, in large part, of the stray light. The electronic circuit adjusts the sensitivity of the photomultiplier tube inversely to the light intensity it receives. A simpler set-up is based on the use of a semitransparent mirror and two connected photodiodes. The signal corresponds to the voltage required to maintain a constant detector response (principle of a *feedback system*). These instruments are characterized by a fast scanning speed (30 nm/s) and the capability of measuring absorbances of several units.

To calibrate the wavelength scale of the spectrometer, a 4% m/v solution of holmium oxide (Ho_2O_3) in 10% v/v perchloric acid is used. The exact positions of the various spectral maxima of this solution are known. The values depend somewhat on the adjustment of the spectral bandwidth. The absorption bands of the spectrum are not symmetrical and the spectral bandwidth selected for scanning the spectrum has an influence on the energy reaching the detector, which in turn has repercussions on the intensity of the signal and finally on the recorded spectrum, e.g. in passing from a slit size of 0.1 nm to 3 nm, the absorption maximum at 536.47 nm is displaced to 537.50 nm.

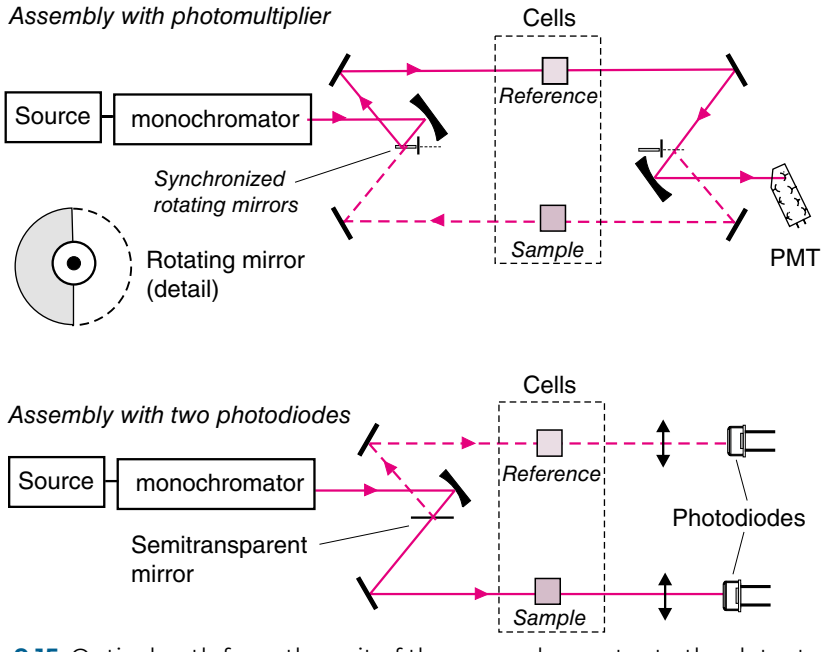


Figure 9.15 Optical path from the exit of the monochromator to the detector for two double-beam instruments (a model with two rotating mirrors and a model with a semitransparent mirror). The arrangement of the apparatus with rotating mirrors is similar to that of IR spectrophotometers, except that the light beam emanating from the source first passes through the monochromator before it hits the sample. This minimizes the photolytic reactions that could occur owing to overexposure to the total radiation emitted by the source. A more compact optical assembly with a single split beam associated with two detectors offers a simpler set-up. A semitransparent, fixed mirror replaces the delicate mechanism of synchronized, rotating mirrors.

Double-beam spectrophotometers allow differential measurements to be made between the sample and the analytical blank. They are preferable to the single-beam instruments for cloudy solutions. The bandwidth of high-performance instruments can be as small as 0.01 nm.

Fluorescent compounds

When the compound studied is fluorescent, the absorption spectrum is less intense because the re-emitted light is added to the true absorbed light. Observed using a reversed-optics set-up, the fluorescence of the sample (exposed to the total radiation of the source) reduces the absorbance in the region of the spectrum where the emission is situated. Alternatively, with a traditional optical set-up, the fluorescence appears when the monochromator selects the region corresponding to the excitation: the absorbance will be less intense in this part

of the recorded spectrum. However, the problem should not be exaggerated, as the fluorescence is emitted in all directions, and the photons taking the optical path for the absorption experiment constitute a very small part of the emitted light.

9.8 MEASUREMENT CELLS AND DEVICES

On the optical path of the spectrometer, a free slot is set aside to place a cell or other device adapted to the physical state of the sample being studied (gas, liquid or solid) and to the spectral domain in question.

For the two major types of analyses (recording of a spectrum or absorbance measurement for an assay), the compound is present in solution. As a function of the desired level of precision, a variety of cells are available, from the simplest small cylindrical tube to microwell plates and rectangular cuvettes, for very precise measurements (Figure 9.16).

Three types of materials are used in their manufacture:

- Different varieties of transparent silica (SiO_2) at 200 nm or above.
- Optical lenses (silicate, borosilicate, etc.) usable at 320 nm or above.
- Various plastic materials for disposable tubes and cuvettes, usable at 300 nm or above in the case of Plexiglas or polystyrene among others.

However, there are many applications requiring cells with specific shapes. Optical paths may then range from several micrometres (HPLC detection cells) to several hundred metres (cells for gas and open-path measurements).

For surface examinations, as is the case in the infrared region (see Chapter 10, Section 10.9.3), there are specular reflection devices (Figure 9.16). Lastly, to conduct measurements remotely and continuously, for example to monitor the concentration of a compound on a manufacturing line, there are immersion probes. Placed in the compartment reserved for conventional cells, this device has two optical fibres to conduct and recover the light after absorption in the medium being studied. There are two types: transmission for clear solutions and attenuated total reflection (ATR, see Chapter 10, Section 10.9.3) for very absorbent solutions.

9.9 QUANTITATIVE ANALYSIS: LAWS OF MOLECULAR ABSORPTION

9.9.1 Beer–Lambert Law

The UV-Visible range is widely exploited in quantitative analysis and, for the visible region, this has been the case for quite some time. The measurements are based upon the Beer–Lambert Law, which, under certain conditions, links the absorption of light to the concentration of a compound in solution.

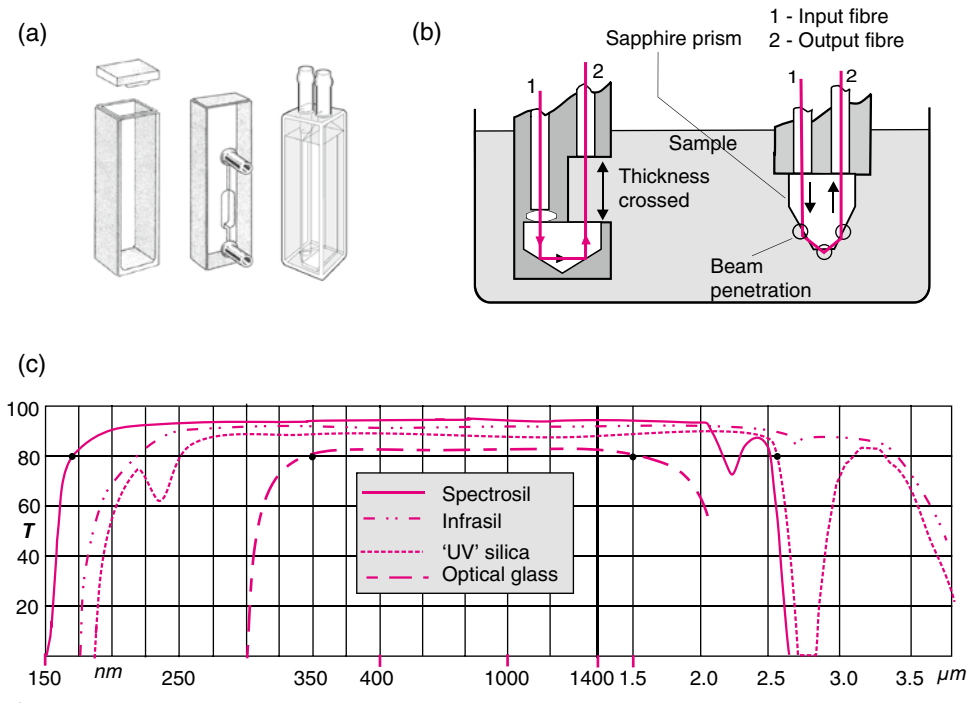


Figure 9.16 Cells and main sampling devices. **a)** Standard cuvette and circulation tank for the study of solutions. **b)** Immersion probe model. Ends of a conventional probe and of an ATR-type probe. Monochromatic light coming out of a spectrophotometer is guided towards an immersion cell and is then returned to the detector. The reference path is also ensured by another fibre optic. The sapphire prism has a refraction index greater than that of the solution. **c)** Transmittance curve of four materials used in the UV-Visible region as a function of wavelength.

The origins of this law are found in the work of the French mathematician Johann Heinrich Lambert who laid down the basis for photometry in the eighteenth century. Later, August Beer, a German physicist of the following century, proposed a law that led to the calculation of the quantity of light transmitted through a given thickness of a compound in solution in a nonabsorbing matrix.

The result was the *Beer-Lambert Law*, which is presented here in its current form:

$$A_\lambda = \varepsilon_\lambda \cdot l \cdot C \qquad (9.4)$$

A represents the absorbance, a dimensionless optical parameter accessible with a spectrophotometer, *l* is the thickness (in cm) of the solution through which the incident light passes, *C* the molar concentration, and ε_λ the molar absorption coefficient ($L \cdot mol^{-1} \cdot cm^{-1}$) at the measurement wavelength λ. This parameter, also called the absorbance factor, is characteristic of the compound being analysed and

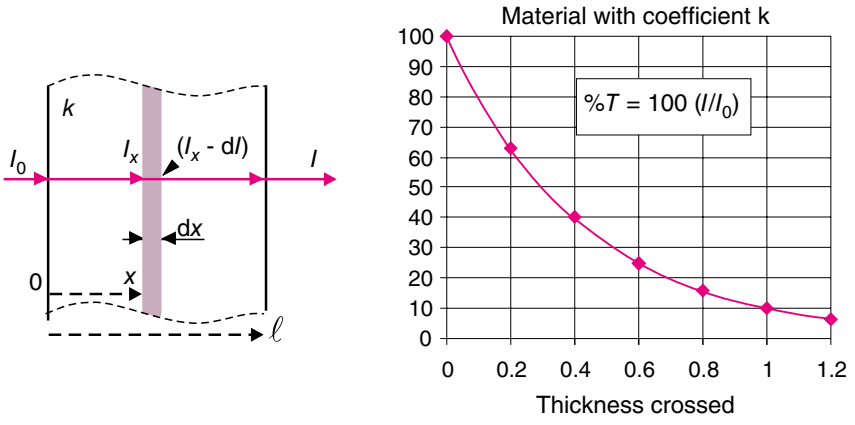


Figure 9.17 Absorption of light by a homogeneous material and representation of percentage transmittance as a function of the material's thickness. The light reaching the sample can be transmitted, refracted, reflected, diffused or absorbed. Here, only the absorbed part is taken into account.

depends among other things on the temperature and the nature of the solvent. Generally, its value is given at the wavelength of the absorption maximum. This can vary over a wide range from zero to over 200,000.

According to Lambert's hypothesis, the intensity I of *monochromatic radiation* is decreased by dI (negative value) after passing through a thickness dx of a material whose *absorption coefficient* is k for the chosen wavelength x (Figure 9.17):

$$\frac{-\mathrm{d}I}{\mathrm{d}x} = kI_x \tag{9.5}$$

If I_0 is the light intensity of the incident radiation before it passes through the medium of thickness l, Eq. (9.5) can be written in integrated form giving the transmitted intensity I (Eqs. (9.6)–(9.8):

$$\left[\ln I_x\right]_{I_0}^{I} = -k\left[x\right]_0^{l} \tag{9.6}$$

$$\ln = \frac{I}{I_0} = -k \cdot l \tag{9.7}$$

$$I = I_0 e^{-kl} \tag{9.8}$$

In 1850, Beer proposed Eq. (9.8) for the case of a solution of low concentration dissolved in a transparent (nonabsorbing) medium, by writing that k is proportional to the molar concentration C of this compound (Figure 9.18):

$$k = k' \cdot \mathrm{C} \tag{9.9}$$

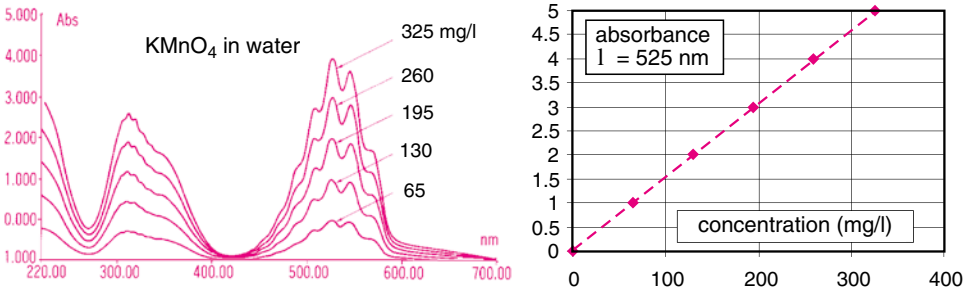


Figure 9.18 Illustration of the Beer–Lambert law. Spectra of aqueous solutions of increasing concentration in potassium permanganate. Graph of the corresponding absorbances measured at 525 nm, showing the linear growth of this parameter.

By replacing k with $k' \cdot C$ in Eq. (9.8), a new relation is obtained, which is better known in the form of Eq. (9.4), in which the *absorbance* A is represented by one of the comparable equations:

$$A = \log\frac{I_0}{I}. \text{ or } A = \log\frac{1}{T} \text{ or } A = \log\frac{100}{T\%} \text{ or } A = 2 - \log T\% \qquad (9.10)$$

The Beer–Lambert Law, originally considered to be a postulate, has subsequently been the subject of many interpretations and demonstrations based on statistical, kinetic or simple logic hypotheses.

This law, which concerns only the absorbed light fraction, can be verified by application of the following conditions:

- the light used must be monochromatic;
- the concentrations must be low;
- the solution must be neither fluorescent nor heterogeneous;
- the solute must not give rise to photochemical transformations;
- the solute must not form variable associations with the solvent.

In general, when the absorbance is to be measured at a single wavelength, the chosen wavelength is generally the absorption maximum. For the absorbance to reflect the concentration correctly, the spectral band $\Delta\lambda$ reaching the detector and preselected by the instrument parameter called *spectral bandwidth* (SBW), must be much narrower (10 times) than the width of the absorption band measured at mid-height.

9.9.2 Additivity of Absorbances

The Beer–Lambert Law is additive (Figure 9.19). This means that if the absorbance A is measured for a mixture of two compounds, 1 and 2, in a solvent and placed in a cuvette of thickness l, the same total absorbance will be obtained if the light had

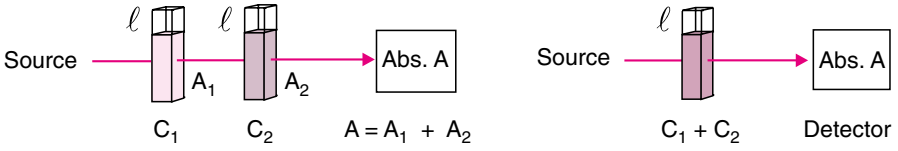


Figure 9.19 Additive nature of absorbances. For all wavelengths, the absorbance of a mixture is equal to the sum of the absorbances of each component in the mixture (assuming the same molar concentrations in the two experiments).

successively passed through two cuvettes of thickness *l*, placed one after the other, one containing compound 1 (Abs. A_1) and the other compound 2 (Abs. A_2). Of course, the concentrations and the solvent must remain the same as for the initial mixture (compound 1 is labelled 1 and compound 2 is labelled 2):

$$A = A_1 + A_2 = l\left(\varepsilon_1 C_1 + \varepsilon_2 C_2\right) \tag{9.11}$$

This principle is illustrated by the study of isobestic points. Consider a compound A, which is transformed by a first-order reaction to compound B. Assume that the absorption spectra of A and B, recorded separately in the same solvent and at the same concentration, intersect each other at a point *I* when one is superimposed on the other (Figure 9.20). Consequently, *for the wavelength at point I*, the absorbances of the two solutions are the same and by corollary the coefficients ε_A and ε_B are

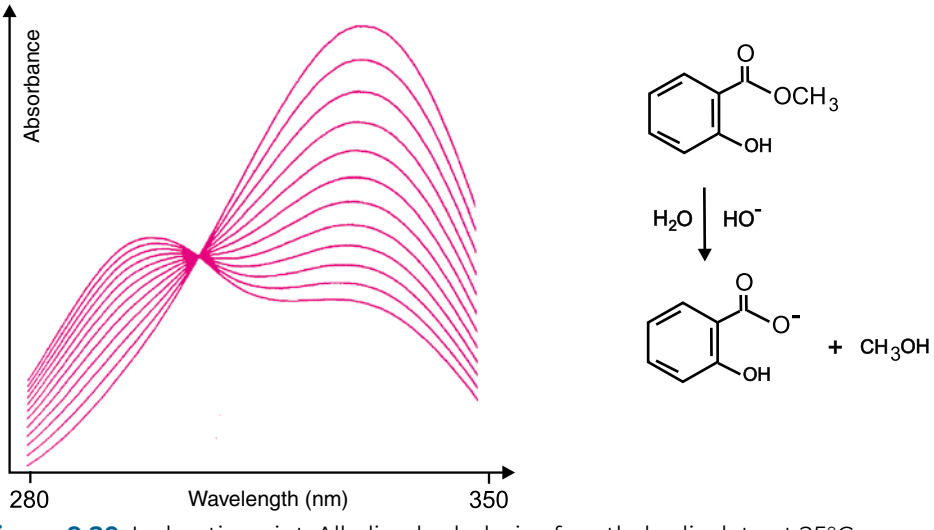


Figure 9.20 Isobestic point. Alkaline hydrolysis of methyl salicylate at 25°C. Superimposition of the successive spectra recorded between 280 and 350 nm at 10-min intervals.

equal ($\varepsilon_A = \varepsilon_B = \varepsilon$). In this transformation, compound A is initially pure and ultimately compound B is also pure. For all intermediate solutions, the total concentration for mixtures of A and B does not change ($C_A + C_B = C$ste), which can be given by the following equation:

$$A_I = \varepsilon_A l C_A + \varepsilon_B l C_B = \varepsilon l \left(C_A + C_B \right) \tag{9.12}$$

All of the spectra of the various mixtures of A + B formed over time will pass through point **I**, called the isobestic point, where the absorbance *A* will always have the same value.

This network of concurrent curves is observed when studying coloured indicators as a function of pH, or during studies on reaction kinetics.

9.10 METHODS IN QUANTITATIVE ANALYSIS

9.10.1 Colorimetry

When a compound does not absorb light, it can nevertheless become the subject of a photometric measurement if it can be transformed beforehand to a derivative that has an exploitable chromophore. Using this method, it becomes possible to quantify all kinds of chemical species that have no significant (i.e. very weak) absorption, or because it lies in a region of the spectrum where other interfering absorptions coexist. To counteract this, the absorbance measurement is preceded by a chemical transformation which must be specific, total, rapid, reproducible, and lead to a stable derivative in solution. This is the principle of *colorimetric tests*.

The term colorimetry arose from the fact that the initial measurements in this field, well before the invention of the spectrophotometer, were carried out with natural light (white light), by direct visual comparison of the colouring of the sample with that of a control of known concentration. Apart from the eye, no other specific apparatus was used.

The two situations that can be encountered most frequently are:

- *The analyte to be measured is present in a matrix whose other constituents also absorb in the same spectral range*: the direct measurement of the absorbance due solely to this compound is therefore impossible (Figure 9.21, curve a). To overcome this difficulty, the compound is transformed specifically into a derivative whose absorption band is situated in a region free from interference by the matrix (Figure 9.21, curve b).
- *The analyte to be measured has no exploitable chromophore in its raw form* (Figure 9.21, curve c): once again this problem is overcome by the creation of a chromophore replacing the initial species, in accordance with the same principle as above (Figure 9.21, curve d).

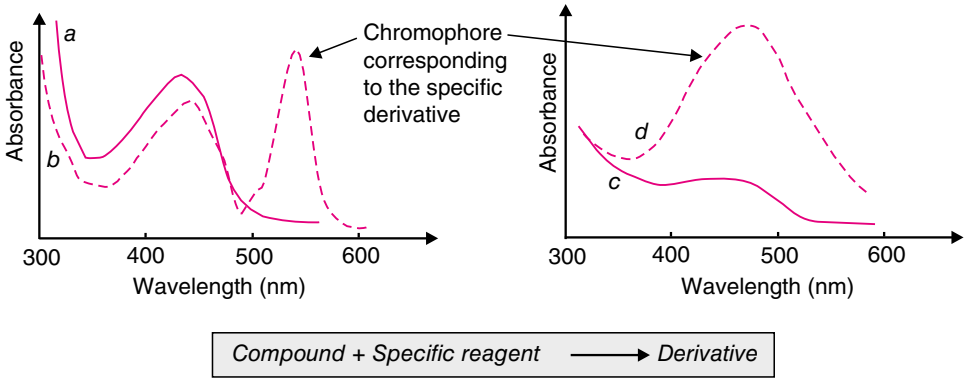


Figure 9.21 Illustration of two frequently encountered situations. A compound whose absorption is masked by the components of the mixture can otherwise be measured using colorimetry by employing a chemical modification which transforms the molecule into a derivative unaffected by any interference.

In colorimetry, it is preferable that the measurement is made on a chromophore whose absorbance is situated toward longer wavelengths, as this reduces the risk of superimposition of the individual absorptions of the various compounds. In addition, when measurement is preceded by a chemical reaction, we rarely know the exact structure of the coloured derivative whose absorbance is measured. Nevertheless, if it is assumed that the reaction involved is quantitative, its molar absorption coefficient can be determined from the molar concentration of the reference compound.

9.10.2 Analysis of a Single Analyte

In practice, the first step is the construction of a calibration curve $A = f(c)$ from solutions of known concentration of the compound to be measured, subjected to the same treatment as the sample. This curve, which is most often a straight line for dilute solutions, allows determination of the concentration C_x of the unknown sample.

Sometimes only a single standard solution is prepared. Its concentration C_R is such that its absorbance A_R is slightly greater than that estimated for the unknown solution A_x (Figure 9.22).

The following formula can then be applied to calculate C_x:

$$C_x = C_R \frac{A_x}{A_R} \tag{9.13}$$

9.10.3 Confirmatory Analysis

Calculation of the concentration of an analyte by Eq. (9.13) can lead to an erroneous result if the sample contains an impurity (absent from the reference solution) which

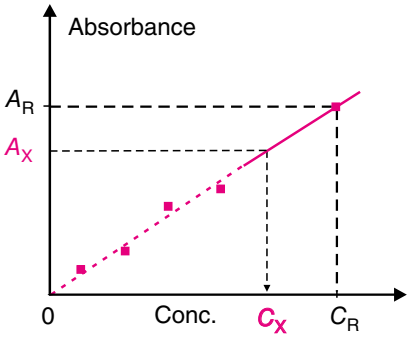


Figure 9.22 Calibration curve. If a single reference solution is prepared, the graph will be a straight line that passes through the origin. The accuracy of the result will be precise when the unknown concentration is close to the reference concentration (the result is determined by interpolation, not by extrapolation).

also absorbs at the wavelength of measurement. Therefore, a method commonly named *confirmatory analysis* is used.

For a given pure compound, the ratio of the absorption coefficients, determined at two wavelengths, is constant and characteristic of this compound. Thus, if one or two extra measurements are performed, offset by a few nanometres with respect to the original measurement, the ratios of absorbance of the sample solution can be calculated and compared with those established in the same way using a pure reference solution. If these ratios are different, it will be presumed that an impurity is present in the sample. Another wavelength has to be chosen to ensure that the analysis is reliable.

Starting from this assumption, it is possible to control the homogeneity of the elution peaks in liquid chromatography on the condition that a UV detector undertaking simultaneous measurements of the absorbance at several wavelengths is present (such as a diode array detector). During the course of the elution of a component, any deviation from the ratio of the absorbances at two wavelengths suggests that a mixture is eluting from the column instead of a pure compound (Figure 9.23). This method will fail, however, in the case of perfect coelution of two compounds.

9.10.4 Multicomponent Analysis (MCA)

When we analyse a mixture of compounds whose individual absorption spectra are known, we can then determine the mixture's composition. According to the law of additivity (Eq. (9.11)), the spectrum of the mixture to be measured corresponds to the weighted sum of the spectra of each of the individual constituents. The classical method of calculation is reviewed here for revision purposes only, since it is no longer done long hand but rather is incorporated into computer software.

Basic algebraic method

Consider a mixture of three components a, b, and c in solution (concentrations C_a, C_b, C_c). Firstly, the absorbances of each of the three compounds are measured, using reference solutions, at three wavelengths λ_1, λ_2, and λ_3 in order to calculate the

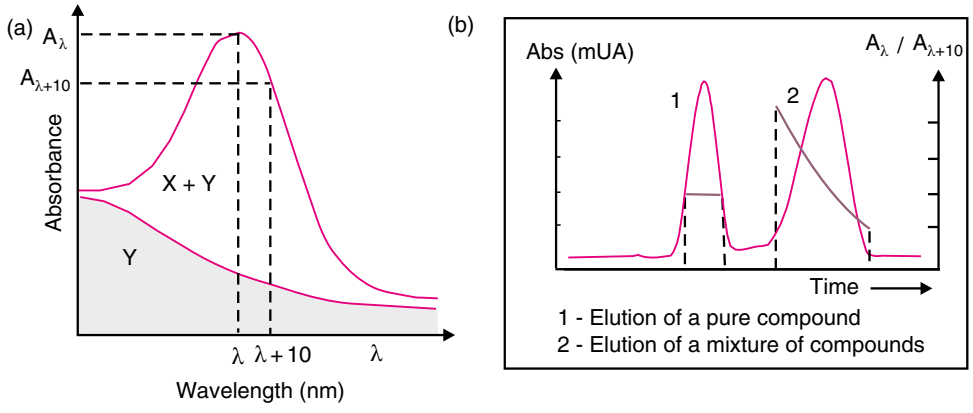


Figure 9.23 Confirmatory analysis. (a) Spectrum of a mixture (X + Y) and spectrum of Y in the same range (grey part); (b) diagram illustrating a chromatogram with two peaks, the first of which is due to a single compound and the second to the elution of two compounds slightly shifted in time. The evolution of the ratio of absorbances during elution enables control of the purity of the eluted compounds. These variations are usually revealed by software as coloured areas.

corresponding absorbance coefficients (nine values in total from ε_a^1 to ε_c^3). Then the absorbances of the mixture are measured at these three wavelengths, i.e. A_1, A_2, and A_3. Through application of the additivity law of absorbances, the following system of three equations can be written (it is assumed that the optical path of the cells is 1 cm):

$$\begin{aligned} &\text{at } \lambda_1 \quad A_1 = \varepsilon_a^1 C_a + \varepsilon_b^1 C_b + \varepsilon_c^1 C_c \\ &\text{at } \lambda_2 \quad A_2 = \varepsilon_a^2 C_a + \varepsilon_b^2 C_b + \varepsilon_c^2 C_c \\ &\text{at } \lambda_3 \quad A_3 = \varepsilon_a^3 C_a + \varepsilon_b^3 C_b + \varepsilon_c^3 C_c \end{aligned} \qquad (9.14)$$

The resolution of this mathematical system, which corresponds to a [3 × 3] matrix, leads to the three concentrations C_a, C_b, and C_c.

$$\begin{bmatrix} C_1 \\ C_2 \\ C_3 \end{bmatrix} = \begin{bmatrix} A_1 \\ A_2 \\ A_3 \end{bmatrix} . \begin{bmatrix} \varepsilon_a^1 & \varepsilon_b^1 & \varepsilon_c^1 \\ \varepsilon_a^2 & \varepsilon_b^2 & \varepsilon_c^2 \\ \varepsilon_a^3 & \varepsilon_b^3 & \varepsilon_c^3 \end{bmatrix}^{-1} \qquad (9.15)$$

This approach yields good results when the compounds lead to significantly different spectra; otherwise, it loses precision when the spectra are in close proximity, as a small measurement error can lead to a large variation in the result. To avoid this risk, instruments with diode arrays use several dozen data points. The system to be resolved (Eq. (9.15)) becomes overdetermined, but this leads to better results.

Multiwavelength linear regression analysis (MLRA)

The analysis of mixtures has given rise to the development of a variety of methods, made possible with quantitative analysis software that can process a large number of data points extracted from the spectra of the sample under study and of calibration solutions. Described below is a method of linear regression that leads to the neutralization of the background noise and therefore to an improvement in the results. This method is applied for the analysis of a two-component sample (Figure 9.24).

The instrument uses three recordings: a spectrum of the sample (which contains the two compounds whose concentrations are presently unknown) and two spectra from the same spectral region for each of the pure compounds (reference solutions of known concentration).

For any wavelength, the absorbance of the mixture of the two analytes a and b (which do not interact with each other) is given by the equation:

$$A = \varepsilon_a l C_a + \varepsilon_b l C_b \tag{9.16}$$

For each of the two reference spectra, supposing that the thickness of the measuring cell is l = 1 cm, the following equations can be written:

$$\text{For compound a } A_{\text{ref.a}} = \varepsilon_a \cdot C_{\text{ref.a}} \tag{9.17}$$

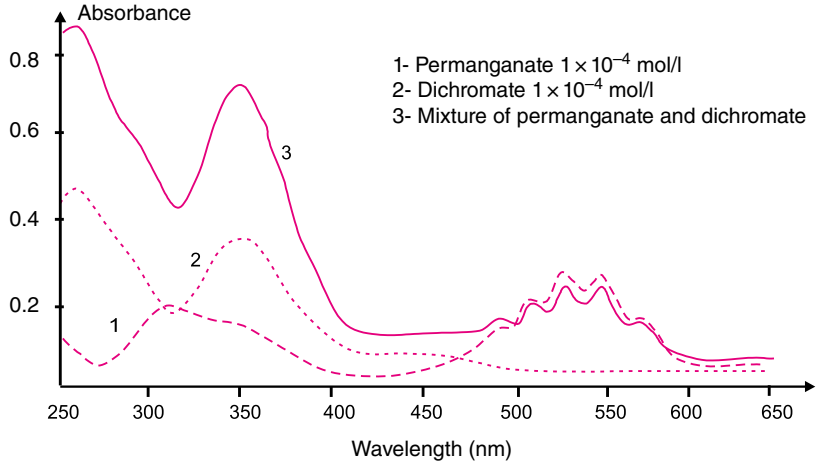


Figure 9.24 Multicomponent analysis. Spectra of a 1 × 10^{-4} M solution of potassium permanganate, a 1 × 10^{-4} M solution of potassium dichromate, and a solution containing 1.8 × 10^{-4} M of dichromate and 0.8 × 10^{-4} M of permanganate. (Source: Based on M. Blanco, H. Iturriaga, S. Maspoch, and P. Tarin, A simple method for spectrophotometric determination of two-components with overlapped spectra, *Journal of Chemical Education* (1989), 66(2), 178. DOI: 10.1021/ed066p178.)

$$\text{For compound b } A_{\text{ref.b}} = \varepsilon_{\text{b}} \cdot C_{\text{ref.b}} \tag{9.18}$$

These two equations lead to the calculation of the molar absorption coefficients ε of each pure constituent, at each wavelength considered. Therefore, Eq. (9.16) can be rewritten as:

$$A = \frac{A_{\text{ref.a}}}{C_{\text{ref.a}}} C_{\text{a}} + \frac{A_{\text{ref.b}}}{C_{\text{ref.b}}} C_{\text{b}} \tag{9.19}$$

By dividing the first element by $A_{\text{ref.a}}$, we obtain for each wavelength:

$$\left(\frac{A}{A_{\text{ref.a}}} \right) = \frac{C_{\text{a}}}{C_{\text{ref.a}}} + \frac{C_{\text{b}}}{C_{\text{ref.b}}} \cdot \left(\frac{A_{\text{ref.b}}}{A_{\text{ref.a}}} \right) \tag{9.20}$$

The first member of Eq. (9.20) is therefore a linear function of the ratio of absorbances found in the second member. The calculated values are on a straight line whose slope ($C_{\text{b}}/C_{\text{ref.b}}$) and *y*-intercept ($C_{\text{a}}/C_{\text{ref.a}}$) enable us to calculate C_{a} and C_{b}. The precision of the results increases with the number of data points used.

9.10.5 Deconvolution

Many data processing software programs can be used to determine the composition of mixtures from their spectra. One of the best-known methods is the Kalman filter algorithm. It automatically finds, through successive approximations, the spectrum of the sample solution by adding saved spectra of each of the compounds using weighted coefficients (additivity law of absorbances). These are automated methods better known by their initials: PLS (*partial least squares*), PCR (*principal component regression*), or MLS (*multiple least squares*) (Figure 9.25).

9.11 METHODS OF BASELINE CORRECTION

Some samples, such as biological fluids, contain micelles or particles in suspension, which by light scattering (Tyndall effect) induce additional absorption. The latter varies in a regular manner with wavelength (Figure 9.26).

There are several methods that seek to correct this phenomenon by subtracting from the measured absorbance the portion that is not due to the analyte. These are in fact corrections made to the baseline from measurements undertaken at various wavelengths.

9.11.1 Modelling by Polynomial Function Fitting

We assume that baseline variations can be represented by a function $A = k\lambda^n$. To find k and n, a region of the spectrum is located in which the analyte does not absorb, so that the software can use this zone to calculate these coefficients by the least-squares

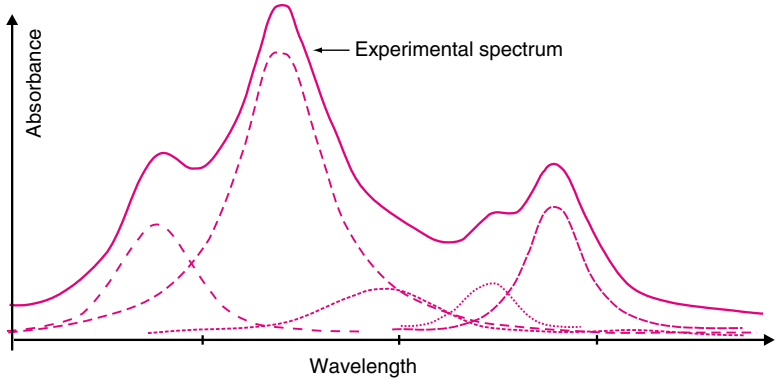


Figure 9.25 Deconvolution of the spectrum of a five-compound mixture. From the experimental spectrum of a mixture of five compounds, the software is able to find the proportion of each one (it is assumed that the spectra of the individual components are known).

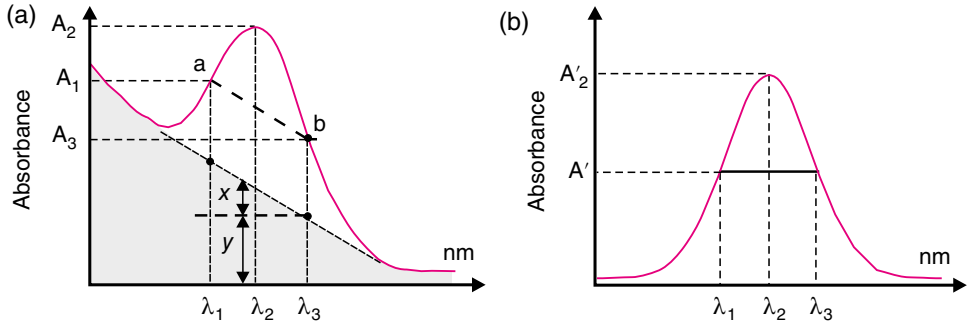


Figure 9.26 Illustration of the concepts behind the Morton–Stubbs calculation. (a) Spectrum of the sample. (b) Spectrum of the pure analyte.

method ($\log a = \log k + n \log \lambda$). Then, whatever the wavelength of measurement, it will be possible to remove the absorbance corresponding to the baseline.

9.11.2 Morton–Stubbs Three-Point Correction

The Morton–Stubbs method (as well as the method using derivative curves, see Section 9.13) leads to an efficient correction of the baseline, provided that the underlying absorption varies linearly in the measurement zone. In a situation such as that represented in Figure 9.26a, in order to quantify the compound the absorbance A_2 read at the maximum λ_2 must be corrected by removing the absorbance corresponding to the values noted as x and y.

To bring this about, a reference spectrum of the pure compound must be established in order to choose two wavelengths λ_1 and λ_3 whose absorbance coefficients are the same (Figure 9.26b). As on the figure, at λ_1 and λ_3, this compound has the same absorbance A' and a maximum absorbance A'_2 at wavelength λ_2. The ratio R between these two values is calculated ($R = A'_2/A'$). Next, on the sample spectrum, the ratio of the absorbances is calculated at these same two wavelengths. On the condition that the baseline is a straight line with the same slope as the segment ab, the value of x can then be calculated (see Figure 9.26a):

$$x = (A_1 - A_3)\frac{(\lambda_3 - \lambda_2)}{(\lambda_3 - \lambda_1)} \tag{9.21}$$

From R and the ratio A_2/A_3 of the sample, y can be deduced, since the ratio of the absorbances, after correction for x and y, has the value R (Figure 9.26):

$$R = \frac{A_2 - (x + y)}{A_3 - y}$$

thus,

$$y = \frac{RA_3 - A_2 + x}{R - 1} \tag{9.22}$$

Subtracting x and y from A_2 leads to the calculation of the corrected absorbance:

$$A = A_2 - x - y$$

This method simply requires having a spectrum of the pure compound. It is independent of the concentration of the reference solution and is used in UV spectrum processing software.

9.12 RELATIVE ERROR DISTRIBUTION DUE TO INSTRUMENTS

For many contemporary spectrophotometers, it is possible to measure absorbance up to four or five absorbance units. These high values correspond to transmitted intensities that are extremely weak ($I/I_0 = 10^{-5}$ for $A = 5$), and thus the results could be less reliable. Therefore, it is important to consider the technical limits of the equipment.

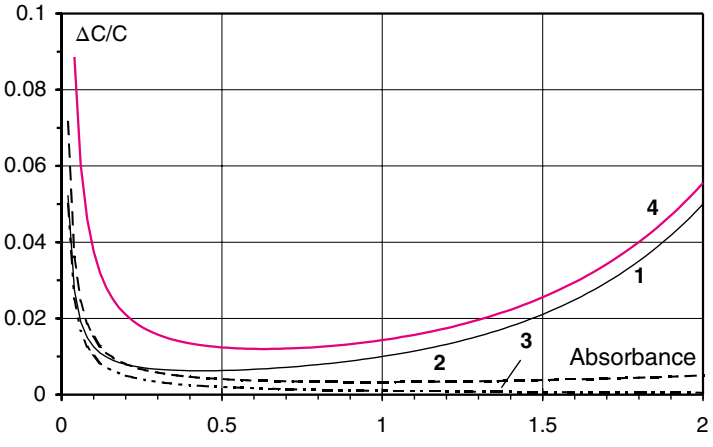


Figure 9.27 Curves representing the average of each of the errors (1 to 3) in absorbance measurements, as well as the relative standard deviation (RSD) of concentration determination as a function of absorbance resulting from their sum (4).

For transmittance measurements, three types of instrument error are recognized. They are independent in origin and thus potentially cumulative (Figure 9.27):

- *The background noise of the light source*. This term ΔT_1 is considered to be constant and independent of T: $\Delta T_1 = k$ (curve 1).
- *The background noise of the photomultiplier tube*. This term ΔT_2 varies as a function of T following a complex relationship as represented by the following equation (curve 2):

$$\Delta T_2 = k_2 \sqrt{T^2 + T} \tag{9.23}$$

- *Stray light* on the optical path of the instrument. This term is proportional to T: $\Delta T_3 = k_3\ T$ (curve 3).

As these sources of error are additive, the relative error in the concentration corresponds – if we use the three equations above – to a curve whose y-coordinate at each point is the sum of the y-coordinates of the three individual errors discussed above. This curve passes through a minimum generally located around $A = 0.7$. For quantitative measurements, it is therefore advisable to adjust the dilutions so that the absorbances are situated in this optimal range.

From the Beer–Lambert Law, it is also possible to link relative error on the measurement of concentration C to the relative error made on the transmittance T (curve 4).

$$\frac{\Delta C}{C} = \frac{1}{\ln T} \cdot \frac{\Delta T}{T} \tag{9.24}$$

9.13 DERIVATIVE SPECTROSCOPY

The principle of derivative spectroscopy consists of subjecting the original spectrum (called *zero-order spectrum*) to a mathematical treatment in order to replace it with its derivative of order *n* ($1 \leq n \leq 6$). This transformation completely changes the visual appearance of the graph by amplifying slight variations in slope, which show peaks or characteristic points, where before there were simple shoulders on the spectrum (Figure 9.28). Many current spectrometers are equipped with useful software in

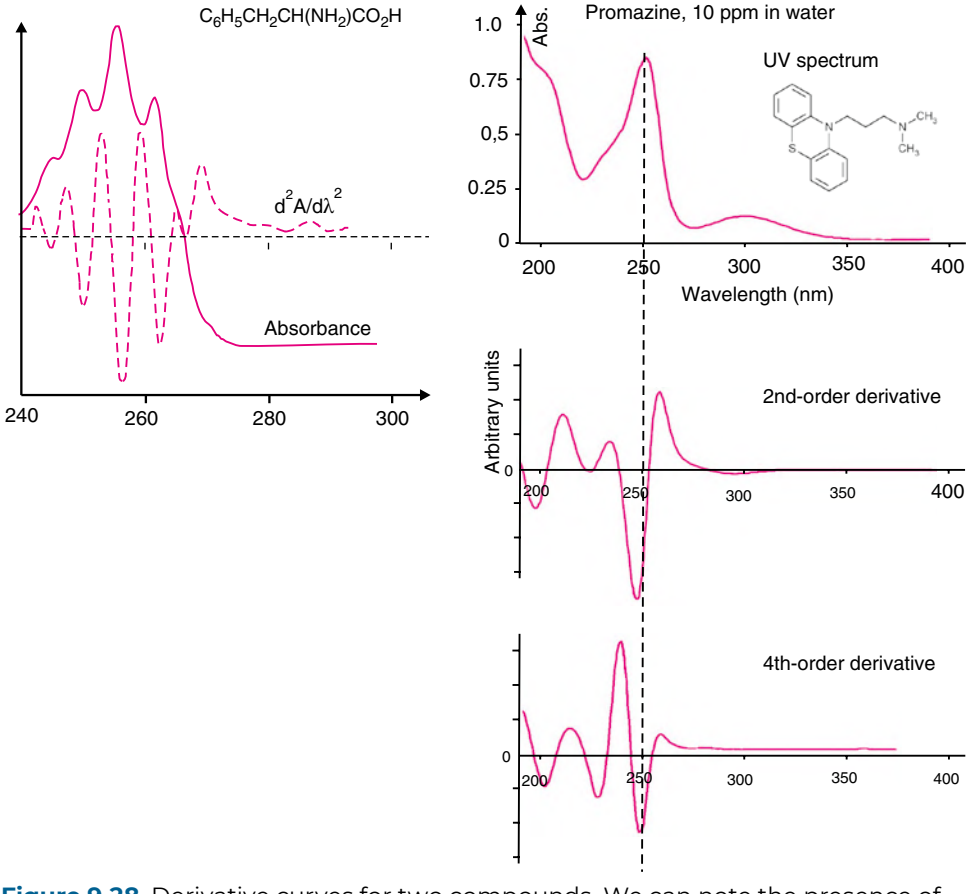


Figure 9.28 Derivative curves for two compounds. We can note the presence of a minimum of the second derivative at the position where there is a maximum on the zero-order spectrum, while for the fourth derivative, an almost complete inversion of the second-order graph can be found. (Source: Composed from a document by Joanna Karpinska (2012). Basic Principles and Analytical Application of Derivative Spectrophotometry, ISBN: 978-953-51-0664-7.)

qualitative mode to more specifically identify monotonous spectra or in quantitative mode to assay an analyte when its spectrum blends into the overall spectrum of the mixture in which it is found (presence of other compounds or an interfering matrix). For example, the second derivative shows a minimum where a maximum is located on the original spectrum and where the fourth derivative has a maximum.

In derivative spectroscopy, the Beer–Lambert Law is still applicable: the derivative values depend on the concentration and their additivity is conserved.

By taking the differential of this equation as a function of the wavelength, we obtain the first derivative:

$$\mathrm{d}A / \mathrm{d}\lambda = \left(\mathrm{d}\varepsilon / \mathrm{d}\lambda\right) lC \tag{9.25}$$

and more generally, we can obtain the *n*th derivative:

$$\frac{d^n A}{d\lambda^n} = \frac{d^n \varepsilon}{\left(d\lambda^n\right)} lC = D_x^n \tag{9.26}$$

where D_x^n represents the value of the *n*th derivative for compound x at wavelength λ.

Therefore, for a mixture of two compounds A and B, the derivative of the mixture is the sum of the derivatives of its two components:

$$D_{mixture}^n = D_A^n + D_B^n \tag{9.27}$$

Manually, the composition of the mixture can be calculated from the derivatives of the mixture at two different wavelengths at least (see Section 9.10.4), but the processing software, differing from one instrument to the next, will generate the result via its own specific method.

The amplitude between the maximum and the minimum values of the nth derivative curve is proportional to the absorbance value of the solution. The calibration curve is established from several standard solutions of various concentrations, to which the same computer processing is being applied, as well as to the sample solution. The choice of derivative order depends on the compounds that may interfere with the analyte.

While the zero-order spectra remain unchanged, whatever the instrument used, the same does not go for the different derivatives, since the calculation modes may vary. This process is of interest in two situations in which the absorbance is altered:

- If the spectrum of the sample in solution is shifted upwards because of a uniform baseline absorption, the derivative curve, which is sensitive only to variations in the slope of the zero-order curve, will not be affected.
- If light scattering occurs in the sample solution (which produces an absorption baseline that grows moderately as one approaches shorter wavelengths), again the derivative curves will be little affected (Figure 9.29).

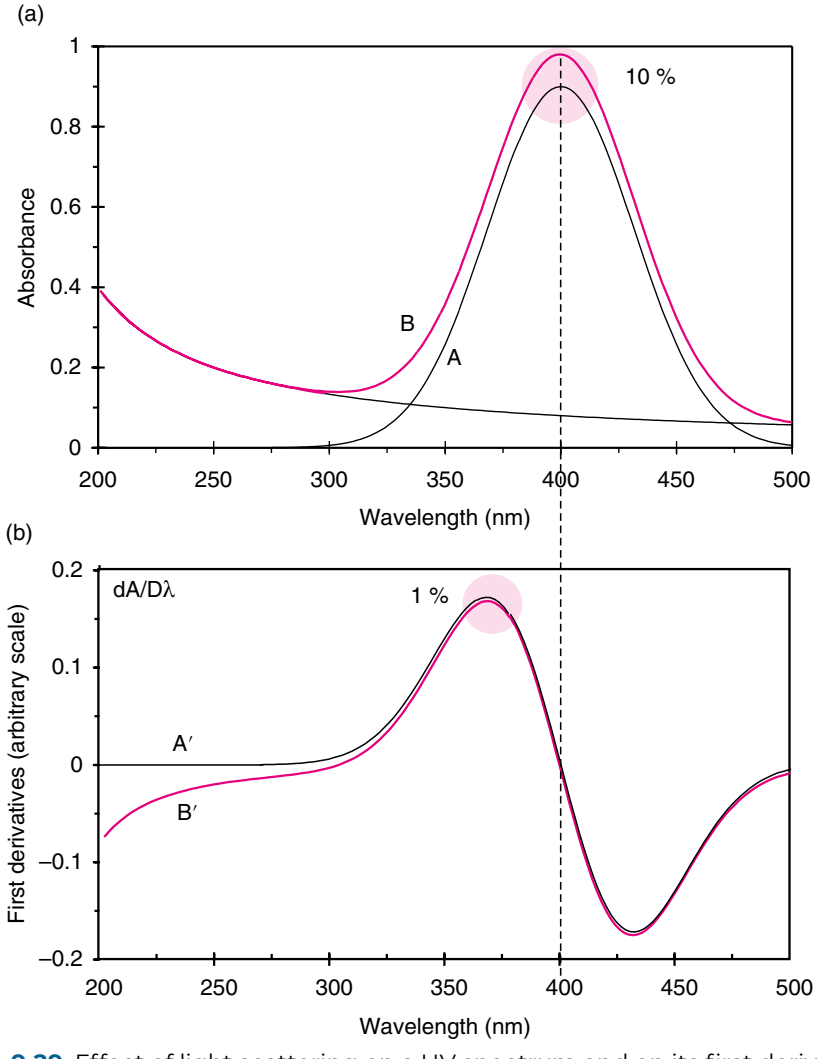


Figure 9.29 Effect of light scattering on a UV spectrum and on its first derivative curve. Comparison of the derivative graphs corresponding to the spectrum of a compound in solution without light scattering (a) and the same spectrum in the presence of scattering (b). Scattering can be seen to disrupt up to 10% of the absorbance on the regular graph, yet only around 1% of the value on the derivative curve (spectra modelled from functions corresponding to Gaussian curves).

The approximate modelling of a simple UV spectrum using a Gaussian curve demonstrates the expected appearance of successive derivatives. However, from the graph obtained from a real spectrum, notable differences can be observed as the graph varies depending on the calculation methods, which in turn depend on the software used.

9.14 VISUAL COLORIMETRY BY TRANSMISSION OR REFLECTANCE

Visual colorimetry has long been used as a simplified form of instrumental absorption spectrometry. Given its modest cost and its frequently astonishing precision, it is therefore widely employed.

The simplest colorimeters, used for routine measurements, are visual comparators (Figure 9.30) similar to Nessler tubes, in use for many years. These selective, ready-to-use tests require no instrumentation and complement established methods. Owing to their rapidity, they help fill the need for semiquantitative analyses of all kinds.

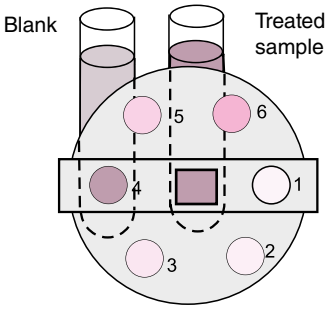
Nessler tubes consist of a series of graduated flat-bottomed glass tubes filled with different concentrations of known standard solutions of the same compound to be assayed, mixed with a derivatizing agent. The solution to be analysed is placed with the same agent in an identical tube and inserted into the series of previously mentioned standards. For this purpose, these tubes are placed over a source of white light, to be observed by transparency. The expected concentration is intermediate between those of the two standards whose colours are closest to that of the tube containing the analyte.

The innumerable tests involving strips that take on a more or less intense colour when soaked in the appropriate medium are among the many routine applications of colorimetry. However, the result, derived from a visual examination of the reflected light (reflectance), is more associated with reflectometry rather than transmission colorimetry. Techniques using the reflection of light by the sample (specular reflection and diffuse reflection) will be addressed when we examine the near and mid-infrared regions.

KEY POINTS OF THE CHAPTER

1. In the near ultraviolet or visible ranges, photon energy disturbs the electrons engaged in the bonds of the compounds being studied, whether these compounds are organic or inorganic. More specifically, when the electrons in a bond absorb a photon, the bond enters an excited state (orbital transition).
2. As there are a limited number of bond types, 'electronic' spectra in the UV-Visible region present absorption bands that are both wide and not very numerous for samples being studied in the condensed phase (liquid or solid). These are continuous spectra, whose absorption maxima are characteristic of structural particularities.





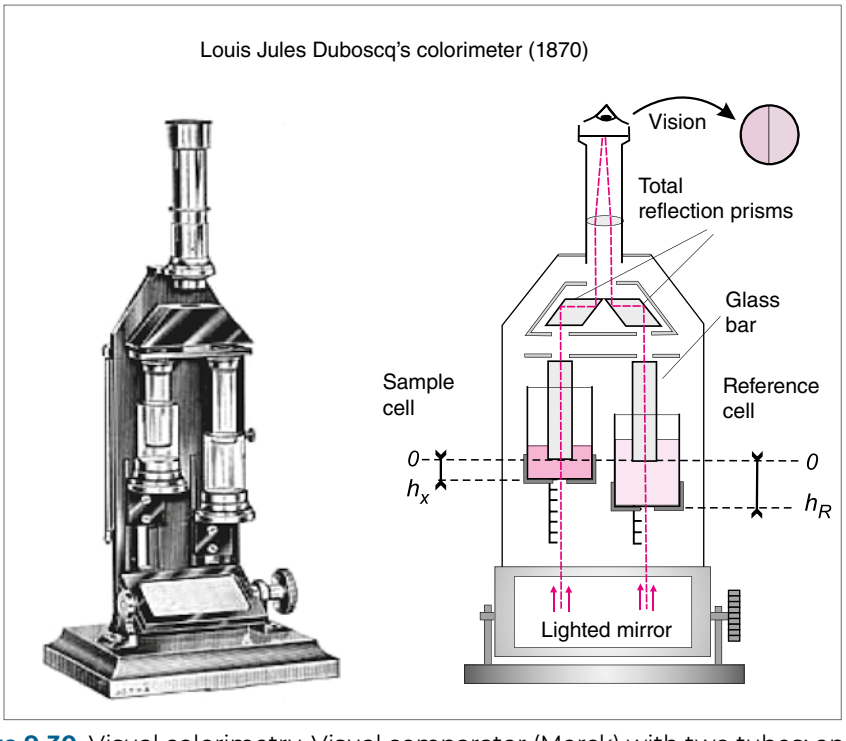


Figure 9.30 Visual colorimetry. Visual comparator (Merck) with two tubes: one for the sample, the other for the blank. In front of the latter, a disc representing the evolution of the hue as a function of analyte concentration is rotated until the two windows have the same hue. The Duboscq colorimeter was used until about the 1950s. Using a system of prisms, light intensities that pass through the two cuvettes (one of which contains the sample to be quantified and the other a reference standard) can be compared in the visual range. This instrument yields the expected concentration by use of the Beer–Lambert Law with comparison of the thicknesses. (Source: Reproduced courtesy of LaMotte Company.)

3. A UV-Visible spectrum leads to little structural information about compounds. However, a single absorption band is enough to proceed with a quantitative analysis. The absorption intensity is proportional to the number of absorbing molecules per unit volume and to a specific coefficient for each compound.
4. As this spectral region is very easily accessible from an equipment standpoint, it is used extensively for quantitative analyses based on the Beer–Lambert Law, which establishes a relation between absorbance and concentration and can be used for low concentrations.
5. The law of additivity of absorbance values governs the case of mixtures. Several methods, based either on basic corrections of the spectrum or on derivative spectroscopy, improve and expand the field of application of these assays.
6. An analyte with no absorption spectrum may still be quantified if, prior to the measurement, it is transformed into an absorbent derivative through a suitable quantitative chemical transformation. There are many applications in the fields of chemistry, biology, the environment, etc.
7. The UV-Visible spectral region has given rise to the largest number of instruments with varied designs in order to measure light intensity.
8. Derivative UV spectroscopy improves the identification of compounds by clearly highlighting poorly defined shoulders, inflection points or maxima. The plots obtained have zero-crossing points, whose precise values facilitate the tracking of singularities on the original spectrum.

PROBLEMS

1. A small amount of coffee (just enough to still see the bottom of the mug) is poured into a cylindrical mug. Then, water is added to the mug, with the bottom still visible. Does the hue become lighter, or does it remain the same?
2. The UV spectrum of a compound in solution is characterised by its absorption factor measured at two wavelengths: ε = 8,500 L·mol^{-1}·cm^{-1} at λ_1 and ε = 4,900 for λ_2. A solution containing this compound has an absorbance of 1.05 at λ_1 and 0.65 at λ_2. Does this solution contain only this compound in the pure state?
3. Exterior paints and varnishes must be protected from the effects of solar radiation which accelerate their degradation (photolysis and photochemical reactions).
   Given that: M = 500 g/mol; ε_{max} = 15,000 L·mol^{-1}·cm^{-1} at λ_{max} = 350 nm, what must be the concentration (expressed in g/l) of a UV additive M such that 90% of the radiation is absorbed by a coating with a thickness of 0.3 mm?
4. We want to determine the concentrations (mol/l) of two salts, $Co(NO_3)_2$ (A) and $Cr(NO_3)_3$ (B), in an unknown sample in aqueous solution. A spectrum is recorded in the visible region for each of these two compounds taken separately in aqueous

solution as well as for the sample solution to be analysed. The optical path of the cuvettes is 1 cm.

The absorbance values measured at 510 and 575 nm on the three spectra are as follows: compound A (1.5×10^{-1} M) presents an absorbance of 0.714 at 510 nm and 0.0097 at 575 nm. Compound B (6×10^{-2} M) presents an absorbance of 0.298 at 510 nm and 0.757 at 575 nm. The solution to be assayed has an absorbance of 0.40 at 510 nm and 0.577 at 575nm.

1. Calculate the four molar absorptivity values: $\varepsilon_{A(510)}$, $\varepsilon_{A(575)}$, $\varepsilon_{B(510)}$ and $\varepsilon_{B(575)}$.
2. Calculate the molarities (M) of the two salts A and B in the sample solution.

5. The concentrations (M) of $KMnO_4$ and $K_2Cr_2O_7$ in an aqueous solution are determined using the MLRA method.

Two reference solutions are prepared: one of $KMnO_4$ (1×10^{-4} M) and the other of $K_2Cr_2O_7$ (1×10^{-4} M). Spectra of the unknown solution and the two reference solutions are recorded between 250 and 400 nm. The optical path of the cuvettes is 1 cm.

Based on the five results found in the table below:

a. Find the equation of the regression line

$$A_{sample} / A_{permanganate} = f\left(A_{dichromate} / A_{permanganate}\right).$$

b. Calculate molar concentrations (M) of $KMnO_4$ and $K_2Cr_2O_7$ in the sample solution.

λ (nm)	Abs. MnO_4^- (ref.)	Abs. $Cr_2O_7^{2-}$ (ref.)	Sample
266	0.042	0.410	0.766
288	0.082	0.283	0.571
320	0.168	0.158	0.422
350	0.125	0.318	0.672
360	0.056	0.181	0.366

6. Fructose F reduces molybdenum (VI) into molybdenum (V) (molybdenum blue) as indicated below:

$$F + Mo(VI) \rightarrow Mo(V).$$

To determine the partial 'α' order of the reaction with respect to fructose, a large excess of ammonium molybdate is added to a fructose solution. The kinetics of this reaction are monitored by measuring, at various moments, the absorbance of the mixture at 720 nm, a wavelength at which only Mo(V) absorbs.

The following results are obtained:

t(min)	0	10	20	30	50	Infinity
A	0	0.175	0.302	0.384	0.494	0.600

Verify that the reaction is of the first order with respect to fructose, and calculate the apparent velocity *k'*.

SOLUTIONS

1. The hue remains the same (theoretically!) because, for a cylindrical form, the product $l \cdot C$ remains constant. For example, if we double the volume with water, the concentration will be cut in half, but the height will have doubled.
2. The ratio of the two ε values for the pure product is equal to 1.735 (8,500/4,900). As the ratio of the two absorbances is different (1.05/0.65 = 1.615), the solution therefore does not contain only the compound.
3. If 90% of the radiation is absorbed, $T = 0.1$.
   Therefore, A = 1 and $C = 2.22 \times 10^{-3}$ M.
   For M = 500 g/mol, $C = 1.11$ g/l.
4. **a.** Values of ε ($L \cdot mol^{-1} \cdot cm^{-1}$): $\varepsilon_{A(510)} = 4.76$; $\varepsilon_{B(510)} = 4.967$; $\varepsilon_{A(575)} = 0.647$; $\varepsilon_{B(575)} = 12.617$.
   b. $C_A = 1.2 \times 10^{-1}$ M and $C_B = 2 \times 10^{-2}$ M.
5. **a.** $y = 1.784x + 0.815$ where

$$y = A_{sample} / A_{permanganate} \text{ and } x = A_{dichromate} / A_{permanganate}.$$

   b. $C_{permanganate} = 8.15 \times 10^{-5}$ M and $C_{dichromate} = 1.78 \times 10^{-4}$ M.
6. As there is a great excess of molybdate, its concentration can be considered constant, the law of velocity becomes: $-\frac{d[F]}{dt} = k'[F]$ if we assume first-order kinetics with respect to fructose, i.e. $ln\frac{[F]}{[F]_{\dot{a}}} = -k' \cdot t$, which can be expressed in absorbance as follows: $\ln\frac{A_\infty - A_t}{A_\infty} = -k' \cdot t$. The values of k' are practically equal, which confirms the first order, and the average value of k' is 0.0348 min^{-1}.

Chapter 10

Infrared and Raman Spectroscopy

Infrared absorption spectroscopy is based on the study of electromagnetic radiation between 1 and 50 micrometres. This spectral range, divided into near, mid, and far infrared, is used to identify or quantify chemical compounds present in all sorts of samples. Raman spectroscopy is based on the scattering of a very small part of the light used to excite the sample. It provides comparable and additional information to that obtained from the mid or far infrared range, but the very low intensity of the

observed signals prevented the spread of this technique for a long time. These two nondestructive methods are accessible with a variety of versatile spectrometers and analysers designed to study target compounds. Many fields use these two techniques widely, ranging from industrial process analyses to field analyses as well as multipurpose laboratories.

Objectives

Correlate absorption bands and molecular vibrations
Describe a mid-IR spectrum
Give an overview of IR sources and detectors
Present several types of spectrometers
Interpret a mid-IR spectrum
Demonstrate the interest of harmonic bands in the near IR
Explain the principles of Raman scattering
Indicate a few applications with portable instruments

10.1 THE ORIGIN OF LIGHT ABSORPTION IN THE INFRARED REGION

In the near and the mid-infrared, the absorption of light by matter originates from the interaction between the radiation from a light source and the chemical bonds in the sample. More precisely, if the atoms situated at the two ends of a bond are animated by vibrational movement with respect to each other and are different from each other, they form an electric dipole that oscillates at a specific frequency. If such a nonsymmetrical bond is irradiated by a monochromatic light source whose frequency is the same as the dipole, then an interaction will occur with the dipole moment of the bond. In other words, the electrical component of the wave can transfer its energy to the bond, on the condition that the mechanical frequency of the bond and the electromagnetic frequency of the radiation are the same (Figure 10.1). This simplified approach can be used to explain that in the absence of a permanent dipole, which is the case with compounds having nonpolar bonds, such as oxygen (O_2), nitrogen (N_2), and chlorine (Cl_2), there will be no coupling with the electromagnetic wave, and therefore no absorption of energy will take place. In the mid-IR, these bonds are considered to be 'transparent'.

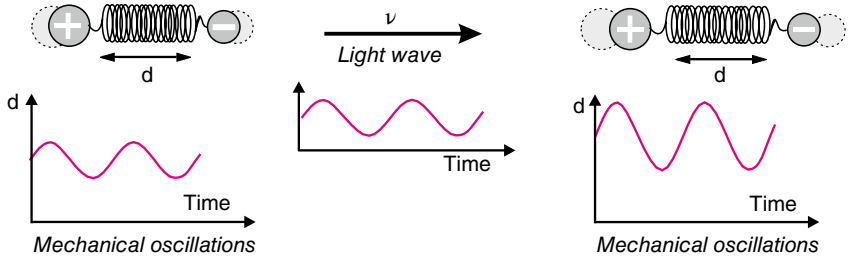


Figure 10.1 Mechanical interpretation of the interaction between a light wave and a polar bond. The mechanical frequency of the wave is not altered by the absorption of a photon, only its amplitude of oscillation increases.

10.2 PRESENTATION OF ABSORPTION IN THE INFRARED REGION

Infrared absorption of the sample, which varies with the radiation wavelengths selected, is generally presented in the form of a *spectrum*, obtained with a spectrometer. The *y*-axis of the graph shows the ratio of the transmitted intensities, i.e. with and without sample, calculated for each wavelength marked on the *x*-axis. This ratio is called the *transmittance* (*T*). On the graph, it is often replaced by the percent transmittance (%*T*) or by *absorbance* (*A*), i.e. the base-10 logarithm of the reciprocal of the transmittance: $A = \log(1/T)$. If the analysis has been conducted using reflected or diffused light, then *pseudo-absorbance* units are used (see Section 10.9.3). Finally, wavelengths are frequently replaced by their equivalent expressed in *wavenumbers* $\bar{\nu}$ whose units are in cm^{-1} or kaysers (Eq. (10.1)). Figure 10.2 corresponds to a spectrum recorded in the mid-IR, between 2.5 and 25 μm.

$$\bar{\nu}_{cm^{-1}} = \frac{1}{\lambda_{cm}} \tag{10.1}$$

10.3 ROTATIONAL–VIBRATIONAL BANDS IN THE MID-IR REGION

Except when the temperature is 0 Kelvin, molecules are in perpetual motion and each one of them possesses three *degrees of freedom*, which refer to the three classic Cartesian coordinates. All of these movements confer a combined mechanical energy upon each isolated molecule. The theory is postulated that the total molecular energy arises from the sum of the independent, though quantified,

terms named rotational energy E_{rot}, vibrational energy E_{vib}, and electronic energy E_{elec}:

$$E_{tot} = E_{rot} + R_{vib} + E_{elec} \tag{10.2}$$

The values of these energies are very different and, according to the Born–Oppenheimer principle, they can vary independently of each other.

In the mid-IR region, a source emitting a radiation of 1,000 cm^{-1} corresponds to a photon energy $E = h\bar{c}\bar{\nu} = 0.125$ eV. If such a photon is absorbed by a molecule, its total energy will be increased by this energy. Theoretically, the term E_{vib} will be modified, but the term E_{elec} will not change.

In practice, as ordinary samples are usually in the form of condensed liquids or solids (pure or in solution) and not in the form of isolated species, numerous dipole–dipole interactions are produced between the species present, which perturb both the energy levels and the absorption wavelengths. Thus, a spectrum is always in the form of broadened signals called bands, possibly extending over tens of cm^{-1} and which common apparatuses are unable to separate into individual transitions (Figure 10.2).

Only small diatomic or triatomic molecules sampled in the gaseous state and at low pressures can lead to spectra containing narrow regular bands whose positions enable us to determine the various energy states predicted by theory (Figure 10.3).

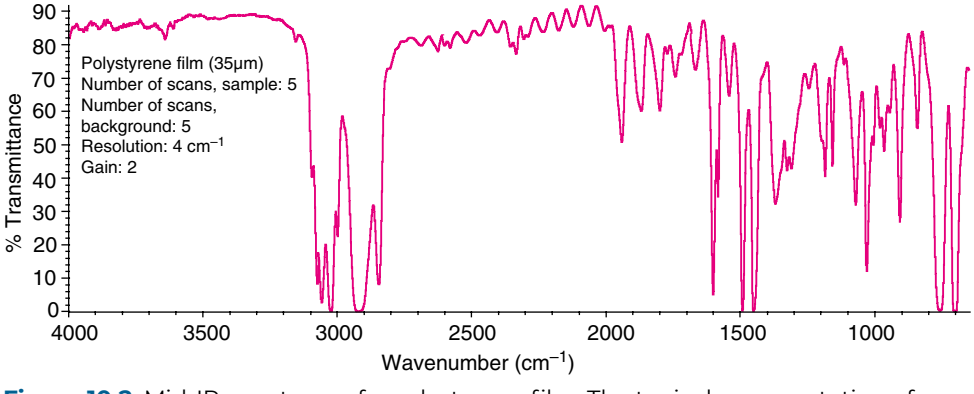


Figure 10.2 Mid-IR spectrum of a polystyrene film. The typical representation of an infrared spectrum with a linear-scale *x*-axis in cm^{-1} (see Eq. (10.1)) on the entire spectral range and with percent transmittance on the *y*-axis (for easier observation of the right part of the spectrum, a scale change is often made at 2,000 cm^{-1} (see Figure 10.9). Transmittance is sometimes replaced by absorbance *A* ($A = -\log T$). The scale, in cm^{-1} (or kaysers), is linear in terms of energy ($E = hc/\lambda$) and will decrease from left to right (i.e. from high to low energy).

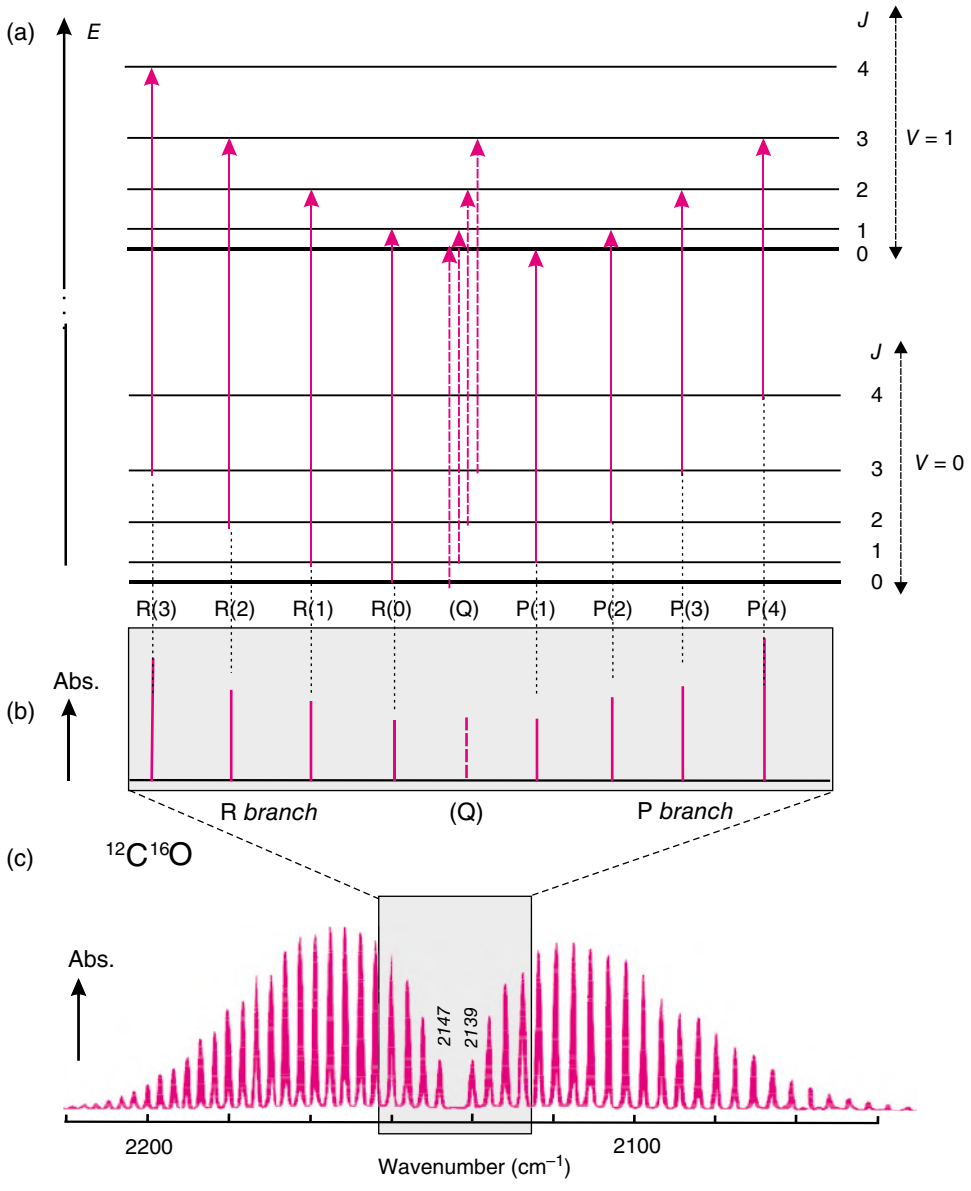


Figure 10.3 Rotational/vibrational levels of a diatomic molecule and corresponding spectrum. Rotational/vibrational band of carbon monoxide. *V* and *J* are the quantum numbers of vibration and rotation. The fundamental vibration corresponds to $\Delta V = +1$ and $\Delta J = +1$. (a) A rotational–vibrational band corresponds to all of the allowed quantum transitions. If the scale of the diagram is in cm^{-1}, the arrows correspond to the wavenumbers of the absorptions. (b) The *R* branch corresponds to $\Delta J = +1$ and the *P* branch to $\Delta J = -1$. They are situated on either side of band *Q* (transition $\Delta J = 0$), absent from the spectrum because it is forbidden for diatomic molecules. (c) Vibrational–rotational band of carbon monoxide (pressure of 1,000 Pa). The various lines are in agreement with the selection rules. The interval between successive rotational lines is not constant due to the anharmonicity factor.

10.4 MECHANICAL MODEL FOR VIBRATIONAL INTERACTIONS BETWEEN ATOMS

To model the vibrations between two atoms, we use a harmonic oscillator, which comprises two masses m_1 and m_2 that are able to oscillate without friction around their equilibrium position along the internuclear axis. The centre of mass is fixed, as the two masses vibrate with respect to this point, as if linked by a spring (Figure 10.4). If the two masses are displaced by a value x_0 with respect to the equilibrium distance R_e and then the system is released, it will begin to oscillate with a period that depends on the stiffness constant of the spring, named *force constant k* (N/m), and on the masses involved.

It can be demonstrated that this model is equivalent to a mass μ (reduced mass of the system in kg) vibrating around a fixed point, with a frequency given by Hooke's law:

$$\nu_{vib} = \frac{1}{2\pi}\sqrt{\frac{k}{\mu}} \tag{10.3}$$

where

$$\mu = \frac{m_1 m_2}{m_1 + m_2} \tag{10.4}$$

The vibrational energy of this simple model can vary in a continuous fashion. Following a short elongation x with respect to the equilibrium distance R_e, E_{vib} becomes:

$$\frac{dE_{Vib.}}{dx} = -k \cdot x \quad \text{or} \quad E_{Vib.} = \frac{1}{2}kx^2 \tag{10.5}$$

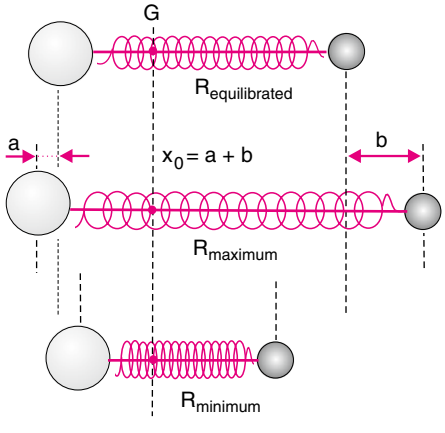


Figure 10.4 A diatomic molecule represented in the form of a harmonic oscillator. The term harmonic oscillator comes from the idea that the elongation is proportional to the force exerted, while the frequency ν_{vib} is independent of it.

Two terms compose it: the potential energy and the kinetic energy of the system, whose sum remains constant over time.

For a bond between two atoms, the wavenumber of a vibration will be written as:

$$\bar{\nu} = \frac{1}{2\pi c}\sqrt{\frac{k}{\mu}} \tag{10.6}$$

where

$$\bar{\nu} = \frac{1}{\lambda} = \frac{\nu}{c} \tag{10.7}$$

The above model of the chemical bond linking two atoms is applicable on the condition that the quantum theory governing species of atomic dimensions is employed. According to quantum mechanics, the vibrational energy of a harmonic oscillator is quantified according to the vibrational quantum number V (V = 0, 1, 2, …, positive whole number):

$$E_{\text{vib}} = h\nu\left(V + \frac{1}{2}\right) \tag{10.8}$$

Before absorption, a very large majority of molecules are found in the ground vibrational state V = 0 (of the ground electronic state E_0), i.e. $E_{(\text{vib})0} = 1/2h\nu$ (energy at 0 K).

The difference between two successive vibrational levels (ΔV = +1) is consequently equal to ν_{inc}, which indicates the frequency of the incidental radiation:

$\Delta E = h\nu_{inc.}$

This difference corresponds to a 'single quantum' transition. This is an allowed transition, which means that the bond may absorb a photon of this same energy $h\nu_{\text{inc}}$, thus leading to an increase in the vibrational energy of the molecule by this quantum (Figure 10.5).

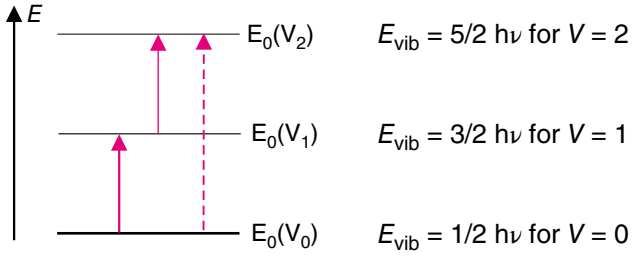


Figure 10.5 Diagram of the vibrational energy levels of a bond. The transition V = 0 to V = 2 corresponds to a weak harmonic band. Bearing in mind the energies of photons in the mid-IR, it can be calculated that the first excited state (V = 1) is 10^6 less populated than the ground state. Harmonic transitions are exploited in the near IR region.

In other words, if the frequency of the incident light radiation ($\nu_{inc.}$) is identical to the vibrational frequency of the bond, the incident wave and the molecule will be coupled, which will then lead to the absorption of the energy transported by the electromagnetic wave.

For different types of bonds, this energy difference corresponds to the energy transported by infrared radiation located between 4,000 and 500 cm^{-1} (2.5–20 μm).

In addition to the ordinary transition $\Delta V = +1$, the transition corresponding to $\Delta V = +2$ ('forbidden' by this theory) appears weakly when the fundamental absorption band is particularly strong (for example, in the case of elongation vibration in organic compounds containing a carbonyl bond C=O (e.g. ketones or aldehydes). Furthermore, no valence vibrational mode is situated higher than 3,960 cm^{-1} (H–F). This is the wavenumber that determines the boundary between the mid- and the near infrared regions. However, for reasons of utility, many current spectrometers accommodate both the near and mid-IR, thus enabling coverage up to 8,000 cm^{-1}.

10.5 REAL COMPOUNDS

The harmonic oscillator model does not take into account the real nature of chemical bonds, which are far from being perfect springs. The force constant k decreases if the distance between the atoms is stretched but increases greatly if attempts are made to superimpose the atoms. More precise models employ correction terms related to the anharmonicity of experimentally observed oscillations.

Furthermore, in the mid-IR, photon energy is sufficient to modify the quantified terms E_{vib} and E_{rot} in Eq. (10.2). The absorption bands obtained thus result from transitions involving both vibrational and rotational energy levels of a given electronic state. These are therefore vibration–rotation spectra; in other words, several dozen distinct rotational transitions accompany each vibrational transition.

For the simplest molecules, it is possible to interpret specific aspects of the absorption bands. Experience and theory have enabled rules of the allowed transitions to be drawn up. Small molecules such as carbon monoxide and hydrogen chloride (Figure 10.3) have been intensely studied from this point of view.

Equation (10.9) leads to the definition of possible absorptions:

$$\Delta E_{VR} = \left(E_{VR}\right)_2 - \left(E_{VR}\right)_1 = \Delta E_{vib} + \Delta E_{rot} \qquad (10.9)$$

A polyatomic molecule of n interacting atoms can be defined by $3N$ degrees of freedom (see Section 10.3). Since three degrees of freedom are required to define its centre of gravity (translational movement of the molecule), the remaining $3N - 3$ are internal degrees of freedom. In general, three are required to define the rotation of the entire molecule. The remaining $3N - 6$ movements correspond to the ground vibrations of the molecule (reduced to 3N – 5 for linear molecules, in which a rotation

around the molecular axis does not constitute a degree of freedom). A molecule containing 30 atoms will have 84 normal modes of vibration, happily not all active, since there must first be variation in the dipole moment. Such a high number of energy levels, to which are added combinations of levels and harmonics, necessarily leads to complex and unique spectra for each type of compound.

10.6 CHARACTERISTIC BANDS FOR ORGANIC COMPOUNDS

The above rules have been applied to all kinds of compounds in order to exploit the use of infrared spectroscopy as a method of structural analysis.

Empirically, a correlation has been observed between the positions of the absorption maxima of certain bands and the presence within a molecule of organic functional groups or of specific structural features within the species examined (Table 10.1). This property comes from the fact that each organic functional group corresponds to a standard set of several atoms. Spectral characteristics shared by all molecules with this function can be seen.

For a given bond, the force constant k (Eq. (10.3)) does not vary significantly from one molecule to another (typically for a single bond, $k \approx 500$ N/m, for a double bond $k \approx 1{,}000$ N/m, and for a triple bond $k \approx 1{,}500$ N/m). Moreover, calculation reveals that the reduced mass of a given functional group tends toward a limit value as the molecular mass increases (see Eq. (10.4)).

The best-known molecular movements are stretching vibrations (symmetrical and asymmetrical) and angular bending vibrations (Figure 10.6).

The absorption bands related to stretching vibrations, shared by all molecules with related organic functional groups, are mainly found in the 1,500–4,000 cm^{-1} zone. The study of this region has led to an initial approach to functional analysis of a molecule, i.e. to describe all chemical groups constituting said molecule.

In contrast, the part of the spectrum below 1,500 cm^{-1}, where the stretching bands of single bonds are found, corresponds mainly to the spectral range of very many angular bending bands of bonds and backbones, which differ for each compound, and it is rarely possible to interpret them all.

For instrumental reasons, the spectral range that extends from 1 to 2.5 μm (4,000–10,000 cm^{-1}, though it has no formal limits) was for a long time poorly understood. However, progress with both detectors and transparent fibre optics in the near IR region, coupled with advances in chemometric methods, has enabled the development of this region.

Near infrared (*NIR*) spectra are often presented as curves without clearly defined peaks. Nevertheless, each recording is representative of the sample and therefore can be used as a fingerprint. For organic compounds, the absorption bands in the near IR

Table 10.1 Correlation chart in the mid-infrared region between functional groups and absorption bands.

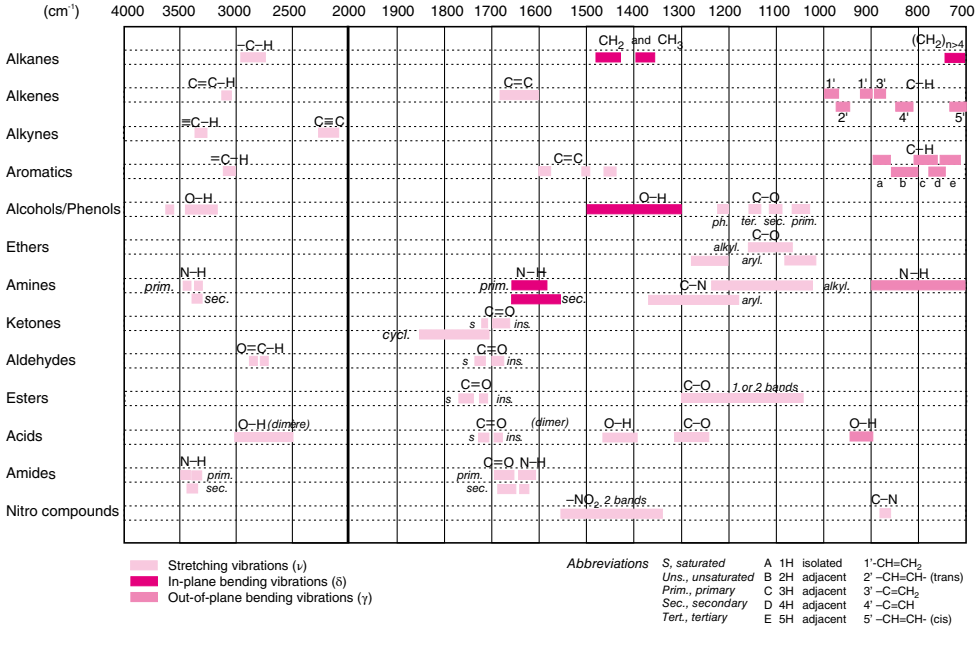
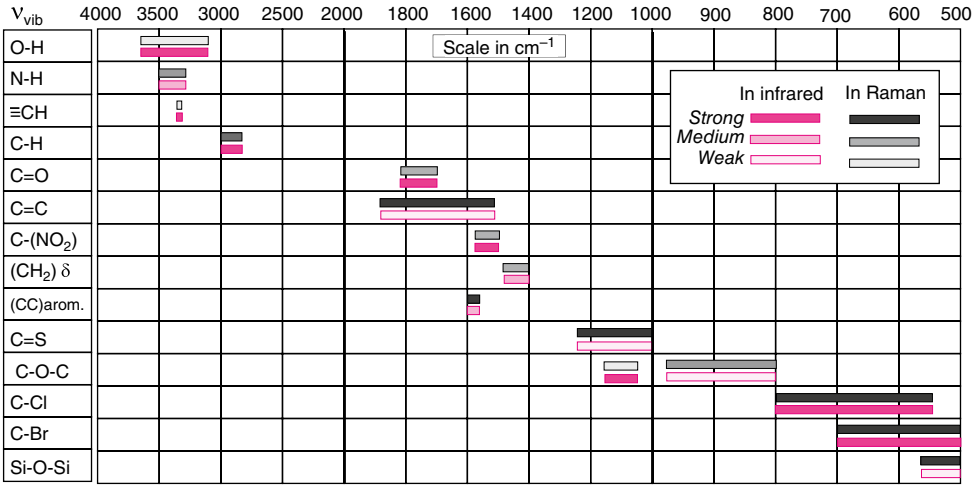


Table 10.2 Comparison of absorption (IR) and scattering (Raman) intensities for several chemical bonds.



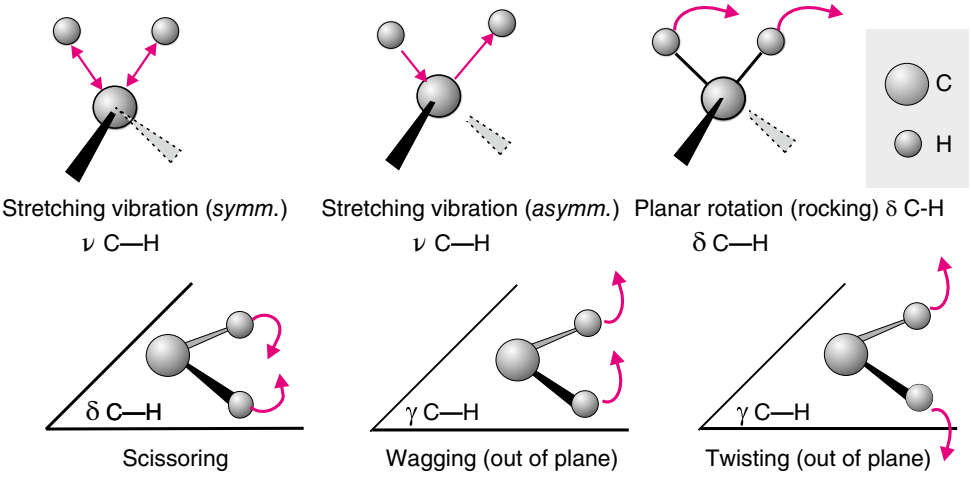


Figure 10.6 Molecular vibrations of CH_2. Characteristic stretching and bending vibrations in and out of the plane (*oop*). In IR spectroscopy, the position and the intensity of the bands are modified by associations between molecules, solvent polarity, etc.

originate from either harmonic bands or a combination of the fundamental vibrations of C–H, O–H and N–H bonds (which are situated around 3000–3600 cm^{-1}).

For carbonyl compounds, the first two harmonics of the fundamental vibration of the C=O bond ($\overline{\nu_0}$ = 1,700 cm^{-1}) appear around 3,400 ($2\overline{\nu_0}$) and 5,100 ($3\overline{\nu_0}$) cm^{-1}, respectively. The combination bands result from the interaction of two or more modes

of vibration for a given functional group and are superimposed upon the preceding bands. The transition energy is approximately the sum of those which have given rise to it.

Chloroform, which absorbs at 3,020 cm^{-1} and 1,215 cm^{-1}, has one combination band at 4,220 cm^{-1}, which approximately represents the sum of the two preceding values.

10.7 INFRARED SPECTROMETERS AND ANALYSERS

These instruments can be divided into two categories: *Fourier transform spectrometers*, which conduct a simultaneous analysis of the whole spectral region from interferometric measurements, and numerous *specialized analysers* for the second category. *Dispersive-type spectrometers* are also used for the near IR.

The first category uses a Michelson-type interferometer or similar device, coupled with a specialized microprocessor to calculate the spectrum digitally, while the second category uses filters or a monochromator installed in specific line-ups as a function of the spectral range studied (Figure 10.7).

For a long time, mid-IR spectrometers were constructed on the principle of double-beam design. This fairly complex optical assembly, which remains in use for the UV/Visible, progressively yields the spectrum of the sample in real time, through the measurement of the transmittance, calculated for each wavelength, by comparing the intensity of light arising from two distinct optical paths, one serving as reference and the other passing through the sample. To this end, the light emanating from the source is separated by a series of mirrors into two beams, one of which passes through the sample compartment and the other through the reference chamber. For each small wavelength interval, defined by the monochromator, the light following each of the two paths arrives at the detector alternatively, thanks to a mirror which rotates about ten times per second. The ratio of the signals obtained, at practically the same instant, corresponds to the transmittance for the wavelength in question.

10.7.1 Fourier Transform Infrared (FTIR) Spectrometers

An FTIR spectrometer contains a single-beam optical assembly with, as an essential component, an interferometer – often of the Michelson type – located between the source and the sample (Figure 10.7a).

The radiation emanating from the polychromatic source impacts on a semi-reflective plate known as the *beam-splitter*, inclined at 45° with respect to the direction of incidental radiation propagation and placed at the centre of the assembly. It is made of a semitransparent film of germanium deposited on a potassium bromide (KBr) support. This device divides the original beam into two halves, one of which is directed towards a fixed mirror and the other towards a moving mirror whose distance from the beam-splitter varies. Recombined, these two beams follow the same optical path passing through the

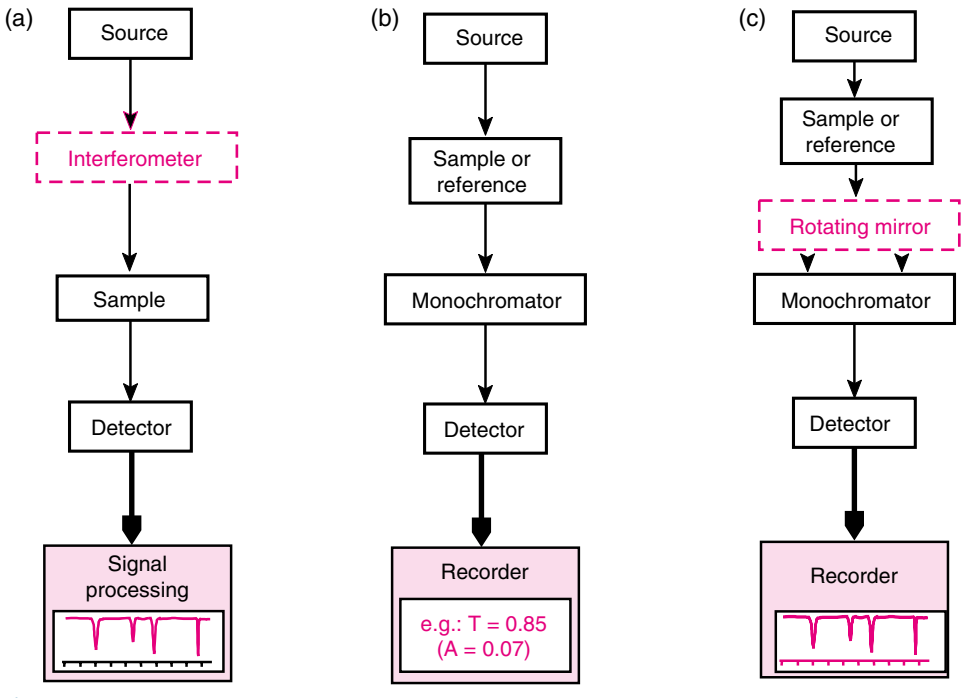


Figure 10.7 Configurations of spectrometers and analysers in the infrared region. (a) Single-beam Fourier transform model. (b) Single-beam analyser containing a fixed monochromator or a filter used when a measurement at a single wavelength will suffice. (c) Dispersive spectrometer, double-beam system. In contrast with spectrophotometers in the UV/visible range, the sample, located prior to the monochromator, is permanently exposed to the full radiation of the source. The energy of the photons in this region is insufficient to break the chemical bonds and thus degrade the sample.

sample before reaching the detector, which measures the overall light intensity received. This is a multiplexing process applied here to the field of optical signals.

The heart of the Michelson interferometer is a moving mirror, the sole moving component, which oscillates between two extreme positions. When its position is such that the pathways travelled by the two beams have the same length, the light composition of the beam leaving the interferometer is identical to that which enters. In contrast, when the moving mirror changes from this particular position, the light leaving has a spectral composition that depends upon the phase difference between the two paths: the signal transmitted over time by the detector is recorded in the form of an *interferogram*:

$$(\text{Total Int.}) = f(\delta)$$

where δ represents the path difference between the two beams (Figure 10.8).

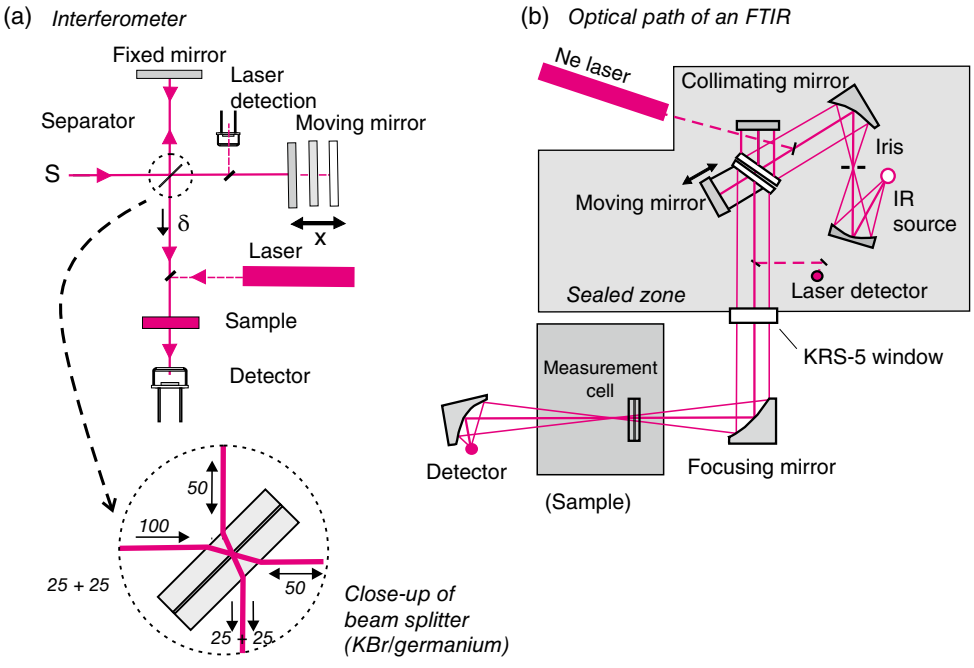


Figure 10.8 The optical assembly of a Fourier transform apparatus. (a) 90° Michelson interferometer with, below, some details of the beam-splitter. (b) The optical diagram of a single-beam spectrophotometer (picture of Shimadzu model 8300. Source: Shimadzu Corporation). A low-power He/Ne laser is used as an internal standard (632.8 nm) in order to locate with precision the position of the mobile mirror by means of an interference method (this second sinusoidal interferogram follows the same optical pathway).

Management of the optical set-up and data acquisition are controlled by a specific electronic interface. During the displacement of the mirror, an ADC converter, linked to the detector, samples the interferogram in the form of thousands of data points. Each of these values corresponds to a mirror position and represents the overall intensity that passes through the sample. This is the second term of a linear equation in which the unknowns correspond to the amplitudes of the *n* different wavelengths (assumed to be a finite number) for the mirror position in question and after absorption by the sample. From these thousands of values, a specialized microprocessor executes the calculation of a giant matrix, using a specific, rapid Fourier transform algorithm, known as the Cooley algorithm, leading to the amplitudes of each wavelength of the spectral band studied.

Bearing in mind a resolution factor imposed by the method of calculation, the classical representation of the spectrum is obtained, $I = f(\lambda)$ or $I = f(\bar{\nu})$. According to Nyquist's theorem, at least two points per period are required in order to calculate a given wavelength of the spectrum. To obtain the spectrum of a sample equivalent to

that obtained with a double-beam spectrometer, two spectra of transmitted intensities are recorded: the first *without the sample* (absorption background) and the second *with the sample*. The conventional spectrum, in % *T*, is obtained from the point-by-point calculation of the ratio of emittances of these two spectra (Figure 10.9).

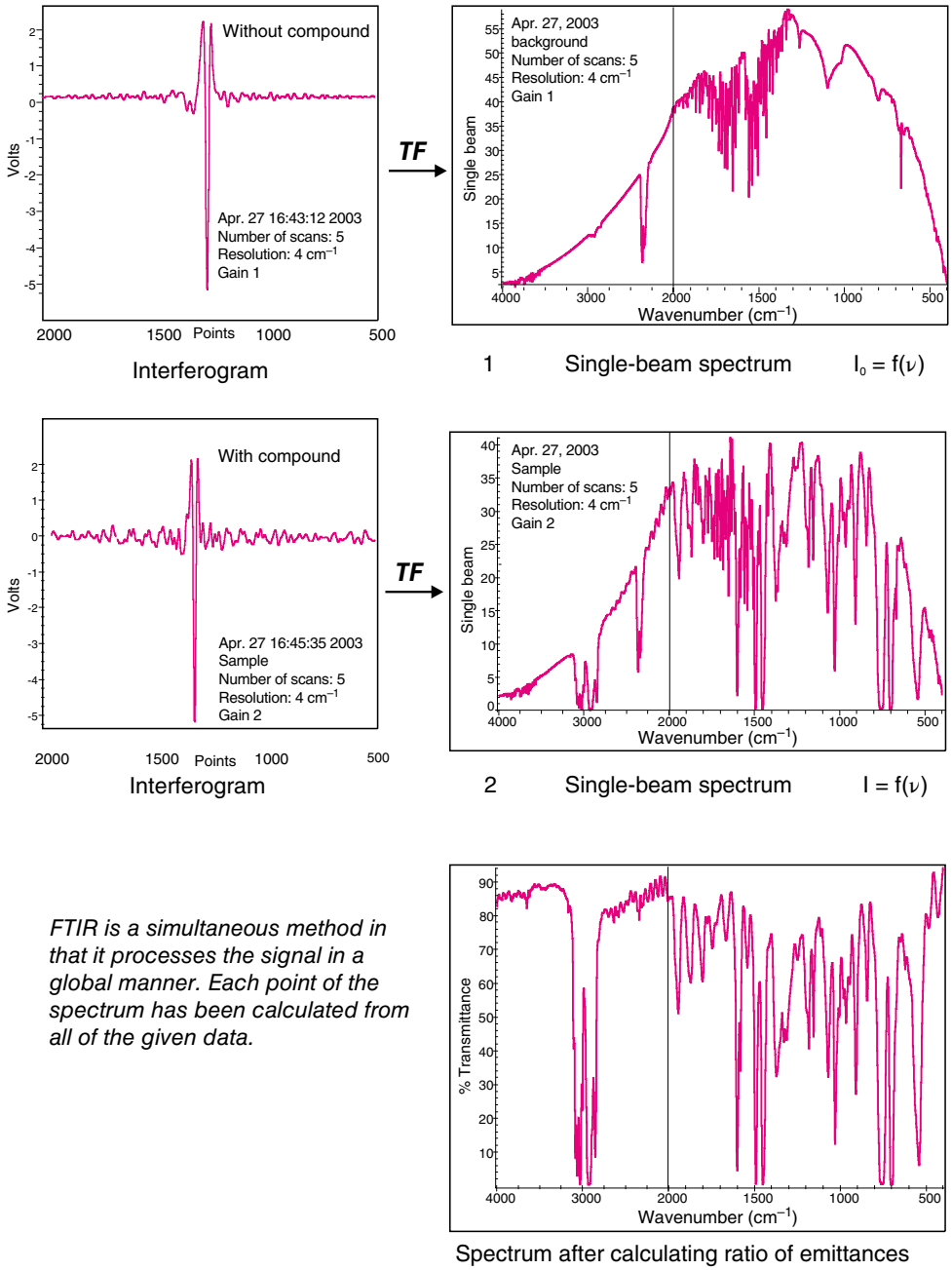
> The French mathematician J.-B. Fourier (1768–1830) could never have imagined how famous he would become with the advent of the computer. The principle of his calculations, published in a treatise concerning the propagation of heat in 1808, is applied in many scientific software packages for the treatment of spectra (acoustical and optical) and images. Legend has it that he developed these calculations when Napoleon's army asked him to improve the dimensions of their cannons. In general terms, a transform is a mathematical operation that permits data to be changed from one type of measurement to another (for example, from time to wavelength).

If two comparable instruments (source, detector, resolution, etc.) are being considered, multiplexed spectroscopy, illustrated here by Fourier transform MIR (mid-infrared) spectroscopy (also adapted to near infrared), presents many advantages:

- Improvement of the signal/noise ratio (*Fellget's advantage*): the multiplexing resulting from the use of an interferometer leads to a much higher signal/noise ratio than for a grating spectrometer. Moreover, considering the extremely short acquisition time as compared to a dispersive system, this ratio can still be improved by accumulating a large number of spectra.
- Increase of the light flow (*Jacquinot's advantage*): the light flow of an interferometer is 20–300 times more intense (depending on the wavenumber in question) than that of a dispersive spectrometer, thanks to a larger optical aperture available at the interferometer inlet. This provides a better signal to the detector, which receives more energy.
- Better precision in wavelength (*Connes' advantage*): the wavenumbers are calculated by self-calibrating from the stable and precisely known wavelength of the He–Ne control laser emitting at 632.8 nm (15,802.7 cm^{-1}). This calculation confers very good reproducibility to the system, which makes it possible to compare spectra.
- Reduction of the parasitic light (relative to any unexpected radiation coming from the detector), which is very low in an interferometer.
- Resolution is constant and better across the entire region under study.

10.7.2 Mid-Infrared Analysers

Many small stand-alone instruments, used for analyses relating to personal protection and other specific targeted applications (e.g. breathalyser tests) are based upon absorbance measurements in the mid-IR or near IR region. These



FTIR is a simultaneous method in that it processes the signal in a global manner. Each point of the spectrum has been calculated from all of the given data.

Figure 10.9 Sequence for obtaining a pseudo-double-beam spectrum with a Fourier transform infrared spectrometer. The apparatus records and memorizes two spectra, which represent the variations of I_0 (blank) and I (sample) as a function of the wavenumber (these are emission spectra 1 and 2). Next, the conventional spectrum, identical to that of a double-beam instrument, is calculated from the ratio $T = I/I_0$ for each wavenumber. Atmospheric absorption of CO_2 and H_2O (stretching and deformation bands observed in spectrum 1 with no compounds) is eliminated in this way. The illustrations correspond to the spectrum of a polystyrene film.

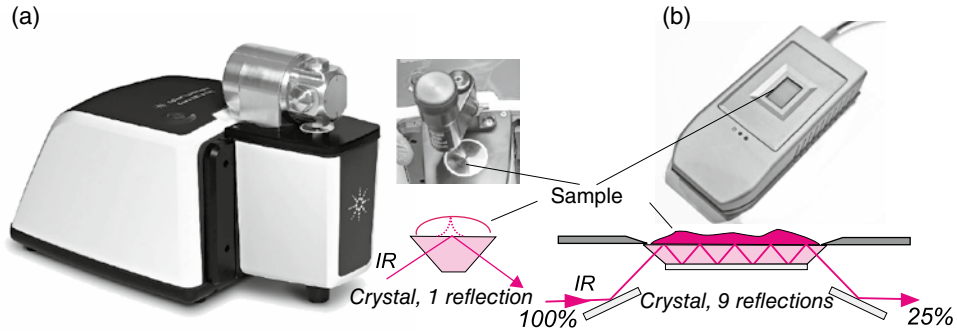


Figure 10.10 Portable FTIR analyser. Small size apparatus allowing the study of liquid samples or powdered solids with the ATR device (see Section 10.9.3). (a) Cary 630 model; single reflection on diamond crystal. (Source: courtesy of Agilent Technologies.) (b) Midir Viavi model; nine reflections on SZn crystal. (Source: Courtesy of Allied Scientific.)

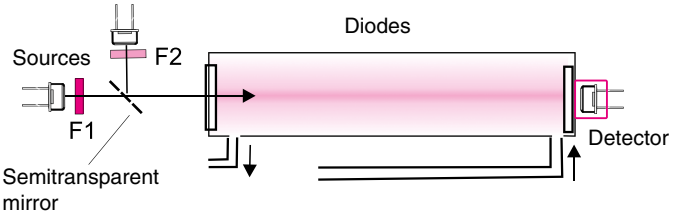
instruments have various designs, ranging from filter instruments to portable FTIR devices (Figure 10.10). They are suited for measuring very low absorbances (1×10^{-5}) and can reliably measure concentrations on the order of ppm for hundreds of gases such as carbon monoxide and dioxide, ammonia, methanal, ethanol, methane, and many others in military, customs or police applications. Measurements of carbon monoxide and dioxide in motor car gas exhaust are carried out with this method. Carbon monoxide is measured by recording the absorbance at 2,170 cm^{-1} (Figure 10.3) while carbon dioxide is measured at 2,350 cm^{-1} (Figures 10.10 and 10.11). For ethanol, the characteristic absorbance of primary alcohols at 1,050 cm^{-1} is used.

The source can be a laser diode, which emits at a precise wavelength, or a simple filament in a silica sheath associated with an interferential filter. The detector is pyroelectric. These *infrared colorimeters* are competitive with instruments based upon other principles (e.g. amperometric detection, see Chapter 20).

10.8 SOURCES AND DETECTORS USED IN THE MID-IR REGION

10.8.1 Light Sources

For near or mid-infrared spectrometers, continuous radiation sources consist either of a QTH bulb (quartz envelope, tungsten filament, and halogen atmosphere, about 3,000 K), reserved for the near infrared domain, or of ceramics or metal oxides



Mid-IR

Gas	cm^{-1}
CO_2	2349
CH_4	3067
NH_3	971

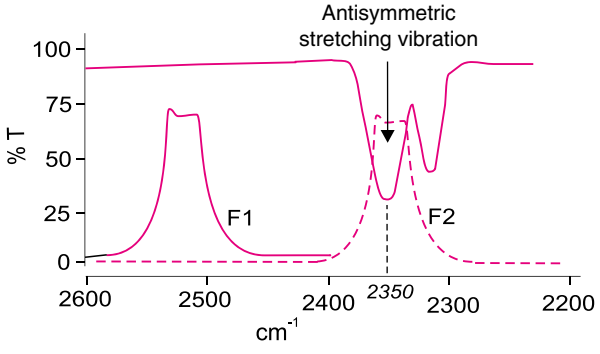


Figure 10.11 Nondispersive gas analyser. The cell containing the gas to be quantified is alternately traversed by two beams of different wavelengths (semitransparent mirror) defined by the choice of two filters, one (F1) located in a zone with no absorption by the gas (reference path) and the other (F2) corresponding to a specific wavelength of the gas. The absorbance obtained from the two measurements leads to the desired concentration. Illustration with the carbon dioxide spectrum on which the bandwidths of the two filters F1 and F2 are superimposed.

brought to temperatures between 1,000 and 2,400 K for the mid-infrared region (Figure 10.12).

These include the Globar source, which is progressively being abandoned; it is composed of a silicon carbide bar (SiC, carborundum), used as a resistor and heated up to 1,500 K by Joule effect. There are also oxidized metal coils, used to improve their emissivity compared to black body radiation. Heated to 1,300 K, without any protective envelope (Nernst lamps), they emit in a large spectral domain spanning from visible to about 25 micrometres, and even reaching an intensity maximum whose approximate wavelength is given by the Wien law, $\lambda = 3000/T$ (λ in μm and T in Kelvin).

In addition to the previously cited sources, there are others, intended for many field instruments, various analysers, and instruments on production lines, many of which make measurements at a single wavelength. These are lower power sources made up either of emissive diodes (LED) or IR lasers with fixed wavelength or which can scan a range of several micrometres, such as carbon dioxide lasers or quantum cascade lasers (QCL). These new sources, which simplify the optical part of the instruments, are increasingly used in industrial installations.

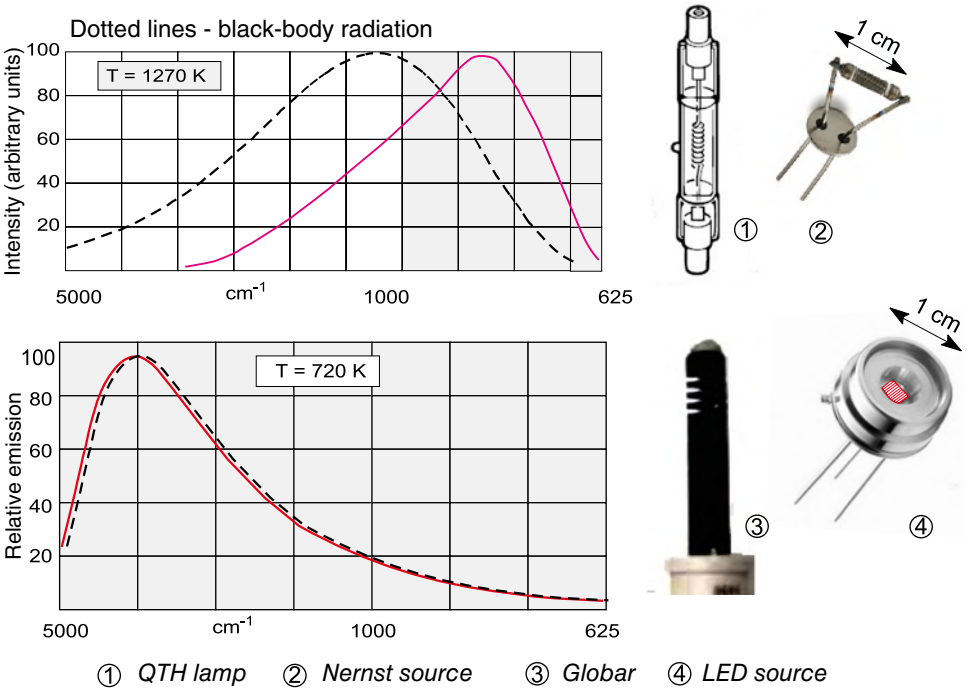


Figure 10.12 Some sources used in the near and mid-infrared region. Operating at less than 1,000°C, the emission curve is nearly perfect when compared to that of a black body at the same temperature. These sources, with or without a reflector, may be protected by a window made of a material that is transparent in the IR region (e.g. BaF_2).

10.8.2 Detectors

The detection of photons in the infrared region was difficult to achieve for a long time and a major cause of this was the mediocre sensitivity of the first spectrophotometers. Depending on the type of application or instrument, either thermal detectors, which are sensitive to heat released by photon absorption (the simplest in design are thermistors, thermocouples and thermopiles) or quantum detectors, which are directly sensitive to photons, are used. In general, the latter respond quicker than thermal detectors. Other sensors, sometimes cleverly designed, may be used for specific applications.

For FTIR instruments, the detector, which must be able to monitor quick changes in light intensity, is the *pyroelectric crystal* or a photodiode type *semiconductor* (Figure 10.13). Lightweight with a small footprint, they have a low thermal inertia and an instantaneous and linear response.

The *pyroelectric detector* contains a monocrystal of deuterated triglycine sulfate (DTGS) or lithium tantalate ($LiTaO_3$), sandwiched between two electrodes, one of which

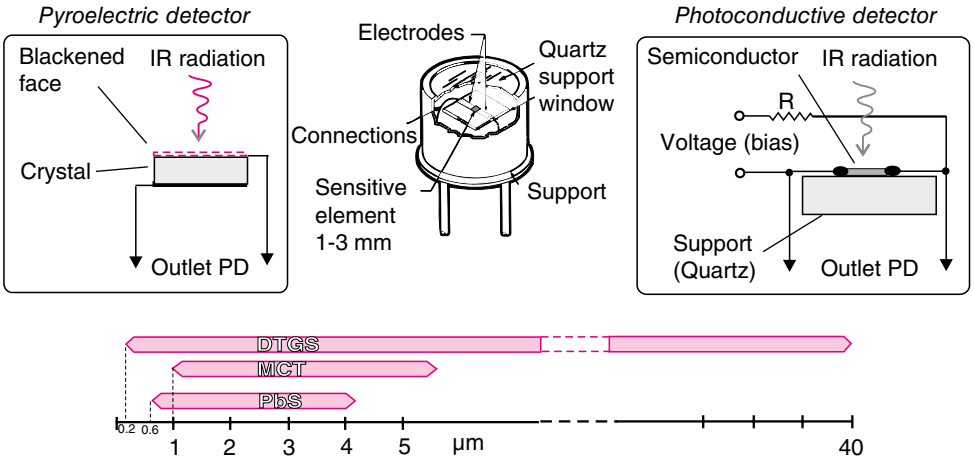


Figure 10.13 Detectors in the infrared region. The operating principles of pyroelectric and semiconductor detectors. Centre, general view of a detector and its housing. Below, range of uses for the main detectors.

is semitransparent to radiation and receives the impact of the optical beam. The crystal is polarized proportionally to the radiation received. It acts as a capacitor and responds only to temperature variations.

The *photoconductive detector* contains a photodiode, which is used either in photovoltaic mode (no bias) or in photoconductive mode (with bias) and contains a P–N junction, which, under the effect of radiation, releases electron/hole pairs that create a PD measurable in an open circuit. This type of detector comprises, for the mid-IR region, a ternary alloy of mercury–cadmium telluride (MCT) or indium antimonide (InSb) deposited on an inert substrate and for the near IR region of lead sulfide (PbS) or a ternary alloy of indium/gallium/arsenic (InGaAs). Sensitivity is improved when these detectors are cooled down to the temperature of liquid nitrogen (77 K).

10.9 SAMPLE ANALYSIS TECHNIQUES

Spectra are acquired from samples either by transmission or reflection. The latter process, now common in the infrared region, is the basis for many qualitative and quantitative techniques in the analysis of all kinds of solid or gaseous samples and aqueous solutions.

10.9.1 Optical Materials

Traditional optical materials used in the visible or the near IR regions become opaque in the mid-IR. Other crystalline or amorphous materials, each having its own particular spectral range, must be used for cuvette walls or detector windows. There are more

than a dozen of these. The most common are sodium chloride (NaCl) and potassium bromide (KBr), both of which have already been mentioned. To replace these fragile, water-soluble materials, caesium iodide (CsI is transparent up to 200 cm^{-1}), silver chloride (AgCl), KRS-5 (thallium bromoiodide), and even diamonds can be used. The latter materials are hard and insoluble, but unfortunately more expensive. AMTIR (Amorphous Material Transmitting Infrared Radiation), a glass composed of germanium, arsenic, and selenium (e.g. $Ge_{33}As_{12}Se_{55}$), is also a good choice.

10.9.2 Transmission Processes

These processes are used when the sample is sufficiently transparent, i.e. for thinner samples.

- *Gases* have low densities and thus require cells with long optical pathways of at least several centimetres (Figure 10.14) but sometimes reaching several hundreds of metres through a complex arrangement of mirrors giving multiple reflections of the infrared beam inside the cell. The cell volume then reaches several decilitres.
- *Liquids* are analysed differently depending on whether it is a qualitative or quantitative analysis. To obtain the sample's spectrum, cells with removable walls are used. They include two discs of a transparent material for the spectral range being analysed (Figures 10.14 and 10.15) between which a sample droplet is placed. For quantitative analysis, however, a cuvette of precise optical path is used in which the analyte is analysed in solution in a transparent solvent over the full extent of the spectral range in question (Figure 10.15).

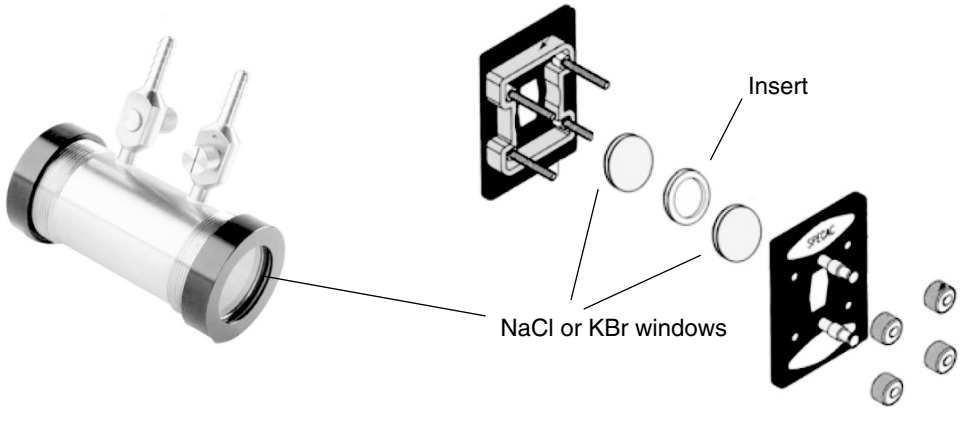


Figure 10.14 Cells in the mid-IR region. View of the direct optical pathway for a gas cell (10 cm) and of a classical cell for liquid films. (Source: Reproduced courtesy of Spectra Tech.) This cell for quantitative analyses is removable, which makes it possible to quickly change the nature of the windows.

Crystal	CaF_2	NaCl	KBr	CsI	AMTIR*	KRS-5**
Limit (μm)	10	15	20	50	12	40

* $Ge_{33}As_{12}Se_{55}$ **TlBr +T II 60:40%

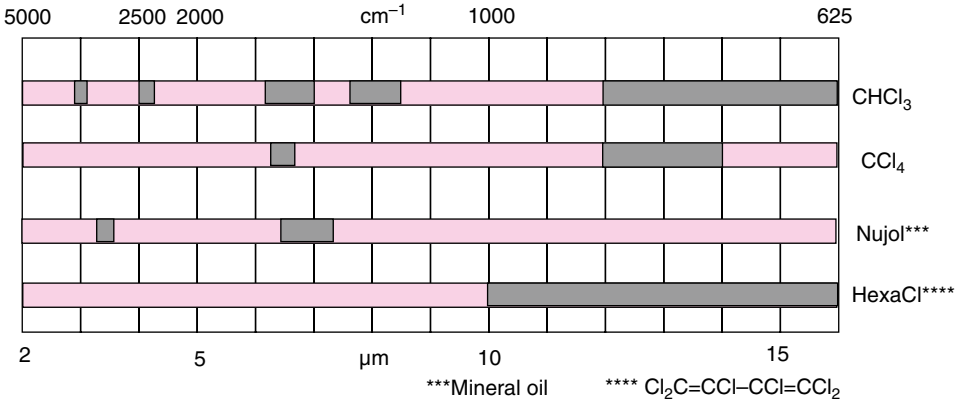


Figure 10.15 *Materials and solvents in the MIR region*. The main crystals or glasses used to create cell walls can be employed below the indicated limits. Some compounds used as solvents in the specified ranges are also indicated (pink zones).

- *Solids* are more difficult to study because of the multiple forms under which they may be presented (bulk, powders, rough or flat surface, variable thickness). If the solid can be dissolved in solution, then the same method as for a liquid is followed. If the solid can be ground down to a fine powder (ideally to particles of less than 1 μm), a few milligrams are dispersed in a medium that will play the role of matrix, such as a mineral oil (called NUJOL™), or anhydrous potassium bromide (KBr). In the latter case, a mixture of solid analyte and KBr is crushed in a mortar and is then compressed into a small translucent disc using a hydraulic press (pressure: 5–8 tonnes/cm^2). The matrix leads to a reduction of the strong optical scatter caused by the particle/air interface, while maintaining the solid in the optical path of the instrument. The finer the powder is ground, the less light will be lost through diffusion.

10.9.3 Reflection Processes

Obtaining spectra by reflection is an alternative to the processes described above. The corresponding devices are based upon *attenuated total reflection, specular reflection* or *diffused reflection* and are only usable with FTIR, as the spectra obtained by reflected

light must be corrected by computer software to render them comparable with transmission spectra. Figure 10.16 summarizes these three techniques.

When a light beam impacts the surface of a medium whose refractive index is different, it can undergo, depending upon the angle of incidence and the variation of the refractive index, either a total reflection, such as with a mirror, or an attenuated reflection after having penetrated in part into the second medium (just a few micrometres for mid-IR). The spectral composition of the reflected beam depends on the compound studied and on the way in which the refractive index of the material varies with the wavelength.

Attenuated total reflection (ATR)

ATR consists of imposing on an optical beam one or more reflections at the interface between the sample and a transparent material in the chosen wavelength range. This material must possess a high refractive index n, such as germanium ($n = 4$), AMTIR ($n = 2.5$), diamonds ($n = 2.4$) or KRS-5 ($n = 2.4$) (Figures 10.16b and 10.17).

If the angle of incidence is greater than the critical angle, a small fraction of the incident light penetrates the sample by a few tenths of a micrometre. This depth of penetration depends on wavelength, angle of incidence, and the difference in the refractive indices of the crystal and the sample. A more complete explanation of this phenomenon would require using the concept of *evanescent wave* from wave optics. The succession of several *total but attenuated* reflections of this type leads to an effective optical path comparable to that which would have been obtained through transmission. The spectrum is nevertheless corrected to take into account the depth of penetration, which varies with the wavelength. It is currently the most widely used process, thanks to its versatility, for solids, powders or liquids (even those containing water).

Optical fibre probes. ATR is used for the study of samples at a distance from the optical bench of the apparatus through the use of transparent optical fibres in the infrared region, over a distance of several metres. They can be used as immersible probes plunged into the product to be analysed, enabling rapid control analyses in difficult or aggressive environments.

Specular reflection

This accessory measures *reflectance*, i.e. the light reflected by the sample in a direction symmetrical to that of the incident light (Figure 10.16a). It is reserved for samples that reflect at least a minimum amount of light (polymer films, varnishes, and some coatings) and which have a very low surface roughness compared to the wavelength of the incident radiation.

The reflected light mainly translates the variations in the refractive index of the compound as a function of wavelength. By comparing, for each wavelength, the specular reflection I of the sample with the total reflection I_0 obtained by replacing the

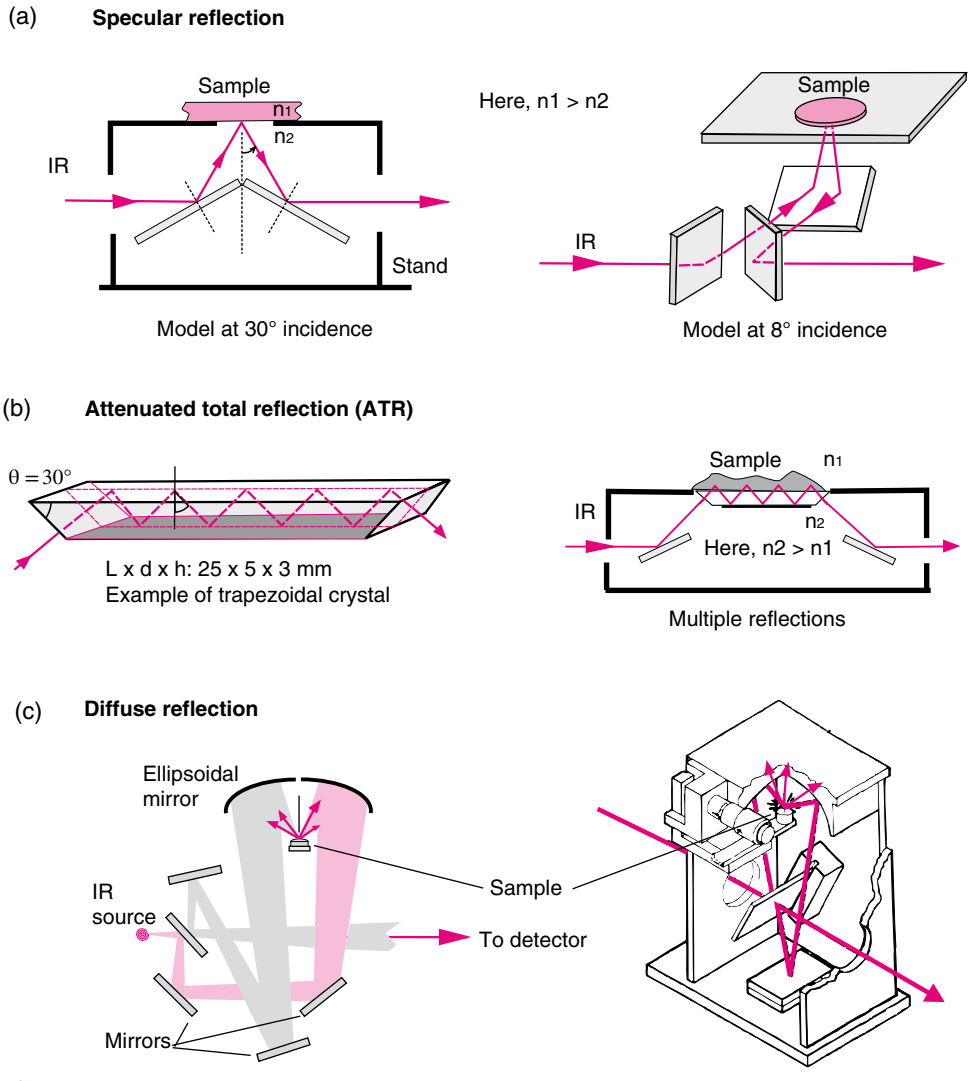


Figure 10.16 Three types of reflections used in the MIR region. (a) Specular reflection device. Optical pathway of the apparatus at a fixed angle of 30° for highly reflective samples and another assembly at 8°. (b) Attenuated total reflection (ATR) instrument with a pyramid-shaped crystal for multiple reflections. (c) Optical set-up for a diffuse reflection device and diagram of a Spectra Tech model.

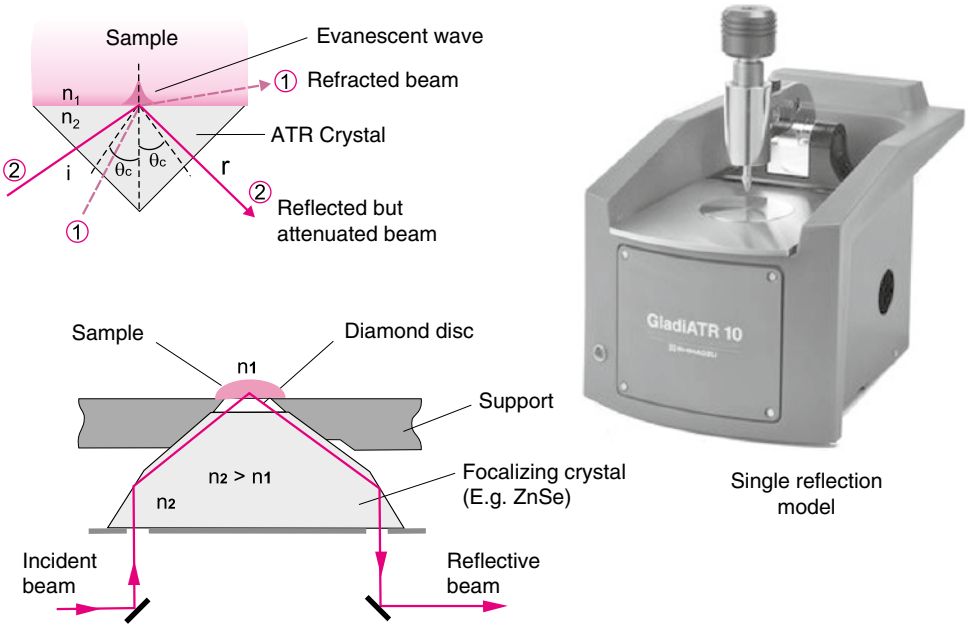


Figure 10.17 Critical angle and evanescent wave. Comparison of the path, whether reflected or refracted, of two beams according to their angle of incidence. Instrument using attenuated total reflection (ATR) with a pyramid-shaped crystal which only allows for one reflection. Commercial model of this type (GladiATR-10 from Shimadzu). The contact between the sample and the crystal protected by a diamond blade (n = 2.42) is facilitated by crushing. (Source: Shimadzu Corporation.)

sample with an aluminium mirror, the instrument can calculate the reflectance $R = I/I_0 = f(\lambda)$. Finally, using a Kramers–Krönig (K–K) mathematical transformation, it calculates a pseudo-absorbance spectrum, which is virtually identical to the one obtained by transmission (Figure 10.18a).

Diffuse reflection

This is a device containing a series of mirrors arranged to collect part of the light diffused by the sample finely dispersed as a powder in KBr (Figure 10.18b). By comparing the diffuse reflection with that of pure KBr, followed by a Kubelka–Munk mathematical correction, a result resembling the classic transmission spectrum is obtained. This very time-consuming technique is however seldom used.

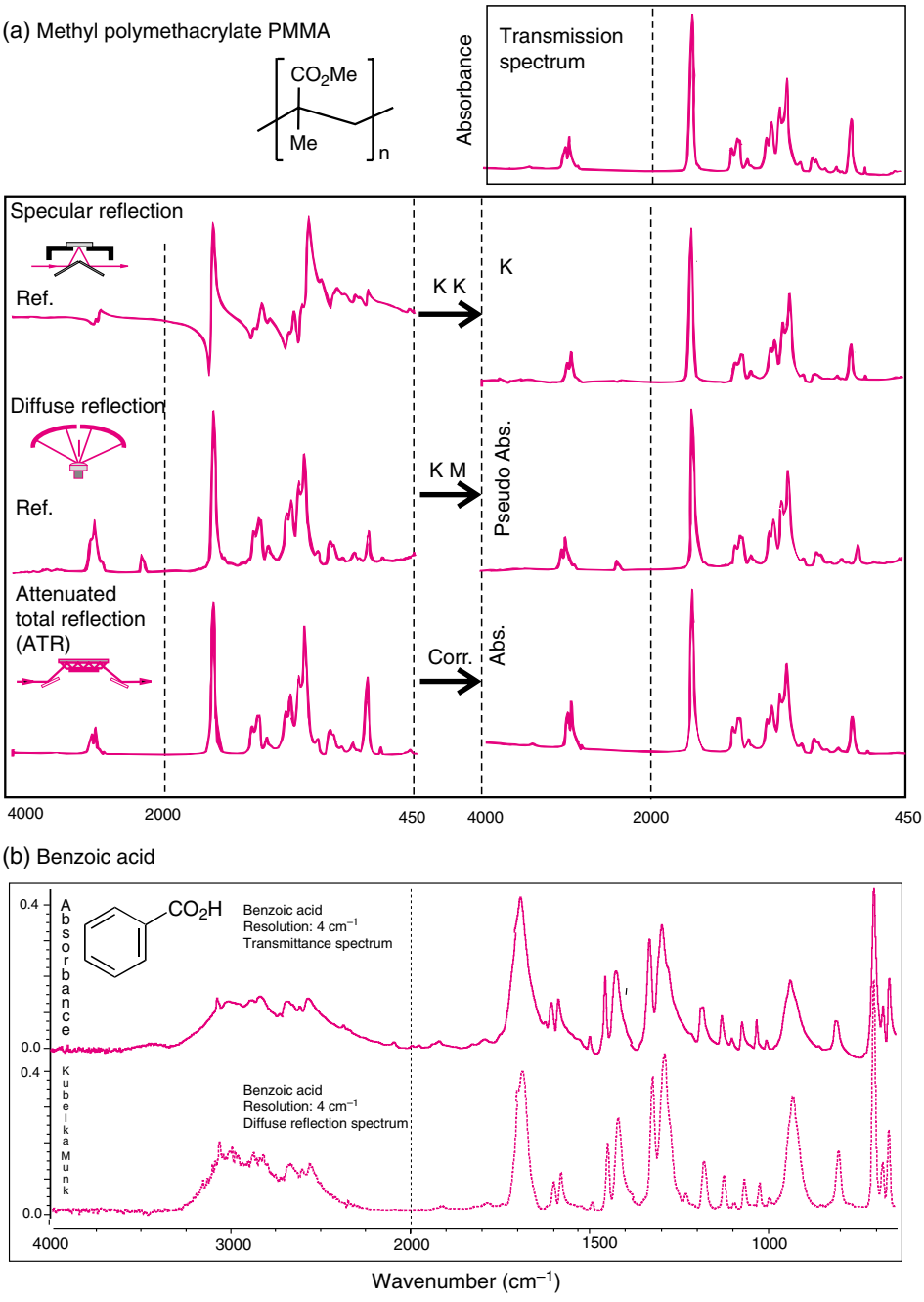


Figure 10.18 Reflection spectra. (a) Spectra from a sample of Plexiglas obtained by use of three types of reflection. Left, raw spectra; right, spectra after correction. Top, raw signal of specular reflection and the result in units of K following application of the Kramers–Kronig calculation (transformation of reflectance). Middle, spectrum obtained in diffuse light: comparison of the raw spectrum with the Kubelka–Munk correction. Bottom, spectrum obtained by ATR, the latter requiring a fine correction to reduce the absorbance at higher wavelengths, which would be overestimated. (b) Comparison of two spectra of benzoic acid, one obtained through transmission, the other by diffuse reflection and subsequent K–M correction.

10.10 COUPLED TECHNIQUES

10.10.1 Chemical Imaging Spectroscopy

The high sensitivity of detectors and the wide range of available lenses allow the study of small samples by transmission or reflection. Coupling a microscope with an FTIR spectrometer thus makes the assessment of 10–30 μm samples possible. The lower limit is related to the laws of diffraction and to the signal/noise ratio. Spatial resolution may reach the order of magnitude of the analysis wavelength.

By extension, this technology helps obtain a mapping linked to the composition of samples with a microstructure.

The instrumentation includes a spectrometer coupled with an observation microscope (Figure 10.19). An optical transfer assembly deviates the infrared radiation from the interferometer to the microscope.

Many industrial or natural compounds are heterogeneous solid mixtures. Traditional methods of analysis of such materials giving mean values for the atomic composition are sometimes insufficient. Chemical imaging spectroscopy conducted one time lets us know about the distribution of the various components and the nature of molecules at each point of the composition. For example, it is possible to study the dispersion of the active ingredient of a medication in its excipient, or to evaluate the impurities in cattle feed. Statistical methods and chemometric approaches such as partial least squares (PLS) or principal component analysis (PCA) are thus very useful in this context.

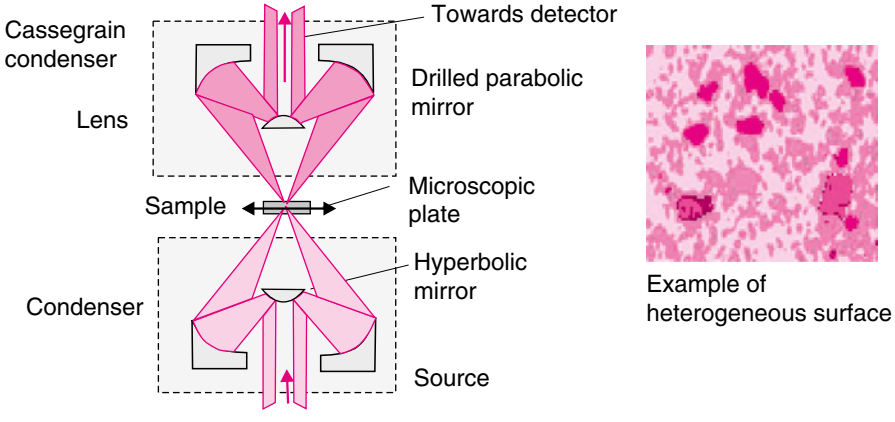


Figure 10.19 Optical path of an IR microscope. The sample can be examined in transmission (left) or in specular reflection modes (right). This accessory is installed in a spectrometer whose beam is deflected. The Cassegrain optics, which are appropriate for the study of small objects, have the advantage in IR that the light is reflected at the mirror surface rather than having to pass through the optical lenses.

10.10.2 GC/FTIR Coupling

Cited for information purposes only, this coupling is not very common due to its lack of sensitivity and its complex layout. Compounds after separation in the column go through a heated transfer line before passing through a small volume, thread-like cell that is also heated. The latter is integrated into a specially made spectrometer module. To increase the optical path of the cell, whose dimensions are 100 × 1 mm (*light pipe*) and thus avoid a loss of resolution, its internal wall is metallized to produce multiple reflections. Finally, FTIR, a single-beam technique requiring an accumulation of several interferograms to increase the quality of the spectra produced, leads to additional difficulties when a succession of compounds are processed, following each other more or less quickly at the chromatography column outlet.

10.11 COMPARISON OF SPECTRA

The identification of compounds can be made easier with *spectral libraries* as compiled by various companies or publishers (e.g. Aldrich, Sigma, Sadtler, and Hummel), whether general or specialized (polymers, solvents, adhesives or organic molecules classified by functionality). For the comparisons to be reliable, these data collections should have been obtained from compounds in the same physical state as the compound to be identified (in the gaseous phase for example, when the method is GC/FTIR). The archived spectra are usually normalized by assigning an absorbance value of one to their most intense band.

The spectrum to be compared must first be adjusted to conform (in terms of absorbance and wavelengths) with the model of the spectra in the library. Then the procedure consists of a mathematical comparison of the unknown compound's spectrum with all those included in the spectral library in order to establish, by the end of the research session, a list of spectra that most resemble that of the unknown compound, each having an index of reliability. The analyst must examine these results carefully, recalling that the objective of this exercise is to help and not to substitute for the analyst's judgement, because if the comparison algorithm is changed, the final listing will probably be different. The best methods of identification are associated with interactive programs in which the analyst contributes information in order to narrow the field of investigation, by defining *filters*.

The algorithm using the absolute difference is the simplest: for each of the j spectra in the library, the sum S_j, for the n defined points along the x-axis, is calculated as the absolute difference between the corresponding absorbance of the unknown and that of spectrum j in the library (Eq. (10.10)). Then the j number of sums S_j are classified in increasing order. The results are presented along with a correlation index.

$$S_J = \sum_{i=1}^{n} \left| y_i^{\mathrm{ref}(J)} - y_i^{\mathrm{inc.}} \right| \qquad (10.10)$$

Other expressions can be chosen: replacing the differences above by their squares or by the differences of these squares or taking the incremental difference between two consecutive points. Each algorithm may minimize or increase the value of particular factors (differences of intensity, background noise, difference in gradient, etc.).

10.12 QUANTITATIVE ANALYSIS

The precision of absorbance measurements and the possibilities of reprocessing the spectra have favoured the use of infrared spectroscopy in quantitative analysis. The method is widely used for the ease with which the absorption bands of a particular compound can be located even in a mixture and because efficient methods of statistical analysis are available for the near IR region.

In the early ages of FTIR, some analysts considered that double-beam dispersive instruments were preferable for quantitative analysis, as they are the only ones to compare the intensities transmitted by the two pathways (reference and sample) at the same instant.

10.12.1 Quantitative Analysis in the Mid-Infrared Region

Practically all molecular species absorb in this spectral domain. The great specificity of the spectra may also be especially interesting when dealing with mixtures of compounds with similar chemical structures. However, quantitative analysis in this wavelength range is unfortunately harder to implement when compared to the UV-Visible. The complexity mainly comes from the fact that spectra have a high number of absorption bands and from the difficulty in being able to control the reproducibility of essential analytical conditions for the development of calibration sets (transmittance and optical paths of cells, and comparison between reference and sample spectra).

For liquid samples, it is nearly impossible to have a pair of identical cells for the solvent and the sample. In fact, cell windows, which are generally polished salts (NaCl, KCl), are highly hygroscopic and deteriorate in the presence of moisture in the atmosphere and in samples. The optical path length l is often very small (<1 mm) to minimize the absorbance of the solvent, since no solvent is truly transparent in this spectral domain. A periodic calibration of the cells' optical path is therefore essential in limiting the uncertainty on value l.

The exact measurement of the optical path in cells of low thickness is made by interferometry. The transmittance of the empty cell is measured for an interval between two wavenumbers $\bar{\nu}_1$ and $\bar{\nu}_2$. Figure 10.20 shows that the beam S_2 has undergone a double reflection from the internal walls of the cell, such that for a normal incidence, it has travelled an additional distance of $2l$. For example, if $2l = k\lambda$, the two light intensities will be additive (the two beams S_1 and S_2 are in phase). As a function of the wavelength, a change in the main beam S_1 by a few percentage is

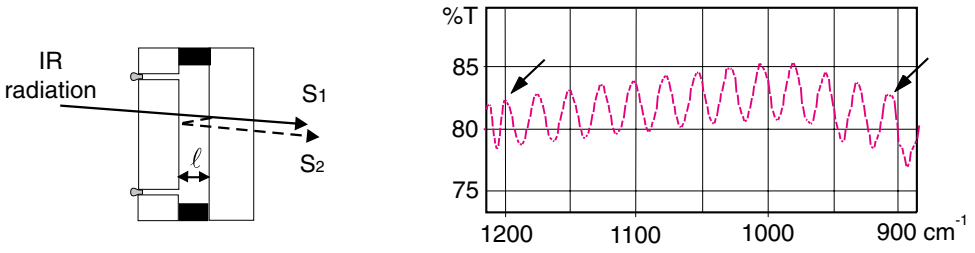


Figure 10.20 Measurement of cell thickness by the method of interference fringes. Left, reflections on the inner wall of a cell (for further clarity, the angle of incidence of the beam on the cell is shifted from normal incidence by a small angle). Right, part of the recording obtained from an empty cell. Twelve fringes are counted between the two arrows. The calculation (Eq. (10.11)) using this data leads to a cell path length of l = 204 µm.

observed. After calculation, if N represents the number of interference fringes counted between $\bar{\nu}_1$ and $\bar{\nu}_2$ (in cm^{-1}), then:

$$l_{(cm)} = \frac{N}{2(\bar{\nu}_1 - \bar{\nu}_2)} \qquad (10.11)$$

For solid samples dispersed in a KBr disc, whose thickness cannot be measured precisely, a compound is introduced as an internal reference (calcium carbonate, naphthalene, sodium nitrite), in an equal quantity to all of the standards as well as to the sample.

The background absorption, which is often visible for a substantial part of the spectrum, also has to be taken into account. In particular, it depends upon the manner in which the sample has been prepared. As absorbances are additive, the absorbance not due to the compound under study must be subtracted from the total absorbance. This is evaluated by the tangent method (Figure 10.21).

However, should these parameters be determined with precision and should it be possible to isolate one or more of the characteristic bands of the analyte, quantitative analysis methods, such as those described for molecular absorption in the UV-Visible domain (see Chapter 9), may be implemented.

Near Infrared

The near infrared region corresponds to the part of the electromagnetic spectrum where wavelengths are between 1,000 and 2,500 nm (i.e. 1–2.5 µm). Though for a long time it had no specific applications, because the direct study of spectra observed in this region did not provide reliable information on the samples analysed, the recent arrival of chemometrics has led to numerous analytical applications. This field is called *Near Infrared Spectroscopy* (NIRS).

10.13 ANALYSIS IN THE NEAR INFRARED REGION

In the region between the visible and mid-infrared, the spectra present large bands, often of low absorbance, corresponding to harmonics and combinations of absorption bands found in the mid-infrared region. Thus, vibrational movements of the C–H, C–N, N–H, C–O or O–H bonds in organic molecules are responsible for several harmonic absorptions which, while specific, are difficult to locate individually (Figure 10.22). The C–C and C–H absorptions, specific for each carbon backbone, conceal or displace the expected absorptions. In this manner, the exploitation of these spectra in analysis involves chemometric methods, which model all absorptions measured at different wavelengths, whatever the origin: analytes and matrix. To extract the information needed, the study of a single absorption alone is not of any interest; the overall spectrum of the sample, which is its digital fingerprint, is what we want to look at. This approach leads to a compositional analysis using software, which

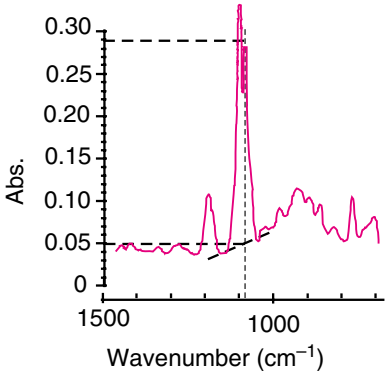


Figure 10.21 Correction of the absorption baseline. If we assume, for the given example, that the measurement of the concentration is based upon the absorbance of the band at 1,070 cm^{-1}, then it will be necessary to deduct from the total absorbance (0.29 AU) the baseline absorption of 0.05 AU, which leads to 0.24 AU.

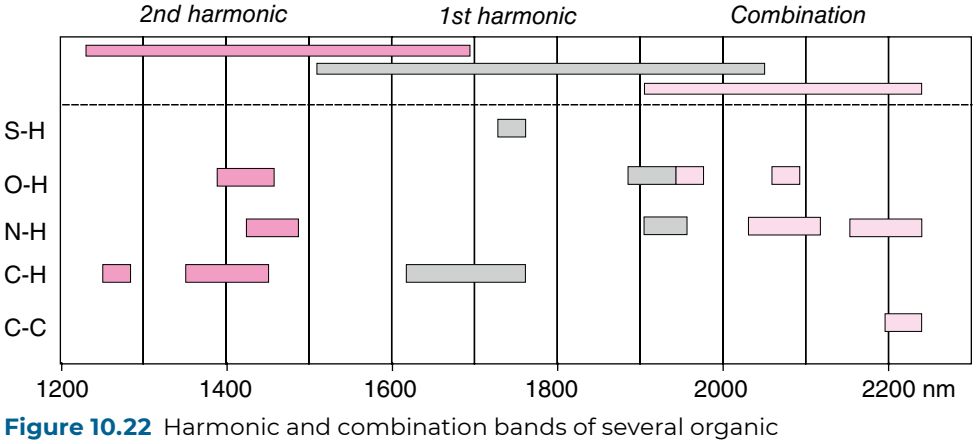


Figure 10.22 Harmonic and combination bands of several organic compound bonds.

guides the operator throughout the calibration phase, based on comparisons between the sample and a selection of references that cover the entire zone of values to be predicted. Therefore, it is not rare to establish correlations with several dozen references, whose composition we know by means of reference analyses (Kjeldhal, for example, for protein assays). This is the field of *correlation analysis*, essential for any quantitative study.

These software programs use multivariate analysis methods such as *Multiple Linear Regression* (MLR), *Principal Component Regression* (PCR) or even *Partial Least Squares* (PLS), enabling the statistical treatment of a large number of data points in order to establish a calibration equation for the assay in question. By using derivative plots, precision is improved (Figure 10.23).

The band that appears around 1,940 nm (or 5,154 cm^{-1}, see Figures 10.22 and 10.23) for water results from the combination of the asymmetrical vibration situated at 3,500 cm^{-1} and the deformation vibration (shearing, see Section 10.6) at 1,645 cm^{-1}. These simple calculations (3,500 + 1,645 = 5,145) are an imperfect approach, which conceals a certain number of factors of the physical state of the compounds, to which infrared is sensitive. Similar approximations can be made for the combination bands of the C–H, N–H and C–O bonds in the near IR region.

10.13.1 Sequential, Spectrograph and FT-NIR Instrument Types

Several types of instruments share the market in this field:

- Nondispersive sequential spectrometers with filters for basic models. Many NIRS analyses are also conducted with custom devices that are not spectrometers. The classic source is replaced by a laser diode selected according to its emission wavelength, which simplifies the optical layout of the instrument.
- Sequential spectrophotometers with a UV-Visible monochromator can cover the entire NIR range, thanks to good selection of their components (sources, networks, detectors).
- Spectrographs with diode array detection using an InGaAs semiconductor adapted to this spectral region.
- There are still few FTIR instruments similar to existing models in the MIR region, because at shorter wavelengths the location of the moving mirror must be known precisely, as even the smallest of vibrations of the interferometer negatively impacts the quality of the sampling, which must be precise (consequence of the Nyquist theorem).

The reference optical materials are quartz (SiO_2, hardness 7 on the Mohs empirical scale) or, even better, sapphire (Al_2O_3, hardness: 9 Mohs), which is transparent in the entire NIR region. Measurements by transmission are used less often than

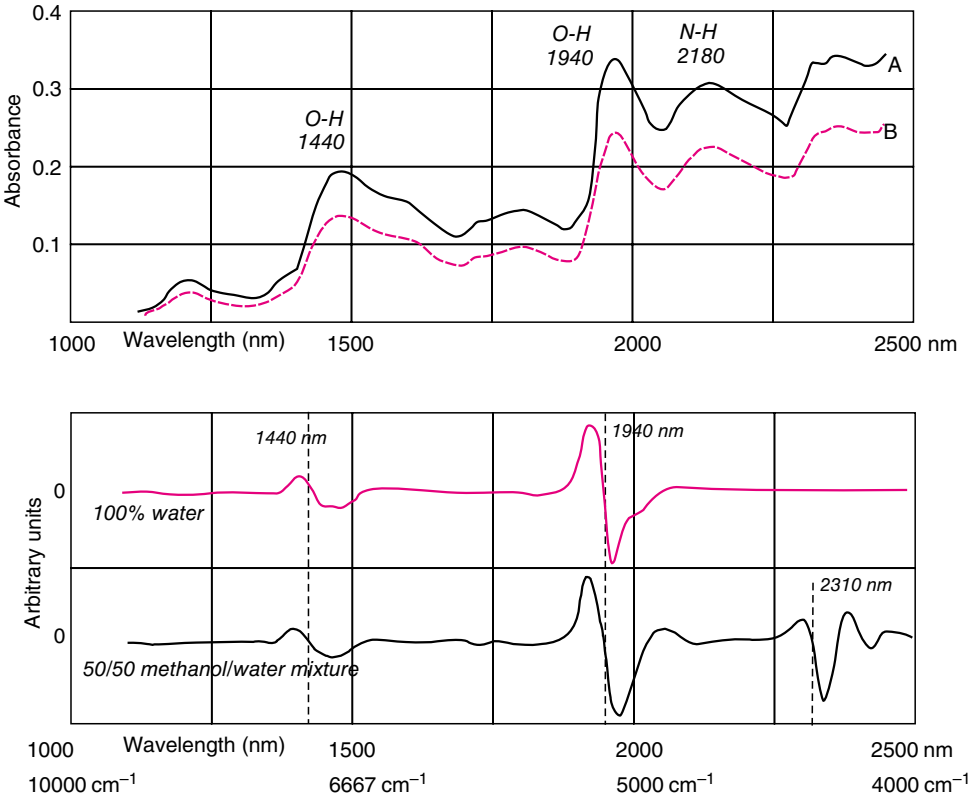


Figure 10.23 Effect of particle size distribution on the spectrum of flour and derivative curves. Top: Comparison of spectra of a given flour sample according to its particle size distribution (Osborne et al.) **A,** coarse grind, **B,** fine grind. If the measurement was conducted at a single wavelength (e.g. 1,940 nm), it would seem that the protein contents are different, but if we refer to the entire spectrum, it is the same product. Bottom: Comparison of the second-derivative spectra of pure water and of a 50/50 mixture of water/methanol in the near IR region. Note that it is possible to measure the methanol concentration from the sigmoidal curves situated around 2,300 nm, without interference from the water signal located between 1,440 and 1,940 nm.

measurements by reflection (specular, attenuated, and diffuse) as seen previously (Figure 10.16). The *integrating sphere* is the most highly developed instrument in diffuse reflection when fine particles are being analysed, at the origin of randomly oriented refractions and diffractions (Figure 10.24). There are two types, one with a single inlet for FTIR (same optical path for reference and sample), while the other has two inlets (two different paths for the reference and sample). In either case, the integrating

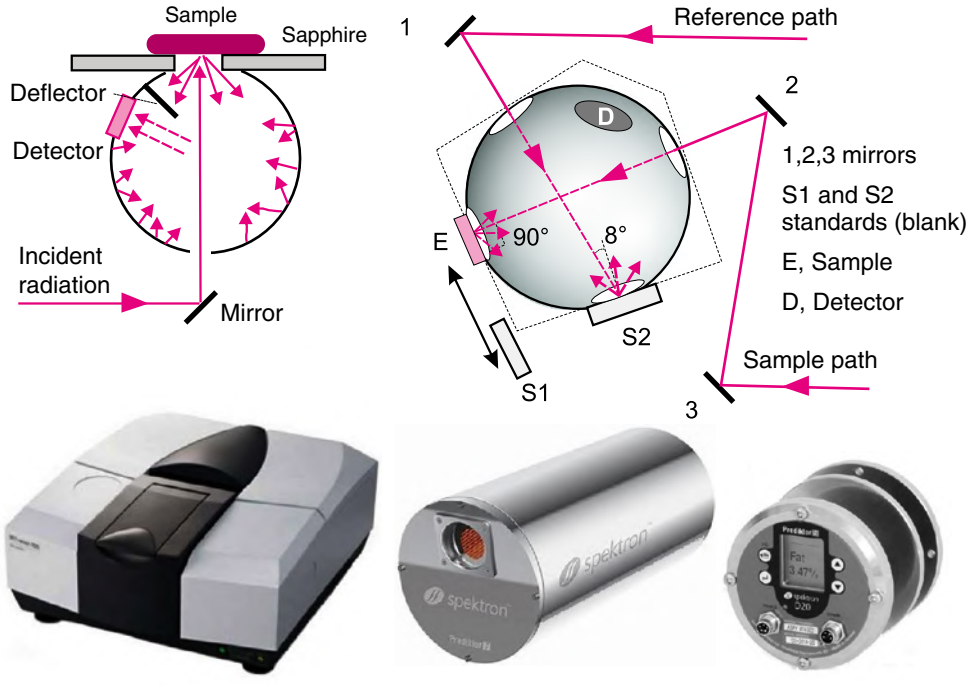


Figure 10.24 Integrating sphere and commercial instruments for NIRS. Left: Diagram of a single-path integrating sphere (diameter: a few cm) used on FTIR spectrometers. The reflecting internal wall (gold or Teflon) is used for multiple reflections to generate a uniformly diffuse luminance, whatever the observation angle, with a deflector protecting the detector (e.g. InGaAs) in order for it not to receive any direct light coming from the sample. Right: integrating sphere for a sequential instrument. Spectrophotometer IRTracer-100 (Shimadzu) and real-time online spectrometer: Prediktor (Spektron), 960-1,700 nm (the light source is outside the instrument). (Source: Shimadzu Corporation.)

spheres have their own detector, located around the wall to retrieve as much of the scattered light as possible. The spectrometer must be able to support this type of measurement cell.

10.13.2 NIRS Applications

NIRS helps to quantify the number of specific chemical bonds in a compound (O–H for water, N–H for proteins or C–H for fats), which, by comparison with databases of reference compounds, enables calculation of the concentrations in question.

NIRS, intended for routine analyses, is especially adapted to homogenous series. The procedure is quick, inexpensive and nondestructive; samples do not have to undergo any preparation. However, if we only have a few samples to process, the technique



Figure 10.25 Portable NIR spectrometer. Left: NIR spectrometer for the analysis of dairy products and grains (StellarCase model from StellarNet Inc.). Right: device operating in diffuse reflectance mode for the analysis of raw materials. (Source: Micronir Viavi, courtesy of Fondis.)

is no longer viable. Transmission cells are not often used. The advantage of the low absorbances encountered is a shallow penetration of the incident radiation in the sample, which makes the study of powders or polymers possible, as long as the surface composition is identical to that of the centre of the solid analysed. However, NIRS cannot be used for mineral substances or to assay traces, since absorptions are too weak.

Applications are mainly found in industrial process analyses:

- In the food industry: for example, proteins, fats in seeds, cellulose or starch, moisture content, etc. can be measured from the O–H, N–H or C–H bonds for fats (certain bonds lead to two or three harmonics whose intensity varies with the physical characteristics of the sample) (Figure 10.25).
- In the pharmaceutical industry: NIRS is used to control raw materials on production lines, manufacturing stages or to conduct quality checks.

Raman Spectroscopy

Raman scattering, which is named after its discoverer, has been known for almost a century. It provides comparable and complementary information to that supplied by the mid- or far infrared region. Long considered difficult to implement, this method of analysis of materials owes its use in industry (production control) and in analytical laboratories to the progress made in instrumentation. Raman spectrometers now constitute a family of instruments which thanks to their specificities complement other instrumental methods. The scope of application is progressively becoming larger.

10.14 PRINCIPLE OF THE RAMAN EFFECT

When a sample composed of identical molecules is irradiated with an intense, monochromatic source (laser), whose photons have a different energy than that corresponding to an electronic or vibrational transition, there is no specific interaction produced (the sample is transparent), except for a very small portion of these molecules (about one billionth of them). The latter in fact interact with photons during two types of collisions:

- *Elastic collisions* with no energy exchange. They are translated by simple diffusion of the incident radiation, without change of frequency, in any direction. This radiation constitutes the intense Rayleigh scattering line, whose wavelength is the same as that of the exciting source.
- *Inelastic collisions* for which there is an exchange of energy between photons and molecules (Figure 10.26). In this case, molecules pass into a *virtual*, transition state of energy, before emitting new photons, which constitute the (weak) Raman scattering, formed by Stokes and anti-Stokes lines, which frame the Rayleigh scattering line.

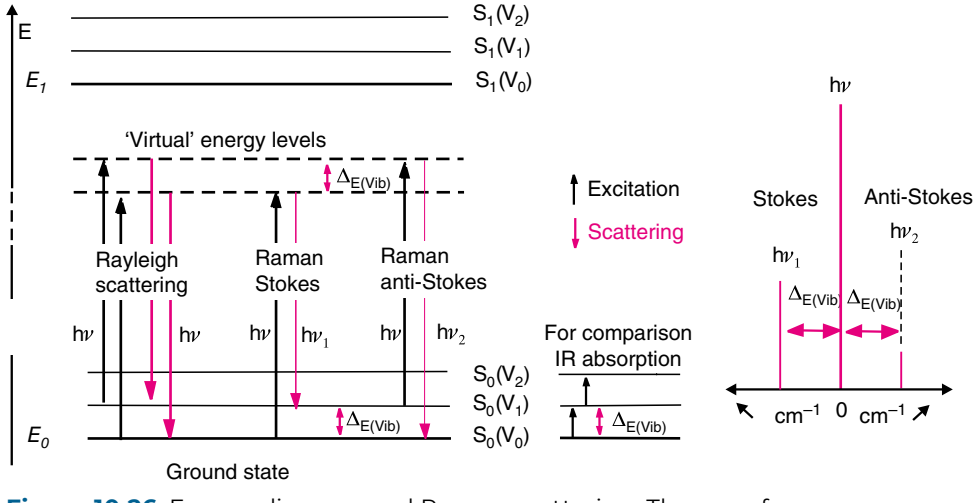


Figure 10.26 Energy diagram and Raman scattering. The use of an energy diagram (Jablonski type) to represent the evolution of the energy of a laser-stimulated bond demonstrates the Raman scattering phenomenon and enables the insertion of virtual energy levels. A polyatomic molecule has many degrees of freedom, some of which are active in infrared spectroscopy (if the dipole moment varies) and others in Raman spectroscopy (if polarizability varies).

Stokes lines. If the molecule is initially at rest, in its ground state $S_0(V_0)$, it may conserve a part of the incident photon's energy to pass into, for example, the first excited vibrational state $S_0(V_1)$ by capturing a vibrational quantum called a *phonon*. In that case, it re-emits a photon with less energy, which leads to a Stokes line (Figure 10.26).

Anti-Stokes lines. As predicted by the Boltzmann distribution equilibrium (see Chapter 15, Eq. (15.5)), a very small proportion of molecules is initially found in the first vibrational state $S_0(V_1)$. After the excitation step, these will be able to return to the ground state $S_0(V_0)$, by emitting a photon of greater energy, whose value corresponds to the energy of the incident photon to which the energy of the vibrational quantum is added, thus ensuring its return to the ground state (destruction of a phonon). This results in a not very intense anti-Stokes line.

In summary, the Raman technique is used to analyse vibrational transitions from the inelastic scattering of light.

As can be seen on the diagram, the energy gap between the Rayleigh line and a Stokes or anti-Stokes line corresponds to the energy of the absorption peak of the IP spectrum, if the vibration mode involved is at least active in both cases. In fact, to observe a Raman line, the polarizability of the bond must be clearly changed by the effect of the electrical field component of the incident electromagnetic wave. A bond with a weak dipole moment (inactive in the IR) has greater polarizability in general and will thus be active in Raman spectroscopy. Several exclusion rules are defined by the theory.

Raman scattering corresponds to an emission of photons bringing the irradiated compound to a lower energy level. Unlike fluorescence, in which the emission of the photon corresponds to a specific wavelength that is different from the one which caused the excitation, Raman peak positions differ depending on the excitation wavelength.

By convention, the Raman spectrum of an analyte reproduces the Stokes lines only. To make its use easier, the same presentation as for an infrared spectrum, whose energy decreases as one moves to the right of the spectrum, is used (Figures 10.27 and 10.28). On the *y*-axis, the intensity of the lines is plotted and, on the *x*-axis, the difference in cm^{-1} between the Rayleigh line and the different Stokes lines is plotted, which amounts to taking the value (in cm^{-1}) of the Rayleigh line as the origin.

To increase the intensity of Stokes lines, the analyte is irradiated with radiation whose energy is close to that which leads to the first electronic transition. Under these conditions, the Raman intensity may be multiplied by a factor of 10,000. This technique, called Resonance Raman Spectroscopy, is used to study analytes at very low concentrations. Another solution consists in irradiating the analyte for much longer, which then reduces its fluorescence ('bleaching' of the compound).

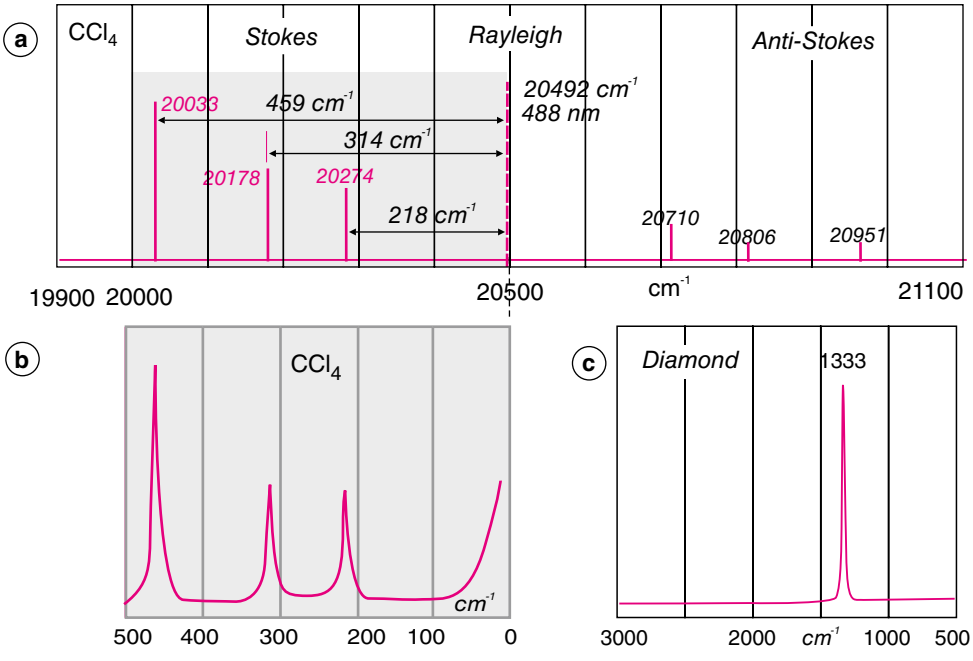


Figure 10.27 *Raman spectrum.* (a) The three main Stokes and anti-Stokes lines of the carbon tetrachloride molecule (excitation: ionized argon laser, 488 nm). (b) Raman spectrum of the same compound. (c) Real spectrum of the 1,333 cm^{-1} diamond 'line'. By using a longer wavelength, the Raman spectrum is obtained without any interference from fluorescence.

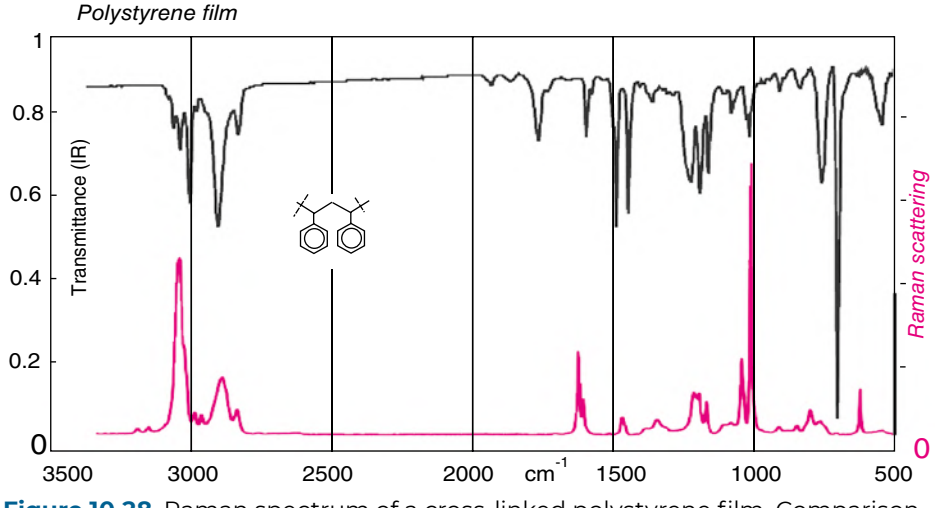


Figure 10.28 Raman spectrum of a cross-linked polystyrene film. Comparison with the infrared spectrum, in black on the figure, recorded in transmittance in the same spectral region.

10.15 INSTRUMENTATION

Raman spectrometers, often called *IR-Raman*, have a source, an analyser, and a detector. Current excitation sources are lasers: neutral-atom or ionized gas lasers, the main ones being helium/neon (632.8 nm), argon (514.5 nm), and krypton (647 nm) lasers, and solid-state lasers, the most common being the neodymium (Nd) laser: composed of an yttrium aluminium garnet (YAG) crystal containing neodymium ions (1,064 nm) for fluorescent compounds.

These sources are associated with optical filters to eliminate secondary emissions from the source as well as most of the Rayleigh scattering. The beam may be focused for micro-Raman applications, i.e. the study of microsamples. Observation is done under standard conditions at a 90° angle.

The analyser system is either a conventional dispersive system or a Michelson interferometer in the case of an FTIR spectrometer.

Depending on the chosen technique, the detector is either a photomultiplier or a CCD sensor.

Miniaturized Raman spectrometers have given rise to many commercial products (Figure 10.29). These portable instruments are used especially for varied applications in the identification of chemicals, raw materials or finished products or even in quality control. Measurements are quick on all sorts of samples, even through plastic or glass packaging.



Figure 10.29 Miniaturized Raman spectrometer. A field device using an excitation laser (1,064 mm) (Source: Progeny ResQ, courtesy of Rigaku Analytical Devices). Examples of spectra obtained with this instrument (200–2,500 cm^{-1} range).

10.16 FIELDS OF APPLICATION

Raman spectra have fewer peaks than those in the mid-infrared region, which creates less interference and thus facilitates assays (Table 10.2). Samples do not require any special preparation, and aqueous solutions may be analysed in glass cells.

> With a helium/neon (633 nm) excitation source, it can be calculated that Stokes lines observed in a 4,000 cm^{-1} range correspond to lines emitted between 633 and 847 nm, a range in which water or glass are transparent.

The fluorescence of certain compounds impedes the observation of Stokes lines (Figure 10.26). However, this type of interference can be circumvented by choosing a source emitting in the near infrared region among the great variety of exciting lines available. In this way, we avoid provoking the fluorescence of the studied compound. Moreover, Raman scattering bands 'move' with the wavelength of the exciting source, unlike fluorescent bands which are generally situated in the visible region. Please note, however, that Raman lines become less intense as the wavelength increases.

Raman spectroscopy and mid-infrared spectroscopy have the same fields of application, which concern the identification of compounds or the characterization of minerals, rocks, and materials in general.

Raman spectroscopy, used in microscopy, leads to a representation of objects with a heterogeneous surface.

KEY POINTS OF THE CHAPTER

1. In the mid-IR region, each molecule leads to a unique spectrum, which enables its identification, for example, if it is identical to the spectrum of an already known compound. If this molecule is still unknown, the study of its spectrum and the existence of known structural correlations may lead to the elucidation of its structure, *in abstracto*.
2. The IR domain, from 2 to 25 micrometres, corresponds to the frequency range between 1.5×10^{14} and 1.2×10^{15} Hz (application of $\nu = c/\lambda$). These frequencies are of the same order as those describing periodic movements between atoms within molecules, which helps correlate molecular vibrations with this infrared range.
3. By subjecting a molecule to a continuous source of near or mid-infrared light, an energy coupling arises, as materialized by absorption of certain frequencies (or wavelengths) by the molecule, whose energy is then increased.
4. The harmonic oscillator model of classical mechanics is a simplified approach to approximately account for vibrations between atoms, by combining force constants and atomic masses, at least for low-amplitude oscillations. Only the quantum approach helps to understand the existence of the harmonics responsible for the main absorptions of the near IR region.

5. To cover the mid-infrared region (3–25 μm), spectrometers include continuous heat sources. The selection of these sources results from a compromise between temperature (Wien's Law) and emissivity (Stefan–Boltzmann Law). Source temperatures are around 1,500 K.
6. FTIR was originally developed to obtain spectra much more quickly than with dispersive models. The novelty resides in the introduction of an interferometer on the optical path of the instrument, at least for conventional spectrometers.
7. The interferometer is a way to measure all frequencies simultaneously. Each signal point recorded according to the position of the moving mirror (the whole forming an interferogram) displays a value in which all frequencies coming from the source intervene. Then, a mathematical calculation (Fourier transform), made easier by advances in computing, helps us to move from the interferogram established according to the mirror's position ($I = f(d)$) to the conventional spectrum ($I = f(\lambda)$).
8. The near IR region leads to spectra that are not very informative when examined visually, but it is still very useful thanks to various regression methods, which enable the extraction of the desired information in order to quantify one or more components by comparing them with a set of known mixtures.
9. The near IR region presents technical advantages over the mid-IR region: inexpensive to use, it enables quick analyses on solid or liquid samples, uses glass cells, and does not require any pretreatment. It is often used for quantitative or compositional analysis in process analyses, in the field, but remains unsuitable for residue assays.
10. Raman spectroscopy is complementary to IR and uses the interaction between the material and light, which slightly changes the frequency. This shift in frequency between the incident light and that which is re-emitted helps, as in IR, to determine the vibration energies, and therefore their frequencies, whatever the excitation λ may be.
11. As Raman lines have very low intensity, the technique long remained confined to research laboratories before finding many current applications in the field, following progress with sources and detectors.

PROBLEMS

1. **a.** What is the energy transported by radiation of wavenumber 1,000 cm^{-1}?
   b. Transform $\lambda = 15$ μm into units of cm^{-1} and then into m^{-1}. What is the wavelength corresponding to a wave number of 1,700 cm^{-1}?
2. If chloroform (trichloromethane) exhibits an absorption band at 3,018.5 cm^{-1} due to the C–H bond, calculate the wave number of the absorption band corresponding to the C–D bond in deuterochloroform (experimental value: 2,253 cm^{-1}).

3. The bond between the two atoms of a diatomic molecule is characterized by a force constant of 1,000 N/m. This bond is responsible for an absorption at 2,000 cm^{-1}. Accepting that the energy of incident radiation is transformed into vibrational energy, estimate a value for the increase in maximum stretching of this bond.
4. Knowing that the fundamental frequency of carbon monoxide is 2,135 cm^{-1} when the spectrum is recorded with carbon tetrachloride as the solvent, calculate the force constant of the bond of this molecule under these conditions.
5. Here is the incomplete raw formula of a compound: C_3H_9OX
   From the absorption band observed at 3,320 cm^{-1} on its IR spectrum, it is possible to determine the nature of unknown atom X in this raw formula, knowing that the force constant of the bond involved in this vibration is 607 N/m. As a result:
   a. Calculate the reduced mass μ of the bond related to this absorption band.
   b. One of the two atoms making up this bond is a hydrogen atom; therefore, deduce the nature of the unknown atom X. Verify this hypothesis, knowing that the molar mass of this compound is M = 75.1 g/mol.
   c. What is the molecular vibration type related to this absorption band?
6. A ZnSe crystal (n = 2.4), with a parallelepiped section, used in a horizontal ATR accessory, has the following dimensions L x w x d = 24 x 8 x 3 mm. The angle of incidence is 45°. A compound with an index of 1.4 is deposited on the entire upper surface of the crystal.
   a. Calculate the critical angle of this crystal when it is coated with the compound.
   b. Calculate the number of reflections on the upper surface.
   c. Deduce the total equivalent thickness at 4,000 cm^{-1} and at 400 cm^{-1}, if we suppose that for each attenuated reflection, the light penetrates a distance equal to a wavelength.
   d. What correction should be done to make the spectrum equivalent to a transmission spectrum?
7. The following experiment is used to determine the percentage of vinyl acetate (VA) in an ethylene/vinyl acetate (EVA) commercial packaging film.
   The absorption band at 1,030 cm^{-1} is used. The absorbance is determined by the tangent method. Results concerning four EAV films of known composition and an unknown EVA film, with thicknesses (in μm) designated by d, are listed below.

EVA	%VA	A_{1030}	A_{720}	d(μm)
1	0	0.01	1.18	56
2	2	0.16	1.55	80
3	7.5	0.61	1.49	82
4	15	0.36	0.45	27
Unknown	?	0.7	1.54	90

a. Taking into account the film thickness *d*, determine, from the data in the table, the equation $A_{1,030} = f(\%\ VA)$, using linear regression, for a film thickness of 1 μm.

b. Explain why the polyethylene band at 720 cm^{-1} (due to CH_2) may be chosen as an internal standard. Then calculate the A_{1030}/A_{720} ratio for the four known films.

c. Using both of the above methods, calculate the vinyl acetate content (%) of the unknown EVA film.

8. To calibrate the internal optical path of measurement cells or to calculate the thickness of transparent films, with index *n* (unless these are reflective), we use an interferometric method, which consists in recording the transmittance of the empty cell (or film) between two wavenumbers $\bar{\nu}_1$ and $\bar{\nu}_2$. The graph obtained appears as a wavy line (Figure P10.1).

a. Write an equation to calculate the thickness of the path in question (cell or film).

b. Calculate the thickness of the polystyrene film, whose spectrum is represented below. Use an average refraction index $n = 1.59$.

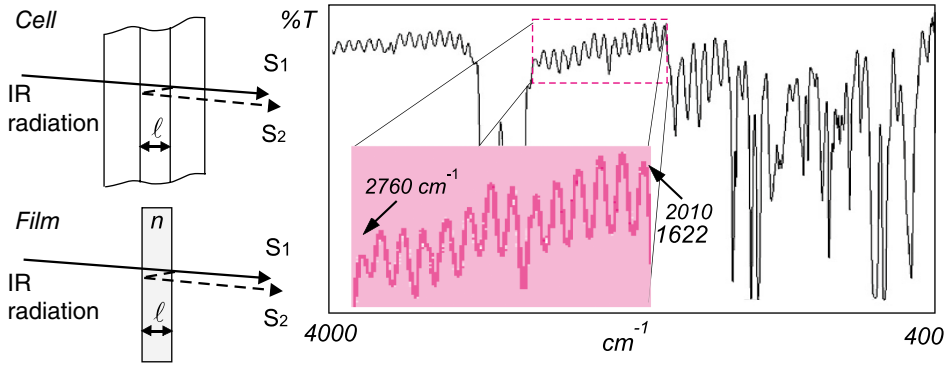


Figure P10.1

SOLUTIONS

1. **a.** $E = 1.99 \times 10^{-20}$ J. For one mole, this corresponds to 12,000 J (i.e. 2.87 kcal/mol).

b. $\bar{\nu} = 666.67\ cm^{-1}$ (or 66,667 m^{-1}); $\lambda = 5.88$ μm.

2. If we assume that the force constant k remains the same for the two species (these are isotopes of the same element), $\bar{\nu}_D = 2{,}215\ cm^{-1}$ (deviation less than 2% from the experimental value).

3. A photon corresponding to 2,000 cm^{-1} carries an energy of

$$E = \frac{hc}{\lambda} = 3.972 \times 10^{-20}\ \text{J}.$$

This energy is found in the form of mechanical energy: $E_{tot} = E_{kin} + E_{pot}$. At maximum stretching Δx, $E_{kin} = 0$ because the speed is null. Energy *E* of the photon is therefore

entirely found in the form of potential energy: $\Delta E_{pot} = 1/2k.(\Delta x)^2$; ΔE_{pot} = E provided by the photon. $(\Delta x) = 8.91 \times 10^{-12}$ m, i.e. about 6% of a bond whose length is 0.15 nm.

4. The approximate reduced mass is 1.138×10^{-26} kg; $k = 1{,}845$ N/m.

5. **a.** By application of Eq. (10.6), $\mu = 1.55 \times 10^{-27}$ kg.

b. Based on Eq. (10.4), $m_2 = (\mu.m_1)/(m_1 - \mu)$ or, by setting m_1 = 1 u, (1.66×10^{-27} kg) and after calculations, $m_2 = 2.34 \times 10^{-26}$ kg, which corresponds to 14 u. Consequently, this is a nitrogen atom (N = 14). Knowing masses H = 1, C = 12 and O = 16 g/mol, the calculation of the molar mass then leads to M = 75 g/mol, a value close to that given in the exercise.

c. Stretching vibration of the N–H bond.

6. **a.** For a relative index of 2.4/1.4 = 1.714, the critical angle α is such that $\sin \alpha = 1/1.714$, i.e. $\alpha = 35.7°$.

b. Number of reflections on the upper surface: 4 (if we assume that the incident beam is perpendicular to the inlet surface).

c. For 4,000 cm^{-1}, i.e. $\lambda = 2.5$ µm, equivalent thickness $4 \times 2.5 = 10$ µm (for 400 cm^{-1}, 100 µm).

d. The absorbance must be multiplied by a factor whose value increases with the wavelength.

7. **a.** $y_1 = 0.0009x + 0.0003$, with $y_1 = A_{1,030}/$µm of film and x = %VA.

b. $y_2 = 0.0531x + 0.0047$, with $y_2 = A_{1,030}/A_{720}$ and x = %VA.

c. %VA = 8.4 based on a) and %VA = 8.46 based on b).

8. **a.** For a cell of internal optical path length l, and under a normal incidence, the S_2 beam is subject to a double reflection on the internal walls, so much so that it goes an additional distance of $2l$ with respect to S_1. If $2l = k\lambda/n$, the two beams S_1 and S_2 are in phase, which will translate into the addition of the two light intensities. For example, $2l = n_1 \cdot \lambda_1/n$ and $2l = n_2 \cdot \lambda_2$. If $N = n_2 - n_1$:

$$N = 2l \cdot n/\lambda_1 - 2l \cdot n/\lambda_2$$

If $\bar{\nu}_1 = 1/\lambda_1$ and $\bar{\nu}_2 = 1/\lambda_2$, we get the equation $l_{(cm)} = \dfrac{N}{2n(\bar{\nu}_1 - \bar{\nu}_2)}$

b. By counting 12 fringes between wavenumbers 2,780 and 2,030 cm^{-1}, the calculation leads to:

$$l_{(cm)} = \frac{12}{2 \times 1.59 \times (2{,}780 - 2{,}030)} = 0.0052\,cm$$

Chapter 11

Fluorescence and Chemiluminescence Spectroscopy

Some compounds emit light when they are excited by photons from the visible or the near-ultraviolet regions. This phenomenon, known as *photoluminescence*, is at the origin of molecular fluorometry, a selective and highly sensitive method for a wide range of measurements. The intensity of the fluorescence is directly proportional to the concentration of the analyte and the measurements are made with the aid of ratiometric fluorometers or spectrofluorometers. In this chapter, some other applications are also addressed, including fluorescent markers in biology as well as

chemiluminescence, which comprises the emission of light during certain chemical reactions. Several applications in analytical chemistry are described for these properties.

Objectives

Describe the origin of fluorescence
Compare several fluorescent compounds
Use fluorescence as a concentration control
Differentiate between Rayleigh scattering and Raman scattering
Compare the different types of fluorometers
Give an overview of a few applications
Distinguish between chemifluorescence and chemiluminescence

11.1 ORIGIN OF FLUORESCENCE

Many compounds, when excited by a light source in the visible or near-ultraviolet regions, absorb energy that is nearly instantaneously re-emitted in the form of radiation. Some present the ability to re-emit this energy nearly instantaneously at a longer wavelength than that of the original excitation light. These are known as *fluorescent compounds*.

11.1.1 Absorption and Relaxation Process

Under standard conditions (1 atm. and 25°C), all molecules are practically in the minimum energy state, as their electrons, in even number, are paired off in their various molecular orbitals. This state is referred to as S_0, or the *singlet state of electron spin* (Figure 11.1).

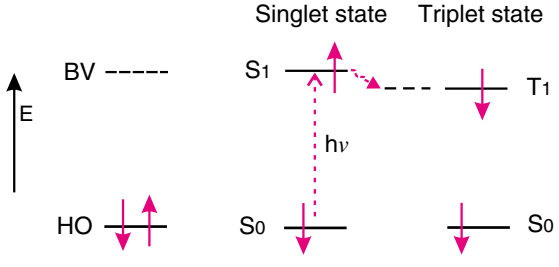


Figure 11.1 Representation, as an energy diagram, of the absorption of a photon by a molecule. Transfer of an electron from an occupied orbital (HO) to a vacant orbital (BV) with appearance of a singlet state evolving into a more stable triplet state (e.g. n → π^* transition). The small arrows designate an electron whose spin is indicated by the arrow direction.



Absorption

When a photon is absorbed, there is an increase in the molecule's energy, which is represented by the passage of one of its electrons into the first excited state (or much less frequently into the second state). This process is very brief (10^{-15} s) with respect to vibrational movements, so much so that there can be no shift of atoms at that instant. During this transfer, the electron conserves its spin. This is also a *singlet state*, called S_1 (or possibly S_2).

Internal conversion

Just after the photon absorption, the molecule is then found in one of the many vibrational levels V_1 related to the excited electron state S_1. If the molecule is not in the isolated state but in solution, the excess vibrational energy is almost instantaneously transferred to the neighbouring solvent molecules, under the effect of intermolecular collisions. However, the positions of the atomic nuclei remain identical to how they were in the ground state (this is the Franck–Condon principle). This nonradiative *vibrational relaxation* very quickly returns the electron to the V_0 level of the S_1 state (10^{-12} s).

Fluorescence

Several relaxation processes are then possible, including one which characterizes fluorescence, represented on a *Jablonski* energy diagram (Figure 11.2). From the V_0 level of the S_1 state, the promoted electron may lose its excess energy by emission of a *fluorescence* photon (10^{-11} to 10^{-6} s). The molecule returns to the S_0 state in one of the related vibrational levels. The fluorescence spectrum is offset from the absorption spectrum towards the longest wavelengths due to the loss of energy caused by vibrational relaxation. It does not depend on the excitation wavelength, except for its intensity.

> After the fluorescence step, various vibrational relaxation processes may still occur if the molecule conserves a part of the energy received during excitation. This excess vibrational energy is dissipated either via collisions or other nonradiative transformations or via the emission of lower-energy photons, which are at the origin of a fluorescence in the mid-infrared region.

Phosphorescence

A third type of nonradiative transition called *intersystem crossing* (ISC) may also occur after the absorption step. It is due to the reversal of electron spin by a 'forbidden' process, because under quantum theory it concerns two electron states of different multiplicities. This transition, called $S_1 \rightarrow T_1$, leads to a triplet state, known as T_1, which is thermodynamically favoured, because its energy is slightly lower than S_1 in accordance with the Hund rule (Figures 11.1 and 11.2). This change will continue: the triplet state will first stabilize in the lowest vibrational state of T_1 (internal conversion). Then, there is either a nonradiative deactivation $T_1 \rightarrow S_0$ or, more rarely, a return with an emission of light known as *phosphorescence* (10^{-4}–1 s). The deactivation of the triplet state generally prevails, because its longer lifespan increases the probability of effective collisions. The

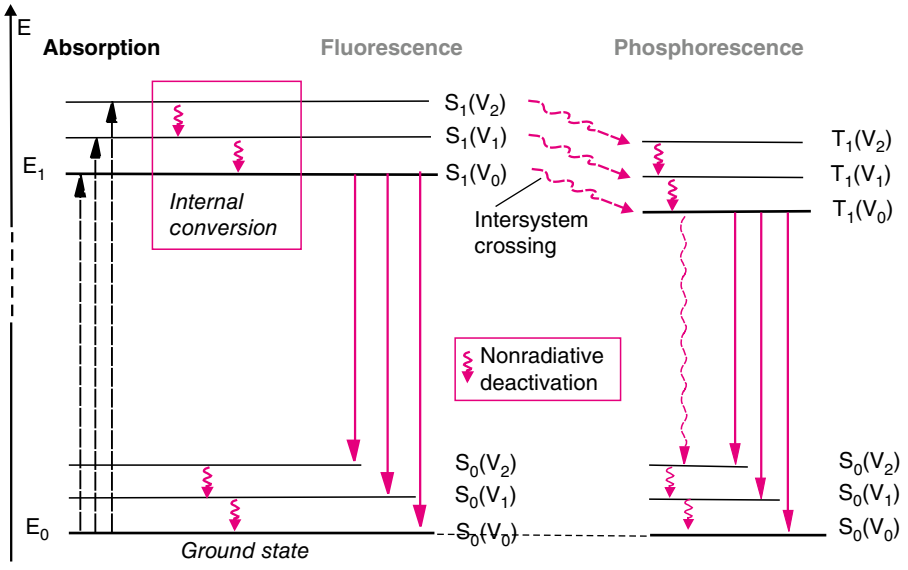


Figure 11.2 Jablonski diagram. According to quantum theory, fluorescence results from transfers between spin states of the same multiplicity, while phosphorescence results from transfers between states of different multiplicity. The T_1 state produces a delay in the return to the ground state of up to several hours. The Stokes shift corresponds to the energy dissipated in the form of heat (vibrational relaxation) during the lifetime of the excited state, i.e. prior to photon emission. However, the real situation is more complex than this simplified Jablonski diagram suggests. At the macroscopic scale, a compound can be both fluorescent and phosphorescent because, at the molecular scale, individual species do not all exhibit the same behaviour. The wavy arrows correspond to vibrational relaxations without photon emission.

speed constant of phosphorescence is very much lower than that of other transitions, as the process involves a new reversal of electron spin. The light emission of a phosphorescent compound may last for several minutes. It can be shorter if the temperature increases following thermal agitation.

11.1.2 Lifespan of Fluorescence

The light intensity due to fluorescence decreases extremely rapidly according to an exponential law. Equation (11.1) relates the intensity of fluorescence I_t to the time t elapsed since excitation:

$$I_t = I_o \cdot \exp\left[-kt\right] \tag{11.1}$$

The *lifespan* of fluorescence τ_0 is defined from k: $\tau_0 = 1/k$. At instant $t = \tau_0$, intensity I_t is equal, according to Eq. (11.1), to 36.8% of the starting intensity I_0. In other words, a fluorescent



compound corresponds at the microscopic scale to a population of individual species, 63.2% of which have returned to a nonemissive state after this time. τ_0 may vary from several nanoseconds to several milliseconds. To facilitate fluorescence measurements, common instruments operate in stationary regime, i.e. by keeping the excitation source on, with the obligation of differentiating the source light from the fluorescent light.

This law of decay also applies to *phosphorescence*. The difference comes from the constant *k*, which is much greater for *fluorescence* than for *phosphorescence* because the latter decays much more slowly.

Time-resolved fluorescence

The development of very short pulsed light sources (e.g. picosecond lasers) has made it possible to plot graphs of fluorescence decay as a function of time. The analysis of the resulting lifespans is used to obtain precious information on the fluorescent species and its environment. However, the complexity and cost of the instruments necessary for such studies limit their use to research and development laboratories.

11.1.3 Stokes Shift

During a fluorescence transition between two energy levels, only one of the emission lines is the same as those of the absorption diagram. It corresponds to the transition between the ground vibrational levels of the electron levels involved and is called a *resonance line*.

On Figure 11.3, which shows the absorbance spectrum and the emittance spectrum on the same graph with a dual scale, resonance lines are related to the overlap zone of the two

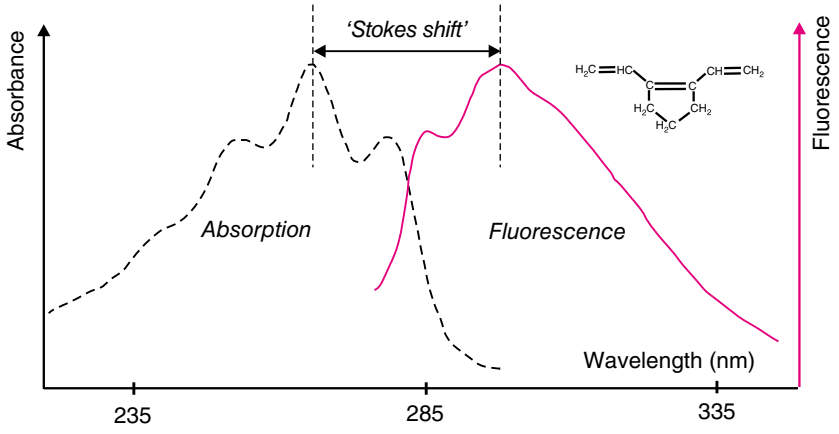


Figure 11.3 Representation on the same graph of the absorbance and fluorescence spectra of divinylcyclopentene. The fluorescence spectrum, which resembles the mirror image of the absorbance spectrum, shows the *Stokes shift*, which can be interpreted by considering the energy diagrams (Figure 11.2). (Source: Adapted from Wolde, A. ten, Dekkers, H. P. J. M., & Jacobs, H. J. C. (1993). Synthesis and fluorescence of a configurationally locked Z-hexatriene: 1,2-divinylcyclopentene. *Tetrahedron*, **49**(27), 6045–6052. doi:10.1016/s0040-4020(01)87189-1.)

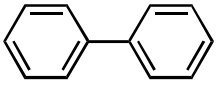
spectra. The high number of levels involved in the transitions accompanying fluorescence explains the appearance of spectra as band spectra, composed of a multitude of lines, whose resolution is often difficult to observe at ambient temperature.

The other fluorescence bands are composed of lines whose wavelength is greater than that of the absorbed radiation responsible for their excitation. This shift to higher wavelengths is called a *Stokes shift*. It explains the apparent mirror symmetry of the absorbance and fluorescence spectra of many compounds (Figure 11.3).

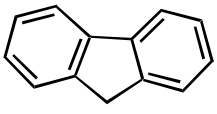
11.2 FLUORESCENT COMPOUNDS

As previously observed (see Section 11.1.1), only the return of the molecule from an excited energy level to a lower energy level through a radiative process is the source of its fluorescence. Not all molecules present this phenomenon. Only molecules whose radiative relaxations occur more rapidly than the nonradiative ones will be fluorescent. The fluorescence quantum yield Φ_f (see Section 11.3) illustrates this competition. A very fluorescent species will have a quantum yield close to one, while this value will practically be null for a nonfluorescent species.

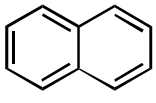
Among organic compounds, rigid cyclic molecules possessing π-bonds produce the most intense fluorescence (Figure 11.4). This fluorescence is increased by the presence of electron donor groups and decreased with electron acceptor groups.



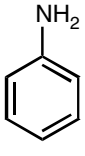
Biphenyl (0.2)



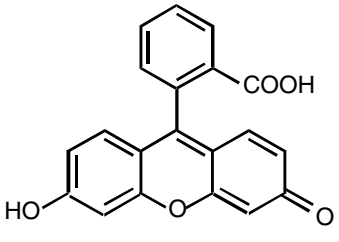
Fluorene (1)



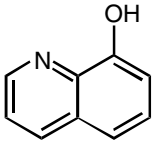
Naphthalene (0.55)



Aniline (0.6)



Fluoresceine (0.92 in NaOH 0.1 M and 0.65 to pH =7)



8-Hydroxyquinoline

Figure 11.4 Fluorescing aromatic compounds. The compound names are followed by their fluorescence quantum yield Φ_f (see Section 11.3), whose values are obtained by comparison with compounds of known fluorescence. The measurements are made at 77 K. 8-Hydroxyquinoline is representative of various molecules forming fluorescent chelates with certain metal ions.

In general, the fluorescence of a molecule may be directly correlated to its structural rigidity. The nonradiative relaxation speed is in fact reduced by the rigidity of the molecule, which makes radiative emission the majority form. An increase in fluorescence is also observed when the fluorescent species is adsorbed on a solid substrate. Moreover, the emission of certain chelating agents increases once they have formed a complex with a metal ion.

In contrast, nonrigid molecules readily lose their entire absorbed energy through decay and vibrational relaxation.

The fluorescence intensity also depends on concentration (see Section 11.3), on pH, on the solvent, and on temperature. Therefore, the quantum yield decreases at high temperatures because the frequency of impacts between molecules increases, which tends to favour nonradiative processes. The same trend occurs when the solvent's viscosity decreases.

11.3 RELATIONSHIP BETWEEN FLUORESCENCE AND CONCENTRATION

At each point of the solution, fluorescence intensity is different because part of the exciting radiation is absorbed before reaching the point under consideration and because part of the fluorescence is trapped before it can exit the cell. Overall, the fluorescence received by the detector thus corresponds to the sum of the fluorescence emerging from each of the individual small volumes constituting the space delimited by the entrance and exit slits (Figure 11.5). This is why it is difficult to calculate the fluorescence intensity (emittance I_f) for the sample. The phenomenon of radiation damping, called *internal quenching*, is due to the partial overlap of the absorption and emission spectra (*colour quenching*), and is increased by transfers of energy from excited species to other molecules or ions through collisions or formation of complexes (*chemical quenching*). That is why the presence of oxygen can cause an underestimation of fluorescence.

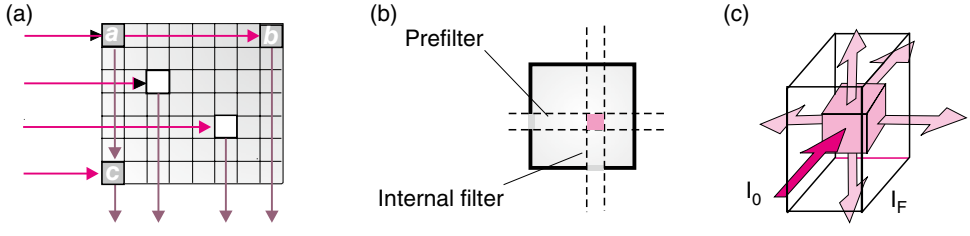


Figure 11.5 Fluorescence intensity. Depending on the site in the solution where the fluorescence is emitted, a variable intensity of light reaches the detector. It is possible to evaluate the re-absorption of the light of fluorescence (comparison between a and c) and the absorption of the incident light (comparison between a and b). In practice, the presence of an iris means that only the light arriving from the central region of the cell is collected.

For solutions, the fluorescence quantum yield Φ_f (between 0 and 1 inclusive) is defined by the ratio of the number of photons emitted to the number of photons absorbed. This is equivalent to the ratio of fluorescence intensity I_f over absorbed intensity I_a:

$$\Phi_f = \frac{\text{number of photons emitted}}{\text{number of photons absorbed}} = \frac{I_f}{I_a} \tag{11.2}$$

If we accept that $I_a = I_0 - I_t$ (where I_t represents the transmitted light intensity), the following reasoning (Eq. (11.5)) allows us to relate I_f to the concentration C of the compound (via Eqs. (11.3) and (11.4)):

$$I_f = \Phi_f \left(I_o - I_t\right) \text{ that is } I_f = \Phi_f \cdot I_o \cdot \left(1 - \frac{I_t}{I_0}\right) \tag{11.3}$$

Since absorbance A is equal to $\log I_0/I$, Eq. (11.3) becomes:

$$I_f = \Phi_f \cdot I_o \cdot \left(1 - 10^{-A}\right) \tag{11.4}$$

where

$$10^{-A} = 1 - 2.303A + \frac{\left(2.303A\right)^2}{2!} - \cdots + \cdots$$

If the solution is diluted, the term A is close to 0 and the term 10^{-A} is therefore close to $1 - 2.3A$. Equation (11.4) can then be simplified as follows:

$$I_f = 2.3 \cdot \Phi_f \cdot I_o \cdot A \text{ thus } I_f = 2.3 \cdot \Phi_f \cdot I_o \cdot \varepsilon \cdot l \cdot C \tag{11.5}$$

where I_0 is the intensity of the exciting radiation, C the molar concentration of the compound, ε its molar absorption coefficient, l the cell thickness, and Φ_f the fluorescence quantum yield.

Equation (11.5) reveals that the fluorescence intensity depends upon the concentration (C), the experimental conditions (l, I_0), and the compound (ε, Φ_f). In practice, during an assay, the parameters for the compound under study and for the instrument are fixed. If they are all factored into a single constant K, then the following equation can be used for weak concentrations ($A < 0.01$):

$$I_f = K \cdot I_0 \cdot C \tag{11.6}$$

Measurements by fluorometry make use of several classic calibration methods employing one or more standards (single-point calibration or calibration curve, or methods of addition), but to obtain the best results, the solutions must be very dilute.

Above a given limit, fluorescence is no longer directly proportional to concentration. This arises because of the nonlinearity of the Beer–Lambert law. The excitation is proportionally weaker and association complexes appear between excited molecules and those that stay in the ground state. This leads to the apparently paradoxical result that the fluorescence can diminish, even though analyte concentration increases. The curve in Figure 11.6 is a graphic illustration of Eq. (11.7) (resulting from Eq. (11.4)) following the introduction of reasonable values for these different parameters.

$$I_f = \Phi \cdot k \cdot I_0 \cdot 10^{-(\varepsilon_1 \ell_1 + \varepsilon_2 \ell_2)C} \cdot \left(1 - 10^{-\varepsilon \ell C}\right) \qquad (11.7)$$

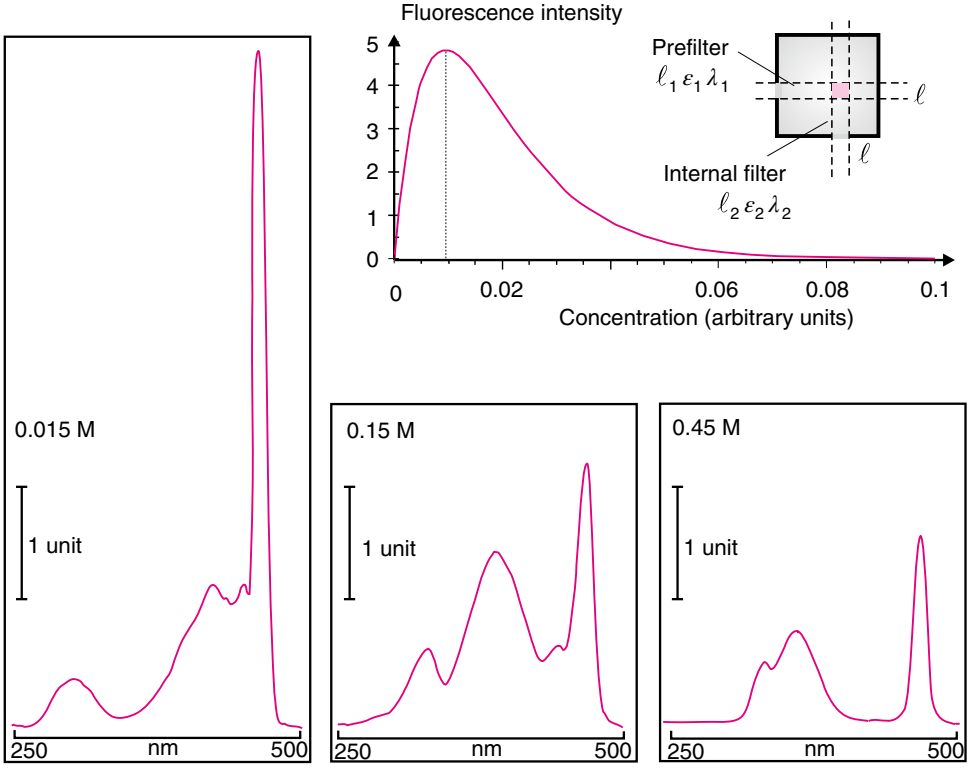


Figure 11.6 Fluorescence intensity and concentration. Modelling of Eq. (11.7) reveals the effect of the concentration upon fluorescence intensity. A fluorescence maximum is observed beyond which fluorescence decreases with a continuing increase in concentration. After the maximum, the more concentrated the solution is, then the weaker the fluorescence is – a kind of *roll-over* or *self-quenching*. The illustrations correspond to three recordings, at the same scale of diacetyl solutions in tetrachloromethane. The curve records the light collected from a small volume with sides equal to *l* situated at the centre of the solution as indicated on Figure 11.4 (parameters ε and *l* chosen arbitrarily).

11.4 RAYLEIGH SCATTERING AND RAMAN SCATTERING

When the wavelengths of excitation and of emission are close together, other bands, near the fluorescence of the sample, can be seen coming from scattering phenomenon due to the matrix or the solvent in which the compound is found (Figure 11.7).

11.4.1 Rayleigh Scattering

The *elastic* Rayleigh scattering is the re-emission, by the matrix or solvent in which the compound is dissolved, of a small fraction of the absorbed excitation light in all directions at the *same wavelength* (see Chapter 10, Section 10.14).

11.4.2 Raman Scattering

On either side of the Rayleigh scattering line, there are satellite lines due to *inelastic scattering* effects. These come from the vibrational interactions between molecules of the solvent and the exciting radiation. The shift between the Raman lines and the Rayleigh line corresponds to the vibration frequencies of the molecule (see Chapter 10, Section 10.13).

Compounds shifted towards longer wavelengths with respect to the Rayleigh line are called *Stokes* lines and those shifted towards shorter wavelengths are called *anti-Stokes* lines.

The difference in energy between the absorbed photons and the re-emitted photons is constant. Therefore, it is possible, by modifying the excitation wavelength, to shift the position of the Raman peak (Table 11.1 and Figure 11.7).

The Raman scattering of water serves as a sensitivity test for fluorometers. This consists of measuring the signal/noise ratio of the Raman peak with a cell filled with water, for example at 397 nm (25,191 cm^{-1}) if the excitation wavelength is fixed at 350 nm (28,571 cm^{-1}), as a result of the specific shift of 3,380 cm^{-1} for this solvent (corresponding to the wavenumber of the stretching vibration ν_{OH}).

The phenomenon of fluorescence is not only used for chemical analyses but is equally applied to substances such as washing powders, which contain a fluorescing agent that

Table 11.1 Position of the Raman peaks, calculated for four commonly used solvents and five excitation wavelengths from a mercury lamp.

Excitation (nm)	254	313	365	405	436	Shift (cm^{-1})
Water	278	350	416	469	511	*3,380*
Ethanol	274	344	405	459	500	*2,920*
Cyclohexane	274	344	408	458	499	*2,880*
Chloroform	275	346	410	461	502	*3,020*

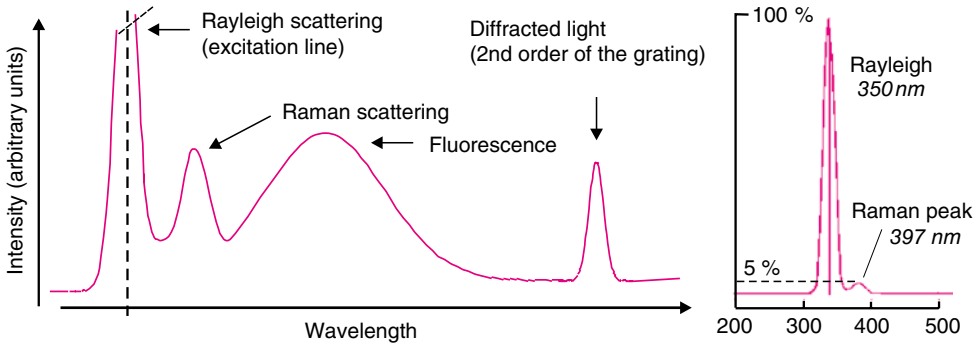


Figure 11.7 The various components of a fluorescence spectrum. The position of the Raman peak depends upon the wavelength of the excitation line and of the solvent. Right, a sensitivity test for a fluorometer.

sticks to the textile fibres. This type of compound absorbs solar radiation in the nonvisible part of the spectrum and re-emits at longer wavelengths, in the blue spectral region of the spectrum, visible to the eye.

11.5 INSTRUMENTATION

A fluorescing compound behaves like a source that emits light in all directions. For weakly absorbing compounds, the detector is placed perpendicular to the source in order to be free from the transmitted component, which would mask the fluorescence and would degrade the detector by saturating it. For strongly absorbing solutions, fluorescence is observed in the continuation of the incident beam, while for opaque or semi-opaque samples, a frontal measurement at a variable angle is preferable (Figure 11.8).

The excitation source is often a xenon arc lamp with a power of 150–800 Watts for simpler instruments. A continuous or pulsed laser (dynamic fluorescence or time-resolved fluorescence) is used for more complex set-ups. The measurement of the light intensity is carried out using a photomultiplier tube or a photodiode.

The solvent, temperature, pH, and concentration are the main parameters which affect the intensity of fluorescence.

Some gases present as traces in the Earth's atmosphere are quantified from their retro-scattered fluorescence, induced following excitation by a very short pulse (1 μs) from a strong monochromatic laser, which can be tuned to a specific wavelength for the compound in question (device known as LiDAR (light detection and ranging)).

Two broad categories of commercial instruments are currently proposed by manufacturers: ratio fluorometers and spectrofluorometers.

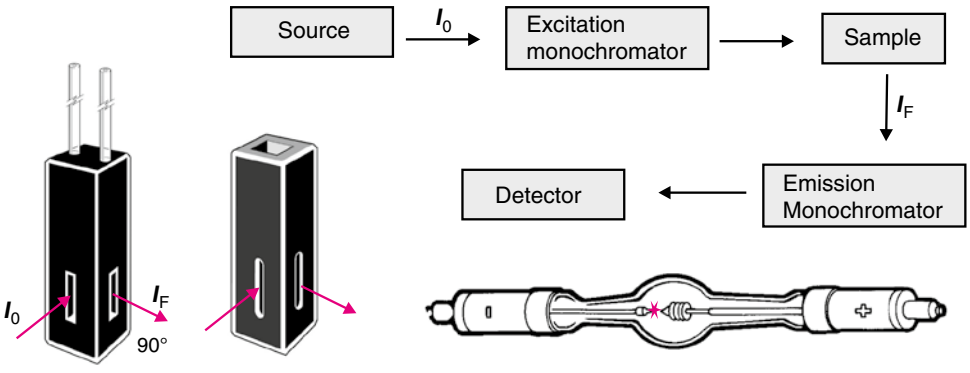


Figure 11.8 Block diagram of a spectrofluorometer with a xenon arc lamp. The fluorescence is measured, in contrast to studies on fluorescence dynamics, under steady-state conditions by maintaining the primary excitation source. Circulation microcuvette and 90° standard cuvette. A diagram of a xenon arc lamp. The pressure of xenon in the lamp is around 1 MPa. These arc lamps, made from a fused silica envelope and without a filament, are sources of white light. The cathode is the finer of the two electrodes.

11.5.1 Ratio Fluorometers (Ratiometric Method)

The light emitted by the primary source initially passes through the excitation monochromator, which allows a narrow band of wavelengths to be selected (15 nm) to induce fluorescence of the molecules of interest in the sample solution. A part of the fluorescence emitted by the compound is collected by the emission monochromator in a direction either perpendicular or parallel (depending upon the model) to the direction of the incident beam. The selected radiation then reaches the detector.

Many manufacturers have commercialized models fitted with filters (Figure 11.9), which replace one of the monochromators (often both even). These instruments are single-beam devices. These basic instruments contain a turret compartment for the calibration solutions, the sample, and a compound to be used as the fluorescing standard. By alternating the calibration solutions, the sample, and the fluorescing standard in the same optical path, the fluorescence ratios can be measured. Thus, a calibration curve can be plotted and the measurement results can be determined. In this manner, possible fluctuations from the source and a large number of settings of the instrument can be eliminated. The commonly used standards in the ratiometric method are quinine sulfate, rhodamine B or 2-aminopyridine.

The use of a xenon flash lamp as the pulsed source enables the study of fluorescence after the source has been switched off. This method is used in immunology, in which fluorescing complexes obtained with lanthanide salts have a fluorescence lifetime of around 1 ms, which is sufficient for very sensitive measurements (in the picomolar range).

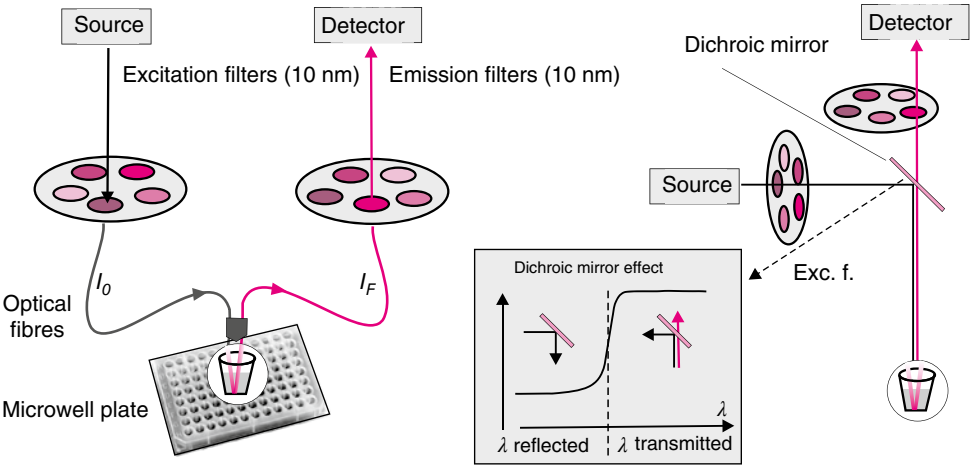


Figure 11.9 Diagram of the optical path of a fluorescence microplate reader. Left, one optical fibre carries the excitation radiation to the wells chosen for analysis and a second fibre recovers the light emitted by the sample via a forward geometry. Right, a model (Synergy-2 from BioTek) with a dichroic mirror in order to improve sensitivity.

11.5.2 Spectrofluorometers

Spectrofluorometers enable a more complete study of fluorescent compounds, notably by recording both excitation and emission spectra (Figure 11.10). They have two motorized monochromators, both of which are able to scan a spectral band. It is possible to obtain the entire emission spectrum while maintaining a fixed excitation wavelength or to record the entire excitation spectrum while maintaining a constant emission wavelength.

Spectrofluorometers have software which can automatically determine the excitation–emission wavelength pair to get the best results (Figure 11.11).

The process can be done manually using the 'one factor at a time' method:

1. The UV spectrum of the compound is recorded with a UV/Vis spectrophotometer.
2. The excitation monochromator is set to the value corresponding to the maximum of the absorption spectrum.
3. The fluorescence spectrum is recorded by scanning the spectral domain of interest by rotating the emission monochromator.
4. The emission monochromator is set to the wavelength of maximum fluorescence and the excitation wavelength is varied. We thus obtain the excitation spectrum allowing the best final choice of the exciting radiation (which may be different from the UV absorption maximum).

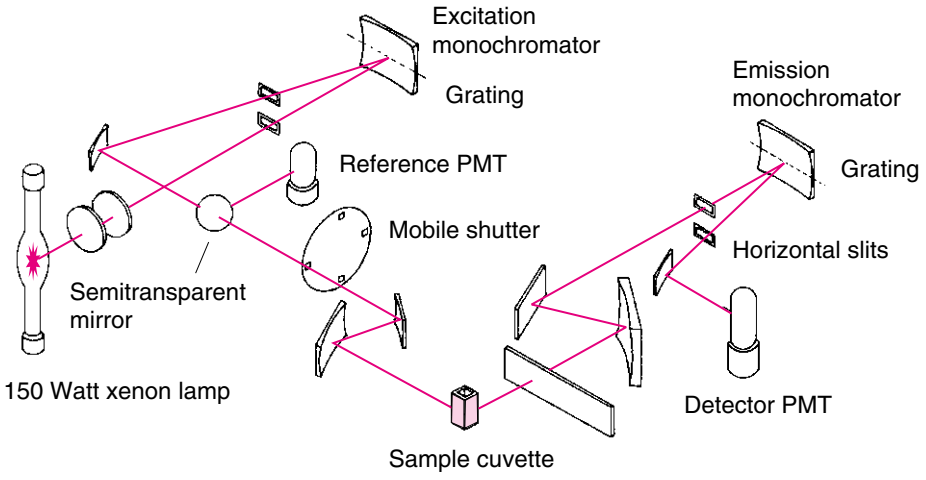


Figure 11.10 Diagram of a Shimadzu F-4500 spectrofluorometer. A fraction of the incident beam, reflected by a semitransparent mirror, reaches a reference PMT. A comparison of the signals from the two detectors compensates for any drifting of the source. This process, for single-beam instruments, gives approximately the same stability as with a double-beam spectrometer. However, the spectrum of a given solution will often present minor differences when recorded on different instruments (reproduced courtesy of Shimadzu).

11.5.3 Time-Resolved Spectrofluorometers

Technological advances with these instruments now help us measure the lifespans of fluorescence, even though these are very short. To do so, there are several methods based either on recording the decay curve of the light emitted when the excitation has stopped or on comparing the modulation of fluorescence as a function of the quick modulation of excitation.

Knowing that the light intensity decays very quickly, the first method requires the use of a pulsed source at frequencies of several megahertz (for example, laser diode or dye laser). The pulse period may vary, depending on models, and according to the application, from several milliseconds (10^{-3} s) to several nanoseconds (10^{-9} s) or even femtoseconds (10^{-15} s). The second method uses a high-frequency modulated source.

During these successive cycles, when the source is switched off, the detector measures the time elapsed until the random moment when a photon of fluorescence reaches it. Then, this event is saved in the memory channel reserved for the corresponding time. After having accumulated a large number of photons, the graph of the decay is finally constructed by creating a histogram of the contents of all of the memory channels (Figure 11.12).

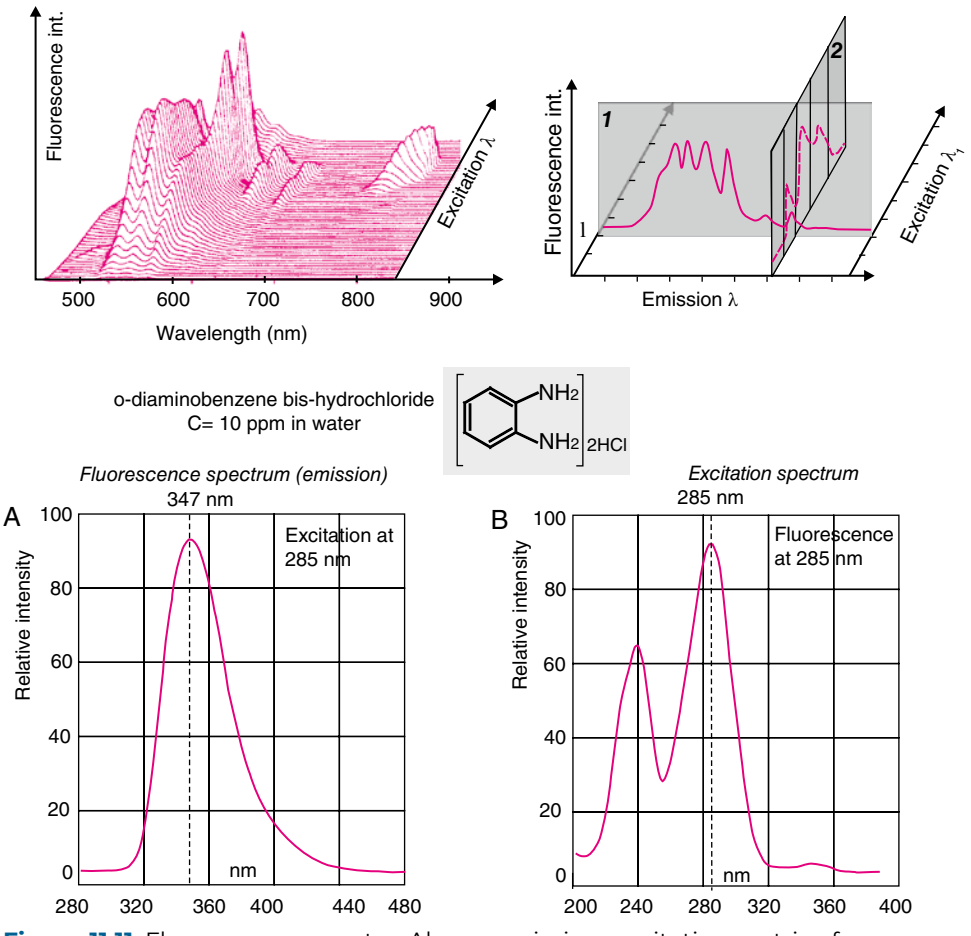


Figure 11.11 Fluorescence spectra. Above, emission–excitation matrix of a mixture of two fluorescent ions. This type of recording leads to the optimum excitation wavelength. Above right, (1) corresponds to a fluorescence spectrum (excitation λ_1). (2) corresponds to the fluorescence observed at λ_2 when the excitation wavelength is varied. Below, emission–excitation spectra A and B.

11.6 SPECIFICITIES AND APPLICATIONS

Fluorometry is very sensitive because a light emission is quantified, rather than a fraction of absorbed light, established by comparing two intensities, as in UV/visible spectroscopy. Sensitivity in fluorometry is often 1,000 times greater than that of UV/visible absorption. It is also improved by the use of powerful sources (xenon arc), whereas in absorption, there is no effect on the ratio of transmitted intensities.

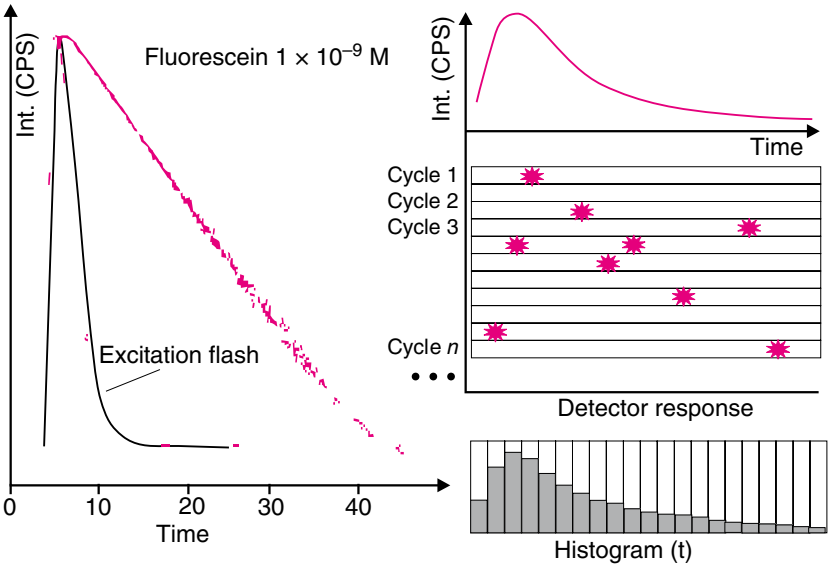


Figure 11.12 Representation of fluorescence decay and the principle of measurement. On the graph, each point represents the contents of a memory channel (time/number of photons). The exponential curve of decay appears here in the form of a straight line. This is due to the choice of a logarithmic scale for the *y*-axis. On the right, the illustration indicates how this representation is reached.

Apart from molecules that fluoresce by excitation (less than 10% of all existing molecules), many can be indirectly made fluorescent by chemical modification or by association with a fluorescent molecule. For example, a *fluorophore reagent* can be grafted to an analyte by chemical reaction (7-hydroxycoumarin can be used to this effect). This is called *fluorescence derivatization*, reminiscent of the process employed in colorimetry.

For the determination of metal cations, chelates are created with oxine (8-hydroxyquinoline), alizarin or benzoin, and then extracted by organic solvents. In biochemistry, fluorescence has numerous applications for the quantification of proteins or nucleic acids by means of reagents that can affix to these compounds with specificity.

There are many, extremely varied applications. These concern the bioanalytical (fluorescence readers for in vitro diagnosis), food (food control), pharmaceutical (analysis of active ingredients), and environmental (toxicity test) industries among others, but may also extend to materials, photonics or optoelectronics.

Chemifluorescence (not to be confused with chemiluminescence, see Section 11.7) is also a particularly sensitive method for the detection of specific proteins. A specific antibody for this protein is required for this. This antibody carries a conjugated enzyme, such as a phosphatase (Figure 11.13). If mixed with a substrate such as a phosphate derivative of fluorescein, the latter compound is released and then revealed by its fluorescence, which is provoked by a suitable source of excitation.

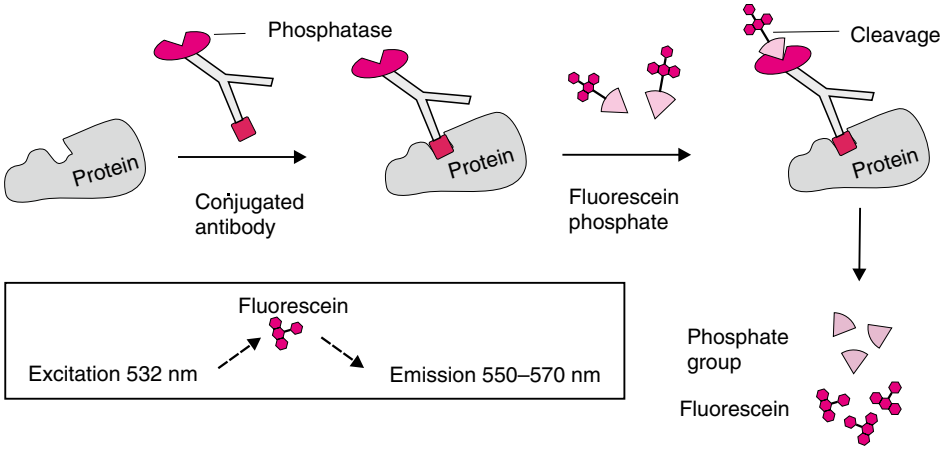


Figure 11.13 Protein assay by chemifluorescence in biochemistry. The diagram depicts the reaction sequence involved. To perform this assay, a protocol, in which the different reactions are considered as distinct steps, must be followed.

To improve analyses by HPLC, amines can be fluorescently labelled (with other coumarins), which makes detection thresholds very low, i.e. on the order of an attomole (10^{-18} M).

The current applications of fluorescence include measurements of polycyclic aromatic hydrocarbons (PAH) in drinking water. In this case, the detector is equipped with a fluorescence flow cell installed at the column outlet. This mode of detection is particularly well adapted for attaining the very low threshold standards imposed by legislation. The same process can be used to measure aflatoxins (Figure 11.14), as well as numerous other organic compounds such as adrenaline, quinine, steroids, and vitamins.

11.7 CHEMILUMINESCENCE

Chemiluminescence corresponds to photon emission coming from the de-excitation of unstable species formed during certain chemical reactions. The formation of excited states occurs without photons, but de-excitation is of the same nature as that which gives rise to fluorescence or phosphorescence. The kinetics of these reactions is variable and subsequent light emission is more or less brief, such that the exploitation of chemiluminescence in quantitative analysis requires being able to integrate the light emitted during the entire time it is produced. Several examples are shown in Figure 11.15.

Luminol and diphenyl oxalate are two compounds very frequently used in a variety of classic applications from nonelectric emergency lighting to the glowsticks and glow necklaces sold at fairgrounds. The reactions that cause emissions are all oxidations by hydrogen peroxide.

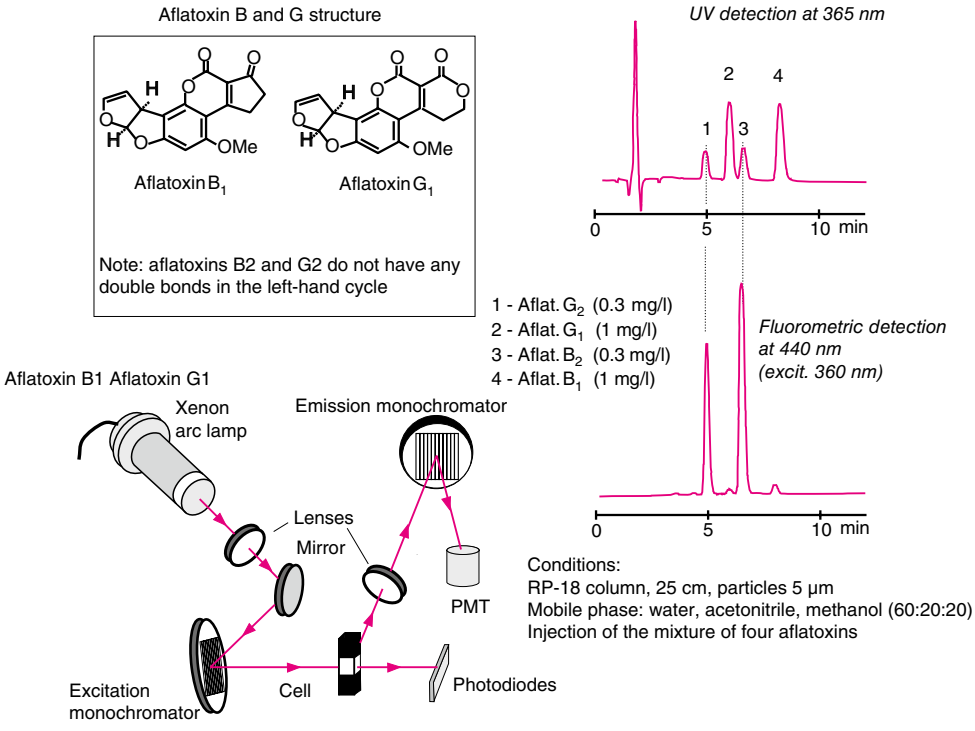


Figure 11.14 Comparison of UV and fluorescence detection following a chromatographic separation. Aflatoxins are carcinogenic contaminants found in certain batches of grains. They can be analysed by HPLC. With UV detection, peak intensities vary with the concentrations of the four compounds. In contrast, detection by fluorescence is much more sensitive to G_2 and B_2 than is UV detection (reproduced courtesy of SUPELCO). Below left, diagram of the different components of a detector based upon fluorescence. This detector is able to find rapidly, for each compound eluted, the best excitation/emission pair without interrupting the chromatographic process. The chromatograms obtained are examined after analysis. (Source: Courtesy of Agilent Technologies).

These reactions are the basis for numerous measurements of great sensitivity. While the use of luminol to measure ferrous iron or hydrogen peroxide may seem anecdotal, there are a certain number of significant applications for chemiluminescence in chemical analysis, such as those that make use of ozone as an oxidizing agent. This strong oxidizing agent enables the measurement of nitrogen monoxide (oxidized to nitrogen dioxide) with the aid of an automatic instrument that includes an ozone generator (Figure 11.16). The scope of use for this measurement is broader than it

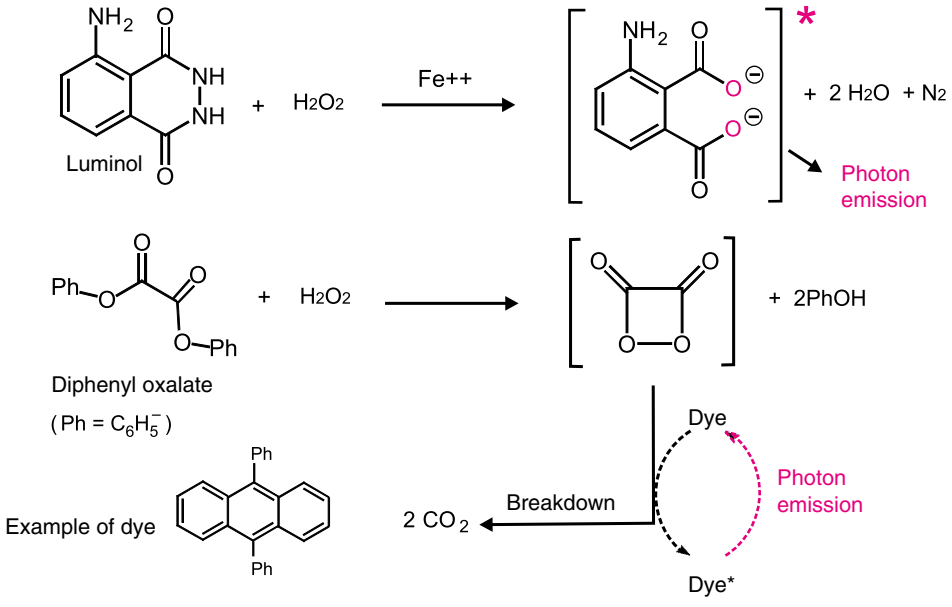


Figure 11.15 Examples of chemiluminescence reactions. The nature of the reaction products is in some cases not well known. Luminol emits an intense 'electric blue' light, while diphenyl oxalate, depending upon the dye used, leads to the emission of a great variety of colours. The aromatic derivative employed as the dye is used to generate the blue colour of glowsticks.

might seem, ranging from studies of polluted atmospheres (nitrogen dioxide can be transformed quantitatively into nitrogen monoxide) to the measurement of the *total nitrogen* in an organic sample, on the condition that the nitrogen has undergone a previous transformation by combustion to nitrogen dioxide.

The same principle can be applied to measuring elemental sulfur, which is first transformed to sulfur dioxide (SO_2) by combustion in the presence of oxygen (Figure 11.17), then reduced to hydrogen sulfide (H_2S) before finally being re-oxidized by ozone (chemiluminescence step).

In gas chromatography, some detectors, selective for the elements sulfur and nitrogen, use this sequence of reactions.

Inversely, ozone can in its turn be measured by chemiluminescence in the presence of ethylene (Figure 11.16).

Luminol, whose chemiluminescence in the presence of hydrogen peroxide can readily be catalysed by horseradish peroxidase (HRP), has found an application in the development of enzyme-linked immunosorbent assays (ELISA) (see Chapter 17, Section 17.5). The light intensity which appears is stable for a few minutes, making this method extremely sensitive over an extended range of concentrations, which is unusual for this kind of test.

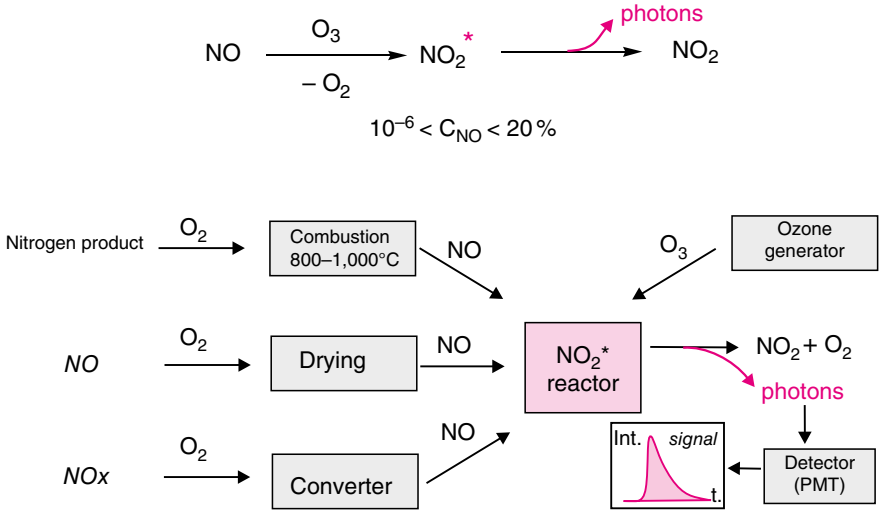


Figure 11.16 Nitrogen analyser using chemiluminescence. Stoichiometric reaction of ozone on nitrogen monoxide. After combustion of the nitrogen-containing sample or after the catalytic reduction of NO_x (e.g. NO_2) to NO (use of a dedicated catalytic converter), this luminescence reaction is used in many measuring instruments (luminometers), some of which give a near-instantaneous response (automotive emissions checks). This total nitrogen measurement can be used in place of the Kjeldahl method (see Chapter 21, Section 21.8).

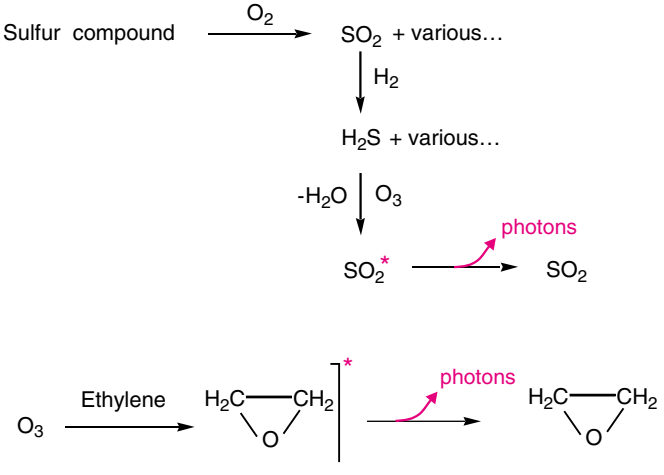


Figure 11.17 Chemiluminescence reactions using ozone. The transformation of a compound containing sulfur dioxide in the excited state, is conducted with ozone in several instruments dedicated to this type of assay. Below, the final reaction can be applied to the measurement of ozone just as readily as for ethylene.

Bioluminescence, which is chemiluminescence of the living world, comes from a transformation cycle that involves luciferin, whose oxidation into oxyluciferin by molecular oxygen is accompanied by a brief emission of light. Two additional compounds are necessary: luciferase, a specific enzyme that catalyses the reaction, and adenosine triphosphate (ATP), which provides energy to the cell. The reaction cycle ends with the regeneration of luciferin. This reaction is used in many measurements in the food industry and in the life sciences, which are outside the scope of this book.

KEY POINTS OF THE CHAPTER

1. Many compounds (10%) are fluorescent if, after having been irradiated by a light source, they nearly instantaneously re-emit radiation with a longer wavelength than that used for their excitation.
2. The impact of a photon on an electron involved in a bond instantaneously propels it into a vacant orbital, while conserving its spin. Several relaxation processes are possible. These will cause the molecule in question to lose its energy received via the impact of the photon. This can be done by nonradiative collisions or by photoluminescence, i.e. by photon emission.
3. Two emissive phenomena must be distinguished, often summarized by the Jablonski diagram: *fluorescence*, near-instantaneous return to the ground state, and *phosphorescence*, where in a preradiative phase, there is a change in electron spin; the electron passes into a *triplet* state, whose lifespan may reach several seconds. The return to the ground state (phosphorescence) involves a triplet/singlet transfer, which is slow.
4. For a fluorescent (or phosphorescent) sample, formed from a population of identical individual species, light intensity decreases exponentially over time:

$$I_t = I_o \cdot \exp\left[-kt\right]$$

5. The lifespan of fluorescence is time τ_0, such that 63.2% of atoms that passed into an excited state have returned to the ground state.

$$\tau_0 = 1/\mathrm{k}.$$

6. Fluorescence is mostly found in compounds with a rigid structure. It can be characterized by the fluorescence quantum yield.

$$\Phi_f = \frac{\text{number of photons emitted}}{\text{number of photons absorbed}} = \frac{I_f}{I_a}$$

7. Nonrigid compounds easily lose all of the absorbed energy by nonemissive vibrational relaxation. Many factors due to the medium in which the compound is found disturb fluorescence (*quenching*): pH, solvent, temperature and viscosity.

8. Rayleigh scattering and Raman scattering. *Rayleigh scattering*, generally intense, is due to the solvent which re-emits a part of the incident intensity (elastic scattering). *Raman scattering* is weak and consists of photons coming from vibrational interactions between the exciting radiation and the solvent.
9. At the macroscopic scale, the continuous irradiation of a fluorescent compound produces an emission that is also continuous, which makes it possible to conduct quantitative measurements. This method, called fluorometry, relies on the relationship between fluorescence and concentration, which can only be exploited for very diluted solutions. K is evaluated by calibration: $I_f = K \cdot I_0 \cdot C$
10. Instruments used for fluorescence measurements are divided into two categories: *filter fluorometers* (excitation and emission), using a ratiometric method, require a fluorescence standard to establish the calibration curve. *Spectrofluorometers* are used to obtain absorption and emission spectra by comparison (the best pairing of excitation and fluorescence wavelengths).
11. Some nonfluorescent compounds may purposely be subjected to a chemical transformation intended to give them this property. This fluorescence derivatization is obtained by grafting on a fluorescent reagent or by creating a series of reactions, which must all be quantitative.
12. *Chemiluminescence* is a photon emission during a chemical reaction going through an unstable state. This property has received industrial applications (ozone, ethylene, nitrogen oxide measurements, etc.).

PROBLEMS

1. Ferrous iron catalyses the oxidation of luminol by hydrogen peroxide. The intensity of the resulting chemiluminescence increases linearly with the concentration of Fe(II) as it rises from 10^{-10} M to 10^{-8} M. To measure a solution of unknown Fe(II) concentration, 2 ml of the solution is extracted and then 1 ml of water is added, followed by the addition of 2 ml of hydrogen peroxide and finally 1 ml of a basic solution of luminol. The luminescence signal is emitted and integrated over a period of 10 seconds, giving a value of 16.1 (arbitrary units). In a second test, 2 ml of the unknown Fe(II) solution is again extracted. In this case, 1 ml of a 5.15×10^{-5} M solution of Fe(II) is added, and the same quantities of hydrogen peroxide and luminol are introduced as before. The signal is measured as 29.6 (same units).
   Calculate the molar concentration of Fe(II) in the unknown solution.
2. An aqueous solution of 9-aminoacridine leads to a fluorescence intensity, which, when measured at 456 nm, is 60% when compared to an external fluorescence control. A sample of this compound, whose concentration is 0.1 ppm in the same solvent, leads to a fluorescence of 40%, under the same conditions (water has negligible fluorescence).
   Calculate the concentration of the sample in ppb.

3. 3,4-benzopyrene is a dangerous aromatic hydrocarbon frequently present in polluted air. It can be measured by fluorometry in a dilute sulfuric acid solution. The excitation wavelength is 520 nm, and the measurement is done at 548 nm.
10 l of contaminated air is bubbled in 10 ml of dilute sulfuric acid. The fluorescence of 1 ml of this solution, measured with a single-beam fluorometer at 548 nm, is 33.33 (arbitrary units).
Two standard solutions, one containing 0.75 μg and the other 1.25 μg of 3,4-benzopyrene per ml of the same dilute sulfuric acid solution, lead to values of 24.5 and 38.6 (same unit scale) for extractions of 1 ml. A sample not containing 3,4-benzopyrene gives a value of 3.5 (same unit scale) by fluorometer under the same conditions.
Calculate the mass of 3,4-benzopyrene per litre of air.
4. Fluorometry is the method selected to measure the concentration of quinine in a commercially available drink. The excitation wavelength is 350 nm, and the wavelength for the measurements is 450 nm. Starting from solution A, which is quinine at
0.1 mg/l, a series of standards are prepared to establish a calibration curve.
A sample of 0.1 ml of the drink is taken and then diluted to 100 ml with 0.05 M sulfuric acid. The value of the signal for this solution is 113.
   a. Using linear regression, calculate the equation of the calibration line.
   b. What is the concentration of quinine in g/l and in ppm?
5. A test for the sensitivity of a fluorometer is to measure the fluorescence intensity of the Raman scattering peak of a cell filled with water and with an optical path length of 1 cm.

	Volume of A	0.05 M H_2SO_4	Fluorescence (arb. units)
Standard 1	20	0	182.0
Standard 2	16	4	138.8
Standard 3	12	8	109.2
Standard 4	8	12	75.8
Standard 5	4	16	39.5
Blank	0	20	0

If the excitation wavelength is set at 250 nm, at what wavelength must the measurement be made (the Raman shift of water is 3,380 cm^{-1})?

SOLUTIONS

1. This is an addition method. The concentration of the Fe(II) solution is 3.07×10^{-5} M.
2. The concentration of the sample solution is (60/40) × 0.1 = 0.15 ppm, i.e. 150 ppb.

3. Firstly, the values read for the three solutions are corrected by subtracting the value for the analytical blank. $x = 1.062$ µg/ml. This value corresponds to the mass of benzopyrene found in 1 l of air.
4. **a.** $I_f = 1.773 \times 10^6 C + 2.22$;
   b. $c = 6.25 \times 10^{-2}$ g/l, i.e. 62.5 ppm.
5. A wave number of 40,000 cm^{-1} corresponds to 250 nm. The Raman peak of water will thus be located at 40,000 – 3,380 = 36,620 cm^{-1}, which corresponds to a wavelength of 273.1 nm.

Chapter 12

X-Ray Fluorescence Spectroscopy

X-ray fluorescence spectroscopy is a chemical analysis technique used to obtain information about the elemental composition of many types of samples. The principle of operation involves irradiating the sample either with an X-ray beam or by bombardment with particles. Target atoms, passing into the ion state, may return the energy received via several mechanisms, especially as short-wavelength radiation made up of characteristic X-rays, which makes their identification possible. The technique encompasses a wide variety of

instruments, both dedicated and multifunctional, ranging in size from portable analysers to high-resolution spectrometers, not to mention X-ray probes fitted to microscopes to conduct specific analyses or to map the elements present.

Objectives

Describe the basic principle
Classify X-ray sources
Review detectors
Compare the two instrument categories (ED- and WD-XRF)
Measure the linear absorption coefficient
Prepare samples to measure
Describe several applications

12.1 BASIC PRINCIPLES

Several phenomena occur when a material used as the target is irradiated with high-energy radiation (the term radiation can refer to electromagnetic radiation or to a particle) of between 5 and 100 keV. A part of the radiation is diffused (elastic Rayleigh scattering or inelastic Compton scattering) while the rest is absorbed and provokes an electron transition. If the incident energy is high enough, the atom moves to the ionized state.

In the case of electromagnetic radiation, the ejected electron is a *photoelectron* (the photoelectric effect). However, when the primary radiation is made up of electron beams, the ejected elections are called *secondary* (Figure 12.1).

Following the departure of the electron, the atom relaxes while a higher-level electron fills the vacant quantum space resulting from this departure.

When the incident radiation energy is large, the emitted photon also has a large energy. It belongs to the domain of X-rays. This phenomenon is the source of X-ray fluorescence.

This X-ray photon may be recaptured by the atom itself and cause the ejection of a peripheral electron (this is *Auger emission*).

> The modes of excitation that can provoke X-ray fluorescence of atoms are varied: photons or particles (fast electrons, protons, α radiation, etc.). Whatever the chosen method, the radiation produced nevertheless yields the same X-ray fluorescence radiation.

How this can be applied in chemical analysis is obvious: it consists of comparing the X-ray fluorescence emitted by the unknown sample with reference spectra whose compositions are known or with pure elements ranging from boron ($Z = 5$) to uranium ($Z = 92$).

The spectra do not depend much on the chemical combinations of elements, because the transitions that occur involve electrons that are not participating in chemical bonds. This simplifies the preparation of the corresponding samples.

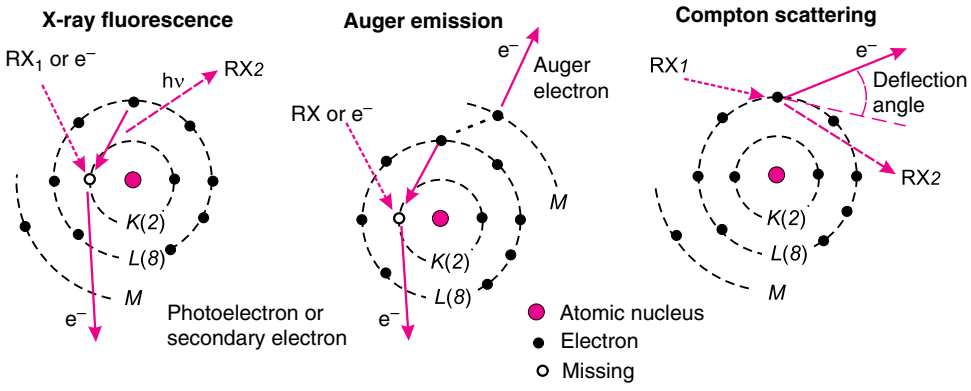


Figure 12.1 X-ray fluorescence, Auger emission, and Compton scattering. X-ray fluorescence and Auger emission: the ionizing radiation ejects an inner electron from the atom, which then creates a gap that is filled by a more outer electron (of greater energy). The excess energy is found either as an emitted photon (X-ray fluorescence), or is eliminated without radiation, by ejection of an electron from the atom (Auger emission). Compton scattering: the incident X-ray photon ejects a peripheral electron ('recoil electron'). The incident photon deviates even more when its energy is weaker.

Although semiquantitative analysis of the elements present does not cause any major difficulties, quantitative analysis is not quite so simple, because a non-negligible part of the radiation emitted far from the sample surface is re-absorbed before leaving the sample. If the influence of the matrix is properly taken into account, especially when the test sample is not homogeneous, the results can be as precise as those obtained by atomic absorption or emission.

12.2 THE X-RAY FLUORESCENCE SPECTRUM

X-ray fluorescence of an isolated atom results from a two-step process (Figure 12.2). The first step is the **photo-ionization of the atom.** The incident outer photon or particle leads to the stripping of an inner electron from the atom, such as a *K* electron if the photon has enough energy. The atom is then ionized as a consequence of the missing inner electron.

The energy of each photoelectron is equal to the difference between the incident energy and that of the energy level initially occupied by the ejected electron. The photoelectron energy spectrum is the basis of a method called ESCA (Electron Spectroscopy for Chemical Analysis), which analyses the kinetic energy of emitted photoelectrons.

The second step is the **stabilization of the ionized atom**. It corresponds to the re-emission of all, or part, of the energy acquired during excitation. Almost instantaneously

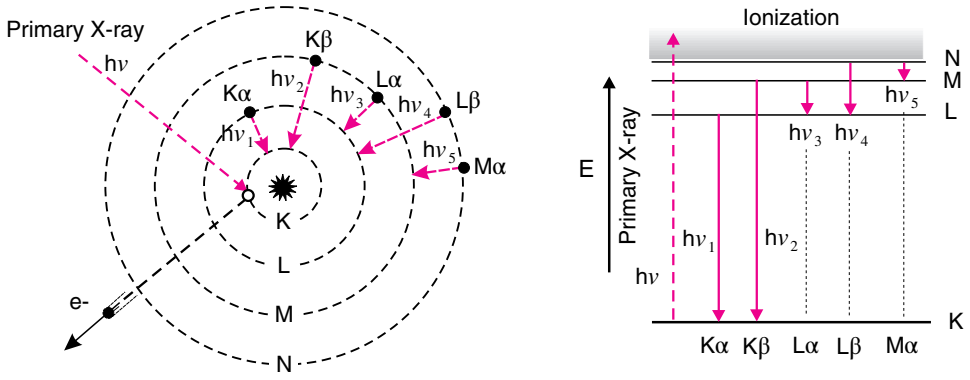


Figure 12.2 Simplified diagram of the source of some fluorescence transitions. For this classic image of an atom of medium atomic mass, the different possible electron reorganizations are displayed when an electron in the K shell is ejected from the atom. These transitions are not significantly affected by the nature of the chemical combination in which the atom is found. In this way, the $K\alpha_1$ line of sulfur changes from 0.5348 nm for S^{VI} to 0.5350 nm for S^0. This difference of around 1 eV is comparable to the natural line width for X-ray radiation.

(in 10^{-16} s), the electrons located in the different levels of this ionized atom reorganize to fill the resulting vacancy. This very quickly brings the atom back to a low-energy state. Cascade rearrangements are observed for heavy atoms, in contrast with lighter elements whose electrons are distributed over a smaller number of ground states.

This reorganization produces the emission of fluorescence photons. If E_1 is the energy of the initial electron and E_2 the energy of the electron which comes to fill the created hole, a *fluorescence photon* may appear, with a probability between 0 and 1, and is characterized by a frequency ν, such that:

$$h\nu = |E_2 - E_1| \tag{12.1}$$

The interpretation given above is simplified, since fluorescence is not the only process by which the atom can lose its excess energy. The spectrum obtained is made more complex by the presence of a continuous emission background due to the X-ray source.

Each atom, starting from $Z = 3$, leads to a set of specific radiations obeying the quantum selection rules of its electrons ($\Delta n > 0$, $\Delta l = \pm 1$). The elements of the first period (hydrogen and helium) do not have X-ray fluorescence spectra as they only possess electrons at level K. From beryllium to fluorine, a single transition of type Kα is noted. Then as the atoms become heavier, the number of possible transitions increases (75 for mercury), though the probability of some of them is very small. Fortunately, only a few intense transitions are required to characterize an element (five or six at most).

For all elements, fluorescence appears in the broad energy range from 40 eV to more than 100 keV (wavelengths from 31 to 0.012 nm).

The precise designation of the possible electron transitions depends upon the orbital energy levels of the atom under study, for which a simplified nomenclature due to Siegbahn is used. For example, for the element iron, Fe$K\beta_2$ refers to a transition corresponding to the transfer of an electron to the level *K*, coming from a neighbouring level. β indicates the M–K transfer ($\Delta n = 2$) and 2 refers to the intensity relative to the transition in the series (1 being more intense than 2). The $K\beta$ transitions are approximately six times less likely to occur (thus less intense) than the corresponding $K\alpha$ transitions (where α marks the shortest 'distance'). For the multiple *L* and *M* levels (3 for *L* and 5 for *M*), the notation according to Siegbahn is not always sufficiently explicit.

Taking account of the expression $E = hc/\lambda$ linking the energy *E* of a photon to its wavelength λ, the radiation emitted can be equally characterized by either its wavelength λ (nm or angstroms) or by its energy *E* (eV or keV). The three most commonly used expressions of conversion are:

$$\lambda_{(\text{nm})} = \frac{1240}{E_{(\text{eV})}} \quad \text{or} \quad \lambda_{(\text{nm})} = \frac{1.24}{E_{(\text{keV})}} \quad \text{or} \quad \lambda_{\text{Å}} = \frac{12.4}{E_{(\text{keV})}} \tag{12.2}$$

12.3 EXCITATION SOURCES IN X-RAY FLUORESCENCE

To induce the X-ray fluorescence of elements in a sample, a source of photons or particles of sufficient energy is required. Generally, these sources are produced by X-ray tubes of variable power (1 W to several kW) or by low-power radioisotope sources. If X-ray fluorescence is understood in the broader sense of X-ray emissions, other excitation processes employing particles (e^-, α) can be included.

12.3.1 X-Ray Generators

In an enclosed space maintained under vacuum, a beam of thermionic electrons, accelerated through a potential difference (PD) of up to 100 kV, hits a target serving as the anode (called *anticathode*), which is made of a metal with an atomic number Z between 25 and 75 (Figure 12.3). It then emits an X-ray, which is composed of a continuous fraction due to the electrons that penetrate the target to a depth on the micrometre scale and whose abrupt deceleration produces polychromatic radiation (white or braking radiation, also known as *bremsstrahlung*) on the one hand, and fluorescent beams characteristic of the metal forming the anode on the other. The wavelength end point λ_0 of this spectrum, which depends on the PD, can be calculated using Eq. (12.2).

By proper choice of the anode and the accelerating voltage of the electrons, various sources can be created, which differ in the X-ray energy they emit: rhodium anode (Kα line at $\lambda = 0.061$ nm; $E = 20.3$ keV) or tungsten anode (Kα line at $\lambda = 0.021$ nm; $E = 59$ keV) among others.

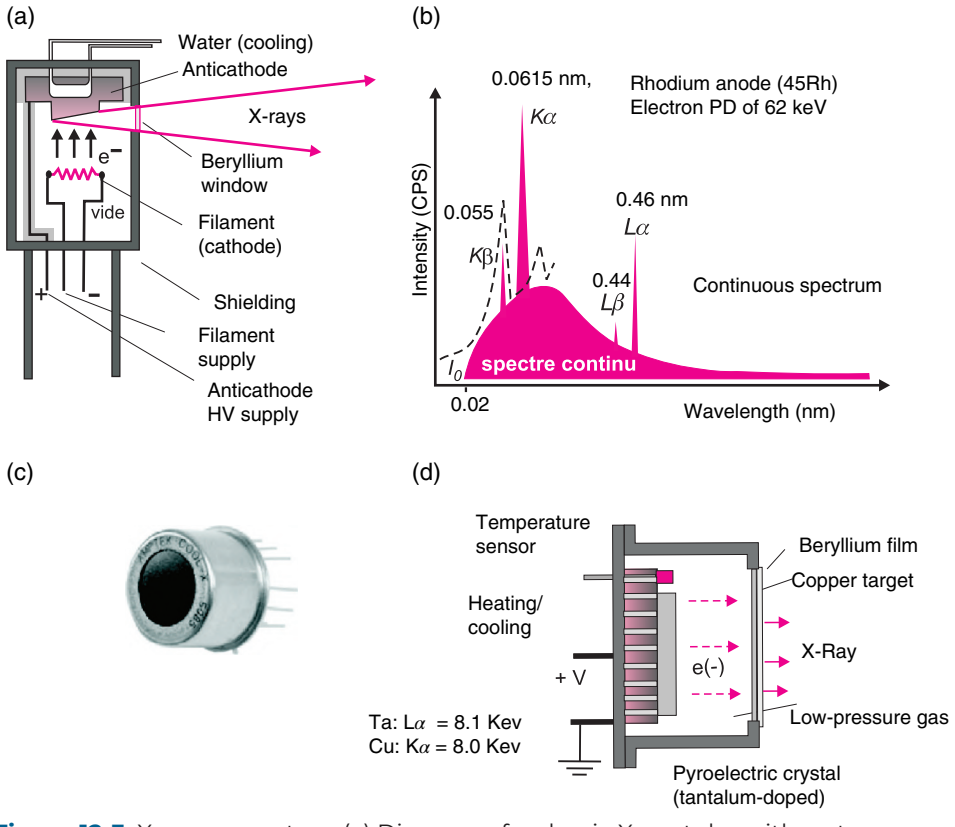


Figure 12.3 X-ray generators. (a) Diagram of a classic X-ray tube with water cooling, essential if the tube is high powered (1–4 kW). (b) Spectrum emitted by the anode of an X-ray tube. The continuous spectrum, whose extent depends on the voltage applied, is distinguished from the line spectrum of the anode. To provide monochromatic sources, filters whose absorption spectrum and thickness isolate a single line (dotted line) are used. (c) Model of a miniaturized generator. (d) Generator using a pyroelectric crystal, with a very small footprint (diam. 15 mm). The production of X-rays on the diagram corresponds to the crystal cooling time (Cu line). The heating/cooling cycle is about 3 min.

Nowadays, X-ray detectors are extremely sensitive and consequently the power of the generators used in chemical analysis tends to be reduced (Figure 12.3). They have been miniaturized to make portable instruments, known as *handheld devices*; their power is of the order of a few watts (Figure 12.12).

Electrons necessary to excite the anode may also be generated from a pyroelectric crystal enclosed in a sort of capsule, as represented in Figure 12.3. The crystal (doped with tantalum) is oriented such that its top surface depolarizes when heated and repolarizes when cooled. In this way, when the temperature increases, the upper face gets positively charged and attracts electrons from the ionized filler gas. Continuous radiation appears,

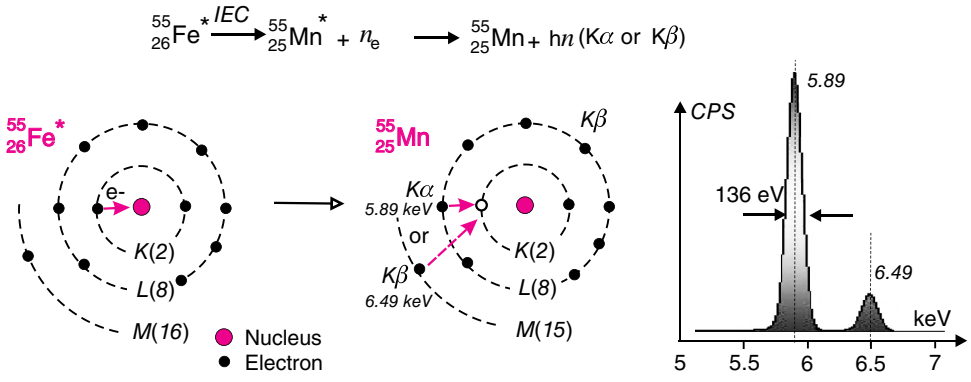


Figure 12.4 Radioactive source ^{55}Fe. Emission spectrum obtained by placing a ^{55}Fe source in the sample compartment of an energy-dispersive spectrometer. The signal corresponds to the X-ray fluorescence of ^{55}Mn, i.e. the daughter nucleus of ^{55}Fe. On this spectrum, the resolution of this spectrum, measured at full width at half-maximum (FWHM) of the main peak, is around 136 eV.

on which tantalum fluorescence lines ($L\alpha$ of Ta) are superimposed. On cooling, the upper face of the crystal becomes negative. The electrons then are accelerated in the reverse direction towards a copper film, which is used as an anticathode. The radiation now includes *K* lines (α and β) of copper, with virtually the same energy as the tantalum lines. The production of X-rays is therefore done alternately.

12.3.2 Radioisotope Sources of X-rays

Radionuclides, evolving by means of inner electron capture (IEC) or *K-capture*, constitute another category of X-ray sources. The passage of one of the two *K*-shell electrons into the nucleus generates the element that precedes it on the periodic table. The missing electron is quickly replaced by another electron coming from an outer shell accompanied by emission of an X-ray fluorescence photon from the daughter nucleus Y (Figure 12.4). This transformation is summarized as:

$$^{A}_{Z}X \xrightarrow{EC} {}^{A}_{Z-1}Y^{*} \xrightarrow{h\nu} {}^{A}_{Z-1}Y$$

There are several known radionuclides of this type, which have sufficiently long periods that they can be used as energy sources (Table 12.1). The activity of these isotopic sources is generally a few millicuries (mCi), generating a flux of 10^6 to 10^8 photons per steradian per second. They are very small in size, but they require special precautions for transport or replacement since they emit continuously, unlike more classic generators.

A β^- radionuclide may also be associated with a second element to be used as a target, which functions as the anode of a conventional X-ray tube. For example, the ^{147}Pm/Al pairing (τ = 2.6 years) emits continuous exploitable X-ray radiation between 12 and 45 keV.

Table 12.1 Several radioisotope X-ray sources.

Transition	Half-life (years)	X-ray	λ (nm)	E (keV)
$^{55}Fe \rightarrow {}^{55}Mn$	2.7	MnKα	0.21	5.9
$^{57}Co \rightarrow {}^{57}Fe$	0.7	FeKα	0.19	6.4
$^{109}Cd \rightarrow {}^{109}Ag$	1.3	AgKα	0.056	22.0

12.3.3 Other Sources of Excitation

α Emitters

To generate X-rays, α-emitting radionuclides such as americium, ^{244}Am (τ = 430 years), or curium, ^{244}Cm, can also be used. During the α collisions, inner electrons are ejected, in turn exciting the X-ray fluorescence of the target. The intensity is on the order of several dozen mCi. This source also produces γ radiation of 60 keV.

Fast electrons

X-ray fluorescence emission can be induced by electrons, and therefore it is not surprising that chemical analysis can be performed using scanning electron microscopes (SEM).

When the object is 'illuminated' by highly energetic electrons, which are necessary to generate SEM images, various interactions are produced within a small pear-shaped volume surrounding the impact zone of the beam on the material (Figure 12.5). An X-ray emission characteristic of the atoms appears, coming from the subsurface

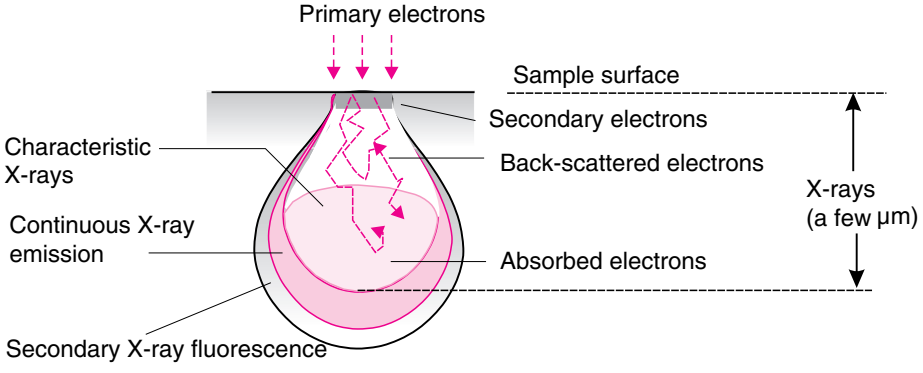


Figure 12.5 Pear-shaped interaction of the electron beam with the material. Different phenomena are produced within the material. A complex emission results, in which the origins of the different radiations collected can be determined. If the energy is high, then the X-rays are formed at a greater depth within the material and, consequently, it will be more difficult to detect them.

zone being subjected to the electron bombardment. In this case, the sample plays the role of the anode in an X-ray tube.

This is why many electron microscopes are equipped with accessories that can analyse this X-ray fluorescence. The energy of the electron beam is adjusted to between 20 and 30 keV, as a compromise to allow the appearance of the characteristic *K* or *L* lines of the elements. This quasi-instantaneous analysis, called *X-ray microanalysis*, relates to a sample volume of approximately 1 μm^3.

12.4 DETECTION OF X-RAYS

In X-ray fluorescence, detectors convert photons into electric pulses. The intensity of the beam is digitally recorded as the number of counts per unit of time (*cps: counts per second*). Depending on the instrument, which may be wavelength dispersive (WD-XRF) or energy dispersive (ED-XRF), the appropriate detector is selected as a function of its characteristics (efficiency, saturation, noise, count rate, etc.). Among the three types of detectors in use, each corresponding to a different technology, two are used for WD-XRF instruments, while the third is reserved for ED-XRF instruments. For the latter, it is essential, due to their design, for the signal intensity to be proportional to the energy of the incident X-ray photon (Figure 12.6). The two most common types are:

- ***Proportional gas counter.*** This is a Geiger–Müller counter. Two types exist: one is a gas-flow type (optimized for energies under 8 keV) and the other is a sealed-gas type, whose sensitivity is better for the 5–15 keV range.

  Each X-ray photon entering the detector causes the ionization of atoms by an inert gas (Xe, Kr, Ar, Ne or He). Thus, for argon, it is considered that an electron/ion pair is created for an incident energy of 26 eV. This gas is often associated with a second gas called the *extinguisher gas*, whose purpose is to quickly bring the ionized gas back to its neutral state. A pulse of current corresponds to each photon, recognized as a *count*.

- ***The scintillation counter*** gives better performance for shorter wavelengths (0.04–0.15 nm; 8–30 keV). It is generally composed of a sodium iodide crystal activated by thallium (NaI/Tl) associated with a photomultiplier tube. This crystal plays the role of a solid scintillator, by producing a light flash with each impact of an X-ray. A portion of the photons formed is collected by the photocathode of the photomultiplier, which then converts them into a more or less intense electronic pulse, also counted as a *count*.

- ***Semiconductive transducer.*** Instead of ionizing a gas, the X-rays strike a semiconductive crystal in which electron/hole pairs are created in proportion to the energy of each incident X-ray photon. For that purpose, the semiconductor, most often a lithium-doped silicon Si(Li) diode, has an active zone whose conductivity

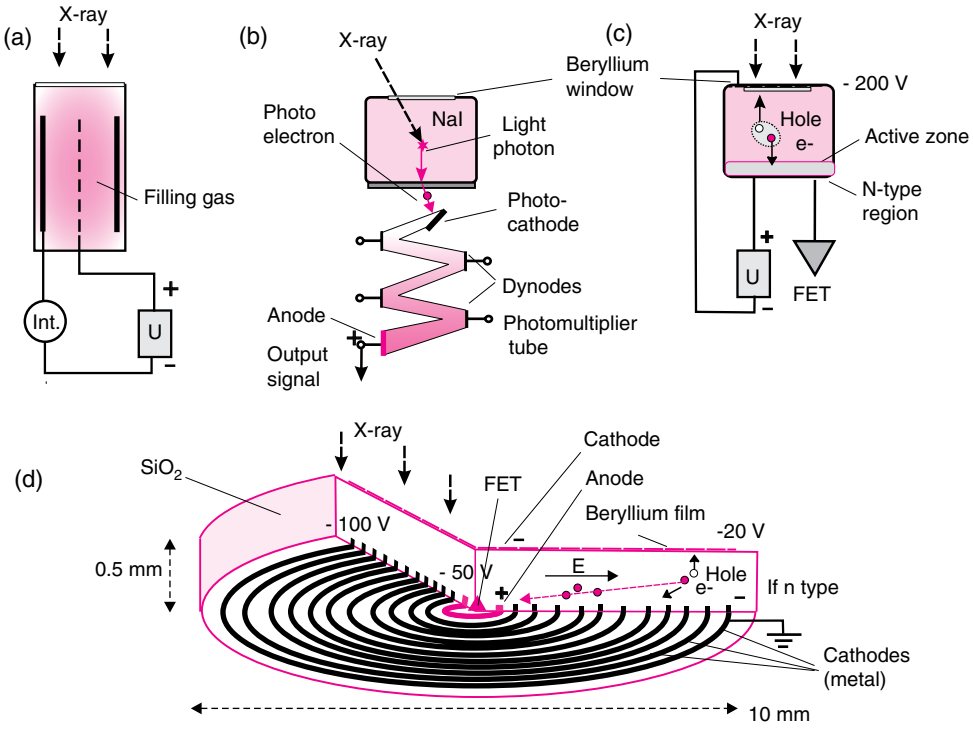


Figure 12.6 Various detectors used in X-ray fluorescence spectrometry. (a) Proportional counter. (b) Scintillation detector (crystal and photomultiplier tube). (c) Semiconductive transducer with a significant active zone. (d) SDD-type pulse counter, whose series of annular electrodes creates a transverse field that makes the electrons converge on the central anode.

increases with the appearance of one electron for approximately 3.6 eV of incident energy. The pulse intensity at the detector outlet (the *count*) is therefore proportional to the energy of the incoming X-ray photon. The background noise is reduced if the sensor is maintained at low temperature by thermo-electric effect (Peltier effect). The incoming surface is protected by a beryllium film with thickness of a few μm (transparent for $Z > 11$). This type of detector, used for ED-XRF analyses, has a good resolution for photons whose energy falls between 0.2 and 40 keV. Several more efficient variants have appeared over the years, including the Silicon Drift Detector (SDD), which is favoured for compact and light field instruments (Figures 12.6 and 12.12). In this variant, a series of annular electrodes creates a transverse radial field, whose direction is such that it entrains released electrons in the silicon semiconductor towards a small central anode. The collected charge is then converted into voltage. By means of various corrections of the signal, this is a quick, sensitive detector with a very good signal/noise ratio.

To classify the photons with better precision, their number is limited to approximately 10,000/s by reducing, if necessary, the source intensity. The instrument's data system should accurately identify the energy of the photons, which arrive as a pack. The resolution of these instruments is given by the full width at half-maximum of the Kα line of manganese, emitted by a ^{55}Fe radioactive source (Figure 12.4). Energy-dispersive instruments have a resolution around 100 eV, which is much greater than the natural width of the spectral lines.

12.5 DIFFERENT TYPES OF INSTRUMENTS

X-ray fluorescence spectrometers are classified into two categories depending on how the spectrum is obtained. The first type uses diffraction phenomena of X-rays by crystals (wavelength dispersive, or WD-XRF), while the second type uses the sensitivity of some detectors to the energy of the photons emitted by the sample (energy dispersive, or ED-XRF) (Figure 12.7). The first category includes a particular group of instruments fitted with filters (single-channel type) and dedicated to specific measurements.

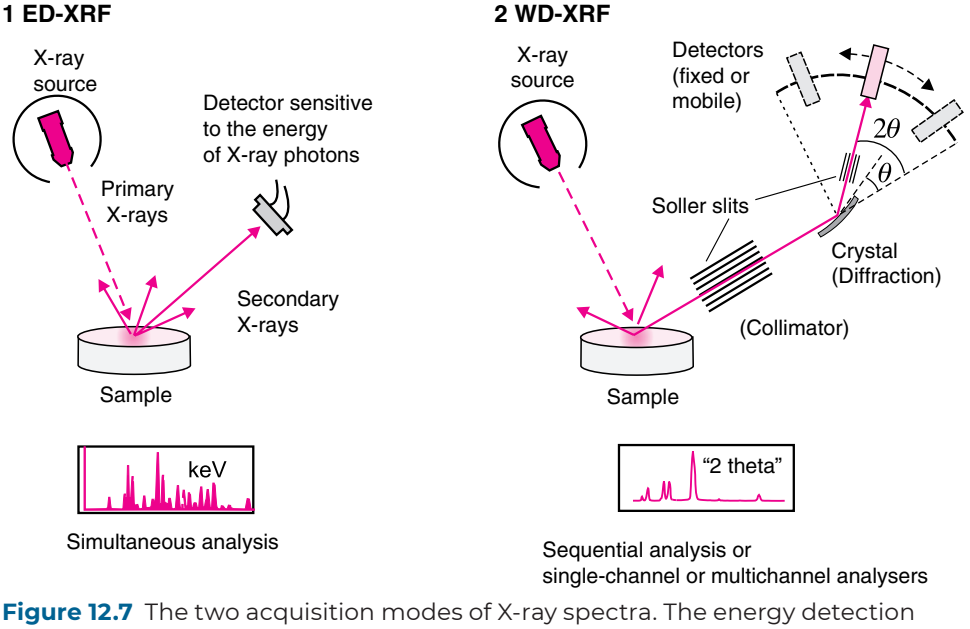


Figure 12.7 The two acquisition modes of X-ray spectra. The energy detection method (ED-XRF) includes a detector which can distinguish each X-ray photon according to its energy. The sequential method yields a wavelength spectrum (WD-XRF) from a goniometric set-up with crystal and detector, whether mobile or fixed (e.g. multichannel devices).

12.5.1 Energy-Dispersive X-Ray Fluorescence Instruments (ED-XRF)

Small energy-dispersive instruments are reserved for both qualitative analysis and routine measurements (Figures 12.8 and 12.12). The spectrum is obtained by use of a detector installed quite close to the sample, which distinguishes the energy of each of the fluorescence photons captured. These instruments are equipped with either a low-power X-ray tube (around 10 W) or a radioactive source for field instruments.

After having defined an acquisition width between 10 to 20 eV, the multichannel analyser (e.g. 2,000 channels) will count the pulses caused by photons received during the several minutes of the measurement. Each photon will be classified in an energy channel corresponding to an interval of a few electron volts (eV). The resulting spectrum is a histogram (Figure 12.4). These devices can be used for simultaneous analyses over the whole spectral range. The results will be better if the acquisition of signals is done during a longer period of time t (sensitivity increases with $\sqrt{t}$). Radiation energy calibration is often performed using an external reference such as the Kα transition of cobalt.

12.5.2 Wavelength-Dispersive X-Ray Fluorescence (WD-XRF) Spectrometers

In this second class of spectrometers, known for their excellent spectral resolution, the X-ray fluorescence radiation emitted by the sample passes through a collimator made of long metal wafers (Soller slits), before hitting a crystal which is cut in such a way

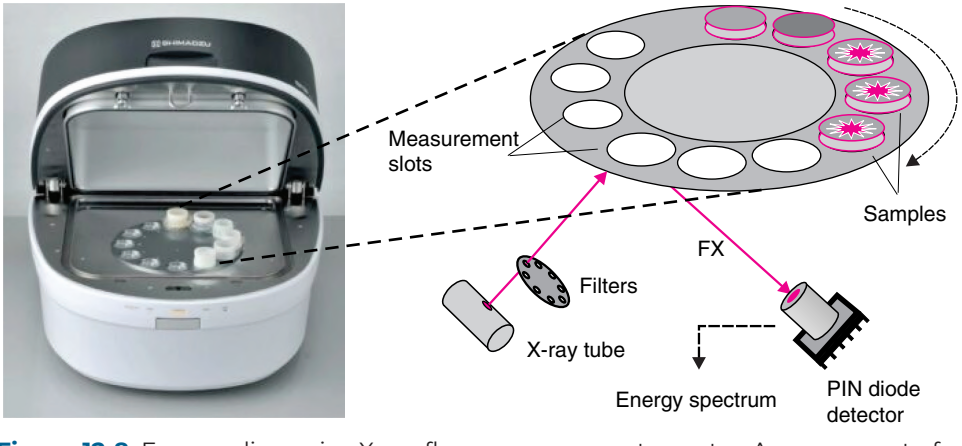


Figure 12.8 Energy-dispersive X-ray fluorescence spectrometer. Arrangement of the different sections in the EDX-7000. (Source: Courtesy of SPECTRO Analytical Instruments GmbH.)

Crystal	Plane (Miller index)	d(Å)
Topaz	303	1.356
Lithium fluoride	200	2.01
Silicon	111	3.14
Graphite	001	6.69
Oxalic acid	001	5.85
Mica	002	9.96
Lead stearate		51

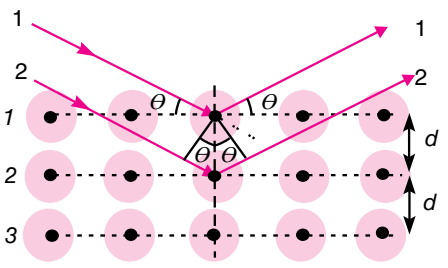


Figure 12.9 Reflecting crystals used in goniometers of wavelength-dispersive spectrometers and the Bragg condition. For the two rays 1 and 2 to be in phase, the difference in their optical path must be a multiple of λ. When this condition is met for planes 1 and 2, it will also be met for all of the other planes, and the overall effect will therefore be reinforced. This is the principle of constructive interference. Each crystal may be suitable for a range of wavelengths. The broader this range is, the greater the interplanar spacing must be for the crystal chosen, but the smaller its angular dispersion will be.

that its constitutive atoms form planes parallel to the surface (Figures 12.7 and 12.10). These planes, separated by a distance d, behave like a stack of parallel mirrors when the angle of incidence θ is the same as the angle of observation of the reflected light (Figure 12.9). The combination of these mirrors has the same effect as a grating: the only radiation observed will be that for which the wavelength satisfies the Bragg condition (n being a whole number called the order of diffraction).

$$n\lambda = 2d\sin\theta \tag{12.3}$$

For construction reasons, θ can generally vary from 5° to about 80°. A greater value of d leads to longer wavelengths; however, the resolution of the instrument is related to the power of dispersion $\mathrm{d}\theta/\mathrm{d}\lambda$, which is calculated by differentiating Eq. (12.3) and which is inversely proportional to d:

$$\frac{\mathrm{d}\theta}{\mathrm{d}\lambda} = \frac{n}{2d\cos\theta} \tag{12.4}$$

These instruments can be classified as:

- ***Sequential analysers***. This includes a goniometric assembly that lets the detector and the crystal undergo synchronized rotations θ and 2θ, to a precision of 1/1,000th of a degree of an angle (Figure 12.10). These instruments are reserved for nonroutine elemental measurements.
- ***Fixed channel apparatus***. The Bragg condition shows that by choosing a crystal (therefore d) and by setting the angle of detection (and so the value of θ), the radiations of wavelength λ that will satisfy the Bragg condition can be

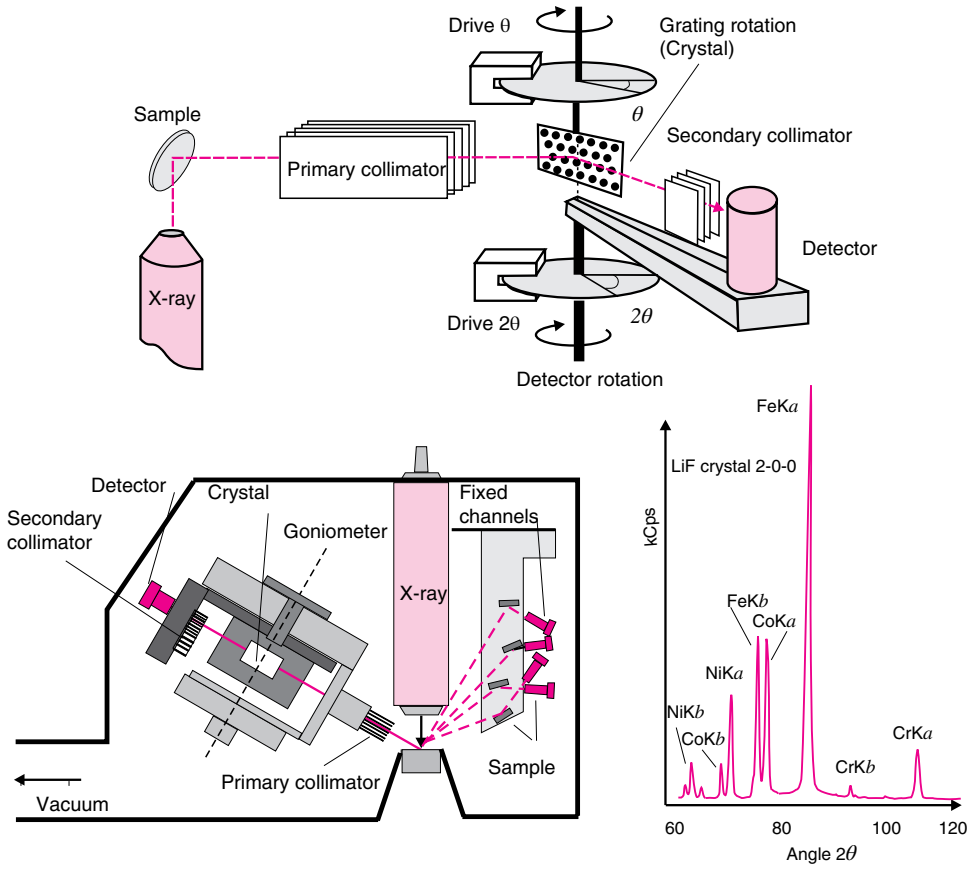


Figure 12.10 Crystal-based sequential spectrometers. Above, a diagram inspired from a Siemens SR300 instrument. The primary collimator channels the X-ray beam emitted by a high-power source. The secondary collimator serves to eliminate all of the diffracted light not parallel with the direction 2θ in which the detector is found. These components are composed of parallel metal sheets. Below is a diagram inspired by the ARL 9800 model, combining a goniometric assembly associated with fixed channels (crystal/detector pairs) positioned for predefined elements; right, a typical spectrum recorded as a function of the angle 2θ obtained with this type of instrument.

isolated. Starting from this principle, several crystal-detector arrangements (up to 28 usually) can be positioned around the sample, with each pair (d, θ) defining a unique wavelength that is highly specific to a predefined element. These instruments are able to undertake simultaneous measurements of several elements.

Low-energy transitions for elements up to phosphorus ($Z = 15$) require operating conditions in a vacuum. The resolution (in eV) can be as low as a few electron-volts. There are models of this type which are adapted for scanning electron microscopes.

12.5.3 Filter Instruments (Fixed Single Channel)

These are robust instruments often installed on site for the continuous monitoring of industrial production. To measure an element on a manufacturing line from a single, characteristic transition, the method consists in linking the concentration of interest to the difference between two counts. The first is obtained by installing a *transmission filter* between the sample and the detector to isolate the characteristic radiation of the element in question, and the second by fitting an *absorption filter* to stop this same radiation. This will enable us, for example, to quantify copper from its $K\alpha$ line by using two filters, one made of nickel and the other of cobalt. The fluorescence originating from the filters themselves is a limiting factor in this method, which is reserved for routine measurements.

12.6 SAMPLE PREPARATION

In qualitative analysis, it is recommended for the sample surface to be as flat as possible and for its composition to be as homogeneous as possible. As these parameters can only rarely be controlled for test samples, a prior treatment is often necessary. Semiquantitative or quantitative analysis also requires taking matrix effects into account, as these constitute sources of error.

> Absorption by the matrix can provoke either an underestimation of the result by optical quenching, or an overestimation of the result when some of the X-ray fluorescence causes a secondary excitation of other elements present in the sample. For example, the presence of iron in aluminium provokes an increase in the latter's fluorescence because the fluorescence of iron also excites that of aluminium.

Preparation methods thus depend on the physical state and on the nature of the material.

- *Liquid samples* do not require special preparation prior to the analysis. A small volume of sample is placed in a type of vial whose bottom is replaced by a film of polypropylene or mylar (polyester) that absorbs X-rays very little. The depth of absorption can be as much as 1 cm for hydrocarbons.
- *Solid samples*, especially if the matrix is not well known, need transformation prior to the measurement. In fact, the measured fluorescence of these bulk materials only concerns a thickness of a few micrometres beneath the surface. This thickness depends on the composition of the solid and on the angle of

incidence of the primary X-rays: this can range from a few angstroms (if the incident angle is acute) up to half a millimetre. Any surface heterogeneity can cause significant variations in the result. This is the main reason for surface treatment prior to analysis.

The two techniques used for sample preparation of a solid are fusion (more complex, but it also gives better results for heterogeneous solids) and pellet formation. Fusion involves mixing a little of the sample, reduced to a powder, with lithium tetraborate ($Li_2B_4O_7$) and various additives. The glass obtained by fusion in an induction crucible furnace, called a bead, constitutes a matrix of light elements and is therefore transparent to X-rays. Pellet formation, using a hydraulic press, is an alternative to fusion. To ensure the cohesion of the pellet, a binder is added. The most common binders are wax (an organic polymer formed from light elements) and micro-crystal cellulose.

12.7 X-RAY ABSORPTION – X-RAY DENSITOMETRY

The penetration of X-rays into the material has an indirect effect on analysis by X-ray fluorescence. Therefore, it is useful to review the law governing this absorption.

The intensity I of the radiation that emerges after having passed through a material with a thickness x (cm) and a *linear attenuation coefficient* called μ (cm^{-1}) for the wavelength considered can be calculated with respect to its initial value I_0 and for an angle of penetration of 90° as follows:

$$I = I_0 \cdot \exp\left[-\mu x\right] \tag{12.5}$$

This equation, which corresponds to integration of the expression $dI = -\mu I dx$, is comparable to that discussed in colorimetry (Beer–Lambert law, see Chapter 9, Section 9.9). The linear attenuation coefficient μ can be calculated for any material if we know ρ, the density of the medium (g/cm^3), and its composition. Individual attenuation coefficients μ_{Mi} (cm^2/g) are available from tables for each element. Next, the weighted mass attenuation coefficient μ_M must be calculated for the material based on its elemental composition.

$$\mu_M = \sum_i x_i \mu_{Mi} \tag{12.6}$$

where x_i refers to a mass fraction of each element in the material. Finally, the density of the material is used, giving $\mu = \mu_M \rho$.

Figure 12.11 shows that a film of beryllium, such as those serving as windows for X-ray detectors, is opaque to the radiation emitted by the lightest elements ($E < 1$ keV for $Z < 10$). The linear attenuation coefficient of a material decreases as the wavelength

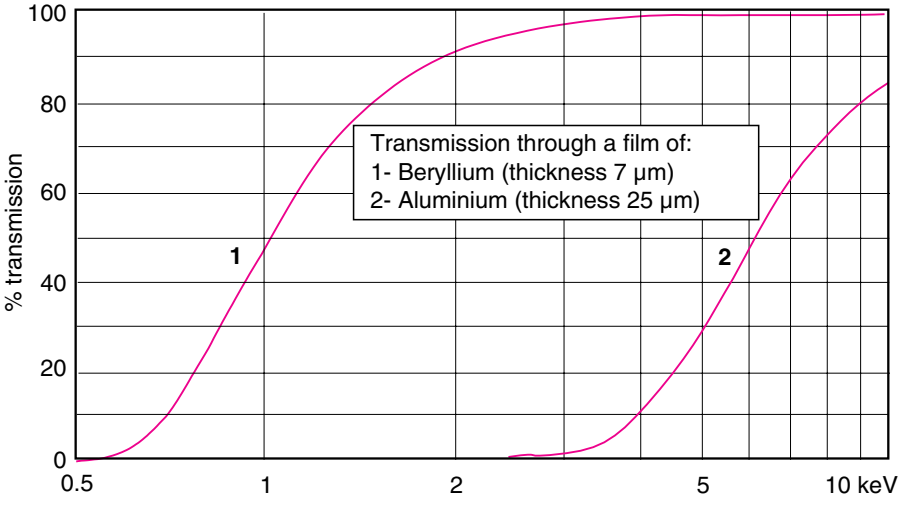


Figure 12.11 X-ray densitometry –percentage transmission for two films made of light elements. A 7-µm film of beryllium is often used to make windows for energy detectors. For a radiation of 1 keV (e.g. Na $K\alpha$), it is noteworthy that the attenuation achieved by this film is still 50%.

decreases. For lighter elements, measurements are made under an atmosphere of helium, while for solids a vacuum is required.

Equation (12.5) estimates that household aluminium foil with a thickness of 12 µm (ρ = 2.7 g/cm^3) absorbs 57% of the intensity of the Ti Kα transition (μ_M = 264 cm^2/g) but only 1% for the Ag Kα transition (μ_M = 2.54 cm^2/g).

12.8 QUANTITATIVE ANALYSIS BY X-RAY FLUORESCENCE

There is a complex relationship between the mass concentration of the element to be analysed and the measured intensity of one of its characteristic spectral lines. For trace analysis, several mathematical models have been developed to correlate fluorescence with the atomic concentration. A series of corrections must be introduced to account for interelement interactions, preferential excitation, self-absorption, and the fluorescence yield (the heavier atoms relax by internal conversion without photon emission). All of these factors require the reference samples to have practically the same structure and atomic composition as the sample under investigation, for all of the elements present. It is mostly because of these reasons that good quantitative analysis by X-ray fluorescence is difficult to obtain. When we work with a solid sample, a perfectly clean,

preferably polished surface is important, since the analysis concerns the composition immediately close to the surface.

This method is well adapted in qualitative analysis for the automatic identification of the spectral lines, often accompanied by very sophisticated visual displays against a multicoloured periodic classification background. In semiquantitative analysis, software can give an approximate composition of the sample without the need for reference standards. However, in quantitative analysis, problems are more serious, with calibration requiring matrices at all comparable points. Many corrections (called ZAF) are required: atomic number (Z), nature of the isotope (A), and fluorescence (F).

12.9 APPLICATIONS OF X-RAY FLUORESCENCE

The X-ray fluorescence method is nondestructive and requires little or no sample preparation beforehand. It can be used to analyse all chemical elements; its only restriction is that elements lighter than boron ($Z < 5$) or heavier than uranium ($Z > 92$) are nondetectable. Moreover, it has a large dynamic range. For example, it is possible to measure two elements present in the same sample in which the concentrations are 50% for one and a few ppm for the other. Progress in the detection of high-energy photons has improved the precision for analyses of heavy elements, which can be obtained from their K spectral lines, but not from their L lines, which are situated in the region of the spectrum where background noise levels are high.

Initially, X-ray fluorescence was used in industries treating metals or alloys and more generally in heavy industry (iron and steel industry, cements, ceramics, glassware).

TXRF or total reflection fluorescence

In this method, which is useful to analyse the surfaces of samples (films deposited on a substrate), the incidence angle of the monochromatic X-ray beam is practically null: 0.1 degrees for example. The beam is not absorbed very much, and the Si/Li detector is placed at 1 or 2 mm above the impact zone.

Thanks to the convenience of current apparatus (Figure 12.12), the scope for this technique is quite broad, both in research and development and for quality control in production. It extends from routine qualitative analyses to measurements that can attain the high level of precision of classic, wet-chemistry methods. Semiquantitative analyses are possible, without reference standards, using high-performance computer software.

X-ray microanalysis, for which samples must ideally be electrical conductors, allows element mapping in heterogeneous samples, observed with a scanning electron microscope.

The list would be very long if all of the analytical applications of X-ray fluorescence were recorded: photography, paper industry, semiconductors (silicon impurities), and petrochemicals (S, P, Cl) among others. There are also many applications in medicine and biology (Ca, K, P, Cl and S in blood serum, haemoglobin content by Fe assay),

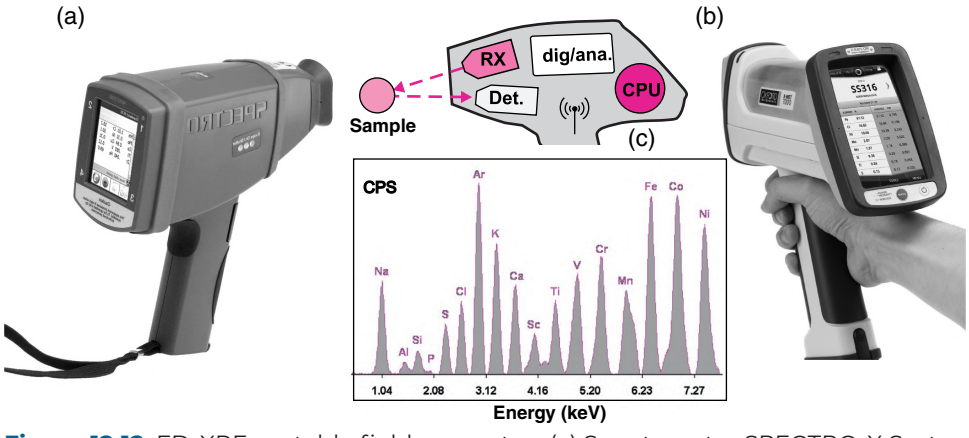


Figure 12.12 ED-XRF portable field apparatus. (a) Spectometer SPECTRO-X Sort from Ametek. (b) Tracer Spectrometer from Oxford Instruments. (c) Conception and example of results that can be obtained with this type of spectrometer.

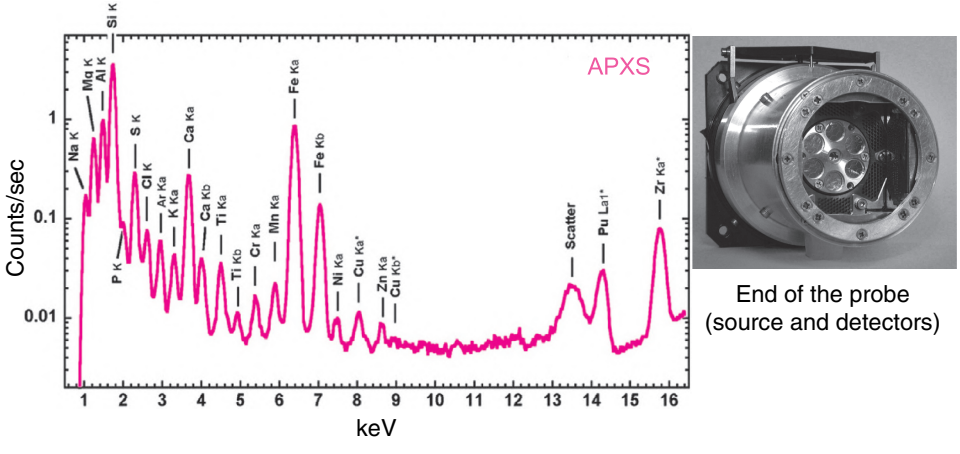


Figure 12.13 Spectrum of a surface sample from Mars, obtained by the Mars Rover Spirit in 2004. (Source Documents courtesy of NASA.)

geology, toxicology, the environment (dust, combustion fumes, soil pollution), waste and emissions management (heavy elements such as As, Cr, Cd or Pb), and analysis of ultra-light elements (nitrogen).

Finally, a current application is the use of X-ray fluorescence to measure lead concentrations in the home (paints and wall coverings). To conclude this list with a less common application, the robotized vehicles sent to Mars, the Sojourner (1996), the Spirit (2003), the Opportunity (2003) and the Curiosity (2012), were all equipped with an alpha-particle X-ray spectrometer (APXS) with a ^{244}Cm radioactive source, to analyse the composition of Mars' soil using X-ray fluorescence (Figure 12.13).

KEY POINTS OF THE CHAPTER

1. When a material is irradiated with high-energy X-rays or particles (electrons, ions, etc.), the constituent atoms of the material pass into the ion state and may re-emit all or part of the energy received as: Rayleigh or Compton scattering, Auger emission, or short-wavelength fluorescence (X-rays).
2. **X-ray fluorescence.** Through the impact of a photon or another particle on an atom, it loses an inner electron, close to the nucleus, which transforms it into an ion that stabilizes through the near-instantaneous reorganization of electrons at different levels of this atom. These energy jumps between atom levels are accompanied by the emission of short-wavelength photons designated by a special nomenclature and which follow the equation:

$$h\nu = \left|E_2 - E_1\right|$$

3. In this region, radiation is indicated either by wavelength or by energy. With the usual units, this leads to the numerical equation for conversion:

$$\lambda_{(nm)} = \frac{1240}{E_{(ev)}} \quad \text{or} \quad \lambda_{(nm)} = \frac{1.24}{E_{(eV)}}$$

   Therefore, an element can be characterized if the fluorescence photon energy it emits is known.
4. **X-ray sources**. Depending on instruments and their uses, there are several types of sources.
   - *X-ray tube*. An electron beam extracted from a heated (e.g. with PD of 100 kV) filament and then accelerated strikes a metal target called the anticathode (or anode). The latter emits radiation, composed of a continuous background level, which results from a slowing down of electrons penetrating into the target (*bremsstrahlung* or braking radiation), on which fluorescent lines characteristic of its composition are superimposed. Filters are necessary if monochromatic sources are to be acquired.
   - *Radioisotope sources*. The decay of certain isotopes (^{55}Fe or ^{57}Co), followed by the emission of X-rays, is used by the internal electron capture (IEC) mechanism as follows:

$$^{A}_{Z}X \xrightarrow{IEC} {}^{A}_{Z-1}Y^{*} \xrightarrow{h\nu} {}^{A}_{Z-1}Y$$

   - *Radioactive sources*. Sealed sources contain an alpha emitter, which causes the X-ray fluorescence of an internal target. These are permanent sources whose intensity progressively decreases; they require certain safety constraints.

5. **X-ray detectors** are used to measure the flow (for WD-XRF instruments) or the energy (for ED-XRF) of emitted photons.
   - *Proportional counter*. Used when resolution must be high (WD-XRF spectrometers). Optimized Geiger–Müller type; pulse intensity is proportional to the energy of the incident photon, which makes it possible to eliminate parasite pulses (background noise and harmonics).
   - *Scintillation detector*. Composed of a photomultiplier tube coupled with a NaI crystal within which X-ray photons generate photons, which after conversion on a photocathode, produce an output signal.
   - *Semiconductive transducer*. Developed more recently, these detectors are composed of an Si(Li) semiconductor that gives a response proportional to the energy of the incident X-ray photon. An electron is produced for an energy jump of about 3.6 eV. This enables the detection of elements located after number 14 (Si).
6. **Two instrument categories.**
   - *Energy-dispersive instruments* (ED-XRF). A large choice of handheld, portable, mobile, and fixed models, generally with a small footprint, can be adapted to routine measurements. In general, single-channel devices include both a transmission filter and an absorption filter.
   - *Wavelength-dispersive instruments* (WD-XRF). Designed on the principle of optical spectrometers whose grating here is a crystal carved so that its planar faces behave like a series of mirrors separated by distance *d*. Only radiation of the wavelength defined by the Bragg equation below will be observed:

$$n\lambda = 2d\sin\theta$$

   Goniometric set-up with synchronized crystal/detector rotation of $\theta/2\theta$ whose resolution is related to the dispersive power $\frac{d\theta}{d\lambda} = \frac{n}{2d\cos\theta}$.

   Some models include several fixed detectors positioned to measure pre defined elements.
7. **Linear absorption coefficient.** For any medium, we can calculate the linear absorption coefficient μ (in cm) from which the intensity transmitted after crossing thickness *x* of the material can be found:

$$\mathbf{I = I_0 \cdot \exp\left[-\mu x\right]}$$

   μ can be calculated from the weighting of the different mass coefficients μ_M present (using a table) and from the density ρ of the material ($\mu = \mu m.\rho$) with

$$\mu_M = \sum_i x_i \mu_{Mi}$$

8. **Applications**. Samples must undergo a preparation depending on their physical state; however, the chemical combination in which the elements are present does not change results.

PROBLEMS

1. **a.** Why does the table of X-ray fluorescence emission lines only start at lithium?
   b. To measure elements whose emitted radiation energy is less than 3 keV, it is necessary to purge the air from the instrument and to replace it with helium. Why?

2. An X-ray tube with a tungsten anticathode serves as the source in an X-ray fluorescence spectrometer containing a goniometer equipped with a crystal of ethylenediamine tartrate. The plane of reflection corresponds to an interplanar distance d = 4.404 Å.
   a. Calculate the angle of deviation measured with respect to the direction of the incident radiation in order to collect the Lβ fluorescence line of bromine (λ = 8.126 Å) emitted from a sample of sodium bromide (it may be assumed that the observations recorded are in first-order reflection).
   b. Knowing that the wavelength of the Kα line of the tungsten anticathode occurs at 0.0209 nm, calculate the minimum accelerating voltage of the electrons from the X-ray tube, to make this line appear.

3. Is it possible to protect yourself from X-rays by wrapping yourself in aluminium foil with a thickness of 12 μm?
   The transmission percentage will be calculated (density of Al = 2.66 g/cm^3):
   - For the *K*α line of titanium (4.51 keV, μ_MAl = 264 cm^2/g),
   - For the *K*α line of silver (22 keV, μ_MAl = 2.54 cm^2/g).

4. Sulfur presents two *K*-type lines: Kα_1 = 0.537216 nm and Kα_2 = 0.537496 nm. **a.** What is the energy difference, expressed in eV, between Kα_1 and Kα_2?
   b. If it is known that the full width at half-maximum of these lines is 5 eV, what conclusion can be drawn?
   c. If the position of the Kα_2 line increases by 2 × 10^{-4} nm when passing from S^{6+} to S°, show that this has no impact on the location of this line expressed in terms of energy.

5. To determine the % mass of manganese (Mn) present in a rock, the element barium (Ba) is used as an internal reference. Two solid calibration solutions in the form of borax beads yield the following results:

Solution no.	% Mn by mass	Mn/Ba ratio
1	0.250	0.811
2	0.350	0.963
unknown sol.	**?**	**0.886**

Determine the % mass for Mn in the unknown solution.

6. A wavelength-dispersive X-ray fluorescence instrument is used to determine the mass percentage of carbon between 2% and 4% in cast iron. While the attenuation of the $K\alpha$ line of carbon is significant in iron, a linear calibration can be conducted if the matrix (cast iron) has nearly the same composition as the standards. Seven standards are used for this calibration. The carbon composition of these standards is carefully determined by a conventional reference method. The intensity in counts per second (cps) of the $K\alpha$ line of carbon and the concentrations of the standards are provided in the following table:

% m/m	2.32	2.93	3.45	3.89	2.87	3.80	3.46
Cps	158	209	243	274	204	262	237

a. Determine the relationship between the line intensity in cps and the composition of the standards in mass % by linear regression.

b. The carbon line intensity, for a cast iron of unknown composition, is determined to be 233 cps. Calculate its mass percentage of carbon.

7. *Measurement of aluminium using X-ray fluorescence spectrophotometry.*
Powdered ink is an important component in printers, fax machines, and photocopiers. The composition of and absence of impurities in this ink are very important for print quality and equipment life. That is why the aluminium present is measured by energy-dispersive X-ray fluorescence.
The instrument used is equipped with a rhodium tube and a high-resolution detector. The sample is compacted and introduced as a pellet in the sample compartment. The conditions of analysis are provided in the following table:

Element	kV	μA	Filter	Medium	Duration (s)
Al	6	900	None	Helium	60

Under the conditions of analysis, the sample compartment is purged to replace the air with helium. Knowing that under the conditions of the experiment, the density of air is equal to 0.54 g/l, that its composition in % volume is 80% molecular nitrogen and 20% molecular oxygen, and that the mass attenuation coefficients of nitrogen and oxygen are respectively equal to 17.7 and 27.2 cm^2/g at 6 keV, calculate:

a. the mass percentages of nitrogen and oxygen in air;

b. the weighted mass attenuation coefficient of air;

c. the transmitted energy percentage per 10 cm of air;

d. the transmitted energy percentage per 10 cm of helium at 300 K under 1 atmosphere (M = 4 g/mol, μ_m = 0.395 cm^2/g). Draw a conclusion.
The calibration is conducted by introducing aluminium into a matrix close to that of the sample, but totally free of aluminium at first. The following results are obtained:

Al (% m/m)	0.10	0.30	0.46	0.60	0.90
cps	24	37	48	56	75

e. Give the relationship between the intensity of the emitted line in cps and the % Al in the standards.

f. Knowing that the ink studied under the same conditions gives an emitted line intensity equal to 38 cps, calculate the composition of the studied sample in terms of % mass of Al.

8. Calculate the linear coefficient of a lead glass that has a density of 5.1 g/cm^3, obtained from a blend of 65% silica (SiO_2) and 35% lead oxide (PbO). The following is obtained:

Element	μ_M (cm^2/g) for line α-RhKL	Molar mass (g/mol)
Si	4.26	28.1
O	0.8	16
Pb	84.1	207.2

9. To conduct a lead check, a portable energy-dispersive X-ray fluorescence instrument (ED-XRF) must be used, enabling us to choose between two possible sources of excitation: Cadmium-109, 460-day period (emits X-rays of 22-26 keV and γ photons of 88 keV) or Cobalt-57, 272-day period (emits γ photons of 122 and 136 keV). The characteristic emissions of lead are 88 keV (*K*α transition); 15.9, 15.2 and 13 keV (*L*α transitions).

a. The intensity of these two sources is nearly 12 mCi. Convert this value into Becquerels.

b. The manufacturer recommends the Co-57 source for lead. Why?

c. What fraction of the 13 keV line is absorbed if a 1-mm thick layer of old wallpaper is placed between the lead and the detector? (The linear absorption coefficient is μ = 2 cm^{-1}).

d. The relatively short period of these sources decreases the sensitivity of this type of instrument over time. For the Co-57 source, calculate the correction factor that must be applied for a reference sample measured every three months to give the same intensity, knowing the measurement time is the same.

SOLUTIONS

1. a. There must at least be one electron in state $n = 2$ (L 'level').

b. Because the components of air significantly absorb the low-energy radiation of X-ray fluorescence. Helium is practically transparent.

2. a. $2d\sin\theta = k\lambda$ (here $k = 1$); $\sin\theta = 0.9226$, i.e. $\theta = 67.31°$. Deviation $2\theta = 134.62°$.

b. $E = hc/\lambda$, i.e. $E = 9.517 \times 10^{-15}$ J or 59,480 eV.

3. For the $K\alpha$ line of titanium, 57% of the radiation is absorbed by the aluminium film. For the $K\alpha$ line of silver, only 1% of this higher-energy radiation will be absorbed.
4. **a.** $\Delta E/E = \Delta\lambda/\lambda$, i.e. $\Delta E = E\Delta\lambda/\lambda = 1.9284 \times 10^{-19}$ J or 1.2 eV.
   b. These two transitions of the sulfur atom cannot be distinguished from each other.
   c. A variation of 2×10^{-4} nm corresponds to a difference in energy of less than 1 eV, which will remain invisible on the spectrum.
5. $x = 0.30\%$.
6. **a.** $I = 69.867\%\ m/m - 0.056$ ($r = 0.995$).
   b. 3.336%
7. **a.** N_2: 77.78%; O_2: 22.22%.
   b. $\mu_M = 19.81\ cm^2/g$.
   c. 89.85%.
   d. 99.93%. Conclusion: helium transmits much more energy than air, that is why the instrument was purged to replace air with helium.
   e. $I = 63.64x + 17.96$ where x is the mass percentage of Al ($r = 0.9997$).
   f. 0.315%.
8. Firstly, the percent composition of the three elements present is calculated:

$$Si\% = \frac{28.1}{28.1+32} \times 65 = 30.4\%, Pb\% = \frac{207.2}{207.2+16} \times 35 = 32.5\%, O\% = 100 - 30.4 - 32.5 = 37.1\%$$

   Then, the weighted coefficient $\mu_{glass} = (4.26 \times 0.304) + (84.1 \times 0.325) + (0.8 \times 0.371) = 28.9\ cm^2/g$
   I.e. for linear coefficient $\mu = 28.9 \times 5.1 = 47.5\ cm^{-1}$
9. **a.** The curie, former measurement of radioactivity, corresponds to 3.7×10^{10} d.p.s. or Bq. The source corresponds to $0.012 \times 3.7 \times 10^{10} = 444 \times 10^6 = 444$ MBq.
   b. Unlike the Co-57 source, the Cd-109 source can only excite the lines corresponding to the *L* or *M* transitions of lead but cannot validly provoke *K* transitions (minimum 88 keV). This is important because if an old white lead or minium (red lead) paint is present, for example, under a thick layer of wallpaper, the *L* lines will be partially reabsorbed, which will lead to an underestimation of the quantity of lead. Only the Co-57 source can provoke *K* transitions, which are higher in energy and may emerge on the surface.
   c. This calculation involves finding the residual activity A of an X-ray which crosses a thickness x of a material with a linear absorption coefficient μ for the line in question: $A = A_0 e^{-\mu x}$, i.e. $A/A_0 = e^{-2\times0.1} = 0.82$. Therefore, 18% of the intensity of this transition is reabsorbed by the material.
   d. Since the period of this radioactive element is $\tau = 272$ days, its radioactive constant $\lambda = 1n2/\tau$, i.e. $\lambda = 0.693/272 = 0.0025\ day^{-1}$. After three months (91 days), $A/A_0 = e^{-0.0025 \times 91} = 0.797$. The correction factor to apply is $1/0.797 = 1.25$.

Chapter 13

Atomic Absorption Spectroscopy

Atomic absorption spectroscopy (AAS) is the basis for a quantification method for most elements (metals and metalloids, generally predefined) that must first be in the free atom state. For this purpose, analytes are heated up to 2,000°C, using a burner or an electrical furnace, except for some elements that lead to refractory combinations, in which case a hydride generator is used. The Beer–Lambert Law is then applied over a short range and varies depending on the elements. Many interferences, both

chemical and physical, require a number of corrections and thus give rise to specific assemblies. These analyses are mainly used in fields related to food, soil, water or plants.

Objectives

Explain the effect of temperature on an element
Understand the concept of line reversal
Apply this principle to measurements using AAS
Describe the corresponding instrumentation
Choose the type of light source
Explain the role of a hydride generator
Correct for physical or chemical disturbances
Take into account nonspecific absorptions

13.1 THE EFFECT OF TEMPERATURE ON AN ELEMENT

The principle of atomic absorption and flame emission is best understood through an experiment performed by Kirchhoff, around 1860, in which he demonstrated that incandescent gases absorb at the same wavelengths as those that they are able to emit.

When the light of an electric arc (serving as a source of white light) is dispersed by a prism, a continuous spectrum is obtained (Figure 13.1(1)). If this radiation source is replaced by a Bunsen burner onto which a few crystals of sodium chloride are sprinkled, then an emission spectrum of sodium is obtained with the well-known yellow doublet located at 589 nm, among other lines (Figures 13.1(2) and 13.2). This part of the experiment illustrates *flame emission* (see Chapter 14). Finally, if the two previous sources, the electric arc and the flame of the Bunsen burner are placed in series along the same optical path, a spectrum will be obtained which, in contrast to Figure 13.1(1), contains dark lines in place of the emission lines of sodium (Figure 13.1(3)). This line reversal results from the presence in the flame of a large proportion of ground-state sodium atoms that can absorb the same frequencies as the sodium atoms emit when excited. This is a manifestation of *atomic absorption*.

This experiment illustrates the concept of potential energy states defined for all atoms as a function of their electron configuration. When an atom in the free state is brought to high temperature or is irradiated by a light source in the near-UV/Visible region, one of its outer-shell electrons is promoted from the ground state to an excited state *e*. This electron transfer corresponds to an absorption of energy.

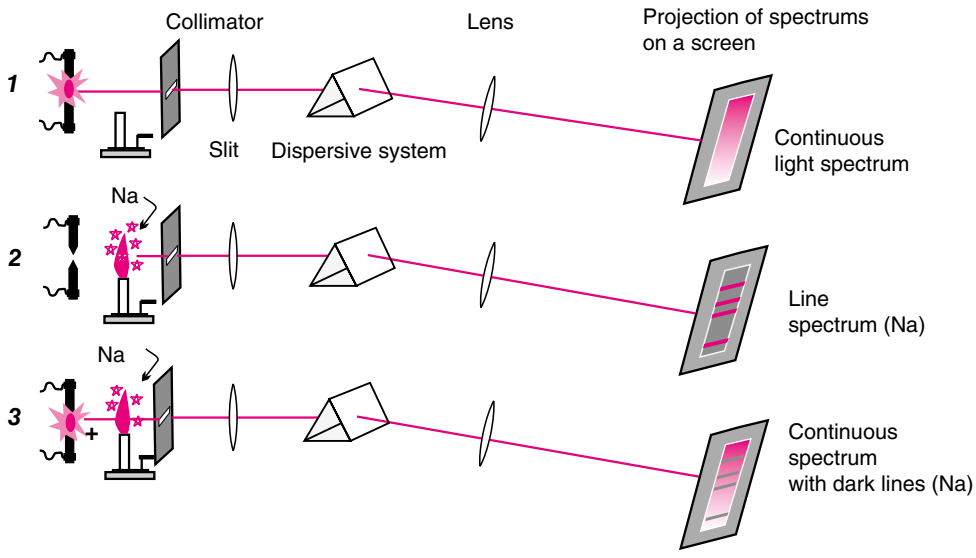


Figure 13.1 Kirchhoff 's experiment on the reversal of lines. The conventional diagram for the optical set-up (collimator, lens) has been simplified for reasons of clarity.

Conversely, when the atom returns spontaneously to its ground state, it can re-emit this excess energy in the form of one or more photons. Thus, in the previously described experiment, the flame induces the most likely transitions of the sodium atom (Figure 13.2).

The Maxwell–Boltzmann distribution function enables a calculation of the effect of temperature on each electron transition. By designating N_0 as the number of atoms in the ground state and N_e as those in the excited state, one obtains:

$$\frac{N_e}{N_0} = g.\exp\left[-\frac{\Delta E}{kT}\right] \qquad (13.1)$$

where T is absolute temperature in Kelvin, g is the ratio of the statistical weights of the ground (0) and excited (e) populations of the element in question (whole number), ΔE is the energy difference (in joules) between the ground (0) and excited (e) state populations, and k is the Boltzmann constant ($k = R/N = 1.38 \times 10^{-23}$ J/K).

If ΔE is expressed in eV instead of joules, Eq. (13.1) becomes:

$$\frac{N_e}{N_o} = g.\exp\left[-11600\frac{\Delta E}{T}\right] \qquad (13.2)$$

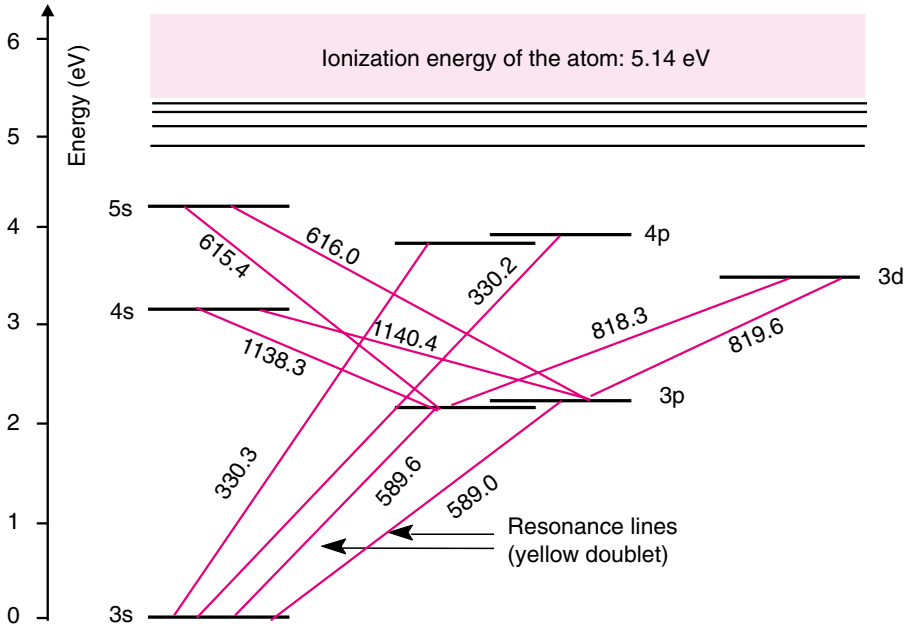


Figure 13.2 Some energy levels of the sodium atom. Simplified representation of the excited states of the sodium atom. Origins of the different emission lines, according to the selection rules. The values are indicated in nanometres.

Each atomic transition leads to an emission or an absorption of energy, which corresponds on the spectrum to a very narrow interval of wavelengths. The uncertainty around the calculated value constitutes the *natural bandwidth of the spectral line*. This bandwidth is temperature dependent and can range from 10^{-5} nm under ideal conditions to about 0.002 nm at 3,000 K. In reality, imperfections in spectrometers are responsible for a much greater widening of the image of lines.

13.2 APPLICATIONS TO MODERN INSTRUMENTS

To assay an element by one of these two methods, it must be in the form of free atoms. The sample is thus heated to a temperature of at least 2,000°C, in order to dissociate all chemical combinations in which the element under study is engaged, as well as the rest of the sample. This pyrolysis leads to the total concentration of the element without distinguishing between the different chemical structures in which that element may have been found in the original sample (this is therefore the opposite of a *speciation analysis*).

Two thermal devices co-exist: one consists of a burner fed by a combustible gas mixture, the other is a type of small tubular electric oven. In the first assembly, used for the majority of elements, an aqueous solution of the sample is nebulized and then introduced into the flame at a constant rate. In the second, the sample is deposited in a small graphite hollow rod open at both ends, where it is volatized. This more expensive assembly has a greater sensitivity for refractory elements (V, Mo, Zr). In both methods, the optical path from source to detector passes through the region containing a cloud of atoms in the free state.

Various instruments exploiting the principle of this phenomenon are used to measure traces of mercury in a polluted atmosphere. A device similar to a colorimeter uses atomic absorption to conduct the measurement. The source is a mercury vapour lamp and the measuring cell is a transparent tube filled with the atmosphere to be monitored. If mercury vapours are present in the optical path, there will be absorption of the radiation emitted by the lamp, a consequence being a decrease in light intensity proportional to the mercury concentration.

Common values of the different parameters of Eqs. (13.1) and (13.2) are shown in Table 13.1 for four elements. Examination of the data reveals that practically all of the atoms remain in their ground state, especially when the energy difference ΔE is large and the temperature is relatively low.

Therefore it would seem preferable, in all circumstances, to base the measurements upon atomic absorption instead of flame emission, since the absorption spectra are simpler than the emission spectra. However, the matrix in which the element is found can be the cause of interferences, chemical interactions, instability of the excited states, and other phenomena that occur at high temperatures (Figure 13.3) and which make absorbance measurements difficult. With the most modern detectors containing a photomultiplier, reliable measurements can be obtained as long as the N_e/N_0 ratio is greater than 10^{-7}. Experience reveals that flame emission is in fact preferable for five or six elements. This is why the alkaline earths, which are elements giving coloured flames, are easily measured by emission (Table 13.1).

Not all of the excited atoms N_e return to their initial state by emitting photons. They can lose their excess energy in other ways as well. The higher the temperature is, the more complex the emission spectrum becomes, mainly due to the emergence of lines originating from

Table 13.1 Ratio of N_e/N_0 for several elements at various temperatures.

Element	λ (nm)	E (eV)	g	2,000 K	3,000 K	4,000 K
Na	589	2.1	2	1.0×10^{-5}	6.0×10^{-4}	4.5×10^{-3}
Ca	423	2.93	3	1.2×10^{-7}	3.6×10^{-5}	6.1×10^{-4}
Cu	325	3.82	2	4.8×10^{-10}	7.3×10^{-7}	3.1×10^{-5}
Zn	214	5.79	3	7.3×10^{-15}	5.7×10^{-10}	1.5×10^{-7}

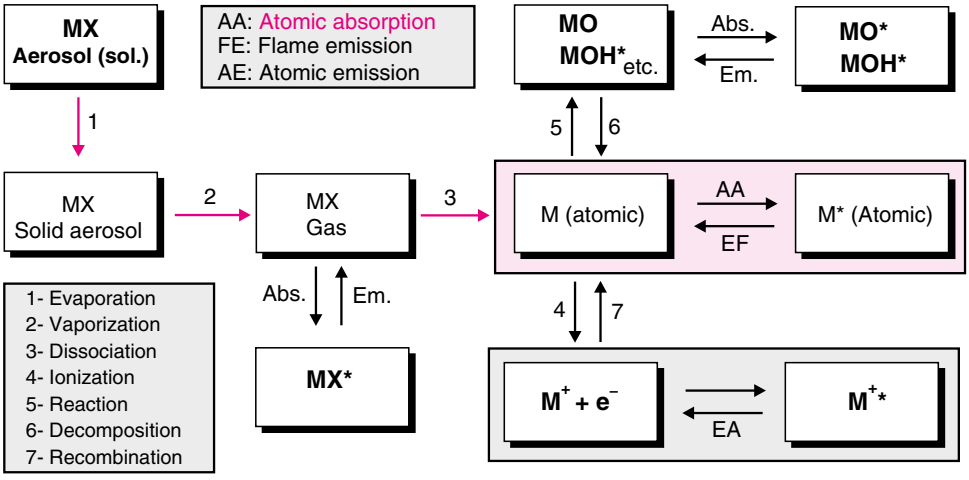


Figure 13.3 Summary of the possible evolution of an aerosol solution in a flame.

ionized atoms, since the spectrum of an atom is completely different from that of its ions (Figure 13.3). To study these complex spectra, instruments that possess optics of very high quality are required. These are known as atomic emission spectrometers (see Chapter 14).

13.3 MEASUREMENTS BY AAS

The quantification of elements by these two methods implies that a relationship exists between the concentration and the intensity of the corresponding light absorption or emission. They make use of classic protocols that comprise a calibration curve from synthetic solutions with growing concentration of the analyte.

The absorbance of the element in the flame depends upon the number of ground-state atoms N_0 remaining on the optical path. Measurements are made by comparing the unknown to the standard solutions:

$$A = k \cdot C \tag{13.3}$$

where A is absorbance, C is the concentration of the element, and k is a specific coefficient for each element at the given wavelength.

The method relies on the Beer–Lambert relationship, but the molar coefficient of absorption ε is not calculated in this case. The instrument yields the absorbance by determining the ratio of the transmitted intensities in the absence and then in the presence of the sample. Linearity is only observed for low concentrations (typically below 3 ppm), or for media in which the matrix effect is negligible (Figure 13.4). If the matrix is complex, the medium must be reconstituted for the calibration solutions or else methods using standard additions have to be used to ensure the linearity of absorbance response.

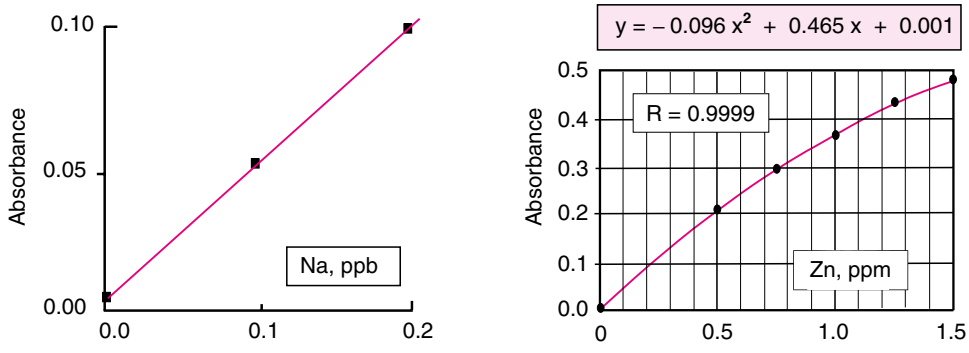


Figure 13.4 Examples of calibration graphs in AAS. Left, a straight calibration line at sub-ppb concentrations obtained with an instrument equipped with a Zeeman effect device (see Section 13.8.2) for the quantification of sodium. Right, a quadratic curve for the measurement of zinc at concentrations in the ppm range with a burner-type instrument. This second graph reveals that when concentrations increase, the absorbance is no longer linear. The quantitative analysis software for AAS provides several types of calibration curves.

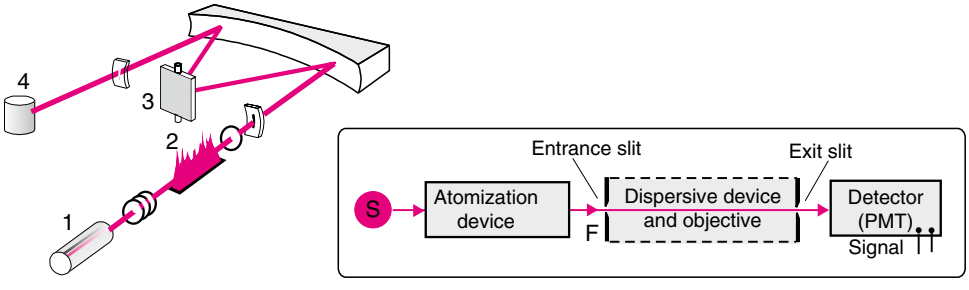


Figure 13.5 The various components of a single-beam atomic absorption apparatus. Model IL 157 built during the 1980s. **1**, source (spectral lamp); **2**, burner flame; **3**, monochromator grating; and **4**, detector (photomultiplier). The source illuminates a slit situated at the entrance of the dispersive system. The exit slit is close to the detector window. It selects a narrow bandwidth of the spectrum, ($\Delta\lambda$ of 0.2–1 nm), which must not be confused with either the width of the exit slit or with that of the image of the entrance slit.

13.4 BASIC INSTRUMENTATION FOR AAS

In its simplest form, an AAS resembles a single-beam spectrophotometer. The optical scheme is illustrated in Figure 13.5 and contains four main parts. The light beam coming from the *source* (1) passes through the *atomizer* (2), in this case the flame, in which the element is brought to its atomic state before being focused upon the entrance slit of

the *monochromator* (3), which selects a very narrow wavelength interval. The optical path ends at the entrance slit of the *detector* (4).

If the element corresponding to the wavelength range selected by the entrance slit of the dispersive system is not present in the flame, the detector will receive all of the light intensity I_0 emitted by the source within this spectral interval. Alternatively, if the element is present, the detector receives a reduced intensity I.

13.4.1 Sources

Since absorption lines of elements are 100 times narrower than the bandwidth of the best monochromators (0.1 nm), the solution retained from the start of the commercialization of atomic absorption spectrometers was to use spectral lamps. These are selected so that they emit only the radiation that can be absorbed by the analyte. More recently, several models have appeared with a xenon arc lamp emitting a light continuum covering the entire effective spectral range. The obvious interest is to conduct the analysis of several elements without changing the source, unlike instruments equipped with lamps dedicated to one, two or three elements.

Hollow cathode lamps (HCL)

Hollow cathode lamps (HCL) are discharge lamps. The anode is made of zirconium or tungsten, and the cathode is made of or coated with the element whose emission spectrum we want to obtain. Neon or argon, under low pressure, is used as filling gas (Figure 13.6). For each analyte being quantified, there is a corresponding lamp, which is a drawback of the method. This is why there are close to a hundred different lamps made from pure elements as well as alloys or fritted powders for multi-element lamps.

When a potential difference of around 300 V is applied between the electrodes, the electrons provoke the ionization of the filling gas. These ions (Ar^+ or Ne^+) acquire enough kinetic energy to strip atoms from the cathode, which becomes equivalent, at the surface, to an atomic gas. If M(Cat.) refers to the element M in its metal state (cathode) and M(Gas) refers to it in the atomic state, emission corresponds to the following sequence:

$$\mathrm{M(Cat.)} \xrightarrow{\mathrm{Ne^+}} \mathrm{M(Gas)^*} \rightarrow \mathrm{M(Gas)} + \mathrm{photon}$$

Each HCL lamp has several emission lines of different intensities. The spectrum emitted by the lamp corresponds to the superimposition of radiation emitted by the cathode and by the filling gas. The width of the emission lines, which depends upon different effects (Doppler, Stark (ionization) and Lorentz (pressure)), is narrower than the corresponding absorption band. The choice of line for a measurement depends,

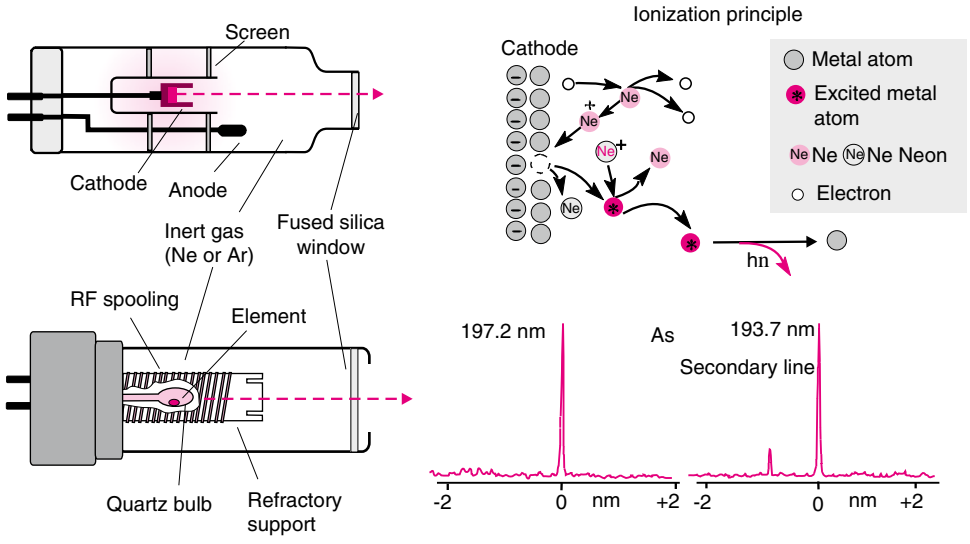


Figure 13.6 Two types of sources in AAS. The cathode is a hollow cylinder whose central axis corresponds to the optical axis of the lamp. Right, pictorial representation of cathode atoms being excited by the impact of neon ions (Ne^+). EDL lamp. The main emission lines of arsenic. These set-ups are not adapted to sodium (melting point too low) or mercury (liquid state), for which we use metal vapour lamps.

amongst other things, on the concentration of the analyte, knowing that in AAS the precision of the measurement diminishes fairly quickly with concentration (dynamic range of 1 to 100).

Electrodeless discharge lamps (EDL)

These lamps emit intense radiation, which improves the measurement of volatile elements such as arsenic (As), mercury (Hg), antimony (Sb), bismuth (Bi), and phosphorus (P). In addition to the filling gas, they contain the selected element in its metal form or as one of its salts. The excitation is obtained by an intense electromagnetic field of radiofrequencies or by microwaves (Figure 13.6).

Xenon lamp

This arc lamp emits a continuous spectrum, of variable intensity, from the near-UV to the near-IR (190–900 nm), which makes the analysis of close to 67 predefined elements possible without having to change the source.

The combination of these light sources with sophisticated correction algorithms continuously compensates for deviations and fluctuations, which reduces the preheating time and ensures stability. The precision of the wavelengths is on the

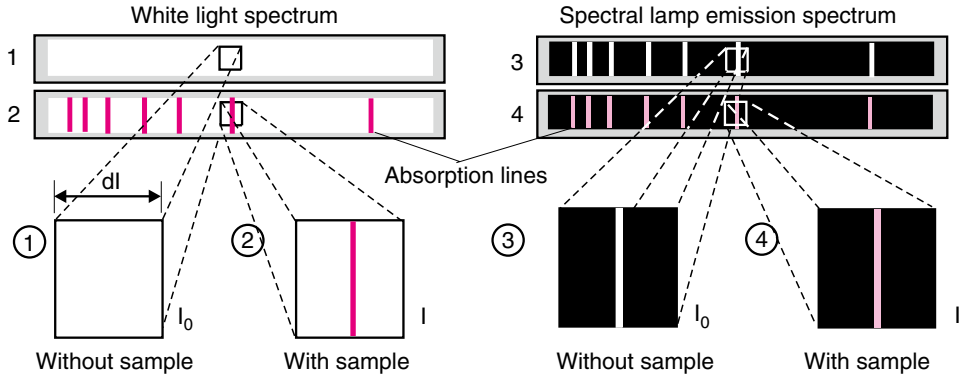


Figure 13.7 Comparison of transmitted intensities in AAS with a continuum light source (1 and 2), and with a lamp that emits spectral lines (3 and 4). The rectangular region describes the wavelength interval 'seen' by the PMT. The PMT signal is proportional to the white parts in the rectangles. In this way, 'the resolution is in the source', as expressed by Walsh, who is considered to be one of the pioneers of atomic absorption.

picometre level when using a neon diode, whose many known reference lines are distributed across the entire spectrum.

It is interesting to compare the intensities transmitted to the detector (I_0 and I), depending on whether one or the other type of source is used. With a line spectrum lamp selected to emit only radiations that the element is able to absorb, the ratio I_0/I could be much smaller than 1, leading to a more reliable measurement since photomultipliers are extremely sensitive (Figure 13.7). If the source emits a *continuum of light*, the ratio I_0/I will always be close to 1 because the absorption lines are very thin (1×10^{-3} nm). Technological progress has now made their use possible.

13.4.2 Thermal Devices for Obtaining Atomic Vapours

Atomization using a flame – burner and nebulizer

This robust mechanical assembly, called the burner, is supplied by a combustible gas mixture and produces a laminar flame. It has a rectangular base of about 100 mm in length by 1 mm in width. The optical axis of the instrument is aligned with the longest dimension of the flame (Figure 13.8). The sample in aqueous solution is nebulized and introduced into the gaseous mixture.

The flame is a complex medium in equilibrium, characterized mainly by its chemical reactivity (related to its oxidative or reductive nature, depending on its fuel/oxidizer ratio), its temperature (Table 13.2), and its emission spectrum in the near UV, owing to the presence of free radicals. These can interfere with the measurement of some

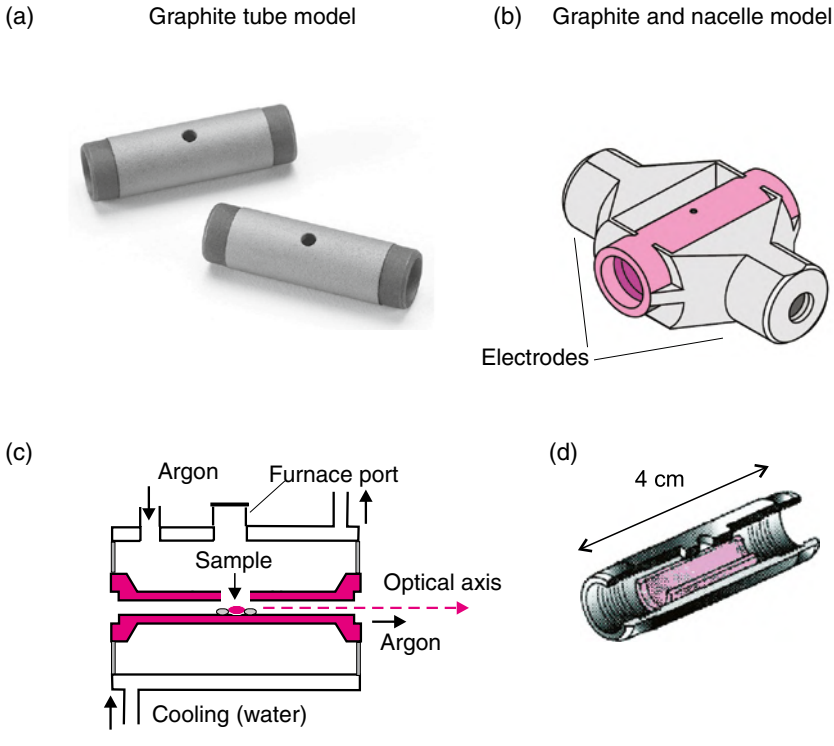


Figure 13.8 Burner and thermoelectric atomization system. (a), (b) Graphite tube models. (c), (d) Graphite furnace: graphite rod placed between two electrodes, in transverse electrical connection. (e) Temperature programming graph showing absorbance as a function of time. The first two steps of the temperature program are conducted under an inert atmosphere (argon sweeping).

Table 13.2 Upper temperature limits for some gas mixtures.

Fuel/oxidizer mixture	Max. temperature (K)
Butane/air	2,200
Acetylene/air	2,600
Acetylene/nitrous oxide (N_2O)	3,000
Acetylene/oxygen	3,400

elements. That is why not just any flame can be used for any element. The flame has two main zones: the middle zone and the outer zone, which is the hottest zone of the flame (about 4 cm from the burner). As a result, the chemical reactivity of the flame is not homogeneous. Therefore, the burner position must be adjusted with respect to the optical path of the instrument. This adjustment, which also depends on the element, is done automatically by many current instruments. The same position must be conserved for calibration and measurement.

Generally, an air/acetylene flame is chosen. To attain higher temperatures (refraction element analysis), the air is replaced by nitrous oxide, N_2O (Table 13.2).

Thermoelectric atomization

The above device with flame and nebulizer may be replaced by a *graphite furnace* comprising a tube of graphite carbon with a small cavity that can hold a precise quantity of sample (a few mg or μl) (Figure 13.8). This carbon rod, whose central axis coincides with the optical axis of the spectrophotometer, behaves as an ohmic resistor that can attain 3,000 K by virtue of the Joule effect (8 V and 400 A). The graphite tube is surrounded by a double sleeve. One contains an inert gas, such as argon, that circulates to protect the elements from oxidation, while the other cools the entire device using water (Figure 13.8). The heating cycle generally comprises three steps. To avoid loss through projections, the temperature is gradually increased, first to dry, then to decompose, and finally to atomize the sample. In this last step, the temperature gradient can reach 2,000°C/s, and thus the sample is brought to the atomic vapour state in three or four seconds. Optimizing the various stages of the thermoelectric program is essential for each measurement.

Compared with the burner, this instrument is much more sensitive, even though atoms remain for only a very short time in the observation zone. The furnace in fact causes atomization of the entire sample, hence a higher density of atoms on the optical path and thus a stronger signal.

Hydride generator

Some elements, such as arsenic (As), bismuth (Bi), tin (Sn) or selenium (Se), are difficult to reduce to atoms in a flame when in higher oxidation states. To measure these elements, the sample is reacted with a reducing agent composed of sodium

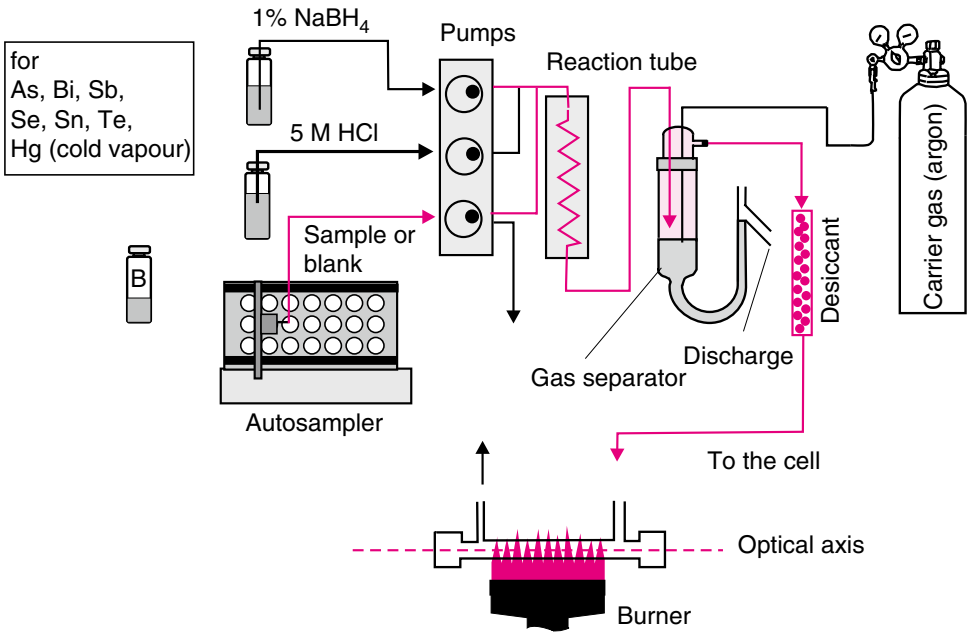


Figure 13.9 Diagram of a hydride reactor. Reserved for certain elements, this automatic sampling device houses a mixing tube where the hydride of the metal (or nonmetal) is formed by reaction with sodium borohydride. An argon flow extracts the metal hydride formed (gas separator) carrying it to a fused silica tube heated to between 800 and 1,000°C in the flame.

borohydride or tin chloride in an acidic medium, just prior to analysis (Figure 13.9). A volatile hydride of the element is formed; this is swept up by a make-up gas into a quartz cell placed in a furnace, most often electrically heated or carried into the burner flame. For arsenic, the transformation is as follows:

$$As^{+++} \xrightarrow{NaBH_4} AsH_3 \xrightarrow{H^+\,(800^\circ C)} As + \frac{3}{2}H_2$$

The metal hydrides, which are easily thermolyzed at around 1,000 K, liberate the corresponding elements in the atomic state. An electrodeless lamp is preferably used as a light source.

Mercury is not transformed into a hydride but rather remains in the metal state (Hg^0). Consequently, a special cell that does not have to be put into the atomizer is used. This is called the 'cold vapour' method and requires specialized instruments (reduction by $SnCl_2$).

13.4.3 Monochromators

The monochromator's role consists of isolating the wavelength of the analysis, while eliminating parasitic light. The selection is generally conducted through a dispersive system with a mobile diffraction grating for single-element systems and a grating/prism combination for multi-element systems (these are called polychromators). In the latter case, the beam is then sent to a multichannel detector (see Section 13.4.4).

The line intensities are very different (Figure 13.8). In single-element systems, the most intense line is generally selected in order to obtain the best sensitivity possible. In multi-element systems, one of the secondary lines of an element may be selected when it is associated with other elements present at very low concentrations to minimize handling operations and successive dilutions.

The second role of monochromators is to eliminate any parasitic light, which essentially comes from the source, whose radiation is actually very complex. The emission lines of the element constituting the cathode are accompanied by other lines coming from the filling gas as well as lines from any impurities. The atomizer is also a source of parasitic light: by its own emission (flame, furnace wall) and from elements in the sample other than the one being analysed, which may be excited and re-emit light.

13.4.4 Detectors

The measurement of light intensities is generally conducted by a photomultiplier tube in single-element systems. These instruments are replaced by multichannel detectors with diode arrays in multi-element systems, thus enabling simultaneous measurement of intensities of the analysed elements at various wavelengths.

13.5 PHYSICAL AND CHEMICAL DISTURBANCES

Many interactions may lead to erroneous analytical results by overestimation or underestimation of the real concentration. These can be classified into three categories.

13.5.1 Spectral Interferences

These are also called *nonspecific absorptions*. They are related to parasitic emissions resulting from the atomization source. These come from a superimposition of lines, from the presence of molecular absorption bands or from light diffusion by the particles present in the atomizer.

Superimposition of lines

The atomization instrument may lead to a parasitic emission (e.g. graphite furnace tube walls). Compounds in the matrix may also lead to other absorptions. Therefore, the superimposition of two radiations cannot be totally avoided: for example, the one chosen for the analysis with a secondary line belonging to another element. In atomic absorption, confusions are rare. However, it may be recommended to conduct a second measurement by changing the wavelength.

Superimposition of absorption and emission of the same element

A nonnegligible fraction of the atoms of certain elements pass to the excited state thermally. These atoms emit photons of the same energy that those remaining in the ground state are able to absorb. To correct the measurement, the lamp voltage is pulsed, which permits differentiation between the emission signal, which is constant, and the absorption signal, which is pulsed.

Molecular absorptions

This type of nonspecific absorption is mainly encountered in the analysis of volatile elements that do not require a high flame temperature in the presence of a refractory matrix. It comes from molecules present in the sample matrix and is composed of large and intense absorption bands, which can considerably disturb analyses.

Diffusion of incident light

Solid or liquid particles resulting from the decomposition of the sample or from the recombination of atoms and molecules can diffuse the incident light, giving rise to continuous absorption, which is superimposed on the absorption of the element being analysed.

In these last two situations, corrections can easily be conducted by working with a blank aspirated into the flame when the interfering components come only from the gaseous mixture. When the parasitic scattering or absorption comes from the sample matrix, either the analytical parameters (flame temperature, fuel/oxidizer ratio) can be changed or an excess of the interfering substance can be added to the measured solutions (called *spectrochemical buffer*). This excess must imperatively be much greater than its concentration in the matrix in order to make the interference negligible.

13.5.2 Chemical Interferences

When atomic absorption is used to search for trace amounts of elements, it is important to take into account the matrix in which these are present. Well-established protocols must be followed to suppress ionic or chemical interferences.

The interferences, also called the *matrix effect*, change the atomic vapour density and/or its speed of formation. They are caused by a reduction in the number of atoms during the decomposition stage and by the formation of poorly dissociable compounds with interfering substances. These compounds are difficult to dissociate owing to their refractory nature or low volatility. To correct for this, mineral salts or organic reagents serving as releasing agents R are often introduced into the solutions to be nebulized. This treatment will effectively release element M from its combination MX, especially if compound RX is more stable than MX:

$$R + MX \rightarrow M + RX$$

Thus when calcium is to be measured in a matrix rich in phosphate ions or in refractory combinations containing aluminium, strontium or lanthanum chloride is added. The desired effect is to liberate calcium and to increase the volatility of the matrix to ensure a more efficient elimination during the decomposition step. In the same way, to measure sodium or potassium, it is usual to add a small amount of Schinkel's solution ($CsCl/LaCl_3$).

With apparatus containing a graphite furnace, ethylenediaminetetraacetic acid (EDTA) is introduced to form 1:1 complexes with bivalent ions, or ammonium nitrate is added if the matrix contains a high concentration of sodium (Figure 13.10).

Finally, the use of very hot flames can provoke partial ionization of certain elements, which decreases the concentration of free atoms in the flame. This phenomenon is corrected by addition of an ionization suppressor (or ionization buffer) in the form of a cation whose ionization potential is lower than that of the analyte. A potassium salt at about 2 g/l is often employed for this purpose.

This variation of the ionization can occur more or less spontaneously when the matrix, by nature, contains one or more alkaline elements. To avoid these random errors, an ionization buffer based upon a potassium or sodium salt is added systematically to the solutions. An alternative consists in preparing a series of standards in a medium close to that of sample.

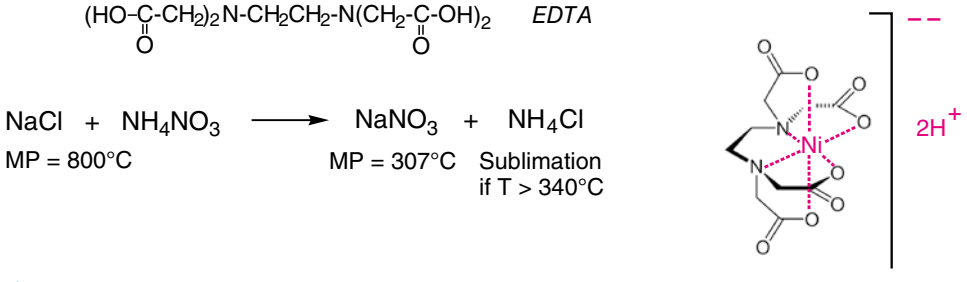


Figure 13.10 Matrix modification. Ammonium nitrate or EDTA increases the volatility of certain elements. A volatile complex of 1:1 type between a molecule of EDTA and a Ni^{2+} ion.

13.5.3 Physical Interferences

These are mainly represented by *transport* interferences, which are often linked to differences in viscosity between the standard solutions and samples and more generally to physical properties of studied solutions.

13.6 CORRECTIONS OF NONSPECIFIC ABSORPTIONS

As in visible or infrared spectrophotometry, it is necessary to carry out baseline corrections to eliminate fluctuations coming from the lamp and nonspecific absorptions due to spectral interferences.

Instruments with a burner generally have only low background noise in the signal because diluted aqueous solutions are being nebulized.

However, with graphite furnaces, samples deposited in the raw state (solid or liquid) can produce parasitic absorptions resulting from incomplete atomization due to the matrix. This gives a baseline of constant absorption (fumes) within the interval defined by the monochromator. To correct this effect, manufacturers propose improved instruments that make it possible to apply various methods. However, for this type of sample, it is not possible to have the matrix alone without the analyte. It would therefore be useless to manufacture instruments with a second heat device that is perfectly identical to the first and placed on a second optical path used as a reference.

13.6.1 Background Correction Using a Deuterium Lamp

Atomic absorption instruments often have a second continuous source, made up of a deuterium lamp (Figure 13.11), intended to assess the parasitic absorptions when the sample is nebulized in the flame. The atomizer, whether a flame or oven, is continuously lit by two sources (semitransparent mirror). The rotating mirror either lets through the beam which either passed through the flame (a) or not (b). The bandwidth is one hundred times larger than the selected absorption line. First, the monochromator is adjusted on the line selected for the measurement of the element. When the sample is nebulized in the flame, the detector alternately receives beams from paths (a) or (b) and therefore measures parasitic absorptions and the absorption due to the element. In the absence of the element, it only measures the background absorption. Since the absorbances are additive, the difference between the two measurements will yield the absorbance of the element alone.

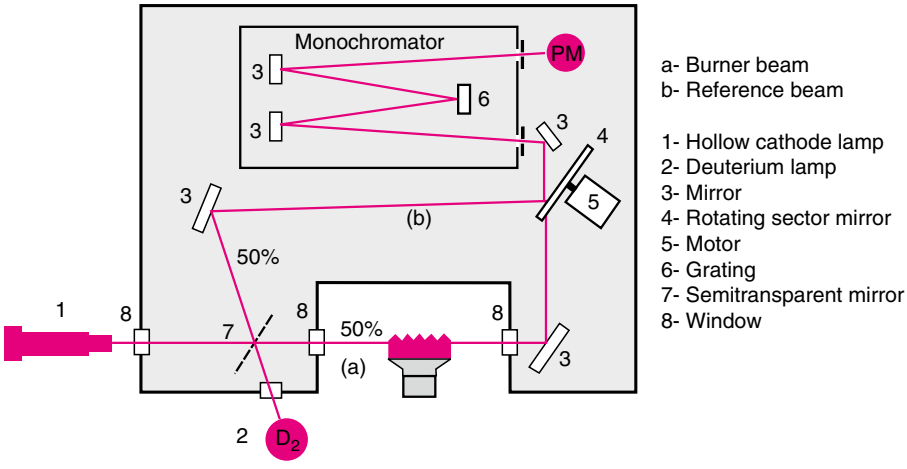


Figure 13.11 Diagram of an AA spectrometer showing background correction with a deuterium lamp. This double-beam assembly is used to compensate for the lamp's drift when it is not yet stabilized. The optical path of the deuterium lamp is superimposed, using a semitransparent mirror, upon the spectral lines emitted by the HCL. The light reaching the detector comes alternately from the reference path (b) and the sample path (a). The instrument measures the ratio of the intensities transmitted by the two beams. The domain of correction is limited to the spectral range of the deuterium lamp, i.e. 180–420 nm. (Source: Reproduced from an optical diagram of the Spectra AA-10/20. Varian Medical Systems, Inc.)

13.6.2 Background Noise Correction using the Zeeman Effect or a Pulsed Lamp

The action of a magnetic field on an atom causes a disruption in the energy states of its electrons through a rise in degeneration of the excited state. This phenomenon, called the *Zeeman effect*, modifies the corresponding emission (or absorption) spectrum of the element, at least as long as the field value reaches a minimum of 1 Tesla. However, not all elements respond in the same way. In the simplest case ('normal' Zeeman effect), each absorption line, observed without a field, is split into three new polarized lines. One of these lines, called component π, remains in the initial position (without a field), while the two others, named satellite components σ, are symmetrically shifted on each side of the π component (a few picometres in a 1-Tesla field). The directions of polarization of the π and σ lines are perpendicular. If a suitably oriented polarizer is placed in the optical path, the absorption component π due to the element can be erased or not. Contrary to the atoms of the element, particles and fumes in suspension during the measurement are not affected by the Zeeman effect (Figure 13.12).

Known as the most effective correction mode, it has led to various configurations on the market. These are differentiated first by the position of the magnet, which may be

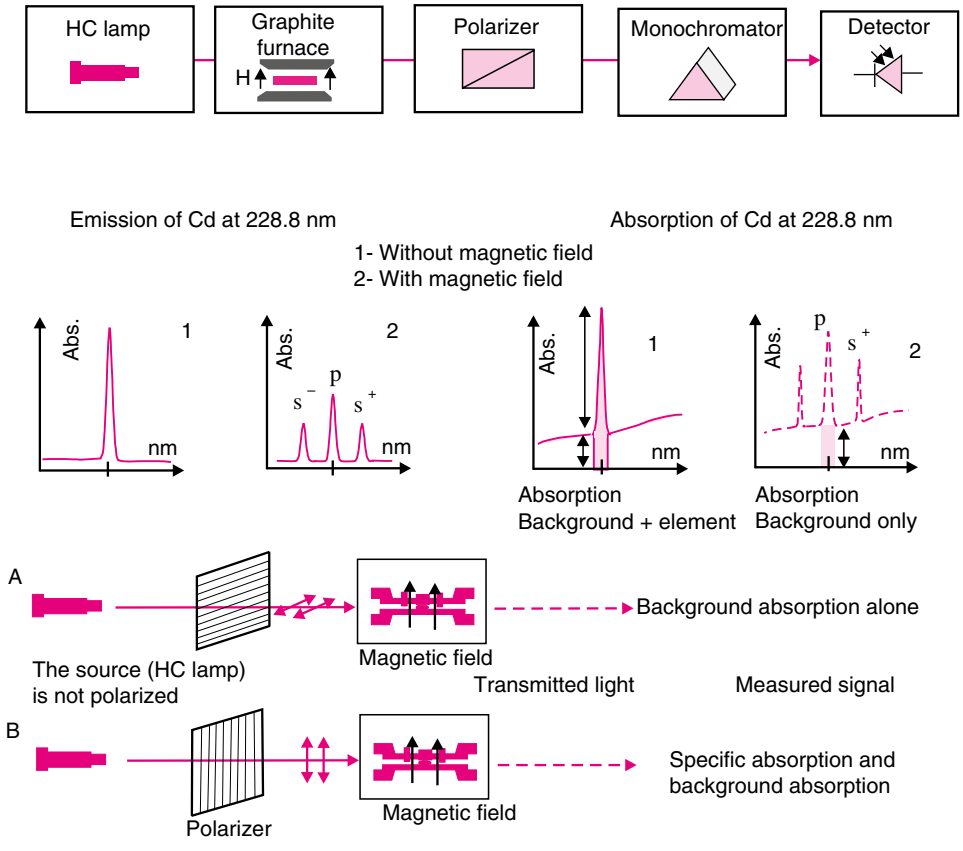


Figure 13.12 Normal Zeeman effect. Pictorial explanation of the constant inverse/transverse assembly from absorption of cadmium at 228.8 nm. In 1, the polarization plane is perpendicular to the magnetic field; in 2, the polarization plane is parallel to the field.

located around the source (*direct* assembly) or the atomizer (*inverse* assembly). Then, each of these configurations may be associated with an assembly that is either *longitudinal* (direction of observation parallel to the magnetic field) or *transverse* (direction of observation perpendicular to the magnetic field), with a constant or alternative field and finally with a fixed or rotating polarizer. The most frequent set-up is inverse–transverse–constant. It consists in operating with a fixed field and in placing a rotating polarizer between the lamp and the atomizer. The signal oscillates between two values, which correspond either to the background absorption plus the absorption of the π component of the element when the direction of the wave polarization is parallel with that of the field (Figure 13.12 B), or to the background absorption alone, when the polarization sends polarized light perpendicular to the magnetic field direction (Figure 13.12 A).

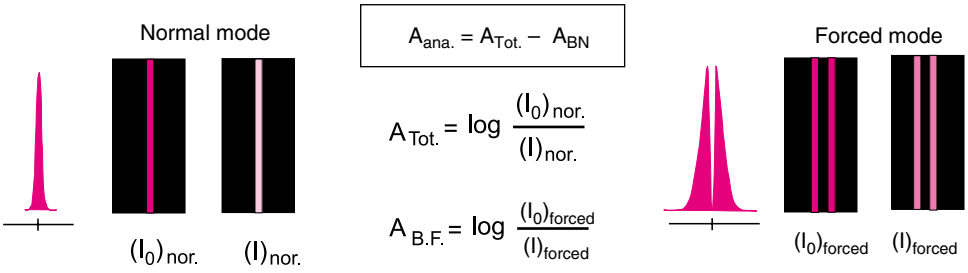


Figure 13.13 Correction using a pulsed lamp. Profiles of an emission line in normal and forced modes. Black rectangles illustrate the wavelength interval coming out of the monochromator, thus impacting the detector for these two situations. I_0 corresponds to the position without a sample in the light path. The absorbance of the analyte (A_{ana}) is equal to the total absorbance ($A_{tot.}$) minus the background noise absorbance (A_{BN}).

Before the Zeeman effect correction became the reference in conducting background noise measurement, another process was developed: correction using pulsed light, known as the Smith–Hieftje correction, which relies on reversal of lines.

For volatile elements, when the intensity of a hollow cathode lamp is increased, the profile of the lines is broadened with a hollow zone in the middle due to evaporation of atoms on the surface of the cathode, whose temperature rapidly increases (Figure 13.13). These atoms reabsorb a fraction of the light, at the specific emission wavelength of the cathode, leading to self-absorption. Moreover, the lifespan of the lamps may be shortened.

Implementation thus consists in making a quick succession of alternating measurements: lamp in *normal mode* (e.g. 10 mA) and then in *forced mode* (500 mA). In normal mode, with the sample in the atomizer, total absorbance (background absorption and analyte absorption) is measured. In forced regime, only the absorbance of the absorption background is obtained, since the lamp practically no longer emits at the specific wavelength of the analyte. The difference between these two values can then be used to calculate the absorbance of the analyte.

13.7 SENSITIVITY AND DETECTION LIMITS IN AAS

The lowest concentration that can be quantified in a sample depends upon many factors. In spectrometry, *sensitivity* is defined for an element as being the concentration expressed in μg/ml, which, in aqueous solution, leads to a 1% decrease (A = 0.0044) in the transmitted light intensity. When possible, a calibration curve should be established with concentrations in the range of 20–200 times this value.

The *detection limit* corresponds to the concentration of the element giving a signal whose intensity is three times the standard deviation of a series of measurements

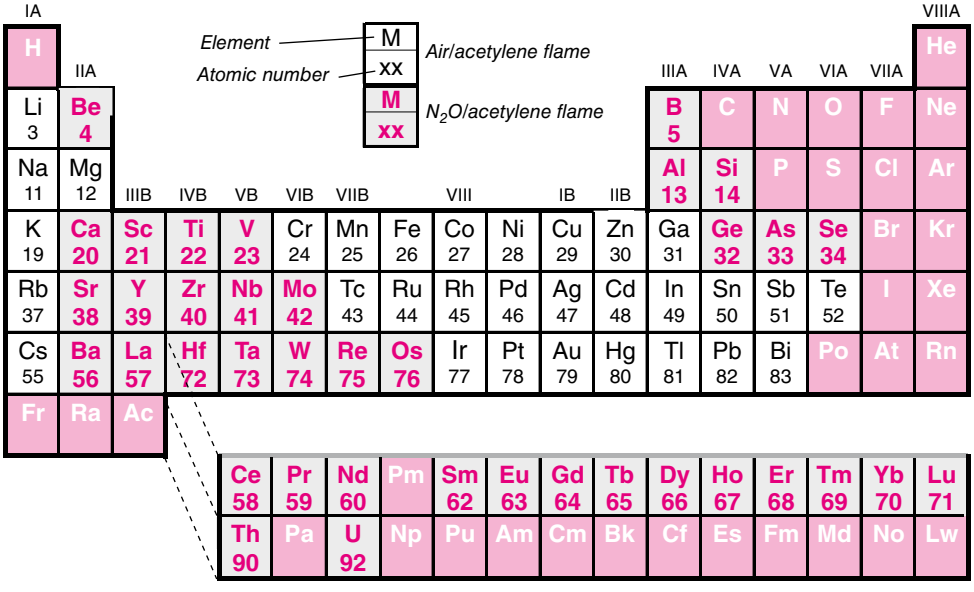


Figure 13.14 Elements measured by AAS.

made on the analytical blank or on a very dilute solution (degree of confidence: 95%). In practice, the concentrations must be at least ten times higher than the detection limit to give reliable measurements (Figure 13.14).

KEY POINTS OF THE CHAPTER

1. The study of the atomic absorption spectra of elements at high temperature is the source of a method used to quantify them, as long as these are in gaseous form made up of atoms, i.e. released from the organic or mineral structure in which they are found in the analysed sample. Therefore, an overall measurement of each element is obtained, i.e. the opposite of a speciation analysis.
2. For elements (metals or metalloids) to pass into a free atom state, the sample is heated to a high temperature (2,000–4,000°C) using a heating device: combustible gas burner (butane, acetylene) or a graphite furnace used as an electric resistor (atomization of solutions or solids).
3. For elements that are hard to make pass into the free atom state (refractory combinations), a hydride generator is used. Thus, for arsenic (As):

$$As^{+++} \xrightarrow{NaBH_4} AsH_3 \xrightarrow{H^+(800°C)} As + \frac{3}{2}H_2$$

4. When a macroscopic set of identical atoms (N_0) is heated to a high temperature, only a very small portion of them pass into an excited state (N_e) before emitting photons when returning to their ground state. Kirchhoff's experiments help to describe emission and absorption phenomena.
5. The fraction that has passed into an excited state (N_e) varies depending on the elements, because it depends on the energy shift between the levels in question. The Maxwell–Boltzmann distribution equation is used to calculate the N_e/N_0 ratio.

$$\frac{N_e}{N_0} = g \cdot \exp\left[-\frac{\Delta E}{kT}\right]$$

6. While nearly all atoms are still in the ground state at 2,000 or 3,000°C, it is difficult to determine the fraction of light absorbed, since measurements are conducted on very narrow absorption lines.
7. To increase the reliability of measurements, we mainly use spectral lamps chosen in order for these to emit only the radiation which can be absorbed by the analyte. These are hollow cathode lamps, which are specific for one or more elements, or else electrodeless discharge lamps.
8. To determine the concentrations of one or more elements selected in advance, a specific absorption is selected for each of them. The absorbance is proportional to the concentration of the element, depending on an equation similar to the Beer–Lambert Law: $A = k \cdot C$. However, the linearity of response is effective only for very low concentrations.
9. The matrix may be the cause of many interferences, either chemical (e.g. molecular absorptions or refractory compounds, corrected by adding a *releasing* agent R that can displace M from the compound MX) or spectral (e.g. particles in suspension, fumes, ionization) justifying the use of an ionization buffer.
10. To correct for the nonspecific absorption of background noise, various changes have been made to instruments (double beam with a deuterium lamp (D_2) or application of the Zeeman effect).

PROBLEMS

1. An AAS method is employed for the determination of lead (Pb) in a sample of adulterated paprika by the introduction of lead oxide (of the same colour). An electrothermal atomic absorption instrument that provides a background correction based upon the Zeeman effect is used.

   0.01 g of the adulterated paprika powder is placed in the tube of the graphite furnace. The area of the absorbance peak is determined at λ = 283.3 nm first in the

absence and later in the presence of a magnetic field. The background-corrected value is 1,220 (arbitrary units). Under the same conditions, 0.01 ml of a 10 g/l lead solution leads to a value of 1,000 in the same units.

Calculate the % mass of lead in the paprika sample under study.

2. The many commercial derivatives of ethylenediaminetetracetic acid (EDTA) used in complexometric titration include a mixed salt of Zn/Na containing one atom of zinc and two atoms of sodium per molecule of EDTA. This mixed salt presents itself in the form of a crystallized hydrate. We want to calculate the number of water molecules of this hydrate by measuring the presence of zinc via atomic absorption.

The sample solution is prepared by dissolving 35.7 mg of the hydrate in 100 ml of water. 2 ml of this solution is then pipetted and diluted again with H_2O to a final volume of 100 ml. Calibration is performed with five solutions of known zinc concentration (mg/l).

Solution	**Zn conc. in mg/l**	**Absorbance**
Blank	0.00	0.0006
Standard 1	0.50	0.2094
Standard 2	0.75	0.2961
Standard 3	1.00	0.3674
Standard 4	1.25	0.4333
Standard 5	1.50	0.4817
Sample	C?	0.3692

a. To perform this measurement, the *slit width* of the atomic absorption apparatus is 1 nm. What does this parameter represent?

b. Knowing that the calibration curve computed by the apparatus indicates that $A = 0.3692$, $C = 0.99$ mg/l, calculate the number of water molecules in the Zn/Na salt.

c. Using the linear regression method, derive an equation for the calibration curve. Calculate the zinc concentration C (in mg/l) in the sample solution.

Data: H = 1.01; C = 12.0; N = 14.01; O = 16.00; Na = 22.99 and Zn = 65.39 g/mol, respectively.

3. To determine the quantity of chromium C_i in an unknown aqueous solution, the standard addition method is used. To do so, five solutions (1–5) are prepared as follows: 10 ml of the unknown solution is added to five 50-ml graduated flasks, then increasing volumes of standard solution E (with a chromium concentration $C_E = 15.7$ mg/l) are added to flasks 2 to 5. Then, the flasks are filled to 50 ml with water.

Absorbance measurements are conducted at λ = 359.7 nm and gathered in the table below:

V_I (ml) unknown solution	V_E (ml) calibration solution	Absorbance in normal power mode of the lamp	Absorbance in high power mode of the lamp
10.0	0.0	0.276	0.075
10.0	10.0	0.367	
10.0	20.0	0.453	
10.0	30.0	0.542	
10.0	40.0	0.629	
q.s. 50 ml			

a. What information does the absorbance measurement in high power mode of the lamp provide?
b. Explain its principle.
c. What other technical resources are commonly used to obtain an equivalent result?
d. What is this interference effect called, and what are its origins?
e. Calculate the chromium content C_I in the sample?

SOLUTIONS

1. The sample introduced in the graphite oven contains $m_x = 1.22 \times 10^{-4}$ g of lead, i.e. 1.22%.

2. **a.** This is the spectral band that is selected by the exit slit and which reaches the detector.
b. The raw formula of EDTA is $C_{10}H_{16}N_2O_8$, that of the mixed salt of zinc and sodium $C_{10}H_{12}N_2O_8Na_2Zn$. In the anhydrous state, the mass of this salt is 399.6 g/mol and the hydrated salt weighs 471.6 g/mol. Therefore, this is a hydrate with four water molecules.
c. $C = 1.14$ mg/l.

3. **a.** The absorbance measurement in high power mode is used to obtain the background noise value.
b. c) d) See text.
e. After removal of the background noise (0.075) for each of the measured absorbances (A), the linear regression calculation leads to the following expression:

$$A = 8.81 \times 10^{-3} V_E + 0.202$$

If A = 0, then $V_E = 22.93$ ml. In light of the concentration of the calibration solution:

$$C_I = 22.93 \times 15.7 \times 10^{-3} = 0.360 \text{ mg/ml or } 360 \text{ ppm}$$

Chapter 14

Atomic Emission Spectroscopy

Atomic (or optical) emission spectroscopy concerns the study of the radiation emitted by atoms subjected to high temperatures, such as those found in plasma. The complexity of the resulting emission spectra requires spectrometers whose dispersive optics are sufficiently high in performance. Complementary to atomic absorption and less complex than mass spectrometry applied to elements, emission spectroscopy is widely used in elemental analysis, though it is not advisable for the analysis of the lighter elements. The high dynamic range

of current instruments enables the analysis in a given sample of several elements with very different concentrations. Less recent, flame photometry involves simpler instruments, but it can be used only to measure a few elements.

Objectives

Present the objective of the method and describe its principle
Compare analyte dissociation processes
Describe the excitation modes of atoms
Review the role of the dispersive optics system
Distinguish between the types of instruments used
Give an overview of performance levels
List a few applications of this analytical method
Compare atomic emission and flame photometry

14.1 OPTICAL EMISSION SPECTROSCOPY (OES)

When a chemical element in the atomic state is exposed to appropriate conditions of excitation, it emits a characteristic radiation. This observation is the basis for a general form of elemental analysis, both qualitative and quantitative. Known as *optical emission spectroscopy* (OES), the *optical emission* of atoms is very complex and leads to spectra with thousands of spectral lines accompanied by a continuous background.

Unlike atomic absorption, atomic emission is used to simultaneously analyse several elements found in the studied sample. The corresponding spectrometers differ based on the excitation state of atoms (plasmas, sparks or light discharge), though all require great optical quality. Their purchase price and their maintenance cost, which are higher than for atomic absorption spectrometers, mean that they are reserved for laboratories with many samples to process.

14.2 PRINCIPLE OF ATOMIC EMISSION ANALYSIS

An atomic emission spectrometer includes (Figure 14.1):

- A device to introduce and dissociate the sample in the atom or ion state.
- An optical system to separate the various optical emissions.
- A detection and analysis system for the emitted radiation.
- A computer system to serve as interface with the user.

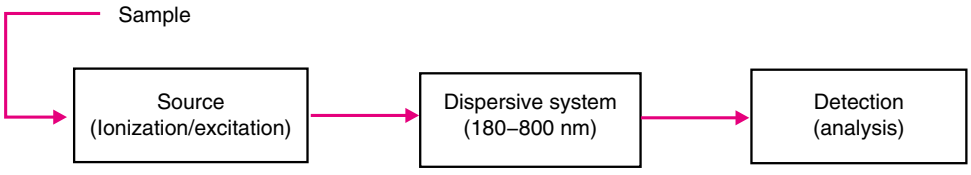


Figure 14.1 Basic design of an atomic emission spectrometer.

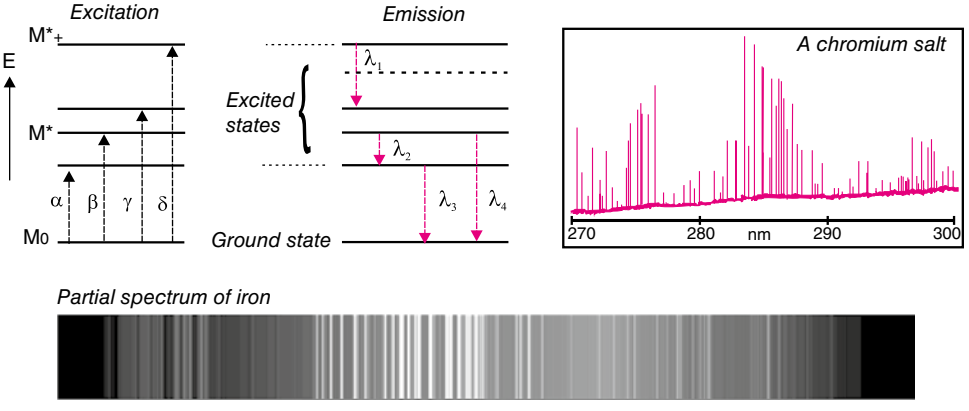


Figure 14.2 Energy transitions in atomic emission. For every atom, there are hundreds of possibilities by which it may return to the ground state, each of these having their own probability. Even the simplest sample contains an enormous number of such atoms, and its emission spectrum will have hundreds of transitions. An example (right) is a partial spectrum of an aqueous solution containing a few ppm of a chromium salt.

As in FES, the light source of the spectrometer is none other than the sample whose atoms have all been simultaneously excited. Qualitative analyses of the composition of even unknown samples can therefore be conducted. This marks an important difference from atomic absorption, in which measurements can only be made on elements for which the instrument has been customized (by selection of the hollow cathode lamp). From a single run, a multi-element analysis can be obtained in just a few minutes, in strict contrast to AAS. However, only elements for which a calibration has been carefully undertaken can really be measured.

Each atom, once in the excited state, can lose its excess energy by emission of one or more photons whose energies can have hundreds of distinct values and characteristics (Figure 14.2). For a sample, the dispersive system thus reveals a multitude of radiations of different wavelengths and intensities. For a given sample, the matrix (all elements present in the solution) will also emit; therefore, the identification and quantification of a particular element that is present only at the ultra-trace level will be obtainable

on the condition that measurements use several wavelengths to construct the sample fingerprint.

Atomic emission analysis requires high-performance instruments capable of locating weak radiation in close proximity to others that are much more intense. This is why preference is given to spectrometers having a broad *dynamic range*.

FES instruments, used for quantifying just a few elements, are much simpler in design (see Section 14.8). For example, when sodium is analysed with a flame photometer whose flame reaches 2,000°C, the sodium atoms are practically the only ones emitting radiation. To measure the emitted light, a simple optical filter placed between the flame and the detector, in the absence of a monochromator, isolates a rather broad spectral band including the radiation emitted by the element.

14.3 DISSOCIATION OF THE SAMPLE INTO ATOMS OR IONS

To convert a sample into free excited atoms or ions, various processes are employed based upon gas plasmas, sparks, lasers or glow discharges.

14.3.1 Gas Plasma Excitation

The most common instruments for atomic emission contain a plasma torch (*inductively coupled plasma* – ICP) that requires a high-frequency voltage or, more rarely, a microwave-generating magnetron.

A plasma is a special medium: considered to be the fourth state of matter, it is formed by isolated atoms in equilibrium between their neutral form and their ionized from (1–2%) and by the associated electrons ($10^{18}/cm^3$), which ensure the overall neutrality of the medium.

To initiate the plasma, a flow of argon, weakly ionized by sparks (Tesla discharge), passes through an open quartz tube situated on the axis of several turns of a water-cooled copper tubing (Figure 14.3). This conductive coil is connected to a radiofrequency generator (typically 27 or 40 MHz) with a power of around 1 kW. The variable magnetic field that is created confines the ions and the electrons to an annular path (appearance of an Eddy current). The medium becomes more and more conductive and behaves like a secondary coil of a short-circuited transformer. The temperature increases considerably by the Joule effect and becomes a plasma whose temperature may exceed 8,000 K at certain points. A second, nonionized argon flow, concentric to the previous one, maintains the plasma isolated from the wall, while cooling the torch's central tube.

Argon is essential to generate plasma. However, in the past few years, new plasma systems supplied by nitrogen have appeared. In this case, the plasma is initiated by a small quantity of argon and maintained by a supply of nitrogen (obtained by air extraction)

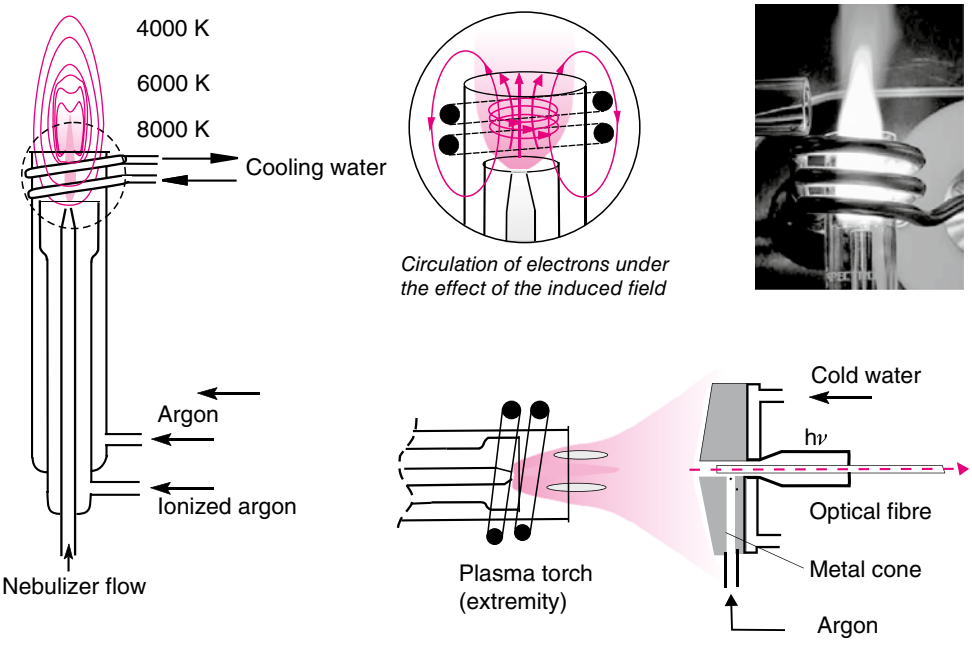


Figure 14.3 Plasma torch obtained by inductive coupling. Plasma torch represented in vertical position (radial viewing) and in horizontal position (axial viewing). The torch is supplied by a radiofrequency current (between 27 and 50 MHz) that induces circulation of electrons in the inert gas. The torch may consume 10–15 l/min of argon, which serves simultaneously as the ionization gas, nebulization gas, and cooling gas (to prevent the torch from melting!). The temperature fluctuates between 2,000 and 9,000 K.

under microwave induction. The use of this easy-to-obtain gas presents the advantage of reducing the cost of operation. However, due to the low temperature of the nitrogen-sustained plasma as compared to argon, the lines studied are different; the excited population in the plasma is smaller, and thus the detection limit is less attractive. The application of this equipment, which enables simultaneous analysis of several elements, can compete with atomic absorption.

The torch also has a small-diameter tube used to introduce the sample. The latter comes as an aerosol mixed with the carrier gas, with a constant flow of a few mg/min. For that purpose, a dual-inlet nebulizer is used: one inlet for the sample in solution, the other for a gas used to break up the liquid into fine droplets and thus generate the aerosol. These two inlets may be concentric, cross-path or parallel path (Figure 14.4). The fine mist of droplets, not exceeding a few micrometres in diameter, passes through a *spray chamber* before arriving in the torch.

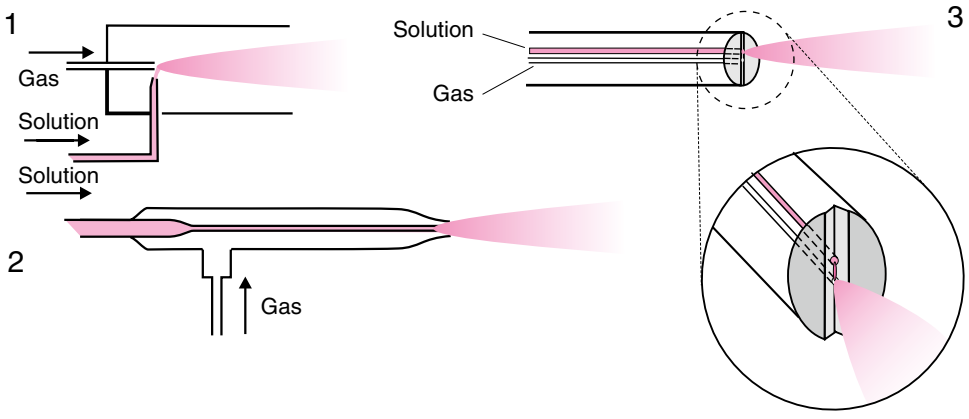


Figure 14.4 Nebulizers. Models: cross-path (1), concentric (2) and parallel path (3). In contrast to models 1 and 2, model 3 is not subject to particulate obstruction by solid impurities in the solution. The solution leaves through a large capillary that drains by gravity along a V-groove design to the outlet where the aerosol is produced. The nebulization chamber at the outlet of the nebulizer is used as a drain to eliminate the larger droplets by gravity.

The position chosen for measurement of light emission in the plasma (either *radial* or *axial*) depends on whether an ionic or atomic spectral line is selected for the assay.

The very high temperature in the torch lets a sizeable portion of atoms pass into the ion state. That is why this mass spectrometry instrument is used for elemental analysis, with the torch in horizontal position (see Chapter 16, Section 16.4).

14.3.2 Excitation by Sparks or Laser

Besides plasmas, which occupy the front rank of atomization and thermal excitation devices, other processes can be used. They involve either electric arcs or sparks for conductive samples (Figure 14.5) or pulsed lasers, which also causes sparks (also plasma) on the surface of the material in question (Figures 14.6 and 14.12). These are called *ablation methods* and much progress has been made with them. They can be adapted to different optical assemblies in current apparatuses (Figure 14.6) and are used in industrial analysis (steel plants, cement works).

14.3.3 Excitation by Glow Discharge (GD)

If the sample is in solid form and is, if possible, an electrical conductor, it can form the cathode of a kind of spectral lamp whose operating principle is identical to that described for a hollow cathode lamp (see Chapter 13, Section 13.4.1 and Figure 14.5). The atoms removed from the surface of the sample are excited by the plasma. This

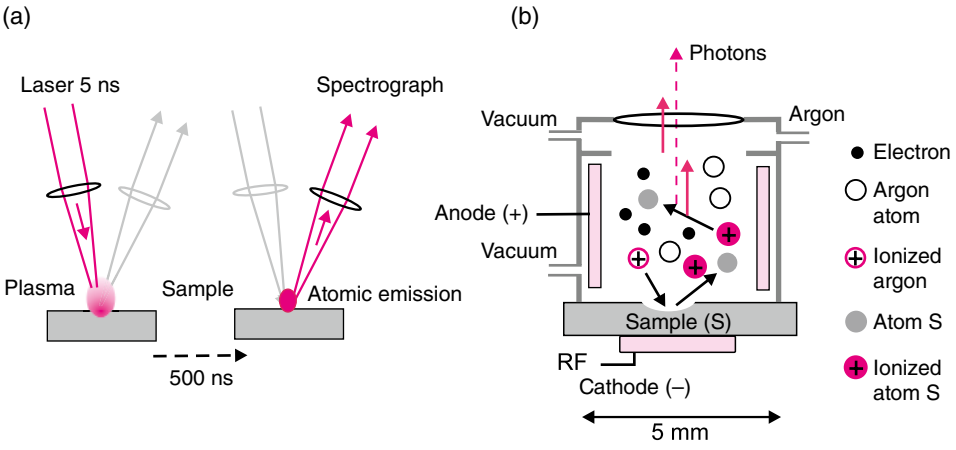


Figure 14.5 Laser and glow discharge excitation instruments. (a) By *laser* (Laser-Induced Breakdown Spectroscopy or LIBS). A laser pulse of a few nanoseconds generates plasma in which atoms of the bombarded sample are ionized. After a short period (e.g. 500 ns), the light emitted by the cooled plasma is analysed by the spectrograph part of the instrument. (b) By *glow discharge*. The sample, originally placed in a small cylindrical cavity with low pressure, is bombarded by argon ions (generated by electron ionization). Atoms detached from the surface of the sample are ionized or excited by impacts of neutral or charged species to form 'cold' plasma. The return of the atoms to their ground state is the source of a photon emission characterizing the sample (and argon). This instrument is the kind used by Horiba.

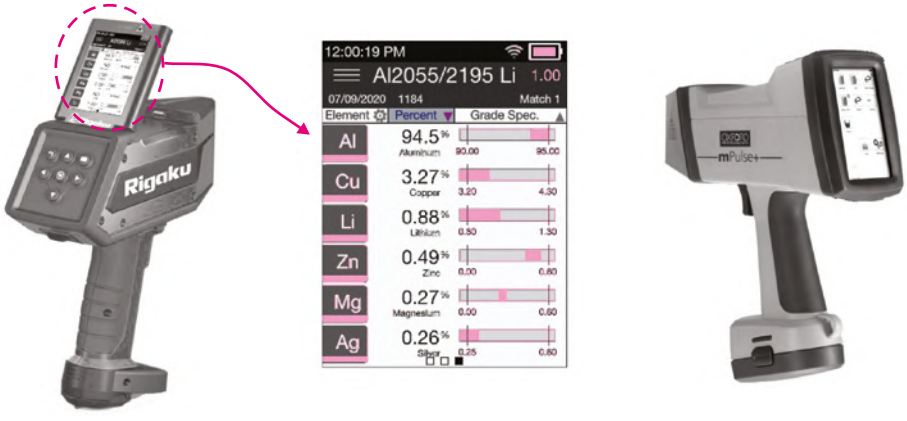


Figure 14.6 Atomic spectrometry using plasma induced by laser. Two models of plasma formation by surface ablation. Left, KT-1005 model from Rigaku with sample results displayed. Right, field spectrometer using a pulsed laser (2.4 GHz) focussed on a microscopic surface of the sample (LIBS technique). (Source: Models reproduced with the authorization of Oxford Instruments and Rigaku Analytical Devices.)

technique is mostly used for surface analyses. It has the advantage of yielding spectra whose emission lines are narrow, since atomization takes place at lower temperatures than with other techniques.

14.4 DISPERSIVE SYSTEMS AND SPECTRAL LINES

The active part of the optical set-up begins with the *entrance slit* onto which the light produced by the sample (the light source) is focused. The width of this slit can be adjusted by a mechanical system and cannot be less than a few micrometres because a minimum amount of light is required to enter. However, its length can reach several centimetres.

The source is thus transformed into a *luminous object* of linear profile and the optical system of the instrument gives as many defined images – the lines – as there are radiations of different wavelengths in the source. Each line corresponds to an almost monochromatic image of the entrance slit of the spectrograph. A single element can generate more than 2,000 lines.

> The lines are distributed in the focal plane as a succession of parallel and very narrow light segments. The width of these lines depends on several independent factors.
>
> Firstly, there are technical reasons characteristic of the instrument, collectively known as 'instrumental parameters': width of the entrance slit, quality of the optics, focal distance, diffraction phenomena through narrow orifices.
>
> However, there are also causes due to quantum mechanics, which ensure that the spectral transitions have a natural width. The radiations emitted by the atoms are not quite monochromatic, in particular with the plasmas, a medium in which the collision frequency is high.
>
> Three effects contribute to the broadening of lines: uncertainty related to the lifespan of the excited states (Heisenberg principle), the Doppler width due to the displacement of the atom with respect to the detector, and the Lorentz width due to the pressure, which increases the rate of collisions between atoms (Figure 14.7). Together, these three effects explain why the width of the lines at 6,000 K can reach several picometres.

To locate the specific radiations of an element to be quantified (the different optical transitions), an optical unit of very high quality is required. These apparatuses can be divided into *spectrographs* and *spectrometers* (see Chapter 9, Section 9.7 and also Figures 9.11 and 14.8). As pointed out above, the source illuminates a slit which becomes a luminous object, which will be analysed by a dispersive system containing a grating (planar, concave or *echelle*).

- In a spectrograph, the entrance slit gives as many images, offset one from each another, as there are different radiations in the source. With a detector composed of a CCD sensor (formerly a photographic plate), a simultaneous recording is obtained.

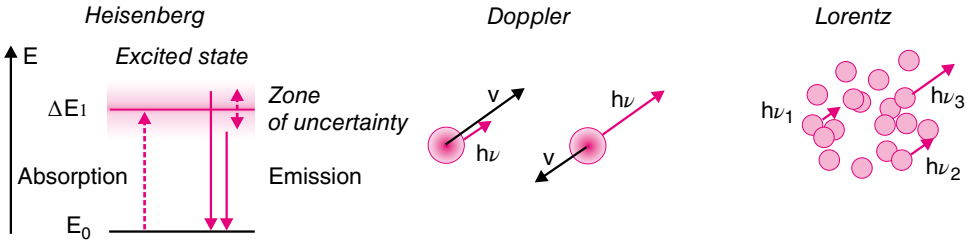


Figure 14.7 Three fundamental causes of atomic line broadening. Representation of a transition between a stable ground level, whose lifespan is infinite, and a very short excited level (Heisenberg); impact of speed (Doppler); impact of pressure (Lorentz).

- In a spectrometer, the detector is fixed and therefore receives only the selected radiation by means of a monochromator (made by assembling an optical grating and a motor). This system is consequently used to measure the intensity of a single line at a time. These devices, used for quantitative measurements, do not provide spectra.
- The monochromator may be replaced by a polychromator, which, in the image focal plane of the entrance slit, has a plate with several outlet windows (or slits), whose positions have been chosen as a function of the elements to be detected. Behind each of these is a detector (Figure 14.8) that collects the selected radiation. The optics do not have any moving parts.

In the beginning, the qualitative study of these spectra was done by visual comparison using spectrographs. Nowadays, these have been replaced by spectrometers capable of resolving line interferences and problems linked to matrix effects. Each element, whether in neutral or ionized form, is responsible for a set of lines of highly variable intensities. The instrument is therefore programmed to both identify and quantify the elements of interest in the sample.

14.5 SIMULTANEOUS AND SEQUENTIAL INSTRUMENTS

Two large families of instruments can be distinguished: one is used to assay several elements simultaneously, while the other measures them sequentially. Present trends favour systems of the first category with fixed optics to undertake predefined applications. The disadvantage of these turnkey solutions is their inability to make an exhaustive analysis of an unknown sample.

Measurements of optical emissions with a photomultiplier tube are possible over a very extended dynamic range. Elements can therefore be quantified at very different concentrations, in other words at very different light intensities of the lines, from a single solution.

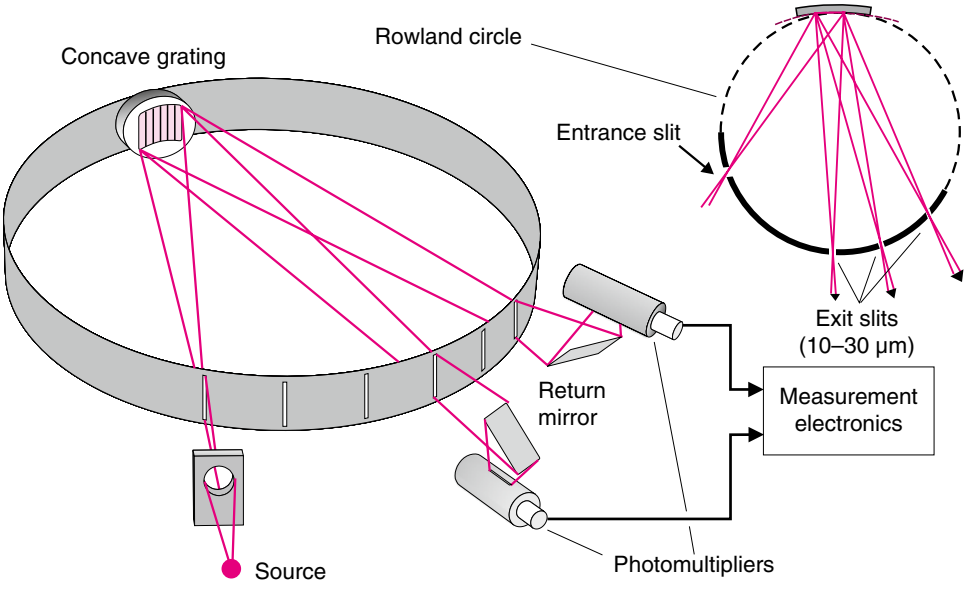


Figure 14.8 Optical diagram of an instrument with a concave grating and a polychromator with long focal distance. Reflecting mirrors can be used when the lines to be detected are too close to one another. Several polychromators are sometimes used together. To the right, the principle of the Rowland circle. The optical part of these instruments must be assembled on a very stable metal support.

14.5.1 Fixed Optic Apparatus Containing a Polychromator

For each selected element, the intensities of one or several spectral lines are measured, which constitute the *analytical lines* or measurement channels. Each channel ends with its own dedicated photomultiplier tube. If the instrument contains a concave grating, the entrance and the exit slits must be situated on the *Rowland circle*, at a tangent to the grating and whose diameter is equal to the radius of the curvature of the grating (Figure 14.8). Technically, owing to a lack of space, it was not easy to install several detectors, so the preferred type of construction has become that described in the next section.

In a single analysis, close to 20 elements can be measured rapidly. Correction of background noise is also possible by the simultaneous determination of several lines belonging to the same element. The linearity of the measurements extends over a broader range than in atomic absorption.

14.5.2 Fixed Optic, *Echelle* Grating Apparatus

Another type of assembly involves the combination of a particular class of grating, called *echelle grating* with a prism focusing device. This leads to a double dispersion of the light, both horizontally due to the grating and vertically by the prism lens. This arrangement of high optical resolution, whose focal distance can attain several metres, is called an *order-separating device*. It enables the simultaneous observation of an extended spectral range, for example 200–800 nm (Figure 14.9).

With an echelle grating, the incident light arrives perpendicular to the short sides of the grooves, which are fewer in number than for a conventional grating (e.g. 100 per millimetre) (Figure 14.9). The observation of the diffracted light is made almost at the same angle, which maximizes the transmission of light (blazed grating). Calculation reveals that radiations of similar wavelengths are found to superimpose in the observation direction as a result of the large interval separating two successive grooves. This multiplexing phenomenon (superimposition of the different orders n of the grating – where n is around 100) is resolved by a prism that uses the second dimension of the focal plane to separate the different images of the entrance slit. The spectrum remains stationary with respect to the optical assembly.

At the end of the optical path, dispersion of the 'lines' is detected on a matrix of sensors whose response furnishes information quasi-simultaneously concerning thousands of spectral spots (Figure 14.10). This means that analyses are much faster. By a series of improvements, the dynamic range of these sensors has become equivalent to that of photomultipliers.

14.5.3 Wavelength-Scanning Instruments (Monochromator Type)

In contrast to the preceding instruments, the dispersive system is movable in order to focus the different wavelengths on the stationary exit slit. A motor allows the preprogrammed selection of the desired wavelengths and the sequential examination of the spectral lines. The light issuing from the source is analysed either by an Ebert assembly (see Chapter 9, Section 9.6), or by a Czerny–Turner dual monochromator in order to obtain very high spectral resolution (a few nm), comparable to the emission line width at this temperature, over a broad spectral region (160–850 nm). Several assemblies of this type can be placed in series to constitute dual or triple monochromators of very high performance without making the assembly too large.

These instruments are notable for their flexibility. A mercury vapour light source serves as a means of internally calibrating the wavelengths in order to rapidly determine the position of the grating with the greatest precision. The optical path passes through an enclosure kept under vacuum. Here again, the proximity of certain spectral lines necessitates a rigid assembly of great mechanical precision.

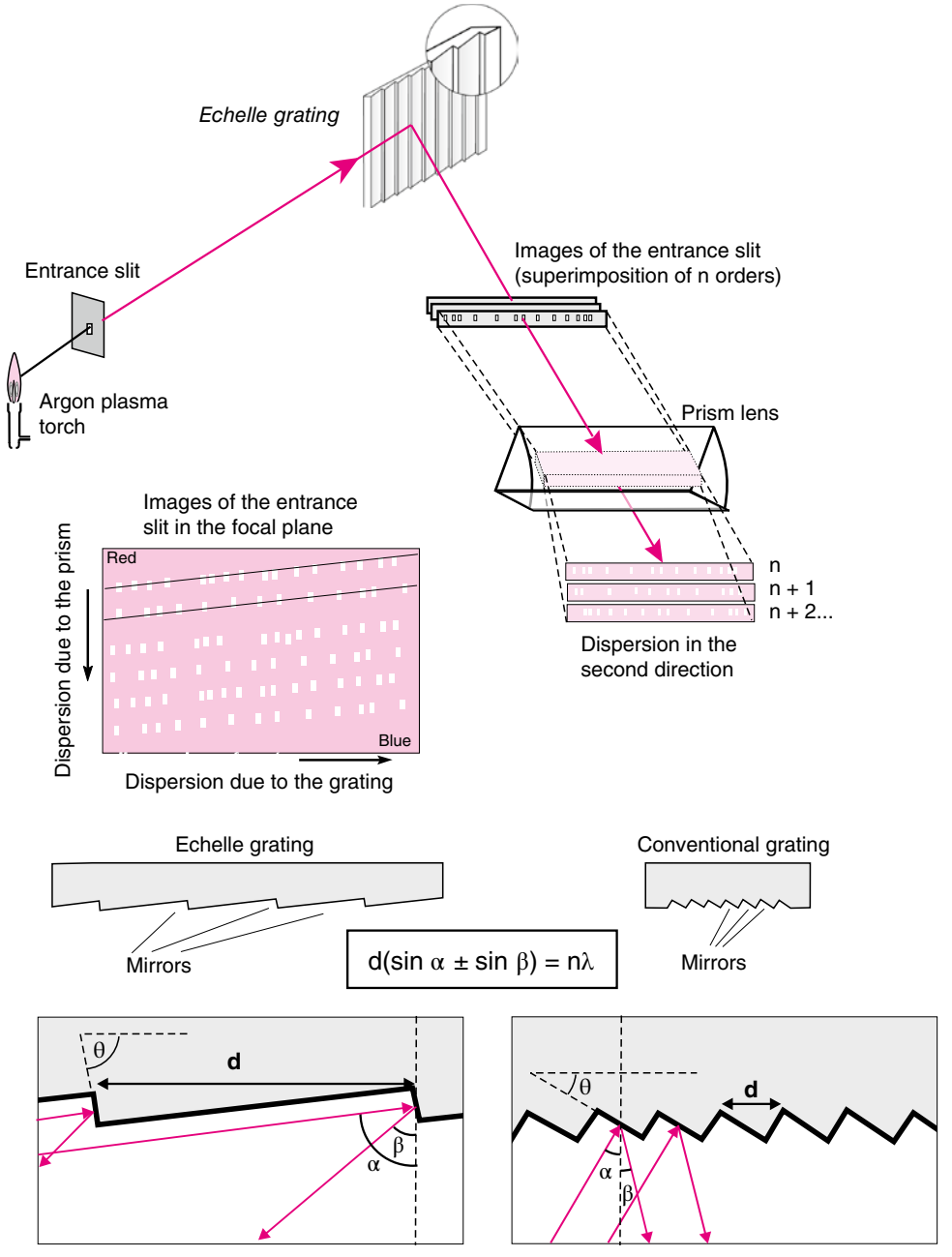


Figure 14.9 Principle of dispersion in the focal plane of an assembly combining an echelle grating and a prism. For clarity, the associated optics (collimating and focusing lenses) have not been represented. Comparison of an echelle grating with a conventional grating. The angle of the grating reflecting the most light is referred to as the blaze angle θ. The basic formula indicates the wavelengths diffracted by the grating at an angle of incidence α, with reference to a line perpendicular to the grating and at an observation angle β (the sign is positive (+) if the observation is from the same side as the incident beam with respect to the perpendicular, and the sign is negative (–) in the opposite case). The resolution can attain that of an apparatus which would elongate the spectrum over a length of 2 m in the focal plane.

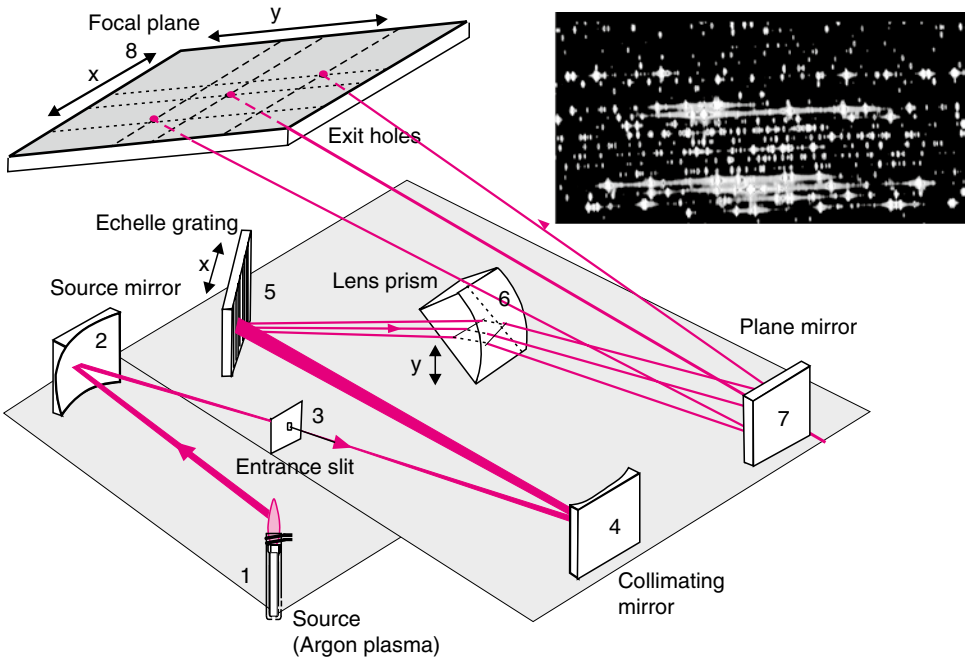


Figure 14.10 Optical diagram for a spectrophotometer with an echelle grating. For greater clarity, only the central section of the beam issuing from source 1 is represented (this beam should cover the whole of mirror 2). Echelle grating 5 separates the radiations arriving from the source in the horizontal plane (in the *x* direction). The prism then shifts the radiation in the vertical plane (in the *y* direction). The path of three different spectral lines is shown. The images of the entrance aperture 2 are in focal plane 8. To detect these radiations, small photomultiplier tubes (PMT) can be installed in specific places, or a matrix sensor with 10^5–10^6 photodiodes (CID or CCD) can be used to cover the entire spectral region from 190 to 800 nm. A full study of the sample is then possible (drawing executed according to a diagram of the 7000 PU model from Unicam). Top right, image of the distribution of light spots generated by a sample on a sensor matrix.

14.6 PERFORMANCE

Rapidity and detection limits, below 1 ppb for many elements, make atomic emission one of the best and most versatile methods of elemental analysis. With instruments containing one CCD-type detector, it is possible in a few seconds to record the whole spectrum in high resolution (e.g. more than 10,000 emission lines).

Some spectrometers more readily accept dominant matrices such as mud or soils, in which the elements silicon (Si), iron (Fe), and aluminium (Al) are in very high concentrations. The device should be chosen as a function of the analyses being performed.

14.6.1 Sensitivity Thresholds and Detection Limits of the Spectrometer

The threshold of sensitivity varies according to the instrument and the element being considered. Numerous comparative tables on detection limit values exist and are continually being updated as a result of the progress made in instrumentation. The sensitivity threshold corresponds to the minimum concentration of the element in solution that will yield an analytical signal, whose magnitude is equal to three times the standard deviation calculated for an analytical blank. This classic definition leads to rather optimistic values that can vary greatly depending on the element. The detection limit of the instrument represents the concentration of an element that allows detection with a confidence level of 95%.

> The detection limit for a method is of more importance than the above detection limit. Generally, measurements are done in a concentration range corresponding to 50 times the detection limit of the instrument. When measuring ultra-traces, it is obvious that sample preparation requires reagents and water of very high purity.

14.6.2 Resolving Power

Spectral resolution can be determined directly through an experimental measurement from a recorded spectrum. This corresponds to the width $\Delta\lambda$ of an emission peak, measured at FWHM using the units of the spectrum. Its value depends upon both the dispersive system and the wavelength.

The *resolving power* of a spectrometer is its ability to separate adjacent spectral lines. This theoretical concept is given by the dimensionless parameter R such that:

$$R = \lambda / \Delta\lambda \tag{14.1}$$

Atomic emission apparatuses have resolving powers on the order of 30,000 up to 100,000. If the exit slit is too wide (bandwidth arriving on the photomultiplier tube), it reduces the resolving power of the spectrometer.

14.6.3 Linear Dispersion and Bandwidth

The optical part of atomic emission spectrometers is characterized by two parameters among others: *linear dispersion* (or its inverse, the *reciprocal linear dispersion*) and *bandwidth*. The reciprocal linear dispersion, the most commonly used value, is expressed in nm/mm (Figure 14.11). It represents the wavelength shift which, in the focal plane, corresponds to an interval of 1 mm. It depends on several parameters, such as the focal distance of the wavelength selector. In general, the better the dispersion is, the greater the physical separation distance between two given wavelengths is.

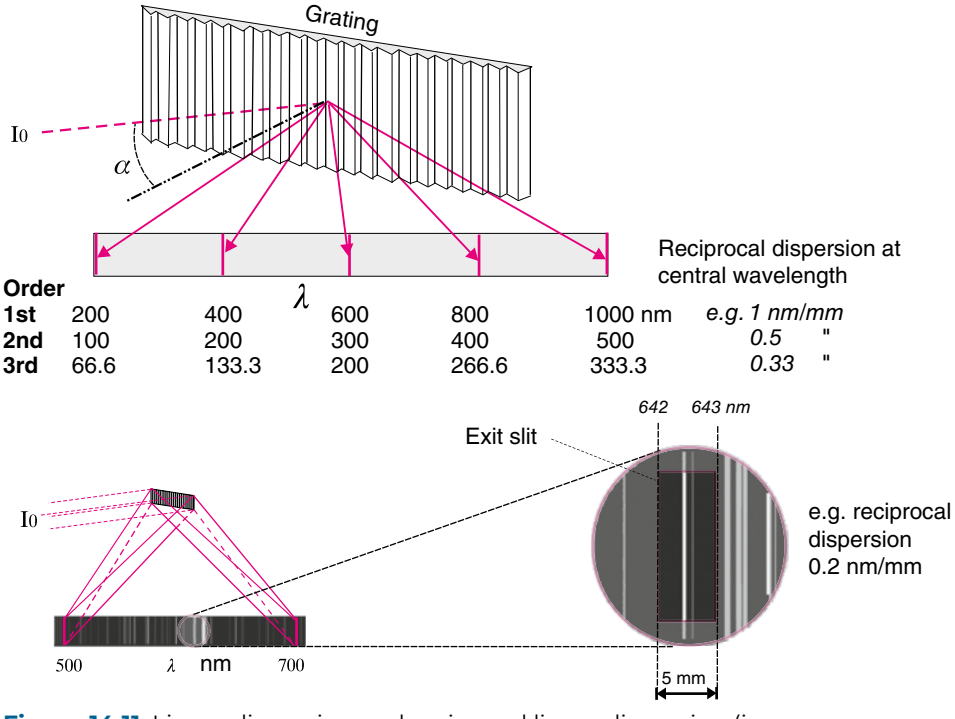


Figure 14.11 Linear dispersion and reciprocal linear dispersion (inverse dispersion). An atomic emission spectrometer has one or more engraved optical gratings, dispersing the various radiations in a focal plane. Reciprocal dispersion can be determined experimentally.

The *bandwidth*, which must not be confused with the width of the exit slit of the monochromator, is the small wavelength interval seen by the detector. It is expressed in picometres and is centred on a nominal wavelength value.

Too narrow a bandwidth will absorb a part of the radiation but will improve the measurement selectivity.

Thus, if the exit slit is 20 μm and if the reciprocal dispersion is 0.2 nm/mm, the bandwidth will be equal to $0.2 \times 0.02 = 4 \times 10^{-3}$ nm, i.e. 4 pm.

14.7 APPLICATIONS OF ATOMIC EMISSION SPECTROMETRY

The large number of elements measurable by atomic emission renders this method of analysis indispensable, as it permits the measurement of elements chosen in advance or the identification of elements present in any sample.

The greatest advantage of this method is being able to conduct analyses on complex matrices with minimal preparation. Moreover, the linearity of response is ensured over a very large range of concentrations, which, from a single solution, enables switching from the measurement of one element to another, even if their concentrations are very different.

Beyond general industrial analytical applications – for metals, alloys, and ores, which are directly analysable without first putting them into solution – the environmental field is without doubt where AES has become essential. Applications include analyses of crop or livestock production (meat, milk), water, air (dust and ash emissions from incinerators), or soils, in which elements are present over a wide range of concentrations. This methodology also has a variety of applications in the forensic sciences or in clinical medicine (analysis of tissues or biological fluids).

GC-AED. This abbreviation refers to the association of a detector based on atomic emission and a gas chromatograph (*Gas Chromatography – Atomic Emission Detector*). The high temperatures generated by gas plasmas can be employed to break down the compounds eluting from the column into their corresponding elements. By choosing one or more specific lines of an element, a selective chromatogram can be obtained of all the eluted compounds that contain this element. Unfortunately, this method of identification of organic molecules implies sample destruction, leaving only information concerning its elemental composition: during this 'chemical butchering', structural information is lost.

14.8 FLAME PHOTOMETRY

Some elements (alkali metals, alkaline earth metals) lead to spectra whose resonance lines (absorption/emission) correspond to energy gaps under 3 eV, which helps to quantify these from their emission spectrum at standard flame temperatures used in atomic absorption spectroscopy.

The technique based on this phenomenon is called *flame photometry*, or *flame emission*. In a flame with a temperature of 2,500°C, a dozen elements at most emit enough traceable radiation without too much interference in order to measure them.

For a population of n excited atoms, the light emission I_e depends on the number dn which return to the ground state during the time interval dt ($dn/dt = k \cdot n$). Since n is proportional to the concentration of the element in the hot part of the flame, the emitted light intensity I_e, which varies by dn/dt, is itself proportional to the concentration:

$$I_e = K \cdot C \tag{14.2}$$

Equation (14.2) is valid only for low concentrations and in the absence of self-absorption. For each photometer model and for each element, the detection range is indicated, which is often much greater than the linear response range.

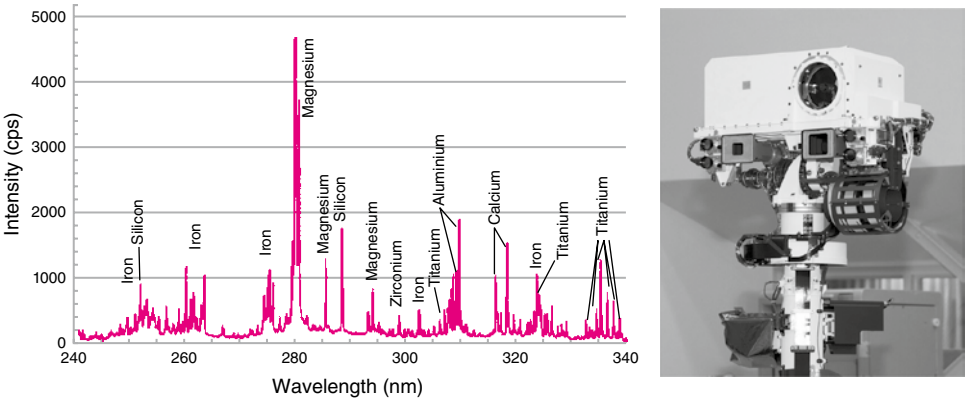
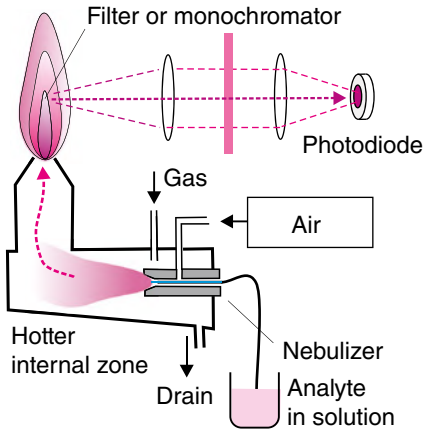


Figure 14.12 Analysis of a sample of Mars soil by atomic emission after laser excitation (LIBS technique). One of the many spectra of Mars soil conducted from data transmitted by the ChemCam, a module on board the Curiosity vehicle (2012). The main peaks between 240 and 340 nm have been identified. (Source: NASA, https://commons.wikimedia.org/wiki/File:20110406_PIA13809_D2011_0404_D036_cropped-full.jpg#/media/File:20110406_PIA13809_D2011_0404_D036_cropped-full.jpg)

Throughout the measurement, the temperature must remain stable, since a variation of 10°C around 2,000°C for sodium, for example, is translated into a 3% variation in the number of excited atoms, therefore 3% of the detection signal (see Chapter 13, Sections 13.1 and 13.2). Flame photometers are simple instruments, with a standard optical set-up (Figure 14.13), whose purchase price is well under that of atomic absorption instruments (which may also be used for this purpose with the source turned off). They are designed in order to quantify five or six elements only, the main ones being sodium, potassium, and calcium. Much older than plasma instruments, they remain very common for routine analyses, especially in the food industry, biochemical laboratories, cement plants, etc. To improve precision, some higher-performance models have two measurement cells in order to compare the emitted light intensity with that of a reference solution.

For the analysis of alkali metals, the hotter the flame is, the more it causes a notable ionization of atoms, thus reducing the number of nonionized atoms available for emission. To resolve this, an *ionization buffer* is added, with a high concentration of an element easier to analyse than the analyte. The increased number of electrons in the flame creates a sort of electron pressure, which decreases the ionization rate of the analyte. For example, to measure sodium, a high concentration of a lithium or caesium salt will be added.

In GC, there is a very sensitive detector for elemental sulfur, based on flame photometry. Therefore, when an organic sulfur compound is pyrolysed in the detector burner, sulfur in the elemental state forms in contact with a reductive air/hydrogen flame, accompanied by a specific emission at 394 nm.



Element	Emission λ (nm)	Colour
Ba	554	yellow-green
Ca	622*	orange
Li	670	red
K	766	purple
Na	589	yellow
Sr	460**	red
Cs	852	(invisible)***

* Formed species: CaO
** Filter usually selected
*** To the eye, but not for the detector

Figure 14.13 Flame photometry. Basic diagram of a flame photometer and list of elements quantifiable by flame photometry (calcium is assayed by its emission at 622 nm, due to calcium oxide, CaO). The dynamic range of these measurements is rather low (two decades).

KEY POINTS OF THE CHAPTER

1. *Atomic emission spectroscopy* (AES) is used to characterize and quantify nearly all elements from their emission spectra when they are brought to the atomic vapour state (atoms and ions) by a sufficiently high-energy source (plasma, laser, sparks). When the stimulation is caused by photons, the technique is named *atomic fluorescence spectroscopy*.
2. Unlike atomic absorption, the instruments do not have a light source: the elements themselves emit the light. The light intensity can then be measured with a spectrometer (these are line spectra).
3. The energy sources used are gas plasmas, sparks, lasers or glow discharges. Photons emitted by atoms during their relaxation to the ground state lead to many precise wavelengths, which only a very dispersive spectrometer can differentiate.
4. Inductive coupling plasma (ICP) torches, capable of reaching 8,000K, are adapted to samples in aqueous solution, which are vaporized in the plasma. Sparks or lasers are adapted to solid samples, for which the quantification of analytes is less precise.
5. ICP-AES is reserved for samples that can be put into aqueous solution and injected into the plasma. The method is known for its great sensitivity (a few ng/ml) and its dynamic linear range over 7–8 decades, which enables the quantification of major and minor elements with a single solution.

6. AES by laser, spark or glow discharge is reserved for solid, conductive samples. The method, which is difficult to adjust and thus long remained qualitative, is considered moderately precise. The operation is similar to that of hollow cathode lamps. It is especially used in the metal industry.
7. To study emission spectra, which have thousands of lines, commercial instruments can be divided into two categories: *sequential*, to analyse several elements with wavelength scanning, or *simultaneous*, with optics including a polychromator and a cross-dispersion device, combined either with multiple detectors or with a sensor matrix.
8. Sequential spectrometers must be able to delimit a very narrow spectral window surrounding the specific line in question. Resolving power (R), linear dispersion, and bandwidth are parameters that define the optical quality of instruments.

$$R = \lambda / \Delta\lambda$$

9. Measurements by flame emission (or photometry), conducted with simpler instruments, are limited to a few elements (alkali metals and alkaline earth metals) for which a more significant fraction of atoms pass into an excited state. The intensity of the light emitted, which is easy to measure, is a function of the element's concentration: $I_e = K \cdot C$.

PROBLEMS

1. When an atom emits radiation, the frequency of the corresponding photon is related to the energy of the levels between which the transition occurs. According to the Heisenberg principle, if the excited state has a lifetime of Δt, then this results in a minimum uncertainty for the energy ΔE, such that $\Delta E \Delta t \geq h/2\pi$. From this equation, calculate the natural width of the transition observed at 589 nm for sodium if $\Delta t = 10^{-9}$ s.
2. The resonance level of the sodium atom is sometimes said to be 16,960 cm^{-1} at ground level. From this value, calculate the corresponding wavelength and the transition energy associated with it.
3. An echelle grating with 100 grooves per mm receives incident light at an angle of 75° with respect to a line perpendicular to the grating. The observation angle varies from 60 to 65°.
   a. In which order *n* must this grating work to observe radiation of 0.35 μm in wavelength?
   b. Calculate the spectral ranges for the value of order *n* found above as well as for (n – 1) and (n + 1).
   c. Determine whether or not the spectral ranges overlap for these three orders.

4. To determine copper traces in sea sediment using the MacLaren method, an atomic emission instrument equipped with a plasma torch is used. Measurements are made at the following wavelengths: 324.754 nm (copper emission), 324.719 nm and 324.789 nm (no emission: correction of background noise), and 324.739 (iron emission) if we presume the presence of iron in the sample.
Three spectra are thus recorded, whose results are shown in the table below (values in μA):

	324.719 nm	324.739 nm	324.754 nm	324.789 nm
Cu calibration	23.1		63.71	8.1
Fe calibration		8.75×10^5	10.5	
Sample	27.5	2.94×10^4	27.49	9.2

The copper calibration was conducted with a solution containing 50 ppm of copper but not containing iron.

   a. Calculate the correction to be made due to the background noise.
   b. Calculate the correction factor to be used to eliminate the overlap effect between iron and copper.
   c. Calculate the quantity of copper (in ppm) in the sample.

5. Five standard solutions were prepared for measuring the lead concentration in two solutions, A and B. The two solutions A and B contain the same concentration of magnesium, which is used as an internal standard. The following data were obtained:

Conc. (mg/l)	Emission signal (arbit. units)	Mg signal
0.10	13.86	11.88
0.20	23.49	11.76
0.30	33.81	12.24
0.40	44.50	12.00
0.50	53.63	12.12
sol A	**15.50**	**11.80**
sol B	**42.60**	**12.40**

From the data, calculate the lead concentration (mg/l) in the two sample solutions, A and B.

6. The potassium in blood serum is analysed by flame emission using the standard addition method. For that purpose, two identical solutions are prepared: 0.5 ml of serum is sampled, and distilled water is added to fill up to 5 ml. In one of these solutions, 10 μl of 0.2 M KCl is added. The values read on the instrument are respectively 32.1 and 58.6. What is the potassium concentration of the serum?

7. Why is a potassium salt, such as KCl, often added when sodium is being measured by flame emission spectrophotometry? Please remember that the first ionization potentials of sodium and potassium are, respectively, 496 and 419 kJ/mol.

SOLUTIONS

1. $\Delta\lambda > 1.84 \times 10^{-13}$ m (1.84×10^{-4} nm).
2. This formula means that the energy increase of 16,960 cm^{-1} corresponds to the resonance line. This unit is used to measure energy ($E = h\nu = hc/\lambda$); λ = 589.62 nm: this is the doublet component called the resonance line (E = 2.102 eV). The second doublet component is not a resonance line.
3. **a.** By applying the formula $d(\sin\alpha + \sin\beta) = n\lambda$, we can calculate two values of n corresponding to two values of the observation angle β given in the problem, (where $\alpha = 65°$), i.e. n_1 = 52.34 and n_2 = 53.49. Therefore, n = 53.
   b. The same formula used to calculate the extreme wavelengths gives the following: for n = 53, λ is between 0.3457 and 0.3533 μm; for n = 52, λ is between 0.3523 and 0.3600 μm and for n = 54, λ is between 0.3393 and 0.3467 μm.
   c. There is an overlap of the ranges.
4. **a.** Correction of background noise = $\frac{A_{324.719} + A_{324.789}}{2} = 15.6$ for the standard and 18.35 for the sample.
   b. Correction due to iron = $\frac{10.5 \cdot 2.94 \cdot 10^4}{8.75 \cdot 10^5} = 0.353$.
   c. C = 9.13 ppm.
5. [Pb/Mg signal ratio] = 8.219[conc.] + 0.345, hence A = 0.118 mg/l and B = 0.376 mg/l.
6. Potassium concentration of the serum: 4.84×10^{-3} mol/l (or 4.84 mmol/l).
7. Potassium ionizes more easily than sodium, which decreases the ionization of sodium after its atomization.

Chapter 15

Nuclear Magnetic Resonance Spectroscopy

Nuclear magnetic resonance (NMR) is a spectroscopic method based on the study of the nuclei of certain elements. It has become irreplaceable in a variety of areas, such as chemistry, biology, the food industry or medical imaging (MRI). It is one of the most powerful methods to obtain structural information on organic or inorganic compounds. A succession of technological advances has led to a growing number of applications for it, whether analytical or not, in industry, ranging

from oil exploration to the identification of food products. This chapter is limited to a simplified study of NMR in solution in the field of organic compounds.

Objectives

Describe the basic theory of NMR
Interpret an NMR spectrum
Measure chemical shifts
Study spin coupling
Analyse the multiplicity of signals
Use a chemical shift table
Describe a few applications

15.1 GENERAL INTRODUCTION

Nuclear magnetic resonance refers to a property of matter that has given its name to powerful method of materials investigation, in particular to identify the structure of organic or inorganic compounds. *NMR spectrometers* are therefore often found in research laboratories, although there are routine applications based on this property, using simpler and more robust dedicated instruments (see Figure 15.31).

Organic chemists quickly understood the interest of NMR, which has accelerated its development and favoured the appearance of many technical improvements. In this chapter, the method will be illustrated using simple examples chosen specifically for

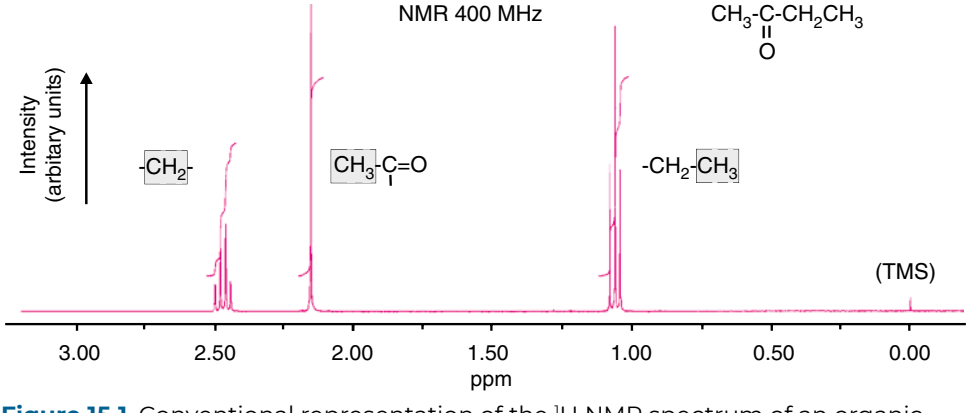


Figure 15.1 Conventional representation of the ^{1}H NMR spectrum of an organic compound. Spectrum of butanone [$CH_3(C{=}O)CH_2CH_3$], with the corresponding integration curve superimposed over the peaks, enabling an evaluation of the relative areas of the main groups of signals identified on the spectrum. The nature of the *y*-axis scale will be explained later on in the chapter.

organic chemistry. In short, this method gathers information concerning interactions between the nuclei of certain atoms present in the sample when they are subjected to an intense magnetic field.

The basic document, delivered by these instruments, is the *NMR spectrum*, a graph representing resonance signals whose position, appearance, and intensity constitute a set of information about the sample under examination, which is much easier to interpret if the compound is pure (Figure 15.1).

To produce these signals, a second magnetic field, around 10,000 times weaker than the first, is overlaid on the main field and is produced by a source of electromagnetic radiation in the radiofrequency region.

To understand the origin of these spectra, which are quite different from conventional optical spectra, nuclear spin, as defined in quantum mechanics, needs to be considered.

The spectrum in Figure 15.1 is an example of one-dimensional NMR (the intensity of the signals is not counted as a second dimension). More sophisticated NMR studies, in two, three, four or more dimensions, enable the position of all of the atoms in the molecule to be located more precisely.

15.2 SPIN/MAGNETIC FIELD INTERACTION FOR A NUCLEUS

All atomic nuclei – as well as all subatomic particles – are characterized by a number of intrinsic parameters, including *spin* $\vec{I}$. This vector quantity introduced by quantum mechanics is used to explain the behaviour of atoms in media where there is a preferential direction. A magnetic field creates just such an orientation for all atoms within this field. The spin of the nucleus $\vec{I}$ has the same dimensions (J · s) as the angular momentum $\vec{L}$ in classical mechanics. The norm of the spin varies from one nucleus to another because it is defined from the *spin quantum number I*, whose value can be zero or a positive multiple of ½ (in natural units h/2π).

An isolated nucleus with a nonzero spin number (e.g. $I = ½$ for the proton) behaves like a small magnet with a magnetic moment $\vec{\mu}$ (J/T) such that:

$$\vec{\mu} = \gamma \cdot \vec{I} \tag{15.1}$$

This *nuclear magnetic moment* $\vec{\mu}$ is represented by a vector colinear to $\vec{I}$, with the same or the opposite direction, according to the sign of γ, which is the *gyromagnetic ratio*.

If $\vec{I}$ is quantified, $\vec{\mu}$ can also be quantified. It can only have values of $2I + 1$, given (in $h/2\pi$ units) in general by the following terms, between $-I$ and $+I$:

$$m = -I, -I+1, \ldots, I-1, I$$

If a nucleus with a nonzero spin number is subjected to a magnetic field $\vec{B}_0$, its potential energy will be changed due to the interaction between $\vec{B}_0$ and $\vec{\mu}$, but it can only have values of $2I + 1$.

If the nucleus is imagined as a microscopic magnet carrying an axis on which $\vec{I}$ and $\vec{\mu}$ are aligned and plunged in the field $\vec{B}_0$, the spin vector $\vec{I}$ of the nucleus is oriented by making an angle θ with $\vec{B}_0$ (Figure 15.2). $\vec{\mu}_z$ represents the projection of $\vec{\mu}$ on the O*z*-axis, which is oriented in the direction of $\vec{B}_0$, therefore:

$$E = -\vec{\mu} \cdot \vec{B}_0 \text{ or } E = -\mu \cos(\theta) B_0 \text{ or, finally } E = -\mu_z \cdot B_0 \qquad (15.2)$$

The quantification of μ_z is the consequence of values permitted for the projection *m* of the spin vector on Oz (*m* is the magnetic spin number).

By combined Eqs. (15.1) and (15.2), the $2I + 1$ allowed energy values are given by the following general equation:

$$E = -\gamma \cdot m \cdot B_0 \qquad (15.3)$$

Therefore, for the nucleus of the hydrogen atom (the proton), where $I = ½$, the two possible values of *E* (in joules) correspond to $m = +½$ and $m = -½$. According to usage, they are referred to as α and β as follows:

$$E_1(\text{or } E_\alpha) = -\gamma \frac{1}{2} \frac{h}{2\pi} B_0 \quad \text{and} \quad E_2(\text{or } E_\beta) = +\gamma \frac{1}{2} \frac{h}{2\pi} B_0$$

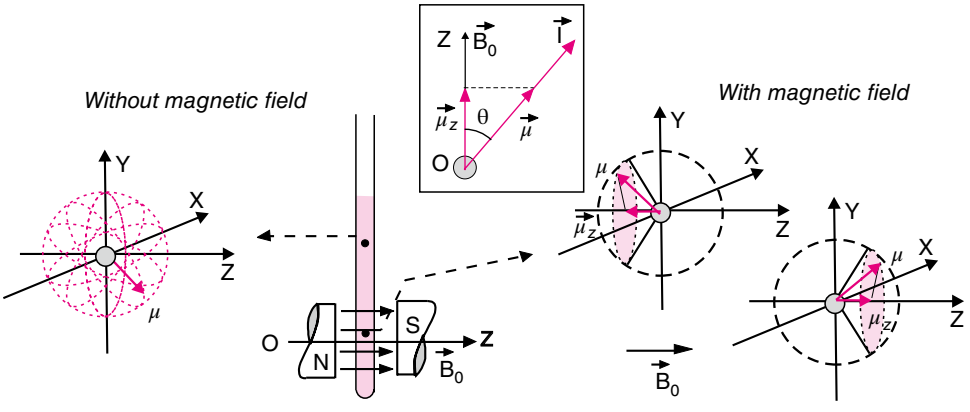


Figure 15.2 Effect of a magnetic field upon a nucleus of spin number ½ for an atom in a compound in solution. If the nucleus is in the upper part of the sample tube, i.e. not subjected to the external magnetic field, $\vec{\mu}$ has no preferential orientation. However, in the central section immersed in the field, $\vec{\mu}$ sweeps the surface of a revolving cone, whose axis lines up with $\vec{B}_0$. The projection of $\vec{\mu}$ is in the same direction as $\vec{B}_0$ or in the opposite direction.

Table 15.1 Values of γ for some nuclei.

Nucleus	Spin	γ (rad/s/T)	Sensitivity	Isotope abundance (%)
^{1}H	½	2.6752×10^{8}	1	99.985
^{2}H	1	4.1066×10^{8}	9.65×10^{-3}	0.015
^{13}C	½	6.283×10^{-7}	1.59×10^{-2}	1
^{15}N	1	-2.7125×10^{-7}	1.04×10^{-3}	0.37
^{19}F	½	2.5181×10^{8}	0.83	100
^{29}Si	½	-5.319×10^{-7}	7.84×10^{-3}	4.7
^{31}P	½	1.084×10^{8}	6.6×10^{-2}	100

15.3 NUCLEI THAT CAN BE STUDIED BY NMR

In a nuclide represented by ${}^{A}_{Z}X$, the nucleus will have a nonzero spin number I if the number of protons Z and the number of nucleons A are not both even numbers. ${}^{1}_{1}H$, ${}^{13}_{6}C$, ${}^{19}_{9}F$, ${}^{31}_{15}P$ have, for example, a spin number $I = ½$, while $I = 1$ for ${}^{2}_{1}H$ (deuterium D) or ${}^{14}_{7}N$. All these nuclei will give an NMR signal. However, the nuclei ${}^{12}_{6}C$, ${}^{4}_{2}He$, ${}^{16}_{8}O$, ${}^{28}_{14}Si$, ${}^{32}_{16}S$, which have a spin number of zero, cannot be studied by NMR. Overall, more than half of the known stable nuclides (at least one isotope per element) lead to an NMR signal, although sensitivity varies enormously depending upon the nucleus. Hence, the proton (the common name for ${}^{1}_{1}H$) or ${}^{19}_{9}F$ are easier to detect than ${}^{13}_{6}C$, which is much less sensitive than the proton and also because it represents only 1% of elemental carbon (Table 15.1).

15.4 BLOCH THEORY FOR I = ½

At the macroscopic scale, even the smallest quantity of a compound is composed of a huge number of individual molecules. With the number of nuclei being so high, the NMR signals reflect their statistical behaviour, as in optical spectroscopy.

Let us consider a collection of identical nuclei whose spin number is $I = ½$. In the absence of an external magnetic field, the individual spin vectors will be randomly oriented and vary constantly with time. From an energy point of view, these nuclei form a population that is considered to be in a *degenerate* state (Figure 15.2). When these nuclei are placed in a strong external magnetic field $\vec{B}_0$ (Oz orientation), an interaction will occur between each small nuclear magnetic vector and this field (see Section 15.2).

Therefore, two groups of nuclei appear according to the direction of their spin vector along the Oz-axis, whose energies correspond to E_1 and to E_2 as defined above (Figure 15.3).

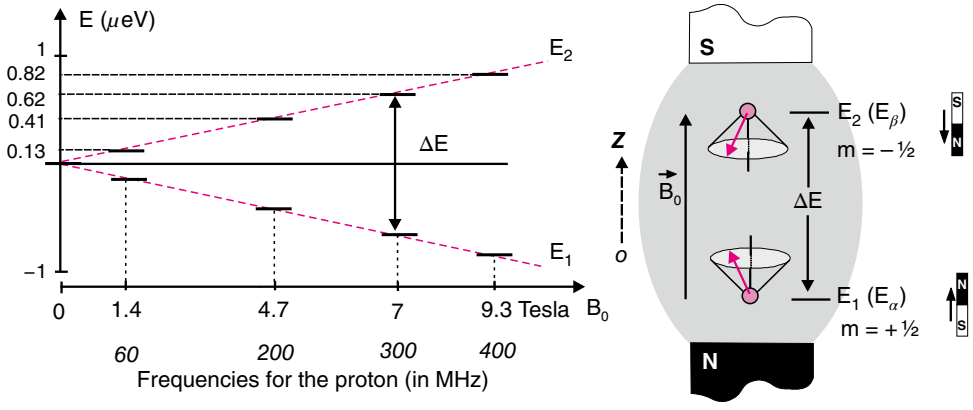


Figure 15.3 Representation of the energy differences for a nucleus with a spin number I = ½ placed in a magnetic field. The four values chosen for the field B_0 correspond, for the proton, to commercial instruments of 60, 200, 300, and 400 MHz respectively. (B_0 represents the magnetic field strength, measured in Tesla: 1 T is equivalent to 10,000 Gauss).

The difference in energy ΔE between the two states is:

$$\Delta E = E_2 - E_1 = \gamma \frac{h}{2\pi} B_0 \qquad (15.4)$$

ΔE is proportional to the field $\vec{B}_0$ (Figure 15.3). Thus, for the proton, if B_0 = 1.4 T, the difference in energy is very small: 3.95 × 10^{-26} J or 2.47 × 10^{-7} eV. The ratio (E_2 – E_1)/B_0 depends only on the value for γ, i.e. on the nucleus being studied (Table 15.1). The population of nuclei located in energy state E_2, is slightly lower than that in the slightly more stable state E_1. Equation (15.5) is used to calculate the ratio of these two populations (Boltzmann distribution equilibrium).

$$R = \frac{N_2}{N_1} = \exp\left[-\frac{\Delta E}{kT}\right] \qquad (15.5)$$

(for ^{1}H, R = 0.999964 if T = 300 K and B_0 = 5.3 T, where k = 1.38 × 10^{-23} J·K^{-1}·at^{-1}).

As a result, it is only the *slight excess* in population E_1 that is responsible for the NMR signal. By increasing the value of the magnetic field, the difference between the two populations will be greater and thus sensitivity will increase. In a graphic representation, the weak magnetization of the sample solution is accounted for by showing all of the individual vectors $\vec{\mu}$ starting from one point. The resulting vector $\vec{M}_0$ is called *macroscopic magnetization* (Figure 15.4b). $\vec{M}_0$ is often used to explain the appearance of signals when nuclei enter resonance.

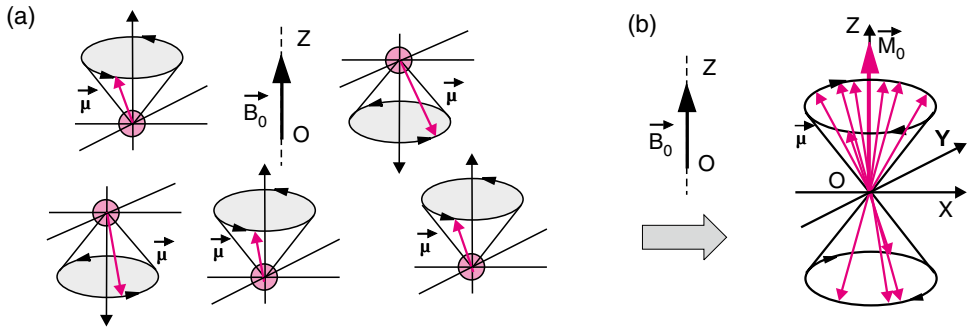


Figure 15.4 Precession and magnetization. (a) A snapshot illustrating the precession movement of the spin vector of five independent nuclei in a magnetic field. (b) The macroscopic magnetization vector which results from the individual orientations of a large number of nuclei. The sample in solution is thus weakly magnetized. $\vec{M}_0$ is an entity to which the laws of electromagnetism can be applied.

15.5 LARMOR FREQUENCY

From an analytical point of view, a nucleus can be identified if ΔE between two populations in a field $\vec{B}_0$ can be measured (assuming $I = ½$), if we know the gyromagnetic constant γ. As in optical spectroscopy, this difference in energy can be determined by creating conditions for passage from one level to the other, i.e. by causing a resonance signal.

To do so, a basic experiment consists in irradiating the nuclei present in a magnetic field with a source of electromagnetic radiation of variable frequency with a propagation direction perpendicular to the external field. Absorption will occur if:

$$h\nu = \Delta E = E_2 - E_1 \tag{15.6}$$

Equation (15.4) leads to the fundamental relation of resonance:

$$\nu = \frac{\gamma}{2\pi} B_0 \tag{15.7}$$

This important and general equation is called the **Larmor frequency**. It links the magnetic field in which the nuclei being studied are located and the frequency of electromagnetic radiation that induces resonance (Table 15.2).

The radiofrequency that induces the exchange between the two levels is equal to the Larmor, or precession, frequency at which the spin vector rotates about the mean O*z* direction. Larmor, an Irish physicist whose work preceded NMR, showed by independent reasoning that ω, the angular rotation frequency of the spin vector around O*z*, has a value of:

$$\omega = \gamma \cdot B_0$$

Table 15.2 Values in MHz of some resonance frequencies for $\vec{B}_0$ = 1T.

Nuclei	^{1}H	^{19}F	^{31}P	^{13}C	^{15}N	^{197}Au
Frequency ν	42.58	40.06	17.24	9.71	4.32	0.729

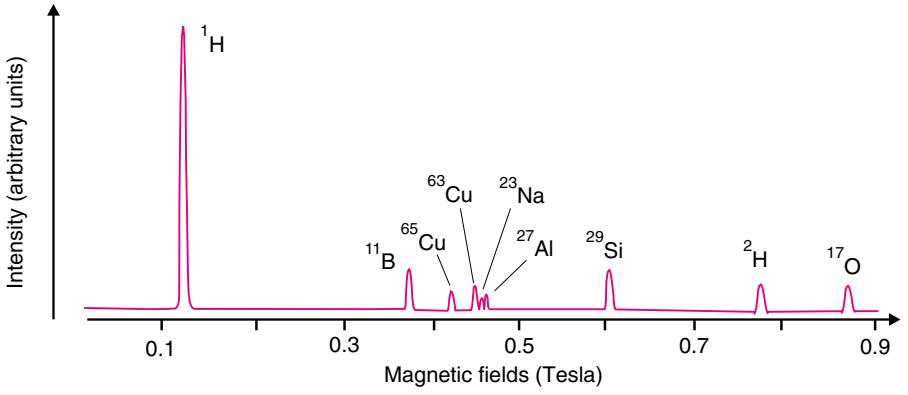


Figure 15.5 NMR spectrum of a water sample placed in a borosilicate glass container, observed at a frequency of 5 MHz (the field is expressed in Tesla; Varian document). There are no commercial instruments of this type because NMR is reserved for more in-depth analyses than the elemental composition of samples.

Since $\omega = 2\pi\nu$, this expression is the same as found in NMR. The two approaches are different: one leads to the frequency of the quantum that separates the two energy levels, and the other to the mechanical precession frequency. These two frequencies have the same value.

The Larmor frequency of a given nucleus increases with B_0. It is situated in the microwave region and varies, for a field of 1 T, from 42.5774 MHz for the hydrogen (^{1}H) nucleus (proton) to 0.7292 MHz for gold (Table 15.2). The instruments are specifically designed for the frequency at which *protons* resonate, even if they are meant for the study of other nuclei.

The construction of the first experimental multinuclei instruments was initiated from these principles. These instruments can scan a wide range of the field while maintaining a fixed radiofrequency. This leads to a qualitative screening of the easiest elements to detect within the sample via their characteristic resonance (Table 15.2 and Figure 15.5).

15.6 SPECTRUM OBTAINED BY PULSED NMR

In the part of the instrument where the sample is introduced, known as the *probe,* the resonance of the nuclei is obtained by superimposing upon $\vec{B}_0$ a weak oscillating field $\vec{B}_1$ that originates from a solenoid powered with alternating current (radiofrequency generator).

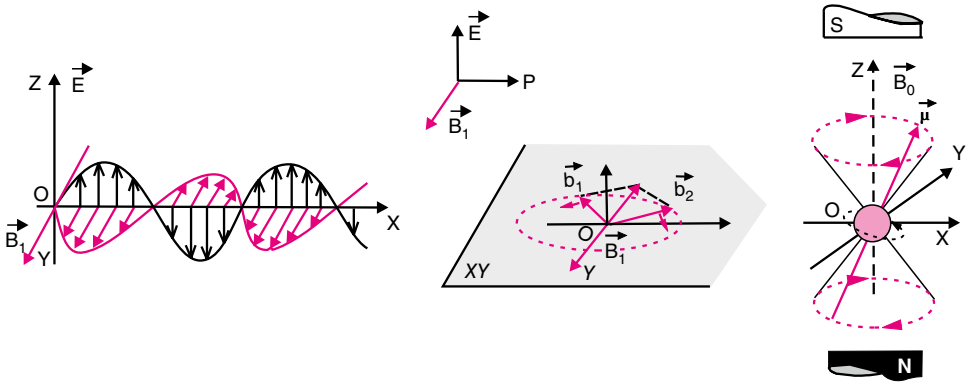


Figure 15.6 Representation of an electromagnetic wave and its effect on a nucleus. In all points on the xOy plane, the magnetic component of the wave can be dissociated into two half vectors $\vec{b}_1$ and $\vec{b}_2$ rotating in phase with opposing velocities. Here, only $\vec{b}_2$ can interact with the nuclei of population E_2.

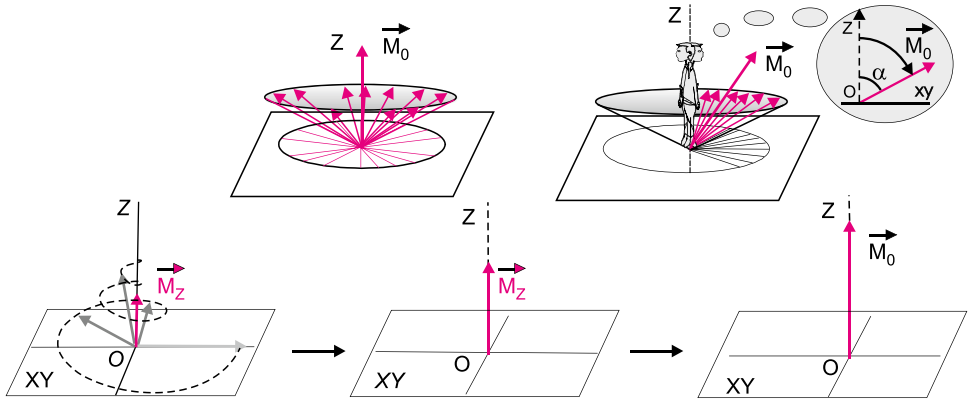


Figure 15.7 Deviation of the magnetization vector $\vec{M}_0$ and return to equilibrium following resonance. Schematic representation in which the individual vectors resulting from the nonequilibrium of the populations are shown (digitally). An observer situated in a rotating frame at the precession frequency would see the magnetization vector tilted by an angle α. Progressive return of the projection $\vec{M}_z$ from $\vec{M}_0$ to its original position.

To understand the interaction of the nucleus with the field $\vec{B}_1$ it is considered that the field results from the composition of two half vectors $\vec{b}_1$ and $\vec{b}_2$ rotating in opposite directions (clockwise and counterclockwise) in the xOy plane with the same angular velocity (Figure 15.6). Only the vector rotating in the same direction as the precession movement may interact with the nucleus. The Lorentz law shows that $\vec{M}_0$ is subject to a force perpendicular to $\vec{B}_1$ and $\vec{B}_0$ (Figure 15.7).

For $\vec{B}_1$ to be suitably oriented, the propagation axis of the radiofrequency needs to be perpendicular to O*z* (Figure 15.7). The transfer of energy from the source to the nuclei occurs when the radiofrequency of the source and the precession frequency are identical. Nuclei in energy state E_1 pass to energy state E_2: in this way, the populations are modified. Before irradiation, the individual vectors $\vec{\mu}$ are out of phase with respect to each other, which is represented by the vector $\vec{M}_0$, aligned in the *Oz* direction (Figure 15.4).

As the resonance condition is reached, all the individual vectors rotate in phase with $\vec{B}_1$. Thus, the direction of $\vec{M}_0$ changes by an angle α, which can be controlled by adjusting the time and the intensity of the applied radiation. Figure 15.7 illustrates this phenomenon with several individual vectors which join together. It is for this reason that $\vec{M}_0$ acquires a $\vec{M}_{XY}$ component in the horizontal plane, which is at a maximum when $\alpha = \pi/2$, and conserves a variable component $\vec{M}_Z$ in the O*z* direction (except when $\alpha = \pi/2$), since the frequency of the rotation of the magnetization vector is then equal to that of the precession movement. The spin of certain nuclei take the second permitted orientation (if $I = ½$).When the irradiation ceases, the system returns progressively to the initial state of equilibrium. A coil detects the component in the O*y* direction (Figure 15.7).

A common interpretation suggests imagining an external observer linked to a frame rotating about the O*z*-axis at the frequency of precession of the concerned nuclei (Figure 15.7). For example, let us assume the case of a single precession frequency. When the radiofrequency has this particular value, the nuclei seem, to the observer, to feel only the effects of the rotating component of the field (i.e. perpendicular to O*z*-axis). The magnetization vector starts to deviate. After the pulse, the individual magnetic moments lose their phase coherence through interaction with the spins of the neighbouring nuclei, faster than they can re-align themselves and return to the original Boltzmann distribution of the populations. Thus, $\vec{M}_0$ loses its xOy-plane component, without having returned to its initial value in the *Oz* direction, which can take a much longer period of time.

Technically speaking, the sample is irradiated by means of a frequency generator ν, sending an intense electromagnetic pulse lasting a few microseconds. This exposes the sample to a frequency range around the value ν, like a flash of white light (to use the example of polychromatic light versus monochromatic light). For example, when working at 300 MHz, in order to irradiate all of the protons, whatever their environment, a frequency range covering at least 6,000 Hz is required. Under these conditions, a small fraction of each type of proton responsible for the NMR signal (but not *all* protons) will absorb the resonance frequency.

It is obviously impossible to find a satisfactory schematic representation for a compound absorbing dozens of frequencies. In such cases, $\vec{M}_0$ can be dissociated into many vectors, each of which precesses around the field direction with its own

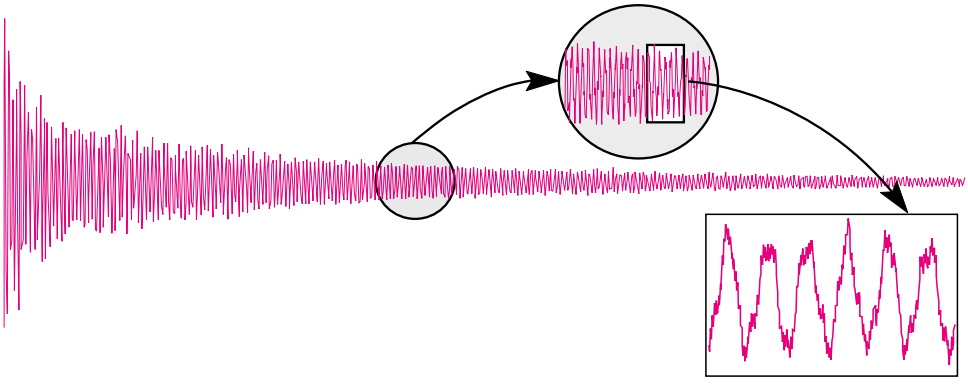


Figure 15.8 FID signal of fluoroacetone (^{13}C) obtained with a pulsed wave apparatus. The signal $I = f(t)$ is converted to a classic spectrum $I = f(\nu)$ by Fourier transform. The interferogram, recorded for a few seconds, can be accumulated several dozen times before being transformed. This FID accumulation is used to improve the signal/background noise ratio.

frequency, which is no longer that of the rotating marker (Figure 15.7, for a simplified situation). During the return to equilibrium, which can take several seconds, the instrument records a complex signal due to the combination of the different precession frequencies present, whose intensity decreases exponentially with time (Figure 15.8).

This signal, which decreases over time, is called *free induction decay* or FID. The signal corresponds at each instant to an overall numerical value that takes into account all the frequencies of the nuclei that have attained resonance at that particular instant. A Fourier transform is used to convert the signal from the *time domain* into a classical spectrum of the *frequency domain*.

This *pulsed wave* technique is used for the study of less sensitive nuclei, such as ^{13}C, and can be computerized.

The current spectrum acquisition process by calculation (emission NMR) did not exist originally. Another method was used. The resonance was searched for by slightly changing B_0 thanks to a coiling of several spires around polar parts of the main magnet (Figure 15.9). Since these were based on field scanning, these instruments generally operated at a fixed frequency (the spectrum is nevertheless monitored in Hz, since the Larmor frequency shows it is possible to make fields and frequencies correspond). This absorption NMR, now almost completely abandoned by manufacturers, corresponded to continuous wave instruments. This process to search for the signal is reminiscent of the sequential recording mode of optical spectra, or, in everyday life, the search for an FM station on a radio. With this principle, the spectrum quality only depends on a very small fraction of the total recording time. Fourier transform instruments make a better use of the measurement time.

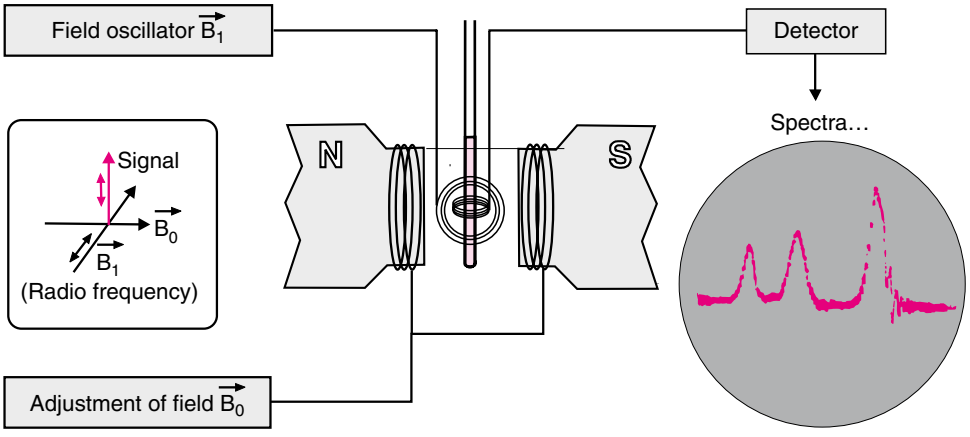


Figure 15.9 CW-NMR apparatus. Arrangement of the different coils around the probe between the poles of the magnet and the orientations of different fields. Right, an old recording (from 1951) of the proton spectrum of ethanol. The three signals correspond to the six hydrogen atoms of CH_3, CH_2, and OH. This will be better understood after reading Section 15.8.

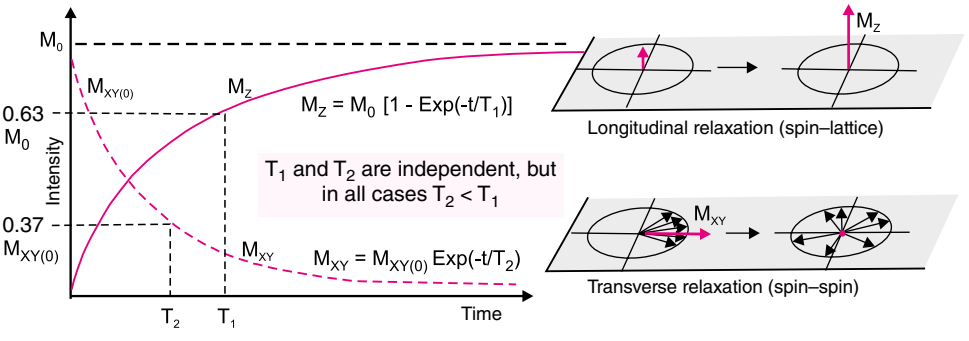


Figure 15.10 The two processes of nuclear relaxation. Evolution of spin–spin and spin–lattice relaxation over time.

15.7 THE PROCESSES OF NUCLEAR RELAXATION

After the radiofrequency pulse, $\vec{M}_0$ will return to its equilibrium value from before irradiation in a variable amount of time, which is dependent upon the medium (Figure 15.10). This relaxation period depends on both the reconstitution of the populations to their initial state (time constant T_1), called *longitudinal relaxation time* (spin–lattice relaxation), and on the loss of phase coherence (time constant T_2), called *transverse relaxation time* (spin–spin interactions). In solutions, T_1 is never longer than

a few seconds (for ^{1}H), while it can be as long as several hours with solids. These two components of $\vec{M}_0$ must therefore be dissociated. Knowledge of the T_1 and T_2 times yields very useful information about the structure of the analysed samples.

15.8 CHEMICAL SHIFT

Generally, within a molecule, each atom has its own particular environment, where the external field value reaching its nucleus is specific to it – at least if there are no specific elements of symmetry in the molecule. These local field variations originate from bond electrons which, by their circulation, create a very weak induced field, which opposes the external field. This is electron diamagnetism. It results in a magnetic screen effect, called *shielding*, which will shift the frequency of resonance when compared to what would be seen in a vacuum (Figure 15.11).

This observation is the basis for the principal exploitation of NMR: rather than observing all of the nuclei present, spread over a wide range of frequencies (several dozen megahertz), the study focuses on a single type of nucleus at a time. In other words, the technique zooms over a very narrow range of frequencies (for example, 1,000 Hz) in order to record the different signals which result from the specificities of each compound.

This shielding effect is quantified by the *shielding constant* σ which appears in Eq. (15.8) and relates the effective field $\vec{B}_{eff}$ that reaches the nucleus to the external field $\vec{B}_0$:

$$B_{\text{eff}} = B_0\left(1-\sigma\right) \tag{15.8}$$

All variations in σ affect the resonance frequency of the corresponding nucleus. This phenomenon is called a *chemical shift*. As many chemical shifts are observed as

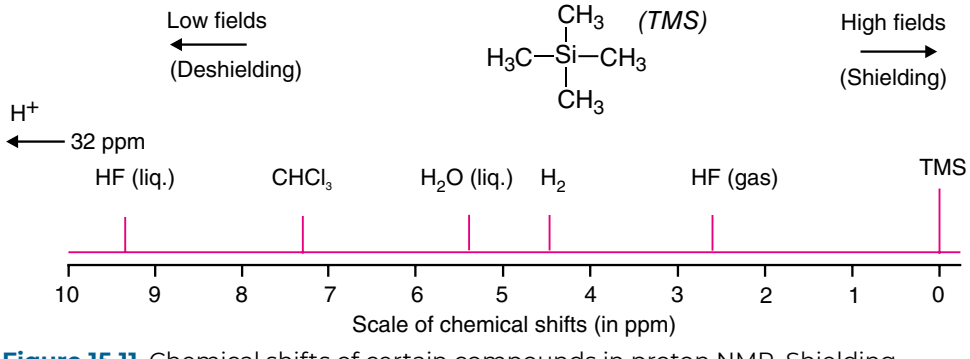


Figure 15.11 Chemical shifts of certain compounds in proton NMR. Shielding effects recorded with a fixed-frequency instrument.

there are molecules containing different shielding constants σ. Consequently, large molecules lead to complex spectra as a result of their numerous σ values.

For a nucleus *i* for which $I = ½$, the Larmor equation is written as follows, with the introduction of the effective field $\vec{B}_{eff}$:

$$\nu_i = \frac{\gamma}{2\pi} B_0 \left(1 - \sigma_i\right) \tag{15.9}$$

Concerning the proton ^{1}H, Eq. (15.9) leads to a difference of 1 Hz for the resonance frequency when the surrounding field varies by 2.3×10^{-8} T, while a variation of 10^{-4} T (1 gauss) induces a shift of 4,258 Hz. For this reason, the magnet's environment must be perfectly controlled in order to maintain the stability of the field $\vec{B}_0$ in the NMR instrument. Thus, the temperature has to be controlled to within 1/100th of a degree. Furthermore, the sample is rotated in a tube with very thin walls to reduce the heterogeneity of the field.

15.9 MEASURING THE CHEMICAL SHIFT

According to the Larmor equation, the slightest variation in the field has repercussions on the resonance frequencies. Thus, it would be inadvisable to try to compare spectra or identify compounds using the absolute signal frequencies obtained with different moments or instruments. For this reason, the chemical shift is referenced by a relative scale $\Delta\nu/\nu_{app}$, which by nature is independent from the instrument. That is why a compound acting as an internal standard, giving off only one signal, is added and can therefore serve as a reference. For the nucleus in question, the different chemical shifts can then be obtained by dividing the frequency difference $\Delta\nu$ between each signal of the compound studied and that of the standard, by the characteristic frequency ν_{app} of the instrument.

The values obtained are expressed in parts per million (ppm). To calculate the chemical shift δ_i (ppm) corresponding to a signal of frequency ν_i with respect to an internal standard (ν_{ref}), there is no need to know the absolute frequencies, only $\Delta\nu$ is necessary.

$$\delta_i = \frac{\nu_i - \nu_{ref}}{\nu_{app}} \cdot 10^6 = \frac{\Delta\nu}{\nu_{app}} \cdot 10^6 \tag{15.10}$$

The internal standard used for both ^{1}H and ^{13}C NMR is tetramethylsilane (TMS). This inert and volatile compound (boiling point = 27°C) yields a single signal in both ^{1}H NMR (the 12 protons are equivalent) and in ^{13}C NMR (for the four magnetically equivalent carbons). The TMS signal is positioned at the origin of the δ scale (Figure 15.11). Almost all organic compounds have positive values for δ, ranging for ^{1}H from 0 to 15 ppm and up to 250 ppm for ^{13}C.

Because of its inductive, silicon-donor nature, the TMS signal is located on the right-hand of the spectrum owing to the shielding effect, thus respecting the convention in spectroscopy that the energy parameter, positioned along the *x*-axis, decreases when moving from left to right on the spectrum.

The δ scale has enabled the generation of empirical correlation tables of the chemical shifts as a function of chemical structure, which can be used irrespective of the NMR instrument (Tables 15.5 and 15.6 at the end of this chapter). Equation (15.10) shows that if we compare the spectra of the same compound, recorded with two instruments whose fields $\vec{B}_0$ make a ratio between them of *k*, then the frequencies of the homologous signals for the two spectra will be in this same ratio *k*, with the values of δ remaining the same. However, these tables are not always sufficient to make a correct assignment of the signals. Software programs have been developed to help in this regard.

15.10 SHIELDING AND DESHIELDING OF THE NUCLEI

Molecules in dilute solution form independent entities without any interaction amongst themselves. However, within a molecule, the electronic and steric environment of each nucleus creates a more or less effective barrier that protects it from the effect of the external field $\vec{B}_0$. This is the source of the chemical shift. The more marked the shielding effect, the more the nuclei are said to be *shielded*: to obtain resonance, the value of the field $\vec{B}_0$ must be increased, at least when using a continuous-wave instrument operating at fixed frequency. Signals situated on the right of the spectrum are said to be resonant at high field and, by contrast, signals observed on the left of the spectrum correspond to *deshielded* nuclei and are said to be resonant at *low field* (Figure 15.11).

15.11 FACTORS INFLUENCING CHEMICAL SHIFTS

Examination of a large number of NMR spectra can reveal the factors that are responsible for predictable and cumulative effects on the chemical shifts.

15.11.1 Inductive Effects

The simple replacement of a hydrogen atom by a carbon-containing group R produces a deshielding of the remaining protons. This effect reaches 0.6 ppm when comparing RCH_3 to CHR_3. This effect can reach 40 ppm for ^{13}C NMR. The state of hybridization of the carbon atoms greatly influences the position of the signals.

Table 15.3 Influence of electronegativity χ of the halogen on δ (TMS ref.)

	CH_3F ($\chi = 4$)	CH_3Cl ($\chi = 3.2$)	CH_3Br ($\chi = 3$)	CH_3I ($\chi = 2.6$)
δ_H (ppm)	4.5	3	2.7	1.3
δ_C (ppm)	75	30	10	-30

These *anisotropic* variations are due to the chemical bonds, that is, to the nonhomogeneous electron distribution around bonded atoms, which, as indicated earlier, has an influence on the shielding constant 'felt' by the nucleus and therefore on chemical shifts.

The chemical shifts of organic compounds are therefore sensitive to the more or less significant delocalization of bond electrons.

Thus, it is understandable that the electron effects that modify the polarity of the bonds, such as the electronegativity of an atom close to the studied nucleus, have an influence upon the chemical shifts. Table 15.3 illustrates the effect of electronegativity of halogens upon the signal position of the methyl group in the halomethane series CH_3X where the chemical shift of the carbon atom signal increases with the electronegativity of the halogen X.

15.11.2 Resonance Effects

The inductive effects are not the only ones responsible for the delocalization of the electron cloud. As Figure 15.12 illustrates, mesomer effects (delocalization of π electrons) also have a significant influence on chemical shifts.

15.11.3 Anisotropy and Induced Local Field Effects

Along with the effects cited above, there is also the effect of small magnetic fields induced by the circulation of π bond electrons. Thus, through circulation, the electrons of an aromatic ring, such as benzene or [18]-annulene, create an induced field, which is opposite to the external field (diamagnetism). This effect is especially marked in the case of [18]-annulene, whose protons inside the ring are fully subjected to the shielding effect related to the induced field (δ = −4.22 ppm), while the protons outside the ring are subjected to the countereffect of this induced field (δ = 10.75 ppm). The same phenomenon is also observed on a smaller scale in the case of isolated double or triple bonds (Figure 15.13). The orientation of this diamagnetism phenomenon depends on the nature of the groups. The field around a double or triple bond is not homogenously distributed (diamagnetic anisotropy). Therefore, acetylenic protons are subjected to a shielding effect, while ethylenic protons are subject to a deshielding effect.

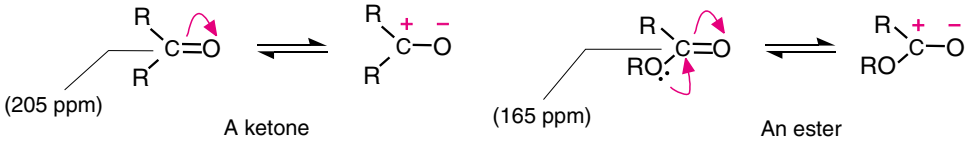


Figure 15.12 Effects of resonance for carbonyl compounds in ^{13}C NMR. If the carbonyl of a ketone is compared with that of an ester, we can note that for the ketone (the more electropositive of the two) the carbon has less protection than the ester. In ^{13}C NMR, the carbonyl signal for a ketone is around 205 ppm, while for an ester it is around 165 ppm.

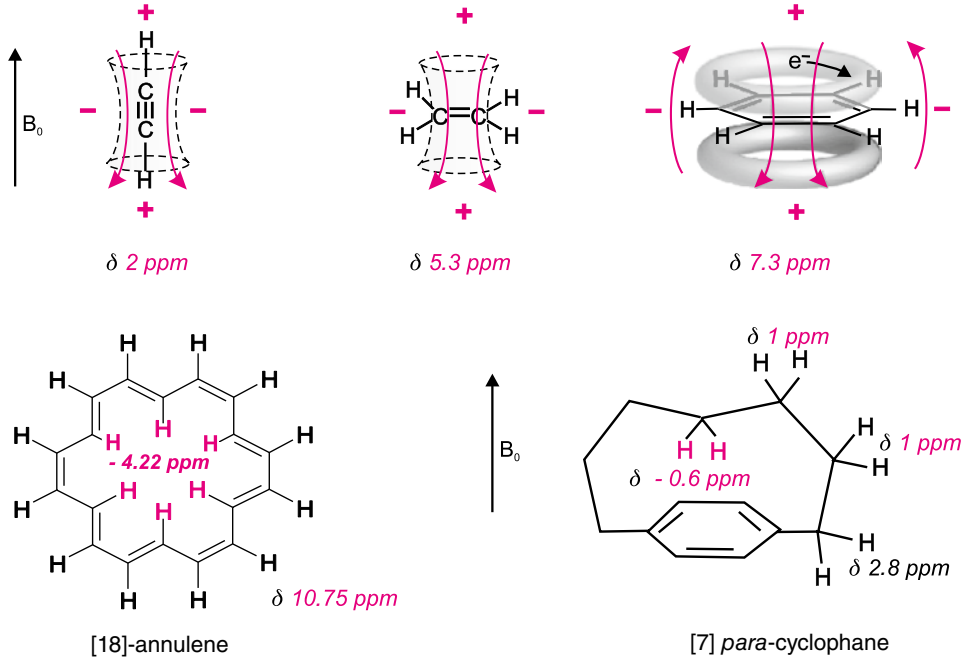


Figure 15.13 Anisotropic effects and induced local fields. The presence of π-bonds can be translated as zones of either a (+) shielding or a (–) deshielding effect. The ethylenic protons are on the outside of a kind of double cone of protection, while the aromatic protons undergo the effect of the electrons moving in two parallel circular clouds.

15.11.3.1 Other Effects (Solvents, Hydrogen Bonding, and Matrix)

The solvents (generally not containing hydrogen atoms) used to dilute organic compounds in ^{1}H or ^{13}C NMR can influence the position of signals. Far more abundant than the solute, the solvent leads to associations with the solute. The stability of such

combinations depends upon their respective polarities. Consequently, correlation tables must provide information about the concentration and the solvent used.

> In ^{1}H NMR, the position of the chloroform signal in chloroform/toluene mixtures varies from 7.23 ppm (90% chloroform/10% toluene v/v.) to 5.86 ppm (10% chloroform/90% toluene). This upfield shift for toluene-rich mixtures is explained by the presence of complexes causing the proton of chloroform to be located in the shielding zone of the aromatic ring.

The most widely used solvent is deuterated chloroform ($CDCl_3$) which is sufficiently polar to dissolve the majority of organic compounds. Also used are acetone-*d6* (C_3D_6O), methanol-*d4* (CD_3OD), pyridine-*d5* (C_5D_5N) DMSO-*d6* (C_2D_6SO), and heavy water (D_2O).

When the studied compounds have labile hydrogen atoms, D ↔ H exchanges can occur in certain solvents. These exchanges will cause alterations in the intensity and position of the corresponding signals. Likewise, hydrogen bonds can modify the electron environment of some protons, making their chemical shift sometimes difficult to predict.

Finally, interactions between neighbouring molecules and also the effect of viscosity will alter the resolution of the spectrum via the spin–lattice relaxation time.

15.12 HYPERFINE STRUCTURE – SPIN–SPIN COUPLING

As a general rule, NMR spectra contain more signals than there are nuclei with different chemical shifts. This phenomenon occurs because the external magnetic field, which influences all of the atoms of the compound in question, causes an alignment of all the nuclear spins and because the orientation taken in the magnetic field by one nucleus affects the neighbouring nuclei, from an energy standpoint, through bonding electrons of this atom. This *coupling* between nuclei decreases greatly with distance. *Homonuclear coupling* (between nuclei of the same type) or *heteronuclear coupling* (between nuclei of different types) gives rise to small shifts in the signal. This hyperfine structure of the spectrum provides additional information about the compound being studied. Homonuclear coupling between protons is quite frequent. The presence of ^{13}C, ^{31}P, and ^{19}F also leads to heteronuclear couplings with the protons.

15.12.1 Typical Heteronuclear Coupling: Hydrogen Fluoride

Hydrogen fluoride (HF) is a simple molecule that allows the observation of a heteronuclear spin–spin coupling between its two atoms, which are linked by a single bond (Figure 15.14). When the ^{1}H NMR spectrum of this compound is examined, two signals of equal intensity are observed, though the HF molecule itself contains only one proton (H). A priori, only one signal should be seen. The reason for this is the following:

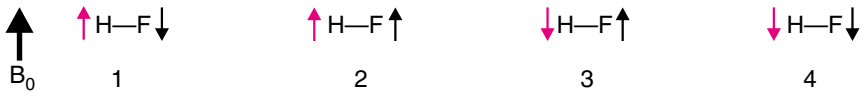


Figure 15.14 Heteronuclear spin-spin coupling of hydrogen fluoride.

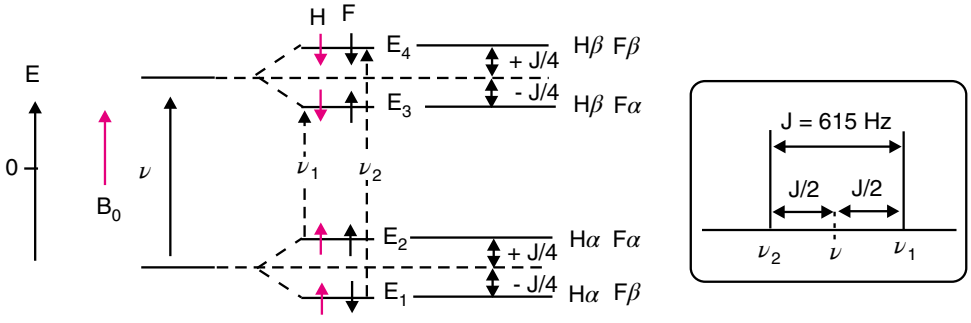


Figure 15.15 Coupling diagram for the HF molecule in ^{1}H NMR. Hypothetical comparison of the situation in which there is no coupling with the fluorine atom and the real situation. The values for ν_1 and ν_2 differ in length by J Hz.

When a sample of this compound is placed in the instrument probe, this is equivalent to simultaneously placing a very large number of individual hydrogen fluoride molecules in the magnetic field $\vec{B}_0$. Knowing that both H and F have the spin number $I = ½$, the molecules are distributed at thermal equilibrium into four populations E_1 to E_4 (Figure 15.15) that correspond to the four possible spin combinations.

If there was no interaction between the H and F nuclei, the populations E_1 and E_2 on one side and E_3 and E_4 on the other would have the same energy. Consequently, in ^{1}H NMR, only one line would be seen. However, in reality, there is an interaction transmitted by the bond between H and F. With respect to the situation with no coupling, a theoretical development leads to the conclusion that the two populations E_1 and E_2 now correspond to two different energy states: it follows that there will be a doubling of the single peak as supposed previously. The energy required to go from E_2 to E_3 is not the same as from E_1 to E_4. The spin orientation taken by the fluorine atom has an effect on the transition energy observed in ^{1}H NMR. The spectrum of this compound will therefore contain two lines corresponding to the transitions situated on Figure 15.15.

The *coupling constants*, expressed in Hertz, are available from the NMR spectrum because, unlike chemical shifts, their value is independent from the instrument's frequency. They are designated by the letter ^nJ with a subscript giving the nature of

the two nuclei in question and, in the top left, the superscript n refers to the number of bonds which separate these nuclei. If we consider the case of HF, the coupling constant can be written:

$$^{1}J_{FH} = 615\,HZ \tag{15.11}$$

The spin orientation of the fluorine atom, which has a resonance frequency very far from that of the proton, will not be disturbed by the resonance occurring in the protons: in a field of 2.35 T, ^{1}H resonates at 100 MHz, while ^{19}F resonates at 94 MHz. The two ^{1}H lines are separated by 615 Hz, irrespective of the intensity of $\vec{B}_0$. The ^{19}F NMR spectrum for this molecule would lead to a similar spectrum. Two signals would be observed, separated by 615 Hz, due this time to the coupling with the proton following one of the two chosen orientations.

When the number of bonds separating the atoms increases, the spin–spin couplings decrease very quickly, unless there are multiple (double or triple) bonds that could propagate the spin effect through the π-bond electrons.

15.12.2 Homonuclear Coupling – Weakly Coupled Systems

Signal splitting is often encountered between adjacent atoms of a given element if their chemical shifts are different.

Nuclei are said to be weakly coupled when the coupling constants are much smaller than the differences in chemical shifts of the nuclei in question (after conversion into Hz). The ethyl group of butanone (Figure 15.1) illustrates this situation. These five protons – three and two that are weakly coupled – give rise to three signals (a *triplet*) around 1.1 ppm and to four signals (a *quadruplet*) towards 2.5 ppm. The origin of these signals in the spectrum can be determined by their chemical shifts (Table 15.4).

Table 15.4 Pascal's triangle and its application to NMR for $I = ½$.

Neighbouring hydrogens	Multiplicity	Intensity						
0	singlet	1						
1	doublet	1	1					
2	triplet	1	2	1				
3	quadruplet	1	3	3	1			
4	quintuplet	1	4	6	4	1		
5	sextuplet	1	5	10	10	5	1	
6	septuplet	1	6	15	20	15	6	1

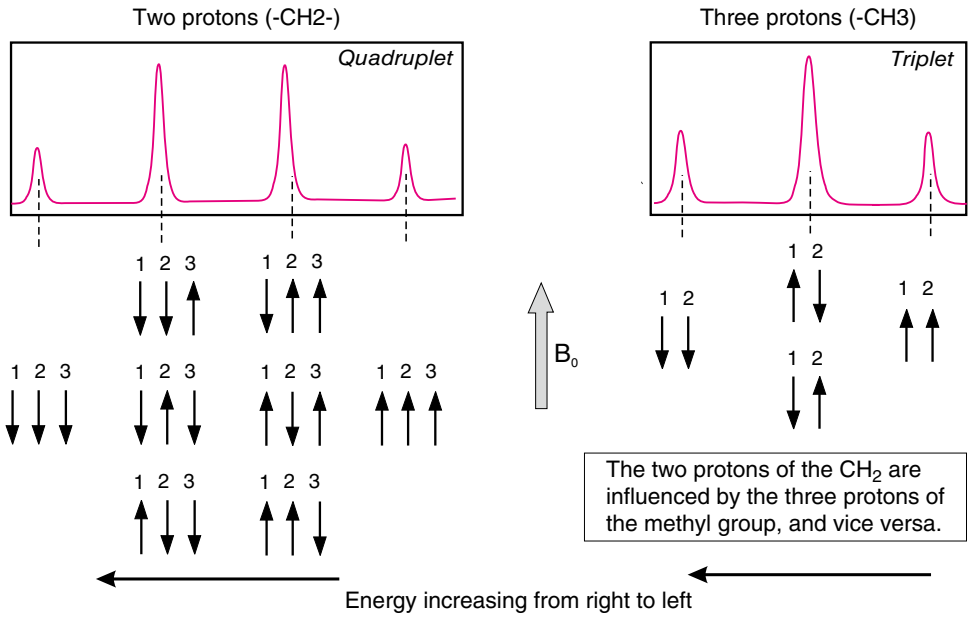


Figure 15.16 Representation of the different spin states for the five protons of an ethyl group. In the same row are found the spin states that produce the same effect on neighbouring nuclei. The sample contains a large number of identical molecules, these are distributed across several populations, each giving a statistically weighted signal as the number of states per row of this diagram.

The triplet is due to the CH_3 of the ethyl group and the quadruplet to the neighbouring CH_2. The relative intensities of the components within each of these multiplets can be deduced by the laws of statistics (Figure 15.16 and Table 15.4).

As a general rule for weak couplings, if n nuclei (spin number I) are placed in the same magnetic environment, influencing in an identical manner one or several adjacent nuclei, then the signals observed for the latter will be formed of $2nI + 1$ regularly spaced peaks – i.e. $(n + 1)$ signals in the case where $I = ½$. The relative peak intensities in a multiplet will follow the same pattern found in each row of Pascal's triangle (Table 15.4). However, if a group of protons is subjected to the effect of neighbouring nuclei whose chemical shifts and coupling constants are not identical, the previous approximation is no longer applicable: the multiplicity of the signals and the intensity of the lines cannot be deduced so easily.

Coupling constants depend, among other things, on the dihedral angle between nuclei, on the nature of the substituents near these nuclei, as well as on the degree of hybridization of the proton-bearing carbons.

15.12.3 Homonuclear Coupling – Strongly Coupled Systems

When the $\Delta\nu/J$ ratio is small, the protons become strongly coupled (where $\Delta\nu$ corresponds to the difference in resonance frequencies of these two protons). The intensities of the multiplet lines are then quite disrupted. New peaks arise from combinations of frequencies, hindering the interpretation of the corresponding spectra. This is one of the reasons for the development of instruments operating at higher frequencies (300 → 800 MHz) with superconducting magnets to obtain the required magnetic fields. Given that $\Delta\nu$ is proportional to the operating frequency, while J remains constant, the Δ/J ratios increase, and once again weakly coupled systems are observed (Figure 15.17). However, there are other phenomena that complicate the spectra when the frequency exceeds 600 MHz.

AB system

When the difference in resonance frequencies between two protons is compared to their coupling constant J, then this system is the simplest strongly coupled system. This system always has two groups of two signals, where each signal is separated by J Hz.

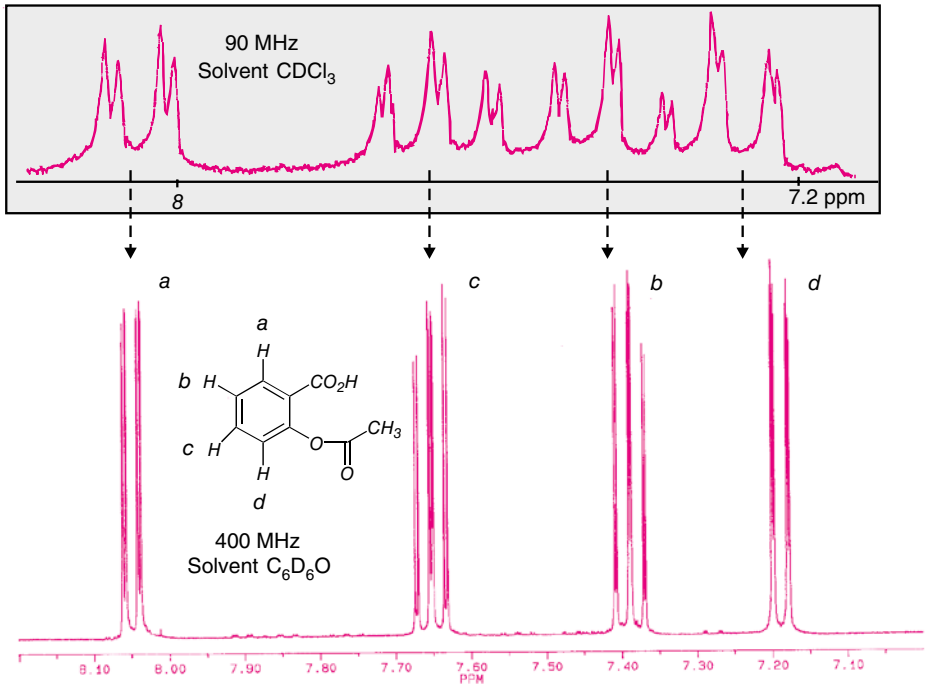


Figure 15.17 Spectrum of the four aromatic protons of aspirin. The figure above reproduces, with a similar scale, the spectrum of aspirin obtained on two different instruments, one running at 90 MHz (solvent: $CDCl_3$) and the other at 400 MHz (solvent: C_3D_6O). The sensitivity of NMR increases with $B_0^{3/2}$.

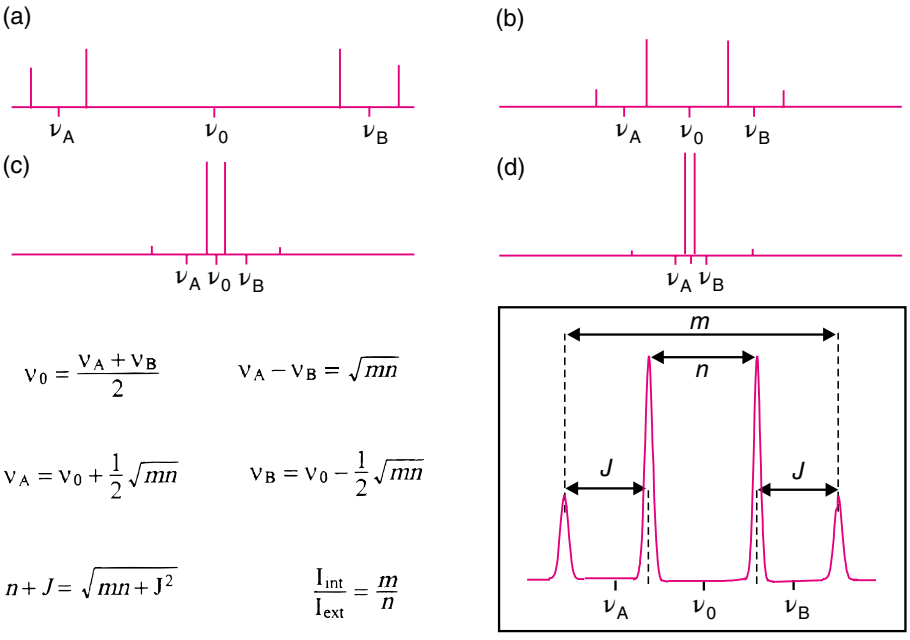


Figure 15.18 Characteristics defining an AB system. (a–d) Gradual modification of a system with two coupled protons as the $\Delta\nu/J$ ratio decreases. Typical appearance and formulae used to analyse an AB system.

However, the intensities are no longer equal, and the chemical shifts, which can no longer be read from the spectrum, must be calculated from the expressions indicated in Figure 15.18.

Nomenclature for coupled systems in ^{1}H NMR. Spectral interpretation for a molecule with many hydrogen atoms is easier when subgroups of signals corresponding to classic situations can be recognized. These particular cases are classified by means of a nomenclature using the letters of the alphabet, selected with respect to the chemical shift (Figure 15.19). If their chemical shifts are identical or very close, the protons are designated by the first letters of the alphabet (AB, ABC, A_2B_2, A_3), while protons with very different chemical shifts are designated by letters that are farther apart, such as A, M, and X.

15.13 SPIN DECOUPLING AND PARTICULAR PULSE SEQUENCES

NMR instruments have accommodated numerous advances to facilitate spectral interpretation. Thus, it is possible to cancel the effect of an existing spin coupling between neighbouring nuclei. This *spin decoupling* technique is based on the fact that a nucleus in resonance does not always conserve the same spin state with time.

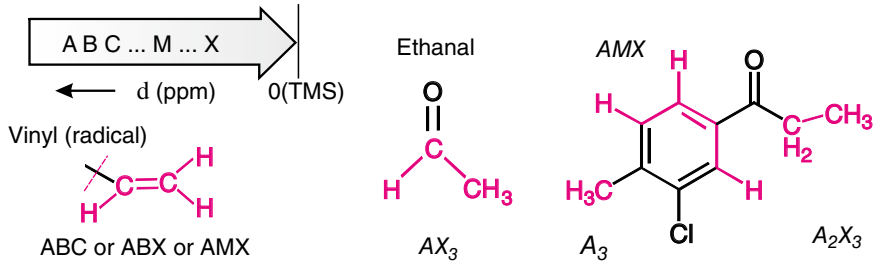


Figure 15.19 Nomenclature of NMR spectra. The ^{1}H spectrum of 3-chloro-4-methylpropiophenone is the complex result of the superimposition of the signals of several independent yet easily recognized subgroups. Thus, the ethyl group (such as ethoxyl) constitutes an A_2X_3 system, ethanal protons form an AX_3 system, while a vinyl group will be an ABC, AMX or ABX system, depending on the example studied.

This uninterrupted and fast transfer between different states induces an average overall effect upon the neighbouring nuclei.

This decoupling sequence is often used in ^{13}C NMR in order to simplify the appearance of the NMR spectrum obtained (Figure 15.20). This technique is used to identify coupled nuclei when it is difficult to interpret the normal spectrum, in particular for superimposed signals.

This decoupling sequence has a second effect on protons spatially close to those irradiated. Quick transitions in energy state caused by the irradiation improve the dipolar relaxation of nearby nuclei.

In this process, a second type of coupling occurs: a dipolar (or direct) coupling. For NMR in solution, this type of coupling cannot be observed.

The improvement of dipolar relaxation leads to an increase in the intensity of signals of nuclei located near the irradiated nucleus or nuclei. This is the *nuclear Overhauser effect* (NOE). It is widely used to obtain structural information on the molecules being studied.

15.14 ^{13}C NMR

In ^{13}C NMR, since the coupling constants $^{1}J_{C\text{-}H}$ fall between 125 and 200 Hz, there are often signal overlaps that make the spectrum less readable. That is one of the reasons why the acquisition of the signal in ^{13}C NMR is accompanied by a proton decoupling sequence, thus leading to thin singlets on the spectrum (Figure 15.25). This simplification of ^{13}C NMR spectra is necessary to improve the signal/noise ratio due to the low abundance (1.1%) and to the low sensitivity of carbon. Due to the nuclear Overhauser effect related to the decoupling of protons, proton-bearing carbon signals are more intense than quaternary carbon signals.

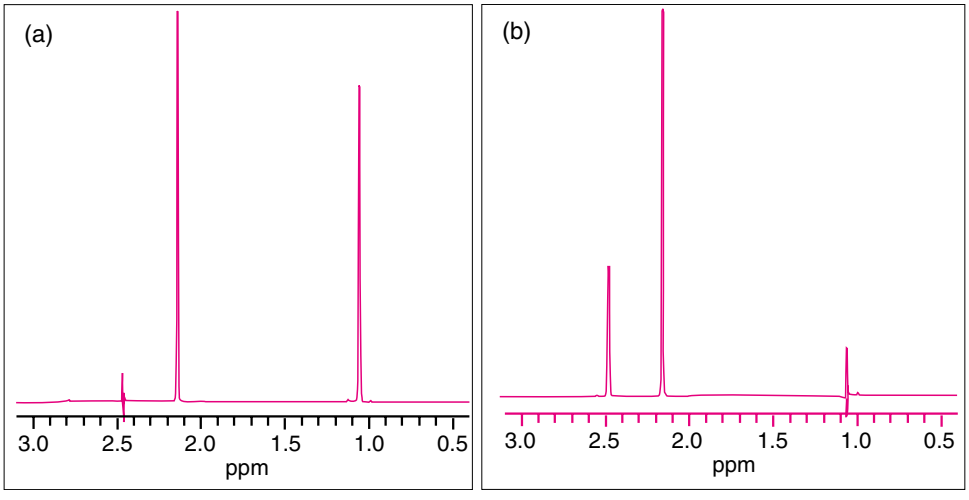


Figure 15.20 Spin decoupling experiment on butanone. Modification of the ^{1}H spectrum by (a) irradiation of the CH_2 group at 2.47 ppm and (b) irradiation of the CH_3 (of the ethyl group) at 1.07 ppm. Compare with the spectrum of butanone in Figure 15.1. For a simple compound such as this, the experiment is only of illustrative interest. On the other hand, a double resonance experiment would allow the precise determination of the different couplings in aspirin (Figure 15.17).

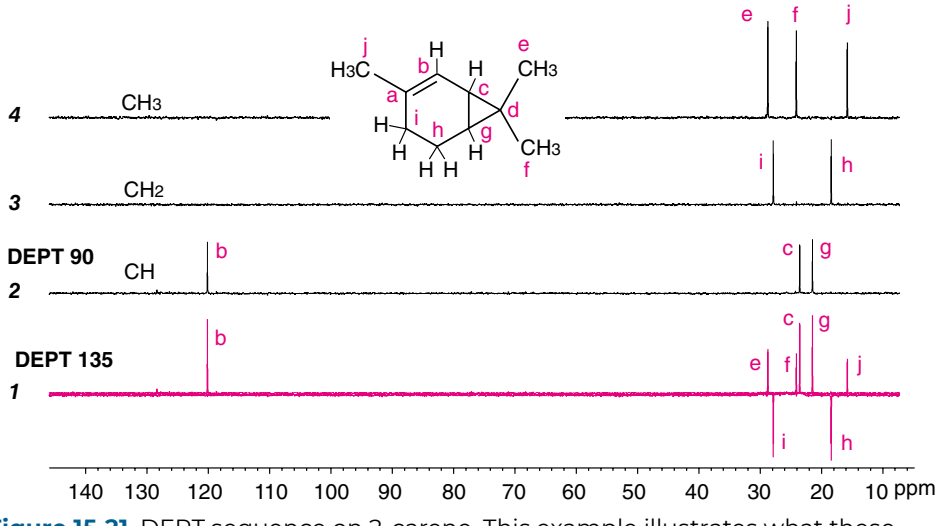


Figure 15.21 DEPT sequence on 2-carene. This example illustrates what these sequences can provide when determining seemingly unknown compound structures.

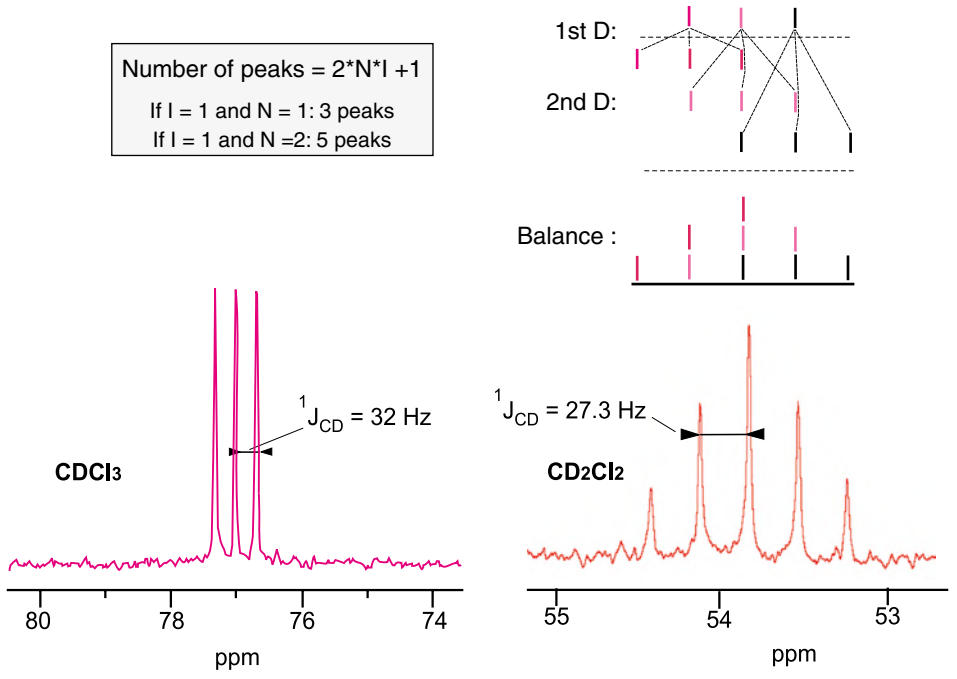


Figure 15.22 ^{13}C spectra of $CDCl_3$ and CD_2Cl_2. Deuterium, for which I = 1, leads to three values of m (–1, 0, 1) and then to three peaks of equal intensity, since there is only one deuterium atom.

The analysis of the ^{13}C spectrum may be accompanied by special sequences called *distortionless enhancement by polarization transfer* (DEPT), which make it possible to differentiate between signals of CH, CH_2, CH_3, and quaternary carbons. Therefore, a DEPT 135 spectrum makes CH and CH_3 signals appear on one side of the plane defined by the baseline, and CH_2 signals are on the other side (Figure 15.21). This sequence, which uses couplings with protons, makes the signals of quaternary (noncoupled) carbons disappear.

On ^{13}C spectra of compounds conducted using deuterochloroform ($CDCl_3$), the presence of three signals with equal intensities (around 77 ppm) can be seen to superimpose with those of the compound. The origin of these signals is the heteronuclear coupling between ^{13}C and deuterium ^2H (or D in the formula) (Figure 15.22).

15.15 TWO-DIMENSIONAL NMR (*2D-NMR*)

NMR is an essential tool for the organic chemist when they want information regarding the structure of complex molecules. To do so, the organic chemist must conduct two-dimensional experiments (*2D-NMR*). There are thus a number of pulse sequences, which are made up of several successive steps (preparation, evolution, mixing, and

detection) and which, for example, help to correlate signals of coupled nuclei. The information that three of the most classic sequences can provide is described briefly in the next section.

15.15.1 Homonuclear Correlation: ^{1}H–^{1}H COSY

The COSY sequence (Correlation Spectroscopy) leads to a spectrum represented in 2D (Figure 15.23) with two frequency axes, with the third being the intensity axis. On the sides of the graph, at the top and on the left, the conventional spectrum appears on a

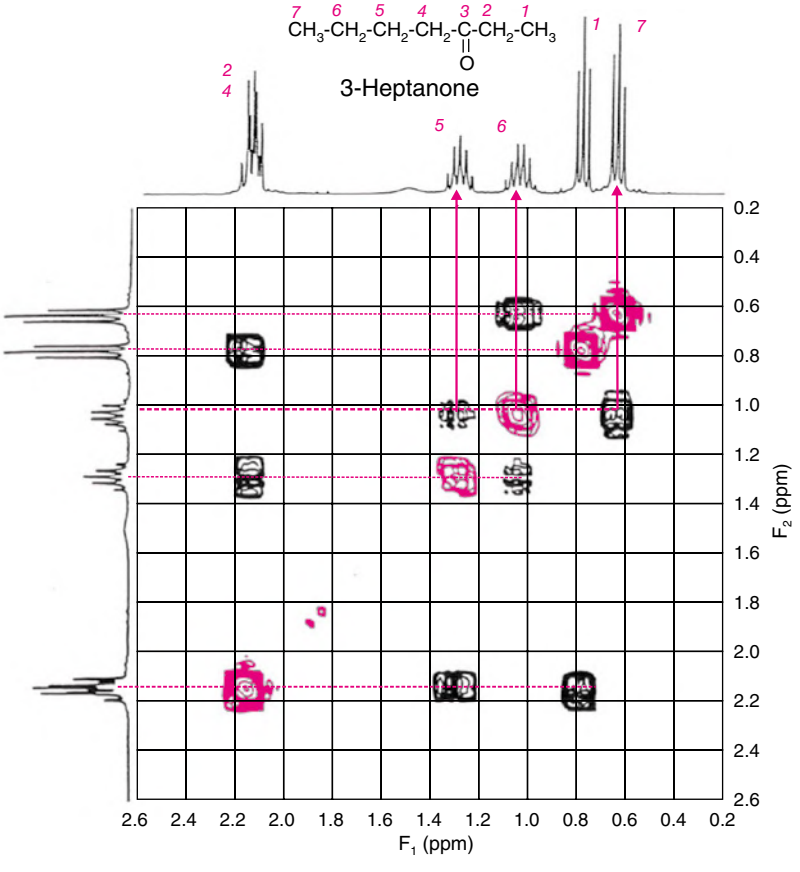


Figure 15.23 COSY spectrum of 3-heptanone. On this representation, on a given horizontal line, protons cause couplings to occur among themselves. Therefore, coupling ^{3}J appears between the protons carried by carbon 6 and the protons carried by carbons 5 and 7. The other existing couplings can be seen on the different horizontal lines.

smaller scale, surrounding the central part, which is a representation in the form of level plots. The spots found on the diagonal line correspond to the spectrum seen from above.

However, the point of this sequence is due to the presence of additional spots found on a given vertical (or horizontal) line. These demonstrate the existence of existing couplings between certain protons, and thus they help identify the corresponding signals. The 3-heptanone spectrum is an example that helps to understand the purpose of this type of sequence.

15.15.2 Homonuclear Correlation: $^{1}H–^{1}H$ NOESY

Just like the COSY sequence, the nuclear Overhauser effect spectroscopy (NOESY) sequence is a $^{1}H–^{1}H$ correlation sequence. Unlike the former, however, it aims to cause nuclear Overhauser effects (NOE) to appear between protons separated by several bonds but still spatially close. The information obtained is complementary to that resulting from the COSY sequence. Nevertheless, the spectrum is slightly more complex. This analysis thus provides information on the configuration and the shape adopted by the molecules being studied (Figure 15.24),

15.15.3 Heteronuclear Correlation: $^{1}H–^{13}C$ HETCOR and HSQC

Many other sequences exist. These are far from being routine experiments and are for experienced users only. The heteronuclear correlation (HETCOR) and heteronuclear single quantum coherence (HSQC) sequences are used to correlate ^{1}H NMR signals with ^{13}C signals (Figure 15.25).

These analyses facilitate the interpretation of ^{1}H and ^{13}C NMR spectra by helping to find a proton signal from the ^{13}C signal or two protons carried by the same carbon. Unlike COSY and NOESY spectra, the number of correlation spots is more limited, as each carbon signal can be related to no more than two magnetically different protons.

15.16 FLUORINE AND PHOSPHORUS NMR

Fluorine and phosphorus are the two heteroatoms in organic chemistry most frequently studied by NMR after hydrogen and carbon.

The fluorine element, consisting of 100% ^{19}F ($I = ½$), can be compared to the ^{1}H proton for its sensitivity. Chemical shifts are distributed on a much greater range (Figures 15.26 and 15.27). As a consequence, in ^{19}F NMR it becomes possible to distinguish between compounds that are chemically very similar. In particular, differences due to stereochemistry are significant, and the $J_{F–H}$ coupling constants can be measured over a greater range compared to $J_{H–H}$ constants (Figure 15.26). On the other hand, relatively few molecules contain atoms of fluorine.

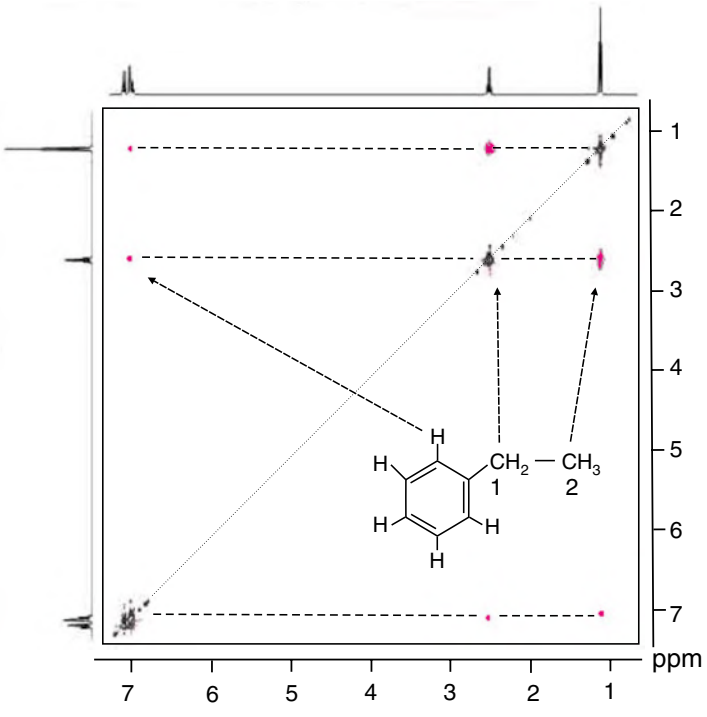


Figure 15.24 NOESY spectrum of ethylbenzene. The result of a NOESY experiment on this molecule provides a simple example, which shows that in addition to the COSY correlation between hydrogen atoms located on carbons 1 and 2, a further correlation spot appears with certain hydrogen atoms located on the aromatic ring.

Phosphorus (^{31}P, $I = ½$), another common element with just one isotope in the natural state, has been studied since the beginnings of NMR. It has great sensitivity, and it is an important element in the composition of numerous biological compounds.

15.17 NMR APPLICATIONS

15.17.1 Structural Analysis

Many natural substances have been identified using NMR. Because structural analysis requires pure compounds, the various natural extracts must thus be separated beforehand into their different components. The large diversity of possible molecules makes this stage especially tedious. The coupling of NMR with a separation method, such as HPLC, is used to overcome this difficulty. Current NMR instruments are sensitive enough to make this coupling a reality (Figure 15.28).

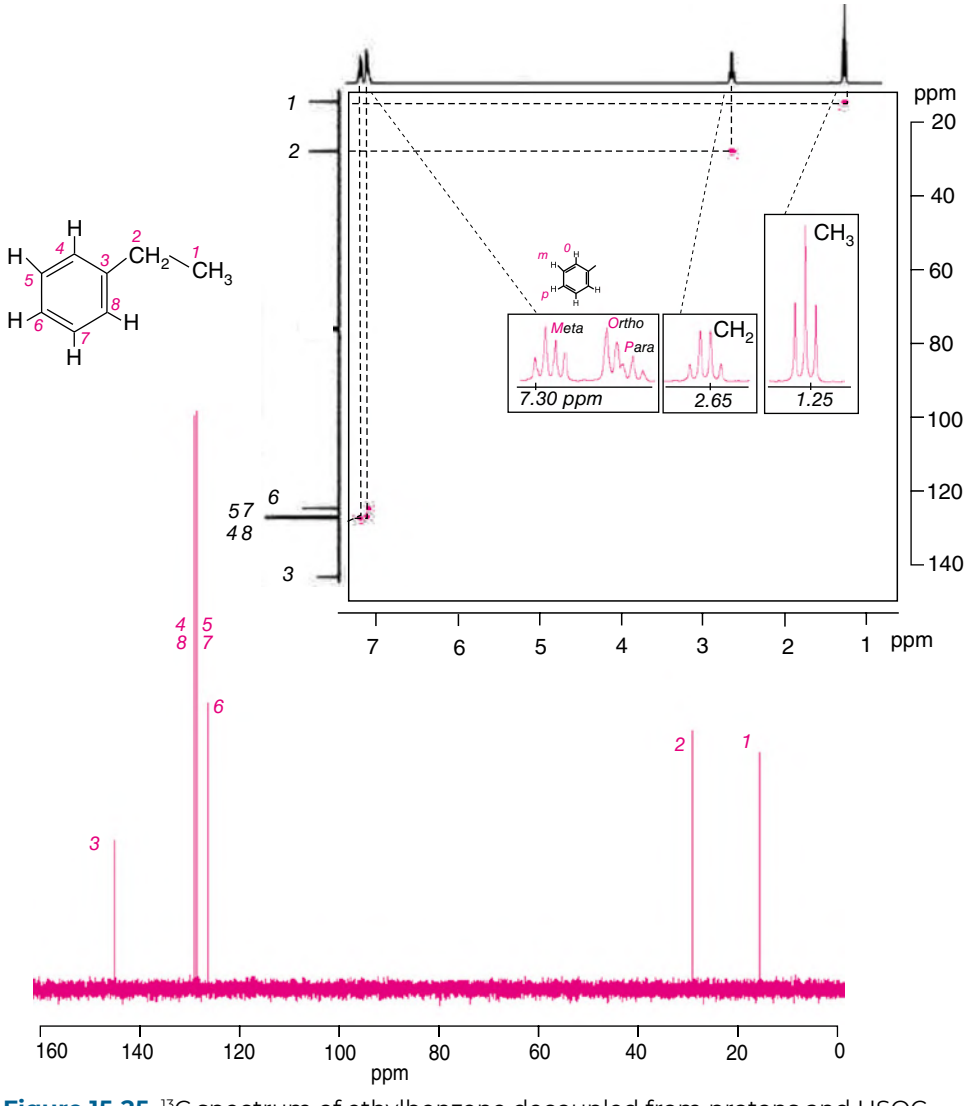


Figure 15.25 ^{13}C spectrum of ethylbenzene decoupled from protons and HSQC spectrum. Each carbon atom gives off a single signal, and thus each signal is represented by a singlet. These simpler, broadband decoupled spectra give less information. The result of an HSQC-type 2D experiment of this molecule has also been reproduced on this figure. Please note that since carbon 3 does not have a hydrogen atom, it does not provide a correlation spot.

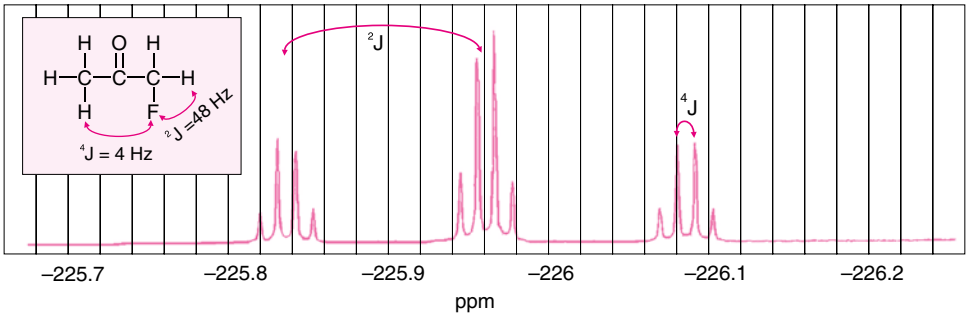


Figure 15.26 NMR spectra of monofluoroacetone. An example of heteronuclear coupling. The single fluorine atom leads to a triplet with CH_2 and to a quadruplet with the methyl group. The resulting signal is thus a triplet of quadruplets (the scale of chemical shifts is positioned with respect to $FSiCl_3$).

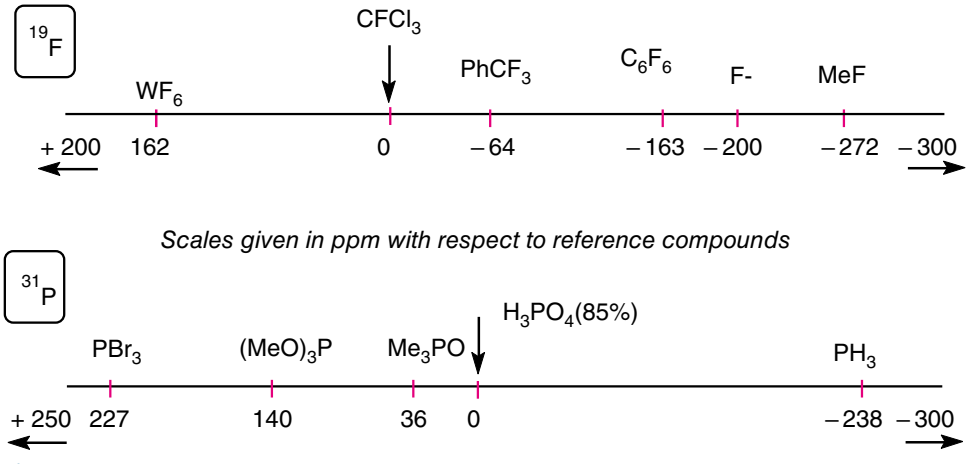


Figure 15.27 Positions of some NMR signals due to fluorine and phosphorus.

The principle consists of taking the mobile phase coming out of the chromatograph detector and sending it to a circulation microcell placed in the NMR instrument. Without interrupting the flow (*on-flow* technique), a large number of NMR spectra are recorded over time and can then be studied to recognize the nature of eluted compounds. However, the obligation to choose a mobile phase compatible with NMR (deuterated solvents) constitutes one of the main limitations on the wider use of this type of technique.

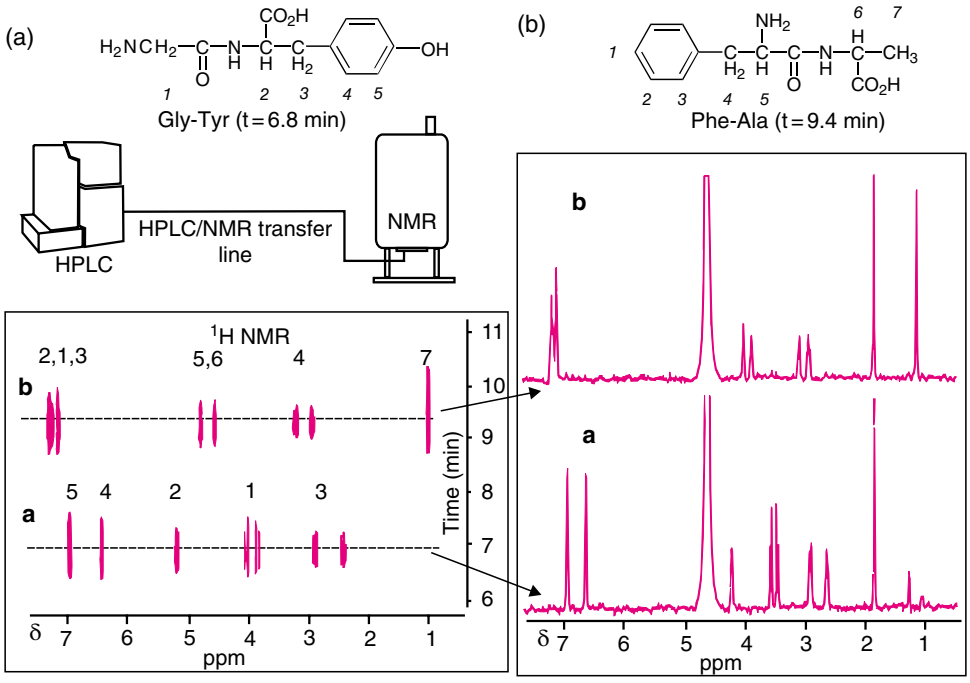


Figure 15.28 Recording obtained from an experiment using coupled HPLC-^{1}H NMR coupling. Separation of two dipeptides (5 µg each) with identification of the signals. For greater clarity, the solvent peaks around 2 and 5 ppm (acetonitrile and water) have been eliminated (Sweeder *et al.*, *Anal. Chem.* 1999, 71(23), 5335).

15.17.2 Quantitative NMR

Although NMR is a method essentially reserved for structural analysis, due to the quality of the information that it can yield, it can also be used to conduct certain measurements in the food industry, medicine, the polymer industry, and the pharmaceutical industry. For the latter, it can be found in most pharmacopoeias, where it is used either to quantify an active ingredient or on the contrary, to measure an impurity.

The method is of interest because it is possible to conduct analyses without complicated preparations or sample destruction, without risk of polluting the instrument, and, unlike many other methods, such as chromatography, without performing a prior normalization stage. In some cases, sample preparation is rapid and easy: it can simply be dissolved and diluted in an appropriate deuterated solvent.

The method relies on the fundamental relationship which shows that the intensity of any NMR signal (generally represented by a number proportional to its area *S*) is directly proportional to the number of nuclei *N* responsible for this signal:

$$S_X = k_{app} \cdot N_X \qquad (15.12)$$

where k_{app} refers to a spectrometer constant, which depends on many parameters. It must remain constant for all resonance signals for the entire spectrum.

The signals selected to identify each component must be well separated. Sensitivity and precision vary depending on the type of nucleus. For the proton, precision on areas is around 1%. For the study of ^{13}C, ^{29}Si, and ^{31}P nuclei, whose relaxation times T_1 are especially long, a paramagnetic relaxation agent is added, such as chromium acetylacetonate, in order to shorten T_1.

In proton NMR, the area S of each signal will thus be directly proportional to the number of protons responsible for this signal and to its molecular concentration. There are several quantification methods.

If all analytes in the sample have been identified and all NMR signals assigned, examination of the NMR spectrum gives near-instantaneous access to molar ratios, as the following example demonstrates.

Consider, for example, a sample comprising a mixture of acetone *A* and benzene *B* dissolved in $CDCl_3$ as solvent. On the ^1H NMR spectrum of this mixture, two signals are observed, one at δ = 2.1 ppm (acetone) and the other at δ = 7.3 ppm (benzene), with respect to TMS. This spectrum corresponds to the weighted superimposition of the two individual spectra (Figure 15.29).

Since the area of *A*'s signal (S_A) is both proportional to the number of protons *a* and to the molar concentration n_A, and since the same goes for the area of S_B for *B* – the signal is proportional to the number of protons *b* and molar concentration n_B – determination of the molar ratio n_A/n_B is nearly instantaneous:

$$\frac{n_A}{n_B} = \frac{\frac{S_A}{a}}{\frac{S_B}{b}} \tag{15.13}$$

If we use the molar masses $\boldsymbol{M}_A$ and $\boldsymbol{M}_B$, the mass concentrations C_A% and C_B% are obtained, knowing that C_A% + C_B% = 100:

$$C_A = 100 \cdot \frac{\frac{S_A}{a} M_A}{\frac{S_A}{a} M_A + \frac{S_A}{b} M_B} \text{ and } C_B = 100 \cdot \frac{\frac{S_B}{b} M_B}{\frac{S_A}{a} M_A + \frac{S_B}{b} M_B} \tag{15.14}$$

In the case of a multicomponent blend (*A, B, …, I, …., Z*) where each component has been identified beforehand, Eq. (15.15) expresses the mass concentration of a component *I*:

$$C_i = 100 \cdot \frac{(S_i / i) \cdot M_i}{(S_A / a) \cdot M_A + (S_B / b) \cdot M_B + \ldots + (S_i / i) \cdot Mi + \ldots + (S_z / i) \cdot M_z} \tag{15.15}$$

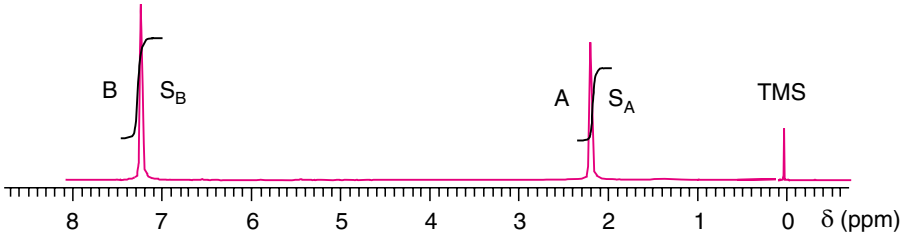


Figure 15.29 NMR spectrum of a mixture of acetone (A) and benzene (B). If S_A = 111 and S_B = 153 (arbitrary units), then knowing that M_A = 58 and M_B = 78 g/mol, we can find that C_A = 35% and C_B = 65% by mass.

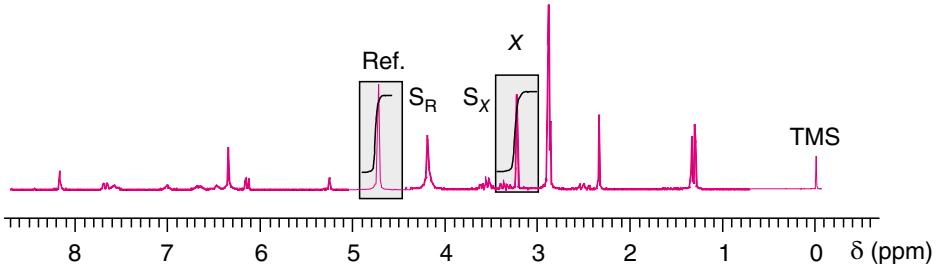


Figure 15.30 Spectrum of a sample to which a reference compound R has been added. The signal S_X belongs to the compound to be measured and the signal S_R to the internal standard.

In the example selected (Figure 15.30), where acetone and benzene have the same number of protons, a = b, Eq. (15.14), after simplification, leads to the following mass percentages:

$$C_A = 100 \cdot \frac{S_A M_A}{S_A M_A + S_B M_B} \text{ and } C_B = 100 \cdot \frac{S_B M_B}{S_A M_A + S_B M_B} \qquad (15.16)$$

With this type of calculation, ^{1}H NMR is frequently used in organic chemistry as a method to find the yield of a reaction $A \rightarrow B$. This is done by recording the spectrum of the raw product after reaction and by identifying a signal belonging to the product formed (*B*) and a signal belonging to the remaining starting material (*A*). The yield of *B* with respect to *A* can be written using the previous notations:

$$R = 100 \cdot \left(\frac{S_B}{b} \right) / \left(\frac{S_A}{a} + \frac{S_B}{b} \right) \qquad (15.17)$$

However, in most cases, only one analyte needs to be quantified, maybe two, in a sample, in which the exact composition is not always known and in which the signals

of the spectrum cannot all be identified. Therefore, just one signal belonging to the compound in question must be identified. Then, as in chromatography, a method analogous to the internal standard is used.

To quantify compound X ($M = M_X$) in a mixture called E, a quantity P_R (in mg) of reference compound R ($M = M_R$) for use as internal standard, is added to quantity P_E (in mg) of the sample E containing compound X, before recording the spectrum. The standard R is chosen such that the signal serving as indicator does not interfere with the signal chosen to quantify compound X (Figure 15.30). Next, on the NMR spectrum of the mixture in solution in a suitable solvent, we identify:

- a signal belonging to compound X (area S_X for x protons);
- a signal belonging to standard R (area S_R for r protons).

If we use n_X/n_R to indicate the molar ratio of X and of R, we can write:

$$\frac{C_X}{C_R} = \frac{n_X}{n_R} \cdot \frac{M_X}{M_R} \tag{15.18}$$

Knowing that

$$\frac{n_X}{n_R} = \frac{S_X / x}{S_R / r} \tag{15.19}$$

and that the concentration C_R, expressed as a mass percentage of S, is equal to:

$$C_R = 100 \cdot \frac{p_R}{p_E + p_R} \tag{15.20}$$

the mass concentration (%) C_X of compound X can be deduced:

$$C_X = 100 \cdot \frac{p_R}{p_E + p_R} \cdot \frac{(S_X / x) \cdot M_X}{(S_R / r) \cdot M_R} \tag{15.21}$$

15.17.3 Analysis via Time-Resolved NMR (TR-NMR)

Low-resolution Fourier transform pulsed NMR (magnet of 0.3–1 Tesla) is used to assess the water or fat content as well as the ratios of solid and liquid phases in a large range of samples, even nonhomogeneous ones. Many applications are used in routine analysis. The principle of this noninvasive method, which can be used on samples without advance preparation, consists of exploiting the information found in the signal, which decreases over time (FID) right after the irradiation pulse (Figure 15.31). This method is known as *Time Resolved-NMR* (TR-NMR). The overall FID, resulting from all protons present in the sample, whatever the compound to which they belong, is

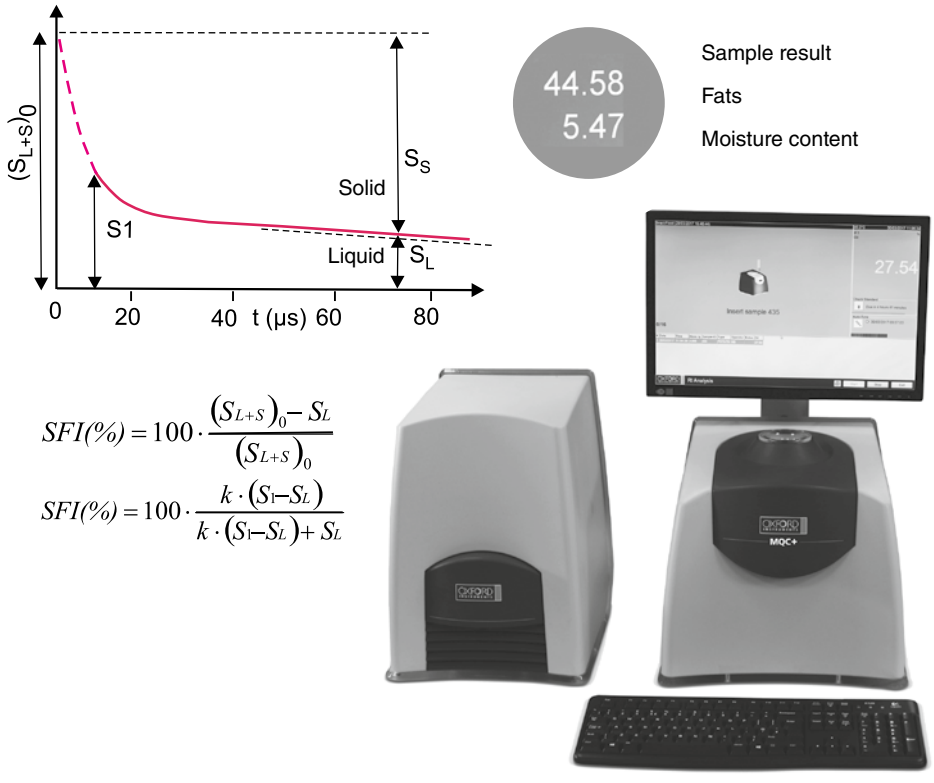


Figure 15.31 Determination of the solid fat index by means of decrease in FID. The decrease in signal after the radiofrequency pulse is used here to determine the solid fat percentage in a sample without preparation. A correction factor is applied to the signal value immediately after the end of the pulse. On the right, a small instrument with a 1 T permanent magnet to quantify water and fats in samples. (Source: MQC model from Oxford Instruments.)

also used for single compounds to extract the two time constants: T_1 (spin–lattice relaxation, or longitudinal) and T_2 (spin–spin relaxation, or transversal), which have already been described (see Section 15.7). These values provide information on the molecular environment (solid or liquid) in which these protons are found. Therefore, in a more or less rigid phase, such as a fat for example, the value of T_1 will be much larger than in a liquid, and that of T_2 will be smaller. Therefore, it becomes possible to distinguish protons engaged in a solid compound from those in a liquid.

This reproducible method is easy to conduct. With each application, a simple calibration is nevertheless necessary with reference samples. The precision is variable and depends on the calculation mode. In complex food systems, the concentration of major constituents can be determined, such as water (free or bound) and fat in seeds, chocolate, French fries, powdered milk, olives, etc., or the *Solid Fat Index* (SFI) (Eq. (15.22)) in animal fats (Figure 15.31). Many other industrial applications, for example in chemistry and pharmaceuticals, are also used to monitor manufacturing processes.

$$SFI = \frac{S_{L+S} - S_L}{S_{L+S}} \times 100 \tag{15.22}$$

To determine the water resources in the subsoil (quantification and depth of water tables), geologists can base their investigations upon proton NMR signals subjected to the Earth's very low magnet field by using a suitably adapted instrument (the Larmor frequency is a few hundred Hz). It is doubtful that there are any dowsers so gifted that they can detect such a weak signal!

15.17.4 Nuclear Magnetic Resonance Imagery

This other application of NMR involves the acquisition of images. All materials containing exploitable nuclei can be studied, but most often it is limited to the nuclei of hydrogen atoms. Applications range from living organisms to geological layers. This significant development of NMR is applied in the medical field under the name of MRI (Magnetic Resonance Imaging). This technique, now very well known, enables a noninvasive exam that is well adapted to soft tissues. The volume examined must be placed in the magnetic field, which requires the manufacturing of very large superconducting magnets for full-body instruments. The signal given by water (in ^1H NMR) is the easiest to observe because biological tissues consist of about 90% water. In other cases, the signal given by the CH_2 groups in fatty tissues is observed. The final image is a mapped representation of the intensity distribution of a given type of signal. The contrast is related to variations in the relaxation time of protons in the selected plane (Figure 15.32). The technical difficulties specific to three-dimensional NMR include focusing (or spatial selectivity), i.e. obtaining the signal from a nearly isolated volume inside the object.

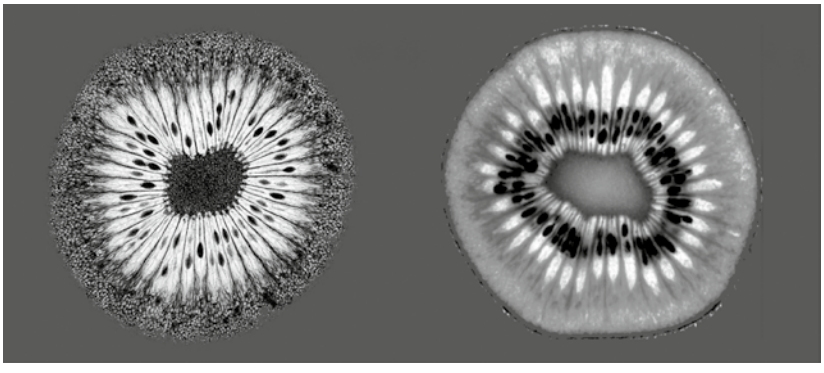


Figure 15.32 Inside view of a kiwi fruit. One of these images is a photo of a kiwi (cut in half) and the other is an image obtained by proton MRI of another kiwi (without having to cut it). The NMR signal varies according to the T_1, T_2, and density characteristics of protons at each point (that photo is on the right).

Table 15.5 NMR chemical shifts of the main types of protons in organic molecules.
(Source: Reproduced by permission of Spectrométrie Spin & Techniques.)

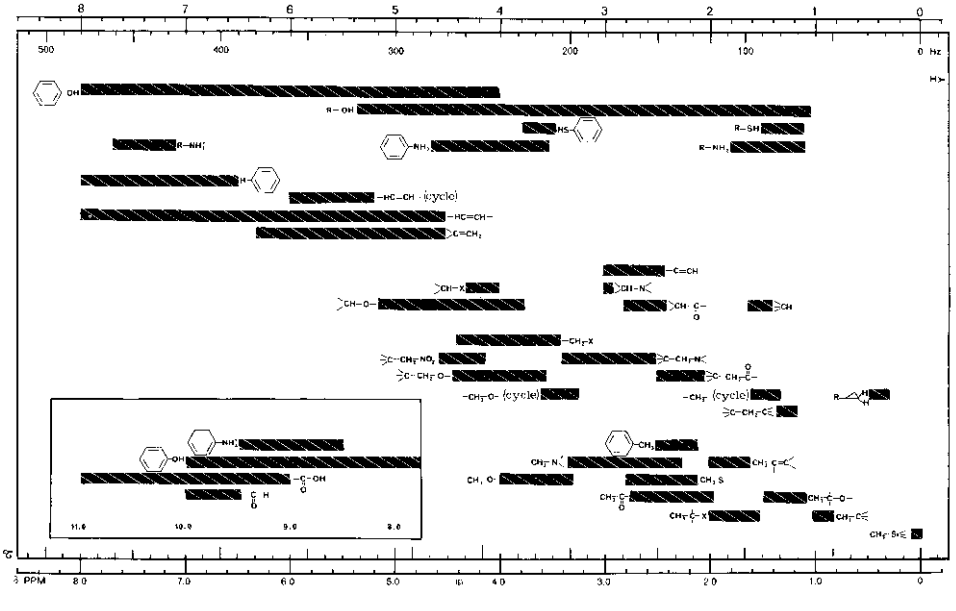
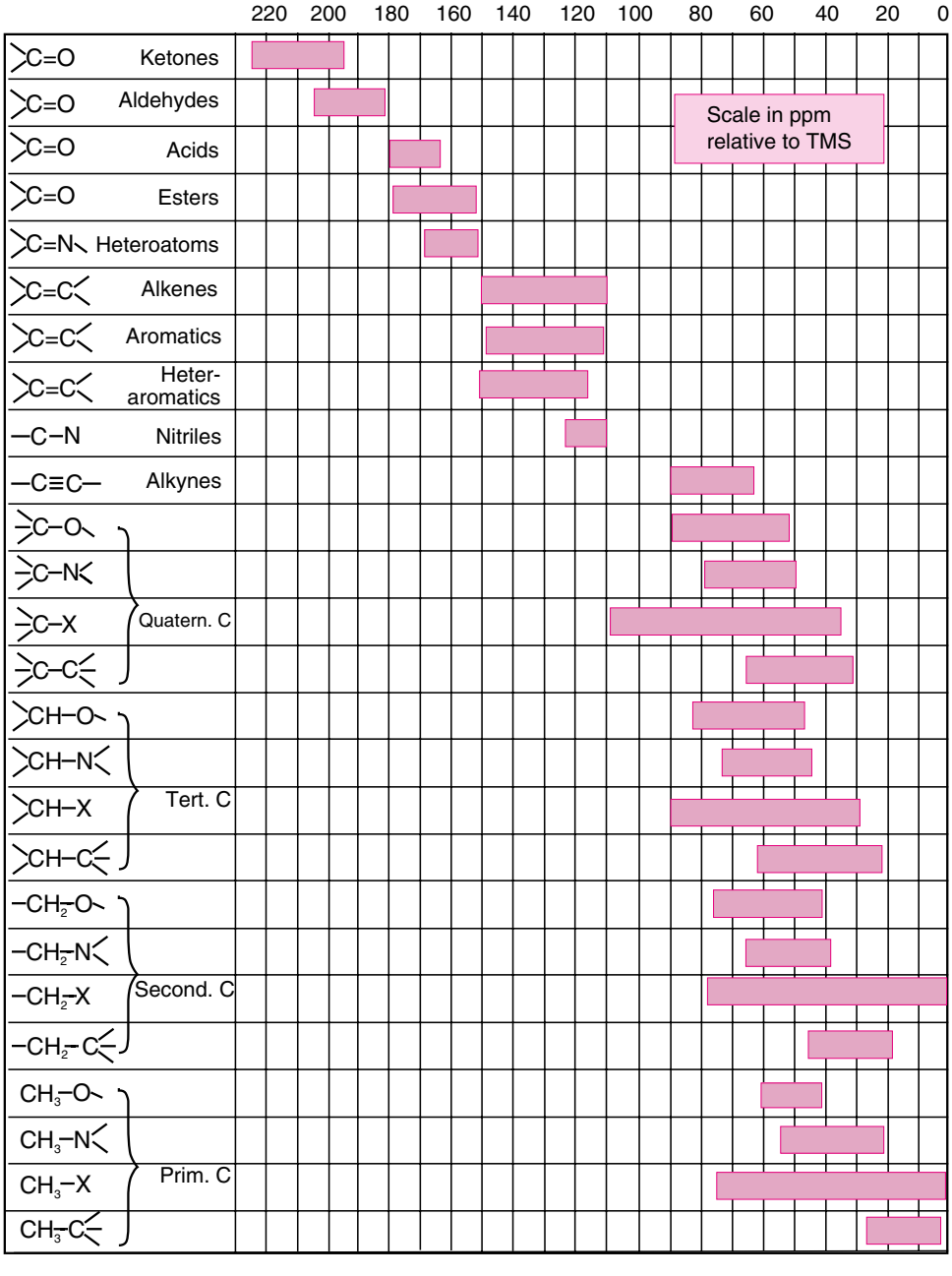


Table 15.6 ^{13}C NMR correlation table of organic functions.



KEY POINTS OF THE CHAPTER

1. NMR relies on the behaviour of the nuclei of certain nuclides when subjected to a powerful fixed magnetic field, B_0 (electromagnet or superconducting magnet, 1–10 Tesla), along with the electromagnetic radiation of specific frequencies, in order to create a field B_1 that is oscillating and orthogonal to B_0.
2. NMR is specific to any nuclide whose spin number I is not zero, thus excluding those whose proton and neutron numbers are both even (in which case I = 0, such as for ^{12}C or ^{16}O).
3. Under the experimental conditions of NMR, absorption signals are obtained, and their positions and intensities, represented as graphs (NMR spectrum), are used to deduce information about the nature and structure of the compound being examined. The hydrogen atom is the most widely studied nuclide. For the most abundant isotope, the nucleus is formed of a single ^1H proton, which is thus referred to as *proton NMR*.
4. In the B_0 field, a nuclide whose spin number is I may occupy 2I +1 levels of energy. Therefore, a homogenous set of protons (^1H), whose spin number is I = ½, will be divided into two populations, where the one with less energy is much less populated. The same goes for ^{13}C, ^{19}F or ^{31}P. This unequal distribution is translated by weak macroscopic magnetization aligned in the direction of B_0 and designated by M_0.
5. **Gyromagnetic ratio**. The energy difference ($E_2 - E_1$) between these two levels is proportional to the B_0 field, and the $(E_2 - E_1)/B_0$ ratio depends only on one constant specific to each nucleus: the *gyromagnetic ratio* γ (for ^1H, if B_0 = 1.4 T, υ is close to 60 MHz). The NMR study of carbon is more difficult, since only ^{13}C isotope (1.1%) has a spin number (I =½) and moreover, γ is about four times smaller than for ^1H.
6. **Larmor frequency**. To identify a nucleus with ½ spin, resonance conditions must be provoked between the two levels. This is achieved if the frequency of the electromagnetic radiation, which generates B_1 perpendicular to B_0, is such that:

$$h\upsilon = E_2 - E_1$$

   This equation, or *Larmor frequency*, establishes a relationship between the magnetic field in which the nucleus in question is found and the action of the brief electromagnetic pulse. Its controlled duration tips M_0 by a predetermined angle, whose return to the initial state (nuclei relaxation) provides information on the structure of the compound in question.
7. **Relaxation**. After the pulse, M_0 returns to its initial position (on the Oz axis) according to an exponential law defined by T_1 (spin–lattice interaction), which is a variable time depending on the medium, on the order of a few seconds. This corresponds to the *longitudinal relaxation,* in order for the magnetization,

according to Oz, to return to 63% of its primitive value. The M_0 component in the Oxy plane quickly disappears (spin-spin interaction). It is defined by the transverse relaxation time T_2, which corresponds to the loss of 63% of its initial value.

8. **Chemical shift**. The exploitation of NMR consists in precisely studying small shifts in the resonance frequency of a given nuclide depending on its local environment. In fact, any nuclide is part of a structure (e.g. a molecule). Its close environment (chemical bonds, other atoms, etc.) modifies the value of the field B_0 which reaches it. This is called *shielding*. The effective field B_{eff}, reaching the nucleus, thus depends on a shielding constant σ which is the cause of the *chemical shift*.

$$B_{eff} = B_0 \cdot (1 - \sigma)$$

9. Chemical shifts are plotted on a relative scale, independent from the instrument, with tetramethylsilane (TMS) as an internal reference, defining point zero on the scale. For organic compounds, there are tables of chemical shifts (scale $\delta = \Delta\nu/\nu_{app}$), which are valid for a type of nucleus (e.g. ^1H or ^{13}C) in relation with the local structure of the compounds in question. The effects that may disturb B_0 include inductive, anisotropic, resonance, solvent and matrix effects.
10. **Spin–spin coupling**. All atoms in the sample with a nonzero spin react to the effect of the B_0 field. In solution, the orientation taken by a nucleus reverberates on neighbouring nuclei via bonds. This magnetic interaction, called homonuclear or heteronuclear coupling (scalar coupling), depending on whether it involves the same type of nucleus or not, leads to finely structured signals (characteristic multiplets) that can be interpreted with simple statistical rules (Pascal's triangle) for weakly coupled systems.
11. The interaction between spins of weakly or strongly coupled systems is defined by the value of the coupling constant J (in Hz), which depends on the nuclei, on the distance separating them and on the structure of the compound, but which is independent from B_0.
12. By means of the various sequences based on the intensity of B_1, NMR spectrometers enable determination of spin-spin and spin-lattice relaxation times and the temporal sequencing of impulses, providing additional information about unknown compounds.
13. **Two-dimensional NMR (2D-NMR)** provides additional information about the structure of molecules through the study of interactions, either through bonds (COSY sequence), through space (NOESY sequence) or between heteroatoms (HSQC sequence).
14. **Applications**. There are many application in laboratories or in industry, in organic chemistry and in biochemistry for NMR in solution (^1H and ^{13}C) with a large selection of instruments ranging from compact and miniaturized field analysers with permanent magnets (1–2 T) to high-field laboratory spectrometers (up to 20 Tesla).

PROBLEMS

1. In tables of nuclides, the values describing their nuclear magnetic moment are often represented as lying upon an axis parallel to $\vec{B}_0$, the applied magnetic field. Therefore, if for a proton, $\mu_z = 1.41 \times 10^{-26}$ J · T^{-1}, calculate its constant γ.
2. If $T = 300$ K, calculate the ratio of the populations N_{E1}/N_{E2} for a proton in an NMR spectrometer where the applied magnetic field $B = 1.4$ T. Make the same calculation for the case where the field $B = 7$ T, given that $\gamma = 2.6752 \times 10^8$ rad · T^{-1} · s^{-1}.
3. In the ^{1}H NMR spectrum from a 200-MHz spectrometer, the scale along the *x*-axis is represented by 1 ppm = 4 cm.
   a. What is the distance between two signals with a separation of 7 Hz?
   b. If it is known that $\gamma_H/\gamma_C = 3.98$, what would happen to the distance calculated above if a ^{13}C spectrum were to be recorded on the same apparatus?
4. a. Calculate the chemical shift, δ (in ppm), of a proton (^{1}H) whose NMR signal is displaced by 220 Hz with respect to TMS (the magnetic induction field of the spectrometer is 1.879 T).
   b. The resonance signal for a proton is displaced by 90 Hz with respect to TMS when measured on a 60-MHz spectrometer. What would happen to this offset if a 200-MHz apparatus is employed?
   c. What would be the corresponding chemical shift (δ, ppm) of the same proton with both of these spectrometers?
5. If the ratio of the gyromagnetic constants γ_F/γ_H is 0.9413, calculate the distance which would separate the TMS signal from that due to a fluorine atom, given that the spectrum scale is represented by 1 ppm = 2 cm. (200 MHz, ^{1}H spectrometer.)
6. Two isomers A and B share the same molecular formula, $C_2HCl_3F_2$.
   The ^{1}H NMR (60 MHz apparatus) spectrum of isomer A comprises a doublet at 6.2 ppm ($J = 70$ Hz and $J = 7$ Hz), while that of isomer B contains a triplet at 5.9 ppm ($J = 70$ Hz).
   a. What is the structure of each of these isomers?
   b. Consider now that a third isomer is present. Under the existing conditions, describe what its spectrum would be.
7. Find the molar mass of salicylic aldehyde from the following information:
   25 mg of vanillin ($C_8H_8O_3$) is added to 20 mg of salicylic aldehyde and the ^{1}H NMR spectrum of the mixture is recorded. On the integration curve associated with the ^{1}H NMR spectrum of the new mixture, it can be observed that heights of levels corresponding to the aldehydic proton of vanillin and to that of salicylic aldehyde are each essentially equal to 20 mm. Remember H = 1, C = 12, O = 16 g/mole.

8. In the circles, place the letters corresponding to the protons of the molecule whose NMR spectrum is represented below (Figure P15.1).

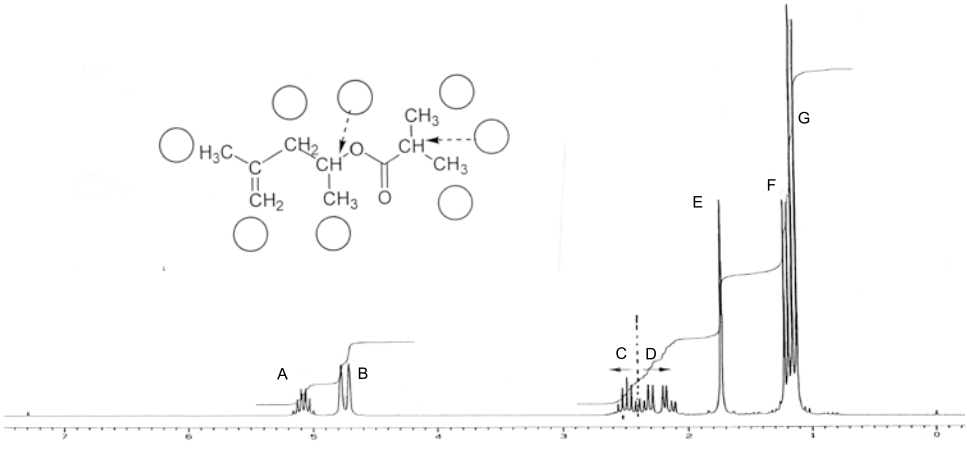


Figure P15.1

9. On the NMR (^{1}H) spectrum of a raw product isolated after chemical reaction, two signals are observed, one of which corresponds to a CH_3 group belonging to the remainder of the initial compound (area: 3,450 in arbitrary units) and the other corresponds to a CH_2 group (area: 12,500) from the product formed. Deduce the yield of this transformation.

SOLUTIONS

1. $\gamma = 4\pi\mu_z/h$; $\gamma = 2.674 \times 10^8$ rad · T^{-1} · s^{-1}.

2. $N_{E(1)}/N_{E(2)} = e^{\Delta E/kT}$; 1.0000095. If B_0 = 7 T, this ratio is equal to 1.0000477.

3. **a.** 7 Hz corresponds to 1.4 mm.

b. If the instrument, for ^{1}H, operates at 200 MHz, for ^{13}C, the frequency is $\nu_C = \nu_H\gamma_C/\gamma_{H,}$ i.e. $\gamma_C = 200 \times 1/3.98 = 50$ MHz; 7 Hz will correspond to 5.6 mm.

4. **a.** δ = 2.75 ppm.

b. 300 Hz.

c. Chemical shifts in ppm are invariable.

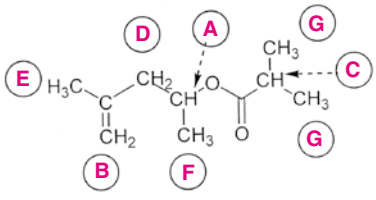
5. The resonance frequency of ^{19}F: 188.255 MHz. Therefore between the two nuclei, there is 11.745 MHz. The distance between signals will be 1,174.5 m.

6. **a.** A: $CHFClCFCl_2$ (dd, J = 70 and 7 Hz) B: CHF_2CCl_3 (t, J = 70 Hz);

b. C: $CHCl_2CF_2Cl$ (t, J = 7 Hz).

7. On the spectrum of the mixture, the two signals are at the same height and each corresponds to one proton, which means that the number of molecules is also the same. Since the calculated molar mass of vanillin is 152 g, by calling x the molar mass of salicylic aldehyde, we get:

$$152/25 = x/20, \text{ thus } x = 121.6\,\text{g}$$



8. By comparing the areas to those of a single proton for the two compounds, we get: starting compound 3,450/3 = 1,150 and product formed 12,500/2 = 6,250. The ratio 6,250 / (1,150 + 6,250) = 0.84 represents the proportion of transformation, i.e. a yield of 84%.

Chapter 16

Mass Spectrometry

Mass spectrometry (MS) is based on the determination of atomic or molecular masses of individual species present in a sample in order to characterize their composition formulae and their structures and to quantify them. For that reason, spectrometers, often placed downstream from a separation technique, subject analytes, after ionization, to electrical and/or magnetic fields under a vacuum, in order to separate the ions formed according to their mass/charge ratio. The scope of this analytical technique is quite vast owing to its versatility

and its extreme sensitivity. Mass spectrometry is an essential investigation method for structured compounds in organic chemistry, biochemistry (peptides, proteins) or inorganic chemistry (ICP/MS technique). Inline GC/MS or HPLC/MS coupling are some of the best analytical methods when there are only very small quantities of complex samples.

Objectives

Review the principle of mass spectrometry
Describe the composition of a mass spectrometer
Explain ionization methods of molecules
Compare vacuum and atmospheric pressure ionizations
Review laws determining the action of electromagnetic fields on an ion
Describe the various designs of MS as pertaining to their performance
Review a few applications of mass spectrometry
Describe the fragmentation principle
Determine the molecular formula of a compound
Identify a compound by means of MS

16.1 BASIC PRINCIPLES

16.1.1 Operating Principle of a Mass Spectrometer

Mass spectrometry is based on the determination of masses of molecules or atoms present in the studied sample. To reach this result, a minute quantity of sample, transformed into a suitable form, is ionized (via bombardment with electrons, atoms, photons, etc.) and the resulting charged species are then submitted, in an enclosed space maintained under high vacuum, to the action of an electric and/or magnetic field, depending on the type of instrument. The study of the forces exerted upon the ions allows the determination of their *mass-to-charge ratio* and thus possibly their nature.

By way of example, the ions resulting from a sample of methanol (CH_3OH) transformed to the gaseous state and then ionized by bombardment with electrons, will include a small amount of positively charged $CH_3OH^{+\bullet}$ ions. These ions, formed in an excited state, possess a surplus of internal energy that will induce an almost immediate fragmentation of many of them. They do not, however, all dissociate by the same pathways, thus leading to different ion fragments whose mass is less than that of the original methanol molecule. Generally these fragments, formed by simple bond cleavage and from subsequent rearrangement reactions, provide information about the structure of the original molecule (Figure 16.1).

The results are displayed as a graph called a *mass spectrum*, presented as the abundances of the ions formed and classified in increasing order of their mass/charge

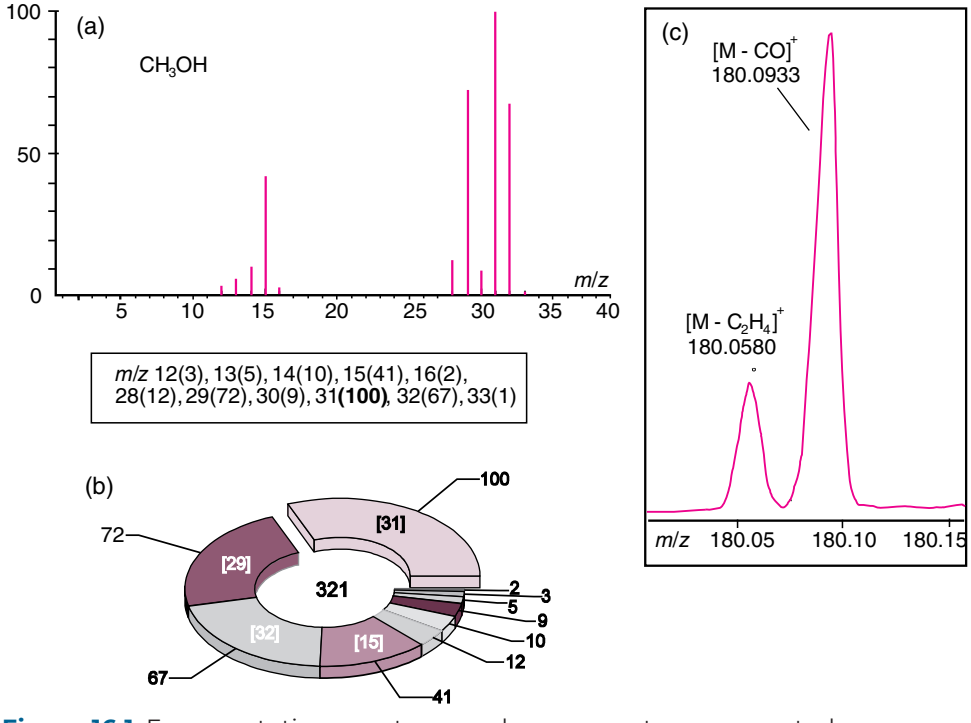


Figure 16.1 Fragmentation spectrum and mass spectrum presented in graphical or tabular form. (a) Fragmentation spectrum of methanol. (b) Unconventional representation of the same spectrum in the form of a circular diagram (hollow pie chart): statistically; for 321 ions formed, there are 100 of mass 31 u (Da), 72 of mass 29, etc. The various ions constitute many different populations. (c) Section of a high-resolution recording of a compound M presenting two ions with very close m/z ratios (one due to loss of CO and the other due to loss of C_2H_4).

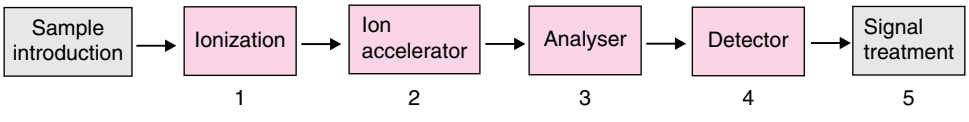


Figure 16.2 Diagram of mass spectrometer components.

ratio (Figure 16.1a). When operating under identical conditions, the fragmentation is reproducible and therefore is a means of characterizing the studied compound.

The sample is subjected to a succession of steps (Figure 16.2).

1. *Ionization:* the sample in the form of gas or vapour is ionized in the *source* of the instrument. Many processes can be used for this first step. At this stage, all molecules lead to a statistical distribution of fragment ions.

2. *Acceleration:* once formed, the ions are immediately extracted from this part of the instrument to be *focused and accelerated* by a series of electronic lenses, in order to increase their kinetic energy.
3. *Separation:* the ions are then "filtered" by the *analyser* according to their mass-to-charge ratio. Some instruments combine several mass analysers in series.
4. *Detection:* after separation, ions terminate their flight by striking the sensor of the *detector* whose signal is proportional to the charges of ions received.
5. *Display of the mass spectrum* obtained from treatment of the signal sent by the detector.

This method only yields the mass-to-charge ratio of the ions, m/q. Logically, to calculate m, q must be known. Since these ions carry a net charge of type $q = ze$ (where e represents the elementary charge of the electron and z is a small whole number), it is said that m can be determined to the nearest factor z. This is why the scale of the spectrum is indicated as m/z (m being in atomic mass units). For molecules of small or medium size *(M < 1,000)*, the ions generally all carry a single charge *(z = 1)*, and thus the increasing order of masses is the same as that of the m/z ratio. This assumes that the spectrum is directly graduated in atomic mass units (u or amu) and whose value is by definition 1/12th of the mass of ^{12}C (1.660538×10^{27} kg). For high masses, which are regularly encountered in biochemistry, masses are often expressed in Daltons (Da), instead of u, even though this unit is not widely accepted by IUPAC. m/z is sometimes expressed in Thomsons (Th).

The expression *mass spectrum* must not lead to confusion: the apparatus does not produce a *spectrum* in the most usual sense of the term, as seen in methods based upon the interaction between a sample and light radiation. The origin of this expression arises most probably from the similarity between the recordings obtained from the first instruments using photographic plates and those of spectrographs, which enabled exploration of the optical spectrum on a screen. Mass spectrometry is not an optical spectroscopy (spectrophotometric) method.

The mass spectrum can be recorded in two very different ways:

- The *continuous spectrum* (or profile) of the selected mass interval. The graph corresponds to a collection of signals in the form of peaks of different widths, depending upon the quality of the instrument (Figure 16.1c). These peaks, distributed as a function of their masses, lead to the determination of the ion masses with a precision reaching 10^{-5} u for the best instruments. The upper mass limit, which is constantly being improved, can exceed 10^6 u.
- The *fragmentation spectrum* (bar spectrum or stick plot). This corresponds to the distribution of all the ions formed, grouped by their *nominal mass* (see Table 16.1) as close as possible to their exact masses and presented in the form of vertical lines. The most abundant type of ion leads to the most intense peak, called the *base peak*, which is given 100% intensity. The intensities of the other peaks are then expressed as percentages of the base peak. This graphical representation of masses distributed by population, whose heights are proportional to their

Table 16.1 Atomic mass of the isotopes of the main elements commonly encountered in organic compounds.

Element	Nominal mass*	Atomic mass (g)	Nuclide (%)	Mass (u)
Hydrogen	1	1.00794	^{1}H (99.985)	1.007825
			^{2}H (0.015)	2.014102
Carbon	12	12.01112	^{12}C (98.90)	12.000000
			^{13}C (1.10)	13.003355
Nitrogen	14	14.00674	^{14}N (99.63)	14.003074
			^{15}N (0.37)	15.000108
Oxygen	16	15.99940	^{16}O (99.76)	15.994915
			^{17}O (0.04)	16.999131
			^{18}O (0.20)	17.999160
Fluor	19	18.99840	^{19}F (100)	18.998403
Sulfur	32	32.066	^{32}S (95.02)	31.972071
			^{33}S (0.75)	32.971458
			^{34}S (4.21)	33.967867
Chlorine	35	35.45274	^{35}Cl (75.77)	34.968853
			^{37}Cl (24.23)	36.965903

(*) **The nominal mass of an element** is the entire mass of its most abundant natural, stable isotope. **The nominal mass of an ion** is the sum of the nominal masses of the atom present in its empirical formula (e.g. HCl^{+} = 36 u).

abundances, is essentially a histogram (Figure 16.1a). Bar spectra such as these are easy to archive and can be used for further comparison and identification. This normalized display, which is incidentally the most commonly used graph in MS analysis, does however have the drawback that two or more ions of different atomic composition can occur at the same nominal mass.

The identification of molecular compounds by mass spectrometry can be achieved from either of the two types of spectral diagrams:

- Attempting to reconstitute the structure of the compound in the manner of a puzzle. This is an exercise which becomes more difficult the higher the mass is. Successive fragmentations of the ions make identification more complex. However, this task becomes simpler
  if several spectra recorded under different conditions are available with appropriate software to help in their interpretation.
- Using a spectral library comprising a large number of fragmentation spectra amongst

16.1.2 Mass Spectrometer Performance Levels

Upper mass limit

All mass spectrometers can determine the *m/z* ratio up to a maximum value. The greatest mass, in daltons, will depend on the number of charges *z* that the ion carries.

For example, if the spectrometer is able to measure *m/z* ratios up to 2,000, it will be possible to detect an ion of mass 80 kDa carrying 40 elementary charges ($q = 40e$).

Sensitivity

The sensitivity of a mass spectrometer is given by the weight of sample consumed per second (e.g. a few pg/s or femtomole/s) in order to obtain a signal of normalized intensity. There is constant renewal of the ions, since the analyte is gradually consumed.

The extreme sensitivity of spectrometers combined with the capacity that detection offers, based on the identification of ions by their mass, has enabled the considerable increase in performance of other analytical methods for organic (GC/MS, HPLC/MS, thermogravimetric analysis/MS) or inorganic (ICP/MS) compounds.

Resolving power

On spectra with Gaussian peaks (obtained with a sector, time-of-flight or cyclotron MS instrument), neighbouring masses become more easily distinguished as the corresponding peaks become narrower. This important property is at the origin of a parameter called the *resolving power R*, for which there are several definitions.

To calculate *R*, the *m/z* value of the peak selected is divided by its width $\Delta(m/z)$ either at the mid-height (Full Width at Half Maximum, FWHM) for an isolated peak or at 5% of the height when two neighbouring peaks are being considered (Figures 16.3 and 16.4).

$$R = (m/z)/\Delta(m/z) \tag{16.1}$$

or

$$R = M/\Delta m \tag{16.2}$$

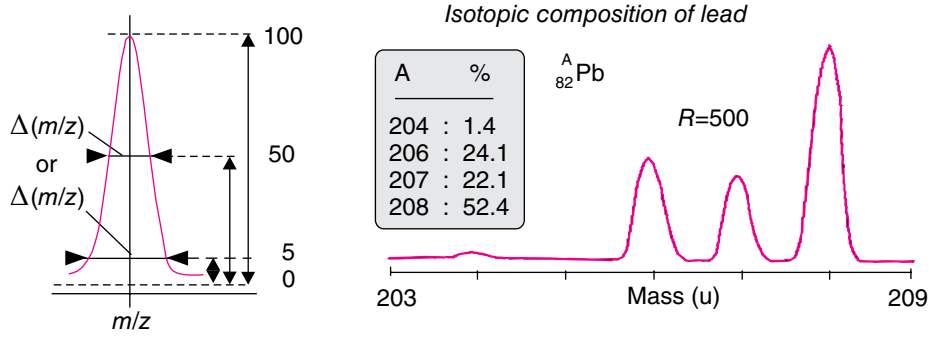


Figure 16.3 Resolving power. Left, definition of this parameter in the case of an isolated peak. Depending upon the manufacturer, the peak width is measured either at 50% or at 5% of its height. Right, example of a low-resolution spectrum of a sample of lead. The value of *R* found depends very much on the compound and the mass chosen.

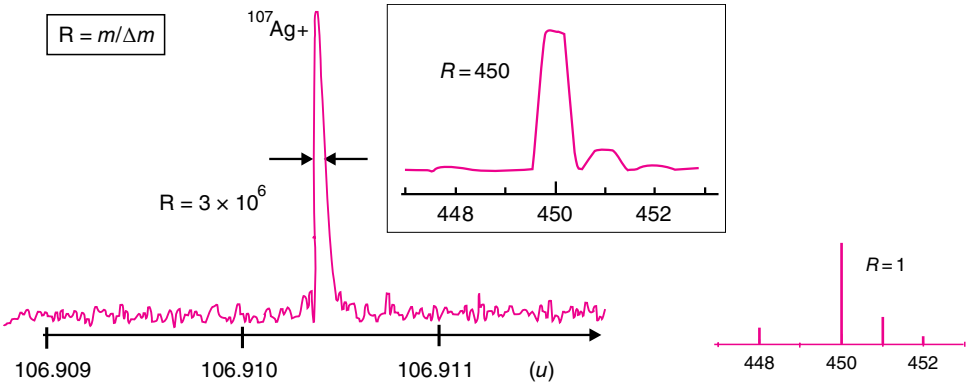


Figure 16.4 Resolving power established using experimental spectra. Left, an example of a recording corresponding to a very high resolving power (ion cyclotron resonance apparatus). Centre, a section of a crude signal for a quadrupole mass filter (R = 450). Right, a more conventional presentation of the same recording (resolution: 1 u).

Resolution is an essential parameter, conditioning the use of the mass spectrometer. *Low-resolution* instruments (R < 5,000) acquire mass spectra in the form of sticks. Furthermore, it can be said that for such a spectrum the resolution (and not the resolving power) is exactly 1 u (for example, distinguishing peak 28 from peak 29, or peak 499 from peak 500, etc.). The raw signal for a compound from an instrument of this category yields a plot constituted of broad, flat-topped peaks (Figure 16.4). However, with *high-resolution* instruments (R > 5,000), which give the mass of ions with a precision of better than 0.001 u, it becomes possible to differentiate neighbouring masses. Therefore, NH_4^+ (18.034 u) and H_2O^+ (18.011 u) ions, whose masses are close to 18, may be identified unambiguously with this type of instrument.

16.2 SAMPLE INTRODUCTION

There are numerous methods for introducing samples into mass spectrometers, both because of the variety of forms that samples can come in and because certain ionization processes have their own specific requirements.

16.2.1 Direct Injection

For gases or liquids, a very small quantity of sample is introduced with a microsyringe into a kind of tank linked to the ionization chamber via a very narrow channel. Under the effect of a high vacuum that is maintained in the tank, the compound is

sucked up and vaporized. This procedure is known as a *molecular leak*. For applications requiring permanent monitoring, the introduction may be done continuously by means of a bypass inlet of the instrument. Gases and volatile compounds (VOC) in the vapour state or dissolved in a liquid such as water can be diffused through a porous membrane.

If the sample is a solid that can pass into a vapour state under vacuum, it is placed on the end of a metal support (heated or not), which slides into the instrument source. For some other modes of ionization, the solid sample is mixed into a matrix (e.g. glycerol or benzoic acid).

16.2.2 Sample Introduction in Coupled Techniques

When an analyte mixture must be quantified by mass spectrometry, it is frequent to subject the sample beforehand to a separation method (chromatography or electrophoresis) in order for analytes to arrive separately in the MS instrument (Figure 16.5).

This type of coupling between two techniques has become very common. The interface is more or less complex, depending on the upstream method selected. In the case of GC/MS coupling, the outlet of the capillary column leads directly to the ionization chamber. The separated analytes arrive in gaseous form, mixed with the carrier gas, whose relatively low flow rate (1–2 ml/min) is compatible with the pumping capacity of the spectrometer, which must maintain the vacuum required for analysis.

HPLC/MS interfacing is more complex, owing to the presence of elution solvents, which often contain water. To maintain a vacuum between 10^{-3} and 10^{-4} Pa in the MS chamber, the solvent must be eliminated, generally by evaporation, which corresponds to a significant gas volume. Therefore, columns with small diameters are favoured. Finally, for compounds of high molecular masses, there are specific interfaces (see Section 16.4).

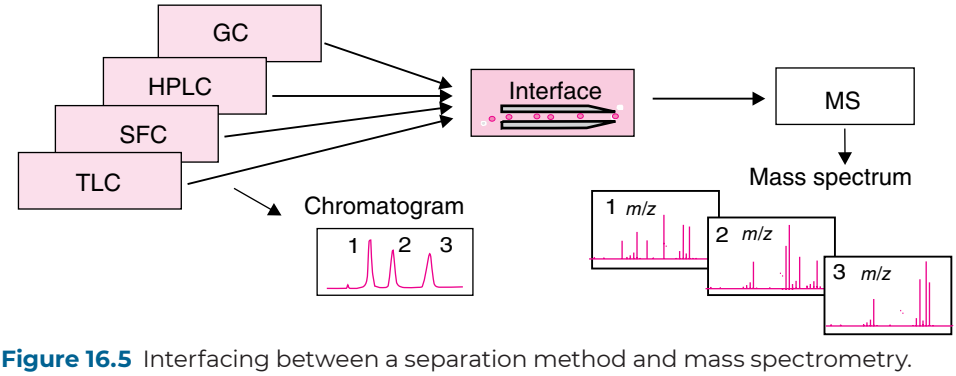


Figure 16.5 Interfacing between a separation method and mass spectrometry.

More rarely used interfaces between thin-layer chromatography or capillary electrophoresis and mass spectrometry (TLC/MS and HPCE/MS) have been addressed previously (see Chapter 5, Section 5.1.4 and Chapter 8, Section 8.3.2).

16.2.3 Pumping Systems

Pumping systems play an essential role in mass spectrometry. They have multiple roles. First and foremost, they ensure a thorough vacuum in the enclosure of the analyser, thus preventing the collision of the analysed ions with the molecules coming from the residual gas. The vacuum in the analyser must be particularly strong when the path followed by the ion (or its lifespan) is long. Therefore, for a path of several metres, a 10^{-4} Pa vacuum is necessary. A strong vacuum also increases the sensitivity of the instrument by limiting the impact of parasitic gas molecules on the detector.

As the path of ions in the analyser is travelled under a strong vacuum, the spectrometer has a combination of two pumps in series to obtain the necessary vacuum in two successive stages (Figure 16.6). The first level ensures a primary vacuum (predraining) to move from atmospheric pressure to a pressure of 1–10 Pa. To do so, a *vane pump* is used.

Then, the second vacuum stage is obtained by adapting a *turbomolecular pump* to the inlet of the previous pump. The typical pump includes a set of rotors, rotating at high speed (1,000 rotations/s), equipped with fins ejecting residual gas molecules towards the inlet of the primary pump (Figure 16.7). These pumps have replaced *oil diffusion pumps*, whose vapours progressively polluted the analyser.

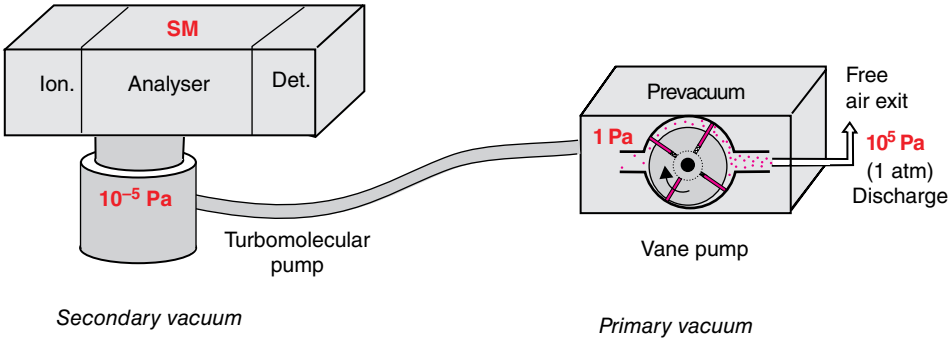


Figure 16.6 Installation of a pumping system of a mass spectrometer. The turbomolecular pump is directly attached to the spectrometer's analyser through its large opening, while the prevacuum pump is generally installed at a distance to eliminate possible vibrations.

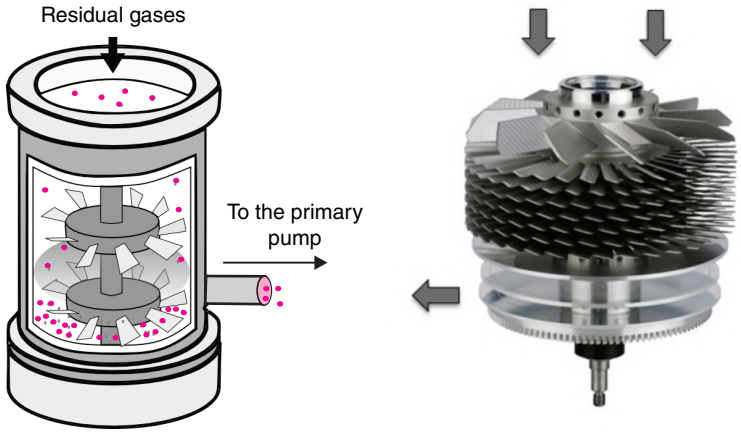


Figure 16.7 Turbomolecular pump. Basic diagram. The pumping effect is obtained by alternating high-speed rotors and stators (fixed). The illustration on the right shows the stacking of the rotors/stators in a pump of this type (Source: Edwards Company).

16.3 MAIN VACUUM IONIZATION TECHNIQUES

Analysis by mass spectrometry begins with the ionization of a compound into individual, entire or fragmented species. The choice of the ionization method depends on the study in question and on the nature of the compound, which could be a small organic molecule, a metal, a biological macromolecule, etc. and upon its state (gas, liquid or solid). The corresponding devices are complex and often installed during assembly of the spectrometer itself, yielding specialized instrumentation.

16.3.1 Electron Ionization (EI)

Electron ionization is the method most commonly used for the analysis of volatile compounds. This is the case for organic molecules. It involves ionization in the gas phase, which occurs in the ion source by the collision of the neutral molecules in the sample and electrons emitted from a filament by a thermionic process (Figure 16.8). The ejection of one of the most weakly held electrons leads to positive ions (formation of a radical cation $M^{+\cdot}$)

This process is reproducible, which helps identify a compound by comparing its spectrum to all those in a spectral library, at least if these contain the spectrum for this compound. Via other mechanisms, negative ions also form in a lesser quantity. To study the latter, the polarities of the electrodes must be reversed, as well as the current direction in the electromagnet, to conserve the trajectory of the ions between the source and the detector.

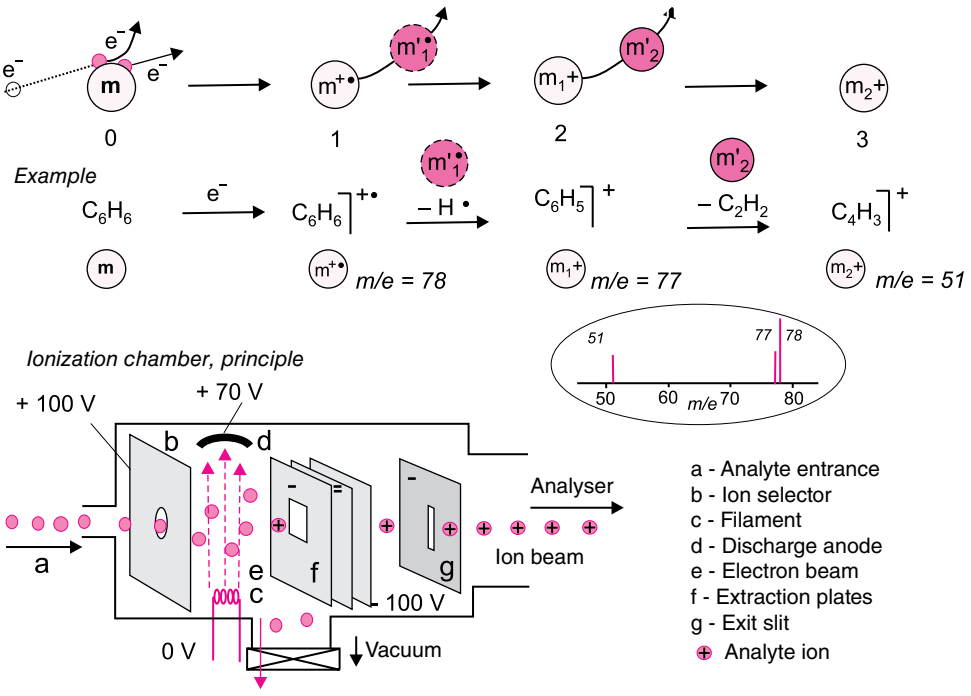


Figure 16.8 Electron ionization. The collision of an electron with a molecule m creates a parent ion and fragment ions m^+_1 and m^+_2 in direct succession. The neutral fragments m'_1 and m'_2 pass undetected. An illustration for the case of benzene is shown. Below, the diagram of an ionization chamber, also known as a collision chamber (or ion source). If a magnetic field is imposed parallel to the electrons, those electrons will follow a spiral movement that enhances the efficiency of this electron gun.

The standard energy of ionization, usually 70 eV, is obtained by accelerating electrons by a potential difference of 70 V. These ionization conditions yield fragmentation spectra presenting the greatest reproducibility and thus a more certain identification by comparing the spectrum found with those in a spectral library.

The efficiency is about one ion produced for every 10,000 molecules. This is a very high-energy process, leading to fragmentation of molecules. To increase the relative intensity of the molecular peak $M^{+}\bullet$, a weaker ionization energy must be selected, such as 15 eV for example (Figure 16.9).

A relatively explicit nomenclature is used to define the origin of the ions formed: monocharged, multicharged, monoatomic, polyatomic, molecular, pseudo-molecular, parent, daughter, fragments, secondary (resulting from filiations after ionization), metastable ions, etc.

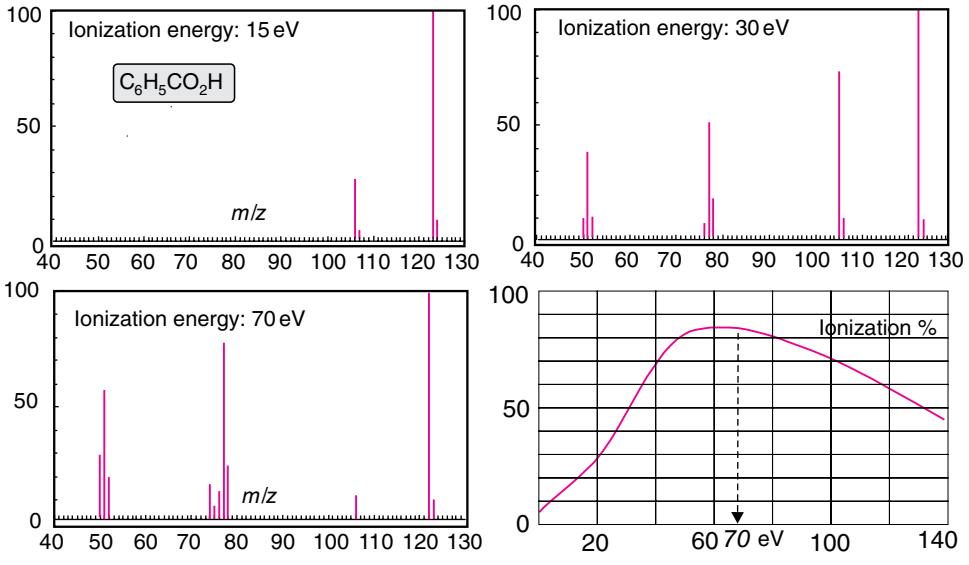


Figure 16.9 Influence of electron energy on fragmentation. An example with benzoic acid is shown here. A graph of the ionization efficiency as a function of electron energy is presented. Note that the spectra are normalized, i.e. the most intense peak takes a value of 100. Some claim that this is an application of Procrustes' method (according to the Greek legend, the brigand Procrustes forced his prisoners to lie on a bed and then, depending on their size, he would either cut their feet or stretch their bodies so they would fit the dimensions of the bed).

16.3.2 Positive Chemical Ionization (CI)

Chemical ionization results from the reaction between the molecules of analyte M and the ions obtained through the electron bombardment of a reagent gas, such as methane, ammonia or isobutane introduced concomitantly with the analyte into the instrument source. The relatively high pressure (a few hundred Pa) that exists in this section of the instrument reduces the mean free path of the molecules, thus favouring collisions. This is a 'gentle' ionization process.

Both positive and negative species are produced. In particular, the formation of the pseudo-molecular $MH]^+$ ion (which is not a radical ion) is observed. Its formation depends upon the affinity of M for H^+. This ion has a lesser tendency to fragment than the molecular ion $M]^{+\cdot}$ obtained by electron impact. This is useful for getting the value of M, for example, with methane (Figure 16.10).

However, for compounds of the RH type, the R^+ ion is also observed (M – 1), even though chemical ionization always leaves a doubt concerning the value of the molecular mass of the compound studied.

Primary reaction: $CH_4 + e^- \longrightarrow CH_4^{+\bullet} + 2\,e^-$ *(Ionization)*

Secondary reactions:

$CH_4^{+\bullet} \longrightarrow CH_3^+ + H^\bullet$

$CH_4^{+\bullet} + CH_4 \longrightarrow CH_5^+ + CH_3^\bullet$ *(Autoprotonation)*

$CH_3^+ + CH_4 \longrightarrow C_2H_5^+ + H_2$

Collisions with M:

$CH_5^+ + M \longrightarrow MH^+ + CH_4$ (ion at M+1)

$C_2H_5^+ + M \longrightarrow MC_2H_5^+$ (ion at M+29)

If M is of the RHtype: $CH_5^+ + RH \longrightarrow R^+ + CH_4 + H_2$ (ion at M – 1)

If M is a tert-butyl cation $(CH_3)_3C^+ + M \longrightarrow MH^+ + CH_2{=}C(CH_3)_2$

Figure 16.10 Chemical ionization. Formation of cationic species from methane and reaction on the molecules M of the compound. The last equation corresponds to the action of the tert-butyl cation, resulting from isobutane through the loss of the tertiary proton. The +• symbol means that it is both a radical (odd number of electrons) and a cation.

NH_4^+ behaves like CH_5^+. Isobutane easily yields the stable $(CH_3)_3C^+$ ion, which leads to the ion (M + 1) by transforming itself into isobutene.

Negative chemical ionization. Other secondary reactions may give rise to negative species, such as the capturing of an electron by means of the analyte's electron affinity, or by reaction between the neutral reagent gas and its ionized form, appearance of CH_3^- or NH_2^-. For example, molecule M will lead to a *pseudo-molecular anion* $[M\text{-}H]^-$: $CH_3^- + M \rightarrow [M\text{-}H]^- + CH_4$

16.3.3 Fast Atom Bombardment (FAB)

This process and the following sections are reserved for polar, low-volatility compounds under vacuum conditions. Ionization is achieved through the impact of fast, nonionized heavy atoms (Ar or Xe) targeting the sample that has been previously dispersed in a nonvolatile liquid matrix (glycerol or diethanolamine) and deposited into the source at the end of a sample probe.

To produce this fast atom bombardment, argon or xenon gas is first ionized by electron impact. These primary ions are then directed towards a collision chamber where they propel slow-moving neutral atoms of this gas (Figure 16.11a) before being eliminated from the main beam by deflection with an electric field.

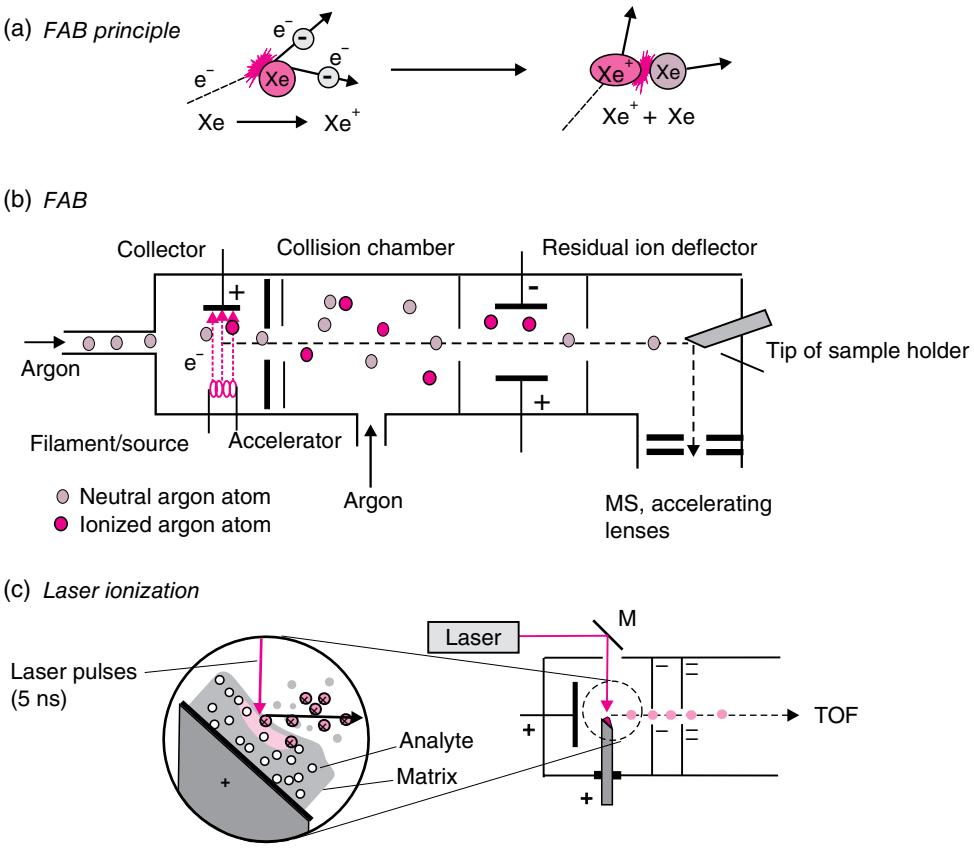


Figure 16.11 FAB and MALDI techniques. (a) The principle of formation of a fast, neutral atom of xenon by collision. (b) The formation of fast atoms of argon in a collision chamber and subsequent bombardment of the sample (*FAB* gun). (c) MALDI (*Matrix-Assisted Laser Desorption Ionization*). The impact of the photon is comparable with that of a heavy atom. Through a mechanism, as yet not fully elucidated, desorption and photoionization of the molecules are produced.

The sample, on the probe, is ionized when bombarded by the fast, neutral atom beam. This FAB (*Fast Atom Bombardment*) technique also causes ionization of the matrix, leading to fairly large background noise, which makes it impossible to study low-MW ions.

Similar to the FAB technique, the bombardment of the target with Cs^+ or Ar^+ ions instead of neutral ions can be referred to as the SIMS technique (*Secondary Ion Mass Spectrometry*). These ions, which can be accelerated, enhance the number of energy fragmentations.

16.3.4 Matrix-Assisted Laser Desorption Ionization (MALDI)

This ionization–desorption mode leads to weak fragmentation. The analyte to be studied is first incorporated into a solid organic matrix (dihydroxybenzoic acid for example) and next deposited upon a sample holder that is irradiated with UV laser pulses (e.g. N_2 laser, λ = 337 nm, 5 ns). The thermal energy received is transferred by the matrix to the compound, which is desorbed and gently ionized in the gaseous state (Figure 16.11c). This process extracts ions in the gas phase from the condensed phase.

Pulsed ionization is particularly well suited to time-of-flight instruments for biomacromolecules, since it yields ions carrying only a single charge, in contrast to atmospheric pressure techniques (see following paragraph).

The analysis of small molecules (M < 500 Da) is limited because of the ions produced as the matrix decomposes. However, other supports based upon porous silica can be used instead of the matrix and therefore make possible the study of these small molecules.

16.4 ATMOSPHERIC PRESSURE IONIZATION (API)

Many new applications of mass spectrometry concern either assays of elements or the study of unstable polar biomolecules (M > 2,000 Da). These applications have become possible with the advent of *atmospheric pressure ionization* (API) techniques applicable to compounds in solution following nebulization.

16.4.1 Spray Ionization

Generally placed at the outlet of a capillary electrophoresis or microcolumn liquid chromatography instrument, these very sensitive devices start by transforming the liquid mobile phase into a fine aqueous mist containing the species to analyse. This mobile phase can have a variable concentration of H^+ ions, depending on the pH, and can also contain cations such as NH^+_4, Na^+, and K^+ (in the case of an electrolyte).

Several techniques are available.

Electrospray ionization (ESI) or ion spray

Very small droplets are formed at the end of a thin silica capillary whose surface has been metallized and which is held at a high positive potential (assuming a study of positive ions). The intense electric field confers a large charge density (z/m) on these droplets (Figure 16.12). Under the effect of a dry gas, the droplets evaporate gradually by losing solvent molecules via complex desolvation and evaporation mechanisms. As the droplets evaporate, the charge density at their surface greatly increases (Rayleigh limit). They explode by liberating nonfragmented ions consisting of analyte molecules carrying a number of variable charges (on average one elementary charge for each 1,000 amu).

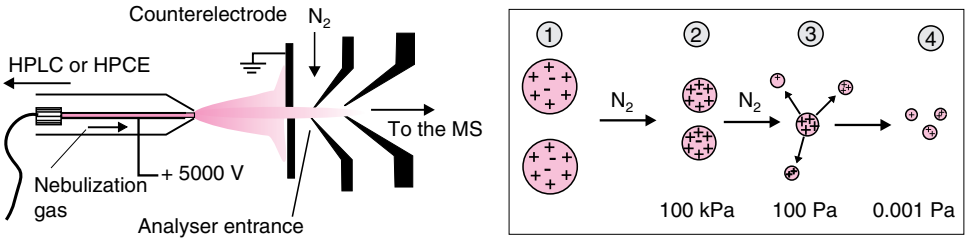


Figure 16.12 Atmospheric pressure ionization by electrospray (ion spray). The outlet capillary is brought to a high potential, leading to a charged mist. The droplets (1), by evaporating, cause an increase in the density of electrical charges (2) to the point that they explode, thus expelling analyte molecules carrying several charges (3). The presence of nitrogen improves the concentration process (4).

Photoionization

This previously described mode of nebulization is associated with photon-induced ionization at around 10 eV. In place of the corona assembly, a UV lamp is used, as in photoionization detection (see Chapter 2, Section 2.7.5). This process, which yields few fragments, is only of interest for molecules of low polarity.

Atmospheric pressure chemical ionization (APCI)

Microdroplets leaving the nozzle are concentrated by a very brief heating up to 350°C before reaching a zone where a gas can be found, such as nitrogen and water vapour, where they are ionized by corona discharge (Figure 16.13). There are many ion/molecule collisions with electron and proton transfer. This technique leads to multicharged ions of the $(M + nH)^{n+}$ type. Unfortunately, this technique is difficult to miniaturize, as the required flow rate is higher than that of electrospray and, as such, is convenient only for volatile and thermally stable molecules of masses less than 1,000 amu.

The advantage of these gentle methods of ionization is to obtain pseudo-molecular, multicharged ions (*z* can be greater than 30), formed *prior to entry* in the spectrometer. With the exception of APCI, these methods are able to extend the range of masses of these instruments up to 10^5 Da (e.g. proteins, polysaccharides, and other polymers).

These processes lead to spectra presenting the phenomenon of an 'ion envelope', which leads to the calculation of the molecular mass (Figure 16.14).

16.4.2 Plasma Ionization

Through nebulization in gas plasma, the sample in solution is sprayed and then atomized. Atoms composing it are subject to collisions with electrons and photons, leading to passage to the ion state carrying a single, typically positive, charge. The ionization

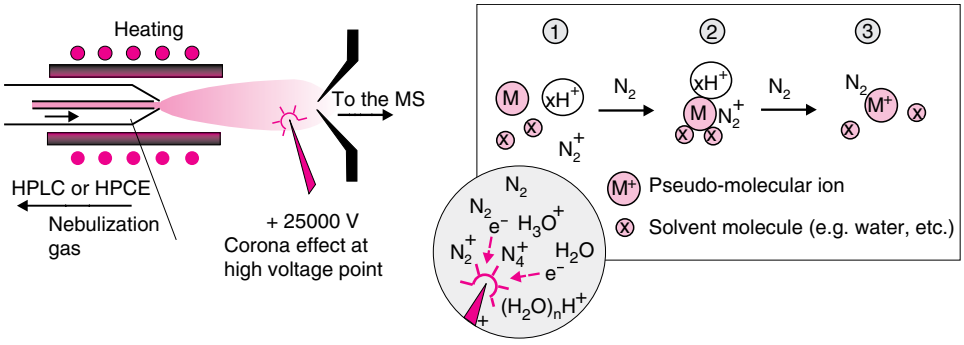


Figure 16.13 Atmospheric pressure chemical ionization. (1) The sample solution is sprayed and mixed with the ions resulting from a reagent gas, such as nitrogen, ionized by a polarized needle at a very high potential (corona effect). (2) Ionized nitrogen or solvent molecules then capture analyte molecules, leading to clusters. (3) These are destroyed by a current of nitrogen, which enables charge transfer to the analyte.

efficiency depends on the plasma's temperature, ionization potentials (between 4 and 15 eV), and the effective section of each element as well as the matrix with which it was introduced into the plasma. It is very low for light elements. This specific ionization mode is used for an analytical technique calling on mass spectrometry to quantify atoms present in the sample (see Section 16.10).

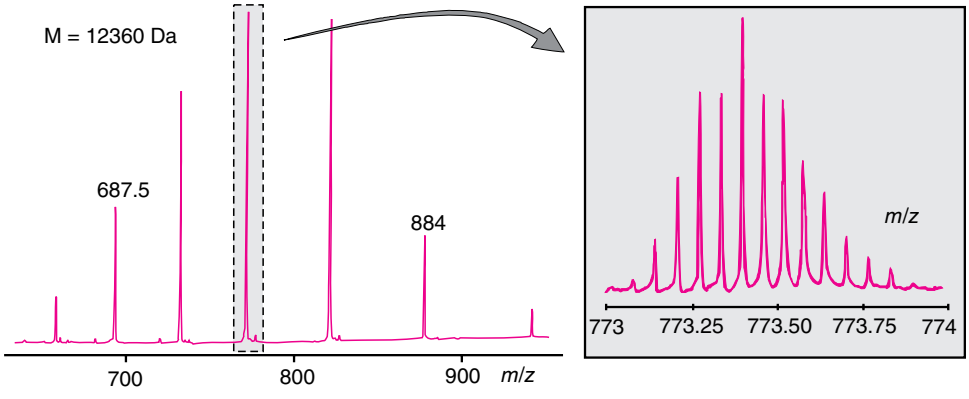


Figure 16.14 Multicharged molecular ions. A spectrum obtained from cytochrome C (horse heart), a protein of molecular weight 12,360 Da, by the electrospray method. Between two consecutive peaks, the difference in the charge on the ion is one unit. Right, the high-resolution cytochrome isotope cluster. From these spectra, it is possible to calculate a value very close to the molecular weight, as well as the number of charges carried by the ions, by two independent methods (see Section 16.8.3).

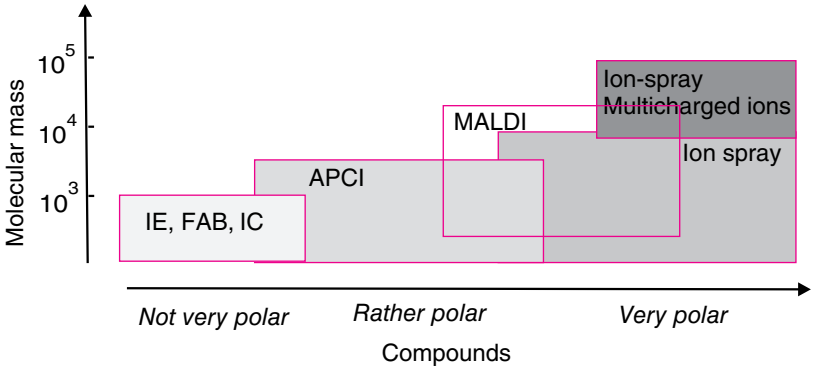


Figure 16.15 Representation of the appropriate ranges for the main ionization techniques.

16.4.3 Choice of Ionization Method

Figure 16.15 provides a recap of the main techniques used to ionize a great variety of molecular compounds or atoms in order to study them with mass spectrometry.

16.5 ANALYSERS

16.5.1 Magnetic Analyser Spectrometer

The idea of determining the mass of ions by subjecting them to the action of a magnetic field led to the first analysers such as those made by Dempster or Bainbridge around 1930 to study isotopes of certain elements (Figure 16.16). These assemblies gave rise to a type of mass spectrometer that is no longer in use. It is, however, of interest to examine them, because their operation can be understood from equations governing the movement of ions in a magnetic and/or electric field.

In these assemblies, the positive ions generated are first accelerated through a voltage difference U. They take on a velocity v that depends upon their mass m (see Section 16.3.1) and are then subjected to a transverse magnetic field $\vec{B}$ responsible for a magnetic flux of density (or field intensity) B. The orientation of the field does not modify the ions' initial velocity but rather forces them on a circular trajectory (Figure 16.16)

The fundamental relationship of dynamics, $\vec{F} = m \cdot \vec{a}$ (where $\vec{a}$ designates the acceleration), applied to ions of mass m on which is exerted the *Lorentz force* $\vec{F} = q \cdot \vec{v} \wedge \vec{B}$ leads, after arrangement, to the following relationship:

$$\vec{a} = \frac{q}{m} \cdot \vec{v} \wedge \vec{B} \qquad (16.3)$$

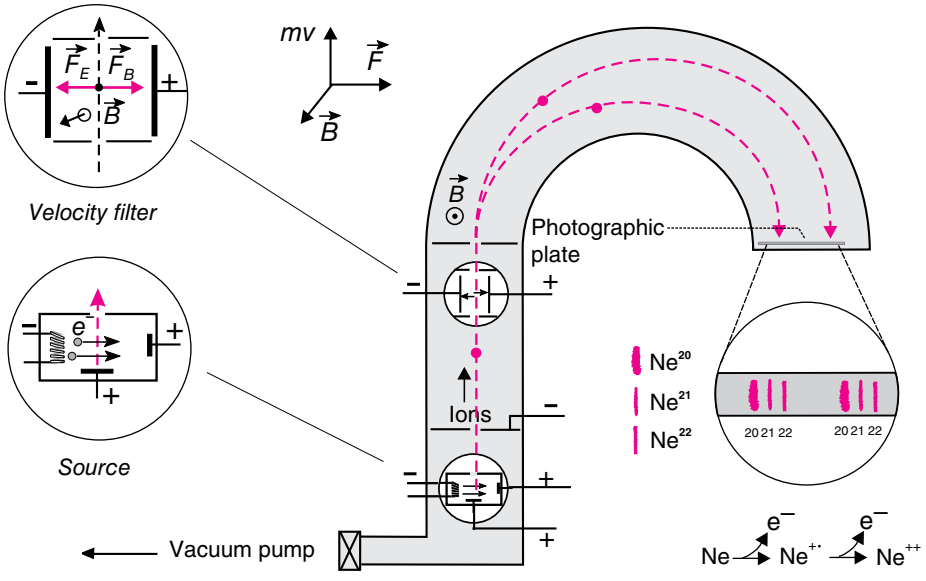


Figure 16.16 A 180° magnetic deflection spectrograph with a velocity filter. The filter eliminates the problem due to the virtual impossibility of having homogeneous, monokinetic beams. The diagram represents a sort of photographic recording of the spectrum of neon. The two series of broken lines represent the removal of one or two of the electrons from the different isotopes of elemental neon (m/z: $^{20}Ne^{+}$, $^{21}Ne^{+}$, $^{22}Ne^{+}$, $^{20}Ne^{++}$, $^{21}Ne^{++}$, $^{22}Ne^{++}$). J.J. Thomson, who around 1913 proved the existence of the isotopes of neon, is considered to be the father of this kind of assembly. Later, work performed by F.W. Aston showed the existence of the isotopes of sulfur and chlorine.

Lorentz Force – The equation of this force is deduced from the general law of Laplace, expressing the force at which a conductor of length $d\vec{l}$ through which current I passes is subjected to magnetic induction $\vec{B}$. The direction of the Lorentz force $d\vec{F} = Id\vec{l} \wedge \vec{B}$ may be found by various recipes, such as the right-hand rule or using the orientations of a direct trihedron.

The orientation of $\vec{B}$ is such that only the centripetal component of the acceleration vector $\vec{a}$ gets involved. The ion trajectory lies in a plane perpendicular to $\vec{B}$ and containing $\vec{v}$. Since $a = \vec{v}^2/R$, by substituting a with its form $a = q \cdot v \cdot B/m$ (where $q = z \cdot e$ is in coulomb, v is in m/s, B is in tesla and m in kg):

$$R = \frac{mv}{zeB} \tag{16.4}$$

The geometry of the assembly is such that ions describe a part of the circumference whose radius R is a function of their velocity and the magnetic field

(e.g. 180 degrees). Therefore, the *m/z* ratio of the ions is only obtained if their speed *v* is known:

$$\frac{m}{z} = \frac{RBe}{v} \tag{16.5}$$

For this, in older instruments, a device situated prior to the magnetic sector, called a *velocity filter* (or *Wien filter*), was employed. An electric field and a magnetic field thus exerted two opposing forces on the ions, such that only the ions remaining on the central trajectory were able to exit the filter (Figure 16.16). Because the net resulting force must be zero, then $qE = qvB$ and therefore:

$$v = E/B \tag{16.6}$$

With a technique known as **electromagnetic separation**, mass spectrometers were employed on a preparative scale (calutrons), notably in the United States where, in 1943, several kg of ^{235}U destined for the manufacture of the first atomic bombs were isolated (the Manhattan Project). This now-dated process has a very low flow rate of under 10^{-3} Pa and is still occasionally in use by other countries.

16.5.2 Multiple Analysers

The principle of these early spectrometers continues in use today for a category of instruments measuring *m/z* ratios very accurately. However, they cannot be used for high molecular masses, owing to the difficulty in producing high-volume magnetic sectors.

As an example, the different parts of an EB-type analyser are described below, including an electrostatic sector (E) between the ion accelerator and the magnetic sector (B). The BE configuration also exists (Figure 16.17).

Ion accelerator

Approximately 5% of the positive ions formed in the ionization chamber or upstream from the spectrometer, depending upon the type of instrument, will finally reach the detector.

The ions must first be accelerated by means of a series of plates of increasingly negative potentials; the total accelerating potential *U* can be as high as 2–10 kV (Figure 16.18). The process must be conducted under sufficient vacuum ($P < 10^{-4}$ Pa) in order to avoid the formation of electric arcs and to minimize collisions between ions. Under these conditions, all the ions carrying the charge $q = z \cdot e$ acquire the same kinetic energy $E_1 = z \cdot e \cdot U$ (Eq. (16.7)). Their velocity after acceleration is therefore inversely proportional to the square root of their mass m_i (Eq. (16.8)).

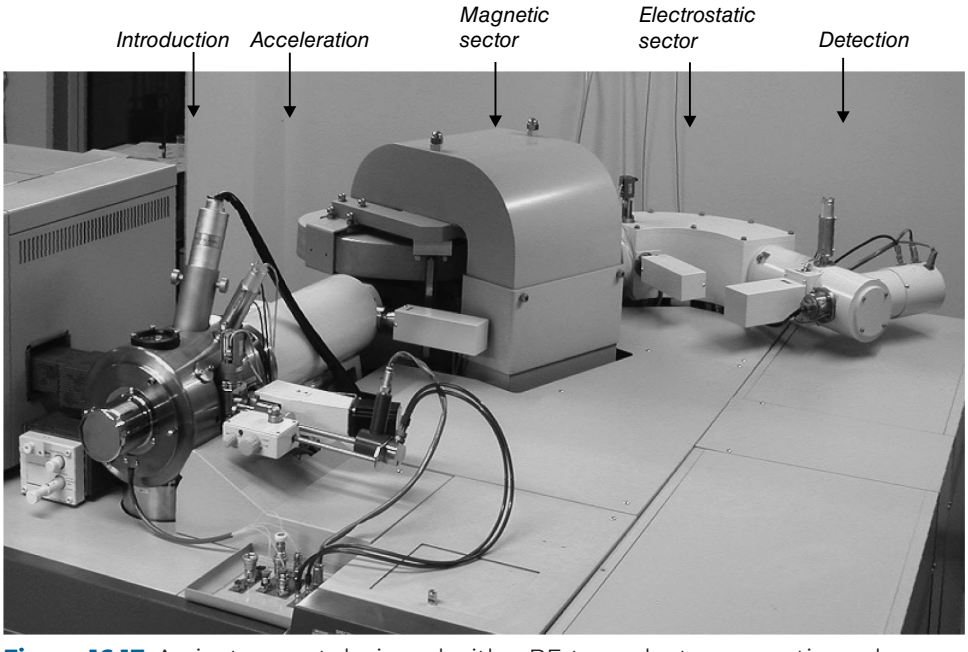


Figure 16.17 An instrument designed with a BE-type electromagnetic analyser. Model JMS 700 from Jeol. In this photograph, the characteristic shape of the electromagnet (magnetic sector) can be seen. The detector is on the far right, and the ion source is on the left. In the configuration presented here, please also note that the instrument is downstream from a GC set-up. (Source: Reproduced courtesy of Jeol, Japan.)

$$E_1 = zeU = \frac{1}{2} m_i v_i^2 \tag{16.7}$$

$$v_i = \frac{\sqrt{2zeU}}{m_i} \tag{16.8}$$

However, since the velocity of the ions before acceleration is nonzero, their total kinetic energy must take into account their initial kinetic energy, E_0, which is small, variable, and unknown:

$$E_{(\text{total})} = E_1 + E_0 \tag{16.9}$$

The precision of the masses obtained will be better when individual ions with the same mass have the same velocity. For this reason, a high accelerating potential U will restrict the range of velocities for ions of the same species (Figure 16.18).

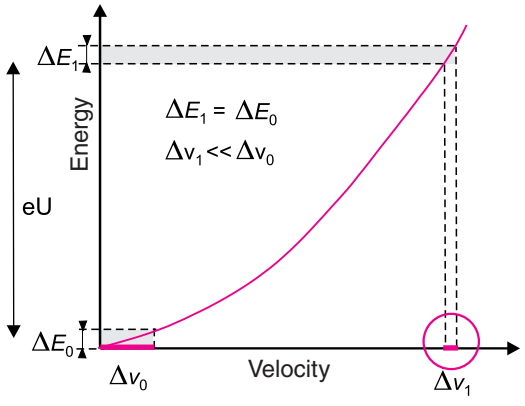


Figure 16.18 Influence of the accelerating voltage upon the range of velocities for ions with the same mass. If we assume that the dispersion of energies before and after acceleration is the same ($\Delta E_1 = \Delta E_0$), the range of velocities after acceleration, Δv_1 is much smaller ($\Delta E_1 = \Delta E_0$ but $\Delta v_1 << \Delta v_0$).

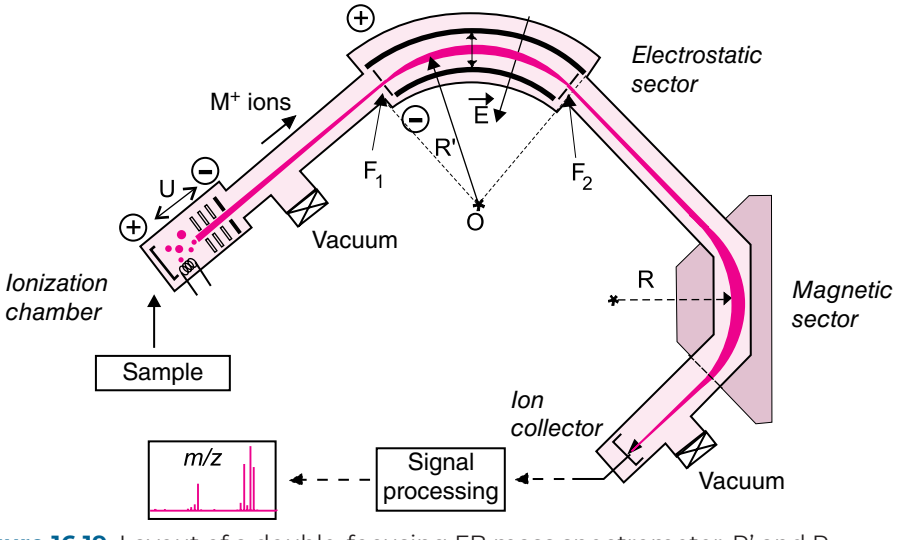


Figure 16.19 Layout of a double-focusing EB mass spectrometer. R' and R measure several decimetres. The laws of physics and equations reviewed in this chapter, though useful in designing these instruments, are not used to calculate masses found by the instrument. All MS devices adopt the comparative measurement principle. Each of them must be calibrated with known masses, such as those of fragmentation ions of perfluorotributylamine.

Electric sector

A radial sector made of two concentric, cylindrical electrodes is used to reduce the non-homogeneity of ion energy and thus further improve the precision of the mass measurements (Figure 16.19).

The Coulomb force $\vec{F} = q\vec{E}$ exerted on the ions is perpendicular to the central trajectory that has an imposed radius of curvature R' (Figure 16.19). Thus, it does not change their energy and is equal to $F = mv^2/R'$. If we replace the term mv^2 by its equivalent in terms of acceleration voltage (Eq. (16.7)), we get Eq. (16.10), which sets the conditions for passage of the ions through this radial filter. When E and U satisfy this equation, only the ions having acquired the proper energy due to voltage U will follow the trajectory of radius R' and be able to leave the sector. This device thus acts as an *energy filter*. Moreover, the electric sector has *direction-focusing* properties: a beam of ions of the same mass and kinetic energy reaching the sector in F_1 with a small angular dispersion α, is re-focused in F_2, which is the mirror image of F_1.If we know the potential drop V between the plates of the sector, which are distance d apart, E can be determined, as can the correlation between U and V:

$$E = \frac{2U}{R'} = \frac{V}{d} \tag{16.10}$$

Magnetic analyser

In the EB configuration, a magnetic analyser (or *sector* or *prism*) is located after the electrostatic sector. The instruments of this type can be divided into various geometries. Figure 16.19 shows one with an assembly whose curvatures imposed onto the ions by the electrostatic sector and then the magnetic field are in the same direction. If we eliminate the velocity *v,* by combining Eqs. (16.4) and (16.7), the *deflection* (or *focusing*) *equation* (Eq. (16.11)) is obtained for a magnetic sector:

$$\frac{m}{z} = \frac{R^2B^2e}{2U} \tag{16.11}$$

Only the ions that have followed the trajectory of radius R, corresponding to the curvature of the instrument's guide tube, will be detected. Hence, to obtain a spectrum within a given mass range, the pulse voltage U or the B value has to be progressively modified. All ions, whatever the m/z value, will successively follow, one after the other, the single trajectory leading to the detector. Such a scan requires around 0.1 second.

Like the electric sector, the magnetic sector focuses the ion beams in direction. At any instant, only ions with a defined mass can follow the radius of curvature R (Figure 16.19). These EB assemblies correct both angular and energetic aberrations of ions (*directional and energy focusing*) without loss of resolution. However, the dimensions and the weight of the magnet limit the study of high masses.

The interdependence of m and U is the origin of a comparative method for precise determination of masses, called *peak matching*. High-resolution instruments are required. We

mix compound X, the precursor of the ion whose mass we want to determine exactly, with a compound such as perfluorotributylamine, or PFTBA ($(C_5F_{11})_3N$, M = 821), whose fragmentation ions have masses known with great precision. An ion whose mass is close to the mass to be determined is chosen as a reference (for example, if the unknown mass is on the order of 200 u, the C_4F_8 ion, which has the mass 199.9872 u, can be taken, and for masses close to 12, the peak of carbon 12 can be used, which is 12 u by definition). A peak comparator composed of an oscilloscope and a series of precision resistors is used to modify *U* in a convoluted manner.

$$m_x / m_{ref} = U_{ref} / U_x \qquad (16.12)$$

16.5.3 Time-of-Flight Analysers (TOF)

The principle underlying *time-of-flight* mass spectrometers is based on the relationship that exists between the mass and velocity of the ions. The instrument, which uses pulsed ionization, measures the time necessary for the masses to travel a distance *L* in a field-free tube under high vacuum (Figure 16.20). The basic relationship of linear-

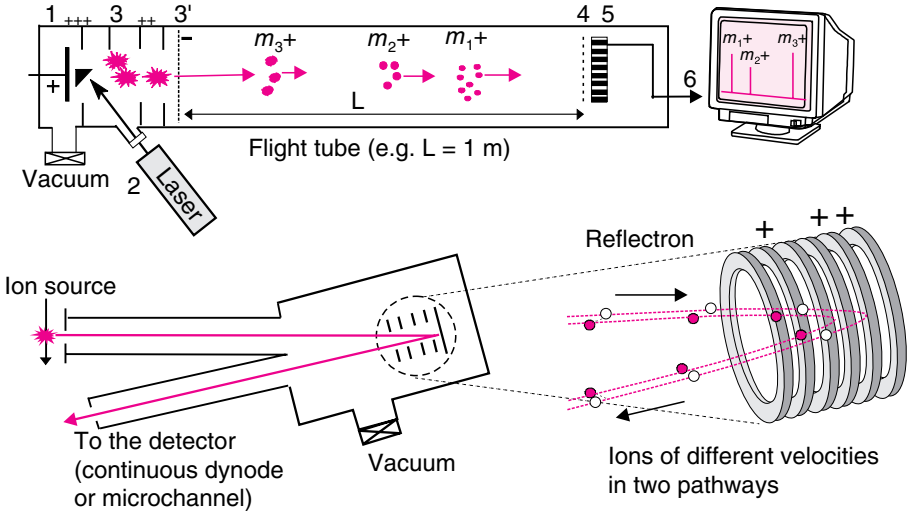


Figure 16.20 A simplified diagram of a time-of-flight spectrometer and the principle of the ion reflector. (1) Sample and sample holder. (2) MALDI ionization device. (3) and (3′) Ions are formed between an acceleration plate and an extraction grid (PD 5,000V). (4) Control grid. (5) Microchannel collector plate. (6) Signal output. Below, a *reflectron*, which is essentially an electrostatic mirror that is used to time-focus ions (several μs) of the same mass but which initially have different energies. The peak widths are of the order of 10^{-9} s and the resolution ranges between 15 and 20,000.

path TOF analysers is obtained by eliminating the velocity v from Eq. (16.7), knowing that $L = vt$:

$$\frac{m}{z} = \left(\frac{2eU}{L^2}\right)t^2 \tag{16.13}$$

In practice, TOF spectrometers enable the analysis of very high masses (300 kDa). To evaluate the masses with precision requires overcoming the initial energy and spatial dispersion of the ions. It can be increased using almost instantaneous ionization methods (a few nanoseconds with a pulsed laser), repeated a great number of times, so that the ions are ejected at the same instant. Many instruments contain a *reflectron* to compensate for kinetic energy dispersion of ions of the same mass (Figure 16.20). With this kind of electrostatic mirror, the most rapid ions will penetrate deeper into the decelerating zone and therefore are more strongly repelled towards the ion collector. This device is used to guide ions of various kinetic energies but of the same mass simultaneously into the detector, thus improving the resolution. In this way, the ions of the same mass are refocused.

16.5.4 Linear Quadrupole Analysers

In addition to the mass spectrometers described above, smaller, lower performance and less costly instruments have been developed for use in applications that do not require high-resolution spectra. This category includes the most widely used instruments based upon the use of quadrupole mass filters. These spectrometers are most often used as mass detectors (GC/MS, HPLC/MS, HPCE/MS or ICP/MS coupled techniques) as well as in a variety of industrial applications for gas and residual atmospheres.

A classic quadrupole is formed of four parallel metal rods (L = 10–20 cm) of hyperbolic cross-section inside the assembly. These four rods are two sets of diametrically opposed electrodes connected electrically and whose contact distance is equal as $2r_0$ (Figure 16.21). These two sets of electrodes are brought to opposite electrical potentials (+U and –U).

The potential Φ_E at any point within the *quadrupole space*, irrespective of z, will be given by:

$$\Phi_E = U \cdot \frac{x^2 - y^2}{r_0^2} \tag{16.14}$$

Φ_E remains between $+U$ and $-U$. The electric potential Φ_E is therefore zero all along the Oz optical axis.

Equation (16.14) implies that in all xOy-planes the points having the same potential are situated on the branches of an equilateral hyperbola whose asymptotes are the straight

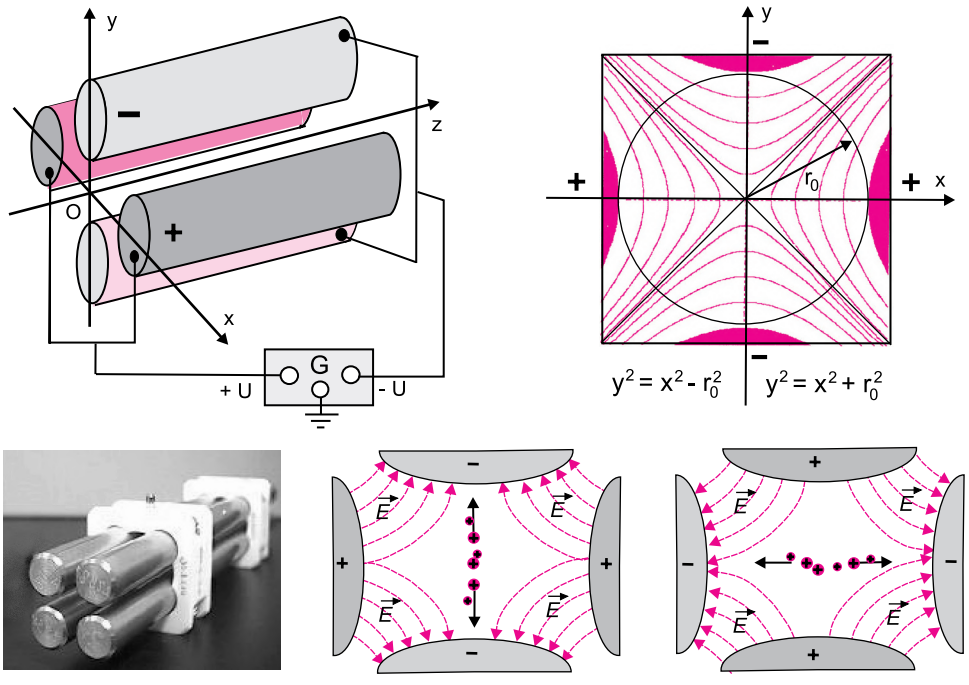


Figure 16.21 Diagram of a linear quadrupole. Notice the connection of bars two by two. These small bars (about the size of a pencil) require specific machining to make them hyperbolic in the central part of the filter. Right, illustration of the field in the quadrupole space from the plot of a series of equipotential hyperbolae. The bars, which are alternatively positive and negative, entrain the ions in a spiral movement.

lines $y = \pm x$. The electric field lines are orthogonal to the equipotential curves for each point M_{xyz} of the space inside the quadrupole. This field has the value:

$$E = -grad\Phi_E$$

A quadrupole may be used in mass spectrometry. When a positive ion with low kinetic energy penetrates via the origin O into the filter maintained under vacuum, the components of its velocity vector in *xyz* space, represented by (*x*, *y*, *z*), will determine its trajectory (Figure 16.21). The central zone behaves like a tunnel along the O*z*-axis, whose walls can attract or repel an ion depending upon the ion's position. The two positively charged electrodes focus it along the O*z*-axis, which corresponds to the bottom of a potential valley (stability zone), while the two negatively charged bars, in contrast, attract it (appearance of a mound of a potential bump in the *yOz*-plane).

To use this assembly as a mass filter, an AC voltage V_{RF} of high frequency ν and of maximum amplitude V_M is superimposed on the DC potential U (ν is of the order of 2–6 MHz and V_M/U is around 6 MHz) (Figure 16.22). The two sets of electrodes are subjected to this alternating voltage, but which is out of phase by π.

$$V_{RF} = V_M \cos(2\pi\nu t) \tag{16.15}$$

The resulting field within the quadrupole is thus now modulated between two extreme values. A potential difference is also created between the inlet and outlet of the quadrupole to move the ions towards the detector. Under these conditions, the ions that penetrate the quadrupole at O are subjected to a force that is variable in both intensity and direction. They will follow complex three-dimensional, generally unstable trajectories in the form of a helix or corkscrew, which will end by hitting either a bar or the detector. Over time, for each point inside the filter, the potential now corresponds to the algebraic sum of Eqs. (16.14) and (16.15):

$$\Phi_{xy} = \left[U + V_M \cos(2\pi\nu t)\right] \frac{x^2 - y^2}{r_0^2} \tag{16.16}$$

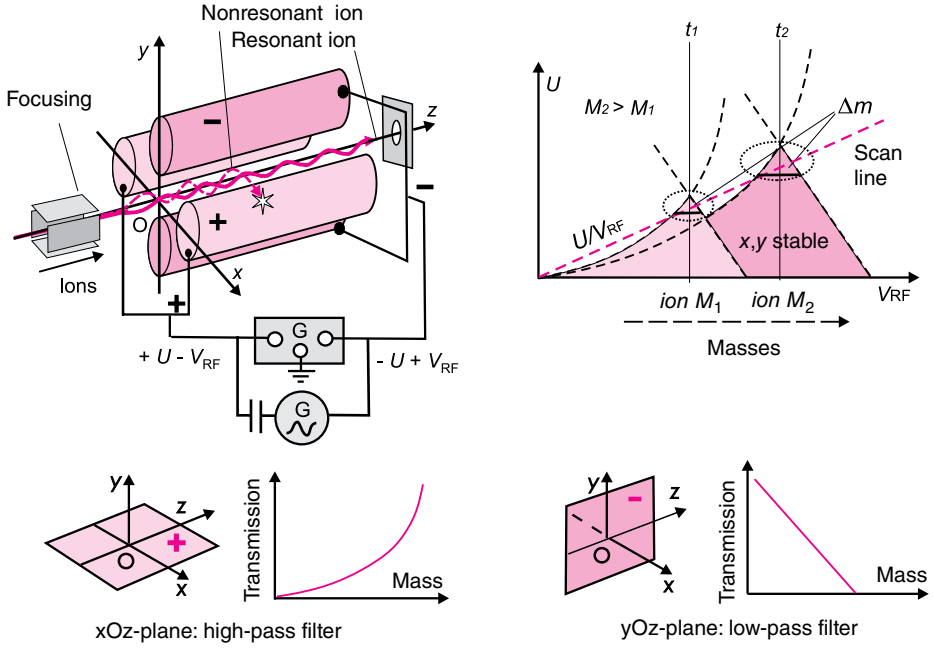


Figure 16.22 Quadrupole filter. Depending upon their mass, the ions respond more or less to the stresses of the variable field. Their xy amplitude must not exceed r_0. Ions enter the filter with a kinetic energy of several dozen eV only. Example of a scan line for a fixed U/V ratio.

To obtain the equations for the movement of an ion, the forces F_X, F_Y, and F_Z exerted upon it must be known (*Mathieu equations*). Calculations show that values of *U*, *V*, and *v* determine the stability or instability of ions in the filter. For example, assuming that an ion has a zero-velocity component along the O*y*-axis, its trajectory will remain in the *xOz* plane. Since the bars *x* are alternately positive and negative ($V_{RF} >> U$), heavy ions, due to their high inertia, cannot follow the variations in field orientation: they will only feel the continuous positive voltage *U* and so will not be perturbed very much. However, lighter ions will be able to oscillate more and more until their amplitude of movement exceeds r_0, in which case they will be lost on one of the rods. The two positive rods *x* constitute what is known as a *high-pass filter*.

Consider now the case of an ion that does not have a velocity component on the O*x*-axis: its pathway will remain in the *yOz* plane. The heavy ions will continue to be inexorably attracted towards the negative bars. Only the light ions can be recovered. The negative rods form a *low-pass filter*.

Quadrupole filters lead to low-resolution spectra. Two options appear. Either the two voltages *U* and V_{RF} are maintained constant and frequency *v* is scanned, or the fixed-value frequency is maintained and the amplitude of voltages *U* and *V* is regularly increased by keeping their ratio constant. The slope of this *operating line* determines the resolution. The greater the slope is, the better the resolution will be and the weaker the sensitivity will be. If Δm is constant (e.g. 1 u), the resolution changes with the *m/z* ratio (Figure 16.23) and many other parameters. For an ion bearing a unit charge *e*, accelerated under a PD of V_z volts at the entrance to a filter of length *L* and radius r_0 using a radiofrequency whose maximum voltage is V_{max}, we get:

$$R = Cste \cdot \frac{L^2 \cdot V_{max}}{r_0 \cdot e \cdot V_z} \qquad (16.17)$$

In the nuclear, semiconductor, and plasma industries, etc., strong vacuums are used and residual gas analysis (RGA) is important. To conduct the analysis, several companies offer quadrupole filters, operating as mass spectrometers, directly connected from the ionization chamber to the enclosures where the vacuum to be analysed can be found (Figure 16.23).

16.5.5 Quadrupole Ion-Trap Analysers

Ion traps used as low-resolution mass spectrometers have many similarities with linear quadrupoles, even though their appearances are totally different. Simple in appearance, yet complex from a fundamental standpoint, these ion-trap detectors are very sensitive and are used for various ionization methods. As their name suggests, the ions are confined between three electrodes (one toroidal and two end-caps), whose particular shape appears to result from a sort of anamorphosis of the four-bar set-up

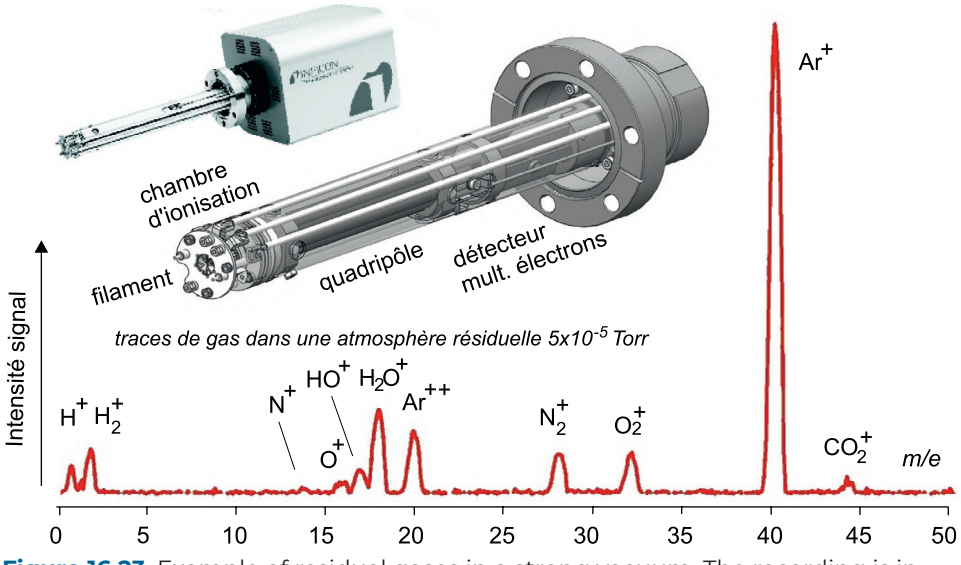


Figure 16.23 Example of residual gases in a strong vacuum. The recording is in constant Δm mode throughout the entire mass range. Quadrupole probe whose source is directly submerged in the vacuum enclosure (information courtesy of Inficon).

of a classic quadrupole (Figure 16.24). They also operate under the effect of a variable electric field. This trap, bounded by the electrodes, constitutes both the source and the analytical and filtration zone of masses leading to their detection. These cheaper analysers are almost exclusively used as mass detectors (GC/MS).

In simple terms, the operating principle can be summed up as follows: the compound is subjected to a very brief electron bombardment, yielding ions that subsequently

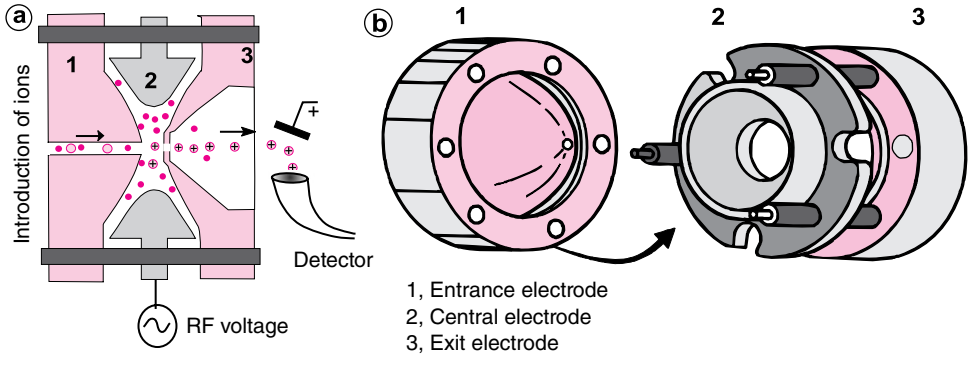


Figure 16.24 Ion-trap spectrometers. (a) Design of the electrodes in an ion trap. (b) Diagram giving a perspective of the end-cap electrodes and the central ring electrode.

penetrate the central part of the filter. A fixed voltage U and a fixed radiofrequency voltage (see Eq. (16.15)) are then applied between the ring electrode and the end-cap electrodes, thus confining these ions to the central space, by forcing them to follow complex yet stable trajectories in the presence of a low pressure of helium, of about 0.01 Pa, introduced to the trap as a form of shock absorber.

To analyse the ions present, the radiofrequency voltage is increased. One by one, in the rising order of m/z ratio, the amplitude of oscillation of the ions grows in the axial direction (z-axis) until they are eventually ejected through a small hole drilled in the terminal, or outlet, electrode. They then reach the detector (Figure 16.25). If a reagent gas is introduced concurrently into the trap, then a chemical ionization of the sample can also be obtained (see Section 16.4.2).

These small, highly sensitive instruments have, however, the drawback of operating over a narrow dynamic range and with a rather mediocre reproducibility of spectra from the same sample, which represents a serious disadvantage in quantitative analysis.

16.5.6 Ion Cyclotron Resonance Analysers (ICRMS)

These mass spectrometers are not widely used and are something of outsiders when compared to the previous ones. Basically a magnetic ion trap, an ICRMS is able to ascertain the mass of ions with very great precision.

The mode of operation can be described as follows: the ions, obtained through one of many existing processes, are directed towards a small cavity situated in the centre of a very intense magnetic field $\vec{B}$ (4–9 T) produced by a superconducting coil. In this cavity, the ions are slowed down by collisions with residual molecules (the vacuum not being perfect) and are trapped through the effects of the repulsion of internal

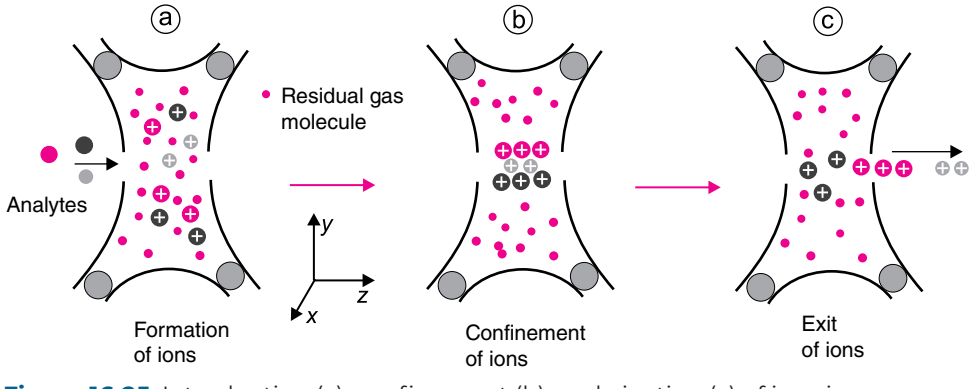


Figure 16.25 Introduction (a), confinement (b), and ejection (c) of ions in a hyperbolic trap.

walls carrying positive potentials. Their trajectories are complex but their projections in the normal *xy*-plane, orthogonal to the field $\vec{B}$, correspond to a circular movement (Figure 16.26). The radius of the circular movement depends upon the kinetic energy of the ion and is given by Eq. (16.4). If the velocity *v* is small and the magnetic field *B* is intense, the circumference will be sufficiently small (a few mm). From the classic expressions $v = \omega R$ and $\omega = 2\pi\nu$, Eq. (16.18) can be obtained. This shows that for each mass there is a particular frequency, *independent from the velocity of the ion*. Hence, at any given instant, there will be as many different frequencies of rotation as there are ions of different *m/z* ratios in the trap.

$$\frac{m}{z} = \frac{eB}{2\pi v} \tag{16.18}$$

or

$$v = 15350\frac{Bz}{m} \tag{16.19}$$

Equation (16.19) gives the frequency ν (in kHz) of an ion of mass *m* (Da) carrying a charge *z* in a field $\vec{B}$ (Tesla). Mass analysis is achieved by precisely measuring the different frequencies of the ions.

To establish the frequencies and consequently the *m/z* ratios of the ions, a brief pulse of an RF band (e.g. 1 ms), commonly called a *chirp*, is sent, covering the entire range of frequencies present. In this way, all of the ions will capture energy. Since their frequencies do not change, the ions follow spiral orbits during the pulse, and thus their velocities increase. Following the pulse, ions with the same *m/z* ratio come together, in phase, which consequently induces a signal (cyclotron resonance) of the same frequency, in the detector plates (Figure 16.26). Overall, the sample leads to a complex signal due to the superimposition of frequencies induced by all the different *m/z* subpopulations. This interferogram, whose intensity [$I = f(t)$] decreases when excitation stops, contains information on the set of all frequencies which it represents. Conditions are gathered for the software to conduct a Fourier analysis. This involves a Fourier transform of $I = f(t)$ to $I = f(\nu)$, and thus in the end to the mass spectrum $I = f(m/z)$. The ions are not destroyed; they may be re-excited or be used for other experiments.

16.5.7 Tandem mass spectrometry (MS/MS)

To obtain more precise information about the structure of fragment ions resulting from the decomposition of molecular compounds, mass spectrometers with at least two analysers in series can be used. They are not necessarily two distinct devices assembled end-to-end but rather they are combined into a hybrid

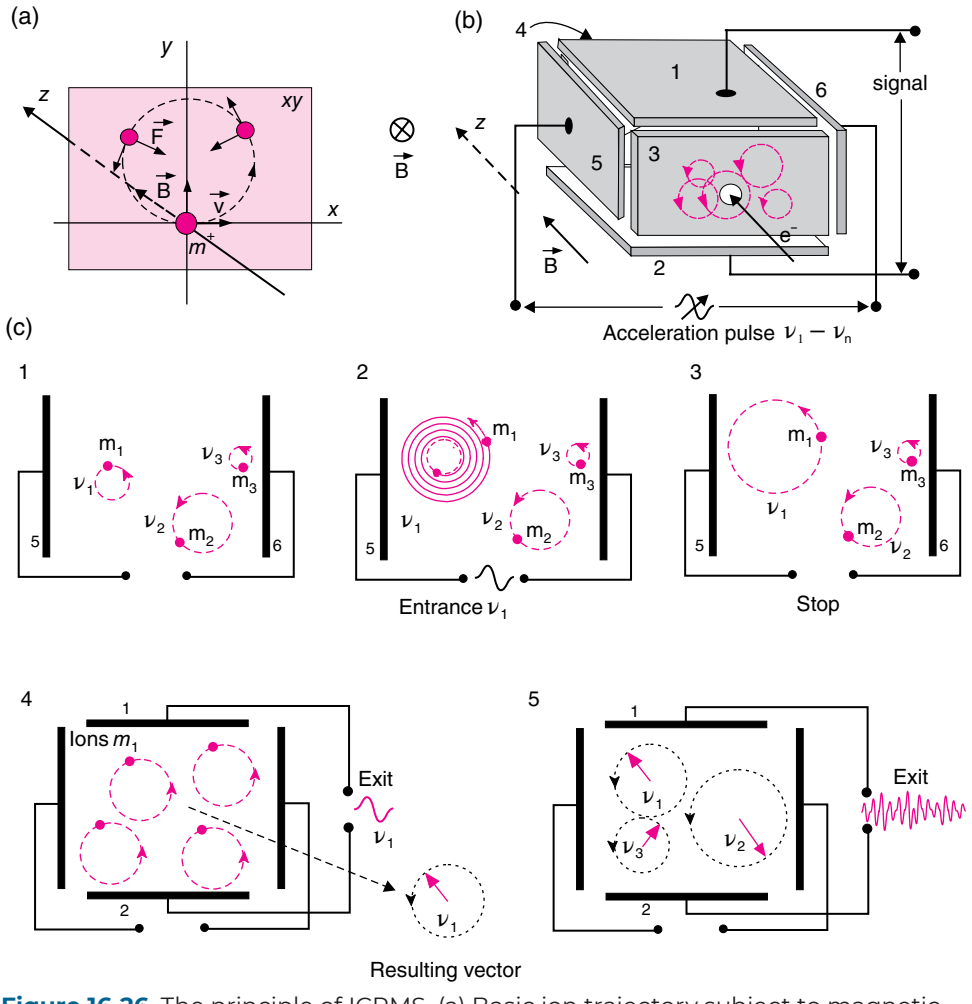


Figure 16.26 The principle of ICRMS. (a) Basic ion trajectory subject to magnetic induction $\vec{B}$ taking account of the orientations chosen. (b) Simplified view of an ion-trapping cell (plates 5 or 6 are excitatory, plates 3 and 4 trap the ions, and plates 1 and 2 are used for detection). The ions are formed either inside the cell or upstream from it. (c) Principle of detection: 1, three ions waiting; 2, pulse of frequency ν_1: only ion m_1 is excited; 3, following excitation, ion m_1 adopts a larger orbit; 4, for a real sample containing a large number of ions m_1, these ions will arrange themselves to orbit in phase. On detecting plates 1 and 2, they produce the effect of an electric field vector rotating at the same frequency ν_1 and whose intensity is a function of their concentration. 5, if the three ions m_1, m_2, and m_3 have been excited simultaneously, the signal at the exit will be an interferogram. A Fourier calculation leads to the usual spectrum.

instrument: a magnetic sector followed by a quadrupole, or two quadrupoles, or a TOF with a second quadrupole (or the reverse). Between the two analysers, there is a collision chamber (Figure 16.27). These instruments can be used in several ways.

- The first method (*descending mode*) consists in choosing ions that correspond to a preselected *m/z* ratio with the first mass analyser, which plays the role of a filter. These ions then cross through the collision chamber where they collide (*CID, collision-induced dissociation*) with a target gas such as helium, argon or nitrogen. This causes further decomposition. The resulting daughter ions continue their course towards the second mass analyser, switched to normal mode. This results in a spectrum that provides structural information about the *m/z* ion selected upstream, to the exclusion of the other ions formed in the ion source (*daughter ion scan*).
- In an alternative method (*ascending mode*), the second mass analyser is set to a selected *m/z* ratio, allowing passage only to those ions, while the first mass analyser performs a scan over the mass range. In this way, all parent ions leading to the same fragment can be identified (*precursor scan*).
- Lastly, a synchronized double scan of both mass analysers is performed with a preprogrammed offset between them, corresponding to a given mass (e.g. a neutral fragment such as CO or C_2H_2). Thus, we identify all the precursor ions which decompose by eliminating the fragment in question *(neutral loss scan)*.

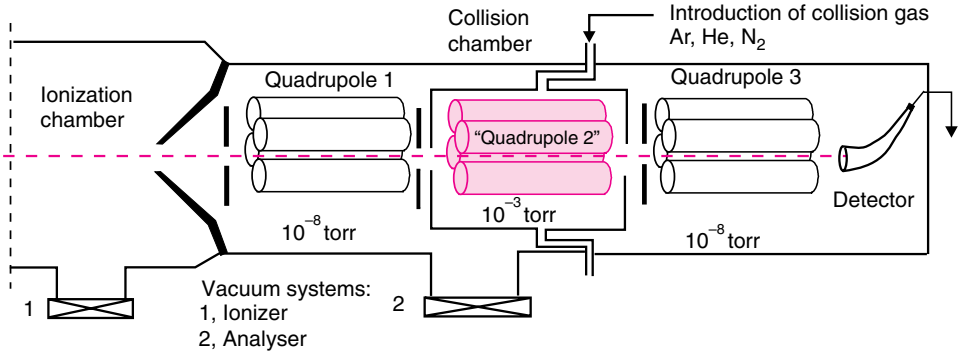


Figure 16.27 A triple quadrupole (MS/MS). In this assembly, called QQQ, the central quadrupole has a simple focusing role. Located in the collision chamber, it is supplied by AC voltage only (absence of DC voltage). The pressure at this spot in the instrument is higher. With these instruments, mixtures can be studied as would be done with the GC/MS technique. There are many applications since the cost of these instruments has greatly diminished.

16.6 ION DETECTION

Detection in mass spectrometry is based upon the measurement of the electric charges carried by ions. Normally, the number of individual ions of the same species is very large and gives an analogue signal output, though certain models have such power of amplification that they are able to detect the impact of a single ion.

To exploit a spectrum quantitatively, it is essential that the number of ions detected reflects the number of ions produced, irrespective of their mass. If the mass spectra are obtained by scanning the magnetic field, the incrementation velocity of the field must not be constant. The deflection formula (Eq. (16.11)) reveals that the m/z ratio varies with the square of the magnetic field.

Major types of detection systems:

- *Electron multipliers with discrete dynodes*. In this device, positive ions strike a conversion cathode, thus liberating electrons which are then accelerated and 'multiplied' via a cascade of up to 20 dynodes (Figure 16.28). This type of detector is extremely sensitive, having a gain of up to 10^8. Alumina-based coatings on the dynodes have improved performance over traditional materials (Cu/Be alloys), which age rather badly in the residual atmosphere of the spectrometers or during nonworking periods (return to atmospheric pressure).
- *Continuous dynode electron multipliers* (*Channeltron*®). The ions are directed towards a collector whose horn-shaped entrance is made of a lead-doped glass which acts as the conversion cathode. The released electrons are attracted towards a positive electrode (Figure 16.28), and their successive collisions against the internal walls give rise to multiplication, as with the separated dynodes. The assembly is usually mounted off-axis to avoid the impact of neutral species as well as photons emitted by the filament, which could also remove electrons.
- *Microchannel plate detectors (MCP)*. They consist of the union of a large number of microchannels arranged like honeycombs. This resembles an electron version of a photographic plate. Each individual detector is formed from a portion of microtube (25 μm diameter) whose interior is coated with a semiconductor material acting as a continuous dynode (Figure 16.28). The avalanche of secondary electrons emitted is retrieved by an anode. This system can record ions of different masses simultaneously.

Amongst the questions to be asked following this brief review of the five categories of mass analysers is whether analyses of the same compound lead to an identical mass spectrum. In reality, there is no universal analyser. Each one of the above has its own potential uses. Some of these instruments analyse the ions as soon as they are formed, while others begin by storing them and absorbing the impact of the collisions, which can result in the abnormal growth of the M + 1 peak. For this reason, mass spectrometry has its own

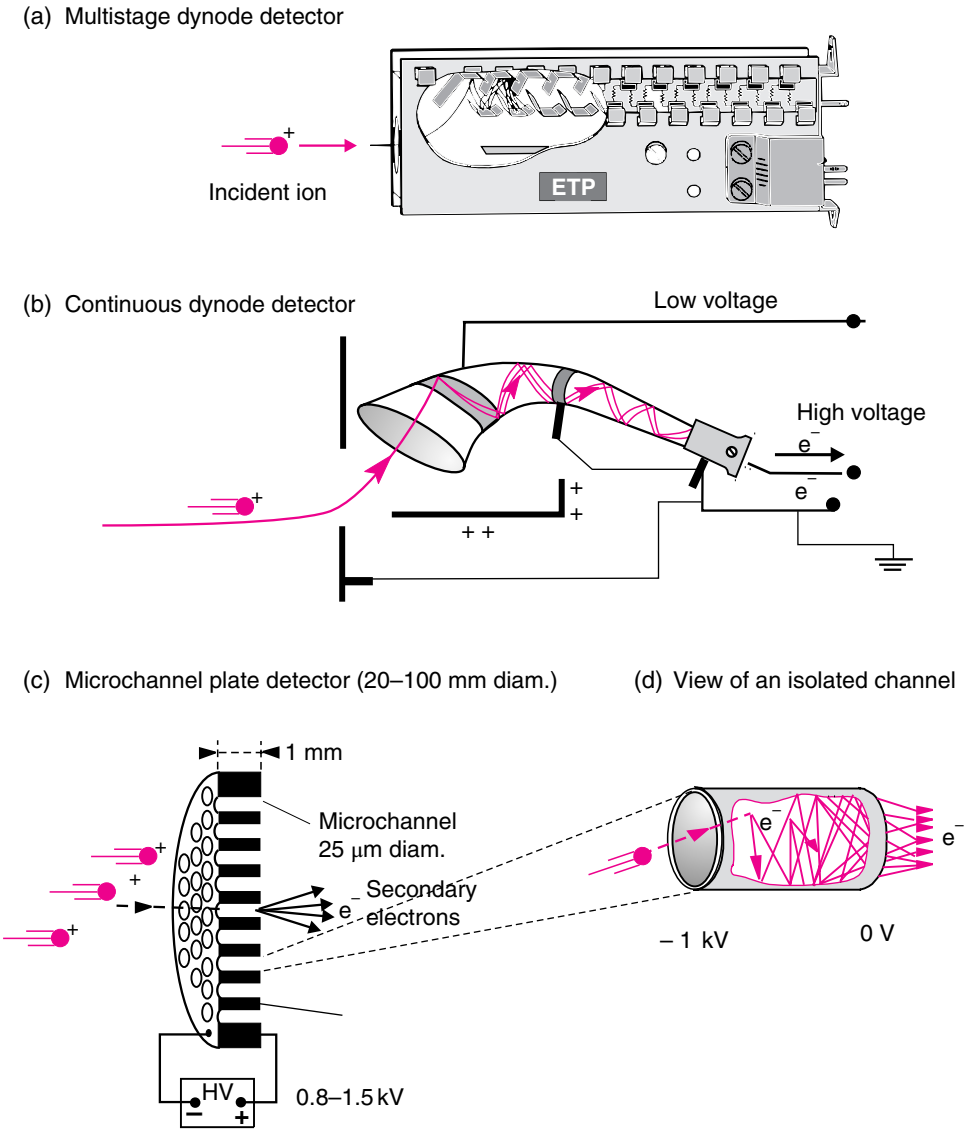


Figure 16.28 MS detectors. (a) Discrete dynode model with active film (Source: Courtesy of ETP Scientific Inc.). (b) Continuous dynode model. Diagram of a Channeltron®: the funnel-shaped cathode permits the recovery of ions coming from slightly different trajectories. The curvature has the effect of restricting the acceleration of the positive ions, which appear via the impact of electrons on the residual molecules, before they strike the cathode and produce further electrons (Channeltron® feedback). (c) Detailed view of the conversion cathode. Multiplication of the electrons in a microchannel. (d) Microchannel plate detector. Multichannel (microchannel) collector system used to localize the impact of ions (Source: Galileo USA).

separate place when compared with optical spectral methods, which are more passive processes. Therefore, to make the spectra comparable, it is crucial to standardize the operating conditions.

Some Applications of Mass Spectrometry

Mass spectrometry has innumerable applications because instruments may be used as simple detectors in combination with separation methods (chromatography, electrophoresis) to conduct elemental and isotopic analyses, find empirical formulae or elucidate structures of organic compounds. The technique is now becoming essential in the fields of genomics and proteomics.

16.7 IDENTIFICATION BY MEANS OF A SPECTRAL LIBRARY

Identification of compounds through comparison of spectra is employed in numerous chemical analysis fields. In mass spectrometry, the method has become essential, especially for all environmental analyses (soil, water, food industry, etc.). For that purpose, many spectral libraries have been created and commercialized. There are general spectral libraries such as Wiley's spectrum collection (close to 800,000 spectra) or the NIST's MS-MS library (with close to 200,000), coded in the JCAMP, mgf or msp formats. Spectra are obtained with several ionization modes, under vacuum or at atmospheric pressure and with TOF spectrometers, quadrupoles, ion traps, etc. Other specialized spectral libraries, containing fewer spectra, target specific categories of compounds: pesticides, proteins, peptides, glycans, and metabolites. Several software packages are used to conduct searches in spectral libraries and thus find the best match for the compound one wants to identify.

In general, the search is divided into several stages: the raw spectrum (continuous plot) of the analyte is transformed into a fragmentation spectrum, then the software searches the spectral library for spectra with peaks at the same positions as in the analyte's spectrum, even if the intensity differs. Finally, the previous selection is redone with a more elaborate algorithm, which, based on selection criteria involving intensity, mass, and rarity of the peak, assigns each spectrum an index ranging from 1,000 for a perfect match down to 0 for no shared peaks. By changing selection criteria, the search can be refined by creating various classifications.

When several analytes must be measured by coupling chromatography and mass spectrometry (GC/MS or HPLC/MS), two specific fragments are associated with them: the *identification fragment* (considered as characteristic of the compound) and the *quantification fragment*. The measurement also requires having a solution of known concentration for these compounds (see Chapter 1, Section 1.15). To do so, there are mixtures with several dozen standards of known concentrations present in the same solution. This process has become common in searching for trace compounds in environmental samples.

16.8 ANALYSIS OF THE ELEMENTAL COMPOSITION OF IONS

16.8.1 Search for the Raw Formula

With a mass spectrometer whose resolving power is at least 10,000, and if we adopt certain hypotheses, it is possible to find the raw molecular formula of an ion or a fragment. For this, the MS system uses exact isotopic masses of the elements believed to be present in order to calculate all the probable molecular formulae that could correspond to experimental mass, while taking into account its margin of error (Table 16.1).

Therefore, the exact masses of atoms provide information on the elemental composition of molecules. For example, molecules of molecular nitrogen and carbon monoxide, while they both have 14 protons and 14 neutrons, differ by 0.0112 u (respective masses of 28.0061 and 27.9949 u).

To select the correct raw formula, the data processing program gives the value of the corresponding *index of hydrogen deficiency* (*I*). This expresses the number of multiple bonds and rings in the corresponding ion (Eq. (16.20)). If the index *I* is a whole number, then it is a radical cation. If *I* is a half number, it is a cation. Generally, an ion with an *odd number* of electrons will fragment with loss of a radical, becoming by consequence a cation with an *even number* of electrons. More rarely, the ion may fragment through a process of rearrangement and thus remain a radical cation.

By associating several other considerations, such as the nitrogen rule (if the compound contains an atom of nitrogen, its molecular mass is an odd number) or the isotopic composition of the elements, the nature of the main ions present on the spectrum can be identified.

$$I = 1 + \left[0.5 \cdot \sum_{1}^{i} N_i \left(V_i - 2 \right) \right] \tag{16.20}$$

N_i represents the number of atoms of the element *i* whose valence is V_i.

16.8.2 Method Based Upon Isotopic Abundances

This is another method to find the molecular formula of the analyte. A low-resolution mass spectrometer will suffice. As explained below, the principle is based upon the isotopic composition of elements with several isotopes. For that reason, any sample of a molecular compound (which consists of an immense population of individual molecules) gives a mass spectrum on which several molecular peaks appear (*isotopic cluster* at M, M + 1, M + 2, etc.), whose intensities reflect the elemental and isotopic composition of the compound.

Therefore, the appearance of this set of signals, presented as sticks and obtained from the compound, must be compared with all sets calculated beforehand from the various empirical formulae, by only using those satisfying the information about the compound to eliminate useless calculations.

Consider a sample of benzene, C_6H_6 (M = 78 u). Knowing that for 100 atoms of carbon, there is approximately 1 atom of ^{13}C, it can be estimated that about 6% of the benzene molecules will include a ^{13}C atom in their backbone and will therefore have a mass of 79 u (the contribution of ^{2}H, or D, 0.01% is negligible). This explains the presence of a small peak observed at m/z = 79 on the mass spectrum of benzene. Similarly, a peak will be observed at m/z = 80, due to the presence of molecules that include either two ^{13}C atoms, two D atoms, or one ^{13}C atom and one D atom.

Equations (16.21) and (16.22) give the relative intensities in percent of the M + 1 and M + 2 peaks for organic molecules containing only the elements C, H, N, O, F, and P (where nC, nN, etc. symbolize the number of atoms of carbon, nitrogen, etc.):

$$(M+1)\% = 1.1nC + 0.36nN \tag{16.21}$$

$$(M+2)\% = nC(nC-1)x\frac{1.12}{200} + 0.2nO \tag{16.22}$$

This method has come into common use, but its efficiency is rather limited, considering the measurement precision on masses M+1 and M+2, generally drowned in the background noise.

16.8.3 Determination of Molecular Masses from Multicharged Ions

Multicharged ions are obtained from macromolecules when they are ionized in the form of a spray (ESI, APCI). Their molecular masses can be determined through an alternative method to that shown on Figure 16.14. A basic spectrum is required to undertake this calculation, which presents a succession of molecular peaks differing by the number of charges carried.

On the spectrum of the compound whose mass is designated by M, two successive peaks M_1 and M_2 can be located whose overall charges, z_1 and z_2, differ by one unit, i.e. $z_2 = z_1 - 1$ (Figure 16.29). Equations (16.23) and (16.24) can be written as:

$$M_1 = \frac{M}{z_1} \tag{16.23}$$

$$M_2 = \frac{M}{z_2} \tag{16.24}$$

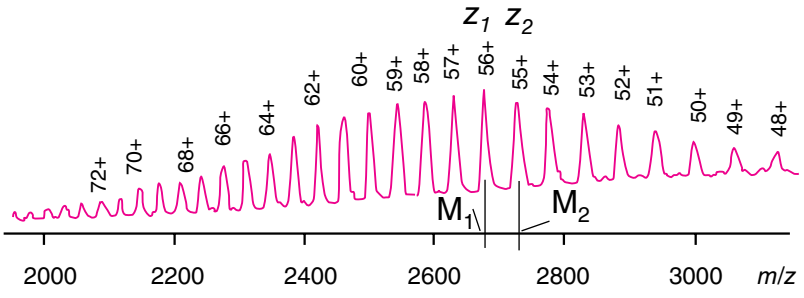


Figure 16.29 Spectral profile of an antibody of high molecular mass. This spectrum has been obtained with an electrospray. Without using the values of the charges reported above each signal, the calculation described in the text leads from M_1 = 2,680 and M_2 = 2,729 to M = 150,080 Da.

Substitution and transposition lead to the following for z_1:

$$z_1 = \frac{M_2}{M_2 - M_1} \tag{16.25}$$

Next, if we know z_1 (rounded to the next whole number), M can be calculated (Da) from Eq. (16.23).

16.9 FRAGMENTATION OF ORGANIC IONS

The interpretation of a mass spectrum – nature and origin of the fragmentation peaks – is a difficult but interesting exercise. Organic chemists find known or envisaged types of ions in order to explain reaction mechanisms in condensed phases. The difference here is that these ions move in a vacuum and do not collide. The very short time intervals between ion formation and ion detection make it possible to observe the existence of very unstable species (lifetime of less than 1 μs).

16.9.1 Basic Rules

Fragmentation by electron bombardment, well adapted to the study of organic molecules, results from the instability of the molecular ion formed initially. The molecule M, whose mass is always an even number (except when there are an odd number of nitrogen atoms), is ionized and becomes a *radical cation* with an odd number of electrons represented by the symbol $M]^{+\bullet}$. For compounds incorporating heteroatoms, ionization will preferentially occur on one of the free doublets (Figure 16.30). At this stage, the molecule M is not yet fragmented:

$$M + e^- \rightarrow M]^{+\bullet} + 2e^-$$

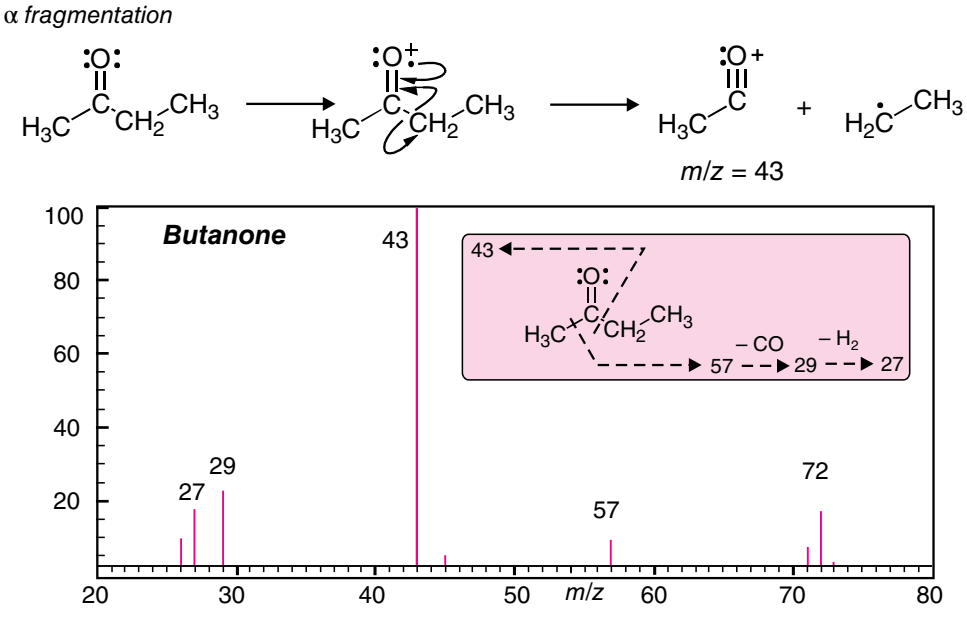


Figure 16.30 Fragmentation spectrum of butanone obtained by electron ionization. The diagram explains the formation of the main ion fragments. The molecular ion leads to the acetyl ion $CH_3CO]^+$ (m/z = 43), which is ten times more intense than ion 57, $CH_3CH_2CO]^+$. The half-arrows symbolize the transfer of a single electron.

In a second step, the radical cation generally fragments. Since a mass spectrum results from the fragmentation of a sample that contains billions of molecules, it will show all possible fragmentations, with an intensity factor that reflects the probability of each.

Several factors control the fragmentation process of $M]^{+\bullet}$. Two basic situations are observed: a cation with an odd mass number is formed (but with an even number of electrons), accompanied by a neutral fragment in the radical state, or else a new radical cation appears accompanied by the loss of a neutral molecule.

- Stable fragments (ionic or neutral) have a greater tendency to form.
- Fragmentations with rearrangements are favoured when they involve a six-membered ring transitional state, often leading to hydrogen transfers at the radical cation site.

16.9.2 Fragmentation of an Ionized σ Bond

In a *hydrocarbon* such as propane, the ionization of a C–C σ bond will lead to the formation of the ethyl cation and a methyl radical, rather than the reverse:

$$CH_3CH_2CH_3\,]^{+\bullet} \rightarrow CH_3CH_2^+ + CH_3^\bullet$$

Similarly, the ionization of isobutane leads principally, by elimination of a methyl radical, to the formation of an isopropyl cation, which is thermodynamically more stable than the methyl cation.

16.9.3 α Fragmentation

In *ketones* (RCOR′), selected as an example, ionization of the keto group by stripping of an electron from one of the lone pairs of the oxygen atom leads predominantly to the homolytic cleavage of the C–C σ bond that is in position α of the site of ionization. Elimination of the larger alkyl chain as a radical is favoured. However, in the general case of a ketone RCOR′, the four fragment ions RCO⁺, R′CO⁺, R′⁺ and R⁺ (Figure 16.30) will be observed.

In ethers (ROR′), the cleavage of the C–C bond adjacent to the oxygen leads to resonance-stabilized oxonium ions (Figure 16.31). In primary amines, an analogous situation is found, with formation of the iminium ion at mass 30: $CH_2{=}NH_2^+$. Elsewhere, the split of the C–O bond in ethers leads to cations R⁺ and R′⁺.

16.9.4 Fragmentations with Rearrangements

Besides these basic examples of fragmentation that occur by simple bond cleavage, ions may rearrange, sometimes by very complex pathways. Fragmentation with the McLafferty rearrangement is a well-known transformation that occurs in molecular ions containing a C=O carbonyl group or a C=C double bond (Figure 16.32). The reaction involves a six-centred mechanism during which there is transfer of a hydrogen atom to the initial ionization site. It continues with the elimination of a neutral ethylene molecule, leading to the formation of a new radical cation of even-numbered mass if the compound does not contain nitrogen.

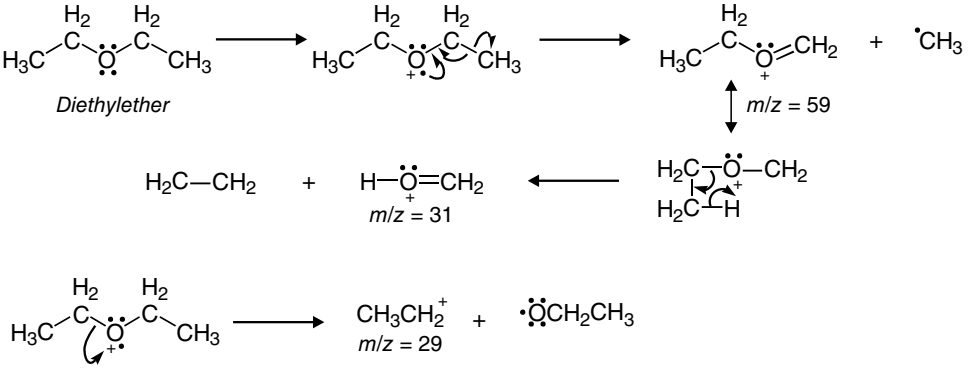


Figure 16.31 Fragmentation modes of an ether, with diethyl ether taken as an example.

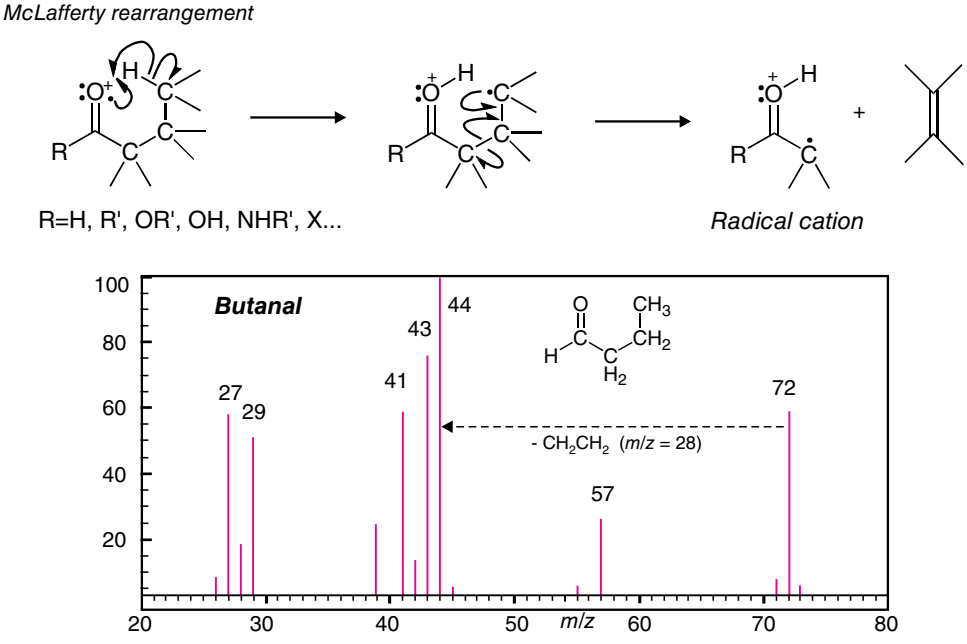


Figure 16.32 McLafferty rearrangement. Situation for butanal C_4H_8O

All these transformations have led to semi-empirical rules that are used in the study of unknown compounds and which are also employed to corroborate the results obtained by automated searches of spectral libraries.

16.9.5 Metastable Ion Peaks

To help identify ions, their instability characteristics can be exploited. A normal ion is sufficiently stable over time to reach the detector. However, if its lifetime is less than a few microseconds, it is a *metastable* ion which can decompose between the source and detector (Figure 16.33).

$$M^{+\cdot} \rightarrow m^+ + m'^{\cdot}$$

Each individual ion has a random lifetime, depending upon its internal energy. However, a great number of identical ions form a population whose decomposition obeys statistical laws. It is described by first-order kinetics. With a double-focusing apparatus, if the daughter ion m^+ is formed before the exit from the electrostatic sector, it will become lost on the internal walls of the instrument because its kinetic energy is insufficient. In contrast, if the decomposition occurs between the exit from the electrostatic sector and the entrance of the magnetic sector, specific adjustments of the instrument will make it

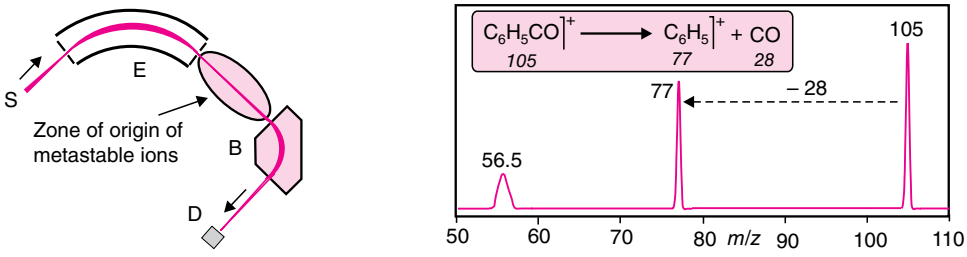


Figure 16.33 Metastable ion peak. Theoretical appearance of the three peaks constituting a metastable transition.

possible to observe the ion m^+ with apparent mass m*, which does not correspond to its real mass. Equation 16.26 relates the positions of the three signals on the spectrum:

$$m^* = \frac{m^2}{M} \tag{16.26}$$

A *metastable* transition therefore leads to three peaks on the spectrum, including the m* peak, also called the *diffuse peak* due to its less clear form and its position, which does not necessarily correspond to a mass close to a whole value (Figure 16.34).

Therefore through *peak matching* and metastable peak analysis, it is possible to determine with great confidence the structure of the ions. However, the latter process has

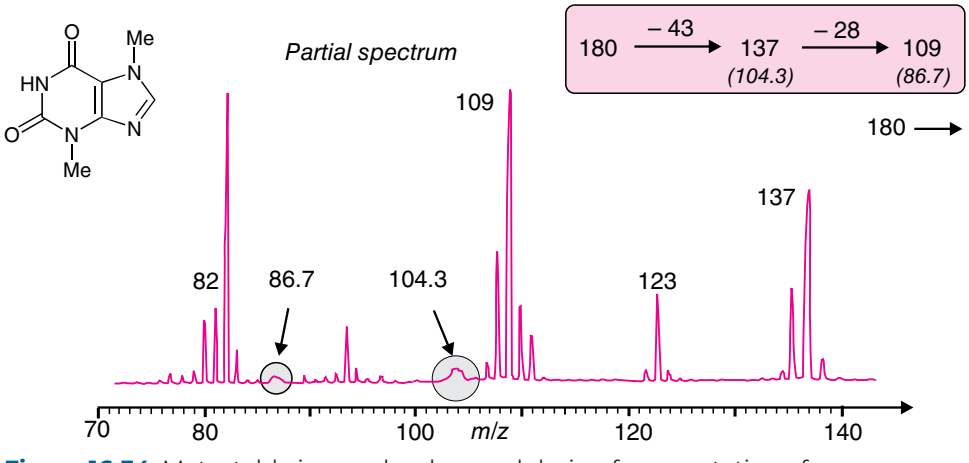


Figure 16.34 Metastable ion peaks observed during fragmentation of theobromine. The molecular ion of theobromine (180 u) gives rise to an ion of mass 137 u through loss of a $CONH^{\cdot}$ radical (43 u). This metastable transition is accompanied by a diffuse peak at 104.3. Furthermore, loss of CO (28 u) from the daughter ion 137 yields a second metastable ion peak, which appears at m/z = 86.7.

become a little out-dated when compared with the new and more efficient techniques (e.g. MIKE or reversed Nier–Johnson geometry) described in specialized monographs.

16.10 PROTEIN ANALYSIS

Without going too far into the complexity of the subject, mass spectrometry has become very useful for biochemists to analyse polypeptides with 10–20 amino acids, resulting from proteins after 'chemical digestion' with specific reagents. It is now possible to find their molecular mass as well as the sequencing of the amino acids composing them. By combining MS with two-dimensional electrophoresis (HPCE) or HPLC as a prior separation method, a very effective tool is available for these studies in proteomics.

A MALDI-TOF or MS/MS configuration is used with ionization at atmospheric pressure (ESI, see Section 16.4.1) to lead, for example, to the fixation of a proton on the polypeptide followed by gentle fragmentation through collisions. The cleavage of peptide bonds by established mechanisms is done in three possible ways, whose likelihood depends on the collision energy (Figure 16.35). Considering the number of peptide bonds that the molecule has, the same type of fragmentation may be reproduced along the entire chain. These fragments are gathered in several series. In a given series, the mass difference between two consecutive ions corresponds to the residue of an amino

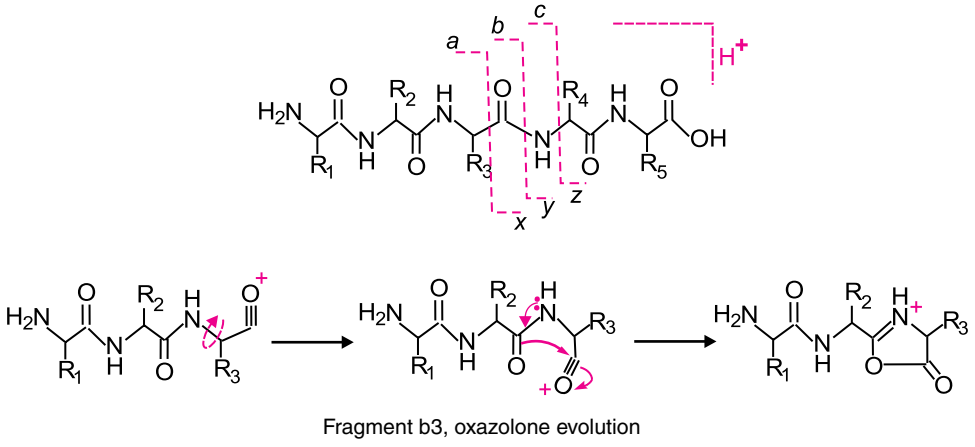


Figure 16.35 Peptide bond cleavage. In this example of a tetrapeptide treated by ESI-type ionization, protonation may fasten on an oxygen atom leading to the formation of many fragments. The letters *a, b, c, x, y,* and *z* refer to the nomenclature of fragments resulting from the same mechanism while moving along the polypeptide chain (on the figure, this would be *a3, b3, c3, x2, y2*, and *z2*), *a, b, c* for terminal N ions and *x, y, z* for terminal C ions. The great number of fragments makes their identification difficult. Software can be of service to users.

acid. Software and databases (e.g. the Protein Data Base) are essential in these *in silico* studies. MS is also used to determine oligonucleotide mass and structure, according to a comparable principle.

16.11 ICP-MS COUPLING

The coupling of a plasma torch and a mass spectrometer, called ICP-MS coupling, is used to track and quantify practically all chemical elements present in samples, except light elements such as carbon, hydrogen, oxygen or nitrogen.

Unlike atomic emission spectroscopy, which relies on spectral line identification, the ICP-MS technique is based on the mass of ions resulting from the torch, knowing that at a high temperature a fraction of atoms transform into cations, which are generally carriers of a single positive charge. Its rather recent development (the first instruments date back to the late 1980s) is in part due to problems posed by the interface between an argon plasma, within which ions are formed at ambient pressure and about 8,000°C, and the mass spectrometer, in which a strong vacuum is maintained (10^{-4} Pa).

ICP-MS interfacing. The sample, which must be in solution, is introduced into the instrument's nebulizer, where it is transformed into a spray, then sent into the horizontally positioned torch to be vaporized, dissociated, atomized, and ionized.

A fraction of this plasma then moves by suction to the top of the water-cooled sampling cone (Figure 16.36). After crossing through a first space connected to a primary vacuum pump, the ions formed pass via a second orifice called the *skimmer* into a second chamber in which the vacuum is even stronger, before entering the mass analyser, most often a quadrupole filter, after having crossed through a set of lenses, also called ion lenses.

The mass spectrum obtained is relatively simple, since peaks correspond to various atoms present in the sample. It is possible to recognize the sample's nature from their atomic masses, which are intrinsic characteristics of each element. However, the existence of isotopes for most elements may make interpretation difficult, since a given mass may correspond to several elements. Also, the isotopic composition of each of these must be used.

> Argon plasmas also contain polyatomic ions at very low concentrations, which arise from the air, from argon or from the sample matrix. These species can be isobaric with the analyte ions and thus could be a source of possible confusion with low-resolution MS instruments. Thus, N_2^+, O_2^+, ArH^+, ClO^+, ArO^+, CaO^+,Ar_2^+, etc., are molecular ions that are isobaric (i.e. have the same *m/z* ratio) as elements such as silicon, sulfur, vanadium, iron, and potassium, which may also be present. In this case, two methods for clarification are necessary: selection of another isotope of the element under consideration (^{54}Fe rather than ^{56}Fe) or use of another spectrometer with a collision chamber between the source and the detector to dissociate the polyatomic ions.

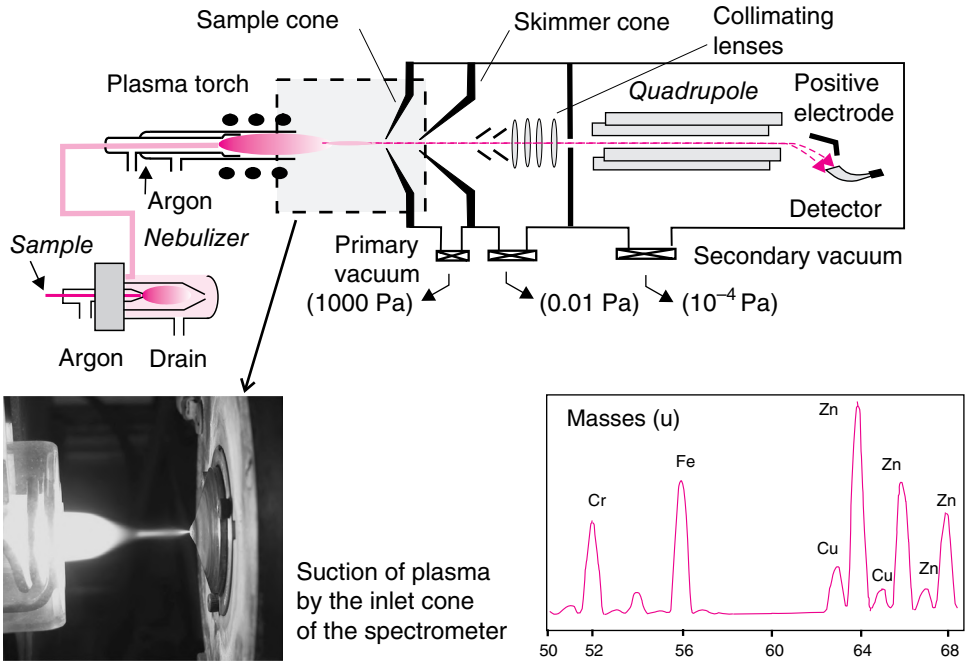


Figure 16.36 Plasma torch ionization and recording obtained by the ICP/MS method. Approximately 10% of the ions generated in the plasma pass into the mass spectrometer. Spectrum of a mixture of some metal elements recorded between masses 50 and 68 u. Note the presence of different isotopes of the elements in question. Here, the resolution is insufficient to distinguish between isobars, i.e. ions coming from different elements but with practically the same mass, which are therefore superimposed on each other on the graph. A space in the form of a collision chamber is sometimes inserted between the introduction chamber and the quadrupole to break down the polyatomic interferences (ArO, Ar_2).

The scope of application is simultaneous, multi-element analysis in practically all fields (metal, soils, water, food, pharmaceuticals, and medicine). The measurement range extends over nearly ten decades. This quick, reliable technique, which is more sensitive than AA, is also used to conduct *speciation analyses* for molecular compounds, when the exit of a chromatography column (GC or HPLC) leads into the torch.

KEY POINTS OF THE CHAPTER

1. The principle of mass spectrometry is to generate from the molecule to be analysed ions carrying electrical charges that are subjected to the action of an electrical and/or magnetic field. Their trajectory in the mass spectrometer will be used to find their mass/charge ratio from the energy acquired and their relative abundance. Positive ions are the ones most commonly studied.

2. A mass spectrometer has three main parts: a space where the sample is transformed into charged particles characteristic of the compound, an analyser that separates ions according to their mass/charge ratio, and a ion flow collection system to transform the ion current into a proportional electrical current, which through the *digital processing of information will lead to conventional representations of the mass spectrum*.
3. Unlike optical spectroscopy, mass spectrometry consumes the sample, which is progressively destroyed. This is the essential point of this method, when mass spectra obtained on different design instruments must be compared, knowing that many parameters influence fragmentation.
4. Mass spectrometry is used to conduct rapid analyses, both qualitative and quantitative, on samples, whatever their physical state (gas, liquid, solid). We can distinguish between direct introduction of a pure compound and introduction after a prior separation (GC, HPLC, HPCE, TLC, etc.).
5. MS sometimes requires us to make a choice between the different sample introduction modes, as well as between the various ionization processes in order to make the most suitable pairing, according to desired performance, for the study of the samples in question.
6. Measuring a sample by MS is done in several stages: isolate the analyte(s) with an upstream separation method (extraction, GC, etc.), obtain mass spectra, use a spectral library (identification peak and quantification peak), and quantify using a test mixture with these same analytes at a known concentration.
7. Performance levels of an MS instrument can be defined by several parameters: accuracy, sensitivity, range of masses, and, especially, resolution. The latter depends on the type of instrument but it decreases when the m/z ratio increases. The best resolutions, exceeding one million, are obtained with cyclotron resonance spectrometers. Their sensitivity makes it possible to detect analytes with concentrations as small as an attomole (10^{-18} M).
8. A tandem spectrometer, i.e. one containing several analysers, can be used in three main analytical modes. For example, the first analyser can be used to select a parent ion, which is fragmented by collisions with an inert gas in the second part of the instrument (collision chamber) before crossing through a second analyser, which will detect the daughter ions formed. The two other modes are explained in the chapter.
9. The development of ionization methods for macromolecules at atmospheric pressure has opened MS to new fields of study in biology, especially proteomics.
10. Finding the structure of a compound from its fragmentation spectrum is often a challenge as difficult as rebuilding an unknown Greek temple destroyed by an earthquake. The solution relies on the knowledge of fragmentation rules.

11. Inductively coupled plasma mass spectrometry enables the simultaneous analysis of elements at trace or ultra-trace levels from samples in solution without prior chemical separation. This application of mass spectrometry to inorganic chemistry has become essential. Only a few elements cannot be measured by this method (H, C, O, N, and halogens).

PROBLEMS

1. Vanadium has two isotopes whose relative abundances are ^{51}V = 99.75% and ^{50}V = 0.25%. To determine the vanadium concentration in a sample of steel, 2 g of the steel is dissolved in an acid medium and 1 μg of ^{50}V is added to the resulting solution.

- **a.** After stirring, an analysis by ICP–MS is performed, yielding a mass spectrum with two peaks centred on the masses 50 and 51, possessing the same surface area.
- **b.** What is the % content of each isotope of vanadium in the unknown steel if the ratio of the areas of the two peaks is the same as that of the masses of the two isotopes?
- **c.** Give a more thorough answer knowing that ^{50}V = 49.947 g/mol and ^{51}V = 50.944 g/mol.
- **d.** Explain why this problem would become more complicated if the steel contained either titanium or chromium.

2. Dibenzosuberone is a ketone whose structure is given in Figure P16.1.

- **a.** Calculate the exact mass from the most abundant molecular peak, and write the isotopic compositions of the different species constituting peak *M* + 1.
- **b.** In the mass spectrum of this compound, among other fragments, one can see two with the same nominal mass, one resulting from the loss of CO and the other from loss of C_2H_4.
  - Explain only the loss of CO from the parent ion and indicate if, owing to the loss of C_2H_4, a positive ion, a radical or a radical cation must result.
  - Find the molecular formula for each of the two peaks.
- **c.** Knowing that the resolution factor of the mass spectrometer is 15,000, is it possible to distinguish the different species constituting the peak *M* + 1?

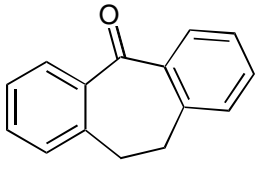


Figure P16.1

3. Calculate the ratio of the intensities of *M* + 1 (M = 721 u) over peak *M = 720* in the isotopic cluster of Buckminsterfullerene (formula: C_{60}), knowing that carbon has two isotopes ^{12}C: 12 u (98.9%) and ^{13}C: 13 u (1.1%).

4. Explain the origin of the different masses observed on the fragmentation spectrum of ethanol by electronic ionization (Figure P16.2). Reminder: O = 16; C = 12; H = 1 u.

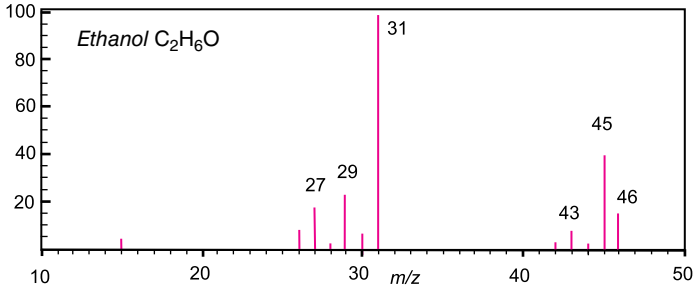


Figure P16.2

5. Based on the mass spectrum of horse myoglobin subjected to the electrospray method (Figure P16.3):

- **a.** Establish the literal expression used to calculate the molecular mass of a compound when there is this type of recording.
- **b.** Apply this expression to calculate the molecular mass of this molecule.
- **c.** How many charges are carried by the peak m/z = 1,060.5 Da?

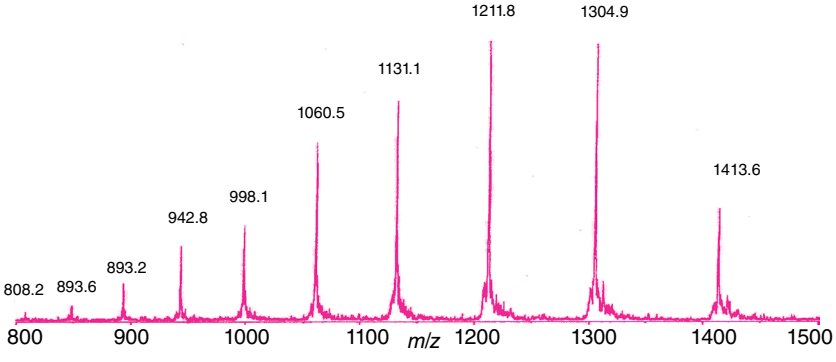


Figure P16.3

6. Insulin has a raw formula of $C_{257}H_{383}N_{65}O_{77}S_6$. Using Table 16.1:

- **a.** Calculate its nominal monoisotopic mass (in Da).
- **b.** Calculate a precise value of this mass.
- **c.** Calculate the molar mass of insulin.

7. The cyclotron resonance frequency is sometimes represented by the following formula: $\nu = 15{,}350 \times \frac{B \cdot z}{m}$ (frequency is in kHz, mass is in Da). Find this expression from the basic relationship.

SOLUTIONS

1. **a.** *Approximate solution:* 0.5 ppm.

b. *More rigorous solution:* 0.513 ppm.

c. If titanium or chromium were present in this steel, there would also be a peak at the nominal mass 51 due to chromium, and the intensity of the mass peak at 50 would be disturbed by the presence of titanium.

2. a. 208.088815 u.

b. Both modes of decomposition correspond to an even mass loss (28 u). The ions formed (m/z = 180) therefore do so according to the general rule for CHO compounds, involving radical cations (Figure P16.4).

- By CO loss: m/z = 180.0939. By C_2H_4 loss, m = 180.05732. On a high-resolution recording, it can be deduced that the peak on the far right corresponds to ions formed from the parent ion by CO loss.
- The raw formula of this ion is therefore $C_{14}H_{12}$. The formula of the second peak is $C_{13}H_8O$.

c. The exact masses of the three isotopomers making up peak M + 1 are calculated (raw formulae given in a), a = 209.09217, b = 209.095054, and c = 209.09321 u. Deviations between these values are much less than the value of ΔM calculated according to the value of R (0.012 u). Under the conditions of this experiment, these three types of molecules thus appear superimposed.

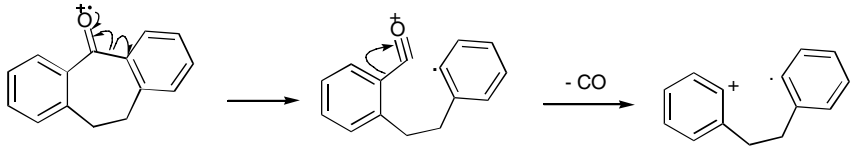


Figure P16.4

3. Peak 720: $I = 0.989^{60} = 0.515$; peak 721: $I = 0.989^{59} \times 0.011 \times 60 = 0.344$; (M + 1)/M = 66.7%.

4. The molecular ion of ethanol (C_2H_6O = 46 u) corresponds to the largest mass on the spectrum. Fragmentations result from the transfer of one or two electrons. They obey many rules. For this molecule, fragmentation diagrams are shown in Figure P16.5.

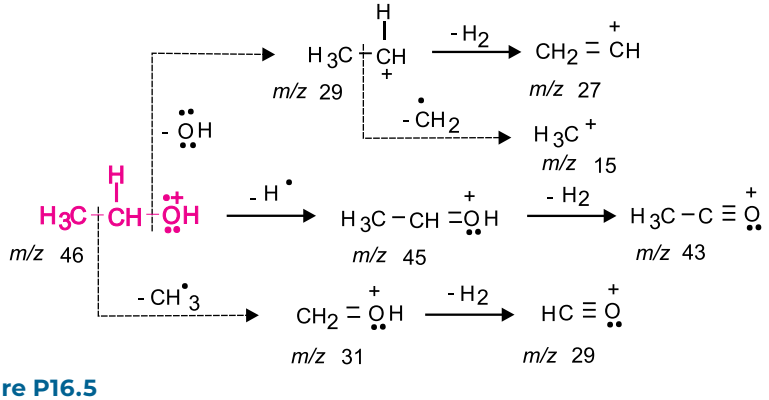


Figure P16.5

5. a. The signals observed all correspond to nonfragmented myoglobin, whose carried charge varies by 1 Da between two successive peaks. Values of *m/z* take into account the increase in mass due to n_1 protons fixed on the molecule. Therefore, by designating as A and B two *m/z* values selected on the recording and as C the number of intervals between the selected peaks (n_1 and n_2 are whole numbers where $n_2 > n_1$):

$$\frac{(M+n_1)}{n_1} = A \qquad \frac{(M+n_2)}{n_2} = B \text{ and } n_2 - n_1 = C$$

or:

$$\frac{M}{A-1} = n_1 \text{ and } \frac{M}{B-1} = n_2$$

$$C = M \cdot \frac{(A-1)-(B-1)}{(A-1)\cdot(B-1)}$$

hence

$$M = C \cdot \frac{(A-1)-(B-1)}{(A-B)}$$

a. By choosing A = 1,413.6, B = 1,060.5, or C = 4, we find M = 16,954 Da.

b. Molecules leading to peak *m/z* = 1,060.5 include z = 16 protons (this number must be a whole number!).

6. a. Nominal monoisotopic mass:

$$M = (257\times12)+(383\times1)+(65\times14)+(77\times16)+(6\times32) = 5{,}801 \text{ Da}$$

b. Precise mass of this ion:

$$(257\times12)+(383\times1.007825)+(65\times14.00307)+(77\times15.99492)+(6\times31.97207)$$
$$= 5{,}803.638 \text{ Da}$$

c. Molar mass of this molecule ($g \cdot mol^{-1}$):

$$(257\times12.0107)+(383\times1.00794)+(65\times14.00674)+(77\times15.99940)+(6\times32.066)$$
$$= 5{,}807.5788 \, g\cdot mol^{-1}$$

7. Based on $\frac{m}{z} = \frac{e\cdot B}{2\pi\cdot v}$, it can be deduced that $v = \frac{e\cdot B\cdot z}{2\pi\cdot m} = \frac{e}{2\pi}\cdot\frac{B\cdot z}{m}$

$$A.num.\, v = \frac{1.602\times10^{-19}}{2\times\pi\times1.66\times10^{-27}\times1{,}000}\times\frac{B\cdot z}{m} = 15{,}350\times\frac{B\cdot z}{m}$$

Chapter 17

Isotopic Analyses and Labelling Methods

Several methods of analysis, now essential, are based on adding tracers to the analyte in order to determine its concentration. These include isotopic dilution methods using either radioactive or nonradioactive isotopes, or species marked with an enzyme, forming the basis for many immunological or radioimmunological assays in heterogeneous or homogeneous phase. Other methods rely on the appearance of radioactive isotopes after irradiation of the sample with particles. In the

appendix to this chapter, several concepts involving radioactive isotopes, marked molecules, their detection and dimensions used in radiochemistry are reviewed.

Objectives

Describe the principle of isotopic dilution
Use a radioisotope in analysis
Use a stable isotope in analysis
Access the isotopic ratio of an element
Describe the enzymatic marking principle
Compare several immuno-enzymatic methods
Review basic concepts of radiochemistry

17.1 PRINCIPLE OF ISOTOPIC DILUTION METHODOLOGIES

The most obvious and ideal method for the determination of the mass concentration of a compound in a mixture would be to totally separate it from all other components of the mixture and then to weigh it. Unfortunately, this can seldom be undertaken, as extraction is neither completely selective nor quantitative. Indeed, it is impossible at a very low concentration. However, analytical chromatography of the sample, a separation method *par excellence*, is an indirect way to achieve extraction and to accomplish this goal. To resolve this problem, there is another method, whose principle is described below.

To measure a particular analyte in a sample, a known quantity of the same compound is added to the sample in a tagged form (e.g. labelled). This *labelling* allows it to be distinguished from the unlabelled form and other compounds present in the initial mixture. After allowing time for mixing, a fraction of the sample is collected and the ratio of both forms of the compound is used to obtain the amount initially present in the sample. The labelling of the compound must not alter its behaviour during the recovery step. This technique should not be confused with the classic methods of standard additions described elsewhere. Additionally, the marking of this analyte must not alter its behaviour in the recovery stage.

> Labelling methods are not only used in chemical analysis. To estimate, for example, the number of salmon in a pond, a certain number of fish are caught, labelled, and then released again into the pond. A few days later, a number of fish (the sample) are removed and from the proportion that are labelled an estimate of the total population can be deduced. For example, if after having labelled and released 500 salmon and if 10 of them are found in a group of 200 fish removed a few days later, then the total number of salmons *(x)* can be estimated at 10/200 = 500/x, thus x = 10,000.

Several analytical methods have been developed from this principle, which the *isotopic analyses* described below are a part of. When measuring a chemical element,

one of its isotopes, either stable or radioactive, is added as a tracer (or indicator). If it is a molecule, the process remains the same by adding its marked counterpart, i.e. the same molecule, if only the isotopic composition of one of its atoms has been modified.

17.2 MEASUREMENT BY ADDING A RADIOISOTOPE

17.2.1 Basic Method

To measure a molecular compound X present in a sample, a quantity m_S^* of the labelled compound X (the reference tracer is made up of the same compound, the only difference being that one atom of the molecule is replaced by a radioactive isotope) is added to a precise quantity of the sample. The specific activity of this tracer is represented by A_s (activity in Bq per unit of mass), while A_x is the specific activity after dilution as calculated from a small quantity of the recovered mixture.

The measured specific activity A_x is less than A_s, because the quantity of the standard m_S^* is mixed with the quantity m_x of the nonlabelled compound. On the other hand, the overall activity is conserved, which is translated by the following expression:

$$A_s m_S^* = A_x \left(m_s^* + m_X \right) \tag{17.1}$$

The quantity m_x of analyte in the sample is obtained by rearranging Eq. (17.1):

$$m_x = m_s^* \left(\frac{A_s}{A_x} - 1 \right) \tag{17.2}$$

If A_x is much smaller than A_s, then Eq. (17.2) can be approximated by the following equation:

$$m_x \cong m_s^* \frac{A_s}{A_x} \tag{17.3}$$

After we have calculated m_x in this way and if we know the mass of sample used, it becomes easy to deduce the concentration of X in the initial sample.

The re-extraction step must involve a quantity sufficient to avoid weighing errors. If the concentration of the compound to be measured is too low, then recovery as a pure compound, even partially, could be difficult.

The preceding method is known as *reverse isotope dilution analysis* when the analyte is the marked compound (therefore radioactive). The principle remains the same: the activity of the analyte (measured from an isolated fraction of the analyte) is determined before and after dilution with the same unlabelled compound. The calculations are identical.

17.2.2 Substoichiometric Method

The above method must be adapted when the analyte is present in very small quantity and impossible to recover in its initial form by employing a traditional procedure. Equations (17.2) and (17.3) suggest that in order to calculate m_x it is not necessary to know A_x and A_s if their ratio is known.

The method consists of making an insoluble derivative of the analyte after mixing in order to be able to recover a small part of it. The same is done with the labelled standard alone (analytical blank) to obtain the same derivative. This method, called *substoichiometric*, pre-dates the immunochemical method for trace measurements.

> For example, let us imagine measuring the quantity of sulfate ion present at only a few percent in an aqueous solution. After having added a known quantity of $^{35}SO^{2-}_4$ as the label, a solution of a barium salt (to form insoluble $BaSO_4$) is introduced in a quantity insufficient to precipitate all of the sulfate ions. By comparing the activity of the recovered precipitate with that of the precipitate obtained from the same quantity of labelled sulfate alone, under identical experimental conditions, the ratio of the activities can be deduced, and thus the quantity of sulfate in the original sample.

17.2.3 Radioimmunoassays (RIA)

The methodology described above, as illustrated by the sulfate example, has been employed since the 1960s for the measurement of insulin in serum. This technique, named *radioimmunology*, is an application of immunochemistry *in vitro*.

The basic idea is, after mixing, to separate the analyte by a means of an ultra-specific reagent (an antibody), which makes it possible to study a single type of molecule amongst thousands of other substances. This approach has led to immuno-measurements that have benefitted enormously from the development of genetic engineering and biochemistry.

As in the previous method, we take a sample containing the analyte and add to it a known quantity of this same analyte in a radioisotopically labelled form (Figure 17.1). Next, to isolate a fraction of the mixture of the two forms, labelled and nonlabelled,

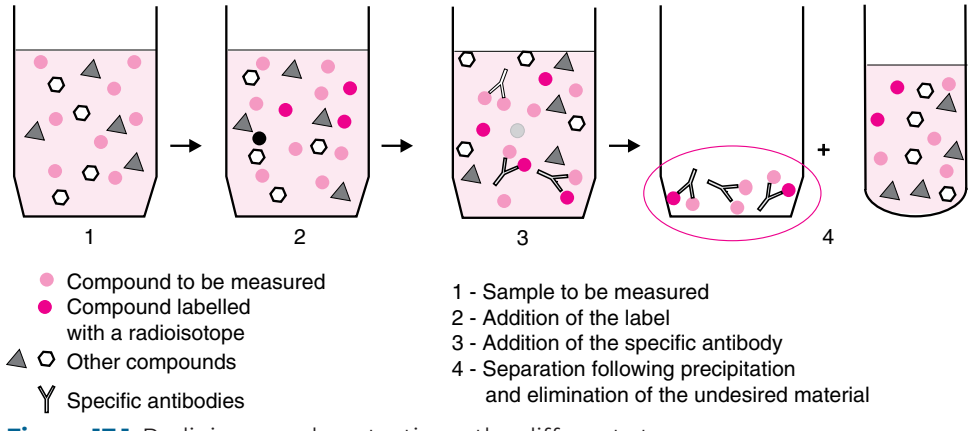


Figure 17.1 Radioimmunology testing – the different steps.

a specific antibody of this analyte is added in a substoichiometric quantity in order to fix both forms. They will be recovered in the same molar ratio as they are in solution. This mixture of complexes will then be separated by precipitation before we assess its activity by comparing it with a reference treated identically.

Radioimmunology has remained a method of choice for certain studies in clinical medicine but has few applications for the assay of low-molecular-weight molecules, because of the use of radioactive substances. However, the method is currently being applied to small compounds labelled with an enzyme (Section 17.5), notably for applications relating to the environment.

17.3 MEASURING BY ADDING A STABLE ISOTOPE

The isotopic dilution method may also be done using stable isotopes. Isotopic concentration measurements are classic applications of mass spectrometry or NMR. There are also *isotopic mass spectrometers* dedicated to these measurements. This method is very sensitive, and therefore it is used for both molecules and elements (about 60 elements have stable, accessible isotopes). Unlike radioactive labelling, in which the assay is based on the detection of atoms that have decayed during the measurement period, here *all* the labelled atoms are taken into account.To measure traces of caffeine in a sample using the HPLC/MS coupled method, we start by making a

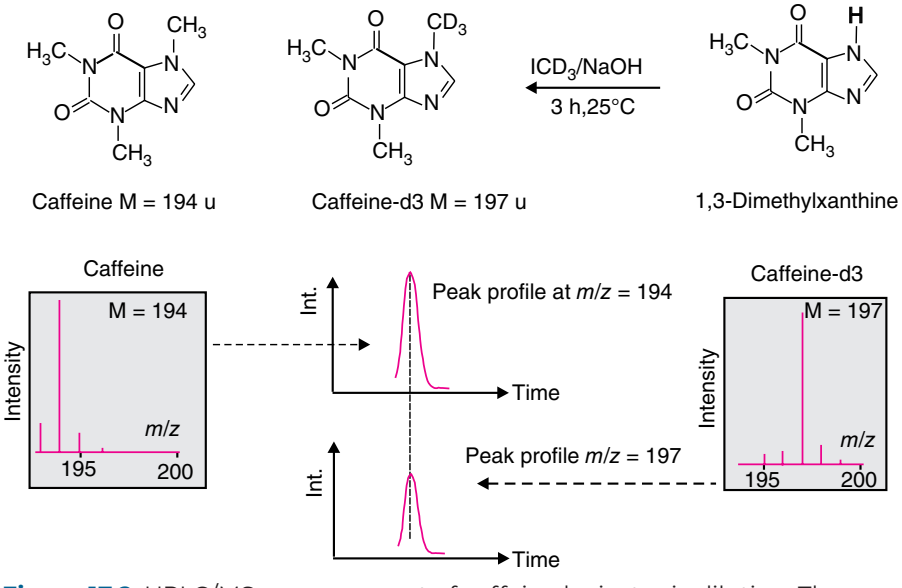


Figure 17.2 HPLC/MS measurement of caffeine by isotopic dilution. The stable isotope used as a tracer is deuterium D. Caffeine-d3 is a product of the methylation of 1,3-dimethylxanthine by ICD_3.

[1] Through synthesis, it is possible, in caffeine (m = 194 u), to replace the three H atoms in the N–CH_3 group of the five-member ring with three atoms of deuterium (D). This produces labelled caffeine of mass 197 u.

solution of this sample in which a known quantity of the d3[1]-labelled caffeine is added as a tracer. In HPLC analysis, these two caffeine samples have the same retention time. During their coelution, the detector alternately measures peak intensities for $m/z = 194$ and $m/z = 197$. Two pseudo-chromatograms (Figure 17.2) are obtained: one for peak 194 and the other for peak 197. Their areas, after calibration, enable us to determine the caffeine concentration in the starting sample.

17.4 MEASURING ISOTOPE RATIOS FOR AN ELEMENT

The analysis of the isotopic composition of many components present in various samples has become rather common in the field of natural substances, because it helps determine their geographical origin. Therefore, it has an economic or regulatory impact. For organic compounds, carbon or hydrogen isotopes are used. The precise measurement of isotopic ratios is also used, with appropriate devices, to improve the accuracy of the various radioactive dating methodologies based upon radioactivity.

In the living world, isotopes of the same element can be differentiated by the biosynthetic pathways they follow. Furthermore, the rate at which a molecule passes through cell membranes will depend on the molecule's isotopic distribution. For example, the $^{13}C/^{12}C$ isotopic ratio of natural vanillin is lower than that of synthetic vanillin; the same effect is observed for glucose, for which the isotopic distribution of carbon and oxygen varies, depending upon the biological cycle being followed by the plant. The detection of adulterated table oils, flavourings, and fruit juices as well as studies of plant metabolism and of numerous medical applications use isotopic abundance as an indicator.

The 'isotopic signature' of natural organic compounds relies on the precise measurement of the $^{13}C/^{12}C$ ratio, which is compared with that of a universal standard. This reference standard is calcium carbonate from Pee Dee (PDB for Pee Dee Belemnite, USA) whose ^{13}C abundance is very high (isotopic $^{13}C/^{12}C$ ratio = 1.12372×10^{-2}). Experimentally, the determination is done by measuring the peak intensities of $^{13}CO_2$ (45) and $^{12}CO_2$ (44) obtained by combustion of the sample. The relative deviation δ (in parts per thousand) of the compound is calculated in this way (Eq. (17.4)). This value of δ is generally negative.

$$\delta_{0/00} = 1000\left[\frac{\left[\frac{^{13}CO_2}{^{12}CO_2}\right]_{sam}}{\left[\frac{^{13}CO_2}{^{12}CO_2}\right]_{ref}} - 1\right] \qquad (17.4)$$

As this measurement involves combustion, the isotopic distribution at the different sites within the compound analysed is not available. For this reason, it is preferred to compare the D/H ratio by ^2H NMR, since this facilitates a comparison at each site of the molecule studied.

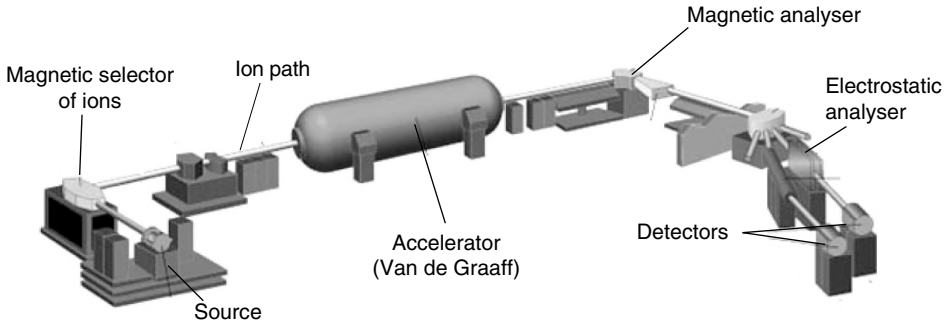


Figure 17.3 Different parts of an accelerator mass spectrometer (AMS). (Source: Courtesy of the Australian Nuclear Science and Technology Organization.)

With a mixture of two compounds A and B, whose isotopic ratio distributions are, respectively, δ_A and δ_B, the experimental value for the mixture δ_M results from the weighted combination of δ_A and δ_B. If *x* represents the B fraction and (1 – *x*) the A fraction, the following equation allows for the calculation of percentages for A and B:

$$\delta_M = (1-x)\delta_A + x.\delta_B$$

$$x = \frac{\delta_M - \delta_A}{\delta_B - \delta_A} \qquad (17.5)$$

There are devices that combine a gas chromatograph, a tubular combustion oven (to generate carbon dioxide), and a low-resolution magnetic field spectrometer with a multicollector of ions for individual tracking of each mass.

Accelerator Mass Spectrometry (AMS)

The precise measurement of the isotopic ratios for very rare isotopes is beyond the ability of conventional mass spectrometers, even if they are specially adapted to such measurements. For isotopes that exist as infinitesimal traces, there exists worldwide a network of around 50 mass spectrometers, each derived from Van de Graaff accelerators, that are used for these analyses (Figure 17.3).

The sample, which must be a solid, is first bombarded by caesium ions. The first mass spectrometer extracts the negative ions corresponding to the preselected mass. Next, to eliminate interferences (isobars), the ions are accelerated by a potential difference of several megavolts towards the positive end of the accelerator. Along the ion beam path, ions collide with a target (a solid or a gas) so that only isolated atoms with several positive charges subsist beyond this point. Concentration measurements are then made by alternately recording isotopic signals arising either from the sample, the standards or the analytical blanks. As such, the dating of ancient objects from their ^{14}C concentration has become much more precise and sensitive, than the classic Libby method, using radioactivity measurements.

17.5 MEASUREMENTS USING ENZYMATIC LABELLING

Another form of labelling consists of using an enzyme. In this case, the label (the enzyme) is enormous compared to the analyte molecule. As concentrations are very low, recovery involves the use of an antibody. This is an immuno-enzymatic assay, the best known example of which are ELISA (Enzyme-Linked Immuno-Sorbent Assay) tests. The amplification factor of the enzyme makes this test very sensitive. The ELISA tests described below require the review of a few concepts.

17.5.1 Antigens and Antibodies

The introduction of a foreign substance (*antigen*) of sufficient molecular weight ($M >$ 5,000 Da) into the organism provokes the production of biomolecules called *antibodies*, which are glycoproteins referred as to immunoglobulins (principally IgGs).

Antigens and antibodies agglutinate with each other. The forces that bind them are principally due to hydrogen bonding and to hydrophobic interactions. Cohesion is also ensured by neighbouring water molecules that exert a *hydrostatic pressure*.

The antigens *induce* not only an immune response but equally *react* with the antibodies which they make appear in the living organism. However, these two properties are different. Small organic molecules such as pesticides are not able to induce the production of antibodies. On the other hand, they will react if suitable antibodies are already present: these molecules are called *haptens*. To generate the antibodies specific to these haptens, the latter must be attached by a covalent bond to transporter proteins (*bovine* or *human serum albumin*, BSA or HSA) or to polysaccharides. Therefore, the hapten must have reactive functional groups, and this structural modification must not influence the specificity of the antibodies that will be produced.

Polyclonal antibodies

The development of antibodies is rather difficult work. If only a single hapten molecule was linked to a carrier protein such as BSA (M = 66,000 Da), the antibody formed would be directed above all towards the epitopes, corresponding to sections of the protein that trigger the production of the antibodies. Thus, several dozen haptens should be bonded. Under these conditions, the majority of the antibodies obtained are specific. For this, successive injections are made on mice, rats or rabbits with the hapten-bound carrier protein (immunogen) accompanied by certain additives. Two weeks later, the same process is repeated with one-fifth of the immunogen concentration. Two weeks after that, blood samples are taken from the animals to verify if the antibodies have appeared. The injections are repeated two or three times in order to create hyperimmunization. The serum of the animals is now a source of the desired antibody, which is of the polyclonal type.

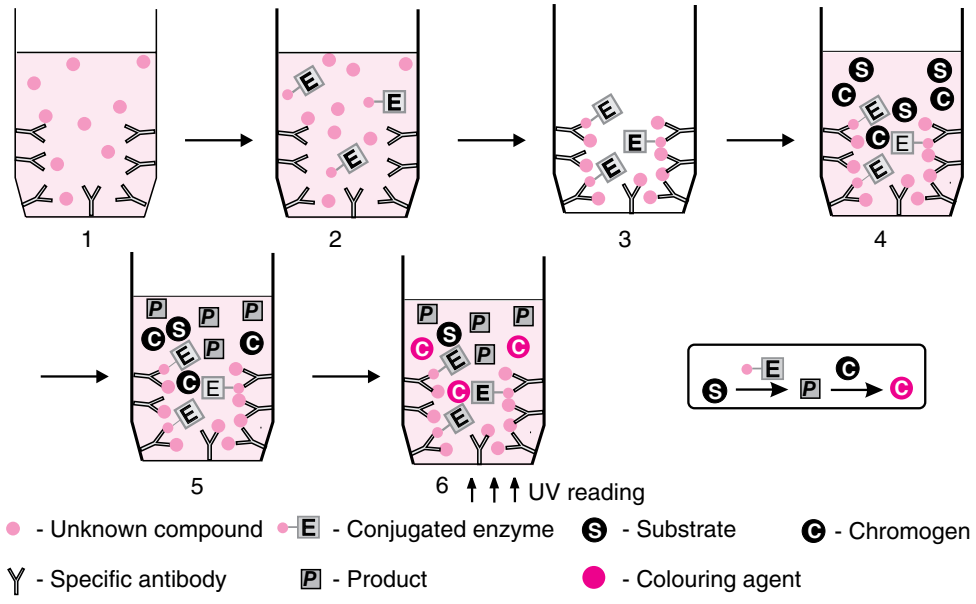


Figure 17.4 The different steps of an ELISA-type immune-enzymatic test with competition (see text). In clinical analysis, there exist numerous measurements of this type.

17.5.2 Enzymatic-Immunoassay (EIA) Method

The assay occurs over several stages:A solution of the test sample (or of a calibration standard) is added to a test tube (or microplate cuvette) with a transparent bottom for detection purposes. The walls have been sensitized by coating them with the appropriate antibody fixed to the surface through adsorption (Figure 17.4).

1. A known volume of a solution of the same compound, which is covalently bound to an enzyme, such as horseradish peroxidase (*enzymatic conjugate*), is added. During an incubation period, the two forms of the compound will enter into competition for binding with the antibodies present on the container walls.
2. Following the incubation period (e.g. 30 min), the container (tube or cuvette) is rinsed several times with water. Only bound molecules will remain in the container. The quantity of the enzyme-linked compound fixed on the walls will be greater when the amount of free unlabelled compound is low in the sample.
3. Then one adds substrate S, specific for the enzyme, as well as chromogen C designed to change colour when acted upon by the reaction product of S on the enzyme (e.g. tetramethylbenzidine).
4. The fixed enzyme will transform a large number of molecules S into species P, which will react with C to give a coloured product. The less there is of the analyte

in the sample, the more there will be of the fixed enzyme and therefore the colour intensity will be stronger.

5. Lastly, the reaction is quenched after a short period by addition of a strong acid, which destroys the enzyme. The test result will then be read with a colorimeter.

Horseradish peroxidase (M = 44,000 Da) is stable and reactive. It has four lysine residues that do not otherwise participate in its activity and are used for its coupling. As its name indicates, a substrate S of this enzyme is hydrogen peroxide, whose decomposition generates molecular oxygen, which reacts with the chromogen (TMB, tetramethyl benzidine). In the presence of luminol, chemifluorescence appears in stable form for several minutes and thus can be used for extremely sensitive measurements (see Chapter 11, Section 11.7).

This classic biological methodology is widely used and applied to fields such as the environment and agriculture in research on pesticides or herbicides (Gaucho, Gramoxone, atrazine, organochlorides, etc.), anabolic steroids and other toxic compounds (aflatoxins, HPA). However, we can only measure compounds for which there exist an appropriate antibody and its enzymatic conjugate.

Absorbance/concentration relationship

N is the total number of anchoring sites on the tube wall, n^* is the number of labelled molecules adsorbed, C^* is the concentration of the labelled species in the solution and C is the concentration of the nonlabelled unknown (the analyte).

The number of bonding sites on the tube wall is very small compared to the number of molecules in solution. The two labelled and nonlabelled species will thus become attached to the antibodies in the ratio of their concentrations in solution (Figure 17.5).

$$\frac{n^*}{N-n^*}=\frac{C^*}{C}=R \tag{17.6}$$

As the absorbance A is proportional to n^*, we get $A = kn^*$ and thus:

$$\mathrm{A}=kN\left(\frac{C^*}{C+C^*}\right)=kN\left(\frac{R}{1+R}\right) \tag{17.7}$$

If we plot concentration along the x-axis and, on the y-axis, the percentage absorbance of the solution relative to a blank (in which all of the antibody sites are occupied by the conjugated enzyme), we can see that absorbance does not vary linearly with concentration. However, if the concentrations are plotted on a logarithmic scale, part of the graph will be linear as a first approximation.

The final absorbance measurement can be replaced by fluorescence detection (this is a fluorescence ELISA test). In this case, the conjugated enzyme chosen is an alkaline phosphatase, which converts the specific substrate into a fluorescent compound.

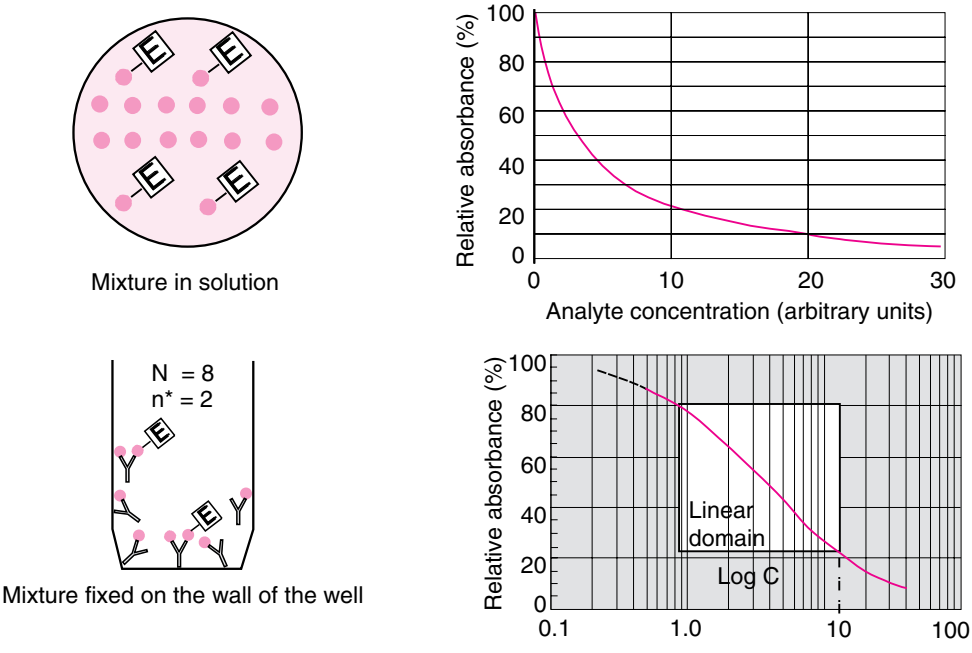


Figure 17.5 Relationship between concentration and absorbance of ELISA assays. The ratio of the two species in solution or fixed to the wall is the same (3 on this figure). Linearity, plotted on a semilogarithmic scale, is only attained over a narrow range of concentrations.

17.5.3 Other Immunoenzymatic Techniques

Several variants of the ELISA technique have been developed using a similar principle. For example, the EMIT (Enzyme Multiplied Immunoassay Test) technique is recognized for quantifying many organic molecules (hormones, drugs, medication, etc.) with detection limits reaching 1 ng/ml. Unlike the ELISA technique, the EMIT technique is done in homogeneous phase.

In the EMIT technique, the analyte (considered as the antigen present in the sample), to which a known quantity of this same analyte labelled with an enzyme (labelled antigen) is added, is put into competition on a limited and insufficient number of antibody sites (Figure 17.6). After an incubation period, there will be two types of antigen–antibody complexes, one with a labelled antigen, the other with the free analyte (nonlabelled antigen). After rinsing to eliminate unbound antigens, there remains, from a catalytic point of view, the free enzyme whose properties remain and the labelled enzyme that has lost its properties (*infected* enzyme). Antibodies will attach even more on the nonlabelled analyte when the latter is more abundant. Consequently, if there are more analyte molecules, there will be more labelled molecules that are not complexed by the antibody. The implementation of the enzyme-specific substrate will lead to even

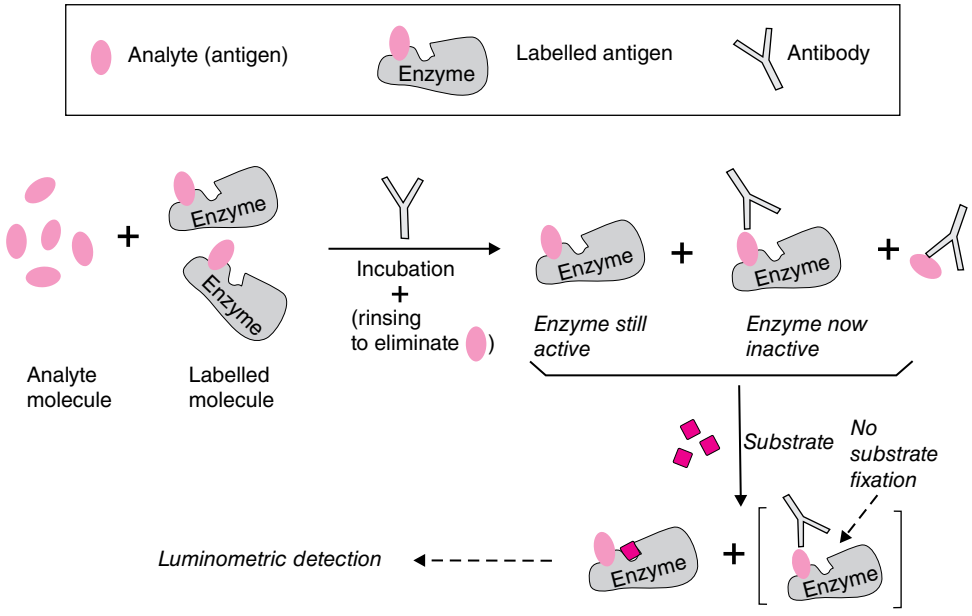


Figure 17.6 The EMIT technique. Enzymes are often used as tracers because they behave as catalysts that amplify the production of detectable species.

more intense colouring when there are more analytes to quantify, and the results of the reactions implemented will depend on the selected enzyme/substrate couple.

The EMIT technique is at the origin of tests used to quickly detect illegal drugs. These tests may come as absorbent strips on which the various ingredients are deposited. A drop of sample is placed on this strip. Its migration by capillarity triggers these reactions until the final colour is obtained. Qualitative results (positive or negative) are obtained with a predefined positivity threshold.

17.5.4 Advantages and Limitations of the ELISA Technique in Chemistry

The various ELISA assays give both precise and reliable results more rapidly than chromatography. They are particularly employed to eliminate all of the nonuseful negative samples, which need not be submitted to the extraction steps and to a subsequent chromatographic analysis. There are, however, several disadvantages:

- The possible recognition of several molecules with different sensitivities can cause scatter in the results. There is therefore a risk of cross-reactivity (a false positive).
- The range of measurement is relatively narrow. As the concentration increases, the measurement becomes less precise (an effect of using a logarithmic scale for the concentrations). A correct dilution for the sample must be found.

- The wells in microtitre plates all have to contain the same quantity of antibodies with the same reactivity. This is, in fact, technically difficult to achieve. The immunization of a laboratory animal is a singular event, which explains that results can differ from one ELISA kit to another.
- These kits must be stored at low temperature. They have a limited lifetime, especially for field analyses.
- The costs of these assays with the duplicate tests and the necessary standards must be kept in mind. They are economical only for a large number of analyses.

17.5.5 Immunofluorescence in Homogeneous Phase by Fluorescent Polarization

A variant of the ELISA technology, both sensitive and rapid, consists of putting the compound to be measured in competition with this same molecule labelled with a fluorophore. The principle of the measurement exploits a specificity of fluorescence. When a small fluorescent molecule in solution is excited by polarized light, the light emitted is not polarized, due to many factors, including the rapid rotation of the molecule around itself. However, if the movement of the molecule is sufficiently slowed down, which occurs when the labelled compound is immobilized by the antibody, then the light emitted is also polarized.

The last step of an immunofluorescent test thus consists of measuring the fluorescence polarization. Therefore, the fluorescence observed will be more polarized when the concentration of the analyte is low, and vice versa.

With a fluorometer adapted to this type of measurement, the data processing software determines the *polarization rate*, which corresponds to the ratio of the difference in fluorescence intensities measured in a parallel plane and then in a perpendicular plane to the polarization plane of the exciting light, over their sum.

The many applications include assays of fungicides, mycotoxins or agrochemicals.

17.6 NEUTRON ACTIVATION ANALYSIS (NAA)

17.6.1 Principle

Around 60 elements can be identified and measured following their transformation into radioisotopes by irradiation with particles. This labelling process is part of the domain of multi-elemental nuclear activation analysis whose sensitivity varies with the element (from 1000 ppm to 1 ppb). Neutrons are the most commonly used particles, because this technique is comparatively easier than those employing charged particles. The capture of one or more neutrons transforms the target nucleus into heavier isotopes (the element remains the same), some of which are unstable and decompose generally by β^- emission (see the following equation and Figure 17.7). The total radio-

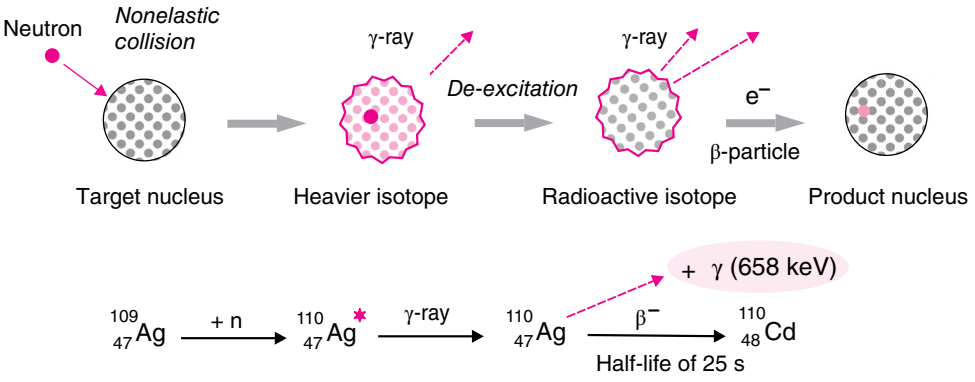


Figure 17.7 Diagram of neutron activation. When a neutron interacts with the target nucleus, a potentially unstable isotope forms. If so, it will almost instantaneously de-excite into a more stable configuration through emission of γ-rays. Then, this radioactive isotope decays by emission of an electron and one or more characteristic γ-rays, at a rate depending on the half-life of this nucleus. Illustration with an ^{109}Ag atom as target.

activity resulting from a sample containing several elements – each consisting of an isotopic family – will of course lead to a complex emission spectrum.

$$ {}^{A}_{Z}X + {}^{1}_{0}n \rightarrow {}^{A+1}_{Z}X^{*} \rightarrow {}^{A+1}_{Z+1}Y + e^{-} $$

The total radioactivity generated in the sample will diminish with time, following a curve corresponding to the envelope of the superimposed activities (the activity of each individual radioelement present follows an exponential law).

Since the β^- spectrum is continuous in terms of energy, the composition of the elements cannot be deduced by a simple analysis of the radiation. However, the key for identification comes from the γ *emission* that accompanies the β^- emission and is characteristic of each radionuclide. This emission spectrum occurs in the same spectral range as X-ray fluorescence.

17.6.2 Sources of Thermal Neutrons

The probability of causing a nuclear transformation depends on the nuclide and on the neutron energy, which together are represented by a parameter called *neutron cross-section*. The known processes to irradiate a sample with neutrons include the use of intense fluxes of 10^{14}–10^{18} neutrons · m^{-2} · s^{-1}, which are available in nuclear reactors of at least 100 kW. With an analyte whose concentration is very low, sufficient levels of activation are reached in several minutes, even if the isotope formed has a long period. The process imposes several restrictions regarding the

sample to be treated: it must be thermally stable and preferably solid. It is enclosed in a tube with a comparator standard of known concentration, before being introduced in a reactor beam port.

Small sources of neutrons – containing an α-emitter (several μg of ^{241}Am or ^{124}Sb) encapsulated in a beryllium envelope – have been developed to avoid these costly logistical restrictions. The nuclear reaction generating the neutron is as follows:

$$^{9}_{4}Be \xrightarrow{\alpha} {}^{12}_{6}C + {}^{1}_{0}n$$

A variant consists in using a few μg of ^{252}Cf (τ = 2.6 years) as a source of fast neutrons (2 MeV) that are slowed by collision with hydrogen atoms (2.4 × 10^6 neutrons · s^{-1} · μg^{-1}). These low-intensity sources release a few hundred million neutrons per second, allowing the measurement of around 20 elements.

17.6.3 Induced Activity – Irradiation Time

The number of radioactive atoms N^* that accumulate in the sample during bombardment tends towards an upper limit: at each instant, the increase in the number of nuclei N^* is equal to the difference between the rate of formation, assumed to be constant (as the number of target nuclei N is very high), and the rate of decay:

$$\frac{dN^*}{dt} = \left(\phi\sigma N\right) - \left(\lambda N^*\right) \tag{17.8}$$

where φ represents the neutron flux, λ the radioactive decay constant, σ the effective cross-section of each target atom, and N is the number of target atoms. This last number is itself related to the mass m of the isotopic fraction f of the element of atomic mass M. If N_A represents Avogadro's number, then:

$$N = \left(m / M\right) N_A f \tag{17.9}$$

The integrated form of Eq. (17.8) allows us to evaluate the number of atoms N^* present after a given irradiation time t_i:

$$N^* = \frac{\varphi \cdot \sigma \cdot N}{\lambda}\left(1 - \exp\left[-\lambda t\right]\right) \tag{17.10}$$

If we consider that the induced activity measured at time t corresponds to $A = \lambda N^*$, then:

$$A = \phi \cdot \sigma \cdot N\left(1 - \exp\left[-\lambda t\right]\right) \tag{17.11}$$

The term in parentheses, called the *saturation factor*, tends rapidly to 1 as t increases. In this way, for $t = 6\tau$ (see Section 17.8), the activity reaches 98% of the limit value. Experimentally, the time of irradiation never exceeds four or five half-lives of the radioisotope. This method should preferably be used for short-lived radioisotopes.

17.6.4 Measurement Principles and Applications

The choice of the detection process depends on the nature of the emitter and the complexity of the emitted spectrum. The most effective solution consists of recording the γ spectrum that accompanies the majority of the β^- emissions of the radionuclides present in the sample following activation (Figure 17.8).

A classic sensor has a NaI(Tl) crystal, which acts by transforming the γ-photon into a luminescence whose intensity is proportional to the energy of the photon, at least if the γ-photon is entirely absorbed by the crystal. The functioning is similar to that of liquid scintillators used for ^{14}C, or if it is a Ge(Li) crystal, it behaves like the fill gas of a Geiger–Müller tube.

If the sample and the indicator are irradiated simultaneously, the specific activity of element X will be the same in the sample and in the standard. Since the total activity is proportional to the mass of X, then:

$$\left(Mass\ of\ X\right)_{sample} = \left(Mass\ of\ X\right)_{std.}\ x\left(Total\ Activity\right)_{sample} / \left(Total\ Activity\right)_{std.} \qquad (17.12)$$

The above equation is valid only if the induced radiation is simple, or when the counting instrument is fitted with a filter permitting the isolation of the radiation due only to the analyte. However, often the matrix itself becomes active and emits

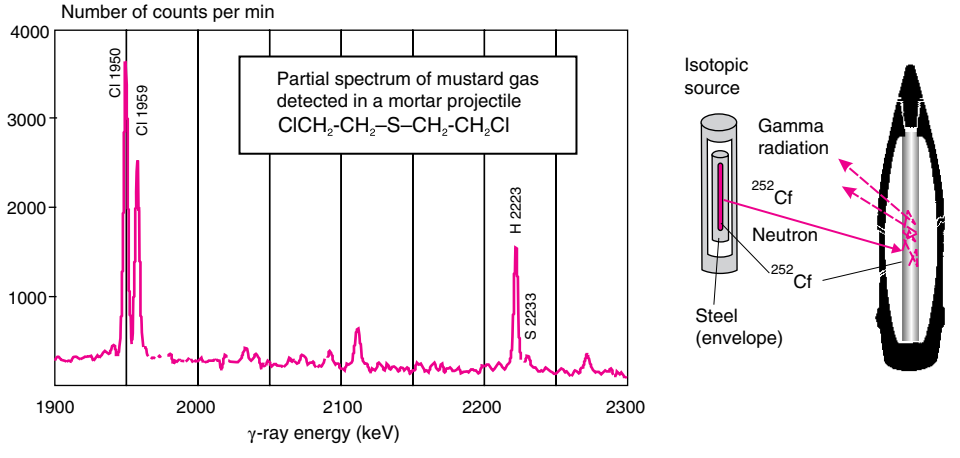


Figure 17.8 A nonconventional application of neutron activation. When it is necessary to destroy a warhead containing chemical weapons, neutron activation can be used to learn about its contents. This diagram, reproduced courtesy of EG & G Ortec, illustrates the general principle of the method. On the left, a small part of the γ spectrum of a warhead containing a chemical agent, obtained in several minutes.

radiation, which comes to be superimposed upon that to be evaluated. For that reason, a certain time gap between the end of irradiation and the start of the measurement must be complied with, in order for the interferences caused by short-lived emitters to be eliminated.

> Suppose, for example, that traces of iron are to be measured in a sample of aluminium. The ^{59}Fe (τ = 46 days) isotope is characterized by γ emission at 1.29 MeV, but during the irradiation time aluminium yields ^{24}Na, which is responsible for a γ-ray situated at 1.37 MeV, through the reaction ^{27}Al(n, α)^{24}Na (τ = 15 hours). To give the short-lived aluminium the time to decay, the measurements are postponed for a few days.

When the sample allows it, the radioelement to be assayed can also be separated using the *entrainment* technique, with a stable isotope of this element. The same treatment conducted on the standard leads to an assessment of the extraction yield. Then, the result is normalized to 100%.

In fact, the neutron activation analysis method is not very common (Figure 17.8). Applications include the characterization of materials (highly pure metals, semiconductors, etc.), the study of the distribution of chemical elements in fossils, ultra-trace analysis in archaeology or in geological and volcanic studies.

Further Information on Radiochemistry

17.7 REVIEW OF RADIOACTIVE ISOTOPES

All chemical elements are generally made up of several isotopes that only differ by the number of neutrons in their nucleus. Some of the natural isotopes of a given element are unstable. These are radioisotopes. Their nuclei have the property of spontaneously decaying and thus yielding a daughter nucleus that differs from the parent nucleus, accompanied by the emission of particles (α and β^-) and by electromagnetic radiation in some cases (γ- or X-rays). The daughter nuclei formed may also emit radiation.

β^- decay corresponds to a nuclear transformation in which the mass number remains constant, while the atomic number varies. β^- emission produces electrons whose energy varies continuously.

$$ {}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{-1}_{0}e $$

17.8 HALF-LIFE τ, RADIOACTIVITY CONSTANT λ AND ACTIVITY A

Any radionuclide, whatever the type of radiation emitted, is characterized by its half-life τ, which is the time taken for half of the corresponding atom population in the sample to decay (from initial time, t = 0). The law of radioactive decay allows calculation

of the number of atoms N present after time t for a population containing N_0 atoms initially. This decrease follows a first-order velocity law, as described by the following equation:

$$N = N_0 e^{-\lambda t} \tag{17.13}$$

where λ represents the radioactive constant of the studied radionuclide. This value, expressed in time^{-1}, is related to the radioactive half-life τ in Eq. (17.14):

$$\tau = \frac{1}{\lambda} \ln 2 \tag{17.14}$$

In fact, what is experimentally accessible is not the number of nuclei present in time t but rather the number of disintegrations per second, called activity A, expressed in Becquerels, or Bq (dps, disintegrations per second), or in Curies, Ci (1 Ci = 3.7 × 10^{10} Bq), which is the more common unit. The activity A is proportional at time t to the number of emitter nuclei N, and therefore to the concentration of the radionuclide, since the proportionality constant is none other than the radioactive constant λ. By saying $A = \lambda \cdot N$ and $A_0 = \lambda \cdot N_0$, the radioactive decay law (Eq. (17.13)) may therefore be rewritten as:

$$A = A_0 \cdot e^{-\lambda t} \tag{17.15}$$

For radioisotopic analyses, the specific activity (A_S) is considered, as this is the activity relative to the totality of nuclei of the same element present (emitters or not). It may be expressed as Ci · mol^{-1} (or as Bq · mol^{-1}), as Ci · g^{-1} (or as Bq · g^{-1}) or as Ci · l^{-1} (or as Bq · l^{-1}) for liquid samples.

17.9 RADIOACTIVE LABELLING OF ORGANIC MOLECULES

A requirement of the isotopic dilution method with radioactive labelling is that the latter must be available. There are suppliers that specialize in the custom manufacture of a relatively large number of organic molecules with single-atom labelling (C, H, S, etc.). Activity can reach about 60 mCi/mmol, if each individual molecule contains a ^{14}C atom at a given site (Figure 17.9). Where possible, ^{14}C is preferred to ^{3}H (tritium), since the latter is easily exchanged and its autoradiolysis creates conservation problems for tritiated molecules.

The ^{14}C isotope used to prepare labelled compounds is obtained in a nuclear reactor by irradiation of solid targets containing atoms of nitrogen (aluminium or beryllium nitride) with low-energy neutrons, known as thermal neutrons, themselves the product of the controlled atomic fission of ^{235}U. The radiocarbon formed is next isolated from

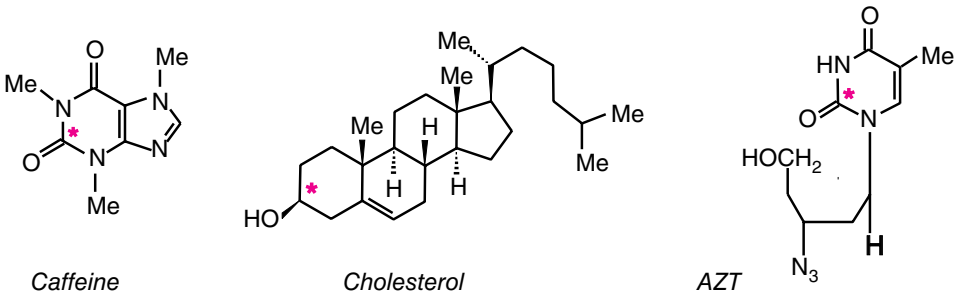


Figure 17.9 Three molecules labelled at a single site with ^{14}C. This radioisotope has a sufficiently long half-life, which means labelled compounds can easily be stored while avoiding corrections to take into account natural decay during the measurement.

the target sample by oxidation to $Ba^{14}CO_3$, the form in which it is delivered to chemists. From $^{14}CO_2$, it is possible to use a plethora of organic chemical reactions to synthesize different compounds in which the radioisotope can be introduced in a specific position.

17.10 DETECTION AND COUNTING OF RADIOACTIVITY

The radiation energies emitted by radioisotopes are quite variable. In the neutron activation method, detection employs the same techniques as those used for X-rays. However for isotopic dilution methods, low-energy β^--emitters are selected (Table 17.1). For this reason, the counting technique consists in dissolving the sample in a solvent (toluene, xylene or dioxane – the latter for water-soluble compounds) that also contains a *scintillator,* also called a *fluor*. The latter transforms the β^--rays into a luminescence whose intensity is proportional to the number o*f* particles emitted.

Table 17.1 Some characteristics of commonly used radioisotopes.

Isotope	Half-life	Type of emission	Energy (MeV)
^3H	12.26 years	β^-	0.02
^{14}C	5,730 years	β^-	0.156
^{32}P	14.3 days	β^-	1.7
^{35}S	88 days	β^-	0.167
^{125}I	60 days	IEC*	0.149

(*) Internal electron capture

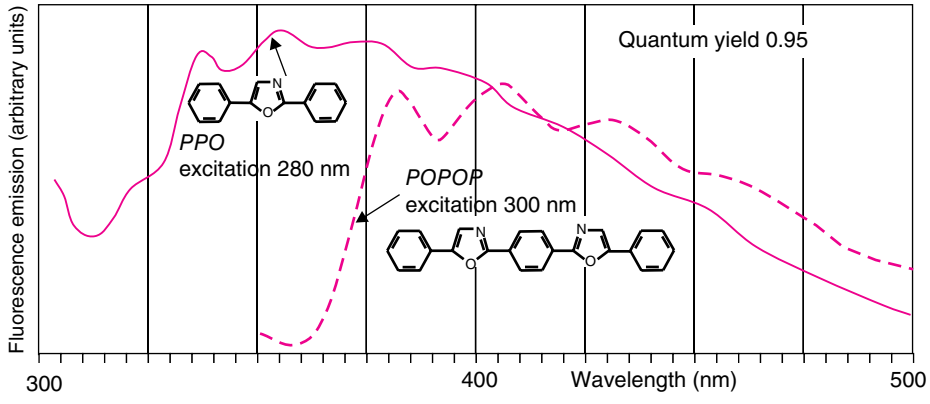


Figure 17.10 Three molecules labelled at a single site with ^{14}C. Fluorescence emission spectra obtained by excitation.

The solvent acts as a relay to transfer energy to the scintillator. Figure 17.10 presents two of the most common scintillators. The scintillation mixture is comprised of, for example, PPO (2,5-diphenyloxazole – which emits in the UV) or POPOP (emitting in the visible domain).

After crossing a filter to limit bandwidth, photons are counted during a given time, using one or two photomultipliers working simultaneously. A single β^- particle produces several hundred photons. Counting only occurs if the two PMTs assembled in opposition provide a signal whose offset never exceeds a few nanoseconds.

17.11 SPECIAL PRECAUTIONS

The methods described above require radioactivity measurements, which are the source of possible contaminations. They require a specific environment, subject to an authorization to have the right to store and handle radioactive substances. Obtaining a license to possess radioactive compounds is granted following an examination of the premises with respect to standards regarding nuclear waste. Moreover, the usual precautions must be taken, so that operators are not exposed to radiation levels exceeding values deemed acceptable by regulations.

Of the most frequently used radioisotopes, only ^{35}S emits high-energy γ-radiation. Nevertheless, a radioisotope, even of weak energy, is very dangerous once it is fixed in the organism. There is no simple and direct relationship between the intensity of radiation (Bq) and the irradiation dose (Sievert or Gray). The latter is a function of the quantity of energy released per mass unit in the irradiated substance.

KEY POINTS OF THE CHAPTER

1. Quantitative analysis by isotopic dilution is one of the labelling methods applicable to all sorts of species (elements or molecules) in a sample. The term *isotopic* is reserved for analyses based on a measurement concerning isotopes, whether radioactive or not.
2. It consists in adding a known quantity of the analyte with a different isotopic composition from the sample, so that it is recognizable, but behaving identically to the analyte. This labelled analyte dilutes the one that needs quantifying. Measurements are conducted using NMR or mass spectrometry.
3. The isotopic dilution method is applicable as long as there is an enriched analyte in the isotope, which will be used as a target to measure it. The result of the analysis relies on the measurement of the ratio of the two signals: before and after dilution. This is a specific, very reliable addition method.
4. In many fields, the abundance ratios of stable isotopes present in an analyte are used. Notation δ (dimension-less) defined by Eq. (17.4) corresponds to the deviation from the value of a standard adopted arbitrarily yet universally.
5. For a mixture of two species whose δ values are known, the δ value of the mixture corresponds to the weighted combination of each species.
6. Labelling may be obtained by immuno-enzymatic or immuno-chemical modification, including ELISA tests classified into several categories. These are competitive methods requiring the use of a tracer, that is analogous to the antigen and which can be tracked as it carries a radioactive atom or enzyme or a chemiluminescent site.
7. In the ELISA technique, the more concentrated the analyte is, the more it will mobilize the antibody and therefore will leave fewer fixation sites for the labelled analyte, which in the revelation stage will lead to less intense absorbance (or fluorescence).
8. The method is not without drawbacks: narrow measurement range, different reactivity of labelled and nonlabelled antibodies.
9. The EMIT method is rather similar but differs in that the antigen conjugated with the enzyme loses its enzymatic properties when it is fixed on the antibody. The final effect is reversed: the more analyte there is, the more intense the colour will be.
10. Neutron activation analysis identifies and quantifies a certain number of elements once they have been made radioactive using neutron irradiation.
11. Induced activity tends towards a limit that is more or less reached after five half-lives of the radioelement that appears (Eq. (17.11)).

PROBLEMS

1. To determine the quantity of penicillin present in a commercial preparation, an isotopically labelled reference of penicillin is used, whose specific activity is 75,000 Bq/g. 10 mg of labelled penicillin is added to 500 mg of sample. After treatment, 1.5 mg of penicillin is recovered with a measured activity of 10 Bq. Calculate the concentration (% mass) of penicillin in the commercial preparation.

2. Patulin ($C_7H_6O_4$) is a compound that is dangerous to human health found in the juice of rotten apples or grapefruits. The current method of measuring this compound resembles that of an immuno-enzymatic ELISA test. A standard solution of pure patulin is prepared in water at a concentration of 1.54 g/l just prior to use.

In the test described, four identical test tubes whose inner wall is covered with a suitable antibody are used. All of the tubes follow the same treatment, differing only in the solution which is introduced in the initial step. The contents are:

- tube 1: a blank, 2 ml of pure water.
- tube 2: 1 ml of pure water plus 1 ml of the undiluted standard solution.
- tube 3: 1 ml of pure water plus 1 ml of the standard solution diluted to 50%.
- tube 4: the sample, 2 ml of fruit juice, filtered and diluted twice with water.

In each tube, the same quantity of conjugated enzyme is introduced, plus all of the other necessary reagents combined.

Following the reaction, the measured absorbance *A* of the tubes is as follows:

- tube 1: $A = 1.03$.
- tube 2: $A = 0.47$.
- tube 3: $A = 0.58$.
- tube 4: $A = 0.5$.

a. Calculate the concentration (in ppm) of patulin in the standard solution.

b. Calculate the % absorbance (also known as the % inhibition) of the solutions in tubes 2, 3, and 4 with respect to tube 1.

c. Explain why the absorbance of tube 1 is greater than that of the other tubes.

d. Calculate the concentration of patulin in tubes 2 and 3 in μg/l.

e. Calculate the concentration of patulin in the fruit juice, in both mg/l and ppb.

3. 1. To measure elemental chlorine present at a very low concentration (of the order of a few ppm) in a sample of steel, using neutron activation analysis (NAA), a reactor producing a thermal neutron flux of 2×10^{16} n $\cdot m^{-2} \cdot s^{-1}$ is required.

a. Using the symbols provided, clearly write the reactions taking place: $^{35}Cl\ (n,\gamma)^{36}Cl$ (β^--emitter, $\tau = 3.1 \times 10^5$ years) and $^{37}Cl\ (n,\gamma)^{38}Cl$ (β^--emitter, $\tau = 37.3$ min.).

b. Explain why the γ emissions of the ^{38}Cl isotope are preferred for the experiment rather than those of ^{36}Cl.

c. Indicate two other methods that could be used to measure the chlorine in steel. Give a precise explanation including reasons, advantages, and disadvantages of the methods chosen as compared to NAA.

2. To eliminate the interference due to manganese (^{55}Mn) present in steel, which leads to ^{56}Mn whose half-life is 2.58 hours and whose radiation is intense, the chlorine is separated by precipitation in the form of silver chloride.

Simplified operating procedure: For five minutes and under the same conditions in the reactor, (a) a sample of the steel to be quantified in Cl and (b) a disc of filter paper onto which 100 μl of a 0.1 g/l chloride ion solution has been previously deposited (after first verifying that the filter paper contains no Cl) are irradiated with neutrons. Then the steel sample is dissolved into solution by boiling with 40 ml of 2 M HNO_3, to which is added 2.00 g of dry KCl. 50 ml of an aqueous solution of 15% (w/v) $AgNO_3$ is added to the resulting solution. The AgCl precipitate is recovered, washed, and then dried. The count results:

Test	Mass	Mass of AgCl	γ-count (1.64 MeV) of ^{38}Cl
Sample	0.51 g of steel	3.726	11,203
Standard	10 μg of Cl		48,600

a. Show that silver nitrate is added in sufficient quantity.

b. Calculate the concentration of elemental chlorine in the steel sample.

Recall that N = 14.007, O = 16, Cl = 35.453, K = 39.098, and Ag = 107.863 g/mol.

4. Insulin concentration can be determined using the radioimmunology (RIA) method with a calibration curve. To do so, standard solutions are prepared as follows: add a constant volume of a labelled insulin solution to insulin solutions of known concentrations so that final insulin concentrations are, respectively, equal to 3, 5, 7, and 9 ng/ml. The total activity of each solution is measured and is equal to 20,000 counts/minute. The same quantity of antibody is added to each solution. An insulin-antibody complex forms. It is isolated and then its activity is counted. The same experimental protocol is monitored on a known volume of a solution of an unknown insulin composition. The following results are obtained:

Insulin Conc.	3.0	5.0	7.0	9.0	Unknown
Activity	13,245	11,111	9,852	9,091	10,100

a. For each solution, calculate the noncomplexed (free) insulin activity. Deduce the free insulin/complexed insulin ratio (*R/R**).

b. Plot the R/R* graph according to the concentration for standards. Deduce the insulin concentration of the unknown being studied.

5. The isotopic factor δ (in %) of the carbon element, measured from the carbon dioxide produced by combustion of natural vanillin is δ = –20, while for synthetic vanillin it is δ = –30.

By using these two values, find the percentage composition in these two types of vanillin in a commercial vanillin sample, knowing that δ = –23.5 was measured.

6. To assess the volume of a large complex-shaped tank, the isotopic dilution method is used, usually with a soluble lutetium salt. This element, composed of two stable isotopes, ^{175}Lu = 97.4% and ^{176}Lu = 2.6%, has a mass of M = 174.97 g/mol.
   1. After filling the tank with water, add 2 g of hexahydrated lutetium trichloride, $LuCl_3 \cdot 6H_2O$, whose mass is 389.42 g. What is the mass of lutetium *X* added to the tank?
   2. After stirring to mix the lutetium salt, sample 1 l of water from the tank. To this sample, add 20 μg of hexahydrated lutetium trichloride prepared from the pure ^{176}Lu isotope ($^{176}LuCl_3 \cdot 6H_2O$ = 390.4 g/mol; ^{176}Lu = 175.94 g/mol).
   a. Calculate the mass of ^{176}Lu, in μg, added to the sample.
   b. Using a classic method, we find that the ratio of the two isotopes $^{175}Lu/^{176}Lu$ is equal to 9. What is the most appropriate instrument for measuring the isotopic ratio?
   c. Present arguments demonstrating that lutetium is the right choice.
   d. Calculate the mass *x* (also in μg) of lutetium in a litre of sampled water.
   e. Lastly, calculate the volume of the tank.
7. To measure iron using a substoichiometric method, in an aqueous solution, the isotopic dilution method is used with ^{59}Fe (β- and γ-emitter, τ = 44.5 days). The concentration of the aqueous solution of iron chloride ($FeCl_3$) used as a tracer is 0.1 M of iron (M = 56 g/mol), a tiny portion of which is composed of ^{59}Fe.

   In centrifuge tube **A**, add 3 ml of water and 1 ml of tracer solution. In a second tube **B**, add 3 ml of water, 1 ml of the solution to measure, and 0.5 ml of the tracer solution. Then, add 6N ammonia solution (NH_4OH) to both tubes in order to precipitate iron as iron (III) hydroxide. A fraction of each of the two precipitates is retrieved. After washing and drying of the precipitate, the γ-activity of iron (III) hydroxide is measured (average values from several measurements) in these two fractions. For **A**: 460 dps for 25 mg; for **B**: 86 dps for 35 mg.

   Calculate the iron concentration of the solution being analysed.

SOLUTIONS

1. A_S refers to the specific activity (per g) of the label, m_S is the mass of this label, A_X is the activity (per g) after recovery, and m_X is the unknown mass of penicillin in the sample. $M_S = 1 \times 10^{-2}$ g; A_S = 75,000 Bq/g; A_X = 6,667 Bq/g; m_X = 0.102 g. In 1 g of the sample there is thus twice as much penicillin, i.e. 0.204 g (20.4%).
2. a. Concentration of the patulin standard solution: 1.54 ppm or 1×10^{-5};
   b. Absorbance (or inhibition) percentage with respect to tube 1: tube 2 = 45.63%; tube 3 = 56.63%; tube 4 (sample): 48.54%;
   c. The absorbance of tube 1 is greater because if no analyte is added to the tube, all antibody sites are occupied by the enzymatic conjugate;

d. The quantity of patulin in tube 2 (2 ml) is $1 \times 10^{-5}/1{,}000 = 1 \times 10^{-8}$ mol, (1.54 μg) i.e. 770 μg /l (where log c = 2.8865). In tube 3, the molar quantity of patulin is two times less, i.e. 0.5×10^{-8} mol (0.77 μg) or 385 μg/l (where log c = 2.5854).

e. The initial solution contains 0.93 mg/l (i.e. 930 ppb) of patulin.

3. 1. a. $^{35}_{17}Cl + {}^{1}_{0}n \rightarrow {}^{36}_{17}Cl^{*} + \gamma$; ${}^{36}_{17}Cl^{*} \rightarrow {}^{36}_{18}Ar^{*} + \beta^{-}$; ${}^{36}_{18}Ar^{*} \rightarrow {}^{36}_{18}Ar^{*} + h\nu$
($\tau\beta_{-}$ = 3.1 × 10^5 years)

$$^{35}_{17}Cl + {}^{1}_{0}n \rightarrow {}^{38}_{17}Cl^{*} + \gamma; {}^{38}_{17}Cl^{*} \rightarrow {}^{38}_{18}Ar^{*} + \beta^{-}; {}^{38}_{18}Ar^{*} \rightarrow {}^{38}_{18}Ar^{*} + h\nu$$

($\tau\beta_{-}$ = 37.3 min)

b. Since the half-life is short, we can make a count over a period of several hours and then with the help of software, the fraction of this isotope may be identified with respect to the constant radiation of emitters with long half-lives (less intense).

c. Atomic absorption, but possible loss of volatile elements during sample treatment. X-ray fluorescence, but that is essentially a surface analysis.

2. a. 2 g of KCl correspond to 0.0268 mol. If AgCl was completely recovered, there would be 3.845 g. However, we found 3.726 g. Therefore, the mass yield is 97%.

b. 50 ml of a 15% solution of $AgNO_3$ (M = 203.868 g/mol) corresponds to 7.5 g of this salt, i.e. 7.5/203.868 = 0.037 mol. It is observed that the silver ion quantity is equal to 0.037/0.268, i.e. 1.38 times the stoichiometric quantity. The γ-count value for total recovery of ^{35}Cl is equal to 11,203/0.97 = 11,549.

The quantity of chloride in the steel is 11,203/48,600 × 10 = 2.38 μg, i.e 4.66 μg/g (4.7 ppm).

4. a. The free insulin activity is obtained by subtracting the activity of complexed insulin from the total activity: 6,755, 8,889, 10,148, 10,909, and 9,900 for the unknown.

b. 0.510, 0.800, 1.030, 1.200, and 0.980 for the unknown; C = 6.6 ng/ml.

5. The composition is 65% natural vanillin and 35% synthetic vanillin.

6. 1. X = 0.899 g of lutetium;

2. a. 9.013 μg of ^{176}Lu was added to the 1 l sample.

b. ICP/MS coupled method;

c. Lutetium, a rare-earth element, is of interest for two reasons. First of all, the quantification of the mass peaks 175 and 176 by the ICP / MS technique is located in a domain where it is very unlikely that the measurements will be disturbed by the presence of isotopes of the same mass but coming from other elements. Moreover, the choice of a hexahydrate, a crystalline solid that is stable, purifiable and water-soluble, facilitates the precise weighing of the small quantities of lutetium 175 and 176 (very rare) required.

d. x = 0.1088;

e. 8,262 l.

7. Specific activity (for 1 g) of hydroxide resulting from adding solution (**A**): $A_S^* = 460/0.25 = 18{,}400$ dps.

Specific activity of hydroxide resulting from solution **B**: $A_x^* = 86/0.035 = 2{,}457$ dps.

This activity comes from 0.5 ml of the addition solution. As the parent solution is 5.6 g/l, the quantity of iron is 2.8 mg. By applying Eq. (17.2): $m_x = 2.8\left(\frac{18{,}400}{2{,}457} - 1\right) =$ 18.2 mg/ml of solution or 18.2 g/l.

Chapter 18

Specific Analysers

Certain analytes, elements, ions or molecules are more frequently measured than others. In response to these specific needs for capacity, cost, and rapidity of analysis, there is a whole range of analytical instruments, used beyond the four walls of the laboratory, consisting of automated analysers, measurement stations or programmable industrial instruments dedicated to specific analyses or process monitoring. These analyses are sometimes based on the specific

and unique properties of the target analytes or on an original instrumental concept.

These specific instrumental analysers are found in a variety of fields, such as the oil industry, metallurgy, agriculture or environmental analysis (water, soil, plant analysis, etc.), the military, the fight against terrorism, sports, and many other areas. This chapter shows a few examples of these instruments.

Objectives

Discover the sector of specific analysers
Know the operating process of some specific analysers
Determine the percentage composition of an organic compound
Perform assays for specific elements (mercury, sulfur, carbon)
Present the basic principle of IMS
Describe the Karl Fischer volumetric titration

18.1 SPECIFIC ANALYSES

When hundreds of analyses per day must be undertaken, traditional laboratory methods and instruments can prove unsuitable (e.g. data throughput too slow or too much data to be processed). One field for which there is increasing demand is that of continuous systems analysis, outside the laboratory, for automation and real-time control of many industrial manufacturing processes. Therefore, in addition to general analytical devices, there are other instruments with integrated systems designed to obtain information quickly about the concentration of specific elements or compounds. The detection part of these instruments has progressed a lot with sensor technology, which has come into widespread use thanks to its diversity and which has made their use easier for a nonspecialist. The species most in demand for such analyses include certain atoms and molecules. Some methods implemented for the measurement of light elements, sulfur and mercury are described below.

18.2 ELEMENTAL ORGANIC ANALYSIS

The very great number of molecular compounds containing carbon (C) and other light elements, such as hydrogen (H), nitrogen (N), and oxygen (O), has led to specific demand to measure these elements. This need is found in a number of industries (petrochemicals, pharmaceuticals or agrochemicals) as well as in basic research in organic chemistry.

$C_{27}H_{27}N_3O_6S$	% calculated	C 62.18	H 5.22	N 8.06
M = 521.5 g/mol	% found	C 62.02	H 5.29	N 8.15

Figure 18.1 Presentation of an elemental analysis.

When molecules are synthesized via reactions or when new compounds are extracted from natural sources, their structure and purity must be determined. For this, a quantitative elemental analysis must be performed. This specific type of analysis (elemental organic microanalysis) enables us to find the percentage elemental composition, in the pure state, of the molecule under study. The measurement of a single element or even two elements (C and H are the most frequent) can be used to verify the proposed molecular formula for a molecule that is not as yet fully described but whose structure has been deduced from spectral studies. In another application, the purity of a compound whose composition and molecular weight are known can be determined by comparing the experimental results obtained from a sample with the theoretical ones (Figure 18.1).

For these light elements, atomic absorption spectroscopy (unusable due to a lack of suitable sources) or X-ray fluorescence spectroscopy (lack of sensitivity due to low-energy radiation) are inappropriate.

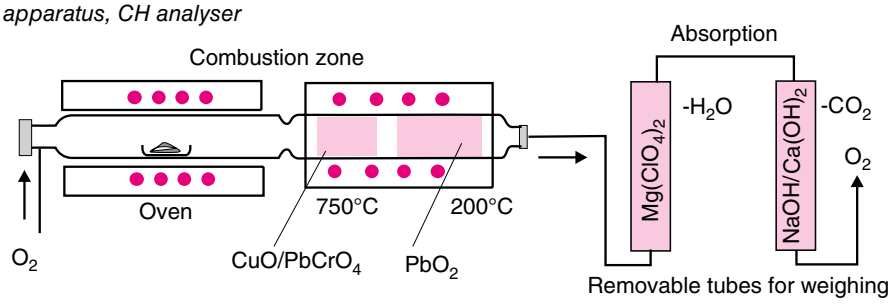
The principle applied for measuring these elements is to burn a small sample (a few milligrams) of the compound by raising it to a high temperature in the presence of molecular oxygen, with custom analysers. The elements hydrogen, carbon, nitrogen, and oxygen are recovered in the form of gaseous oxidation products. This also has the advantage of separating them physically from the matrix.

Elemental organic analysis gives essential information, though insufficient to deduce the molecular formula for a totally unknown compound. There are several reasons for this. Firstly, devices are designed to measure only a small number of elements among all those present. Furthermore, even if all of the elements present are measured, only the relative molecular formula is known. For example, the molecular formula $C_{54}H_{54}N_6O_{12}S_2$ would have the same percentage composition as the results shown in Figure 18.1. To remove this ambiguity, mass spectrometry or NMR are used in most cases to find the raw formula. Finally, it should be noted that for any compound, percentage analysis, even in the case of some experimental error, can be combined with mass spectrometry, for example, to find the compound's exact raw formula, with the atoms always in whole numbers.

18.2.1 The Traditional Methods of Pregl and Simon

Current elemental analytical methods are based on the same basic principle in use since 1830: to find the atomic composition of an organic compound from the gases formed during its combustion. Firstly, the compound undergoes rapid combustion at high temperature in an excess of molecular oxygen. If the elements carbon, hydrogen, and nitrogen are present, the resulting combustion products, carbon dioxide (CO_2), water (H_2O), and a mixture of molecular nitrogen and nitrogen oxides ($N_2 + NO_x$) will be formed. Elemental oxygen is determined separately by combustion in the presence of

Pregl apparatus, CH analyser



Simon apparatus, CHN analyser

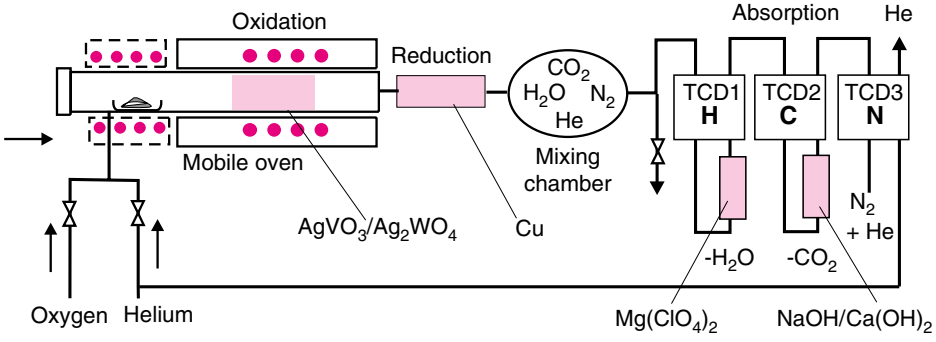


Figure 18.2 Pregl and Simon methods of microanalysis. Generations of students will recall having calculated, with some degree of anxiety, molecular formulae from the masses of carbon dioxide and water trapped in the removable absorbers. In Simon's more recent version, the reaction gases pass over copper powder to reduce the nitrogen oxides to molecular nitrogen. Three thermal conductivity detectors (TCD) signal the presence of H_2O, CO_2, and N_2. The measurement of the three elements requires around 10 minutes.

carbon to form carbon monoxide. The amounts of these gases lead by calculation to the percentage of the elements in the initial sample.

In the early microanalytical devices developed by Pregl (around 1930) for carbon and hydrogen, the combustion of a precise mass of about 1–3 mg was done at 750°C in a current of molecular oxygen, the transformation of CO into CO_2 being completed by passage over a mixture of copper oxide and lead chromate (Figure 18.2). The masses of elements carbon and hydrogen were calculated from the increase in weight of two tubes (previously weighed), one containing magnesium perchlorate (for H_2O) and the other soda lime (for CO_2); precision reached 0.1%, provided that the scale could measure in mg.

Later (Simon, 1960), a second generation of analysers was developed (Figure 18.2) by adding a device to reduce nitrogen oxides into molecular nitrogen in the presence of copper powder. In these CHN analysers, to simplify the operation, double weighing is replaced by measurements of thermal conductivity differences in gas mixtures before and after passage through a selective trap for either water or carbon dioxide.

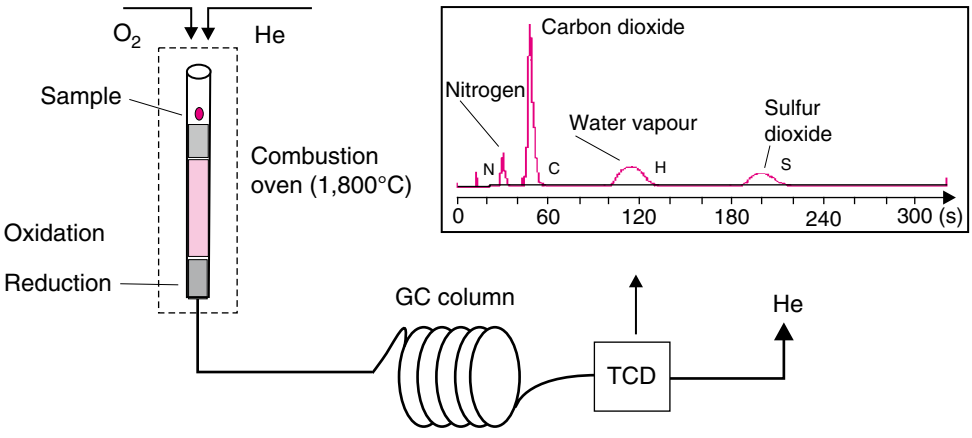


Figure 18.3 Microanalysis apparatus with chromatographic detection. The filled column separates the four gaseous components entrained by the helium carrier gas. By means of a prior calibration, the concentration of each of the four elements in the sample can be deduced from the areas of the corresponding peaks on the chromatogram.

The operation is simpler and more robust than that of chromatographic separation instruments for the gases formed as described above.

18.2.2 Current Elemental Organic CHNS–O Analysers

Other modern analysers retain the sample combustion principle, but the gases formed are separated by a chromatographic method (Figure 18.3).

About 1 mg of substance is pyrolized in an oxygen and helium current. Combustion gases then pass over the copper powder, which retains excess oxygen (formation of CuO) and reduces the nitrogen oxides formed into molecular nitrogen (if this element is present). The elements nitrogen, carbon, hydrogen, and sulfur thus give rise to four compounds in the gas state: N_2, CO_2, H_2O and SO_2, which can be separated in a packed GC column associated with a thermal conductivity detector or with an electron capture detector, which is more sensitive for SO_2.

Some models are also used to quantify oxygen, which is always measured separately, from a second sample, requiring an adaptation of the oven. Pyrolysis must in fact be conducted in the absence of oxygen by using nickel on carbon powder (nickel has a catalytic effect) to obtain carbon monoxide (CO).

18.3 TOTAL NITROGEN ANALYSERS (TN)

Total nitrogen content is used to characterize numerous products and samples in the oil, paper, and coal industries. To conduct these measurements, custom analysers have been developed for this single element and are favoured over the aforementioned

instruments. Therefore in the farming sector, *total Kjeldahl nitrogen* (TKN) has been adopted by a number of countries as the reference method to calculate protein contents, which are assumed to correspond to 6.5 times the nitrogen content. TKN corresponds to the sum of organic nitrogen and ammonia (NH_3).

In the Kjeldahl method, the nitrogen-containing compound is first digested in sulfuric acid in the presence of a catalyst, before being taken up by a mixture of sodium hydroxide, calcium oxide, and calcium hydroxide (soda lime) in order to transform the nitrogen within the substance to a single ammonia-based solution. This sequence ends by steam entrainment to remove the ammonia formed (which is volatile) for measurement by acidimetry. This method can be rendered semi-automatic but there is the drawback of using inherently corrosive reagents and titrated solutions. In addition, it is rather time consuming.

Current instruments are an adaptation of general CHN analysers, based on the pyrolysis of the sample in order to convert the nitrogen element either into molecular nitrogen, as in the old Dumas method (1833), with the difference that the gas volume is no longer measured with a graduated cylinder (Figure 18.5), or into nitrogen monoxide for chemifluorescence or amperometric detection.

The sample (between 100 and 500 mg depending on the nitrogen content), enclosed in a small tin foil container, falls into a vertical oven, where it is mineralized by combustion in oxygen. At this stage, nitrogen oxides form in contact with copper oxide. Then, these oxides are reduced into molecular nitrogen in the presence of tungsten. After passing through a trap to remove water and the combustion gases (Figure 18.4), or after separation on a GC column, the molecular nitrogen mixed with the carrier gas (CO_2 or He) arrives in a detector, where its concentration can be determined.

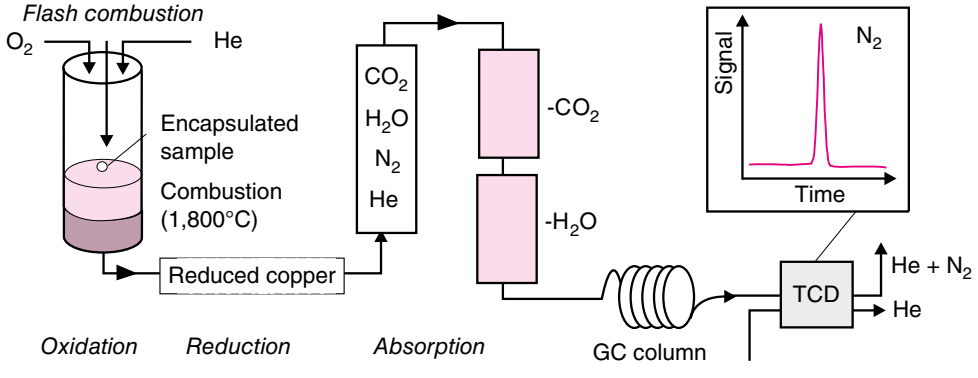


Figure 18.4 Instrument for nitrogen analysis. This instrument is a modern adaptation of the revised Dumas method with a vertical oven and detection by means of a catharometer (He is the carrier gas).



Figure 18.5 A total nitrogen analyser. Model COT/TN (Source: Reproduced with authorization from Skalar). Among others this type of instrument can be used for measuring proteins. Detection by chemifluorescence.

The general detection procedure is based on the thermal conductivity of the gas (catharometer).

In Dumas' original method (1840), the substance was heated to high temperature in the presence of copper oxide in a carbon dioxide atmosphere. Elemental nitrogen was partially oxidized to nitrogen oxides NO_x, which were subsequently reduced on copper. The nitrogen percentage was calculated from the volume of the nitrogen gas released.

Other specialized detectors are based on the chemiluminescence of nitrogen oxides (see Chapter 11, Section 11.7) either with nitrogen monoxide (NO):

$$NO + O_3 \rightarrow O_2 + NO_2^* \rightarrow NO_2 + h\nu$$

or on the oxidation of NO in contact with an electrochemical sensor (see Chapter 20, Section 20.12):

$$NO + 2H_2O \rightarrow HNO_3 + 3H^+ + 3e^-$$

18.4 TOTAL SULFUR ANALYSERS

Sulfur content is often measured for industrial process control by X-ray fluorescence, because the samples usually form part of a series, which may be compared with very similar standards. However, for other types of samples that can be destroyed by combustion in a current of oxygen, the measurement is based upon the volume of sulfur dioxide (SO_2) formed in an instrument similar to those described previously.

There are three main quantification processes where the other combustion gases do not interfere:

- Detection of sulfur dioxide by UV fluorescence, a process that can be summarized as follows, where R–S designates the compound:

$$R-S+O_2 \rightarrow SO_2 + other\ \ combustion\ \ products$$

$$SO_2 + hv \rightarrow SO_2 \rightarrow SO_2^* + hv'$$

- Specific action of iodine on SO_2 in the presence of water, a reaction which is exploited elsewhere for the measurement of water by the Karl Fischer method (Chapter 20). The instrument determines the quantity of current in a coulometric cell needed to generate, from an iodide-containing electrolyte, just the amount of iodine necessary to oxidize SO_2. Since it is known that two electrons are required to oxidize a molecule of SO_2, one sulfur atom will correspond to $2 \times 1.6 \times 10^{-19}$ C.
- Oxidation reaction of sulfur dioxide in sulfuric acid, in contact with the working electrode of an amperometric cell (see Chapter 20, Section 20.8):

$$SO_2 + H_2O \rightarrow H_2SO_4 + 2H^+ + 2e^-$$

18.5 TOTAL CARBON ANALYSERS (TC, TIC, AND TOC)

The domestic or industrial water treatment industry uses a specific type of carbon analyser. For aqueous sediment samples, it is in fact useful to distinguish organic carbon from inorganic carbon. As a result, the three values are respectively expressed as *total organic carbon* (TOC), *total inorganic carbon* (TIC), and finally *total carbon* (TC), which represents the sum of the two previous totals.

For low TOC concentrations, one of the methods consists in exposing a measuring cell containing the water sample to UV radiation (185 nm) in order to form hydroxyl ions (HO^-) from molecular oxygen dissolved in this water. These radicals oxidize organic compounds into carbon dioxide, which changes the conductivity of the water, as measured between two titanium electrodes.

TIC can be measured through the action of an acid such as phosphoric acid (H_3PO_4) (since TIC is essentially present in the form of carbonates), while TC can be determined with carbon analysers by catalytic oxidation of the sample at a high temperature (1,000°C). To assess the quantity of carbon dioxide formed, an infrared absorbance measurement is done using a nondispersive instrument (see Chapter 10, Section 10.7.2), or a coulometric measurement is conducted. In the latter case, the CO_2 entrained by the carrier gas sweeps the coulometric cell containing ethanolamine and a pH-sensitive colorimetric indicator. The variation in absorbance is related to the quantity of CO_2 present.

For gaseous samples, instruments with a flame ionization detector, whose signal is proportional to the TOC concentration, can also be used.

18.6 MERCURY ANALYSERS

Mercury is a dangerous pollutant because of the ease with which it yields highly toxic organic derivatives such as dimethylmercury. This element may be present in air as well as in condensed phases, and thus assays for mercury are performed at a wide variety of sites, including water and wastewater treatment plants as well as industrial incinerators. These methods must be highly sensitive (ppb level) (Figure 18.6).

Depending on the nature of the sample, it is either pyrolysed in the presence of the catalyst in order to make mercury appear in metal form (Hg(0)), or it is first decomposed under heat with an oxidizing mixture made up of acids. Mercury is then transformed into Hg(II) ions that are subsequently reduced to elemental mercury

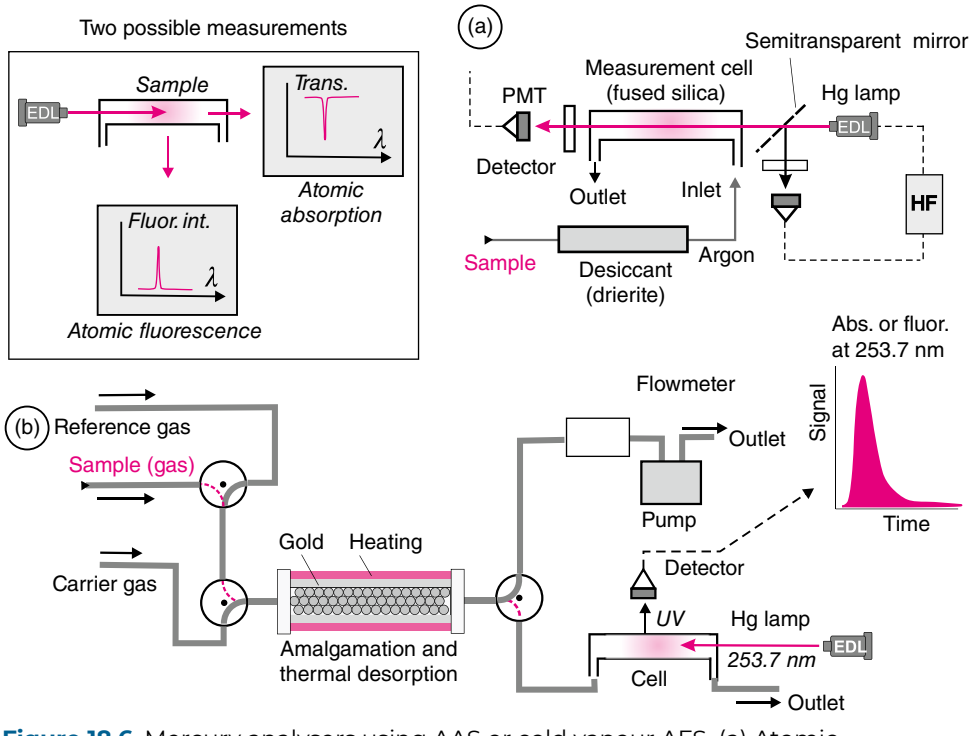


Figure 18.6 Mercury analysers using AAS or cold vapour AFS. (a) Atomic absorption instrument (Source: Reproduced courtesy of Mercury Instr. USA). (b) Atomic fluorescence instrument (Source: Reproduced courtesy of Genesis Laboratory Systems Inc.).

(Hg(0)) with a tin salt (Sn(II)). The vapour pressure of mercury is sufficient for the sweeping of the solution with air to drive it into the measuring cell placed on the optical path of a mercury discharge lamp. The measurement is based either on the atomic absorption line of this element at 253.7 nm, or on the fluorescence of mercury when it is excited by this same type of lamp.

When atmospheric mercury is to be measured, the mercury is concentrated by passing the gaseous mixture through a trap containing gold. This gold amalgamation is then heated to desorb the mercury, which is detected and measured as described previously.

Alongside the method described above, used in many commercially available instruments, there is also a voltammetric method, which is adapted to the assay of mercury (II) salts in aqueous media. It consists in conducting an assay by Anodic Stripping Voltammetry (ASV) (see Chapter 20, Section 20.7). The work electrode has a gold wire on which the mercury is amalgamated.

18.7 ION MOBILITY SPECTROMETRY (IMS)

This method for studying volatile compounds, often mistakenly compared with time-of-flight mass spectrometry (see Chapter 16, Section 16.5.3), should instead be described as 'gas phase electrophoresis'. The principle is based upon the time taken by the ions, formed from compounds ranging from very small volatile molecules to biomolecules such as proteins, to migrate from a source to the detector, in a tubular space maintained at ambient pressure (Figure 18.7). For each ion, this migration depends on many factors, including its shape, its charge, its effective cross-section, and its interaction with the gas. In the environment of the ion source, which consists of a film of ^{63}Ni (β^--emitter) or a UV lamp, the presence of air leads to ionic aggregates (clusters) $[(N_2)_x (H_2O)_y H]^+$ and $[(N_2)_x (H_2O)_y O_2]^-$ that will react with the molecules of the compounds present, depending on their electron affinity. This gentle chemical ionization process (different from *APCI*, see Chapter 16, Section 16.4) forms an aerosol that migrates in a few milliseconds under the effect of an electric field gradient whose value (200–400 V/cm) and polarity can be adjusted.

An ion mobility factor K_{ion} can be attributed to each analyte (equivalent to electrophoretic mobility), which depends not only on the mass, charge, and size of the ion, but also on the temperature and pressure of the air in the instrument. As a result, Eq. (18.1) is established (to be compared with Eq. (8.4), thus providing the electrophoretic mobility of an ion). The threshold of detection can reach a few ppb. By calling the speed of the ion v, the distance L travelled during the time t_M in the field E, $(E = V/L)$, then we obtain:

$$K_{ion} = \frac{v}{E} = \frac{L}{t_M} \cdot \frac{L}{V} \qquad (18.1)$$

Commercial instruments based on this principle may be divided into two categories: small portable field instruments and laboratory instruments often coupled with a time-of-flight mass spectrometer, for example to study biological macromolecules.

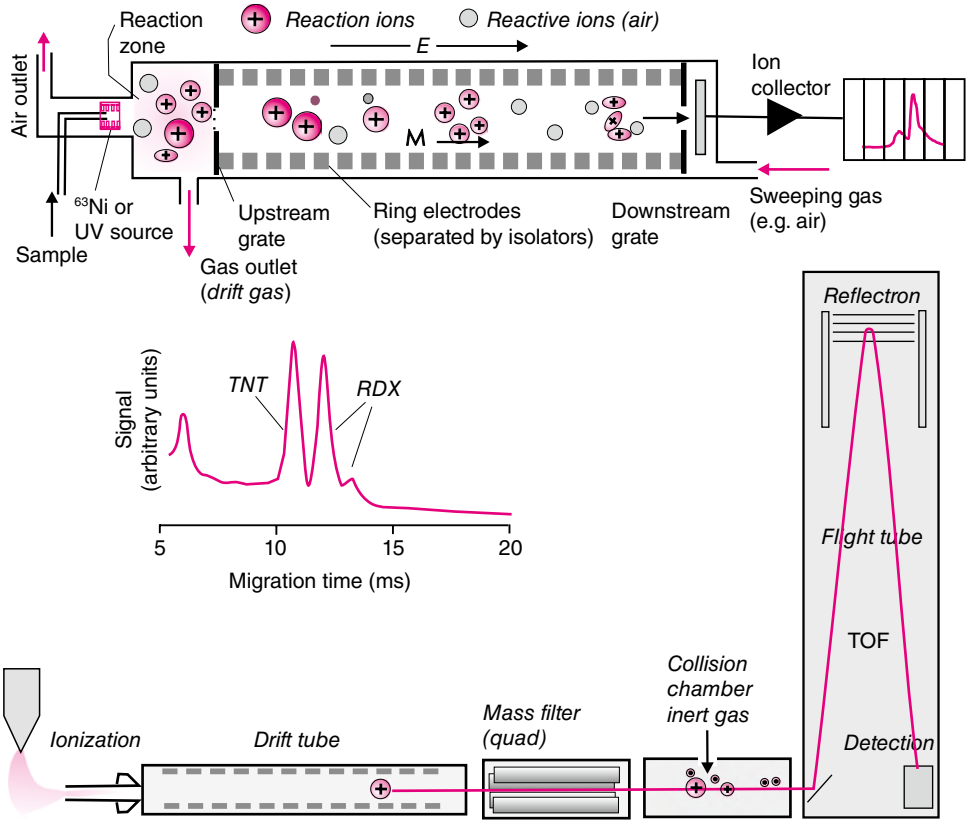


Figure 18.7 Ion mobility spectrometer (IMS). Ions are repetitively admitted into the analyser tube by controlling the polarity of the upstream grid. Example of recording obtained from two explosive devices, thus showing that different profiles can be attributed to each of them. Layout of an installation including an IMS and a TOF.

Field instruments

As the migration of ions is done in ambient air, this technique has received many applications for analyses beyond a laboratory setting thanks to portable instruments. Each ion generates a signal that is specific to it and whose collection constitutes an ion spectrum, a sort of fingerprint of the sample, which may be authenticated using a spectral library integrated into the instrument.

There are many applications: oil products, toxic chemicals and chemical weapons, narcotics, explosives, various pollutants, customs, etc. These 'electronic noses' capable of recognizing several dozen compounds in seconds are now increasingly common.

Laboratory instruments

Laboratory IMS instruments are generally coupled with a TOF-type mass spectrometer, which enables a double separation of ions, first according to their mobilities and then according to their mass/charge ratios.

This nearly instantaneous, two-dimensional method avoids having to use an upstream chromatographic separation. It may also be associated with an additional separation of ions by inserting a quadrupole filter between the two types of spectrometers (Figure 18.7). The performance of these line-ups is measured by the separation factors they are able of reaching. Applications mainly fall in the field of proteomics for protein, peptide or lipid analyses.

18.8 KARL FISCHER VOLUMETRIC METHOD

The water content of a great variety of samples can be determined in one of the most widely used measurements, using a volumetric or coulometric method. The volumetric method is described below (for the coulometric method, see Chapter 20, Section 20.9).

18.8.1 Review of the Reactions Used

In the presence of water, iodine reacts with sulfur dioxide through a redox reaction that is specific to these three compounds:

$$I_2 + SO_2 + 2H_2O \rightarrow H_2SO_4 + 2HI$$

This quantitative reaction can be used to determine water content. This can be achieved if, for example, a base is added to neutralize the acids formed and make the reaction complete. Since iodine is a solid and sulfur dioxide is a gas, an auxiliary polar solvent is used, serving as both a diluent and as a reaction medium. Generally, methanol is used. Under these conditions, sulfur dioxide is not simply dissolved in the solvent but also interacts with it, leading with methanol, to methyl hydrogen sulphite. The latter then reacts with iodine in the presence of water.

$$CH_3OSO_2H + I_2 + H_2O \rightarrow CH_3OSO_3H + 2\,HI$$

In contrast to the first reaction in the presence of methanol, *a single* molecule of water reacts to convert *two* atoms of iodine. Methyl hydrogen sulphite is oxidized to methyl bisulfate, which is converted in the presence of an RN-type base into an ammonium salt. The Karl Fischer reaction can therefore be reformulated as follows:

$$H_2O + I_2 + \left[RNH\right]^+ CH_3OSO_2^- + 2RN \rightarrow 2\left[RNH\right]^+ I^- + \left[RNH\right]^+ CH_3OSO_3^-$$

With pyridine as an RN base, a pyridinium salt $C_5H_5NH^+$ $(CH_3OSO_3)^-$ is formed. This toxic base, with its highly unpleasant odour, is now replaced by other, more stable reagents. Because atmospheric moisture could alter the results, the reaction cell is isolated from the atmosphere by drying tubes. Moreover, since the diluting solvent is rarely anhydrous, owing to its hygroscopic nature, its water content must be determined prior to the measurement. Consequently, the water equivalence of the KF reagent (concentration *T*) is determined. Then, the water contained in the sample is measured. The various operations are then done in a suitable container.

As for many volumetric measurements, it is possible to use either *direct titration* or *back titration*. In direct titration, the end point is reached by gradual addition of the KF (Karl Fischer) reagent. For back titration, an excess of this reagent is introduced, then the excess is titrated using a nonanhydrous solvent whose water equivalence has been determined by pretitration and which has subsequently been placed in the burette.

18.8.2 Preliminary Assay of the Karl Fischer Reagent

By direct titration

In the assay container, volume V_1 of the solvent (e.g. methanol) is introduced. V_2 is the volume of reagent necessary to reach the equivalence point (this is called the 'solvent blank').

Then, in the assay cell, a mass *m* (in mg) of water is added, either with a microsyringe of the type used in chromatography (to the nearest tenth of a microlitre), or as a precise mass of a solid hydrate of known composition: oxalic acid with $2H_2O$ (28.57% water), or sodium tartrate with $2H_2O$ (15.66% water). V_3 is the new volume (in ml) of reagent necessary to reach the equivalence point. The concentration *T* of the reagent (its equivalence in water, normally around 5 mg/ml) is given by:

$$T = \frac{m}{V_3} \tag{18.2}$$

By back titration

Firstly, a sufficient quantity of reagent is introduced to cover the detection electrodes of the cell (this quantity of about 20 ml depends on the cell). Then, solvent-grade methanol is added to reach the equivalence point. After this first pretitration dehydration phase, we obtain a perfectly anhydrous environment in the cell, for use in the other stages:

- Add volume V_2 of KF reagent to the cell, which is then neutralized with volume V_1 of solvent-grade methanol.
- A specific mass *m* of water (in mg), just like for direct titration, is introduced and volume V_3 of the KF reagent is added in excess of the necessary quantity by 2 or 3 ml;
- The excess reagent is neutralized by volume V_4 of solvent-grade methanol.

The concentration T (expressed in mg/ml) is:

$$T = \frac{m}{V_3 - \frac{V_2}{V_1} V_4} \tag{18.3}$$

The concentration of the KF reagent changes over time due to its lack of stability in solution. This is why manufacturers generally sell this reagent in two distinct bottles: one contains a mixture of sulfur dioxide, methanol, and a base; the other contains iodine or an iodine solution. The user mixes the contents of the two bottles before use.

18.8.3 Determination of the Water Content of a Sample

By direct titration

P mg of the sample, either as is or in solution in volume V_5 of the same solvent, is added to the anhydrous solution found in the cell. If volume V_6 of the KF reagent is needed to reach the equivalence point, the mass of water p contained in the test sample is equal to:

$$p = T \cdot \left(V_6 - \frac{V_2}{V_1} V_5 \right) \tag{18.4}$$

However, if the sample is added without a solvent ($V_5 = 0$), the quantity of water in the sample will be given by the simplified formula:

$$p = T \cdot V_6 \tag{18.5}$$

By back titration

P mg of the sample is added to the titration vessel and volume V_7 (*measured out*) of KF reagent corresponding to an excess is added. Finally, this excess is measured as shown above with volume V_8 of methanol.

From these values, the mass of water p (in mg) in the P mg of test sample is:

$$p = T \cdot \left(V_7 - \frac{V_2}{V_1} V_8 \right) \tag{18.6}$$

Finally, the water content (expressed in percentage) is given by the equation:

$$H_2O\% = 100 \times \frac{p}{P} \tag{18.7}$$

KEY POINTS OF THE CHAPTER

1. The quantification of light elements is difficult using spectroscopic methods. That is why old methods using sample combustion have never been fully abandoned but rather have been improved thanks to technological advances.
2. Whatever the adaptations or variant offered by manufacturers, the shared principles of these methods consist of quantifying the gases and vapours (CO_2, N_2, H_2O, SO_2, etc.) resulting from the combustion of the sample in the presence of molecular oxygen, in order to calculate the concentrations of these elements.
3. The presence of carbon in water is an indicator of its contamination by pollutants. According to the equation TIC + TOC = TC, the total carbon (TC) content represents a composite parameter including all of the carbonated structures into a single mass.
4. Several methods are used: TOC (total organic carbon) is generally obtained by chemical or photochemical oxidation of the water sample while TIC (total inorganic carbon) is obtained by thermal oxidation (the CO_2 formed is detected by infrared measurement).
5. The assay for mercury in the vapour state (Hg(0)) results from a fluorescence or absorbance measurement of the emission line at 257.3 nm. In the latter case, it is an application of cold vapour atomic absorption spectrometry (CVAAS). The principle of the method consists of subjecting the sample to a treatment, which depends on the type of sample (chemical reduction or mineralization by hot dry path) to transform organic mercury compounds or mercury salts into metal mercury in the vapour state (Hg(0)).
6. Ion mobility spectrometry (IMS) is a hybrid method between electrophoresis and mass spectrometry. Volatile or semivolatile compounds, entrained by a gas, are first ionized (radioactive source) then subjected to an electrical field in a gaseous atmosphere. Each compound will reach a certain speed depending on its mass, its charge, and its volume. In other words, if there is a prior calibration, its mobility will help identify it, in the best case scenario.
7. IMS may be coupled with mass spectrometry, in which case the m/q ratio is obtained as a second identification method for studied compounds. In an even more sophisticated manner, atmospheric pressure ionization (API) can be set up upstream from the IMS, followed by MS. This leads to an API/IMS/TOF/MS line-up!
8. The water content calculation, according to the Karl Fischer chemical method, is one of the most common assays in the world, conducted with an automated titrator, using either a volumetric method (if water > 100 ppm) or a coulometric method (if water < 100 ppm). It relies on the oxidation of SO_2 by iodine, which is only triggered if water is present:

$$I_2 + SO_2 + 2H_2O \rightarrow H_2SO_4 + 2HI$$

9. This reaction is accelerated by using a base to eliminate hydrogen iodide, a potential source of hydroiodic acid. In the presence of a solvent such as methanol, one mole of molecular iodine reacts with a single mole of water (instead of two as in the equation above). It is sulfite, not SO_2, which is oxidized in this case. The final point corresponds to a slight excess of iodine, detected by a potentiometric method (monitoring of DC voltage).

$$CH_3OSO_2H + I_2 + H_2O \rightarrow CH_3OSO_3H + 2HI$$

PROBLEMS

1. Analysis of an organic compound has revealed that it contains only carbon, hydrogen, and oxygen. From the combustion, in a Pregl apparatus, of 5.28 mg of this compound, 10.56 mg of carbon dioxide and 4.32 mg of water are obtained.

   What is the molecular formula of the compound if mass spectrometry reveals no significant peaks above $m/z = 89$?

   Given: $M(C) = 12$ g/mol; $M(O) = 16$ g/mol; $M(H) = 1$ g/mol

2. An analyser based upon AAS and containing an elemental reduction device using tin (II) chloride is employed to measure mercury in an industrial wastewater sample. The *cold vapour* line-up is used. Three standard solutions and the solution of the test sample give the following results:

Solutions	Conc. (ppb)	Signal (muA.s)*
Standard 1	0.02	16.66
Standard 2	0.05	42.52
Standard 3	0.2	171.2
Test sample	?	48.25

*Integration of the absorbance signal (in milliunits) as a function of time.

   a. Indicate the main advantage and the main drawback of this method.
   b. Calculate the concentration (ppb) of mercury in the sample solution.

3. Many countries have adopted the multiplier factor of 6.38 of the total nitrogen concentration to calculate the protein concentration in common food products. This is an average value. How can this value be justified?
4. The Kjeldahl method is used to calculate the amount of protein in a batch of rillettes from the measurement of total nitrogen.
   1. To check the proper operation of the measurement line-up, a sample formed by mixing 1.0 g of leucine ($C_6H_{13}O_2N$) and 1.0 g of asparagine ($C_4H_8O_3N_2$) is subjected to the protocol for this assay. The final neutralization

requires 21.4 ml of 1.03N HCl solution (1.03 mol/l). Verify by calculating the theoretical value.

2. 20.0 g of rillettes are sampled after the homogenization of a larger batch and subjected to this same protocol. The neutralization requires 31.8 ml of HCL solution. Calculate the official concentration of protein in this sample.

Reminder: C = 12.01, H = 1.0, O = 16.0, and N = 14.01 g/mol.

5. The quantity of water found in powdered milk can be determined using the Karl Fischer method. The reagent is calibrated beforehand as follows:

   In the measurement cell, 15 ml of a solvent composed of methanol and formamide is introduced. 3 ml of KF reagent is added to reach the point of equivalence. Then, 205 mg of oxalic acid dihydrate is introduced. 13 ml of KF reagent must be added to reach the point of equivalence once again. Finally, 1.05 g of powdered milk dissolved into 10 ml of the same solvent is introduced. 12 ml of KF reagent must be added to reach the point of equivalence.

   Calculate the % mass of water in the powdered milk.

6. Calculate the mass concentration of water in a batch of 'anhydrous' ether, knowing that 1 coulomb is needed to reach the point of equivalence in a Karl Fischer assay (coulometric version) conducted with 1 ml of ether. Remember that two iodine atoms are required per water molecule and that the density of ether is 0.78 g/ml.

SOLUTIONS

1. %C = [10.56 × (12/44)/5.28] × 100 = 54.55;

   %H = [4.32 × (2/18)/5.28] × 100 = 9.09;

   %O = 36.36%

   If the compound has the formula $C_xH_yO_z$ and if M = 88 since there is no significant peak in MS over 89, then:

$$12x / \%C = y / \%H = 16z / \%O = M/100; x = 4; y = 8; z = 2.$$

2. **a.** Advantage: the cold vapour instrument is specific to mercury; drawback: only inorganic mercury is measured.
   b. Equation of the least-squares line: [signal] = 858.33[conc.] – 0.46; *C*(ppb) = 0.06.

3. For example, take a protein, which by nature corresponds to the condensation of a large number of varied amino acids, whose formula is known: insulin $C_{257}H_{383}N_{65}O_{77}S_6$ (M = 5,807.6 g).

   The mass of nitrogen is 65 × 14 = 910 g, which corresponds to 910/5,807 = 0.157% of the total mass. Inversely, 1 g of measured nitrogen will come from 1/0.157 = 6.4 g

of protein. An identical calculation using bovine serum albumin (BSA) $C_{3072}H_{4828}N_{816}O_{928}S_{40}$ (M = 69,296 Da) leads to 6.1. Actually, this factor was obtained from a precise analysis of milk. This value of 6.38 is very approximate, as nitrogen does not come only from proteins (but also from urea, amines, phospholipids, etc.).

4. 1. Calculation of the quantity of theoretical nitrogen in the sample:

   Quantity of nitrogen in 1 g of leucine (M = 131.17 g/mol): 14.01/131.17 = 0.11 g. Quantity of nitrogen in 1 g of asparagine (M = 132.12 g/mol): 28.02/132.12 = 0.21 g.

   One mole of HCl is needed to neutralize one mole of ammonia (NH_3), which corresponds to 14.01 g of nitrogen. For a total of 0.32 g of nitrogen, theoretically 0.32/14.01 = 0.023 mole of HCl is needed, i.e. for a 1.03 N solution (0.023/1.03) x 1,000 ≈ 22.2 ml. Therefore, about 97% of the sample's nitrogen is found (21.4/22.2), which corresponds to the expected precision for this assay.

   2. Calculation of the quantity of nitrogen corresponding to 31.8 ml of 1.03 N HCl: 14.01 × (31.8 × 1.03/1,000) = 0.46 g. 0.46 × 6.5 = 2.98 g of protein equivalent, or (2.98/20) × 100 = 14.9 ≈ 15% in the analysed rillettes.

5. *Reagent calibration:* 1 ml of this solvent has a water content of 3/15 ml of KF reagent. In oxalic acid dihydrate (M = 126 g/mol), the water concentration by mass is 28.57%. This data helps us to find the reagent concentration: T = 28.57 × 205 × 1/13 = 4.51 mg/ml.

   Assay: 10 ml of the solvent 'neutralizes' 10 x 3/15 = 2 ml of reagent, therefore 1.05 g of powdered milk reacts with 12 – 2 = 10 ml of this reagent. Therefore, 4.51 × 10 = 45.1 mg of water in the sample, i.e. a concentration of (45.1/1,050) × 100 = 4.3%.

6. In the coulometric version of the KF assay, one water molecule requires two iodine atoms and two electrons. Therefore, 2 × 96,500 coulombs (C) are needed for one mole, i.e. 1.8×10^4 mg of water; 1 C thus corresponds to $1.8 \times 10^4/(2 \times 96{,}500)$ = 0.0933 mg of water. This corresponds to the quantity of water in 1 ml of ether. In each litre, therefore there is 93 mg of water. The concentration is 93/0.78 = 120 mg/kg (120 ppm).

Chapter 19

Potentiometric and Ionometric Methods

A great number of quantitative measurements in chemical analysis are based on electrochemistry. Called electrochemical methods, they can be separated into two categories: those based on measurements of potential differences between two half-cells (potentiometry) and those which exploit measurements of current (voltammetry). This chapter is limited to assays by potentiometry (not including potentiometric titrations), with a review of the general principles based on the passive evaluation of the potential existing between two electrodes, at zero current in the measurement circuit,

and thus without electrolysis. Ionometry helps quantify chemical species (ionic or molecular) in aqueous solution with specific electrodes containing a selective membrane. Two quantification methods are described and several applications are presented.

Objectives

Describe a measurement cell
Distinguish between the activity and the concentration of an ion
Describe the glass electrode (pH)
Classify ion-selective electrodes (ISE)
Explain the role of selective membranes
Choose a quantification method
Compare direct ionometry with an addition method

19.1 MEASUREMENT CELLS

A potentiometric assay involves a measurement cell composed of two electrodes immersed in an aqueous solution containing the analyte in ion form (Figure 19.1). Each of these electrodes constitutes a half-cell:

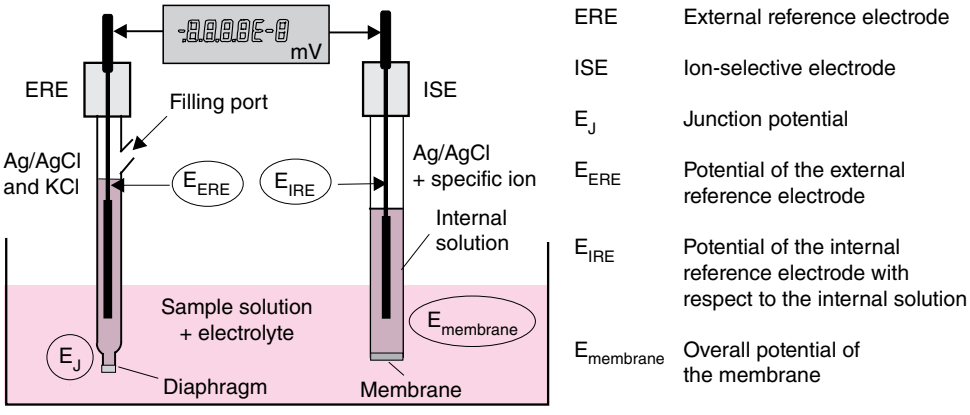


Figure 19.1 Measurement set-up of an ion-selective electrode (ISE). The selected ion-permeable membrane creates a variable potential depending on the concentration of the analyte (ionic) in solution. The other potentials are imposed by construction. The ionic strength of the supporting electrolyte (which must not react with the analyte!) must be even greater than that of the sample containing the analyte. Measurements are made using ionometers. Section 19.4 provides information about the practical aspects of this type of assay.

- The *external reference electrode (ERE)*, whose potential E_{ERE} is practically constant with respect to that of the solution containing the analyte. In general, this is an electrode, either calomel or Ag/AgCl, immersed in a solution of potassium chloride (KCl). Electrical contact with the sample solution is achieved through finely porous fritted glass. This porous wall is at the origin of a low-potential junction E_J that can be minimized by using a saturated solution of potassium chloride.
- The *measurement electrode* (or *indicator electrode*) designed as an internal reference electrode (IRE) containing the solution of a specific ion, is separated from the sample solution by a **selective membrane** that is permeable to the analyte. This constitutes an *ion-selective electrode (ISE)*. The potential difference (PD) between the IRE and the internal membrane surface is constant, as fixed by its design (nature of the reference electrode and the activity $a_{i.reference}$ of the reference solution). In contrast, the PD between the external surface of the membrane and the sample solution (E_{memb}) depends upon the activity of the target ion ($a_{i.solution}$). The potential difference across the membrane is described by the Nernst equation:

$$E_{memb.} = 2.303\frac{RT}{zF}\log\frac{a_{i.solution}}{a_{i.reference}} \tag{19.1}$$

where R is the perfect gas constant, T is the temperature in Kelvin, z the charge of the studied ion with its sign, and F is the Faraday constant (charge carried by a mole of elementary charges = 96,485 C).

The measured PD E_{cell} between the two terminals of the measurement cell (battery) may thus be:

$$E_{cell} = E_{IRE} + E_{memb} - E_{ERE} + E_j$$

As the terms E_{IRE} and E_{ERE} are independent of the activity of the ion to quantify and the junction potential is low and nearly constant, the PD at the cell terminal E_{cell} may be expressed in a simplified manner by Eq. (19.2) where E′ is a constant only related to the measurement unit, without any relation to a standard oxidation-reduction potential.

$$E_{cell} = E' + 2.303\frac{RT}{zF}\log a_{i.solution} \tag{19.2}$$

The ISE potential thus responds to the activity of the analyte ions and not to their concentration. The activity a_i of an ion *i* is linked to its concentration C_i by the relationship $a_i = \gamma_i C_i$, where γ_i is the activity coefficient between 0 and 1, which depends on the total ionic strength *I*, which is a measure of the quantity and charge of all ions present in the medium.

$$I = 0.5\Sigma C_i z^2_i \tag{19.3}$$

For example, for an ion *i*, of charge *z*, Eq. (19.2) becomes:

$$E_{Cell} = E' + 2.303\frac{RT}{zF}\log\gamma_i C_i \tag{19.4}$$

Since the activity coefficient depends on the total ionic strength of the solution, the latter is fixed by adding a supporting electrolyte whose ionic strength is much greater than that of the sample solution. With this buffer, known as the ISAB (Ionic Strength Adjustment Buffer), the activity coefficient does not vary much depending on the quantity of analyte, and therefore a response directly related to its concentration is obtained:

$$E_{cell} = E'' + 2.303\frac{RT}{zF}\log C_i \tag{19.5}$$

Assay or **titration?** A potentiometric assay is based on a potential measurement made on the sample, as well as on one or more calibration solutions, as described later in this chapter. A potentiometric titration, however, consists of continuously measuring the potential difference between two electrodes (reference and indicator) immersed in a solution, as a function of the titrant volume added to detect the point of equivalence and by deducting the concentration of the studied analyte.

19.2 THE PH ELECTRODE

This electrode, also called the *glass electrode*, is specific to H^+ ions and highly selective. Glass, in this case, does not refer to the material of the electrode (which could equally be made of plastic) but rather to the membrane that ensures contact with the solution. This membrane is composed of a thin wall of special glass that has a high sodium concentration (25%). Its surface becomes hydrated in the presence of water and is comparable to a gel, while its inside corresponds to a solid electrolyte.

On a microscopic scale, this glass has an orthosilicate network. This salt of orthosilicic acid $Si(OH)_4$ has an open structure with sodium cations that allow the transfer of charges from one side of the glass membrane to the other (Figure 19.2). The outside of the membrane is in contact with the sample solution while the inside is in contact with the internal electrolyte, which has a constant acidity (pH ≅ 7). The two walls are polarized and are the seat of exchanges between Na^+ and H^+ cations:

$$H^+(\text{solution}) + Na^+(\text{glass}) \rightleftarrows Na^+(\text{solution}) + H^+(\text{glass})$$

If the activity a_{H+} of H^+ ions (or H_3O^+ in aqueous solution) is different on either side of the glass membrane (which is always the case), a *binding PD* will appear between them, whose value is characteristic of the a_{H+} of the solution and thus of its pH (pH = $-\log a_{H+}$). The latter is determined with an electronic millivoltmeter, i.e. the pH meter, which monitors the PD between the glass electrode and an external reference electrode (ERE) made of Ag/AgCl (now favoured, for the environment, over the calomel electrode

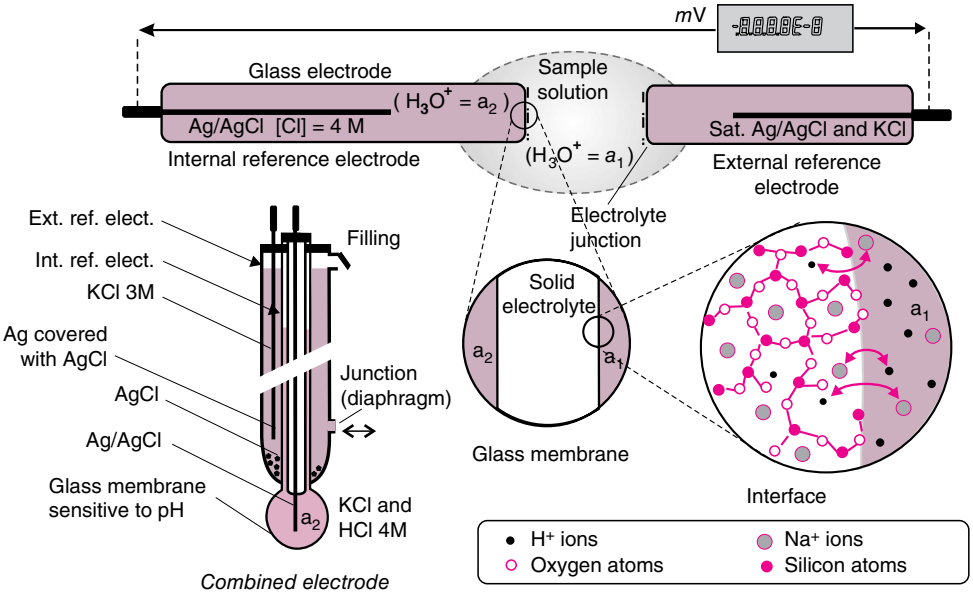


Figure 19.2 Glass electrode for measuring pH. The concentration of H^+ ions is accessible through the potential difference that appears between the glass electrode and a reference electrode (here an Ag/AgCl electrode). Above, a cross-section of the membrane permeable to H^+ ions. When an H^+ ion forms a silanol bond, a sodium ion moves into solution to preserve the electroneutrality of the membrane. A cross-section of a combined electrode in a classical arrangement, where the reference electrode surrounds the glass electrode, except at its end. The junction permits the migration of ions without any mixing of the liquids on each side.

(Hg_2Cl_2)). The instrument, after calibration, directly calculates the pH of the solution. These electrodes are often combined in concentric fashion.

19.3 ION-SELECTIVE ELECTRODES (ISE)

During the early days of potentiometry, it was noted that interferences could occur and affect pH value, depending upon the composition of the glass used for the pH electrode membrane, principally if other basic ions are present besides the sodium ion. This is known as the *sodium, or alkaline, error*, which occurs for pH values above 13. This led to the development of glasses that could respond selectively to monovalent ions such as Na^+, K^+, Li^+, and Ag^+ but, in practice, only the glass electrode for Na^+ is still being used. In all cases, an ionic material is required that will permit the establishment of concentration equilibria with a specific ion. Selectivity depends on the material that serves as the membrane and which separates the solution to be analysed from the inside of the electrode. A PD will appear if the activity (or the concentration) is

different between the two faces of the membrane. Measurement involves only the free ions in the solution. These electrodes must be coupled with a reference electrode, whose electrolyte, generally KCl or KNO_3, depending on the case, slowly migrates through a diaphragm to ensure the junction with the analyte solution (Figure 19.3).

About 20 ion-specific electrodes (ISE) are in routine use, and their replaceable membrane changes in nature with the ion species to be assayed. They serve either for direct ionometry, or as indicator electrodes for measurements involving both titrimetric and complexometric assays performed with the aid of automatic titrators. Three types of membrane coexist.

19.3.1 Crystalline Membrane

The electrode potential is measured through a crystalline material that is used as a membrane. Thus, the specific electrode for the fluoride ion is made up of a disc of lanthanum trifluoride doped with a europium salt to improve its conductivity. Only fluorine ions (F^-) can migrate through the crystalline network (Figure 19.3). The reference electrode is often at a double junction, only one of which directly enters into contact with the solution being analysed.

Other solid membranes, made up of agglomerated crystalline powders, can measure Cl^-, Br^-, I^-, Pb^{2+}, Ag^+, and CN^- ions. The internal electrolyte can be eliminated (dry contact without an electrolyte). However, it is preferable to insert a polymer layer with mixed-type conductivity to ensure the passage of electrons from the membrane towards the rest of the circuit (Figure 19.3).

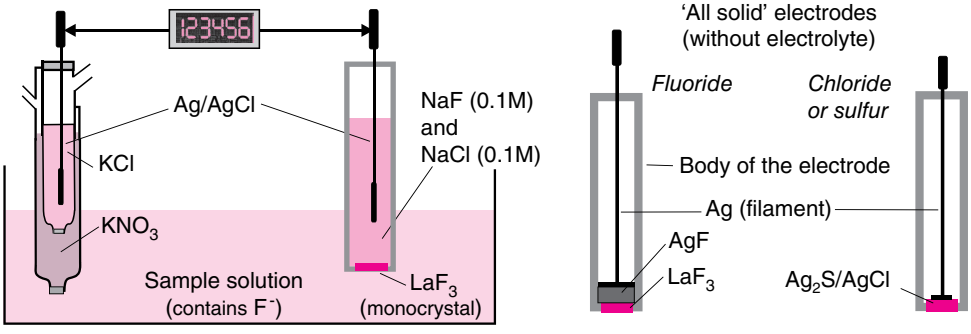


Figure 19.3 Measurement set-up of ion-specific electrodes (ISE). Diagram of a measuring cell with an ISE for the fluoride ion using a double-junction reference electrode. From the two sides of the LaF_3 monocrystal, which is the sensitive component of the electrode, an equilibrium is produced, $LaF_3 \leftrightarrow LaF^+_2 + F^-$. One of the faces becomes positive with respect to the other. To avoid osmosis of KCl into the analyte solution, the reference electrode is surrounded by a second chamber that contains an auxiliary, noninterfering electrolyte. 1 M KNO_3 is often used for F^-, Cl^-, I^-, CN^-, S^- or Ag^+. Actual units, sometimes very compact, are not always easy to visualize. Right, 'all solid' ISE: for chloride ion.

19.3.2 Polymer Membrane

A second category of membrane, considered to be liquid, is made up of a polyvinyl chloride disc made supple by a plasticizer in which about 1% of an *ionophore* is incorporated, i.e. an ion-specific transporter. For example, to measure calcium ions, an aliphatic diester of phosphonic acid $(RO)_2P(O)O^-$ (Figure 19.4) can be used.

> Formerly, to separate the internal and external solutions of the specific electrode, a porous, hydrophobic disc was used. It was saturated with an organic solvent that is immiscible with water in the solutions on either side and contained an *ionophore*.

The membrane, obtained by either one of these processes, behaves as an *immobilized liquid*. The counterion is a soluble molecule in the organic phase.

On each side of the wall, a concentration equilibrium is established. Therefore, for the calcium ion:

$$\left[(RO)_2P(O)O\right]_2 Ca \rightleftarrows 2(RO)_2P(O)O^- + Ca^{2+}$$

A measurable PD will appear if the concentrations of the calcium ions are different on the two sides of the membrane. Electrodes for ClO^-_4, BF^-_4, NH_4^+, and Cu^{2+} are of this type. Several dozen chelating molecules are used for these assays. However, these electrodes are more subject to interferences with other ions.

19.3.3 Hydrophobic Membrane for Gases

The membrane separating the probe from the sample solution is made up of a vinyl polychloride- or polyethylene-based microporous polymer (1 μm pores), impermeable

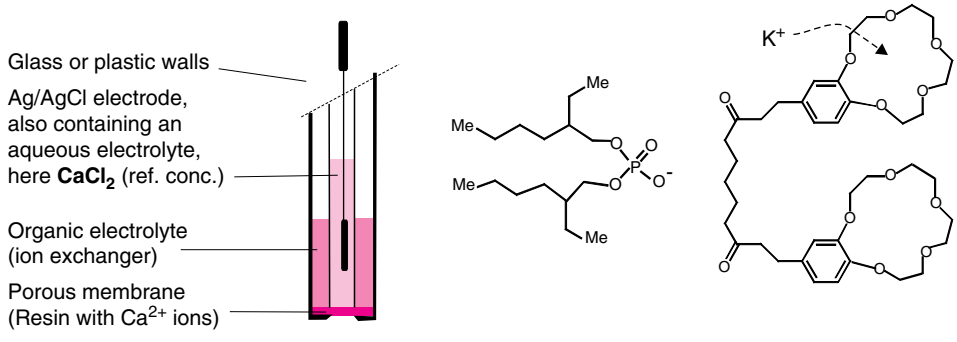


Figure 19.4 Liquid membrane electrode for the Ca^{2+} ion and organic compounds used as ionophores. Two commercial chelating ionophores: dioctyl phosphate, selective for Ca^{2+} ions, and the K2 ionophore, which is selective for the K^+ ion. With its ether-crown substructure, the latter molecule is representative of other macrocyclic compounds used for this purpose.

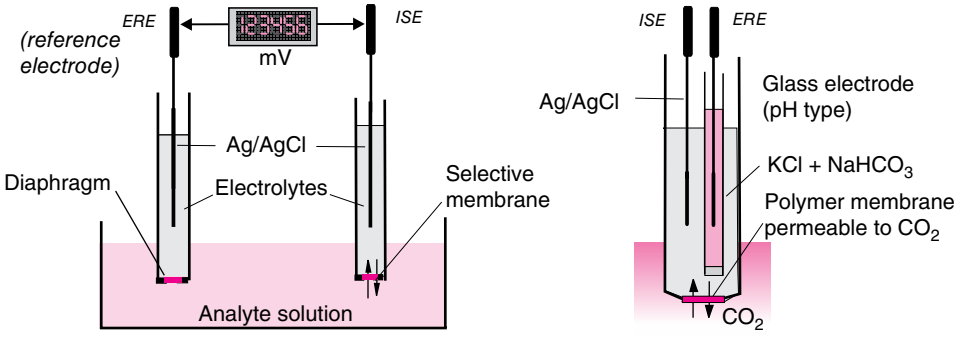


Figure 19.5 Dissolved gas potentiometric sensor. Diagrams with two separate electrodes or a single combined electrode, combining an ISE and an ERE (here, a pH electrode). The unit constitutes an electrochemical cell to measure dissolved CO_2. The measured species is H^+, and thus all acids or bases are potential interferences. The selectivity depends on the choice of membrane.

to water and to ions, but not to gas molecules, which may cross it freely (gas effusion). These probes contain an interior solution able to react with the analyte gas, as well as a pH electrode. These are combined electrodes (Figure 19.5).

For carbon dioxide (CO_2), for example, the internal pH electrode is in contact with a low-concentration bicarbonate solution (0.1 M), itself separated from the sample solution by the membrane. The CO_2 dissolved in the sample crosses the membrane and reacts with the bicarbonate solution.

The pH is therefore modified, which the internal electrode then detects. In addition to carbon dioxide (CO_2), several gases such as ammonia (NH_3), sulfur dioxide (SO_2), and nitrogen oxides (NO_x) may be assayed using this principle. For other gases (HCN, HF, and H_2S), the signal is generated by the modification of the anion concentration around the inner wall of the membrane.

19.4 QUANTIFICATIONS METHODS

As previously shown, the response from the measurement cell is directly related to the concentration of the species to quantify. Traditional quantification methods are thus applicable. However, these require adding an ionic strength adjustment buffer (ISAB) to the sample or even a pH buffer if the anions to be quantified are conjugated bases of a weak acid (therefore, *Total Ionic Strength Adjustment Buffer* or TISAB). The potential presence of interfering ions in the sample must also be taken into account. In fact, as specific as the electrode used can be, it may show a response to certain ions whose charge and size are close to those of the analyte.

As for any quantitative analyses, the concentration must be in the linear response range in log C_i (this range may vary by several decades depending on the studied ion). Once all of these precautions have been taken, traditional quantification methods such external calibration and the multiple addition method may be used.

For a monovalent ion at ambient temperature (298 K), the *slope factor S*, represented by the term 2.303 *RT/F*, is equal to 0.0591 V (Table 19.1). This is also the value of the slope factor per pH unit for a glass electrode (0.059 V per pH unit).

In practice, the calibration curve $E_{cell} = f(C_i)$, using a set of standard solutions, leads to the real slope (i.e. the experimental value of the slope factor), which is an indicator of the performance of the electrode system used.

19.4.1 Direct Ionometry – External Calibration

The protocol of this type of quantification is identical to the one used in all analytical techniques.

From a parent solution of the species to be quantified, a series of standard solutions is prepared, and an identical quantity of the recommended ISAB is added to each. For each solution, the response of the measurement cell is measured in order to establish the equation of the calibration curve (by linear regression). The base-ten logarithm of the concentration of each reference solution is plotted on the *x*-axis and the response of the measurement cell on the *y*-axis (Figure 19.6).

The sample to be quantified is treated in the same way as the reference solutions. The cell's response is measured and the concentration *C* is determined by using the calibration curve equation.

19.4.2 Multiple Addition Method

Though longer than the single addition method, this method leads to more reliable results, as computing helps to easily determine a straight-line equation from several measurements. The protocol is as follows:

- A solution containing a volume V_i of the sample solution of unknown concentration C_i is prepared. Volume *V* of the ionic strength buffer is added to it

Table 19.1 Values of the ideal slopes at 298 K for a few ions.

Valence	Ion	Slope *S* (mV/decade)
–2	S^{-}	–29.58
–1	Cl^{-}, F^{-}	–59.16
+1	Na^{+}, K^{+}	+59.16
+2	Mg^{2+}, Ca^{2+}	+29.58

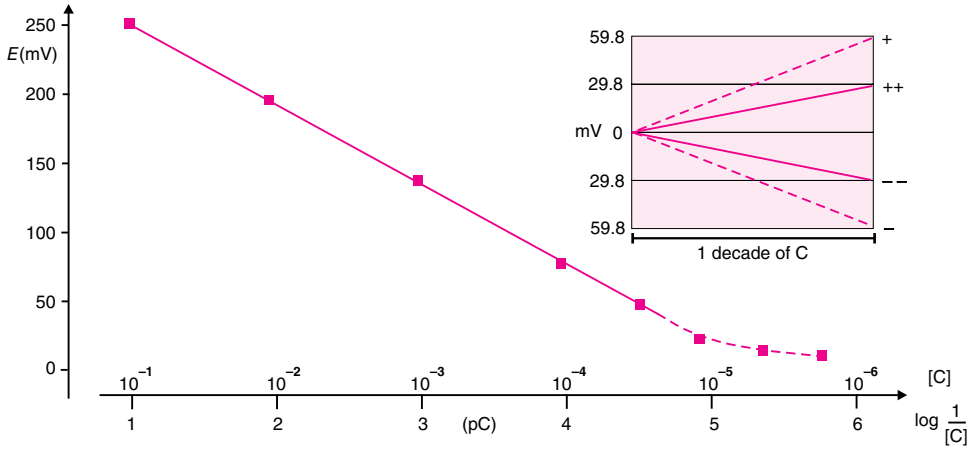


Figure 19.6 Example of an assay by direct potentiometry. The calibration curve of the specific electrode for the chloride ion has almost an ideal slope value. The range of linear response for the different ISEs extends over 4–6 orders of magnitude depending on the ion. The insert shows the theoretical slope factors as a function of ion valence.

and it is filled with the solvent up to volume V_{tot} in a volumetric flask. The value E_0 of the measurement cell response is measured:

$$E_0 = E'' + 2.303\frac{RT}{zF}\log\frac{C_i V_i}{V_{tot}} = E'' + S\log\frac{C_i V_i}{V_{tot}} \tag{19.6}$$

S is the slope factor. E'' gathers several factors. It may be assessed experimentally and it remains constant during the calibration.

- Then, different solutions containing the same volume V_i of the sample solution, the same volume V of the buffer, and a volume V_a, which differs from one solution to the next, of an addition solution of known concentration C_a, are prepared. These are then filled to the same volume V_{tot} with solvent.
- The measurement cell response E is measured for each solution, which is a function of variable V_a. Therefore:

$$E = E'' + S\log\frac{C_i V_i + C_a V_a}{V_{tot}} \tag{19.7}$$

By comparing each component between Eqs. (19.7) and (19.6):

$$\frac{E - E_0}{S} = \frac{\Delta E}{S} = \log\frac{C_i V_i + C_a V_a}{C_i V_i} \tag{19.8}$$

or

$$10^{\frac{\Delta E}{S}} = 1 + \frac{C_a}{C_i V_i} V_a = Y \tag{19.9}$$

Therefore, all that is left to do is to determine the equation of the straight line $Y = f(V_a)$, whose slope is used to obtain C_i.

A variant of this method consists of using volumes V_a that are negligible when compared to $(V_i + V)$. It is thus no longer necessary to complete to volume V_{tot} in a volumetric flask.

19.4.3 Potentiometric Titration

Selective electrodes have a highly variable specificity. However, precision can be increased when they are used as indicator electrodes in potentiometric measurements. The concentration of ions present in solution and the ionic strength vary only slightly during the measurement, relative to the concentration of the ion to be measured. This property can be used for quantification when two different ionic species undergo a stoichiometric reaction. In such cases, the end point in the measurement is characterized by the complete disappearance of one of the species, by the appearance of an excess of one of the species or by the appearance or the disappearance of a secondary species.

To calculate the concentration of aluminium ion, for which no selective electrode exists, the analyte is transformed into insoluble aluminium fluoride AlF_3 by adding sodium fluoride. The equivalence point of this reaction is determined by the presence of an excess of free sodium fluoride, which is detected by a sudden variation in the potential of a fluoride ion selective electrode. In general, a great number of automatic titrators, from competing manufacturers, can be used to conduct assays of this type.

19.5 APPLICATIONS

Direct potentiometric measurements are often used in routine analyses because they are easy to implement. They measure PD values at the terminals of the cell or specific combined electrode and can be used to conduct continuous measurements in real time. Their sensitivity reaches concentrations of the order of a ppm. However, possible interfering elements must be considered because not all ISEs are highly specific.

The selectivity of an electrode with regards to other ions is quite variable. It is defined for each interfering ion by a coefficient $K_{\text{specific ion/interfering ion}}$. For example, $K_{Br/Cl} = 2.5 \times 10^{-3}$ indicates that the selectivity of the electrode for the bromide ion will be $\frac{1}{2.5 \times 10^{-3}}$, i.e. 400 times greater than for the chloride ion.

Some routine applications of this method are:

- in agriculture, the analysis of nitrates in soil samples;
- in foodstuffs, the analysis of ions (such as NO_3^-, F^-, Br^-, Ca^{2+}, etc.) in milk or meat, and carbonated gas in various beverages (wine, beer, fruit juices, etc.);
- in clinical medicine, the measurements of CO_2 in blood;
- in industry, the analysis of chlorides in paper pulp, cyanides in electrolysis baths, chlorides and fluorides in galvanic processes, gases such as CO_2 and NO_x, etc.
- in the biomedical sector, the analysis of sodium and potassium, as an alternative to flame photometry, as well as certain ions in serum, saliva, and gastric fluids.

The lambda sensor, which is found in all motor cars with catalytic converters, is an oxygen probe based on the principle of selective electrodes. This sensor, which looks like a spark plug, has a zirconium oxide (ZrO_2) membrane that behaves as a solid electrolyte. The outer wall is in contact with the exhaust fumes while the inner wall (the reference) is in contact with air. Between these two walls, a PD appears, collected by two electrodes, which is indicative of the difference in oxygen concentrations.

KEY POINTS OF THE CHAPTER

1. The potentiometric assay of an ionic analyte (a metal cation or an anion such as nitrate, etc.) is based on the PD appearing between two electrodes, where each constitutes a half-cell soaking in a solution containing the ion to be measured.
2. The reference electrode conserves a constant potential, while that of the ion-selective electrode (ISE) varies with the activity of that ion, because it contains a solution of this ion separated from the analyte solution by a membrane permeable only to the ion for which it was designed.
3. The membrane separating the two electrolytic solutions constitutes a selective interface. A PD, related to the activity of the ion in question, is established between its two faces (Nernst equation).
4. The overall PD between the two electrodes (measurement and reference) therefore reflects the activity of the ion in solution. When an assay is conducted requiring several measurements (sample and calibration solutions), the same quantity of a buffer (ISAB) is added to all solutions in order to conserve the same ionic strength for all measurements. For each ion, there is an appropriate buffer.
5. The pH electrode is a specific electrode sensitive to H^+ ions, often including, for convenience, an electrode used as a reference (combined electrode). The indicator electrode contains an acid separated from the external solution by a special glass membrane, enabling proton exchanges.

6. The design of an ISE depends on the ion for which it was created. There are several types of membranes: solid ones (crystalline) or those considered to be liquid, involving a polymer impregnated with an ionophore.
7. There are several dozen ions (cations and anions), gases (CO_2, HCN, H_2S, SO_2, etc.) or molecules and biomolecules (e.g. enzymes) that can be measured with specific ISEs.
8. The two quantification methods are direct ionometry and the multiple addition method.
9. The protocol for direct ionometry leads to a calibration curve, whose slope is close to the theoretical slope factor deduced from the Nernst equation (i.e. 59 mV for a monovalent ion).
10. The protocol for the addition method gives more precise results as long as the sample solution can constitute a charged matrix. Of the two suggested methods, the addition method can be used to be free from the usual drift in the ISE, which occurs during prolonged use.

PROBLEMS

1. What is the degree of dissociation and the pH of an aqueous solution of 0.85 M ethanoic acid? (K_A of $CH_3CO_2H = 1.8 \times 10^{-5}$ at 20°C.)
2. Knowing that the standard potential of a Cd/Cd^{2+} electrode *vs.* a standard hydrogen electrode (SHE) is 0.403 V, what happens to this potential if the electrode is plunged into a 0.01 M aqueous solution of $CdSO_4$?
3. 30 ml of a 10^{-3} M aqueous solution of an Fe^{3+} salt is mixed with 20 ml of a 10^{-3} M solution of a Ti^{3+} salt. What is the molar concentration of the iron and titanium salts in the resulting solution?

   (Ti^{3+}/Ti^{4+} = 0.2V *vs.* normal hydrogen electrode (NHE); Fe^{2+}/Fe^{3+} = 0.77 V *vs.* NHE and NSE/NHE = +0.25 V).
4. Find the expression below as used in the single addition method. Please remember that C_X represents the concentration of the analyte ion (Volume V_x of the solution to which a small volume V_R of a reference solution of ion concentration C_R is added). ΔE represents the potential difference between the two measurements, and S represents the real response slope of the electrode (0.0591/z, at 298 K).

$$C_X = C_R \frac{V_R}{V_R + V_X} \cdot \frac{1}{10^{\Delta E/S} - \frac{V_X}{V_R + V_X}}$$

5. To determine the acidity constant of an HA/A^- acid-basic pair, absorbances are measured at 360 nm for three solutions S_1, S_2, and S_3 of this pair with a molar concentration of 10^{-4} mol/l and the pH values are $pH_1 = 2$, $pH_2 = 5.5$, and $pH_3 = 8$

respectively. $A_1 = 0.1$, $A_2 = 0.4$, and $A_3 = 0.7$ by using cells with a 1 cm optical path. If we suppose that only the acidic form is present at pH_1 and that the only the basic form is present at pH_3, calculate:

a. the molar absorption coefficients of the acidic form and the basic form;

b. the pKa of this pair. Are the approximations made justified?

6. A measurement cell [saturated calomel electrode (SCE)//H+ (a = x)/glass electrode] develops a potential difference of +0.2094 V when the solution in the right-hand compartment is a buffer solution of pH = 4.007, and a potential difference of 0.0322 V when the solution in the right-hand compartment is a buffer solution of pH = 7.000.

Calculate the pH of two unknown solutions whose potential differences are, respectively, −0.3011 V and +0.1163 V.

7. Determine the molar concentration of fluoride ions contained in a teacup using the multiple addition method. The selective electrode and the reference electrode are immersed in 25 ml of tea to which 20 ml of an ionic strength buffer of constant pH is added. It is filled with water to the gauge line of a 50 ml volumetric flask. The potential difference at the cell terminals is measured and $U_0 = 98$ mV.

Another 50 ml volumetric flask is prepared, to which is added: 25 ml of tea, 20 ml of buffer, and 1 ml of addition solution of concentration $C_a = 100$ mg /l. It is filled to the gauge line with water. $U_1 = -73.9$ mV. In the same manner, prepare three other volumetric flasks to which is added: 25 ml of tea, 20 ml of buffer, and, respectively, 2, 3, and 4 ml of the addition solution. The following values are found: $U_2 = -91.7$ mV, $U_3 = -102.1$ mV, and $U_4 = -109.5$ mV. The slope is −59.2 mV.

Calculate the molar concentration of fluorides in the tea.

8. To assay calcium (Ca^{2+}: 40.0 g/mol) and magnesium (Mg^{2+}: 24 g/mol) in Gruyere cheese by complexometric titration with an indicator electrode, heat 2 g of grated gruyere in 100 ml of 0.01 M H_2SO_4 and 3 g of urea to 40°C for 10 minutes, then neutralize with dilute sodium hydroxide. Each sample is composed of half of the blend thus obtained, whose pH is brought to 10 by an NH_3/NH_4Cl buffer.

Start by measuring all Ca^{2+} and Mg^{2+} ions with EDTA. The point of equivalence is identified with a Cu^{2+} ion-selective electrode, after having added 1 ml of Cu-EDTA complex, prepared by mixing two solutions of 0.05 M EDTA and $Cu(NO_3)_2$. Then, only Ca^{2+} is measured with EGTA (in this case, we add 1 ml of Cu-EGTA prepared as above). The titration reagent is a 0.02 M solution of EDTA or EGTA depending on the measurement used.

For (Ca + Mg), the point of equivalence is obtained with 11.0 ml of EDTA, and for Ca alone with 10.0 ml of EGTA.

a. Explain the principle for these two measurements.

b. Find the concentrations for Ca^{2+} and Mg^{2+}.

$pK_{dissociation}$ of chelate	Cu^{2+}	Ca^{2+}	Mg^{2+}
EDTA	18.8	11	8.7
EGTA	17.9	11	5.2

SOLUTIONS

1. Where x is the concentration of H^+ ions;

$CH_3COOH \rightarrow CH_3COO^- + H^+$ ($K_a = 1.8 \times 10^{-5}$).

$K_a = x^2/(0.85 - x)$ i.e. $x = 0.0039$, therefore $H^+ = 0.0039$ M, i.e. pH = 2.4.

The degree of dissociation will be $(0.0039/0.85) \times 100 = 0.46\%$.

2. $[Cd^{2+}] = 0.01$;

$E = E_0 - RT/nF\text{Ln}[\text{Red}]/[\text{Ox}]$;

$E = -0.403 - (0.059/2) \times \log(1/0.01) = -0.462$ V.

3. The reaction is as follows: $Ti^{3+} + Fe^{3+} \rightarrow Ti^{4+} + Fe^{2+}$; however, we added 1.5 times the stoichiometric quantity for Fe^{3+} and there will no longer be any Ti^{3+}; $[Ti^{3+}] = 0$; $[Ti^{4+}] = 4 \times 10^{-4}$ M

$[Fe^{2+}] = 4 \times 10^{-4}$ M; $[Fe^{3+}] = 2 \times 10^{-4}$ M (half of the previous value).

4. The first measurement conducted on the sample solution is of the following type: $E_1 = E'' + S\log C_X$. The second measurement is of the following type: $E_2 = E'' + S\log (C_X V_X + C_R V_R)/(V_X + V_R)$. The term $(C_X V_X + C_R V_R)/(V_X + V_R)$ represents the new compound concentration measured when volume V_R of concentration C_R is added to volume V_X of concentration C_X:

$$\Delta E = E_2 - E_1 = S\log\left(C_X V_X + C_R V_R\right) / \left[\left(V_X + V_R\right) C_X\right]$$

i.e. $10^{\Delta E/S} = (C_X V_X + C_R V_R)/[(V_X + V_R)C_X] = V_X/(V_X + V_R) + C_R V_R/[(V_X + V_R)C_X]$. By isolating C_X in the first member of this expression, the result is the proposed formula.

5. **a.** $\varepsilon_{ac} = 10^3$ mol · l^{-1} · cm^{-1}; $\varepsilon_{bas} = 7 \times 10^3$ mol · l^{-1} · cm^{-1};

b. We use material conservation equations [Acid] + [Base] = C and absorbance additivity equations at pH =5.5, i.e. 10^3[Acid] + $7 \cdot 10^3$[Base] = 0.4, which gives us [Acid] and [Base] at pH = 5.5 then pKa = 5.5. For pH = 2, we are at pKa – 3.5: therefore, it can be legitimately said that the pair is present in acidic form. For pH = 8, we are at pKa + 2.5: it can thus legitimately be said that the pair is in basic form. Therefore, the approximations are justified.

6. Knowing that the potential of a glass electrode may be expressed as $E = a - b$pH, we have to find a and b from the two buffers; $b = 0.0592$ V (traditional slope) and $a = 0.4466$ V.

For $E = -0.3011$ V, pH = 12.63 and for $E = 0.1163$ V, pH = 5.58.

7. The measured PD U is in the form $U = A - 59.2\log[F^-]$ where U is in mV. For the solution without addition $U_0 = A - 9.2\log(C_i V_i/V_{tot})$ where C_i is the fluoride

concentration of tea, V_i = 25 ml of tea and V_{tot} = 50 ml, and for a solution with addition: $U_j = A - 59.2 \log[(C_i V_i / C_a V_a) / V_{tot}]$; if we plot $10^{U0-Uj/59.2}$ on the *y*-axis as a function of V_a, a linear slope $C_a/C_i V_i$ is found. By linear regression, this slope is equal to 800. Thus, concentration $C_i = 5 \cdot 10^{-3}$ mg/l.

8. **a.** EDTA and EGTA are commercial tetra-acetic acids, whose titrated solutions are used in complexometric titration. The eight pK_a values (four for each of the two acids) are staggered from 2 to 10. They form *hexadentate* chelates (1:1) with various divalent ions, such as Ca^{2+}, Co^{2+}, Mg^{2+}, and Cu^{2+} whose stability increases in a basic medium. By designating the titrating species as H_2Y^{2-}, the following can be written:

$H_2Y^{2-} + M^{2+} \rightarrow MY^{2-} + 2H^+$ (Figure P19.1).

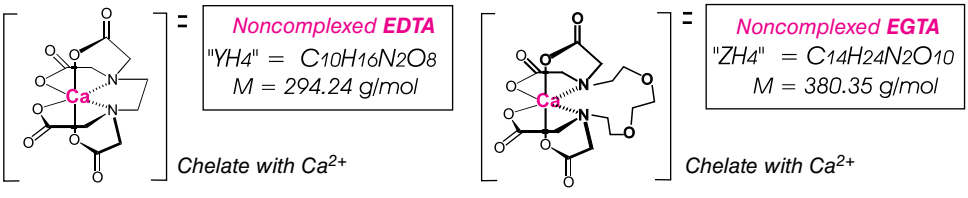


Figure P19.1

When, for the first assay (Ca + Mg), the Cu-EDTA complex is added to the solution being titrated, which contains Ca^{2+} and Mg^{2+} ions, and even though these complexes are not very soluble (see pK_d), the following equilibria appear:

$$Cu-EDTA + Ca^{2+} \rightarrow Ca-EDTA + Cu^{2+}$$

$$Cu-EDTA + Mg^{2+} \rightarrow Mg-EDTA + Cu^{2+}$$

The presence of Cu^{2+} ions in solution is found by the copper-selective electrode. As the titrating reagent (0.02 M EDTA) is added, the Ca^{2+} and Mg^{2+} ion concentration decreases and to finish the Cu^{2+} ions are chelated again, which causes a decrease in electrode potential and therefore marks the point of equivalence. The EDTA consumed is used to chelate the Ca and then the Mg (the two dissociation pK_d values for Ca^{2+} and Mg^{2+} differ by 2.3).

In the second assay, with EGTA, it is the same except for the magnesium chelate, which is less stable, since Ca^{2+} ions are once again chelated before Mg^{2+} ions. Therefore, only Ca^{2+} ions are measured (the two pK_d values differ by 5.8).

b. 1 ml of titrant corresponds to 0.02 × 40 = 0.8 mg of Ca^{2+} and to 0.02 × 24 = 0.48 mg of Mg^{2+}. Therefore, in 1 g of cheese: 10 × 0.8 = 8 mg of calcium, i.e. C_{Ca} = 0.8% (8 g/kg) and (11.0 – 10.0) × 0.48 = 0.48 mg of magnesium, i.e. C_{Mg} = 0.048% (or 480 mg/kg).

Chapter 20

Voltammetric Methods

In contrast with potentiometry, several electroanalytical methods, called voltammetric, apply an external source of electricity to the sample solution in order to induce an electrochemical reaction that would not otherwise spontaneously occur, in full or in part. It is thus possible to measure all sorts of ions or organic compounds that can either be reduced or oxidized electrochemically.

Dropping mercury electrode polarography, the best known of all voltammetric methods, has given rise to highly evolved variants.

Many current applications of voltammetry rely on the use of amperometric sensors integrated into all sorts of measurement devices. This chapter also focuses on an example of coulometric techniques: the water content assay using the Karl Fischer method.

Objectives

Present the three-electrode voltammetric cell
Describe the dropping mercury electrode
Compare the variants of polarography
Know the principle of stripping voltammetry
Situate coulometry among amperometric methods
Apply this information to the coulometric measurement of water
Explain what an amperometric detector is
Describe two redox sensors with two electrodes (CO_2 and O_2)
Describe the biosensor used to measure blood glucose.

20.1 THE VOLTAMMETRIC METHOD

The sample solution is placed in a work cell after an inert supporting electrolyte with a higher concentration has been added to it. This cell has three electrodes in the classic configuration: a *reference electrode*, whose potential remains constant (calomel or Ag/AgCl electrode), a *working electrode*, which leads to the oxidation or reduction of the analyte by contact (vitreous carbon, platinum, gold or mercury micro-electrode), and, finally, the *auxiliary electrode*, made of an inert metal (most often platinum) or carbon and which is used only to make the medium electrically conductive. The auxilliary electrode is not involved in electrolysis (Figure 20.1). A variable potential difference is imposed between the working electrode and the auxiliary electrode in order to avoid an electrolytic reaction around the reference electrode, which would lead to a change in its potential. However, it helps to monitor variations in the intensity of the electrolysis current when the PD is changed between the working electrode and the reference electrode. The PD imposed between the working electrode and the reference electrode is limited by two extreme values depending on the pH of the reaction mixture, the working electrode used, and the solvent used (most often water), which may be oxidized or reduced. For a platinum electrode, for example, these limits are +0.65 V relative to the standard calomel electrode (SCE) (oxidation of water: $H_2O \rightarrow \frac{1}{2}O_2 + 2H^+ + 2e^-$) and −0.45 V (reduction: $H_2O + 2e^- \rightarrow H_2 + 2OH^-$).

20.2 THE DROPPING MERCURY ELECTRODE

The most original working electrode is without a doubt the one containing a microdroplet of mercury as its active part. It has both advantages and disadvantages that make it extraordinary. The voltammetric technique making use of this electrode is called *polarography*.

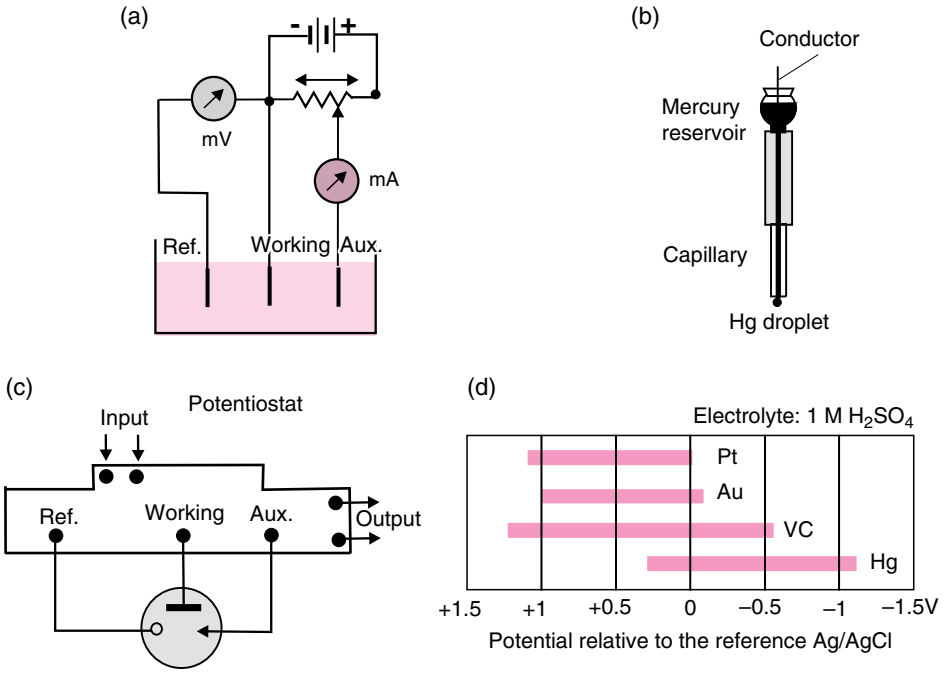


Figure 20.1 Assembly of a voltammetric cell. (a) DC set-up in which no current passes through the reference electrode (a consequence of its high impedance). (b) A model indicator electrode (dropping mercury working electrode). Many metals can be reduced at the surface of the mercury before H^+ is reduced in its turn. (c) Representation of a measuring cell controlled by a potentiostat. This electronic set-up maintains the potential of a working electrode at a constant level with respect to a reference electrode. (d) Ranges of use for the four principal working electrodes (VC stands for vitreous carbon). The mercury electrode can be used over a wide cathodic range with electrolytes such as KCl or NaOH (−2 *V*). Although overlaps are observed with different electrodes, their sensitivities can be quite different for a given species.

This electrode is composed of a glass tube whose central capillary (10–70 μm in diameter) is maintained in a vertical position, enabling the transfer of mercury between a reservoir and the capillary tip. This capillary tip is immersed in an *unstirred* solution containing the target analyte mixed with a supporting electrolyte. The surface of the droplet of mercury increases until it falls (after around 4–5 s). The fall is most often provoked by a device producing a small impact on the electrode. A new identical droplet replaces the previous one, with a fresh, uncontaminated surface. The mercury is quickly limited at positive potentials (+0.25 V as compared to the SCE). Beyond that, it too oxidizes. However, at negative potentials, it can be used up to −1.8 or −2.3 V, depending on whether the supporting electrolyte is acidic or basic. This range provides many analytical possibilities, especially for heavy metals. The mercury

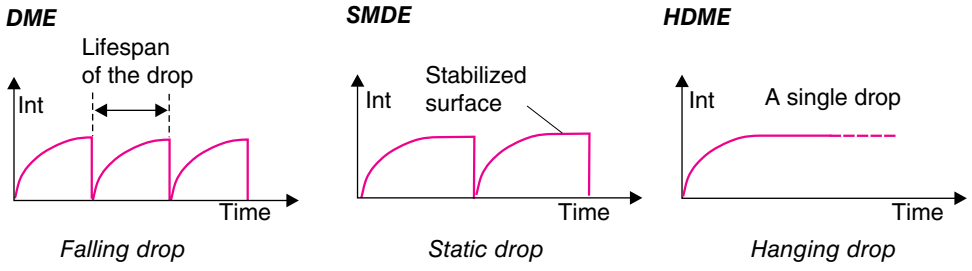


Figure 20.2 Evolution of the mercury drop diameter over time. Fresh-drop DME mode, static-drop SDME, and hanging-drop HDME.

must be very pure (hexadistilled and conserved under nitrogen). However, in light of its toxicity, its use represents a real drawback for this technique. After use, it must be recovered and recycled.

The dropping mercury electrode may be used in three different modes (Figure 20.2).

- DME (Dropping Mercury Electrode) mode: the capillary at the base of which the droplet forms is continuously supplied by the reservoir. The droplet grows and then falls with the help of a slight impact. This mode is not often used these days.
- SMDE (Static Mercury Drop Electrode) mode: the capillary's mercury supply stops as soon as the droplet reaches a certain size. The size of the droplet remains constant for a certain amount of time before it is dislodged by an impact on the capillary.
- HMDE (Hanging Mercury Drop Electrode) mode: the droplet conserves a constant size throughout measurement (for use in stripping mode).

20.3 DIRECT CURRENT POLAROGRAPHY (DCP)

In the basic experimental set-up – now rarely used – a direct, variable potential $E_{working}$ is applied to the mercury droplet. It is linear with respect to time and its incrementation (or scanning speed) is of the order of v = 1–2 mV/s from the initial potential E_i selected as a function of the analyte. So, it can be written:

$$E_{\text{working}} - E_{\text{ref}} = E_i \pm vt \qquad (20.1)$$

The resulting *polarogram*, $I = f(E_{\text{working}} - E_{\text{ref}})$, may be a simple sigmoid curve (polarographic wave, Figure 20.3) or a succession of sigmoids if several analytes must be studied. The first part of the curve, which shows the appearance of a

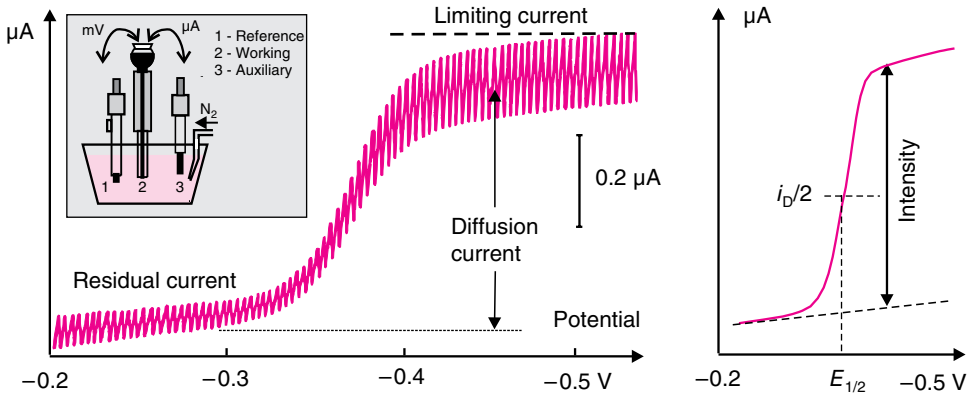


Figure 20.3 A polarographic wave. Polarogram of a solution containing 10 ppm of Pb^{2+} in 0.1 M KNO_3, obtained with a dropping mercury electrode. The median position of the wave (about −0.35 *V*) is characteristic of lead, while the height of the step reflects its concentration. For a better presentation of the graph, the oscillations have been damped. Right, a graph shows the measurement of i_D. Only the polarogram envelope is represented (no oscillations). This polarogram shows a fine, saw-toothed structure, which is produced from the renewal of the mercury droplets.

slight current (residual current), corresponds to an accumulation of charges on the surface of the working electrode (negative if this is a reduction, or positive owing to a lack of electrons if it is an oxidation) and on a solution surface located a few μm from the working electrode (analyte ions and supporting electrolyte ions). There is no electrolysis. As soon as the potential of the working electrode is sufficient (energy provided by a generator of sufficient voltage), the electrolysis takes place and an increase in current is observed (faradaic current, which is limited by the supply of analyte to the working electrode). The curve reaches a plateau (diffusion limit) due to the fact that the analyte's diffusion speed becomes equal to its conversion speed in contact with the electrode.

The appearance of this polarogram is used both for the qualitative analysis, because the potential $E_{1/2}$ (half-wave potential) measured at mid-height of the polarographic wave is characteristic of the analyte (value slightly different from the standard redox potential of the pair in question), and for the quantitative analysis, since the diffusion current intensity (difference between the limit intensity measured at the wave peak and the intensity of the capacitive current extrapolated from the limit intensity measurement) is proportional to the analyte concentration.

The elimination of all traces of dissolved oxygen in the solution is essential, for example by nitrogen bubbling. Otherwise, the polarogram would be unusable, owing to the effect of the reduction of oxygen in water according to the two steps detailed in Figure 20.11.

20.4 DIFFUSION CURRENT

The movement of the analyte in solution is the result of three phenomena:

- *Migration*, whose direction depends on the ion's sign. Between the working electrode and the reference electrode, there is a potential difference and therefore an electrical field. Any charged species placed in an electrical field is subjected to a force that causes its migration towards an electrode. This migration phenomenon may be maintained constant by adding a supporting electrolyte (e.g. potassium chloride) with a concentration well above that of the analyte (of the order of 100 times more concentrated).
- *Convection*, which is a natural movement of any species in solution. This movement may be minimized by forced stirring.
- *Diffusion*, which is the natural movement of chemical species from a region when their concentration is much higher towards a region where this concentration is much lower. This is the case when the concentration of the analyte in the solution and its concentration around the working electrode (lower since the analyte is consumed by electrolysis) are taken into consideration. If this movement towards the working electrode is too low, this will limit the intensity of the electrolysis current. It is useful to recall that the intensity of a current corresponds to the quantity of electricity transported per unit time, i.e. to the number of moles of analyte transformed on the electrode per unit time.

The mean limiting diffusion current, observed on the plateau of the polarographic wave, was modelled by the Ilkovic equation:

$$\bar{i}_D = 607 \cdot n \cdot D^{1/2} \cdot m^{2/3} \cdot t^{1/6} \cdot C \qquad (20.2)$$

The diffusion current $\bar{i}_D$ is expressed in μA with the following units: n is the number of electrons transferred during the electrolysis of the ion, D the diffusion coefficient of the analyte ion (cm^2/s); m, mass flow rate of mercury (mg/s); t, droplet interval (s); C, analyte concentration (mol/l). The coefficient 607 is established for 25°C.

Equation (20.2) shows the proportionality of the limiting diffusion current with the molar concentration of the analyte.

The saw-toothed graph observed on a classic polarogram may be eliminated by measuring the current intensity only at the end of the life of the mercury droplets and by using a DC voltage that increases in steps (Figure 20.4). This is called sampled direct current polarography.

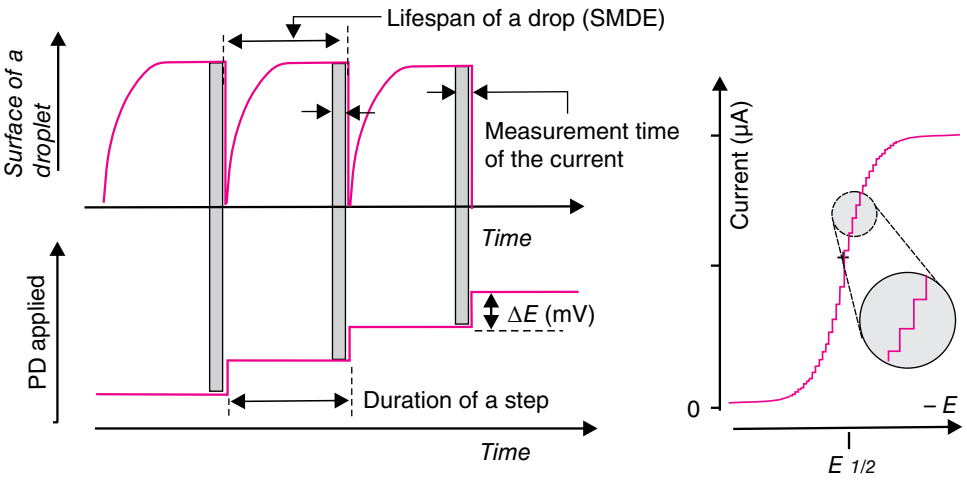


Figure 20.4 Sampling polarography. This technique increases the sensitivity of classic direct current polarography by attenuating the capacitive current. The grey zone corresponds to the instant and to the duration of the current measurement.

20.5 PULSE POLAROGRAPHY

The evolution of polarography is based on the desire to reduce the residual (capacitive) current and thus increase the faradaic current and therefore the limiting diffusion current. Hence, to improve sensitivity and to be able to distinguish analytes whose half-wave potentials differ by only a few tens of mV, the mercury droplet is no longer subjected to a gradually increasing voltage but rather to pulses.

There are three main techniques:

- Normal Pulse Polarography (NPP)
- Differential Pulse Polarography (DPP)
- Square Wave Polarography (SWP).

20.5.1 Normal Pulse Polarography (NPP)

To eliminate the jagged appearance of the classic polarogram, the standard voltage ramp described earlier is replaced by a series of brief pulses (between 50 and 100 ms) of increasing potential. Each pulse begins at the same voltage low enough to avoid any redox reaction of the analyte (Figure 20.5). These pulses are delivered at the same frequency as the renewal of the mercury droplet (between 2 and 4 s). A single current measurement is made just prior to the fall of each droplet. At this instant, the faradaic

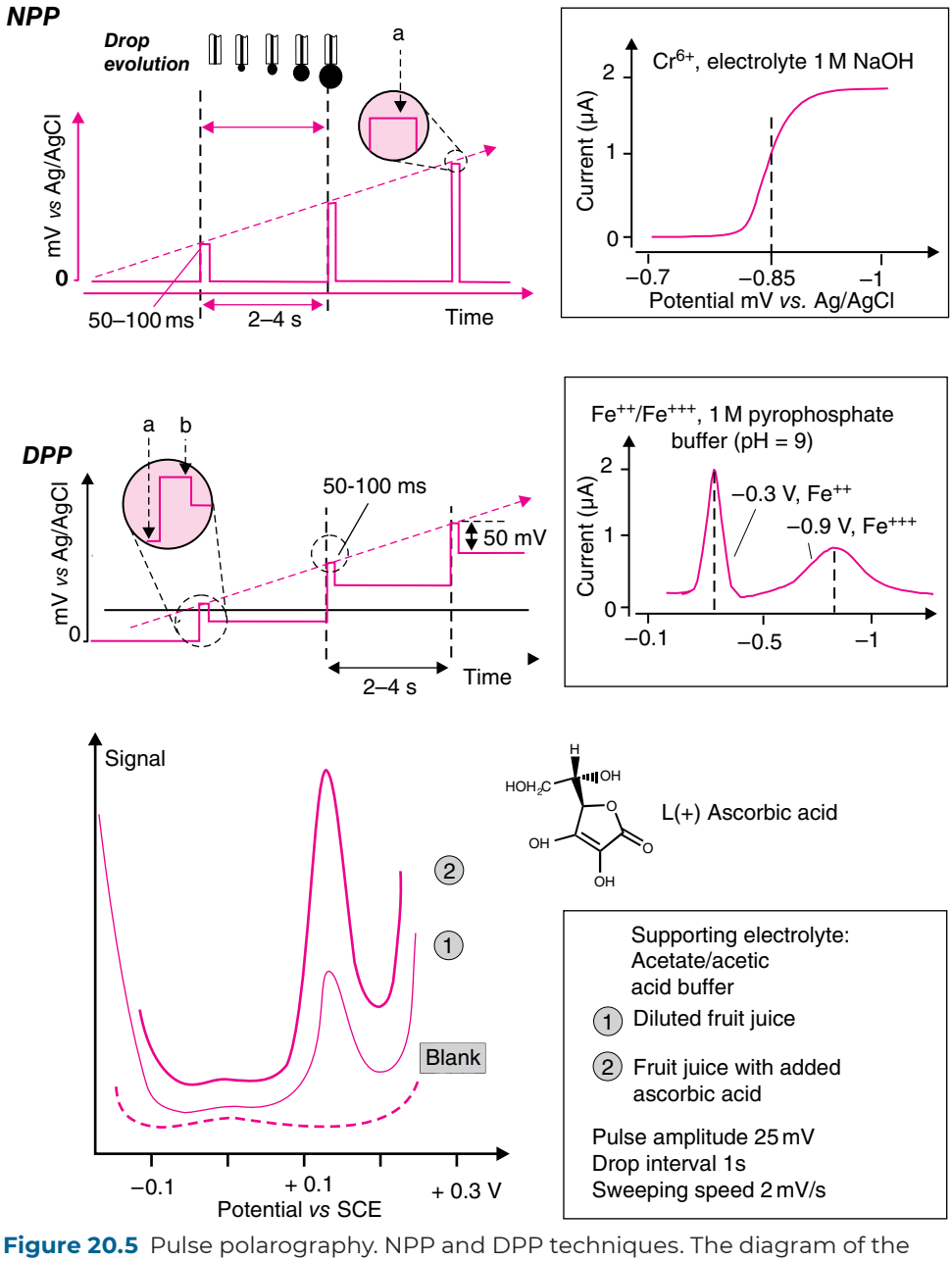


Figure 20.5 Pulse polarography. NPP and DPP techniques. The diagram of the tip of the working electrode with a droplet forming shows at what instant the measurements are made (see the arrows a and b). Examples of measurements. The measurement of ascorbic acid (vitamin C) in a fruit juice (see conditions) corresponds to an oxidation on the working electrode.

diffusion current is stabilized, while the residual current has become negligible. Sensitivity is increased by two to three orders of magnitude over the basic method.

20.5.2 Differential Pulse Polarography (DPP)

A voltage is applied to the working electrode, corresponding for each successive droplet of mercury to a pulse of the same amplitude (50 mV) but starting from a higher base voltage. Each droplet, at the end of life, is the subject of two current measurements, the first just before the pulse and the second just before the droplet falls (Figure 20.5). The graph of the difference of these currents, as a function of the voltage, exhibits a peak for each analyte, whose height is proportional to its concentration.

20.5.3 Square Wave Polarography (SWP)

This technique is a form of sampled direct current polarography (general staircase step increase in the potential applied to the working electrode) on which square wave pulses are superimposed at steps of amplitude ΔE_A, which may go up to 50 mV, in general at a frequency of 125 Hz. The electrolysis current is measured when the size of the mercury droplet is constant at the summit of a square wave pulse and at the base of the following pulse. By calculating the difference between these two currents, for each step, as a function of the potential of that step (excluding the square wave pulse), a peak is obtained, whose current at the summit is proportional to the concentration of the analyte (Figure 20.6).

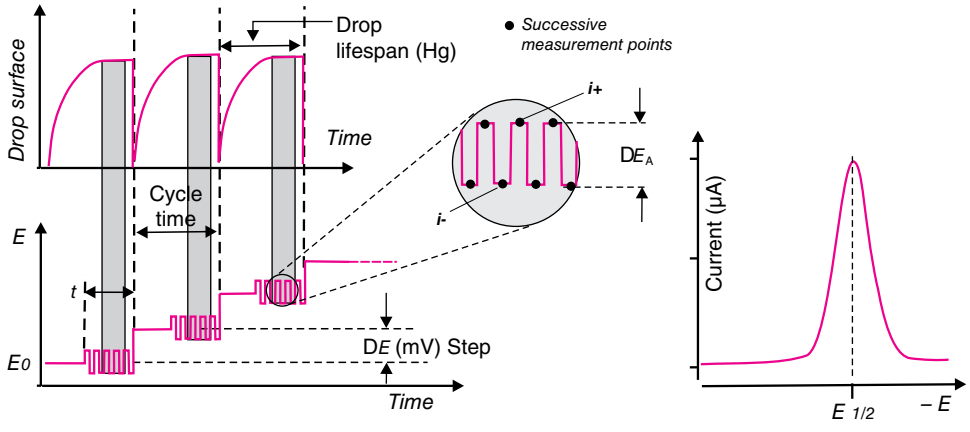


Figure 20.6 Square wave polarography (SWP). A periodic square wave signal with an amplitude of about ΔE_A = 2.5 mV and a period equal to 40 ms is superimposed at each potential level *E*. The difference ΔE (staircase step between each level) is about 12 nV for a period of 0.5–10 s. The current is measured in the grey zone around the high points (i+) and around the low points (i–) of the periodic square wave signal. The difference in current i+/i– is plotted as a function of the staircase step potential. (Source: Courtesy of Metrohm.)

20.6 ALTERNATING CURRENT POLAROGRAPHY (ACP)

This technique is rather similar to the previous one. The difference resides in the fact that instead of sending square wave pulses, an AC voltage of low amplitude (5–20 mV) is superimposed over the DC voltage. Then, in the current measurement circuit, an AC current of variable amplitude is observed. This amplitude is plotted as a function of the AC potential and a peak is obtained whose current at the summit is proportional to the analyte concentration.

20.7 STRIPPING VOLTAMMETRY

This general analytical technique is used to measure traces of metal ions or anions. It is carried out in two stages:

- *Electrolysis*. On a graphite carbon electrode, whose tip is covered with a mercury film, immersed in a stirred solution containing the analyte(s), a deposit by electrolysis of a fraction of the electroactive analytes present is caused in a few minutes. This stage may be a cathodic process, in which case, the metal cations are reduced in contact with the electrode. Alternately, the process may be anodic, and thus anions are deposited on it. The mercury film electrode (MFE) may be prepared in advance or formed in situ during this first stage. In the case of an M^{n+} cation analysis, the metals form amalgams M(Hg), which are deposited on the electrode surface. In the case of an anion analysis, these are detected as mercury salts.
- *Stripping*. Following electrodeposition, in the case of the cathodic process (metal deposit), a potential sweep is conducted in the positive direction by progressively reducing the PD between the working and reference electrodes. The reversibility of redox reactions leads to an electrochemical re-oxidation of the analytes (Figure 20.7). This particular case is known as *anodic stripping voltammetry*. The elements are identified by their redox potential and their concentration is obtained by the standard addition method.

When combined with the DPP technique, *anodic or cathodic stripping voltammetry* constitutes a universally applied, and therefore the most widely used, voltammetric method. Current instruments, whose parameters are controlled by software, enable reproducible conditions and thus offer a cheaper alternative to spectroscopic methods such as atomic emission.

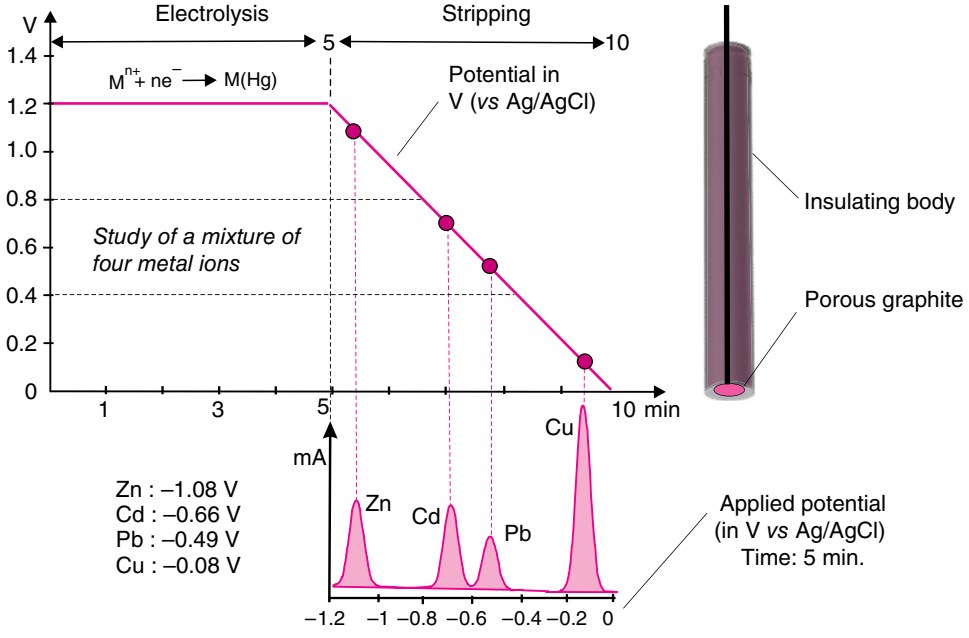


Figure 20.7 Determination of cations with a mercury film electrode by anodic stripping voltammetry. Two-step linear voltage ramp of the working electrode potential in the analysis of metals present in a sample of sea water. The electrode is composed of a small vitreous carbon disc on which a mercury film (10–50 μm) is generated at the moment of analysis by adding 0.2 ml of mercury nitrite (1 g/l) to the sample solution.

20.8 COULOMETRIC MEASUREMENTS

The methods described above correspond to the partial electrolysis of the analytes in contact with the working microelectrode. By contrast, coulometric methods are based upon quantitative conversion of the analytes. Should that be the case, the concentration of an analyte is calculated by applying the theory, without needing to conduct a calibration. This is therefore not a comparative method.

The selectivity is sometimes average and the method is slow. Two types of coulometric methods coexist:

- *Potentiostatic coulometry* involves holding the electric potential (the voltage) of the working electrode constant during the reaction. This avoids parasitic reactions. The current decreases as the analyte disappears from the solution. An electronic integrator used to compute the quantity of current used.

- *Amperostatic coulometry*, in which the current is maintained constant during the measurement by an amperostat until the signal which marks the end of the reaction of the analyte. This is the simplest form of coulometric measurement. There are many commercial devices of this type. A coulometric titrator is equipped with a working electrode of large surface area and an auxiliary electrode separated from the reaction compartment by a diaphragm. This protection of the second electrode provides a barrier between the species formed in contact with it and those formed at the working electrode, thus avoiding an eventual reaction.
- Often, direct and total transformation of the analyte proves difficult because the product formed accumulates in contact with the electrode, causing its polarization. This problem is countered by addition of a precursor in large excess, which is employed for the release of an intermediate reagent that will react with all the analyte. In this way, the analyte does not participate directly in the electron transfer process.

For example, the measurement of Cl^- ions will be possible from Ag^+ ions generated at the surface of a silver anode. The quantity of current used, in coulombs, leads to a precise calculation of the quantity of analyte transformed, provided that the current serves only for the formation of Ag^+ ions. Standard solutions are not required for these measurements. The absolute quantity of ions formed is determined from the current consumed. The quantity of current $Q(C) = i(A) \cdot t(s)$ corresponds to the analyte alone.

20.9 COULOMETRIC TITRATION OF WATER CONTENT

Many manufactured products, as well as solvents, raw materials, and food products are analysed for their water content or 'percentage moisture'. Of all the available methods, the Karl Fischer titration (Chapter 18) is perhaps the most widely used, involving chemical reactions combined with potentiometric detection. Below, the coulometric method, which is very sensitive and adapted to water contents below 1% and down to a few ppm (Figure 20.8), is described. Unlike the volumetric method, which requires several steps, especially to determine *concentration* C of the reagent beforehand, the iodine necessary for the measurement is generated from a precursor iodide via an electrochemical path using electrical pulses applied to the electrodes of an electrolyser. The electrochemical reaction is considered to be quantitative.

The modified KF reagent contains an iodide (the precursor) that is oxidized to molecular iodine in contact with an anode. This electrolysis cell is equipped with a diaphragm between the anodic and cathodic compartments (Figure 20.9). On the anode, the iodide ion is oxidized into iodine ($2I^- \rightarrow I_2 + 2e^-$). The volume of KF reagent, used in the classic method, is replaced by the more precise measurement of

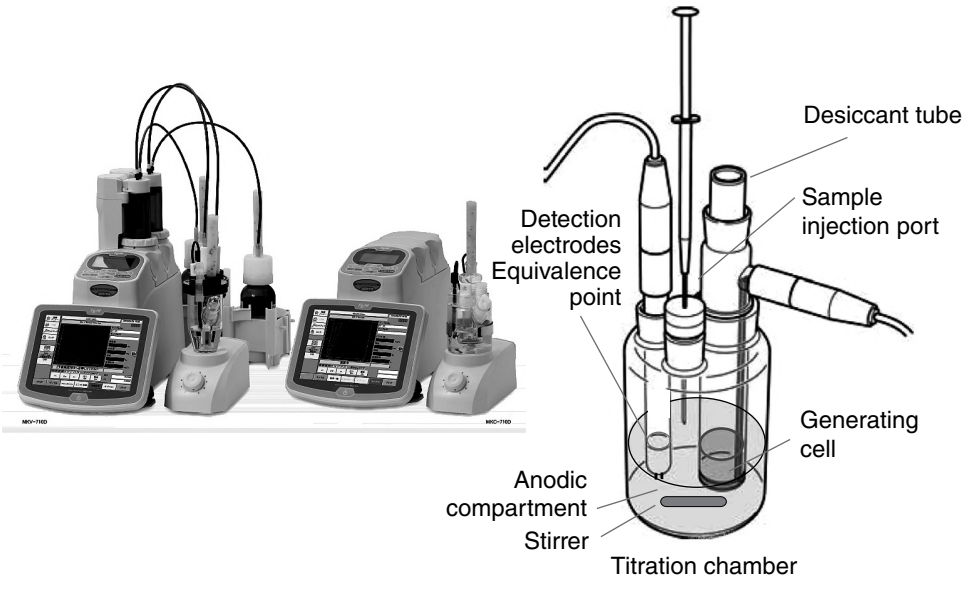


Figure 20.8 Coulometer to determine water content in accordance with the Karl Fischer method. Titrator MKV-710.Vet Coulometer 710.C (Source: Courtesy of Kyoto Electronics Manufacturing Co, Japan).

a quantity of current, knowing that 1 mg of water is equal to 11.72 Coulombs. A mole of water requires 192,974 Coulombs (twice the Faraday constant). This is an absolute determination, since the relationship between the quantity of current and the quantity of iodine generated is quantitative.

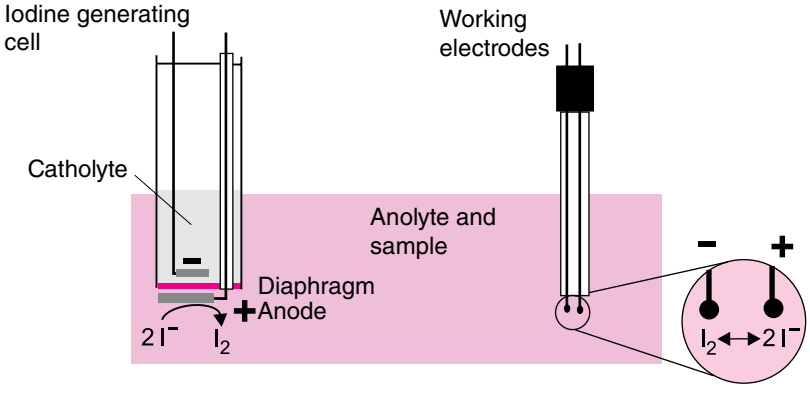


Figure 20.9 Karl Fischer coulometer electrodes. The purpose of the diaphragm is to prevent ions reduced on the cathode (or H_2) from possibly reducing oxygenated compounds present in the sample, which could generate water.

The detection of the point of equivalence is an application of imposed current potentiometry (*dead stop end point*). A voltage generator delivers a constant polarization current (about 50 μA) between two small platinum electrodes, which corresponds to a certain voltage (Figure 20.9). When equivalence is reached, iodine appears in solution and is reduced to iodide in contact with one of the two electrodes (cathode), which is re-oxidized on the neighbouring electrode (anode). In summary, the voltage must decrease to conserve the initially set current. Therefore, the end of the measurement is found.

20.10 VOLTAMMETRIC DETECTION IN HPLC AND HPCE

Voltammetry has been adapted as a highly sensitive mode of detection for electroactive compounds in liquid chromatography (provided that the mobile phase is conductive) and capillary electrophoresis. The miniaturized cell has three classic electrodes, including a working electrode brought to a defined potential with respect to the reference electrode. The detection cell is swept by the mobile phase as it leaves the column (Figure 20.10).

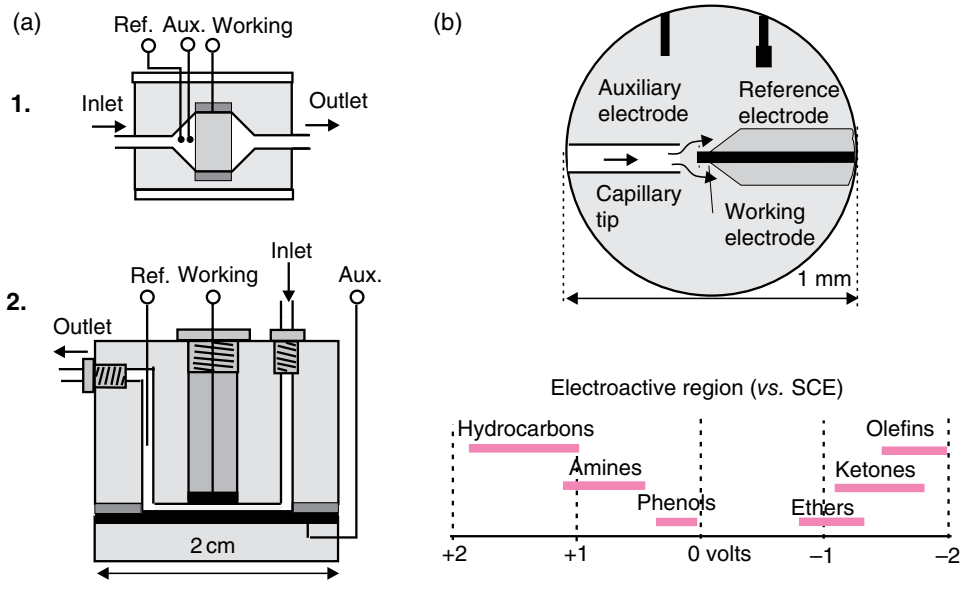


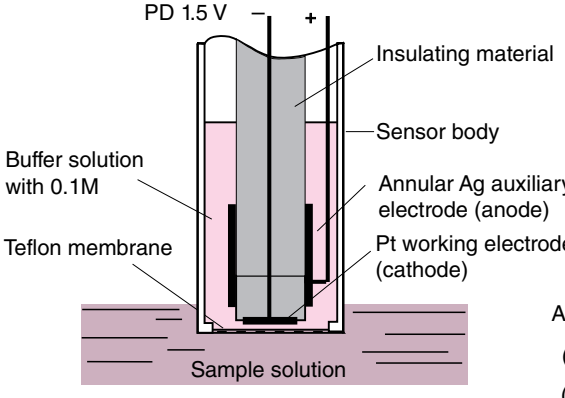
Figure 20.10 Voltammetric detection in HPLC and HPCE. (a) Two models of voltammetric cells. The porous graphite working electrode has a large surface and operates under coulometric conditions. The flow of the mobile phase around the working electrode ensures renewal of the electroactive species. (b) Detail of the tip of a capillary in HPCE. The working electrode receives the ions exiting the capillary. The cell is not materialized. It is included with the cathodic compartment of the instrument. Apart from phenols, aromatic amines, and thiols, few analytically important molecules are electroactive.

20.11 AMPEROMETRIC SENSORS

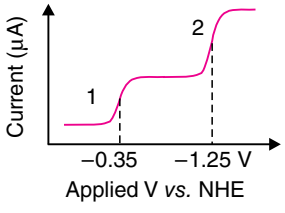
Voltammetric measurements are not simply restricted to analytical laboratories. The applications of these methods are more numerous than is at first obvious. A large number of analytical instruments, whether portable or not, intended to make precise measurements of substrates present in gas mixtures, vapours or solutions are equipped with electrochemical sensors. These devices operate on the principle of the two- or three-electrode amperometric cell. Several sensors of this type are described below.

20.11.1 Clark Oxygen Probe

The electrochemical reduction of molecular oxygen in contact with an electrode provides the basis for several models of sensors, which differ only by the reactions used. The best known of these is without doubt Clark's dissolved oxygen sensor, which contains a silver anode and a platinum cathode (working electrode), both in contact with a KCl-based electrolyte (Figure 20.11). This cell is separated from the sample by a Teflon membrane that is permeable to oxygen. After crossing this membrane, the oxygen gas comes into contact with the cathode, about 20 μm away from the membrane. A potential difference of around 1.5 V is applied between the two electrodes, the cathode being brought to a sufficiently negative potential to reduce all of the oxygen



The two polarographic waves of oxygen



At the anode

$$4\,Ag + 4\,Cl^- \longrightarrow 4\,AgCl + 4\,e^-$$

At the cathode

$$(1)\quad O_2 + 2\,H^+ + 2\,e^- \longrightarrow H_2O_2$$

$$(2)\quad H_2O_2 + 2H^+ + 2\,e^- \longrightarrow 2\,H_2O$$

Balance: $O_2 + 4\,H^+ + 4\,e^- \longrightarrow 2\,H_2O$

Figure 20.11 A Clark sensor for oxygen determination. Cell with two concentric electrodes. The Teflon membrane, permeable to oxygen, must be very close to the cathode, so that the double diffusion process through the membrane and subsequently in the liquid film will lead to a stable signal after about 10–15 seconds.

(reactions 1 and 2). The current in the external circuit is proportional to the quantity of gas migrating across the membrane and consequently to its concentration in solution (Faraday's law).

20.11.2 Redox Sensors for Gas or Vapours

Amperometric sensors have become omnipresent, especially in the field of gaseous substance analysis, such as protective equipment and control instruments (Figure 20.12).

These amperometric gas sensors (AGS) most often include the three conventional electrodes (working, auxiliary, and reference) immersed in an immobilized electrolyte. The assembly looks like an electrical battery. When the sensor is exposed to an atmosphere containing the electroactive compound in question, and once the latter has reached a specific porous Teflon diffusion barrier, it is oxidized (CO, H_2S, NO, H_2, HCN) or reduced (Cl_2, NO_2) in contact with the working electrode, causing an electron transfer via the external circuit. Several specific sensors may be present in a given detector, which then adjusts the electrode's potential as recommended for each of them.

These sensors are very sensitive and have a rapid response time over a wide dynamic range (four orders of magnitude). In contrast, they function over a limited temperature range (–10/+40°C), requiring recalibration. Their lifetime varies because they wear out and gradually become contaminated. Lastly, erroneous results may be generated in the presence of other gases in large concentrations.

For certain gases, two electrodes are enough, as the sensor operates like an electrical battery, without the need for a potentiostat. However, the linear measurement range is narrower. Two examples illustrate the reactions which occur in AGS sensors for molecular oxygen (O_2) (Figure 20.13) and for carbon monoxide (CO) in household detectors (Figure 20.14). The detection cells contain electrolytes that are immobilized by absorbent materials. For O_2 a slightly basic solution (transfer of hydroxyl ions) of potassium acetate is used and for CO a strong mineral acid (proton transfer), such as sulfuric acid, is employed.



Figure 20.12 Amperometric gas sensors (AGS).

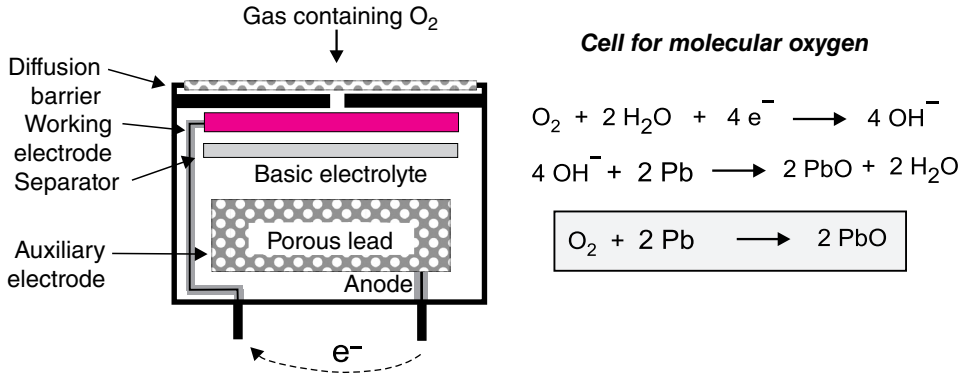


Figure 20.13 Two-electrode cell for oxygen. The lead anode is progressively transformed into lead oxide. The OH^- ions migrate from the working electrode to the auxiliary electrode via the ion conductor, while electrons follow the opposite path using the external circuit. This sensor is a lead/air battery of sorts and wears out over time, like any battery.

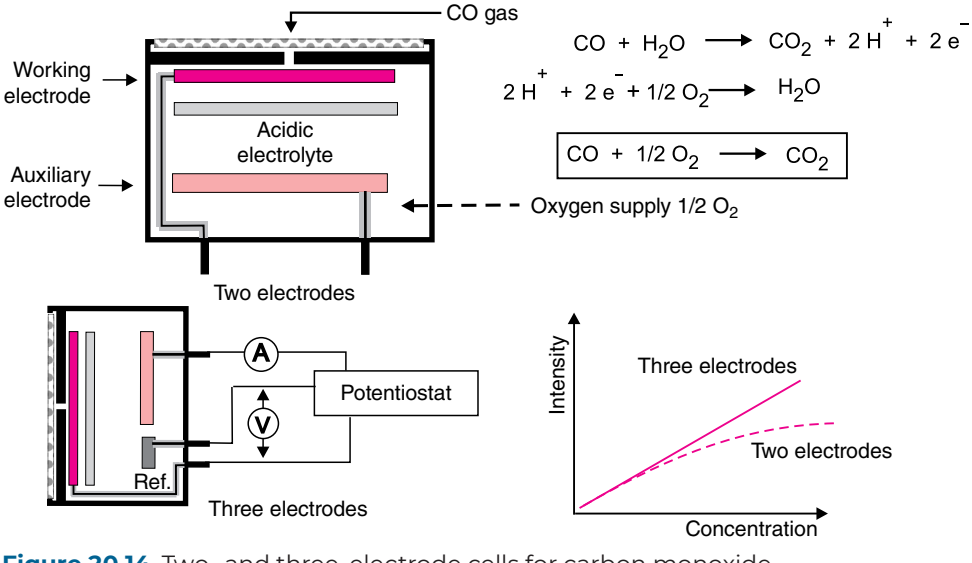


Figure 20.14 Two- and three-electrode cells for carbon monoxide. Electrochemical transformation consumes oxygen, which must be present in the cell. In the assembly with the reference electrode, the potentiostat electronically compensates for the value of the detected current, in order to increase the range of linearity of the sensor.

20.11.3 Biosensors with Amperometric Detection

Applications of voltammetric sensors have been extended to many chemical species (sugars, alcohol, pesticides, cholesterol, etc.) by creating selective membranes with a biological component, such as an oxidoreductase enzyme, immobilized between two permeable films and on which the analyte in question will react specifically. This active part behaves like a filter, in order to measure the analyte alone without prior treatment of the sample.

In general, a *biosensor* is a device with two parts:

An element, usually very small in size, in contact with the sample solution on which a selective chemical transformation will occur, involving specific enzymatic reactions.

A *transducer*, i.e. a detector using a physical measurement (optical, potentiometric, conductimetric or amperometric) to convert the change produced by the chemical reaction into a measurable signal.

Biosensors represent only one category of electrochemical sensors among many others.

Amperometric detection, for example, can be used to measure fructose with fructose-dehydrogenase, alcohol with alcohol-dehydrogenase or organochloride pesticides with cholinesterase. However, the measurement of blood glucose using portable devices is the most widespread application.

D-glucose in solution, essentially in hemiacetal form, is easily dehydrogenated (oxidized) into D-gluconolactone, a cyclic ester that is generally hydrolysed into D-gluconic acid, with which it is in equilibrium. Only the first is a redox reaction, while the second is an electrically neutral hydrolysis. The stoichiometric balance involves the transfer of two electrons and two protons from one glucose molecule to an oxygen molecule, which is reduced to hydrogen peroxide. Each D-glucose molecule thus loses two electrons (Figure 20.15).

All commercial applications use glucose oxidase (*GOx*), an oxidoreductase that is both a catalyst and a reagent when it is associated with a cofactor used as a prosthetic group. This cofactor is flavin adenine dinucleotide (FAD), which serves as a temporary electron trap, moving from its oxidized form (FAD) to its reduced form (FADH2) by capturing two protons and two electrons (Figure 20.16). The re-oxidation of FADH2 into FAD may be done with oxygen, which is the natural transfer agent of electrons in this transformation, but also using various 'artificial' oxidative mediators, such as ferrocene, ferricyanide ion or a ubiquinone, to make the reaction insensitive to oxygen fluctuations in ambient air. The quantity of glucose in the sample is thus accessible by measuring the quantity of current caused by the re-oxidation of the selected mediator in contact with the working electrode (anode). The choice of a mediator that can be oxidized at a lower voltage minimizes the effect of potential interfering ions.

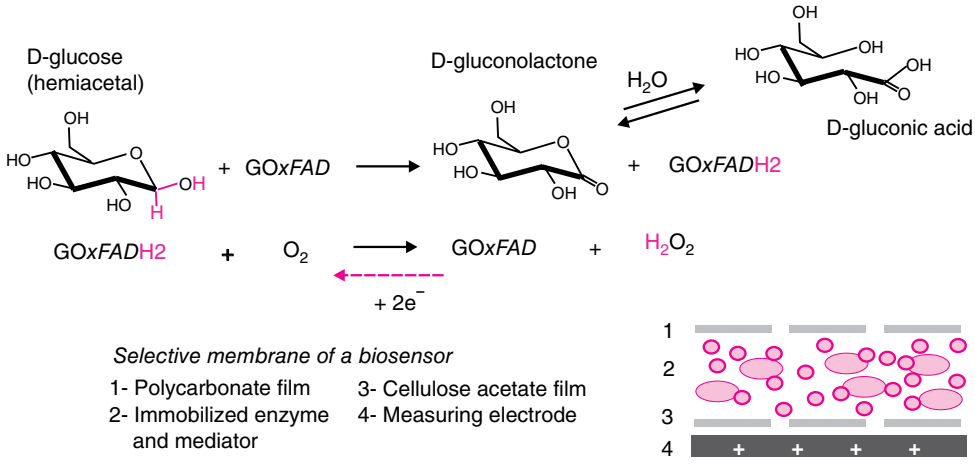


Figure 20.15 Oxidation of glucose by glucose oxidase. The reaction diagram shown here corresponds to the balance of this reaction, in which oxygen intervenes only to regenerate the cofactor (same number of oxygen atoms for glucose and for the lactone). Detail of the active part of a chemical recognition membrane. The polycarbonate film acts as a barrier for macromolecules potentially present in the sample, and the cellulose acetate film lets small H_2O_2 and O_2 molecules pass through.

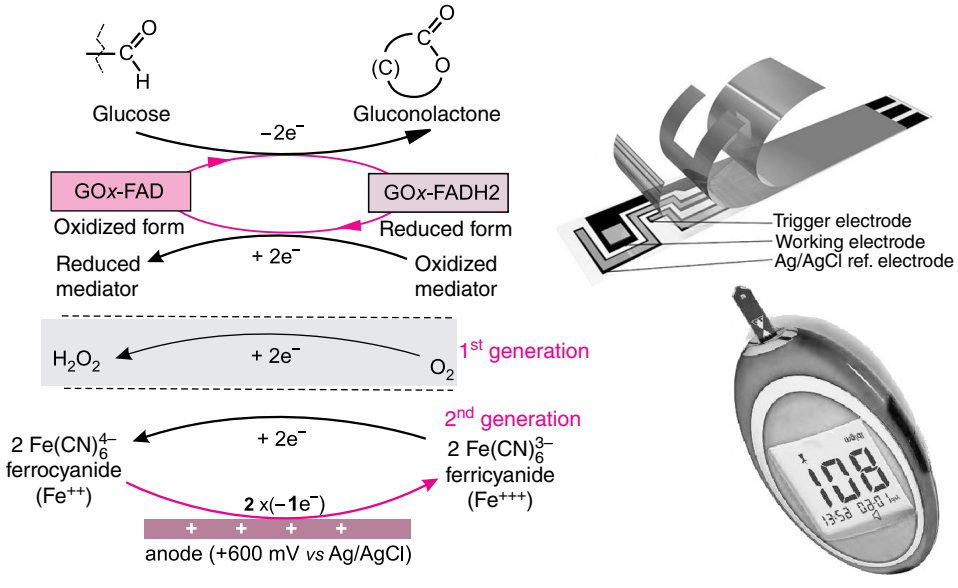


Figure 20.16 Amperometric measurement of glucose. First generation based on the natural presence of molecular oxygen and example of the second generation using ferricyanide ion as a mediator. Example of a glucometer to determine the level of glucose in the blood. The strip, which integrates electrodes and the membrane, is disposable (single use). The volume of blood necessary, of the order of 0.5 µl, is defined by the strip itself using the trigger electrode, which surrounds the reagent cavity.

The precision of this amperometric measurement with a biosensor depends on several major factors: penetration speed of the analyte into the thin enzymatic layer, kinetics of the reaction, conversion yield, chemical conversion speed, and precision of the sample volume, since the final result is expressed in g/l. The measurement thus has no meaning until after the calibration.

KEY POINTS OF THE CHAPTER

1. Voltammetry is based on a measurement of current between two electrodes (working and auxiliary) immersed in a container containing the analyte(s) as well as a supporting electrolyte. A third electrode is used to control the potential of the working electrode.
2. By progressively increasing the PD between the reference electrode and the working electrode, a reduction or an oxidation of the analytes is caused on its surface, which translates into the passage of a current between the working electrode and the auxiliary electrode. This electrolysis only transforms a miniscule part of the analyte present.
3. The most commonly used reference electrode (Ag/AgCl) is associated with a limited selection of working electrodes, the most original of which is the dropping mercury electrode, which is the basis for several techniques: a dropping, stabilized or hanging droplet. Its range of use is large for negative potentials, where as a cathode, it reduces metal cations.
4. This method called polarography has given rise to many high-performance variants for very low concentrations. In the basic experiment with only one active analyte, the current/potential curve is sigmoidal, and the potential value at slope change is characteristic of the reactive ion, and the current is proportional to its concentration.
5. The measurement medium must be free of dissolved oxygen (which may reach 300 μmol/l), since the latter causes two polarographic waves, which may skew results.
6. The diffusion current expressed by the empirical Ilkovic equation expresses the different factors which impact its intensity, including of course the concentration of the analyte.
7. Stripping voltammetry is a two-step method that uses an electrode covered with a mercury film (cathode) on which ions (e.g. M^{2+}) will first be reduced and will be fixed as Hg/M amalgams. In the second step, the PD is decreased, which releases the amalgams (anodic stripping), and the cations successively go back into solution. The current/potential curve makes it possible to identify the ions present, while determining their concentration requires an addition method.

8. Unlike the polarography method, coulometry consists in quantifying an analyte using the quantity of current needed for its electrochemical oxidation or reduction. It is assumed that the electrical current supplied only transforms the analyte.
9. The calculation of water content in a sample using the Karl Fischer coulometric method consists in generating the iodine necessary for the anodic oxidation of an iodide. Two iodide ions are needed to form an iodine molecule, which will then react with a water molecule. The quantity of electricity corresponding to a mole of water is therefore equal to two times the Faraday constant, i.e. 2 × 96,487 = 192,974 C.
10. Voltammetric assays are not limited to metal ions only. Biosensors are used to quantify many molecular species if they can be reduced or oxidized by an enzymatic reaction (oxidoreductase).

PROBLEMS

1. In a polarography experiment, an aqueous solution containing a concentration of 10^{-3} M Zn ($Zn(NO_3)_2$) employs 0.1 M KCl as supporting electrolyte.

   Calculate the ratio of the migration and diffusion currents for the Zn^{2+} ion in the proximity of the cathode.

   Data: ion mobilities ($m^2 \cdot V^{-1} \cdot s^{-1}$): $Zn^{2+} = 5.5 \times 10^{-8}$, $NO^-_3 = 7.4 \times 10^{-8}$, $K^+ = 7.6 \times 10^{-8}$, and $Cl^- = 7.9 \times 10^{-8}$.
2. A dropping mercury electrode is regulated such that one droplet falls every four seconds ($t = 4$ s). The mass of Hg corresponding to 20 drops is 0.16 g. Calculate the average mass flow rate of Hg in mg/s.

   If the flow is proportional to the height of mercury in the reservoir, what would happen to the lifetime of the droplets if the height of Hg was increased threefold?
3. Calculate the mean diffusion current i_D for the Pb^{2+} ion present at a concentration of 10^{-3} M in aqueous 0.1 M KCl at 0°C.

   Data: $m = 2.10^{-3}$ g/s, $t = 4$ s, $D = 8.67 \times 10^{-6}$ cm^2/s.
4. **a.** A sample of 25 ml has a Zn^{2+} concentration of approximately 2×10^{-8} M and leads via electrolysis to a current of 1.5 nA.
   1. Calculate the time necessary to deposit 3% of the Zn present.
   2. Show that the stripping technique is more sensitive.

   b. Find the depletion percentage of a 20 ml solution of 1×10^{-3} M zinc nitrate during a five-minute polarography measurement. The current in the circuit is 15 μA.
5. To measure ferrous iron present in an aqueous solution by amperostatic coulometry, it is preferable to add a small amount of cerium salt (Ce^{3+}). Recall that the Ce^{3+} cation is oxidized at a lower potential than that of water and that it is more oxidative than Fe^{2+}. Write out the reactions in question.

SOLUTIONS

1. The problem involves calculating the ratio of the zinc transport number with regards to the transport number of all other ions present in solution; the calculation leads to:

$t_{Zn}^{2+} = 6.98 \times 10^{-3}$. For 1 Coulomb exchanged, Zn^{2+} will transport 6.98×10^{-3} C and the rest by the solution: $(1 - 6.98 \times 10^{-3}) = 0.993$ C; $i_m/i_D = 7.03 \times 10^{-3}$ or 0.007. The transport of zinc to the electrode is controlled by diffusion.

2. 20 droplets, i.e. 0.16 g, fall in 80 s. The mercury flow rate is:

$$0.16 / 80 = 2 \times 10^{-3} \text{ g/s}.$$

For a height three times greater, the flow rate will be three times higher, i.e. 6×10^{-3} g/s. Since the droplets still have the same weight, they will succeed each other at the rate of $(0.16/20)/(2 \times 10^{-3}) = 1.33$ s.

3. By applying the Ilkovic equation: $i_D = 7.15$ μA.

4. **a.** **1.** $t = 1.93 \times 10^3$ s, i.e. 32.16 min.

2. *Note:* such a current of 1.5 nA is practically undetectable. However, if we use stripping voltammetry, this quantity of zinc, for example, will be stripped in 1 s, which will give an easy-to-detect signal of $i = 2.895$ μA.

b. Two electrons are needed to reduce one zinc ion. The quantity of current used in the experiment is 4.5×10^{-3} C, which will reduce 2.33×10^{-8} mole of Zn: the depletion is 0.12%.

5. As Fe^{2+} is oxidized in contact with the anode, its concentration decreases. As we are operating at constant current, the potential of the electrode will increase. Water will then be oxidized, thus distorting the measurement's result. However, in the presence of a Ce^{3+} salt, this salt will be oxidized before the water into Ce^{4+}. The latter will also oxidize the remaining Fe^{2+}. The relationship between the quantity of current and the Fe^{2+} concentration will not be affected. Nevertheless, an indicator will be added to the medium to show when Fe^{2+} is no longer present.

Chapter 21

Sample Preparation

Most samples require a stage to prepare the analyte (i.e. the species to be measured) in a way that is adapted to the protocol followed for its analysis. This essential stage, which follows sampling, determines the quality of the analytical results. Since more than half of analyses involve the search for traces or ultra-traces, extraction techniques are generally used in order to obtain a sufficient increase in the concentration of the analytes.

This time-consuming stage has led to the development of new processes, combining chemistry and robotics, with the aim of impacting the total analysis time as little as possible.

For the most part, this chapter is dedicated to treatments used to prepare samples for chromatographic techniques.

Objectives

Show the importance of the prior treatment of the sample before analysis
Describe a few processes that are useful prior to a chromatographic analysis
Explain the principle and variants of head-space analysis
Use microwaves to accelerate mineralization treatments
Extract analytes using a supercritical solvent

21.1 THE NEED FOR SAMPLE PRETREATMENT

To start an analysis, a sample is required. This contains the species being studied, called the *analyte*, which is generally mixed with a variable number of other compounds that together constitute what is called the *matrix*. When measuring traces of an analyte in the presence of other compounds thousands of times more abundant, there is a fear that the results will be influenced by the composition of the matrix. These types of interferences between analytes and other compounds require a pretreatment of the sample. Since this additional work is often tedious and time consuming (Figure 21.1), efforts have been made to facilitate or even automate traditional adsorption, extraction, and precipitation methods.

Measurement by atomic absorption at the mg/l level of elements in drinking water or the recording of an infrared spectrum of a pure organic compound, as often performed in a teaching laboratory, are ideally simple situations in which the sample does not need any pretreatment. The same cannot be said for most common samples.

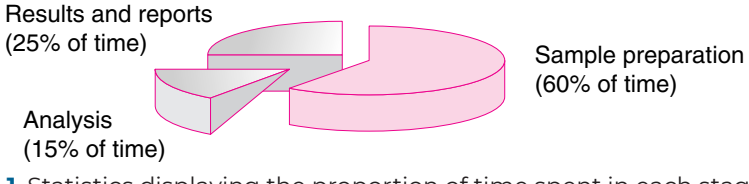


Figure 21.1 Statistics displaying the proportion of time spent in each stage of a chromatographic analysis. Sample preparation generally represents a large fraction of the total time dedicated to the complete analysis.

21.2 SOLID PHASE EXTRACTION (SPE)

SPE is currently used for the purification or concentration of a crude extract prior to the measurement of one or more of its constituents. SPE is increasingly used to replace liquid/liquid extraction, which is generally done in a separating funnel and is comparatively slower and more expensive than SPE.

SPE operates in a small open plastic cartridge, similar to the barrel of a syringe and containing an appropriate solid adsorbent (e.g. 100–300 mg cartridge of an RP-18 phase). A known volume of the liquid sample (Figure 21.2) is passed through this cartridge.

Two methods can be used:

- The most commonly used method comprises an initial step in which the compound of interest is retained by the adsorbent-containing column, followed by rinsing to eliminate as much of the undesired substances as possible from the initial matrix for the subsequent measurement. The analyte is then recovered by elution with a small volume of solvent. The resulting solution has a much higher analyte concentration, which is particularly useful in trace analysis.
- The second method, less frequently used, though very simple, consists of passing the sample through the adsorbent column, which retains the undesired compounds, while the analytes pass freely through. In this case, the analytes are purified, but not concentrated.

The nature of the adsorbents closely resembles that of the solid stationary phases used in liquid chromatography. Bonded silica gels (of reverse phase polarity) with particle sizes between 40 and 100 μm allow percolation at a faster flow rate. Other adsorbents, containing graphite or styrene-divinylbenzene copolymers, with large specific surface areas and functional groups, are more stable for solutions with very acidic pH. There are also extraction discs (0.5 mm in thickness and 25–90 mm in diameter) used in the same way as filter paper in the funnel of a vacuum flask. They retain traces of organic compounds present in large volumes of aqueous solutions, something that is not possible with the cartridges described above. In such a case, there is a concentration effect (the enrichment factor can be as high as 100, which facilitates the analysis by increasing the detection limits).

Typical uses are as follows: nonpolar compounds are isolated from a polar matrix with a cartridge or a disc of C-18 derivatized silica gel; conversely, polar compounds are isolated from a nonpolar matrix with a cartridge filled with a normal phase. Charged species can be isolated on an ion exchange phase. These three situations are well adapted to analysis by reverse-phase, normal-phase, and ion-exchange HPLC.

This sort of chemical filter is single-use only. The process may be automated. In general, it is better for hydrophobic, or nonpolar, compounds rather than for ionic substances.

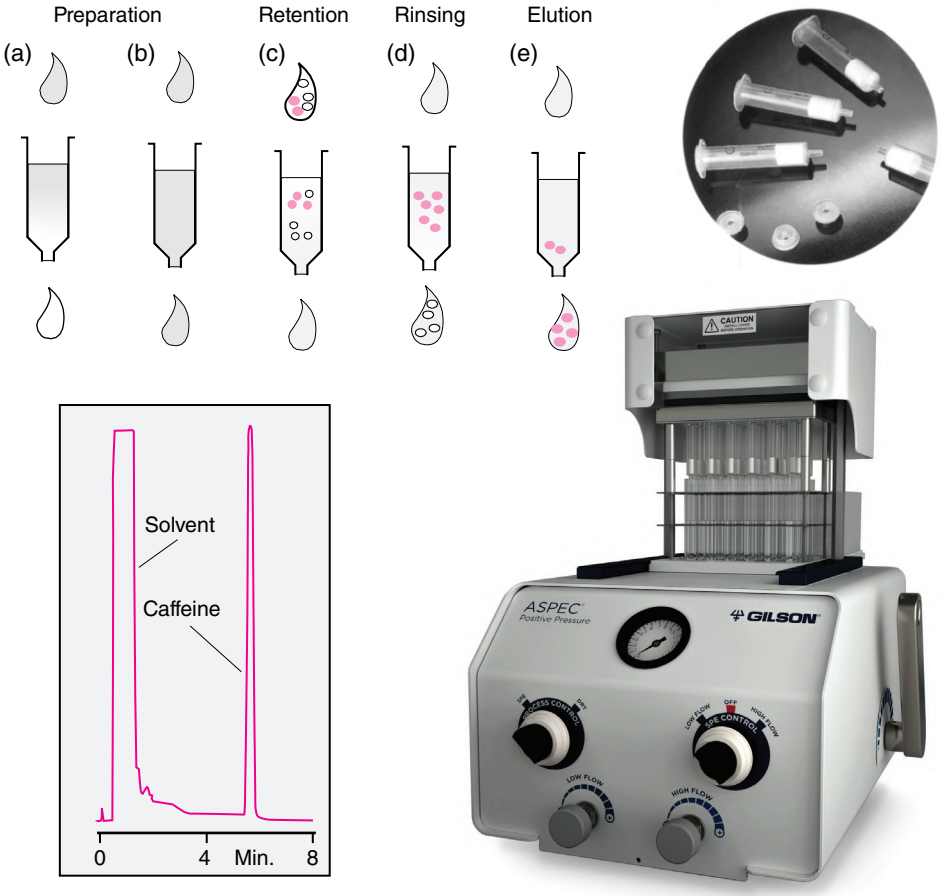


Figure 21.2 Solid phase extraction. The separation of an analyte from the matrix. In this example, the analyte is the only compound retained on the cartridge. Note the following steps: (a), (b) activation and rinsing of the adsorbent prior to use; (c) introduction of a known volume of the sample; (d) elimination of interferences by rinsing; (e) desorption and recovery of the analyte by percolation. On the right, chromatogram (GC) obtained following treatment of 1 ml of coffee using this sequence. Below, on the left, SPE cartridges with reservoirs of 1–3 ml; on the right, automated sample preparation (Aspec model PPM. SoyrceL Reproduced courtesy of Gilson).

Two specific points require attention. During the extraction step, the breakthrough volume must not be reached, as beyond this amount the compound to be extracted will no longer be trapped by the cartridge. Elsewhere, during the percolation step, the recovery of the compound adsorbed is rarely complete. If the measuring technique uses detection by mass spectrometry, this quantification problem can be resolved, prior to extraction, by taking a given volume of the sample and adding to it a known quantity of this same compound

labelled with a stable isotope. Since the behaviour of the two forms of the compound is the same during extraction, the comparison of their respective signals will permit the establishment of their ratios and consequently, the concentration of the unknown. This is the equivalent of the method of internal standards using a standard whose relative response coefficient is 1 (see Chapter 1, Section 1.1.6).

A current research theme concerns the preparation of adsorbents having a marked affinity for a specific type of compound. For this, polymers are synthesized with recognition sites in the form of surface deformations that are appropriate for trapping the target molecules. The principle resembles that of immuno-extraction, which is even more selective, as explained in the next section.

In conventional liquid/liquid extraction, the analyte is diluted in a large volume of solvent. Consequently, when the extracted solution is concentrated, so too are the nonvolatile impurities contained in the solvent. This frequently makes the method unacceptable unless ultrapure solvents are used. For example, if 1 mg of analyte is mixed with 1 ml of 99.9% pure solvent, the latter brings with it a mass of added contaminants equal to that of the analyte.

21.3 IMMUNO-AFFINITY EXTRACTION

Retention of the analytes with classic SPE cartridges is based upon hydrophobic interactions. This method may lack selectivity if the matrix containing trace amounts of the analyte also contains interfering compounds at much greater concentration. As the results could be distorted, extraction materials have been developed to isolate the analyte alone. This molecular recognition is based upon interactions of the antigen/antibody type (see Chapter 17, Section 17.5).

By using covalent bonding to fix an antibody on an appropriate solid support, its surface structure will be modified to selectively retain the corresponding antigen (Figure 21.3).

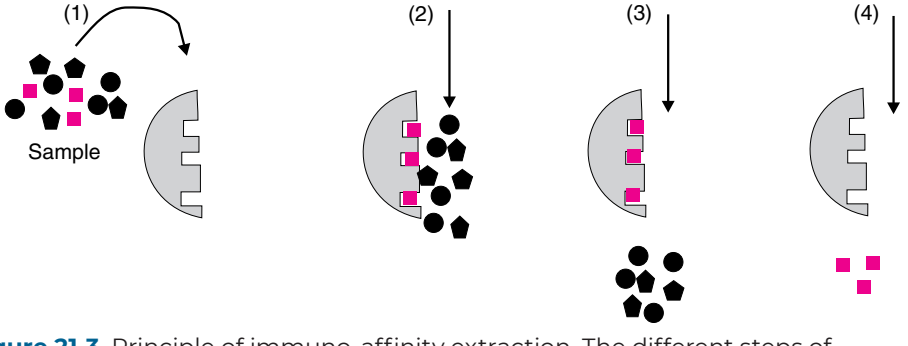


Figure 21.3 Principle of immuno-affinity extraction. The different steps of a classical procedure on a cartridge containing the immobilized antibodies adapted to the square molecules as depicted on this illustration. After the fixing step (2), rinsing (3) and then elution (4) of the analyte are carried out with an organic solvent.

This principle has long been used in the medical field and is now being applied for small organic molecules. There are extraction cartridges containing antibodies known as anti-atrazine, anti-simazine or anti-isoproturon (triazine and phenylurea family). They are used to extract the herbicides of the same names

As already stated above, extraction yield never attains 100%. For immuno-extraction, there can be an insufficient number of active sites. It is therefore indispensable to introduce a known amount of a tracer.

21.4 MICRO-EXTRACTION PROCESSES

21.4.1 Solid-Phase Micro-Extraction (SPME)

Two devices are used for this purpose. The first has a specific syringe, in which a short fibre, affixed to the end of the piston, can be exposed by the needle or can be retracted. This porous, silica fibre coated with an appropriate adsorbent material, such as a C-18 silica gel, is introduced for several minutes into the solution to be extracted (Figure 21.4a). The second device is composed of a small magnetized bar, also coated with an adsorbent, which is stirred in the solution (*Stir Bar Sorption Extraction, SBSE*). In both systems, the extracted compounds are then desorbed either with a solvent (HPLC analysis) or with heat (SBSE-GC-MS). Fibres or magnets can be used several times.

21.4.2 Liquid-Phase Micro-Extraction (LPME)

Liquid/liquid micro-extraction consists in immersing a microsyringe, filled with a very pure solvent, such as hexane, into an aqueous solution containing the products to be extracted (Figure 21.4b). At the end of the needle, a droplet of this solvent appears, which will then absorb dissolved compounds in an aqueous medium for several

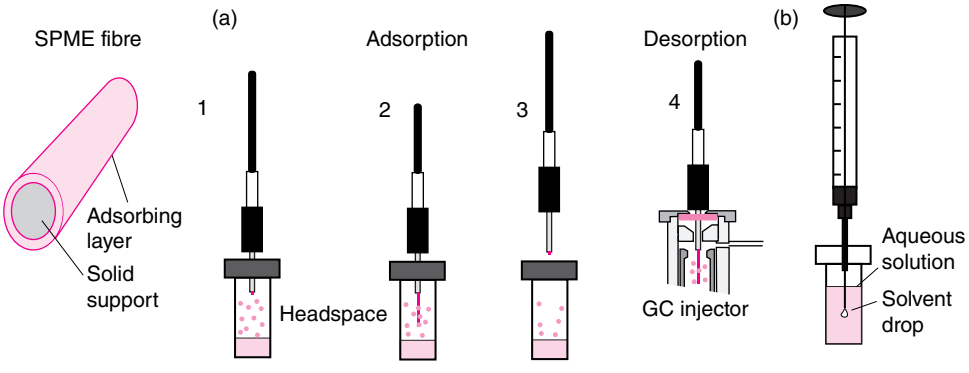


Figure 21.4 Micro-extraction procedures. (a) Micro-extraction using an adsorption fibre; (b) micro-extraction with a single drop of organic solvent.

minutes. Then, this droplet will be suctioned back into the syringe before being injected into a GC instrument.

The purification of small quantities of aqueous solutions traditionally performed using a separatory funnel can be carried out advantageously on a small column containing a porous, chemically inert material that is strongly water adsorbent. The aqueous sample is first added to the column (whose size must be such that all of the solution can be absorbed). At this stage, the water diffuses into the inert material and behaves as an immobile film in the aqueous phase. This enables the neutralization and extraction of the aqueous phase with a nonmiscible solvent.

21.5 GAS EXTRACTION ON A CARTRIDGE OR A DISC

To measure low concentrations of molecular compounds present in gaseous samples in air (industrial hygiene, environmental analyses), the best method consists in trapping them by passing a known volume of the sample through a short tube of adsorbent material or through a disposable SPE cartridge (Figure 21.5).

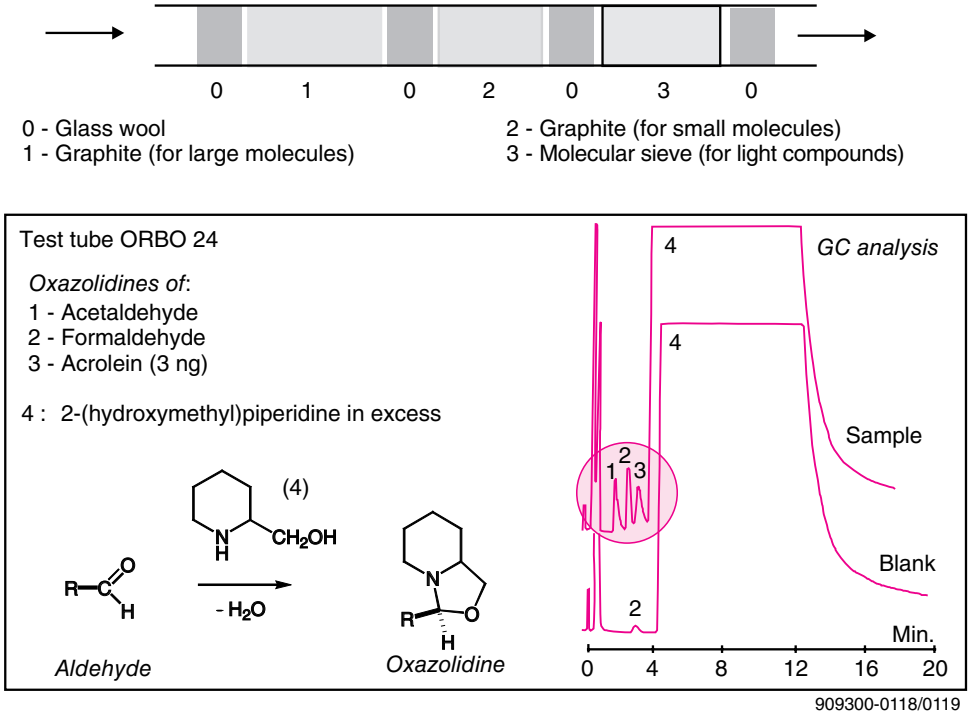


Figure 21.5 Gas extraction. Principle of a gas/solid extraction column. A chemical reaction used to detect an aldehyde by derivatization (test of a polluted atmosphere). (Source: Supelco Inc.)

The composition of adsorbents is variable: molecular sieves, graphite-based carbon black, and organic polymers with functional groups. A given tube can be composed of a series of several adsorbents. A pump is used to suction a preprogrammed volume of gas, as a function of the capacity of the trap (flow rate of 0.1–1 l/min).

The adsorbed compounds are recovered either through extraction by means of solvents (often using carbon disulfide) or by thermal desorption using a carrier gas (which avoids analyte dilution).

This technique can be adapted to gas chromatography. The trap is inserted into a special oven and heated with a temperature gradient able to attain 350°C in a few seconds. The desorbed compounds are directly transferred to the GC injector. As an alternative to these special ovens, there exist extraction tubes of a sufficiently small diameter that can be directly introduced into a modified GC injector.

The recovery of the compounds is considered to be satisfactory when it attains 60%, but it is often almost quantitative with this device.

Purge and trap

The same extraction tube filled with several layers of adsorbents enables the retention of a broad range of molecules with different polarities (Figure 21.5). Each layer protects the next, which is more active. The first layer serves to purge the sample of heavy compounds. Lighter compounds and gases are adsorbed by the subsequent layers. To retrieve (trap) the molecules, the sweeping gas circuit is reversed, so that the high molecular weight compounds retained at the entrance of the tube are eliminated first (see Section 21.6).

A new range of problems has appeared with the increasing demand for trace analyses (more than 50% of all samples treated): adsorption on the walls of the containers, action of atmospheric oxygen, evaporation of the analytes, etc. As a consequence, there is a preference for methods that lead to a concentration effect with a short extraction time.

The same principles are applied for developing *detector tubes*, which contain, along with the adsorbent, a reagent that causes a change in colour of the tube contents. This is a simple method, without extraction, to monitor contaminants without analysis. Breathalyser tests for alcohol are found in this category. There now exist many dosimeter badges used in industrial hygiene for pollution control in workplaces or in the environment. The circulation of air in contact with the badge is affected naturally and not with a pump. The flexibility of this method is due to the wide choice of adsorbent phases that permit selectivity of extraction. It may be necessary to stabilize some molecules, which may decompose around the adsorbent, by transforming them into a specific derivative by a reagent mixed with the adsorbent.

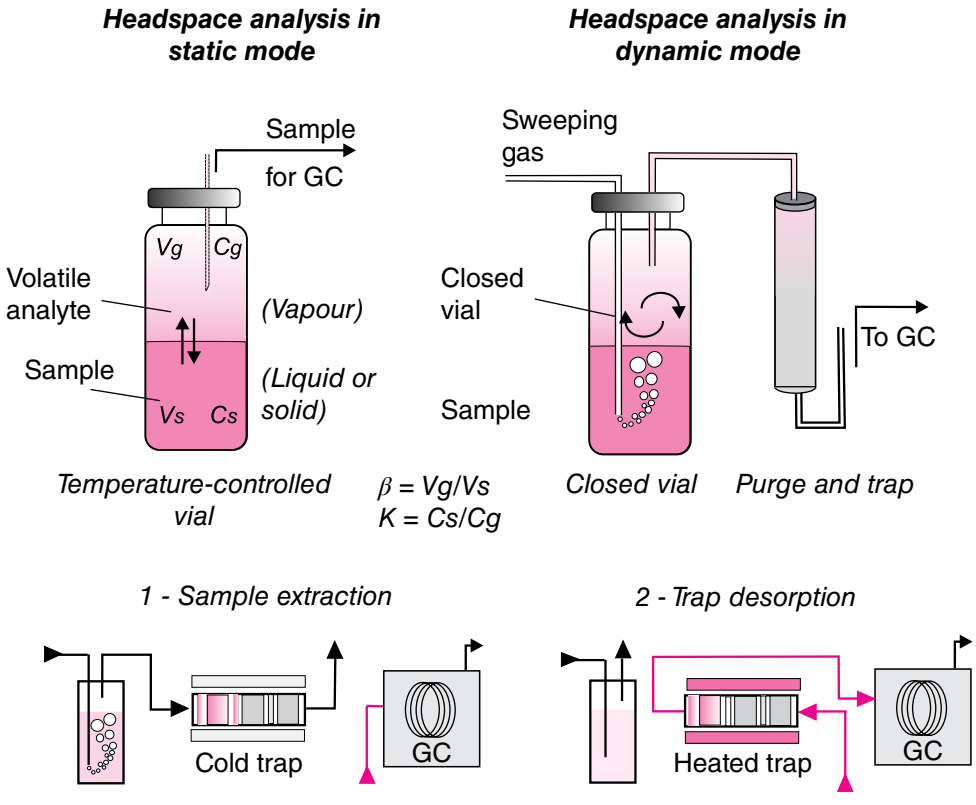
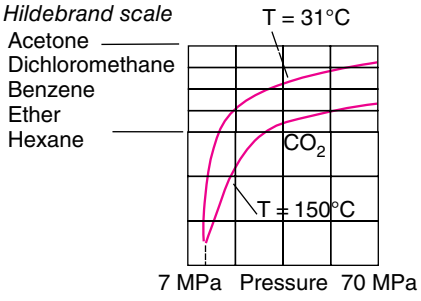


Figure 21.6 Headspace analysis in static mode and in dynamic mode. In static mode, the sample is placed in a pressurized bottle. When the thermodynamic equilibrium is reached, the sampling for GC analysis takes place. In dynamic mode, a carrier gas first entrains volatile compounds towards a trap (purge and trap), then the trap is thermally desorbed to release the compounds, which have been concentrated for GC analysis.

21.6 HEADSPACE

Headspace is a sampling device used in tandem with a GC instrument (Figure 21.6). It is reserved for the analysis of volatile compounds present in a matrix that cannot be directly analysed by chromatography. Basic principles of headspace analysis are illustrated by the two following variants:

- *Static mode*: the sample (solid or liquid matrix) is placed in a glass container capped with a septum such that the sample occupies only a part of the container's volume. After thermodynamic equilibrium is reached between the phases



Supercritical fluid properties

Fluid	Critical point (°C)	Dipole moment (Debye)
CO_2	32	0
N_2O	37	0.2
$CHClF_2$	96	1.4

Figure 21.7 Supercritical fluid extraction. Comparison of the eluting powers of CO_2 with respect to the usual solvents (Hildebrand scale) as a function of temperature and pressure. Extraction in the supercritical phase is an automated method, which makes it a profitable investment as long as the number of samples is high.

(½–1 hour), a sample of the vapour is taken. Under these conditions, the quantity of each volatile compound present in the headspace will be proportional to its concentration in the matrix. After calibration (using internal or external standards), it is possible to match the real concentrations in the sample with those of the vapours injected in the gas chromatograph.

- *Dynamic mode*: a special open container, called a sampler, is used, containing the aqueous phase to be extracted. A carrier gas such as helium is bubbled through it in order to entrain the volatile components toward a purge-and-trap instrument (see above) where they are adsorbed and concentrated (Figure 21.7). The procedure continues with a thermal desorption of the trap using reversed gas flow for injection in the chromatograph. This principle is semiquantitative.

This method is quite reliable for repetitive analyses involving stable matrices. However, if the matrix composition fluctuates, then this disturbs the equilibrium factors and reduces the precision of the results, owing to an irregular calibration. In this case, cartridge-based extraction would be preferred.

21.7 SUPERCRITICAL PHASE EXTRACTION (SPE)

Extraction using a fluid in the supercritical state (CO_2, N_2O or $CHClF_2$ [Freon 22]) is a well-known process employed in the food industry (see Chapter 6, Section 6.1). Analytical extractors operate on the same principle. They incorporate a very resistant tubular container in which the sample is placed (in solid or semisolid form) with the fluid in the supercritical state. There are two modes of operation:

- *Off-line* mode consists of depressurising the supercritical fluid following extraction. By returning to the gaseous state, it leaves the analytes in a concentrated form

with the undesirable impurities on the internal wall of the extraction vessel. The treatment continues with selective extraction using conventional solvents or on a solid-phase cartridge to combine extraction and selectivity.

- *On-line* mode consists of performing the direct analysis of the extract, while it is still under pressure, by passing it through a chromatographic set-up (SFC or HPLC). This process is applied only to those compounds for which matrix-related interferences are not likely to occur.

The method regulates four parameters that affect selectivity: pressure, temperature, extraction time, and choice of modifiers. It is also possible to change the solvating character of the fluid by the introduction of an organic modifier (e.g. methanol, acetone), since the polarity of carbon dioxide gas alone, in the supercritical state, is comparable to that of hexane (at 100 atm. and 35°C). Nonetheless, to separate the analyte from the matrix requires knowledge of the solubility and transfer rate of the solute in the solvent as well as of the physical and chemical interactions between solvent and matrix (Figure 21.7).

The use of microwaves combined with extraction using a conventional solvent in a pressurized container is another very efficient process for the rapid treatment of samples. It may be used as a substitute for supercritical fluids.

21.8 MICROWAVE REACTORS

Atomic absorption spectrometry (AAS), atomic emission spectrometry or mass spectrometry with plasma sources (ICP-AES and ICP-MS) are techniques that have become commonly used, requiring putting samples in solution. Many approaches have been created to decompose inorganic samples or those containing organic materials. The treatment, known as *mineralization*, is done using dry processes (such as oven heating) or by wet treatment (treatment with acids). Classic processes involve fusion with lithium metaborate or attack by a mixture of acids HF–$HClO_4$.

Classic handling in open containers, under a fume hood in an acidic vapour atmosphere, with the ever-present risk of cross-contamination, is a long and difficult operation that fundamentally has changed little over the past century. Beyond the practical problems involved, some sample matrices are quite impossible to treat under these conditions (refractory or volatile materials, certain minerals, charcoal and heavy oils).

In this field, real progress has appeared with the arrival of microwave digesters (Figure 21.8). Microwave digesters have a 50-l oven, containing a set of small Teflon autoclaves with a volume of a few ml, in order to contact samples with the attack solution (necessary for carbonization or oxidation). Under the effect of microwaves, temperatures reached by the acidic solutions under pressure quickly exceed the boiling points observed under normal conditions, so much so that the attack process is greatly



Figure 21.8 Microwave digester. Detailed view of the 15-count carrousel and small autoclaves. Material for microwave digestion (reproduction with authorization of Anton Parr).

accelerated. Under such conditions, digestion by sulfuric or perchloric acids, hydrolysis or Kjeldahl mineralization in oxidizing media can be performed.

21.9 ON-LINE ANALYSERS

Almost all laboratory instrumental methods have been adapted for analytical procedures in the field, from pH measurement to NMR analyses. These instruments are generally less versatile than their laboratory counterparts. Dedicated to specific measurements and suitable for harsh environments or hazardous areas, they must be robust in design. The use of a selective sensor plunged into the centre of a product to be analysed is not always possible. A small fraction of the product is thus sampled for suitable treatment and measurement.

KEY POINTS OF THE CHAPTER

1. For many physico-chemical techniques, measuring an analyte in a sample involves prior treatment. This is the key stage for any good chemical analysis.
2. The purpose is to improve the selectivity of the measurement and/or to concentrate analytes in order to reach the domain of sensitivity of the detector, while taking into account various objectives, such as the use of small volumes of organic solvents (‘green’ approach).

3. This time-consuming stage may include a succession of operations. The on-line approach consists of automatically following the treatment with an analysis.
4. Analytes whose concentration are at the trace level can only be measured after their extraction from the sample matrix, which may contain hundreds of other interfering compounds.
5. Upstream from chromatography analyses, extraction methods are omnipresent, including liquid/liquid extraction (LPE or LPME) or extraction on a column or on fibre (SPE, SPME, SPDE) followed by a thermal desorption.
6. Moreover, in two-dimensional chromatography, the first dimension corresponds to the preparation stage of the analyte(s).
7. The headspace technique, reserved for volatile compounds (GC) only, exists in static or dynamic mode. The purge-and-trap system involves a thermal desorption. Static mode is reserved for volatile compounds and dynamic mode for heavier compounds. The quantification is often based on the multiple addition method using certified references.
8. The measurement of target elements (for ICP/MS and AA) contained in solid matrices involves their solubilization. These techniques include the destruction of the matrix using microwaves as well as dry or wet mineralization.

Chapter 22

Basic Statistical Parameters

In chemical analysis, as in many other sciences, statistical methods are essential. The calibration curve constitutes an everyday application, just as an analytical result can only be ascertained if an estimate of the possible error has been considered. Once a measurement has been repeated, statistical analysis becomes possible. However, the laws of sampling and tests based upon hypotheses must be understood in order to avoid worthless conclusions, or to set up quality assurance. Systematic

errors (user-based, instrumental, etc.) or gross errors that lead to results beyond reasonable limits do not enter into consideration in this chapter. For the tests most frequently encountered in chemistry, only indeterminate errors are considered here.

22.1 CENTRAL VALUE, ACCURACY, AND RELIABILITY OF A SET OF MEASUREMENTS

When *n* measurements are repeated upon the same sample (in the chemical sense of the term) and particularly to a high degree of precision, it is frequently found that individual values differ slightly within the group. In these cases, a preference for an estimation of the final result based upon the arithmetic mean of these *n* measurements, is expressed by the *central value* or *mean* $\bar{x}$. This mean is accepted as being better than any individual measurement taken at random:

$$\bar{x} = \frac{x_1 + x_2 + \ldots + x_n}{n} = \frac{\sum x_i}{n} \tag{22.1}$$

An alternative to the mean or central value, though more rarely used in chemical analysis, consists of determining the median: after having classified the values in increasing order, this is the result situated in the middle of the series, except when the number of the series is even, in which case the average of the two middle numbers is taken. The median has the advantage of not giving any particular weight to any abnormal value. For example if the values are 12.01, 12.03, 12.05, and 12.68, the arithmetic mean is 12.19 and the median 12.04, a value certainly better than the mean as it does not take into account the last value in the example given, which appears to be abnormal (dispersion of 0.67). The median is another type of statistic (nonparametric statistical method, see Section 22.8).

The central value and the individual measurement are random variables, since they are impossible to know in advance. However, it is not by repeating the same measurement a great many times and finding values with little difference between them that the central value is rendered close to the *exact value*, equally called the *true value* x_0 (at this stage unknown), because of the possible existence of systematic errors.

Each measurement *x* must be considered as the sum of the true value x_0 and an *absolute experimental error* ε. The absolute error ε_i of measurement *i* is thus expressed as:

$$\varepsilon_i = x_i - x_0 \tag{22.2}$$

If *n* becomes very large, as in the case of a statistical population, the mean $\bar{x}$ becomes the *true mean* μ and approaches the *true value* x_0 in the absence of systematic errors. This assumes that the measurements follow normal, or Gaussian, distribution (see Figure 22.2).

Table 22.1 Examples of results from an analysis grouping the various parameters of Eqs. (22.1) and (22.2) (the true value is $x_0 = 20$)

Chemist	**1**	**2**	**3**	**4**
Measurement 1	20.16	19.76	20.38	20.08
Measurement 2	20.22	20.28	19.58	19.96
Measurement 3	20.18	20.04	19.38	20.04
Measurement 4	20.2	19.6	20.1	19.94
Measurement 5	20.24	20.42	19.56	20.08
Mean $\bar{x}$	20.2	20.02	19.8	20.02
Median	20.20	20.04	19.58	20.04
Accuracy ε	0.2	0.02	0.2	0.02
Relative error E_R (Equation (22.4))	0.01 or 1% or 10^4 ppm	1×10^{-3} or 0.1% or 1,000 ppm	0.01 or 1% or 10^4 ppm	1×10^{-3} or 0.1% or 1,000 ppm
Variance (s^2)	7.84×10^{-4}	0.1183	0.1772	3.6×10^{-5}
Standard deviation (s)	**0.028**	**0.344**	**0.421**	**0.006**
Comments on the results	Precise, not accurate	Not precise, accurate	Not precise, not accurate	Precise, accurate

If the *true value* x_0 is known (e.g. analysis of a standard of perfectly known composition), the *accuracy* of the result or of the method used is characterized by the *total error* ε calculated from the true mean as follows (Table 22.1, Figure 22.1).

$$\varepsilon = \mu - x_0 \tag{22.3}$$

The relative error E_R of a measurement (or of the *mean value*) corresponds to the ratio of the absolute value of the deviation corresponding to $|\varepsilon_i|$ (or $|\varepsilon|$), over the true value. E_R can be expressed as a percentage or in ppm.

$$E_R = \frac{|x - x_0|}{x_0} \quad (x \text{ for } x_i \text{ or } \bar{x}) \tag{22.4}$$

If the *true value* x_0 is not known, which is usually the case in chemical analysis, then the *experimental error* of measurement i, i.e. e_i, is calculated by replacing x_0 with the mean $\bar{x}$ in Eq. (22.2):

$$e_i = x_i - \bar{x} \tag{22.5}$$

e_i represents the algebraic difference between the mean and the ith measurement. The *mean experimental error* $\bar{d}$ or mean deviation, calculated over n measurements allows an appreciation of the *precision*:

$$\bar{d} = \frac{\sum |x_i - \bar{x}|}{n} \tag{22.6}$$

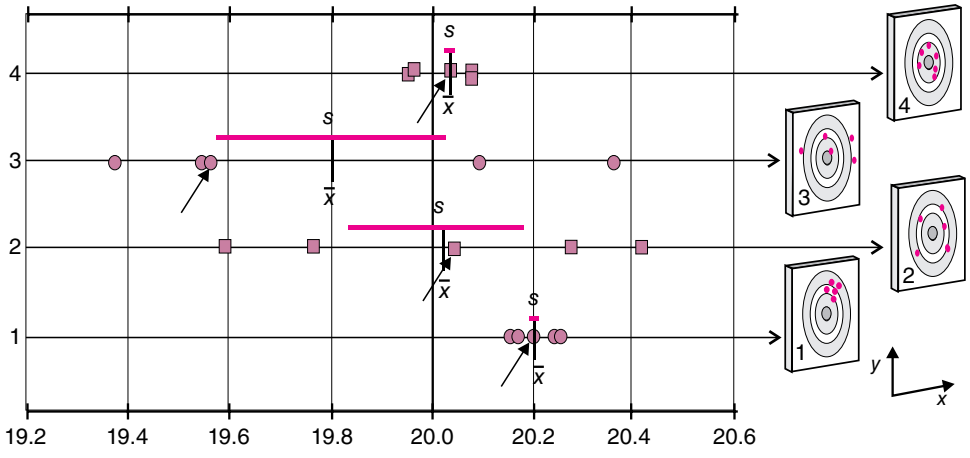


Figure 22.1 Graphic illustration of the data from Table 22.1. To illustrate accuracy and precision, the standard deviations have been calculated using Eq. (22.8). Note the differences between the arithmetic means, indicated on the graph by a short vertical bar, and the corresponding median values, which are shown by arrows. For the results from chemists 3 and 4, the difference is fairly large. Chemist 1 has very probably committed a systematic error. On the right, a classic illustration of precision and accuracy depicted with the aid of a target. This image is less simple than it would appear because there remains an uncertainty for both x and y.

In the last expression, absolute values of the experimental error are taken, because if algebraic values were used this parameter $\bar{d}$ would rapidly tend towards 0 as n increases.

22.2 VARIANCE AND STANDARD DEVIATION

The above precision value (*mean deviation*) however has the drawback that it cannot be interpreted statistically, as both large and small individual deviations, though not being equally probable, have the same weight. The summing of the squares of the differences is preferred, which leads to the most widely used definition for *precision* (reproducibility) in statistics, known as the *variance* s^2, which is estimated for a set of n measurements by:

$$s^2 = \frac{\Sigma(x_i - \bar{x})^2}{n-1} \tag{22.7}$$

The square root of the *variance* is the *standard deviation*, represented either by s (when the number n of measurements is small) or by σ (when a large statistical population of values is available). The standard deviation is a dispersion indicator, expressed in the same units as x.

Calculators and spreadsheets possess the corresponding functions (for an 'infinite' number of measurements, s is replaced by σ and $n - 1$ by n):

$$s = \sqrt{s^2} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} \tag{22.8}$$

To compare results or to express the uncertainty of a method, s is often expressed in a relative manner. Calculations are made therefore of the *relative standard deviation* (or *RSD*), also called the *coefficient of variation* (*CV*) and most often expressed as a percentage:

$$CV = 100 \times \frac{s}{\bar{x}} \tag{22.9}$$

When the result of an analysis arises from a calculation involving many experimental values, each having its own standard deviation, there will consequently be a propagation of errors. The precision of the result is calculated using simple equations that are found in most introductory textbooks on statistics.

22.3 RANDOM OR INDETERMINATE ERRORS

In the absence of systematic errors, accidental errors 'due to chance' exist that cannot be controlled because they are indeterminate. Their direction and amplitude vary in a nonreproducible manner from one measurement to another. A mathematical analysis of the *error curve* leads to the conclusion that the arithmetic mean $\bar{x}$ of the individual values is the best estimate of the *true mean* μ (Figure 22.2).

The symmetry and appearance of this curve show that:

- There are an equal number of errors, positive and negative, with respect to the central value.
- Small errors are more frequent than large errors.
- The most frequently encountered value is the *mean* μ (without error).

The normal distribution law (bell-shaped Gaussian curve) is the mathematical model which best represents the distribution of random or indeterminate errors due to chance:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \tag{22.10}$$

To make this expression universal, the origin of the x-axis is taken to be the true mean μ of the population, and its standard deviation σ is the unit of measurement (Figure 22.2).

If $X = \frac{x-\mu}{\sigma}$, Eq. (22.10) becomes:

$$f(X) = \frac{1}{\sqrt{2\pi}} exp\left[-\frac{X^2}{2}\right] \tag{22.11}$$

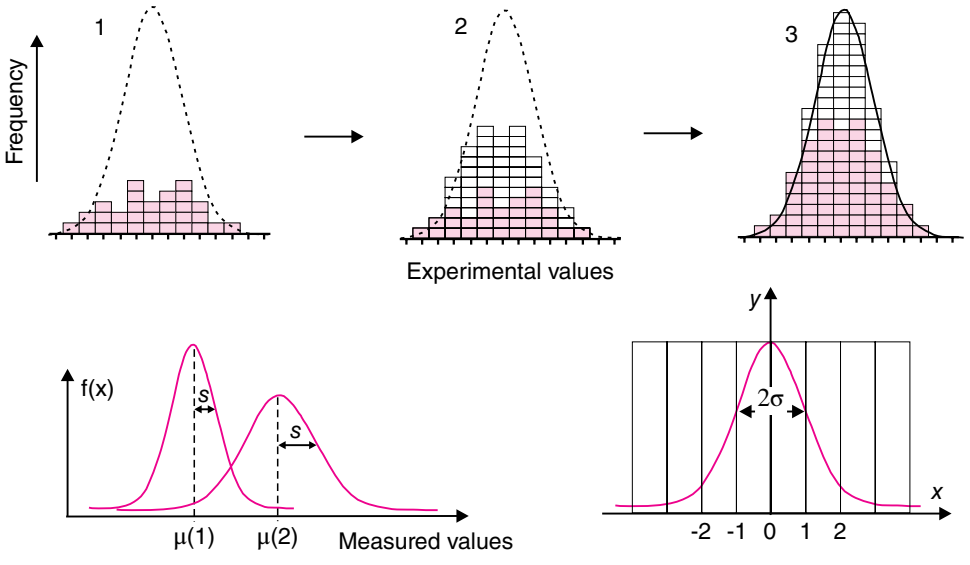


Figure 22.2 Gaussian curves. When the number of measurements increases and if the interval determining the group is narrow, then the shape of the graph (measurement/frequency) will take the form of a Gaussian curve (normal distribution law). Below, two series of results centred on two different means. If the number of measurements is very small, it is not possible to estimate the true mean. At the bottom right, a reduced form of the Gaussian distribution is shown.

If the number of measurements is not very large, it is important to know their distribution. The curve obtained yields information about the reliability *Re* of the results. The reliability of a mean, as an estimate of the true value, increases with the square root of the number of measurements *n*:

$$\mathrm{Re} = K\sqrt{n} \tag{22.12}$$

Thus, when the number of measurements n_1 increases to a larger number n_2, the reliability *Re* is improved by a factor *k* such that:

$$k = \sqrt{\frac{n_2}{n_1}} \tag{22.13}$$

The distribution function is the integral of the function $f(X)$. This function is such that 95.4% of the area lies within an interval of width $\pm 2\sigma$ around the central value. Therefore, the chances are 95.4% that the error associated with a given measurement will be contained within the interval $\pm 2\sigma$. A high value for the standard

deviation σ signifies a broadened error curve. If the number of values is small (a few units), s must be used instead to estimate σ (Figure 22.2). The greater n is, the better will be the match.

Thus, s and $\bar{x}$ are estimates of σ and of the true mean μ, which relate to the total population.

In practice, it is not possible to know either the true mean μ or the standard deviation σ, as there will be a limited number of values n. Thus, s must be calculated. Elsewhere, when a large number of repetitive measurements concerning the same sample is available, the Chi-square test (χ^2 test) is used to ascertain if the frequency distribution (i.e. the number of times a value is recorded) differs in a significant manner from that of a population that follows the normal distribution law. Without using proper software, this test of normality involves a lengthy calculation process.

22.4 CONFIDENCE INTERVAL OF THE MEAN

When the number of the measurements n is small (for example, if n is between 4 and 15), and if there are no systematic errors, the true mean μ can be quite different from the central value $\bar{x}$. An estimation of the true mean must then be made by calculating a confidence interval to which we assign a probability (for example 95%) that the true value will be found. This operation leads to a risk of error.

The *confidence interval*, around the mean $\bar{x}$, that includes the true mean μ (or x_0 in the absence of all systematic errors) is given by:

$$\bar{x} - \frac{t \cdot s}{\sqrt{n}} \leq \mu \leq \bar{x} + \frac{t \cdot s}{\sqrt{n}} \tag{22.14}$$

In Eq. (22.14), Student's coefficient t is a statistical factor that depends on n and on the chosen confidence level. These values are displayed in Table 22.2. The greater n is, then the narrower this interval will be, with s being the standard deviation of the series of measurements.

Student's t test can equally lead to an adjustment of the number of measurements to be conducted in order to attain a result with the preselected level of confidence.

If, as well as the mean $\bar{x}$, the true value x_0 (or the true mean μ) is also known, Eq. (22.14) will permit the calculation of the value of t, according to the degree of confidence chosen (Eq. (22.15) if $n < 20$; otherwise, Eq. (22.16)). A value of t larger than that indicated in Table 22.2, on the line corresponding to the value of n, will be due to a systematic error.

$$t = \frac{|\bar{x} - \mu|}{s} \sqrt{n} \tag{22.15}$$

Table 22.2 Values of the Student's two-sided confidence coefficient *t* (calculation of Student's distribution).

N	90% confidence level	95% confidence level	99% confidence level
2	6.31	12.71	63.66
3	2.92	4.30	9.93
4	2.35	3.18	5.84
5	2.13	2.78	4.60
6	2.02	2.57	4.03
7	1.94	2.45	3.71
8	1.90	2.36	3.50
9	1.86	2.31	3.36
10	1.83	2.26	3.25
11	1.81	2.23	3.17
12	1.80	2.20	3.11
15	1.76	2.14	2.98
20	1.73	2.09	2.86
30	1.70	2.05	2.76
60	1.67	2.00	2.66
120	1.66	1.98	2.62
9,999	1.65	1.96	2.58

or

$$t = \frac{|\bar{x} - x_0|}{\sigma}\sqrt{n} \tag{22.16}$$

This confidence interval is also calculated when a method of measurement is tested with a sample whose corresponding true value x_0 is known, though it remains to be seen if the latter is situated in the calculated confidence interval.

In the formula that leads to the calculation of confidence intervals, the *standard error of the mean*, or the uncertainty of the standard deviation, may appear, with the following relationship:

$$s_{\bar{x}} = \pm\frac{s}{\sqrt{n}} \tag{22.17}$$

22.5 COMPARISON OF RESULTS – PARAMETRIC TESTS

When it is necessary to compare the results of two methods, of two instruments applying the same method of analysis, or of two laboratories on the same sample, it is essential to refer to statistical tests. Two main families exist. The first, known as

parametric tests, suppose that data are distributed according to a normal law (on which the values in Student's table are based), while *nonparametric tests* are based on so-called *robust statistics*, i.e. less sensitive to abnormal values. In analytical chemistry, it is not often that large quantities of data are collected. Therefore, statistical tests must be applied wisely, since they display a certain risk level, traditionally chosen as a percentage, for example 10%, 5% or 1%.

22.5.1 Comparison of Two Variances, Fisher–Snedecor Law

The comparison of the standard deviations s_1 and s_2 of two different sets of results is known as a test for *equality* of *variance*. In this test, the *F* factor, which is the ratio of the two variances, is calculated such that $F > 1$:

$$F = \frac{s_1^2}{s_2^2} \tag{22.18}$$

The *null hypothesis* (statistical terminology) states that if there are no significant differences in the variances, then the ratio must be close to 1. Reference should therefore be made to the Fisher–Snedecor values of *F*, established for a variable number of observations (Table 22.3). If the calculated value exceeds that found in the table, the means are considered to be significantly different. Since the variability s_1^2 is greater than s_2^2, the second series of measurements is therefore the more precise one.

The adaptation of this test to robust statistics (nonparametric tests) requires simply to calculate the ratio $F = R_1/R_2$ of the dispersions (the difference between the two extreme measurements) of the two series under comparison.

Table 22.3 Summary of the *F* threshold values established for a confidence level of 95%.

Number of measurements (denominator)	Number of measurements (numerator of the ratio *F*)						
	3	4	5	6	7	10	100
3	19.00	19.16	19.25	19.30	19.33	19.38	19.50
4	9.55	9.28	9.12	9.01	8.94	8.81	8.53
5	6.94	6.59	6.39	6.26	6.16	6.00	5.63
6	5.79	5.41	5.19	5.05	4.95	4.78	4.36
7	5.14	4.76	4.53	4.39	4.28	4.10	3.67
10	4.26	3.86	3.63	3.48	3.37	3.18	2.71
100	2.99	2.60	2.37	2.21	2.09	1.88	1.00

Amongst the examples in Table 22.1, there is no significant difference between the results from chemists 2 and 3 at the 95% confidence level. For them, $F = 1.5$, and Table 22.3 shows that the two data sets are significantly different with five measurements by both chemists only if $F \geq 6.39$.

22.5.2 Comparison of Two Experimental Means, $\bar{x}_1$ and $\bar{x}_2$

Sometimes it is desirable to compare the means obtained from two series of measurements to determine if they are significantly different, bearing in mind that the true value is unknown. Initially, it must be determined whether there is any significant difference with respect to the precision of the two means (see Section 22.5.1). Next, a calculation using Eq. (22.19) for the *pooled* standard deviation s_p is performed. Then, the corresponding value of t is calculated using Eq. (22.20). It is compared to the value in the table for $n = n_1 + n_2 - 2$ measurements and for the chosen confidence level. If the value of t in the table is greater than the value calculated, then it can be concluded that these two means are not significantly different.

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \tag{22.19}$$

$$t = \frac{|\bar{x}_1 - \bar{x}_2|}{s_p}\sqrt{\frac{n_1 n_2}{n_1 + n_2}} \tag{22.20}$$

22.5.3 Estimation of the Detection Limit of an Analyte

Equations (22.19) and (22.20) are useful for estimating the smallest concentration of an analyte that can be detected, but not quantified, with a chosen level of confidence. One of the two means ($\bar{x}_2$ for example) is considered to be the result of a set of measurements made on the analytical blank. It will be noted as $\bar{x}_b$ with a standard deviation of s_b. If $n_1 = 1$, then Eq. (22.19) simplifies to $s_p = s_b$, leading to Eq. (22.21) (using a t value extracted from Table 22.2 for $n_1 + n_b - 2$):

$$\Delta x = \bar{x}_1 - \bar{x}_b = \pm t \cdot s_b \sqrt{\frac{n_1 + n_b}{n_1 \cdot n_b}} \tag{22.21}$$

If six measurements are undertaken on an analytical blank, leading to a standard deviation of 0.3 μg for the measurement considered, then with Eq. (22.21) and for a confidence level of 99%, a value of 1.31 μ g will be calculated as the detection limit if only a single measurement is made and a value of 0.59 μ g if five measurements are made ($\bar{x}_b = 0$).

Empirically, it can be assumed that the instrumental detection limit corresponds to the concentration which leads to a signal whose relative intensity (central value $\bar{x}$) is twice the standard deviation calculated for 10 measurements with the analytical blank (95% confidence). The limit of quantification is always much higher.

22.6 REJECTION CRITERIA USING THE *Q* QUOTIENT (OR DIXON TEST)

Sometimes, a value within a set might appear aberrant. Although it might be tempting to reject this data point, it must be remembered that it is only abnormal in reference to a given law of probability. There exists a simple statistical criterion for conservation or rejection of this outlier value. This is Dixon's test, which consists of calculating the following ratio (on the condition that there are at least seven measurements):

$$Q = \frac{|\text{value in question} \quad \text{its closest neighbour}|}{\text{largest value} - \text{smallest value}} \tag{22.22}$$

Q calculated in this manner is compared with a table of critical values of Q as a function of the number of data (Table 22.4). If $Q_{\text{calculated}}$ is greater than Q_{critical}, the value in question can be rejected.

Note: The ASTM standard (American Society for Testing Materials) uses a different test for rejection of an outlier, called the reduced central value $e_i = (x_i - \bar{x})$, which has its own table of characteristic critical values. Henry's curve, whose principle is described in more advanced statistical texts, also constitutes a good method for detecting outliers using a visual approach.

Table 22.4 Tabular summary of critical values for *Q* (Dixon's test)

Number of measurements, *n*	95% two-sided confidence level	99%
3	0.94	0.99
4	0.77	0.89
5	0.64	0.78
6	0.56	0.70
7	0.51	0.64
8	0.47	0.59
9	0.44	0.59
10	0.41	0.53

Source: M. Neuilly, *Techniques de l'ingénieur, Analyses et Caracterisation*, 1996.

22.7 CALIBRATION CURVES

Instrumental quantitative analysis is commonly based upon comparative methods. For example, the sample that contains the analyte and a standard that contains the same concentration of this analyte will yield identical signals, if an instrument with the same settings is used. In the majority of cases, a series of standards (or solid specimens) of known concentrations are measured, rather than a single one. To take matrix effects into account, the standard addition method may be used. Several reference values thus obtained will be plotted as figurative points on a graph where the *x*-axis corresponds to the concentration and the *y*-axis to the signal value.

Depending on the hypothesis selected and knowing that the position of each point is tarnished by an error, the calibration curve will be defined. In other words, the signals yielded by the instrument (*y*-axis) are modelled as a function of concentration (*x*-axis). A model is adopted, $Y = F(X)$, allowing an evaluation of *y*, since *x* and the function are known. Therefore, the uncertainty of the results is an accumulation of the uncertainty linked to the measurement and also to the model chosen for the function (which may be too simple or too complex). The interpretation of calibration results calls on statistical methods. Quantitative analysis software uses numerous calculation models. Here only the main results are given concerning the simple linear regression, i.e. the statistical approach most frequently encountered in quantitative analysis (Figure 22.3).

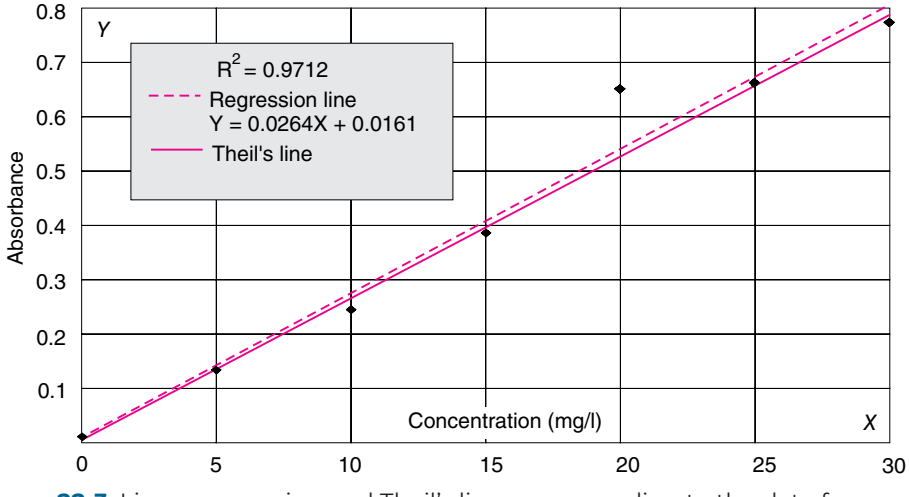


Figure 22.3 Linear regression and Theil's line corresponding to the data from Figure 22.4. Though in close proximity, the two lines lead to different results. For example, for $y = 0.5$, we get $x = 18.85$ (Theil) and $x = 18.33$ (linear regression), a difference close to 3%.

22.7.1 Simple Linear Regression

As the response of many detectors is linear for the measured variable, taking into account the differences due to experimental conditions as well as to the instrument, the goal is to find the parameters of the straight line that best fits with the observations.

What errors were made? Must all experimental points intervene with the same weight? The adjustment by the *least squares method* considers that one of the two variables is without error and that the other is subject to random fluctuations. This is the most commonly applied method. Coefficients a and b of the regression line equation $y = ax + b$, as well as the standard deviation on a and the estimate on y, are represented by a certain number of formulae found in quantitative analysis software.

Moreover, the dimensionless *Pearson correlation coefficient* R gives a measure of the reliability of the linear relationship between the x and y values. A value of +1 or –1 translates into a strong bond between the two variables. This method supposes that the errors in y follow the law of normal distribution. R^2 is the *coefficient of determination*, which conveys information about how the variations of x overlap variations in y.

$$a = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - \left(\sum x_i\right)^2} \tag{22.23}$$

$$b = \bar{y} - a\bar{x} \tag{22.24}$$

$$R = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{\sqrt{\left[n\sum x_i^2 - \left(\sum x_i\right)^2\right] \cdot \left[n\sum y_i^2 - \left(\sum y\right)^2\right]}} \tag{22.25}$$

In classic calculations, the experimental error is considered to affect the y value exclusively and not the concentration plotted on the x-axis. If this is not the case, the data points will not have the same quality to estimate the regression line. Hence came the idea of according less value to data more distant from the line. Through iterative calculations, the equation of a straight line is reached that takes account of the weighting of each point.

To compare whether two methods of analysis are highly correlated, a series of n standards of different concentrations are analysed by two pathways. Each standard is then represented by a point on a graph where the x-axis shows the results of method 1 and the y-axis the corresponding results from the second method. The correlation coefficient R is determined, and Eq. (22.26) is applied. If the t value obtained is greater than the value found in the table for $n - 2$ degrees of freedom, there is a strong correlation between the two analytical methods.

$$t = R\sqrt{\frac{n-2}{1-R^2}} \tag{22.26}$$

22.7.2 Multiple Linear Regression

Analysis by multiple linear regression leads to the prediction of a result (the *dependent variable*) when it depends upon a certain number of factors (the *independent variables*). Software allows this type of calculation in which the contribution of each factor is determined using standards. The prediction of the result is calculated by a general formula, such as Eq. (22.27), where the hypothesis is that there are three factors in x:

$$y = a + bx_1 + cx_2 + dx_3 \tag{22.27}$$

For example, the protein content in cheese, and thus its nitrogen content by percentage (result), can be established from the IR absorbance measured at different wavelengths. This method is also useful for calculating the parameters A, B, and C of the van Deemter equation (Chapter 1, Section 1.10), corresponding to experimental data.

22.8 ROBUST METHODS OR NONPARAMETRIC TESTS

The statistical tests developed above assume that the data follow a normal distribution. However, there are analytical methods whose results reveal different distributions. They are either asymmetrical or symmetrical but 'non-normal'. In some approaches, these distributions are considered to result from the superimposition of a normal distribution and abnormal points. The use of the median (see Section 22.1), rather than the arithmetic mean, is part of this alternative approach. The standard deviation is replaced by the mean deviation *MD*.

$$MD = \sqrt{\frac{\pi}{2}} \cdot \frac{\sum \left| x_i - \bar{x} \right|}{n} \tag{22.28}$$

Under such circumstances, the calibration curve can be reconsidered, as seen in the following example, which illustrates Theil's method.

To estimate the best straight line for a series of seven data points (x, y) in a colorimetric titration, the method consists first in ranking the x values in increasing order (Figure 22.4). Since Theil's method requires an even number of data points, the median value must be rejected in this example.

Then, the calculation is the following. Firstly, we calculate the slopes of three straight lines, the first of which passes through the first point and through the point immediately following the median and so on for the next two lines. The median of the three calculated slope values is taken as the term a of the linear equation. Then, the six b values are calculated from $b_i = y_i - ax_i$ and also ranked in increasing order. The median of these values will become the term b of the equation (Figures 22.3 and 22.4).

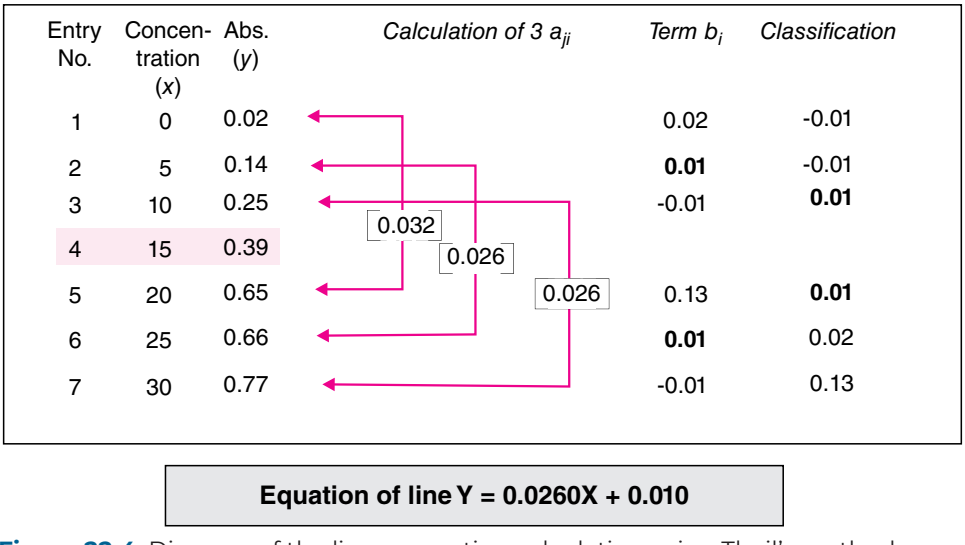


Figure 22.4 Diagram of the linear equation calculation using Theil's method ($y = ax + b$).

In chemistry this method should be more widely used when the number of measurements is too small to be sure of the normality of the distribution.

Three advantages of Theil's method should be highlighted:

- it is not assumed that all of the errors are in *y*;
- it is not assumed that the errors follow the normal distribution law;
- an aberrant data point does not affect the final parameters of the linear equation.

22.9 OPTIMIZATION WITH THE ONE-FACTOR-AT-A-TIME (OFAT) METHOD

When a measurement depends upon a signal (for example, absorbance or intensity of fluorescence) that is itself influenced by several factors, then, in general, it is customary to seek overall conditions that will lead to the maximum signal.

If the factors involved in an analysis are independent from each other, a common practice is to study the influence of each one on the result by using a simple repetitive method, as indicated in the following example.

Imagine that a value given by a sensor depends upon two independent factors *x* and *y*. After having set the factor *x* to the value x_1, the influence of factor *y* on the signal is studied (the signal is measured in the third dimension). For the value *Y*, it is

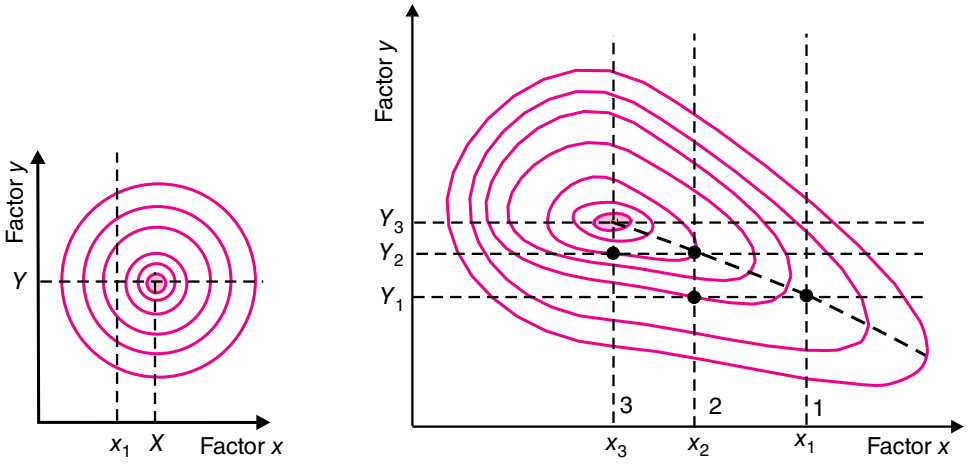


Figure 22.5 The one-factor-at-a-time method. If there is a continuous-response surface, the iso-response curves will lead to a variety of typical situations.

observed that the signal passes through a maximum. This *Y* value is then selected, and then *x* is varied in order to optimize its value, which is *X* (Figure 22.5). Generally, by repeating this procedure a new (*X*, *Y*) pair is found which gives slightly improved values each time.

This repetitive method does not always constitute the best approach to the problem. If the iso-response curves form a complex envelope, showing notably a ridge – the mathematical illustration of the interaction between the two factors – the above method can lead to a false optimum depending upon the parameters selected at the beginning. This different approach has the aim of discovering the optimal conditions using the minimum number of attempts. This is illustrated by the sequential simplex method of optimization and by different designs described in specialized textbooks.

PROBLEMS

1. With a column whose stationary phase is squalane and under differing conditions of use, the following values for the Kovats factor of benzene (recognized mean value = 653) have been determined: 650, 652, 648, 651, and 649.

Is the value of 653 significantly different from the other values calculated? (Consider a confidence level of 95%).

2. **a.** The true value of a measurement is 131.9 μg/l. Four chemists (A, B, C, and D) each repeat the same procedure six times. The individual values obtained are

Table P22.1

Chemist A	130.7	131.6	133.5	132.3	132.6	129.1
Chemist B	125.0	132.3	136.9	137.9	125.9	131.6
Chemist C	136.7	134.5	134.1	135.4	136.0	137.6
Chemist D	130.7	109.9	131.9	115.6	131.3	132.6

displayed in Table P22.1. Comment on these results in terms of accuracy and precision (use the standard deviation, *s*).

b. Let us suppose that two chemists A and B used two different apparatuses. To determine if the precision of these apparatuses is significantly different, apply a test of variance, *F*.

3. Once again return to the results found by chemists A and C in particular (see problem 22.2). If the true value is 131.9 μg/l, calculate and compare the respective values of Student's *t* for these chemists. Can the presence of a systematic error be concluded for the data of either A or C?

4. a. Results of experimental measurements repeated five times are as follows: 24.24, 24.36, 24.80, 24.20, and 24.10. Verify whether the third value, which seems to be high compared to the others, should be considered as a value out of acceptable range.

b. Now consider two complementary measurements: 24.12 and 24.25. Consider again the question posed above and reach a conclusion by applying the calculation of the standard deviation.

5. Measurements have been recorded and then repeated seven times for two different methods, giving means of 42 ($s = 0.3$) and 45 ($s = 0.2$).
Do these two methods yield results that are significantly different?

6. A compound is accompanied by a certificate indicating its purity as being 99% with $s = 0.08$, established over five analyses. Four control measurements performed upon the same compound have led to the following values: 98.58, 98.91, 98.62, and 98.80. Should the original value be rejected?

7. A series of standards used in atomic absorption leads to the following results:

Concentration μg/l	1	2.5	5	10	20	30
Absorbance	0.06	0.19	0.36	0.68	1.21	1.58

a. Can the linear regression method be used with these results?

b. If so, over what range of concentrations?

8. A series of measurements must be repeated five times. Eight successive values for the blank used in the analysis yield the following: 0.3, 0.75, –0.3, 1.5, –0.9, 1.8, 0.6, and 1.2.
Calculate the limit of detection by applying a 99% confidence level.

SOLUTIONS

1. The average value is 650 with $s = 1.581$. For the confidence level indicated, the value is $t = 2.776$. We can then calculate:

$$t.s / (n)^{1/2} = 1.963$$

The results yield a range of 650 ±1.963 in which we have a 95% chance of finding the true average. There is probably a systematic error in these experiments. However, if we set a confidence level of 99% ($t = 4.6$), we would have $s/(n)^{½} = 3.25$ and therefore a range of 650 ±3.25. The value of 653 would be included in this interval and would thus be considered as a viable result.

2. **a.**

Chemist	Average value (x)	Standard deviation (s)	$\in$	RSD%	Conclusion
Chemist A	131.6	1.56	0.3	1.18	Correct and precise
Chemist B	131.6	5.37	0.3	4.08	Correct yet imprecise
Chemist C	135.7	1.33	3.8	0.98	Incorrect yet precise
Chemist D	125.3	9.93	6.6	7.93	Incorrect and imprecise

b. $F = (s_1/s_2)^2 = 11.88$. For two series of measurements, the frontier value is 5.05. Therefore, the precision of the two apparatuses is significantly different.

3. The value of t calculated (with $n = 6$) for chemist A ($s = 1.559$) is 0.471. This value is smaller than those presented in the table of t values: 2.57 (for 95%), and 4.03 (for 99%). There is probably not a systematic error. However, for chemist C ($s = 1.325$), we still find $t = 4.05$, a value which indicates a high likelihood of a systematic error.

4. **a.** The value of $Q = (24.8 – 24.36) / (24.8 – 24.10) = 0.63$ is inferior to that found in the table for five measurements and a 95% confidence level (value 0.64). We should therefore not reject the value 24.8. However, if we recalculate following the addition of the two new values indicated, Q remains unchanged but in the table of seven values we have 0.51. The value 24.8 is now rejected.

b. The s and mean values are very different depending on whether these measurements are included ($s = 0.239$ and mean 24.29) or rejected ($s = 0.095$ and mean 24.21). If we take the median values, we have 24.24 (with) and 24.22 (without). In this case, the median seems preferable to the mean.

5. The pooled standard deviation s_p should be calculated, as follows:

$$s_p^2 = \frac{6 \times 0.3^2 + 6 \times 0.2^2}{14 - 2} = 0.2549^2 \qquad t = \frac{3}{0.2549}\sqrt{\frac{7^2}{14}} = 22$$

In the table $t = 2.2$, therefore the two methods do not produce the same result.

6. The problem is centred around the comparison of two averages. That for the first percentage purity is 99%, while the average value for the four analyses reported is 98.73 with s_{n-1}=0.155. The 'pooled standard deviation' of the two series of values is:

$$s_p = \{(4 \times 0.08^2 + 3 \times 0.155^2)/7\}^{1/2} = 0.118$$

The value of t based upon (5 + 4 – 2) degrees of freedom leads, according to the table, to 2.365 for a confidence level of 95%.

We must next calculate $2.365 \times 0.118 \times \{(4 + 5)/(4 \times 5)\}^{1/2} = 0.187$ and compare this value with the difference in the averages of the two series, 99 – 98.73 = 0.23. The result of this comparison appears rather large and therefore leads to the conclusion that the two averages can be considered to be incompatible. Thus the original value will not be retained.

7. If we consider that the law of variation of absorbance with concentration is a straight line, then this line will have the equation: $A = 0.05$ [conc] + 0.08. The differences are relatively large. A quadratic adjustment would be preferred and across a narrower range of concentrations.

8. For the blank, $s_{n-1} = 0.82$. The value of t calculated for 5 + 8 measurements is 3.17. Therefore $\Delta x = 3.17 \times 0.82 \times [(5 + 8)/(5 \times 8)]^{1/2} = 1.48$. The limit of the detection is around 1.5 mg.

Appendix
Table of Some Useful Constants

Some Physico-Chemical Constants		
Quantity	**Symbol**	**Value (SI units)**
Avogadro's number	*N*	$6.02252 \times 10^{23}\ mol^{-1}$
Molar volume of a perfect gas	V_0	$22.414 \times 10^{-3}\ m^3 \cdot mol^{-1}$
Perfect gas constant	R	$8.3143\ J \cdot K^{-1} \cdot mol^{-1}$
Boltzmann constant (R/N)	k	$1.3806 \times 10^{-23}\ J \cdot K^{-1}$
Planck constant	h	$6.6262 \times 10^{-34}\ J \cdot s$
Reduced Planck constant (h-bar)	$h/2\pi$	$1.0546 \times 10^{-34}\ J \cdot s$
Faraday constant	F	$96{,}487\ C \cdot mol^{-1}$
Speed of light in a vacuum	c	$2.997925 \times 10^{8}\ m \cdot s^{-1}$
Charge of an electron	e	$1.60210 \times 10^{-19}\ C$
Electron rest mass	m_e	$9.10953 \times 10^{-31}\ kg$
Neutron rest mass	m_n	$1.67496 \times 10^{-27}\ kg$
Proton rest mass	m_p	$1.67265 \times 10^{-27}\ kg$
Atomic mass constant	u (amu)	$1.660566 \times 10^{-27}\ kg$

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